CHAPTER

72

ENGINE

(LEAP-1B ENGINES)

BOEING

737-7/8/8200/9/10
AIRCRAFT MAINTENANCE MANUAL

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411	May 15/2024	
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72-21-08		
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72-21-09		
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72-21-10		
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72-21-11		
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O 639	Sep 15/2026	
O 640	Sep 15/2026	
72-22-00		
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807	May 15/2024	

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72-22-00 (cont)		
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R 628	Sep 15/2026	
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R 654	Sep 15/2026	
R 655	Sep 15/2026	

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72-22-02 (cont)		
O 656	Sep 15/2026	
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72-22-02 (cont)		
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804	May 15/2024	

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ENGINE

Subject/Page	Date	COC
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72-23-00		
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O 602	Sep 15/2026	
O 603	Sep 15/2026	
O 604	Sep 15/2026	
O 605	Sep 15/2026	
O 606	Sep 15/2026	
O 607	Sep 15/2026	

Subject/Page	Date	COC
72-23-00 (cont)		
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O 638	Sep 15/2026	
O 639	Sep 15/2026	
O 640	Sep 15/2026	
O 641	Sep 15/2026	
R 642	Sep 15/2026	
O 643	Sep 15/2026	

Subject/Page	Date	COC
72-23-00 (cont)		
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R 646	Sep 15/2026	
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679	Jan 15/2026	

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ENGINE

Subject/Page	Date	COC
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R 697	Sep 15/2026	
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R 698.2	Sep 15/2026	
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O 698.6	Sep 15/2026	
O 698.7	Sep 15/2026	
O 698.8	Sep 15/2026	
O 698.9	Sep 15/2026	
O 698.10	Sep 15/2026	
O 698.11	Sep 15/2026	
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72-23-00 (cont)		
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72-23-01		
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O 802	Sep 15/2026	

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O 804	Sep 15/2026	
O 805	Sep 15/2026	
O 806	Sep 15/2026	
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R 835	Sep 15/2026	
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R 837	Sep 15/2026	
R 838	Sep 15/2026	
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O 841	Sep 15/2026	
R 842	Sep 15/2026	
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O 602	Sep 15/2026	

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O 605	Sep 15/2026	
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72-23-04		
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O 804	Sep 15/2026	
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835	May 15/2024	
836	May 15/2024	
72-23-04 (cont)		
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838	May 15/2024	
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841	May 15/2024	
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843	May 15/2024	
R 844	Sep 15/2026	
O 845	Sep 15/2026	
O 846	Sep 15/2026	
O 847	Sep 15/2026	
R 848	Sep 15/2026	
R 849	Sep 15/2026	
R 850	Sep 15/2026	
R 851	Sep 15/2026	
R 852	Sep 15/2026	
R 853	Sep 15/2026	
R 854	Sep 15/2026	
R 855	Sep 15/2026	
R 856	Sep 15/2026	
O 857	Sep 15/2026	
O 858	Sep 15/2026	
R 859	Sep 15/2026	
O 860	Sep 15/2026	
O 861	Sep 15/2026	
R 862	Sep 15/2026	
R 863	Sep 15/2026	
R 864	Sep 15/2026	
R 865	Sep 15/2026	
O 866	Sep 15/2026	
O 867	Sep 15/2026	
O 868	Sep 15/2026	
O 869	Sep 15/2026	
R 870	Sep 15/2026	
R 871	Sep 15/2026	
R 872	Sep 15/2026	

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72-23-04 (cont)		
R 873	Sep 15/2026	
O 874	Sep 15/2026	
R 875	Sep 15/2026	
R 876	Sep 15/2026	
A 877	Sep 15/2026	
A 878	Sep 15/2026	
A 879	Sep 15/2026	
A 880	Sep 15/2026	
A 881	Sep 15/2026	
A 882	Sep 15/2026	
72-23-06		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
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413	Jan 15/2025	
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416	May 15/2024	
72-23-06		
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606	May 15/2024	
607	May 15/2024	
608	May 15/2024	
609	May 15/2024	

Subject/Page	Date	COC
72-23-06 (cont)		
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645	May 15/2024	

Subject/Page	Date	COC
72-23-06 (cont)		
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648	May 15/2024	
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72-23-06		
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807	May 15/2024	
808	May 15/2024	

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809	May 15/2024	
810	May 15/2024	
811	May 15/2024	
812	May 15/2024	
813	May 15/2024	
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844	May 15/2024	
72-23-06 (cont)		
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72-23-07		
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72-30-00		
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808	May 15/2024	
72-30-00 (cont)		
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R 821	Sep 15/2026	
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823	May 15/2024	
824	May 15/2024	
72-40-00		
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R 606	Sep 15/2026	
607	Sep 15/2024	
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72-50-00		
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606	May 15/2024	
607	May 15/2024	

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609	May 15/2024	
610	May 15/2024	
611	May 15/2024	
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72-50-02		
R 601	Sep 15/2026	
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R 603	Sep 15/2026	
R 604	Sep 15/2026	
R 605	Sep 15/2026	
R 606	Sep 15/2026	
O 607	Sep 15/2026	
R 608	Sep 15/2026	
R 609	Sep 15/2026	
R 610	Sep 15/2026	
R 611	Sep 15/2026	
R 612	Sep 15/2026	
R 613	Sep 15/2026	
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A 615	Sep 15/2026	
A 616	Sep 15/2026	
72-50-02		
A 801	Sep 15/2026	
A 802	Sep 15/2026	
A 803	Sep 15/2026	
A 804	Sep 15/2026	
A 805	Sep 15/2026	
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A 814	Sep 15/2026	
72-50-03		
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72-50-03 (cont)		
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72-59-01		
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608	May 15/2024	

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72-60-00 (cont)		
609	May 15/2024	
610	May 15/2024	
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72-62-00		
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822	May 15/2024	
823	May 15/2024	
72-62-00 (cont)		
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825	May 15/2024	
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827	May 15/2024	
828	May 15/2024	
72-62-03		
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72-62-04		
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409	May 15/2024	
410	May 15/2024	

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72-62-05		
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402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
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72-62-06		
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72-63-00		
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R 803	Sep 15/2026	
R 804	Sep 15/2026	
O 805	Sep 15/2026	
O 806	Sep 15/2026	
O 807	Sep 15/2026	
O 808	Sep 15/2026	
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O 810	Sep 15/2026	
O 811	Sep 15/2026	
O 812	Sep 15/2026	
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O 814	Sep 15/2026	
O 815	Sep 15/2026	
O 816	Sep 15/2026	
R 817	Sep 15/2026	

Subject/Page	Date	COC
72-63-00 (cont)		
O 818	Sep 15/2026	
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O 820	Sep 15/2026	
O 821	Sep 15/2026	
O 822	Sep 15/2026	
72-63-01		
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72-63-01		
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424	May 15/2025	

Subject/Page	Date	COC
72-63-01 (cont)		
425	May 15/2025	
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R 429	Sep 15/2026	
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72-63-03		
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72-63-04		
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72-63-04		
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72-63-05		
401	May 15/2025	
402	May 15/2025	

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403	May 15/2024	
404	May 15/2024	
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72-63-08		
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403	Sep 15/2024	
404	Sep 15/2024	
405	May 15/2024	
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72-64-01		
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403	May 15/2024	
72-64-01 (cont)		
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72-99-99		
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SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
ENGINE - MAINTENANCE PRACTICES	72-00-00		201	PTF ALL
High Pressure Rotor (N2) Turning for Borescope TASK 72-00-00-910-801-G00			201	PTF ALL
Zero-Index of the N2 Rotor Alignment TASK 72-00-00-822-801-G00			207	PTF ALL
Manually Turn the Fan N1 Blades TASK 72-00-00-910-802-G00			215	PTF ALL
ENGINE - INSPECTION/CHECK	72-00-00		601	PTF ALL
Continue-In-Service Limits TASK 72-00-00-800-801-G00			601	PTF ALL
Preparation for Borescope Inspection TASK 72-00-00-290-801-G00			602	PTF ALL
Booster Blades and Vanes Borescope Inspection TASK 72-00-00-290-803-G00			623	PTF ALL
High Pressure Compressor Section Borescope Detailed Inspection TASK 72-00-00-290-804-G00			640	PTF ALL
Combustion Section and High Pressure Turbine Section Stage One Nozzle Leading Edge Detailed Inspection TASK 72-00-00-290-805-G00			666	PTF ALL
High Pressure Turbine Section Detailed Inspection TASK 72-00-00-290-806-G00			698.5	PTF ALL
High Pressure Turbine (HPT) Stage 1 Nozzle Borescope Inspection TASK 72-00-00-290-807-G00			698.62	PTF ALL
High Pressure Turbine (HPT) Stage 1 Shroud Borescope Inspection TASK 72-00-00-290-808-G00			698.67	PTF ALL
High Pressure Turbine (HPT) Stage 2 Nozzle Borescope Inspection TASK 72-00-00-290-809-G00			698.74	PTF ALL
High Pressure Turbine (HPT) Stage 2 Shroud Borescope Inspection TASK 72-00-00-290-810-G00			698.82	PTF ALL

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SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
High Pressure Turbine (HPT) Blade Tip Material Loss using Tip Notches Inspection TASK 72-00-00-290-812-G00			698.88	PTF ALL
Low Pressure Turbine (LPT) Section Borescope Inspection TASK 72-00-00-290-811-G00			698.97	PTF ALL
High Pressure Turbine (HPT) Active Clearance Control (ACC) Air Manifolds Inspection TASK 72-00-00-210-810-G00			698.149	PTF ALL POST SB 737-CFM-LEAP-1B-75-0003
ENGINE - CLEANING/PAINTING	72-00-00		701	PTF ALL
Engine Cleaning Specifications TASK 72-00-00-100-801-G00			701	PTF ALL
Engine Gas-Path Cleaning With Pure Water TASK 72-00-00-100-802-G00			704	PTF ALL
Engine Gas-Path Cleaning With Cleaning Solution TASK 72-00-00-100-807-G00			716	PTF ALL
Engine External Surfaces Cleaning TASK 72-00-00-100-806-G00			728	PTF ALL
Exhaust Tube Support - Cleaning TASK 72-00-00-100-811-G00			729	PTF ALL
Injection Nozzle - Cleaning TASK 72-00-00-100-812-G00			736	PTF ALL
Exhaust Air Tube - Cleaning TASK 72-00-00-100-813-G00			738	PTF ALL
Outer Exhaust Air Tube - Cleaning TASK 72-00-00-100-814-G00			743	PTF ALL
Rear Ventilation Tube - Cleaning TASK 72-00-00-100-815-G00			749	PTF ALL
ENGINE - BSI PLUG D - REMOVAL/INSTALLATION	72-00-00		401	PTF ALL
BSI Plug D Removal TASK 72-00-00-000-801-G00			401	PTF ALL
BSI Plug D Installation TASK 72-00-00-400-801-G00			406	PTF ALL

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ENGINE - REPAIR	72-00-00		801	PTF ALL
Low Pressure Turbine (LPT) Blades Boro - Blending Repair TASK 72-00-00-350-801-G00			801	PTF ALL
FAN STATOR ASSEMBLY - INSPECTION/CHECK	72-21-00		601	PTF ALL
Fan Case Module Inspection TASK 72-21-00-200-802-G00			601	PTF ALL
Confirmed Air Leakage in the Fan Compartment Fan Case Color Inspection TASK 72-21-00-220-802-G00			661	PTF ALL
Low Pressure Compressor Stage 1 Inlet Guide Vane and Rotating Ring Spacer Inspection TASK 72-21-00-220-801-G00			669	PTF ALL
OUTLET GUIDE VANE - REMOVAL/INSTALLATION	72-21-03		401	PTF ALL
Outlet Guide Vane Removal TASK 72-21-03-000-801-G00			401	PTF ALL
Outlet Guide Vane Installation TASK 72-21-03-400-801-G00			423	PTF ALL
OUTLET GUIDE VANE - REPAIRS	72-21-03		801	PTF ALL
Aluminum Outlet Guide Vane Leading and Trailing Edge Blending Repair TASK 72-21-03-320-801-G00			801	PTF ALL
Titanium Outlet Guide Vane Leading Edge Blending Repair TASK 72-21-03-320-803-G00			808	PTF ALL
Outlet Guide Vane Struts 1 and 5 Leading Edge Blending Repair TASK 72-21-03-320-802-G00			814	PTF ALL
Outlet Guide Vanes Coating Local Repair TASK 72-21-03-370-801-G00			820	PTF ALL
Outlet Guide Vanes Paint Primer Local Repair TASK 72-21-03-370-802-G00			841	PTF ALL
Outlet Guide Vane Two-Lug Nut Plate/Assembly Gang Channel Nuts Replacement TASK 72-21-03-960-801-G00			856	PTF ALL

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LOW PRESSURE COMPRESSOR STAGE 1 INLET GUIDE VANE - REMOVAL/INSTALLATION	72-21-04		401	PTF ALL
Low Pressure Compressor Stage 1 Inlet Guide Vane Removal TASK 72-21-04-000-801-G00			401	PTF ALL
Low Pressure Compressor Stage 1 Inlet Guide Vane Installation TASK 72-21-04-400-801-G00			407	PTF ALL
FAN CASE FORWARD ACOUSTIC SHROUD - REMOVAL/INSTALLATION	72-21-05		401	PTF ALL
Fan Case Forward Acoustic Shroud Removal TASK 72-21-05-000-801-G00			401	PTF ALL
Fan Case Forward Acoustic Shroud Installation TASK 72-21-05-400-801-G00			406	PTF ALL
FAN CASE FORWARD ACOUSTIC SHROUD - REPAIR	72-21-05		801	PTF ALL
Acoustic Shroud Repair by Filling Up TASK 72-21-05-300-801-G00			801	PTF ALL
Acoustic Shroud Seal Replacement TASK 72-21-05-300-802-G00			808	PTF ALL
AFT FAN CASE HEAT SHIELDS - REMOVAL/INSTALLATION	72-21-07		401	PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0006
Aft Fan Case Heat Shield No. 1 Removal TASK 72-21-07-000-801-G00			401	PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0006
Aft Fan Case Heat Shield No. 1 Installation TASK 72-21-07-400-801-G00			406	PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0006
Aft Fan Case Heat Shield No. 2 Removal TASK 72-21-07-000-802-G00			407	PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0006
Aft Fan Case Heat Shield No. 2 Installation TASK 72-21-07-400-802-G00			412	PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0006
Aft Fan Case Heat Shield No. 3 Removal TASK 72-21-07-000-803-G00			413	PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0006

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Aft Fan Case Heat Shield No. 3 Installation TASK 72-21-07-400-803-G00			419	PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0006
Aft Fan Case Heat Shield No. 4 Removal TASK 72-21-07-000-804-G00			420	PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0006
Aft Fan Case Heat Shield No. 4 Installation TASK 72-21-07-400-804-G00			429	PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0006
Aft Fan Case Heat Shield No. 5 Removal TASK 72-21-07-000-805-G00			431	PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0006
Aft Fan Case Heat Shield No. 5 Installation TASK 72-21-07-400-805-G00			441	PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0006
Aft Fan Case Heat Shield No. 6 Removal TASK 72-21-07-000-806-G00			443	PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0006
Aft Fan Case Heat Shield No. 6 Installation TASK 72-21-07-400-806-G00			447	PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0006
Aft Fan Case Heat Shield No. 7 Removal TASK 72-21-07-000-807-G00			448	PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0006
Aft Fan Case Heat Shield No. 7 Installation TASK 72-21-07-400-807-G00			452	PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0006
Aft Fan Case Heat Shield No. 8 Removal TASK 72-21-07-000-808-G00			453	PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0006
Aft Fan Case Heat Shield No. 8 Installation TASK 72-21-07-400-808-G00			459	PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0006
AFT ACOUSTICAL PANEL - REMOVAL/INSTALLATION	72-21-08		401	PTF ALL
Aft Acoustical Panel Removal TASK 72-21-08-000-801-G00			401	PTF ALL
Aft Acoustical Panel Installation TASK 72-21-08-400-801-G00			409	PTF ALL

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AFT ACOUSTICAL PANEL - REPAIR	72-21-08		801	PTF ALL
Acoustic Panel Impact Repair by Filling Up TASK 72-21-08-300-801-G00			801	PTF ALL
OUTLET GUIDE VANE PLATFORM - REMOVAL/INSTALLATION	72-21-09		401	PTF ALL
Outlet Guide Vane Platform Removal TASK 72-21-09-000-801-G00			401	PTF ALL
Outlet Guide Vane Platform Installation TASK 72-21-09-400-801-G00			406	PTF ALL
OUTER OUTLET GUIDE VANE PLATFORM - DAMPER REPAIRS	72-21-09		801	PTF ALL
Outer Outlet Guide Vane Platform - Damper Replacement TASK 72-21-09-960-802-G00			801	PTF ALL
FAN DISK SHIELD - REMOVAL/INSTALLATION	72-21-10		401	PTF ALL
Fan Disk Shield Removal TASK 72-21-10-000-801-G00			401	PTF ALL
Fan Disk Shield Installation TASK 72-21-10-400-801-G00			404	PTF ALL
LOW PRESSURE COMPRESSOR ROTATING RING SPACER - REMOVAL/INSTALLATION	72-21-11		401	PTF ALL
Low Pressure Compressor (LPC) Rotating Ring Spacer Removal TASK 72-21-11-000-801-G00			401	PTF ALL
Low Pressure Compressor (LPC) Rotating Ring Spacer Installation TASK 72-21-11-400-801-G00			405	PTF ALL
FAN OVERHEAT SENSORS - REMOVAL/INSTALLATION	72-21-12		401	PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0178
Fan Overheat Sensors Removal TASK 72-21-12-000-801-G00			401	PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0178
Fan Overheat Sensors Installation TASK 72-21-12-400-801-G00			408	PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0178

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FAN OVERHEAT SENSORS - INSPECTION/CHECK	72-21-12		601	PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0178
Fan Overheat Sensor Inspection TASK 72-21-12-200-801-G00			601	PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0178
FAN BOOSTER ASSEMBLY - INSPECTION/CHECK	72-22-00		601	PTF ALL
Flowpath Fairing Inspection TASK 72-22-00-200-806-G00			601	PTF ALL
Shroud Segment Inspection TASK 72-22-00-200-807-G00			603	PTF ALL
Fan and Booster Module Inspection TASK 72-22-00-200-808-G00			605	PTF ALL
FAN BOOSTER ASSEMBLY - REPAIRS	72-22-00		801	PTF ALL
Platform Front Shroud Blend Repair TASK 72-22-00-300-801-G00			801	PTF ALL
Self-Locking Shank Nuts on the Platform Front Shroud Replacement TASK 72-22-00-900-801-G00			806	PTF ALL
FRONT SPINNER CONE - REMOVAL/INSTALLATION	72-22-01		401	PTF ALL
Front Spinner Cone Removal TASK 72-22-01-000-801-G00			401	PTF ALL
Front Spinner Cone Installation TASK 72-22-01-400-801-G00			408	PTF ALL
FRONT SPINNER CONE - INSPECTION/CHECK	72-22-01		601	PTF ALL
Front Spinner Cone Inspection TASK 72-22-01-200-801-G00			601	PTF ALL
FRONT SPINNER CONE - REPAIRS	72-22-01		801	PTF ALL
Front Spinner Cone Attachment Holes Blending and Touch-Up of Sulfuric Anodizing TASK 72-22-01-300-802-G00			801	PTF ALL
FAN BLADES - REMOVAL/INSTALLATION	72-22-02		401	PTF ALL
Fan Blades Removal (Selection) TASK 72-22-02-000-804-G00			401	PTF ALL

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Fan Blades Installation (Selection) TASK 72-22-02-400-804-G00			401	PTF ALL
Fan Blades (Pair) Removal TASK 72-22-02-000-802-G00			401	PTF ALL
Fan Blades (Pair) Installation TASK 72-22-02-400-802-G00			408	PTF ALL
Fan Blades (Set) Removal TASK 72-22-02-000-803-G00			410	PTF ALL
Fan Blades (Set) Installation TASK 72-22-02-400-803-G00			419	PTF ALL
Fan Blade Replacement TASK 72-22-02-960-802-G00			421	PTF ALL
FAN BLADES - INSPECTION/CHECK	72-22-02		601	PTF ALL
Fan Blade Inspection TASK 72-22-02-200-801-G00			601	PTF ALL
FAN BLADES - CLEANING/PAINTING	72-22-02		701	PTF ALL
Fan Blade Cleaning TASK 72-22-02-100-801-G00			701	PTF ALL
FAN BLADES - REPAIRS	72-22-02		801	PTF ALL
Fan Blade Coating Replacement (Local Repair of Anti-Erosion Paint) TASK 72-22-02-900-802-G00			801	PTF ALL
Fan Blade Coating Replacement (Local Repair of Platform Wear-Strips) TASK 72-22-02-900-803-G00			816	PTF ALL
Fan Blade Label Replacement TASK 72-22-02-900-804-G00			824	PTF ALL
Fan Blade Metal Leading Edge Blending TASK 72-22-02-900-805-G00			831	PTF ALL
Fan Blade Repair of the Missing Polyurethane (PU) Film with PU Paste TASK 72-22-02-300-801-G00			843	PTF ALL

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FAN BLADE PLATFORMS - REMOVAL/INSTALLATION	72-22-03		401	PTF ALL
Fan Blade Platforms - Removal TASK 72-22-03-000-801-G00			401	PTF ALL
Fan Blade Platforms - Installation TASK 72-22-03-400-801-G00			406	PTF ALL
FAN BLADE PLATFORMS - INSPECTION/CHECK	72-22-03		601	PTF ALL
Fan Blade Platforms Inspection TASK 72-22-03-200-801-G00			601	PTF ALL
FLOWPATH FAIRING - REMOVAL/INSTALLATION	72-22-05		401	PTF ALL
Flowpath Fairing Removal TASK 72-22-05-000-801-G00			401	PTF ALL
Flowpath Fairing Installation TASK 72-22-05-400-801-G00			409	PTF ALL
FLOWPATH FAIRING - INSPECTION/CHECK	72-22-05		601	PTF ALL
Flowpath Fairing Inspection TASK 72-22-05-200-801-G00			601	PTF ALL
Shroud Segment Inspection TASK 72-22-05-200-802-G00			607	PTF ALL
FLOWPATH FAIRING - REPAIR	72-22-05		801	PTF ALL
Flowpath Fairing Spot Touch-Up of Anodization TASK 72-22-05-300-801-G00			801	PTF ALL
SPLITTER FAIRING - REMOVAL/INSTALLATION	72-22-06		401	PTF ALL
Splitter Fairing Removal TASK 72-22-06-000-801-G00			401	PTF ALL
Splitter Fairing Installation TASK 72-22-06-400-801-G00			406	PTF ALL
SPLITTER FAIRING INSPECTION/CHECK	72-22-06		601	PTF ALL
Splitter Fairing Inspection TASK 72-22-06-200-801-G00			601	PTF ALL
BOOSTER ANTI-ICE DUCT - INSPECTION/CHECK	72-22-08		601	PTF ALL
Booster Anti-Ice Duct Inspection TASK 72-22-08-200-801-G00			601	PTF ALL

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LOW PRESSURE COMPRESSOR SHAFT - REPAIR	72-22-40		801	PTF ALL
Low Pressure Compressor (LPC) Shaft Paint Touch-up Repair TASK 72-22-40-300-801-G00			801	PTF ALL
FAN FRAME MODULE - INSPECTION/CHECK	72-23-00		601	PTF ALL
Fan Frame Module Inspection TASK 72-23-00-211-804-G00			601	PTF ALL
Fan Frame AGB Fittings and Axial Mount Inspection TASK 72-23-00-211-805-G00			698.44	PTF ALL
FAN FRAME - REPAIRS	72-23-00		801	PTF ALL
Strut 1 Outer Fitting - Blending TASK 72-23-00-300-801-G00			801	PTF ALL
Strut 5 Outer Fitting - Blending TASK 72-23-00-300-802-G00			810	PTF ALL
Fan Frame Hub Inserts Replacement TASK 72-23-00-900-801-G00			819	PTF ALL
FAN CASE - REPAIR	72-23-01		801	PTF ALL
Fan Case Abradable Coating Repair TASK 72-23-01-300-801-G00			801	PTF ALL
Fan Case Acoustic Shroud Bracket and/or Attachment Parts Replacement TASK 72-23-01-960-801-G00			810	PTF ALL
Fan Case Counterplates Replacement TASK 72-23-01-960-802-G00			817	PTF ALL
Fan Case Impact Repair on the Trench Filler TASK 72-23-01-960-803-G00			825	PTF ALL
Fan Case Rivetless Nut Plate Assembly Replacement TASK 72-23-01-960-804-G00			838	PTF ALL
Fan Case Fireproof Layer Local Repair TASK 72-23-01-300-802-G00			844	PTF ALL
Fan Case Abradable Liner Cracks Local Repair TASK 72-23-01-300-803-G00			860	PTF ALL

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FAN FRAME MODULE SEGMENT SHROUDS - REMOVAL/INSTALLATION	72-23-04		401	PTF ALL
Fan Frame Module Segment Shrouds Removal TASK 72-23-04-000-801-G00			401	PTF ALL
Fan Frame Module Segment Shrouds Installation TASK 72-23-04-400-801-G00			428	PTF ALL
FAN FRAME MODULE SEGMENT SHROUDS - INSPECTION/CHECK	72-23-04		601	PTF ALL
Fan Frame Module Segment Shrouds Inspection TASK 72-23-04-200-801-G00			601	PTF ALL
FAN FRAME SHROUD - REPAIRS	72-23-04		801	PTF ALL
Fan Frame Shroud - Key-locked Inserts Replacement TASK 72-23-04-300-801-G00			801	PTF ALL
Fan Frame Shroud Repair (PTFE Lubricant Varnish in the V-groove) TASK 72-23-04-300-802-G00			809	PTF ALL
Fan Frame Shroud Fittings Seal - Replacement TASK 72-23-04-960-801-G00			818	PTF ALL
Fan Frame Shroud Segments 1-5 Blending TASK 72-23-04-320-801-G00			827	PTF ALL
Retainer Ring Nut - Replacement TASK 72-23-04-960-802-G00			844	PTF ALL; AIRPLANES PRE SB-CFM-LEAP-1B-72-0374
Retainer Ring Riveted Nuts - Replacement TASK 72-23-04-960-807-G00			850	PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0374
Fan Frame Shroud Segment Coating Replacement TASK 72-23-04-960-803-G00			856	PTF ALL
Variable Bleed Valve (VBV) Duct Nut Replacement TASK 72-23-04-960-804-G00			865	PTF ALL
Shroud Junction Fitting Nut - Replacement TASK 72-23-04-960-805-G00			871	PTF ALL
Shroud Junction Fitting - Nut Plate Two Lugs - Replacement TASK 72-23-04-960-806-G00			878	PTF ALL

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FAN FRAME STRUT PANELS - REMOVAL/INSTALLATION	72-23-06		401	PTF ALL
Fan Frame Strut Panels Removal TASK 72-23-06-000-801-G00			401	PTF ALL
Fan Frame Strut Panels Installation TASK 72-23-06-400-801-G00			412	PTF ALL
FAN FRAME STRUT PANELS - INSPECTION/CHECK	72-23-06		601	PTF ALL
Fan Frame Strut Panels Inspection TASK 72-23-06-200-801-G00			601	PTF ALL
FAN FRAME STRUT PANELS - REPAIR	72-23-06		801	PTF ALL
Fan Frame Strut 1 Fairing Seal Replacement TASK 72-23-06-960-801-G00			801	PTF ALL
Fan Frame Strut Pad Seal Replacement TASK 72-23-06-960-802-G00			810	PTF ALL
Fan Frame Strut 1 and 5 Fairings Blending TASK 72-23-06-300-801-G00			819	PTF ALL
Fan Frame Strut Pad Blending TASK 72-23-06-300-802-G00			833	PTF ALL
Fan Frame Strut Pad Nut Replacement TASK 72-23-06-960-803-G00			845	PTF ALL
FAN FRAME BIFI SUPPORT – INSPECTION/CHECK	72-23-07		601	PTF ALL
BIFI Support Inspection TASK 72-23-07-200-801-G00			601	PTF ALL
HIGH PRESSURE COMPRESSOR SECTION - INSPECTION/CHECK	72-30-00		601	PTF ALL
High Pressure Compressor (HPC) Section Inspection TASK 72-30-00-200-801-G00			601	PTF ALL
HIGH PRESSURE COMPRESSOR SECTION - REPAIR	72-30-00		801	PTF ALL
High Pressure Compressor (HPC) Rotor Blades/Blisks Borescope Blending Repair TASK 72-30-00-350-801-G00			801	PTF ALL

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COMBUSTION CASE - INSPECTION/CHECK	72-40-00		601	PTF ALL
Combustion Case Inspection TASK 72-40-00-200-801-G00			601	PTF ALL
TURBINE SECTION - INSPECTION/CHECK	72-50-00		601	PTF ALL
High Pressure Turbine (HPT) Section Inspection TASK 72-50-00-200-801-G00			601	PTF ALL
Turbine Center Frame (TCF) Aft Engine Mount Lugs (Scheduled Maintenance Requirement) Inspection TASK 72-50-00-200-802-G00			605	PTF ALL
Turbine Center Frame (TCF) Section Inspection TASK 72-50-00-200-804-G00			608	PTF ALL
LOW PRESSURE TURBINE (LPT) ROTOR AND STATOR MODULE - INSPECTION/CHECK	72-50-02		601	PTF ALL
Low Pressure Turbine (LPT) Rotor and Stator Module Inspection TASK 72-50-02-200-801-G00			601	PTF ALL
LPT COOLING MANIFOLD ASSEMBLY LOOP CLAMP AND BRACKET - REPAIR	72-50-02		801	PTF ALL
LPT Cooling Manifold Assembly Loop Clamp and Bracket - Replacement TASK 72-50-02-960-801-G00			801	PTF ALL
TURBINE REAR VANE MODULE-INSPECTION/CHECK	72-50-03		601	PTF ALL
Turbine Rear Vane Module Inspection TASK 72-50-03-200-801-G00			601	PTF ALL
INJECTION NOZZLE - REMOVAL/INSTALLATION	72-59-00		401	PTF ALL
Injection Nozzle Removal TASK 72-59-00-000-801-G00			401	PTF ALL
Injection Nozzle Installation TASK 72-59-00-400-801-G00			405	PTF ALL
INTERNAL EXHAUST AIR TUBE - REMOVAL/INSTALLATION	72-59-01		401	PTF ALL
Internal Exhaust Air Tube Removal TASK 72-59-01-000-801-G00			401	PTF ALL

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Internal Exhaust Air Tube Installation TASK 72-59-01-400-801-G00			411	PTF ALL
ACCESSORY DRIVES - INSPECTION/CHECK	72-60-00		601	PTF ALL
Accessory Drives Inspection TASK 72-60-00-200-801-G00			601	PTF ALL
Accessory Drives Damper Bearings Inspection TASK 72-60-00-200-803-G00			650	PTF ALL
TRANSFER GEARBOX MODULE - REPAIR	72-62-00		801	PTF ALL
Blending and Spot Touch-Up of the Transfer Gearbox Module TASK 72-62-00-300-802-G00			801	PTF ALL
Blending and Spot Touch Up on Specific Area of the Transfer Gearbox Assembly TASK 72-62-00-300-803-G00			814	PTF ALL
Transfer Gearbox Assembly Corrosion Removal TASK 72-62-00-300-804-G00			821	PTF ALL
TRANSFER GEARBOX ASSEMBLY - REMOVAL/INSTALLATION	72-62-03		401	PTF ALL
Transfer Gearbox Assembly Removal TASK 72-62-03-000-801-G00			401	PTF ALL
Transfer Gearbox Assembly Installation TASK 72-62-03-400-801-G00			416	PTF ALL
TRANSFER SHAFT - REMOVAL/INSTALLATION	72-62-04		401	PTF ALL
Transfer Shaft Removal TASK 72-62-04-000-801-G00			401	PTF ALL
Transfer Shaft Installation TASK 72-62-04-400-801-G00			407	PTF ALL
TRANSFER GEARBOX SLIDER - REMOVAL/INSTALLATION	72-62-05		401	PTF ALL
Transfer Gearbox Slider Removal TASK 72-62-05-000-801-G00			401	PTF ALL
Transfer Gearbox Slider Installation TASK 72-62-05-400-801-G00			406	PTF ALL

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RADIAL SHAFT - REMOVAL/INSTALLATION	72-62-06		401	PTF ALL
Radial Shaft Removal TASK 72-62-06-000-801-G00			401	PTF ALL
Radial Shaft Installation TASK 72-62-06-400-801-G00			407	PTF ALL
ACCESSORY GEARBOX MODULE - REPAIR	72-63-00		801	PTF ALL
Blending and Spot Touch-Up of the Accessory Gearbox Module TASK 72-63-00-300-802-G00			801	PTF ALL
ACCESSORY GEARBOX ASSEMBLY - MAINTENANCE PRACTICES	72-63-01		201	PTF ALL
Accessory Gearbox Assembly Oil Replenishing TASK 72-63-01-610-802-G00			201	PTF ALL
ACCESSORY GEARBOX ASSEMBLY - REMOVAL/INSTALLATION	72-63-01		401	PTF ALL
Accessory Gearbox Assembly Removal TASK 72-63-01-000-801-G00			401	PTF ALL
Accessory Gearbox Assembly Installation TASK 72-63-01-400-801-G00			423	PTF ALL
MANUAL CONTROL COVER - REPAIR	72-63-03		801	PTF ALL
Blending and Spot Touch-Up of the Manual Control Cover TASK 72-63-03-300-802-G00			801	PTF ALL
ACCESSORY GEARBOX MANUAL CONTROL COVER - REMOVAL/INSTALLATION	72-63-04		401	PTF ALL
Accessory Gearbox Manual Control Cover Removal TASK 72-63-04-000-801-G00			401	PTF ALL
Accessory Gearbox Manual Control Cover Installation TASK 72-63-04-400-801-G00			404	PTF ALL
ACCESSORY GEARBOX MANUAL CONTROL COVER - INSPECTION/CHECK	72-63-04		601	PTF ALL
Accessory Gearbox Manual Control Inspection TASK 72-63-04-210-801-G00			601	PTF ALL

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CHAPTER 72
ENGINE

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
ACCESSORY GEARBOX MANUAL CONTROL COVER - CLEANING/PAINTING	72-63-04		701	PTF ALL
Accessory Gearbox Manual Control Cover Cleaning TASK 72-63-04-100-801-G00			701	PTF ALL
ACCESSORY GEARBOX LINK - REMOVAL/INSTALLATION	72-63-05		401	PTF ALL
Accessory Gearbox Link Removal TASK 72-63-05-000-801-G00			401	PTF ALL
Accessory Gearbox Link Installation TASK 72-63-05-400-801-G00			409	PTF ALL
ACCESSORY GEARBOX LINK - INSPECTION/CHECK	72-63-05		601	PTF ALL
Accessory Gearbox Links Visual Inspection TASK 72-63-05-200-801-G00			601	PTF ALL
ACCESSORY GEARBOX OIL PRESSURE TEMPERATURE SENSOR SUPPORT ASSEMBLY - REMOVAL/INSTALLATION	72-63-07		401	PTF ALL
Accessory Gearbox Oil Pressure Temperature Sensor Support Assembly Removal TASK 72-63-07-000-801-G00			401	PTF ALL
Accessory Gearbox Oil Pressure Temperature Sensor Support Assembly Installation TASK 72-63-07-400-801-G00			404	PTF ALL
ACCESSORY GEARBOX COIL ASSEMBLY - REMOVAL/INSTALLATION	72-63-08		401	PTF ALL
Accessory Gearbox Coil Assembly Removal TASK 72-63-08-000-801-G00			401	PTF ALL
Accessory Gearbox Coil Assembly Installation TASK 72-63-08-400-801-G00			405	PTF ALL
ACCESSORY GEARBOX ROTATING SEALS - REMOVAL/INSTALLATION	72-64-01		401	PTF ALL
Accessory Gearbox Rotating Seals Removal TASK 72-64-01-000-801-G00			401	PTF ALL
Accessory Gearbox Rotating-Seals Installation TASK 72-64-01-400-801-G00			412	PTF ALL

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ECCN 9E991 BOEING PROPRIETARY - See title page for details

CHAPTER 72
ENGINE

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
ENGINE MODULES - REMOVAL/INSTALLATION	72-99-99		401	PTF ALL
Fan Disk and Booster Spool (Scheduled Maintenance Requirement) Replacement TASK 72-99-99-960-801-G00			401	PTF ALL
Low Pressure Compressor (LPC) Shaft Assembly (Scheduled Maintenance Requirement) Replacement TASK 72-99-99-960-809-G00			401	PTF ALL
HPC Rotor Stage 1 Blisk; Stage 2 Blisk; Stage 3-4 Blisk; Stage 5 Blisk; 6-10 Spool; and Rotating Seal (Scheduled Maintenance Requirement) Replacement TASK 72-99-99-960-803-G00			402	PTF ALL
HPT Rotor Stage 1 Disk; HPT Rotor Stage 2 Disk; the Forward Outer Seal; Mid Seal; and the Aft Rotating Seal (Scheduled Maintenance Requirement) Replacement TASK 72-99-99-960-804-G00			402	PTF ALL
LPT Disks 1-5, LPT Stage 4 Stubshaft Disk and LPT Shaft (Scheduled Maintenance Requirement) Replacement TASK 72-99-99-960-811-G00			403	PTF ALL
HPC Impeller Tube Support (Scheduled Maintenance Requirement) Replacement TASK 72-99-99-960-810-G00			404	PTF ALL
Combustion Case Assembly (Scheduled Maintenance Requirement) Replacement TASK 72-99-99-960-807-G00			404	PTF ALL
High Pressure Turbine Case (Scheduled Maintenance Requirement) Replacement TASK 72-99-99-960-808-G00			404	PTF ALL

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ENGINE - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) High Pressure Rotor (N2) Turning for Borescope

(2) Zero-index of the N2 rotor alignment

(3) Manually turn the fan N1 blades.

TASK 72-00-00-910-801-G00

2. High Pressure Rotor (N2) Turning for Borescope

(Figure 201)

A. General

(1) This task provides the instructions on how to turn the High Pressure Compressor (HPC) rotor with these tools:

(a) Hand operated drive tool

(b) Electronic turning tool.

(2) The HPC and HPT rotor is also referred to as the N2 rotor.

(3) For this procedure the accessory gearbox will be referred to as the AGB.

B. References

Reference	Title
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
72-63-04-000-801-G00	Accessory Gearbox Manual Control Cover Removal (P/B 401)
72-63-04-400-801-G00	Accessory Gearbox Manual Control Cover Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-14866	Turning Tool - Electronic Part #: 9481M92P05 Supplier: 58828 Part #: 9481M92P06 Supplier: 58828 Opt Part #: 9481M92P03 Supplier: 58828
SPL-14867	Adapter - Offset Crank Pad Part #: 9471M47P01 Supplier: 58828
SPL-14868	Adapter - Core Turning Part #: 9471M48P01 Supplier: 58828
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

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E. Prepare for the High Pressure Compressor Rotor Manual Motoring

SUBTASK 72-00-00-860-001-G00

(1) For the left engine, open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

SUBTASK 72-00-00-860-002-G00

(2) For the right engine, open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 72-00-00-860-003-G00

(3) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-00-00-010-001-G00

(4) Do this task: Accessory Gearbox Manual Control Cover Removal, TASK 72-63-04-000-801-G00.

F. High Pressure Rotor (N2) Turning for Borescope

SUBTASK 72-00-00-480-013-G00

(1) If you will turn the N2 rotor manually with a hand tool, do the steps that follow:

NOTE: You can also use the electronic turning tool to turn the N2 rotor. See the applicable steps below.

(a) Make sure that the tool is clean of foreign objects or dirt before you insert it in the AGB manual control pad.

CAUTION

DO NOT USE A TOOL TO PUT THE 3/4-INCH SQUARE DRIVE INTO THE GEAR SHAFT. THIS CAN CAUSE DAMAGE TO THE AGB BEARING.

(b) Attach a 3/4 inch square drive hand tool to the manual control pad.

(c) Turn the handle of the tool in the clockwise direction as seen from forward looking aft.

(d) After you complete the task to turn the N2 rotor, remove the 3/4 inch square drive tool from the AGB manual control pad.

SUBTASK 72-00-00-480-014-G00

(2) If you will use the electronic turning tool, SPL-14866, do these steps:

(a) Make sure that the core turning adapter, SPL-14868 is fully clean of foreign objects or dirt before you insert it into the AGB manual control pad.

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CAUTION

DO NOT USE A TOOL TO PUT THE 3/4-INCH SQUARE DRIVE INTO THE GEAR SHAFT. THIS CAN CAUSE DAMAGE TO THE AGB BEARING.

(b) Install the core turning adapter, SPL-14868 on the AGB manual control pad.

NOTE: Attach the core turn adapter to the Accessory Gearbox (AGB) with the same bolts that attached the AGB manual control cover.

(c) Attach the offset adapter, SPL-14867 to the core turning adapter, SPL-14868.

(d) Use the manufacturer's assembly instruction to set up the turnover kit for use.

NOTE: Make sure that the motor programming software (9471M90P01) is loaded into the electronic turning tool per the manufacturer's instructions.

(e) Install the electronic turning tool, SPL-14866 to the offset adapter, SPL-14867.

(f) Set the motor tool display of the electronic turning tool, SPL-14866 to zero degrees.

NOTE: Use the manufacturer supplied instruction to set the motor tool display to zero degrees.

CAUTION

DO NOT USE THE DRIVE MOTOR AS A BRAKE TO STOP ENGINE MOVEMENT. THE REVERSAL OF THE DRIVE MOTOR WHILE THE ENGINE TURNS CAN BREAK THE TORQUE LIMIT DEVICE OF THE DRIVE.

(g) Operate the drive motor in a clockwise direction as seen from forward looking aft.

(h) After you complete the task to turn the N2 rotor, remove the electronic turning tool, SPL-14866 as follows:

1) Remove the electronic turning tool, SPL-14866 from the offset adapter, SPL-14867.

NOTE: Follow the manufacturer's instructions to remove the electronic turning tool from the offset adapter and store the tool kit.

2) Remove the offset adapter, SPL-14867 from the core turning adapter, SPL-14868.

3) Remove the core turning adapter, SPL-14868 from the AGB manual control pad.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 72-00-00-410-001-G00

CAUTION

MAKE SURE THAT YOU INSTALL THE COVER ON THE GEARBOX. IF YOU DO NOT INSTALL THE COVER, ENGINE OIL LEAKS CAN OCCUR DURING OPERATION AND ENGINE DAMAGE CAN OCCUR.

(1) Do this task: Accessory Gearbox Manual Control Cover Installation, TASK 72-63-04-400-801-G00.

SUBTASK 72-00-00-860-004-G00

(2) For the left engine, remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

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SUBTASK 72-00-00-860-005-G00

(3) For the right engine, remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 72-00-00-860-006-G00

(4) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

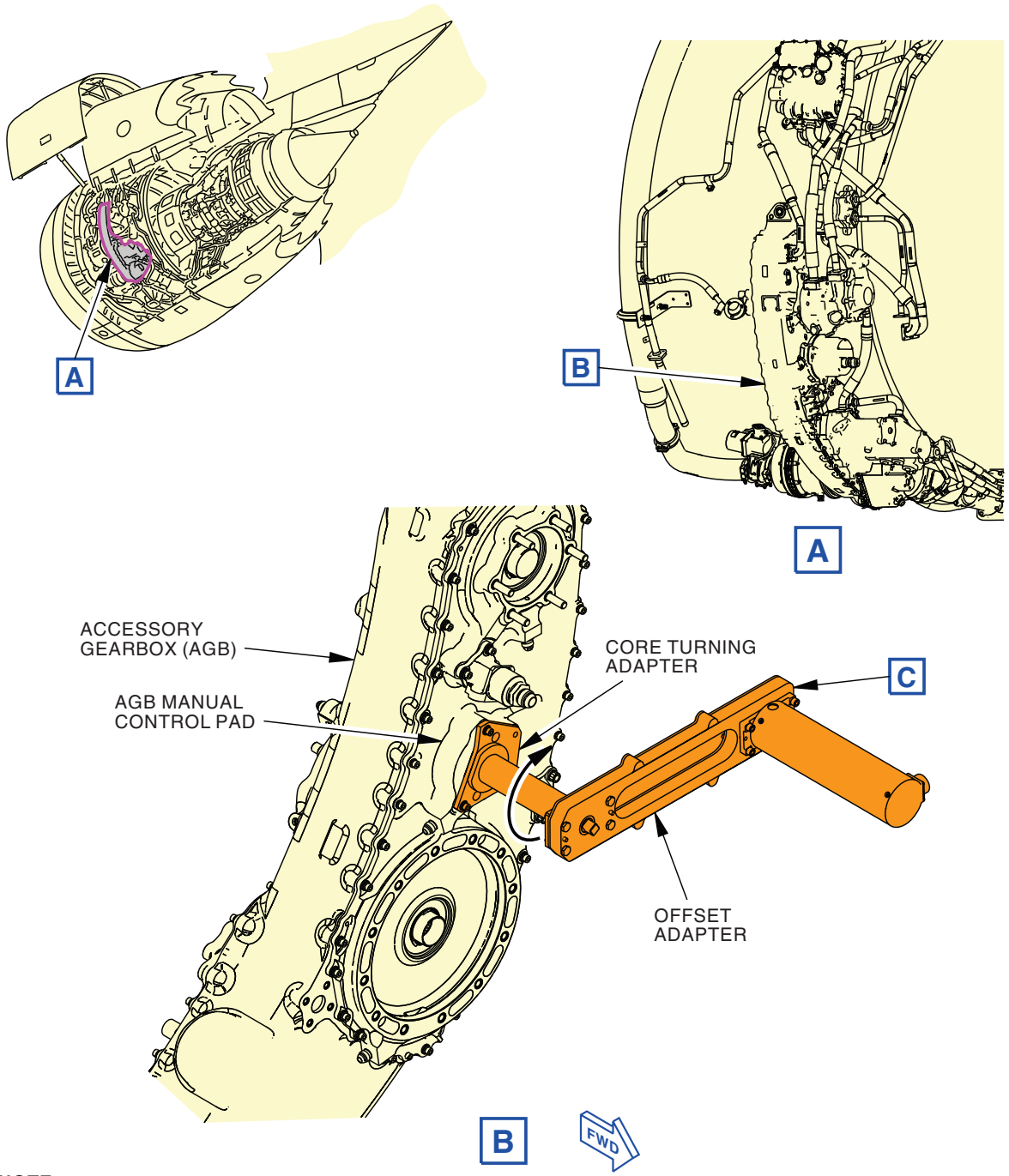
H. High Pressure Rotor (N2) Turning for Borescope Test

SUBTASK 72-00-00-700-002-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOTE:

A HAND OPERATED TOOL CAN BE USED AS AN ALTERNATIVE TOOL FOR THE ELECTRONIC TURNING TOOL.

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N2 Core Turning Tools
Figure 201/72-00-00-990-855-G00 (Sheet 1 of 2)

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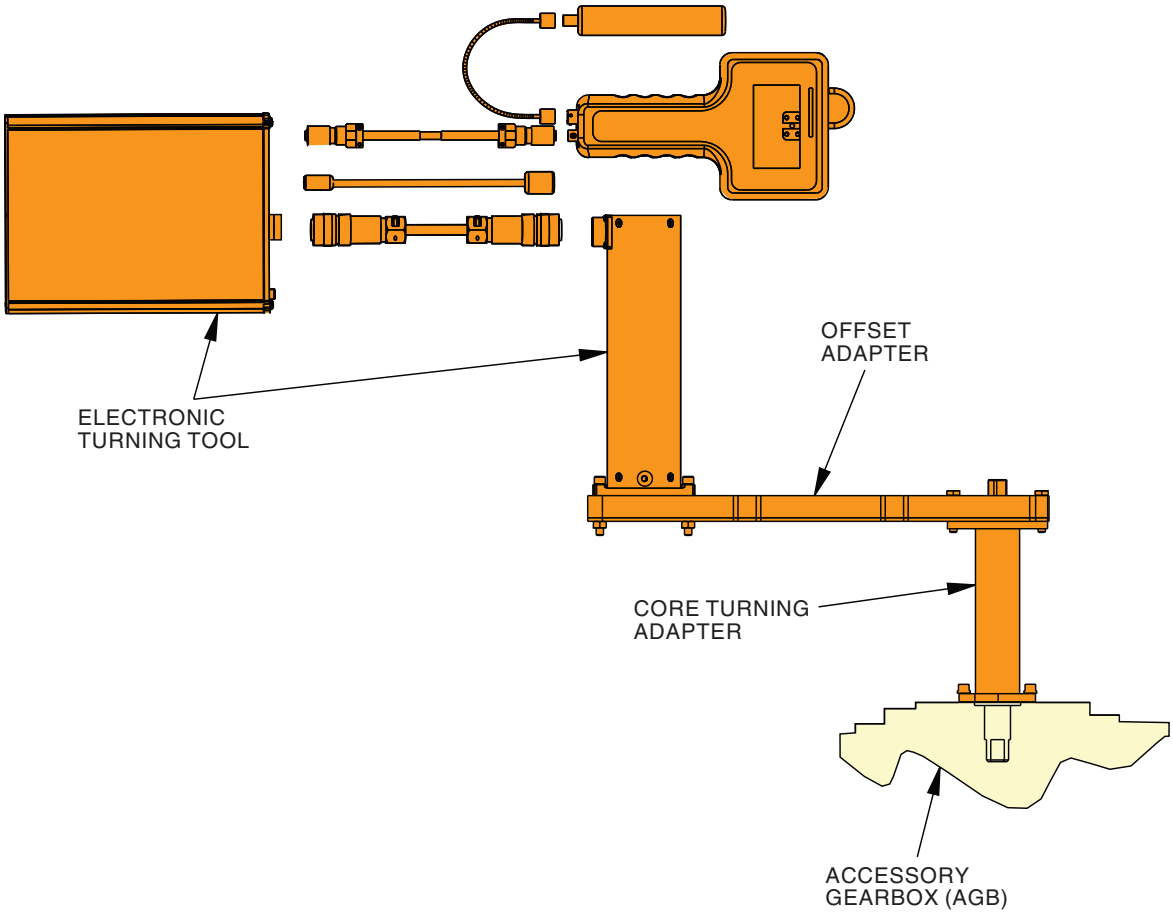

ELECTRONIC TURNING TOOL

C

2485101 S0000584877_V2

N2 Core Turning Tools
Figure 201/72-00-00-990-855-G00 (Sheet 2 of 2)

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TASK 72-00-00-822-801-G00

3. Zero-Index of the N2 Rotor Alignment

(Figure 202, Figure 203, Figure 204)

A. General

(1) This task gives the instructions to align for zero indexing the High Pressure Compressor (HPC) rotor.

(2) You do this task before you can examine each stage of the HPC rotor and High Pressure Turbine (HPT) rotor.

(3) The inspection of the HPC rotor and HPT rotor must start with the HPC rotor at the zero-index position.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
72-00-00-290-801-G00	Preparation for Borescope Inspection (P/B 601)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-14866	Turning Tool - Electronic Part #: 9481M92P05 Supplier: 58828 Part #: 9481M92P06 Supplier: 58828 Opt Part #: 9481M92P03 Supplier: 58828
STD-858	Tag - DO NOT OPERATE
STD-10717	Light Source - Borescope, (Meets General Electric or CFM Standard Practices Manual Specification)
STD-10719	Borescope - Rigid (Meets General Electric or CFM Standard Practices Manual Specification such as: model R857ELSV60)

D. Consumable Materials

Reference	Description	Specification
D50355 [CP2961]	Lubricant - Anti-Seize Compound	D6Y31C2

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1

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(Continued)

Number	Name/Location
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Prepare to Zero-Index the N2 Rotor Alignment

SUBTASK 72-00-00-860-007-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-00-00-010-002-G00

(2) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1

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(Continued)

Number	Name/Location
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

SUBTASK 72-00-00-980-007-G00

(3) Do this task: High Pressure Rotor (N2) Turning for Borescope, TASK 72-00-00-910-801-G00.

H. Zero-Index of the N2 Rotor Alignment

SUBTASK 72-00-00-480-001-G00

(1) Install the electronic turning tool, SPL-14866 (High Pressure Rotor (N2) Turning for Borescope, TASK 72-00-00-910-801-G00).

SUBTASK 72-00-00-010-003-G00

(2) Remove the plug from borescope port L (TASK 72-00-00-290-801-G00).

SUBTASK 72-00-00-820-001-G00

(3) Identify the zero-index point for the HPC rotor:

(a) Turn the light source, STD-10717 to ON.

(b) Insert the rigid borescope, STD-10719 into borescope port L.

(c) Use the borescope to view the rotor blade platform for the 9th stage HPC rotor.

NOTE: The rotor blade platform for the 9th-stage HPC rotor is aft of borescope port L.

(d) Slowly turn the electronic turning tool, SPL-14866 in a clockwise direction to turn the HPC rotor clockwise as seen from forward looking aft.

(e) Look through the rigid borescope, STD-10719 until you see the first locking lug come into view.

(f) Very slowly continue to turn the HPC rotor until you see the second locking lug.

NOTE: The second locking lug is two blades away from the first locking lug.

(g) Align the leading edge of the blade that follows the second blade locking lug with the leading edge of the 9th-stage compressor stator vane.

NOTE: This will be blade No. 1.

(h) If you will use the manual motoring procedure to identify the zero-index point, do this step:

1) Count the number of 9th-stage blades to know when a full turn of HPC rotor is complete.

NOTE: The 9th-stage of the HPC rotor has 64 blades. This includes the 4 locking blades.

(i) If you will use the rotor drive motoring procedure to identify the zero-index point, do this step:

1) Use the manufacturer instructions to set the motoring tool display to zero degrees.

SUBTASK 72-00-00-010-004-G00

(4) Remove the rigid borescope, STD-10719 from borescope port L.

NOTE: All stages of the HPC rotor and the HPT rotor are now zero-indexed in relation to each other.

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SUBTASK 72-00-00-080-002-G00

(5) After the zero-index identification is complete, turn the light source, STD-10717 to OFF and do this step:

(a) Let the blower stay ON for a sufficient time to cool the lamp and case.

SUBTASK 72-00-00-410-002-G00

(6) Install the borescope plug L:

(a) Do a visual inspection of the threads on the borescope port (TASK 72-00-00-290-801-G00).

(b) If it is necessary, repair minor damage to the threads.

NOTE: Minor damage to the threads include burrs and galling. For thread information, refer to Figure 602 in the Preparation for Borescope Inspection, TASK 72-00-00-290-801-G00. If you repair the threads, make sure that no debris enter the engine.

(c) Apply Tiodize T8E-H grease, D50355 [CP2961] on the threads of the inner and outer borescope plugs.

(d) Tighten the inner borescope plug to 76.0 in-lb (8.6 N·m) to 84.0 in-lb (9.5 N·m).

(e) Loosen inner borescope plug one quarter turn.

(f) Tighten the inner borescope plug again to 17.0 in-lb (1.9 N·m) to 19.0 in-lb (2.1 N·m).

(g) Compress the spring and tighten the outer borescope plug to 195.0 in-lb (22.0 N·m) to 200.0 in-lb (22.6 N·m).

SUBTASK 72-00-00-080-003-G00

(7) Remove the manual motoring adapter or rotor drive motoring tool.

SUBTASK 72-00-00-410-003-G00

(8) Install the motoring pad cover (TASK 72-00-00-910-801-G00).

I. Put the Airplane Back to Its Usual Condition

SUBTASK 72-00-00-410-004-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 72-00-00-860-008-G00

(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

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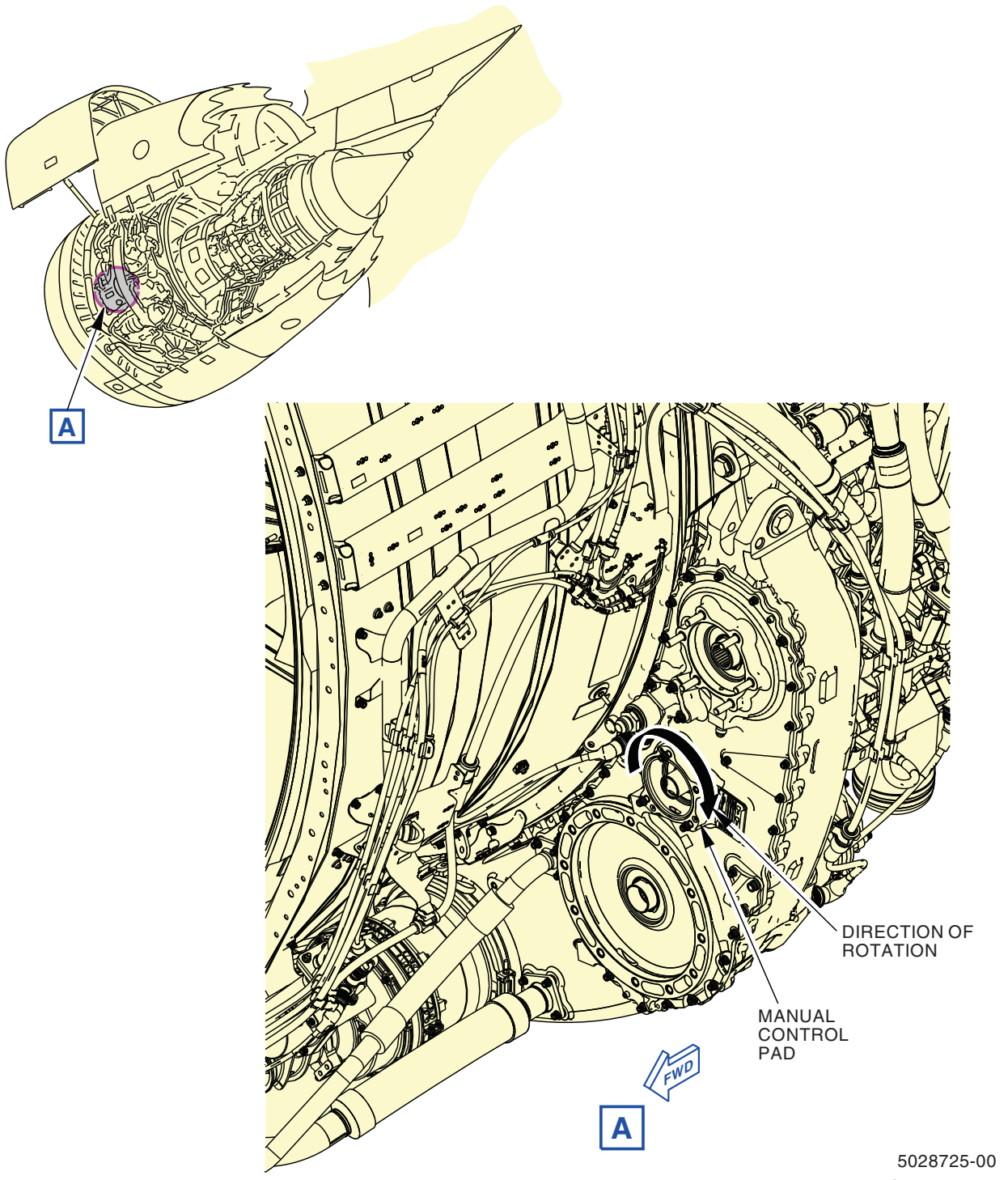

AGB Crank Pad Location
Figure 202/72-00-00-990-856-G00

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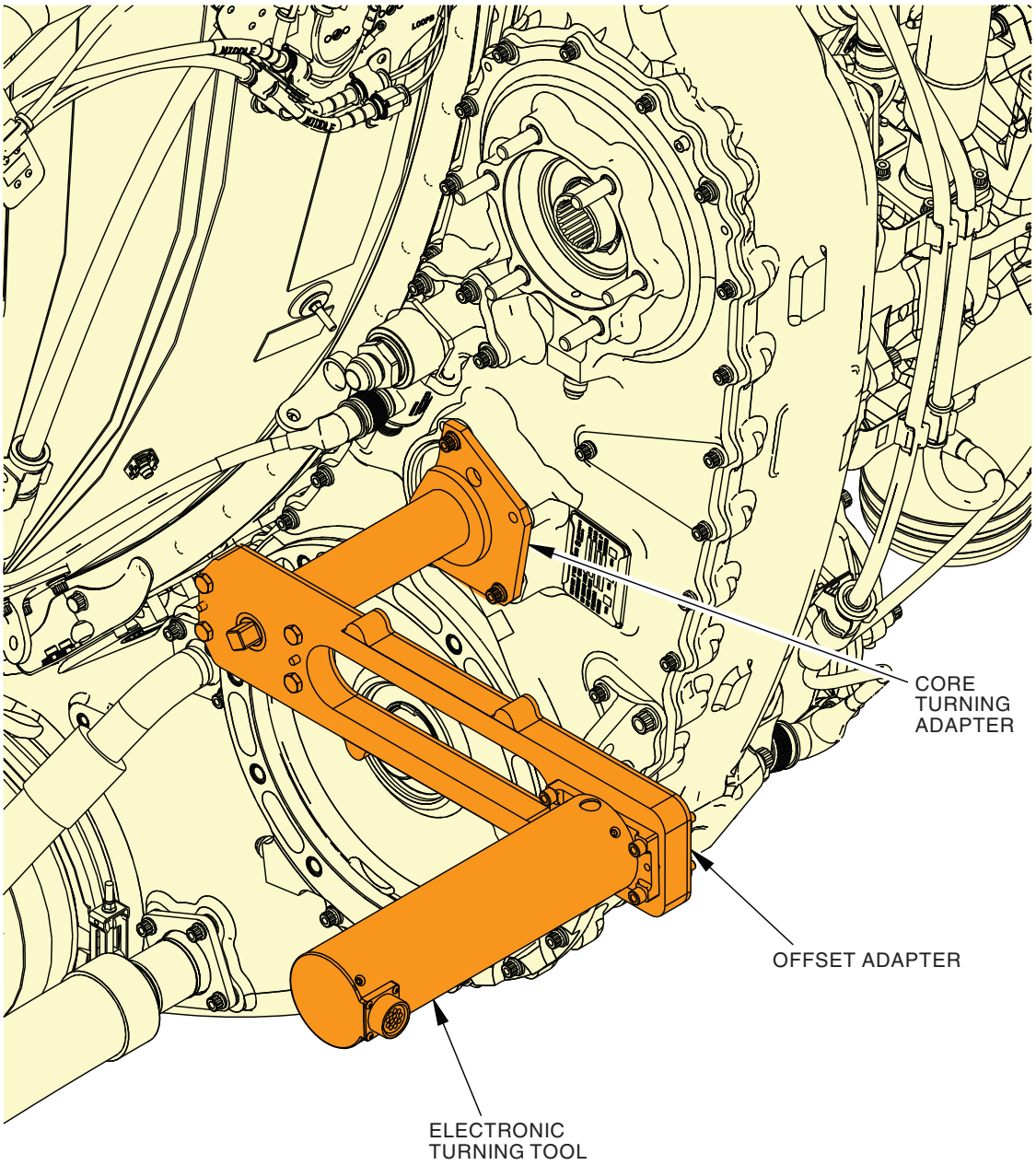

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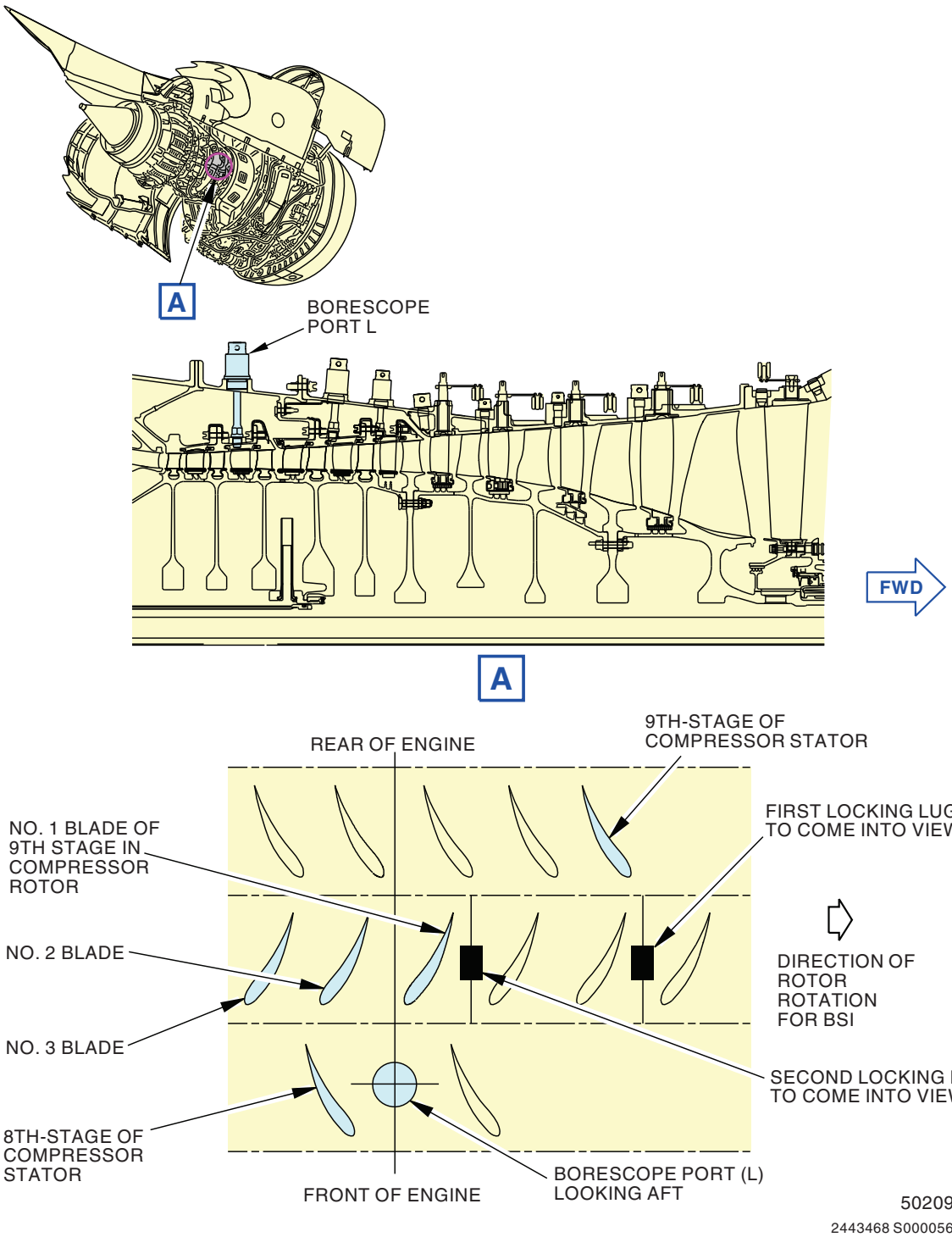
Electronic Turnover Tool Installation
Figure 203/72-00-00-990-857-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Zero Index of the N2 Rotor Alignment
Figure 204/72-00-00-990-851-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 72-00-00-910-802-G00

4. Manually Turn the Fan N1 Blades

A. General

(1) This task is for turning the fan rotor and N1 fan blades manually.

(2) You can use this task to identify and free an N1 rotor that will not turn (locked rotor condition).

B. References

Reference	Title
71-00-00-910-801-G00	Prepare the Engine for Operation (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-2586	Mat - Protective, Engine Inlet Part #: 956A7616G02 Supplier: 58828
STD-585	Mat - Protective, Closed Cell Neoprene, 30-50 duro, Weather and Oil Resistant, 3/8 Inch Min Thickness, Minimum 42x60 Inches with Warning Streamers
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

F. Prepare for the Manually Turn the Fan N1 Blades

SUBTASK 72-00-00-860-080-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-00-00-010-035-G00

(2) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

SUBTASK 72-00-00-860-009-G00

(3) If snow or ice is present in the inlet or exhaust of the engine, do the ice removal and engine warming procedures (TASK 71-00-00-910-801-G00).

SUBTASK 72-00-00-020-001-G00

(4) Put the protective mat, SPL-2586, or protective mat, STD-585, in the engine inlet.

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G. Manually Turn the Fan N1 Blades

SUBTASK 72-00-00-860-010-G00

WARNING DO NOT TOUCH FAN BLADES WITHOUT HAND PROTECTION. FAN BLADES CAN CAUSE INJURY TO HANDS.

CAUTION TURN THE FAN ROTOR WITH YOUR HAND ONLY. USE THE FAN BLADES TO TURN THE FAN ROTOR. DO NOT USE THE LOW PRESSURE TURBINE TO TURN THE FAN ROTOR. DO NOT USE TOOLS OR DEVICES ON THE FAN BLADES TO TURN THE FAN ROTOR. DAMAGE TO THE ENGINE COMPONENTS CAN OCCUR.

(1) Carefully turn the fan rotor blades with the flat of your hands.

NOTE: The fan rotor should turn easily.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 72-00-00-420-001-G00

(1) Remove the protective mat, SPL-2586, or protective mat, STD-585, from the engine inlet.

SUBTASK 72-00-00-410-029-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 72-00-00-860-081-G00

(3) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

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(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

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ENGINE - INSPECTION/CHECK

1. General

A. This procedure has these tasks:

(1) Continue-In-Service limits

(2) Preparation for borescope inspection

(3) Booster blades and vanes borescope inspection

(4) High pressure compressor section borescope detailed inspection

(5) Combustion section and high pressure turbine section stage one nozzle leading edge detailed inspection

(6) High pressure turbine section detailed inspection

(7) High Pressure Turbine (HPT) stage 1 nozzle borescope inspection

(8) High Pressure Turbine (HPT) stage 1 shroud borescope inspection

(9) High Pressure Turbine (HPT) stage 2 nozzle borescope inspection

(10) High Pressure Turbine (HPT) stage 2 shroud borescope inspection

(11) High Pressure Turbine (HPT) blade tip material loss using tip notches inspection

(12) Low Pressure Turbine (LPT) section borescope inspection

PTF ALL POST SB 737-CFM-LEAP-1B-75-0003

(13) High Pressure Turbine (HPT) Active Clearance Control (ACC) air manifolds inspection.

PTF ALL

B. Alternative videoscope or borescope tool sets, that meet the minimum requirements of the GE Commercial Engine Standard Practice Manual, SPM 70-32-84, Borescope Systems, may be used as alternates for the tools listed in the procedure. If you use alternative tools or equipment, it is the operator's responsibility to make sure that they give the same results and are as safe to the parts and personnel as the tools or equipment specified in the procedure.

TASK 72-00-00-800-801-G00

2. Continue-In-Service Limits

A. General

(1) This task provides the instructions for the Continue-In-Service (C-I-S) limits of the engine.

(2) When a defect is found that is more than the serviceable limits, the C-I-S limits may be applicable.

(3) If applicable to the type and quantity of engine damage or wear that you find, the C-I-S limits will permit for a limited amount of engine operation as a one-time use extension or reduced interval repeat inspection, before you must correct the damage.

(4) The purpose of the C-I-S limits is to permit the airplane to be flown with lower interval repeat inspections to a repair facility where you can correct the engine damage.

B. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

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C. Continue-In-Service Limits

SUBTASK 72-00-00-200-001-G00

(1) You must follow these steps when you use the C-I-S limits.

(a) You must complete the necessary inspection or repairs before the end of the C-I-S limit.

(b) You must use the C-I-S limits as designated in the related task.

(c) When the C-I-S limit is specified in flight cycles and flight hours, the parameter that occurs first is the limit.

(d) The C-I-S limits are given for specific engine components and areas.

1) You must use the specific C-I-S limits for the component or area where you find the damage or wear.

2) One time C-I-S limits can only be used for one extension period to permit scheduling of the engine for repair or removal.

3) Repeat, decreased interval limits are applicable if the conditions stay within the stated dimensional limits.

a) The engine can continue-in-service within the limits until the interval expires, and an additional inspection is needed to repeat a C-I-S interval.

——— END OF TASK ———

TASK 72-00-00-290-801-G00

3. Preparation for Borescope Inspection

(Figure 601, Figure 602, Figure 603, Figure 604, Figure 605, Figure 606, and Figure 607)

A. General

(1) This task gives the instructions to do the preparation for the borescope inspection.

(2) You use the borescope procedure to examine the gas path of the primary airstream.

(3) These are the sections of the engine that you can examine with the borescope procedure:

(a) Booster

(b) High Pressure Compressor

(c) Combustor

(d) High Pressure Turbine

(e) Low Pressure Turbine.

(4) There are engine borescope ports that provide access for the borescope inspection.

(a) The borescope ports have threaded plugs that you must remove to do the borescope inspection.

NOTE: The borescope ports A and B for the booster inspection do not use borescope plugs.

(5) The borescope inspection tasks provide the maximum serviceable limits. You must repair the engine if there is damage or wear that exceeds the maximum serviceable limit.

(6) If there are Continue-In-Service (C-I-S) limits, they are included with the maximum serviceable limits.

B. References

Reference	Title
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)

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(Continued)

Reference	Title
27-81-00-860-804	Leading Edge Flaps and Slats Retraction (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
72-00-00-000-801-G00	BSI Plug D Removal (P/B 401)
72-00-00-400-801-G00	BSI Plug D Installation (P/B 401)
72-00-00-822-801-G00	Zero-Index of the N2 Rotor Alignment (P/B 201)
72-00-00-910-801-G00	High Pressure Rotor (N2) Turning for Borescope (P/B 201)
72-63-04-000-801-G00	Accessory Gearbox Manual Control Cover Removal (P/B 401)
72-63-04-400-801-G00	Accessory Gearbox Manual Control Cover Installation (P/B 401)
74-21-02-000-801-G00	Left Igniter Plug Removal (P/B 401)
74-21-02-000-802-G00	Right Igniter Plug Removal (P/B 401)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE
STD-6547	Thermometer - 0 to 350 deg F Range

D. Consumable Materials

Reference	Description	Specification
D00599 [CP2442]	Oil - Engine	
D50017 [C02-071]	Compound - Antiseize And Lubricating - Pure Nickel Special Grade	GE Spec D6Y28C1 or A50TF198
D50034 [CP2619]	Compound - Antiseize - Pure Nickel Special, Never - Seez	
D50068 [CP2544]	Lubricant - Molykote P37, Paste	
D50355 [CP2961]	Lubricant - Anti-Seize Compound	D6Y31C2

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

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G. Prepare for the Borescope Inspection

SUBTASK 72-00-00-860-011-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-00-00-860-012-G00

(2) Do this task: Leading Edge Flaps and Slats Retraction, TASK 27-81-00-860-804.

SUBTASK 72-00-00-860-013-G00

(3) Do this task: Leading Edge Flaps and Slats - Deactivation, TASK 27-81-00-040-801.

SUBTASK 72-00-00-860-014-G00

(4) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING

DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING

OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

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SUBTASK 72-00-00-480-002-G00

(5) Assemble the borescope inspection tools per assembly instructions from the borescope tool manufacturer.

SUBTASK 72-00-00-280-001-G00

CAUTION

MAKE SURE THE TEMPERATURE OF THE ENGINE AND THE ADJACENT AREAS NEAR THE BORESCOPE INSPECTION PORTS ARE LESS THAN 150°F (65.6°C). DO NOT DO A BORESCOPE INSPECTION IF THE ADJACENT AREAS ARE TOO HOT TO TOUCH. DO THE BORESCOPE INSPECTION WHEN THE ENGINE TEMPERATURE IS LESS THAN 150°F (65.6°C). THIS WILL PREVENT DAMAGE TO THE BORESCOPE EQUIPMENT.

(6) Make sure that the internal and external temperature of the engine is less than 150°F (65.6°C) before you insert the borescope probe into the engine.

(a) If the engine has been recently run, measure the temperature of the engine with a metal probe type thermometer, STD-6547.

NOTE: It is recommended to wait 3 to 5 hours after engine shutdown to do the borescope inspection. This will allow the engine to become sufficiently cool.

NOTE: You can dry motor the engine to decrease the engine core temperature.

SUBTASK 72-00-00-290-001-G00

(7) Find the borescope port location for the engine components that you will inspect (Figure 602).

(a) Remove and tag borescope plugs as required for the inspection.

(b) Remove and tag igniter plugs as follows:

1) Do this task: Left Igniter Plug Removal, TASK 74-21-02-000-801-G00.

2) Do this task: Right Igniter Plug Removal, TASK 74-21-02-000-802-G00.

SUBTASK 72-00-00-020-006-G00

(8) Remove the borescope plugs as follows:

(a) Apply a spray of penetrating oil to the ratcheting mechanism and let it soak.

(b) Apply break torque with a wrench to unseat the plug from the case.

(c) Apply another spray of penetrating oil to lubricate the plug threads if necessary and do step (b) again.

(d) Remove the plug.

SUBTASK 72-00-00-020-007-G00

(9) To remove borescope plug D, do this task: BSI Plug D Removal, TASK 72-00-00-000-801-G00.

SUBTASK 72-00-00-840-001-G01

(10) If turning of the N2 rotor is required, accomplish the following steps:

(a) Do this task: Accessory Gearbox Manual Control Cover Removal, TASK 72-63-04-000-801-G00.

(b) Do this task: High Pressure Rotor (N2) Turning for Borescope, TASK 72-00-00-910-801-G00.

(c) Do this task: Zero-Index of the N2 Rotor Alignment, TASK 72-00-00-822-801-G00.

(d) Do this task: Accessory Gearbox Manual Control Cover Installation, TASK 72-63-04-400-801-G00.

SUBTASK 72-00-00-290-002-G00

(11) Find the torque values and thread specifications for the borescope plugs (Figure 602).

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SUBTASK 72-00-00-290-003-G00

(12) When you remove the borescope plugs, do an inspection of the borescope plugs and the borescope ports as follows:

NOTE: Plug threads C and D are lubricated with engine oil, D00599 [CP2442]. Plug threads E, F, G, H, I, J, K, L, and M are lubricated with Tiodize T8E-H grease, D50355 [CP2961]. Plug threads N, O, P, and Q are lubricated with Tiodize T8E-H grease, D50355 [CP2961] or as an alternative, with pure nickel special compound, D50017 [C02-071] or pure nickel special compound, D50034 [CP2619]. Plug threads R, S, T, U and V are lubricated with lubricant, D50068 [CP2544]. All other plugs are coated with pure nickel special compound, D50034 [CP2619].

(a) Do a visual inspection of the threads on the borescope port.

CAUTION DO NOT ALLOW CHIPS OR UNWANTED MATERIAL TO GET TRAPPED BETWEEN THE INITIAL PARTS AND THE REPAIR PARTS. IF THERE IS TRAPPED DEBRIS, THE REPAIR WILL BE UNSATISFACTORY.

1) If it is necessary, repair minor damage such as burrs or galling.

NOTE: For thread specification, refer to Figure 602. If you repair the threads, make sure that you do not allow debris to enter the engine.

(b) Do a visual inspection of the vespel ring (plug C and D).

1) If the vespel ring is missing or partially missing, replace the borescope plug.

(c) Do a visual inspection of the borescope plugs for cracks.

1) Cracks are not permitted.

a) Replace the borescope plug.

(d) Do a visual inspection of the borescope plugs for deformation.

1) Deformation is not permitted.

a) Replace the borescope plug.

(e) Do a visual inspection of the borescope plugs for a broken plug spring.

1) Broken plug spring is not permitted.

a) Replace the borescope plug.

(f) Do a visual inspection of the borescope plugs for wear on the locking device seating face.

1) Wear is permitted if not more than 0.004 in. (0.10 mm) in depth.

(g) Do a visual inspection of the borescope plugs for damaged or bent locking lugs.

1) Damaged or bent locking lugs are not permitted.

a) Replace the borescope plug.

(h) Do a visual inspection of the borescope plugs for damaged threads.

1) One full thread, total (cumulative) can be missing if no less than half of the lead-in thread remains.

a) If there is other thread damage, replace the borescope plug.

(i) Examine the borescope plugs with the spring and tabbed locking ring (anti-rotation feature) as follows:

1) Axially compress the spring by pushing the tabbed locking ring toward the plug body using hand force.

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2) Keep the spring compressed and rotate the plug by hand.

NOTE: You can spray penetrating oil into the locking mechanism to free it. This must be done and the rotation test must be done again before you replace the plug.

3) A plug which does not rotate in the removal (anti-clockwise) direction is permitted.

4) A plug which rotates in the removal direction and produces an audible clicking sound is permitted.

5) A plug which rotates in the removal direction with no resistance and no audible clicking sound is not permitted.

a) Replace the borescope plug.

6) A plug that rotates in the installation (clockwise) direction, with or without resistance or an audible clicking sound, is permitted.

7) A plug that does not rotate in the installation direction is not permitted.

a) Replace the borescope plug.

SUBTASK 72-00-00-420-005-G00

(13) To install borescope plug D, do this task: BSI Plug D Installation, TASK 72-00-00-400-801-G00.

——— END OF TASK ———

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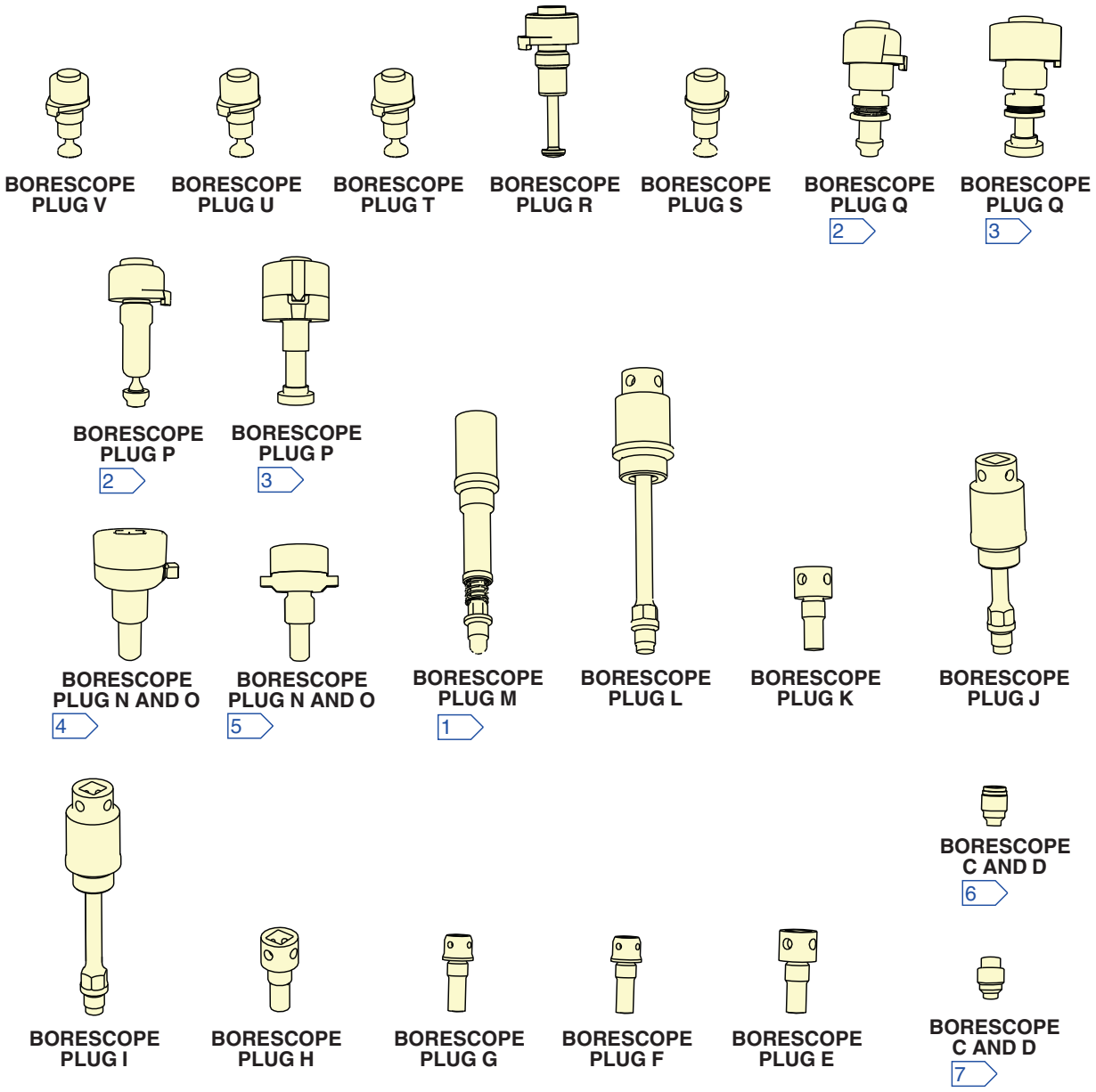

1 PRE LEAP-1B SERVICE BULLETIN 72-0161
2 PRE LEAP-1B SERVICE BULLETIN 72-0188
3 POST LEAP-1B SERVICE BULLETIN 72-0188
4 PRE LEAP-1B SERVICE BULLETIN 72-0157
5 POST LEAP-1B SERVICE BULLETIN 72-0157
6 PRE LEAP-1B SERVICE BULLETIN 72-0179
7 POST LEAP-1B SERVICE BULLETIN 72-0179

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Engine Borescope Plugs Identification
Figure 601/72-00-00-990-838-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1 PORT	ENGINE SECTION	2 RADIAL/AXIAL IN DEGREES	VISIBLE COMPONENTS	THREAD SPECIFICATION	3 PLUG TORQUE in-lb (Nm)
A	BOOSTER	R107/A174	STG 2 BLADE TE STG 3 BLADE LE		
B	BOOSTER	R105/A179	STG 3 BLADE TE STG 4 BLADE LE		
C	HPC	R146/A197	STG 1 BLISK LE	0.5625-18 UNJF-3A	175.0 in-lb (19.8 Nm)
D	HPC	R131/A199	STG 1 BLISK LE (BOREBLEND ONLY)	0.5625-18 UNJF-3A	175.0 in-lb (19.8 Nm)
E	HPC	R132/A204	STG 1 BLISK TE STG 2 BLISK LE	0.4375-20 UNJF-3A	205.0 in-lb (23.2 Nm)
F	HPC	R135/A208	STG 2 BLISK TE STG 3 BLISK LE	0.375-24 UNJF-3A	205.0 in-lb (23.2 Nm)
G	HPC	R141/A211	STG 3 BLISK TE STG 4 BLISK LE	0.375-24 UNJF-3A	205.0 in-lb (23.2 Nm)
H	HPC	R095/A123	STG 4 BLISK TE STG 5 BLISK LE	0.4375-20 UNJF-3A	205.0 in-lb (23.2 Nm)
I	HPC	R097/A215	STG 5 BLISK TE STG 6 BLADE LE	0.375-24 UNJF-3A	**INNER:** 1. 80.0 IN-LB (9.0 NM) 2. LOOSEN ONE QUARTER TURN. 3. TIGHTEN TO 18.0 IN-LB (2.0 Nm) ABOVE PREVAILING RUN-ON TORQUE. **OUTER:** 1. 205.0 IN-LB (23.2 Nm)
J	HPC	R97/A217	STG 6 BLADE TE STG 7 BLADE LE	0.375-24 UNJF-3A	**INNER:** 1. 80.0 IN-LB (9.0 NM) 2. LOOSEN ONE QUARTER TURN. 3. TIGHTEN TO 18.0 IN-LB (2.0 Nm) ABOVE PREVAILING RUN-ON TORQUE. **OUTER:** 1. 205.0 IN-LB (23.2 Nm)
K	HPC	R143/A220	STG 7 BLADE TE STG 8 BLADE LE	0.4375-20 UNJF-3A	205.0 in-lb (23.2 Nm)
L	HPC	R155/A221	STG 8 BLADE TE STG 9 BLADE LE	0.375-24 UNJF-3A	**INNER:** 1. 80.0 IN-LB (9.0 NM) 2. LOOSEN ONE QUARTER TURN. 3. TIGHTEN TO 18.0 IN-LB (2.0 Nm) ABOVE PREVAILING RUN-ON TORQUE. **OUTER:** 1. 205.0 IN-LB (23.2 Nm)
M 4	HPC	R151/A221	STG 9 BLADE TE STG 10 BLADE LE	0.375-24 UNJF-3A	**INNER:** 1. 80.0 IN-LB (9.0 NM) 2. LOOSEN ONE QUARTER TURN. 3. TIGHTEN TO 18.0 IN-LB (2.0 Nm) ABOVE PREVAILING RUN-ON TORQUE. **OUTER:** 1. 205.0 IN-LB (23.2 Nm)
N	COMBUSTOR	R038/A234	COMBUSTOR STG1 NOZZLE	0.500-20 UNJF-3A	105.0 in-lb (11.9 Nm)

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Borescope Port Data
Figure 602/72-00-00-990-839-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1 PORT	ENGINE SECTION	2 RADIAL/AXIAL IN DEGREES	VISIBLE COMPONENTS	THREAD SPECIFICATION	3 PLUG TORQUE in-lb (Nm)
O	COMBUSTOR	R322/A234	STG 1 NOZZLE STG 1 BLADE LE	0.500-20 UNJF-3A	105.0 in-lb (11.9 Nm)
P	HPT	R087/A238	STG 1 NOZZLE STG 1 BLADE LE	0.625-18 UNJF-3A	90.0 in-lb (10.2 Nm)
Q	HPT	R097/A241	STG 1 BLADE TE STG 2 NOZZLE STG 2 BLADE LE	0.750-16 UNJF-3A	90.0 in-lb (10.2 Nm)
R	LPT	R130/A249	STG 2 HPT BLADE TE STG 1 LPT NOZZLE LE STG 1 LPT BLADE LE	0.6250-18 UNJF-3A	53.0 in-lb (6.0 Nm)
S	LPT	R110/A253	STG 1 BLADE TE STG 2 BLADE LE	0.500-20 UNJF-3A	53.0 in-lb (6.0 Nm)
T	LPT	R135/A256	STG 2 BLADE TE STG 3 BLADE LE	0.500-20 UNJF-3A	53.0 in-lb (6.0 Nm)
U	LPT	R115/A259	STG 3 BLADE TE STG 4 BLADE LE	0.500-20 UNJF-3A	53.0 in-lb (6.0 Nm)
V	LPT	R120/A262	STG 4 BLADE TE STG 5 BLADE LE	0.500-20 UNJF-3A	53.0 in-lb (6.0 Nm)

1 BORESCOPE SELECTION: 0.2 INCH (5.0 mm) MAX PROBE DIAMETER; 45 OR 90 DEGREE TIP VIEWING ANGLE; 7.9 INCH (20.0 cm) MIN PROBE LENGTH; 0.5 INCH (12.7 mm) TIP LENGTH.

2 RADIAL DEGREES ARE VIEWED IN A CLOCKWISE DIRECTION, AFT LOOKING FORWARD (ALF). AXIAL DEGREES ARE BY ENGINE STATIONS FRONT TO BACK.

3 PLUG THREADS C AND D ARE LUBRICATED WITH ENGINE OIL, D00599 [CP2442]. PLUG THREADS E, F, G, H, I, J, K, L, AND M ARE LUBRICATED WITH TIODIZE T8E-H GREASE D50355 [CP2961]. PLUG THREADS N, O, P, AND Q ARE LUBRICATED WITH TIODIZE T8E-H GREASE D50355 [CP2961] OR AS AN ALTERNATIVE, WITH PURE NICKEL SPECIAL COMPOUND D50017 [C02-071] OR D50034 [CP2619]. PLUG THREADS R, S, T, U, AND V ARE LUBRICATED WITH MOLYKOTE P-37, D50068 [CP2544]. ALL OTHER PLUGS ARE COATED WITH PURE NICKEL SPECIAL COMPOUND, D50034 [CP2619].

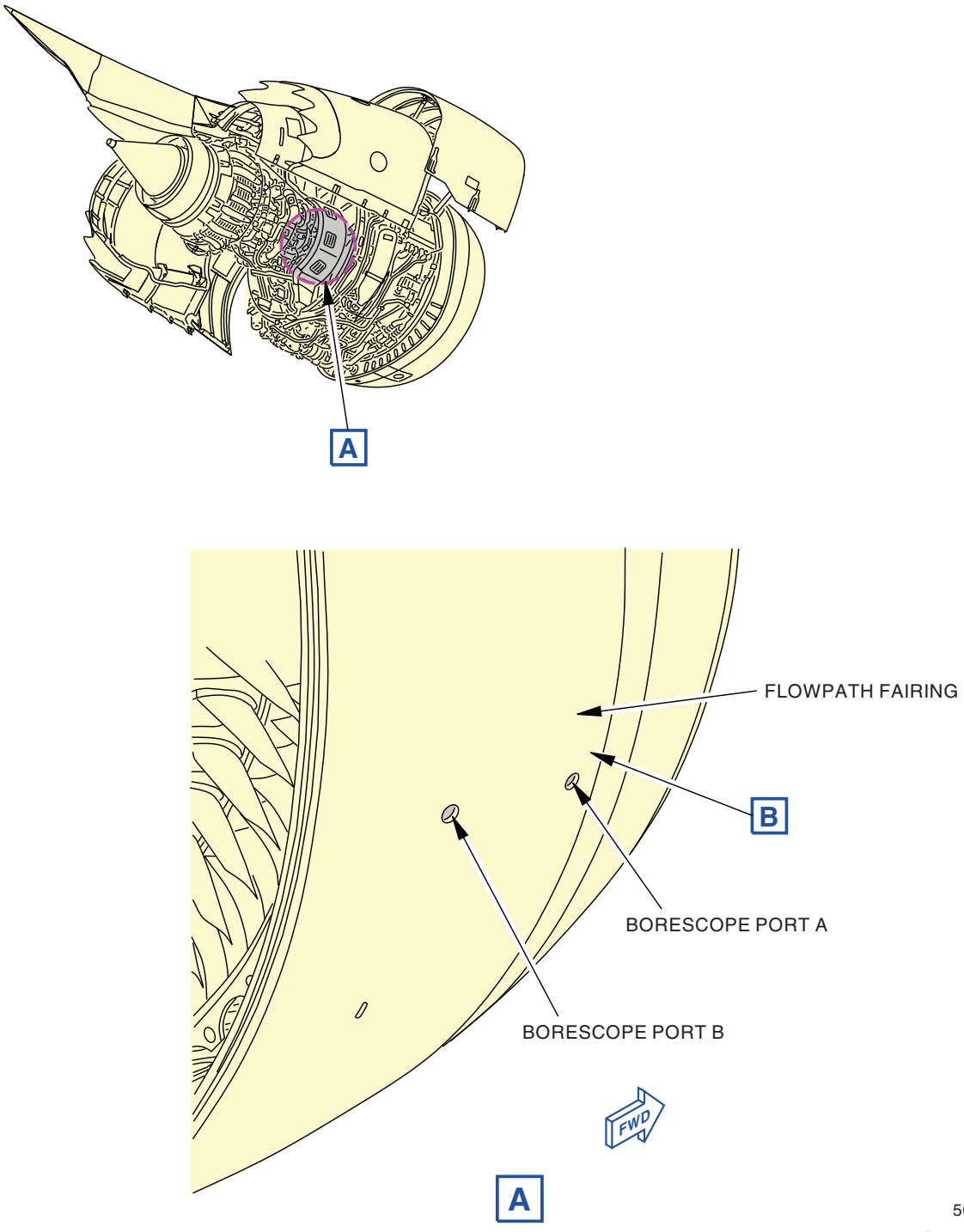
4 PRE LEAP-1B SERVICE BULLETIN 72-0161.

5020989-00

2433183 S0000562752_V9

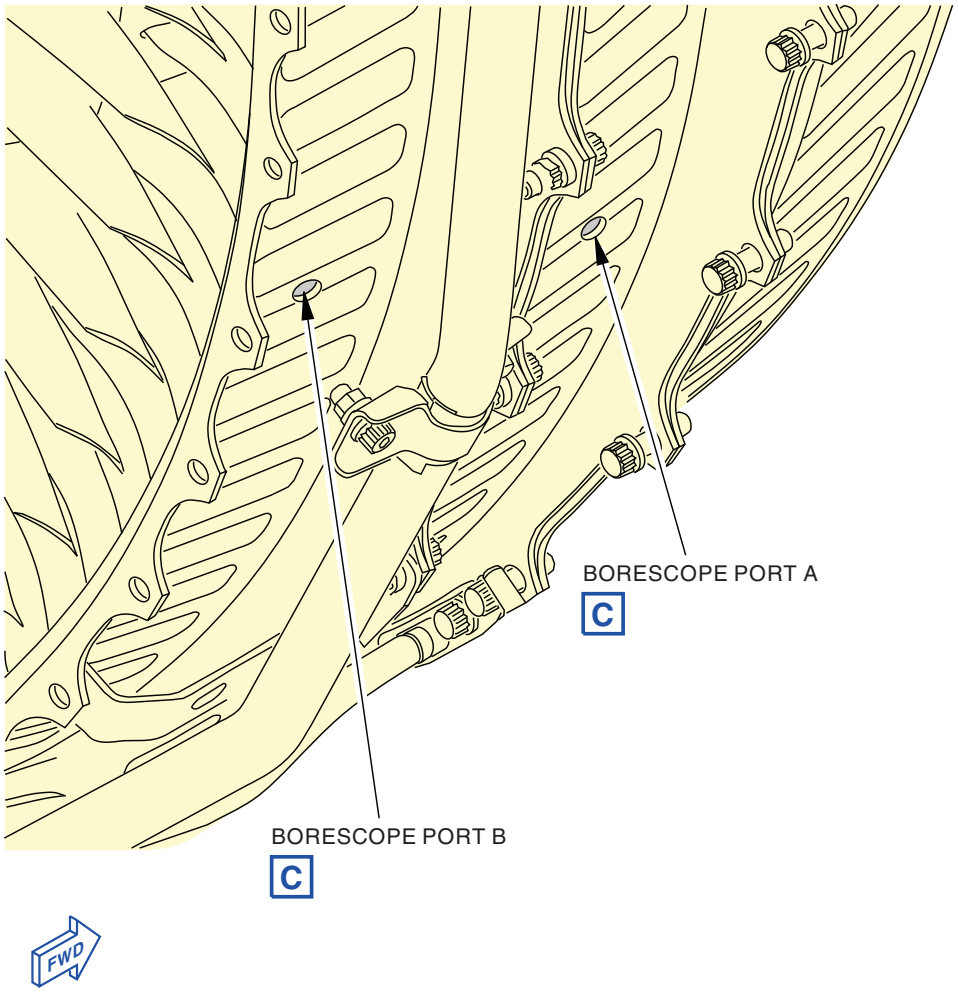
Borescope Port Data
Figure 602/72-00-00-990-839-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Booster Borescope Port Locations
Figure 603/72-00-00-990-840-G00 (Sheet 1 of 3)

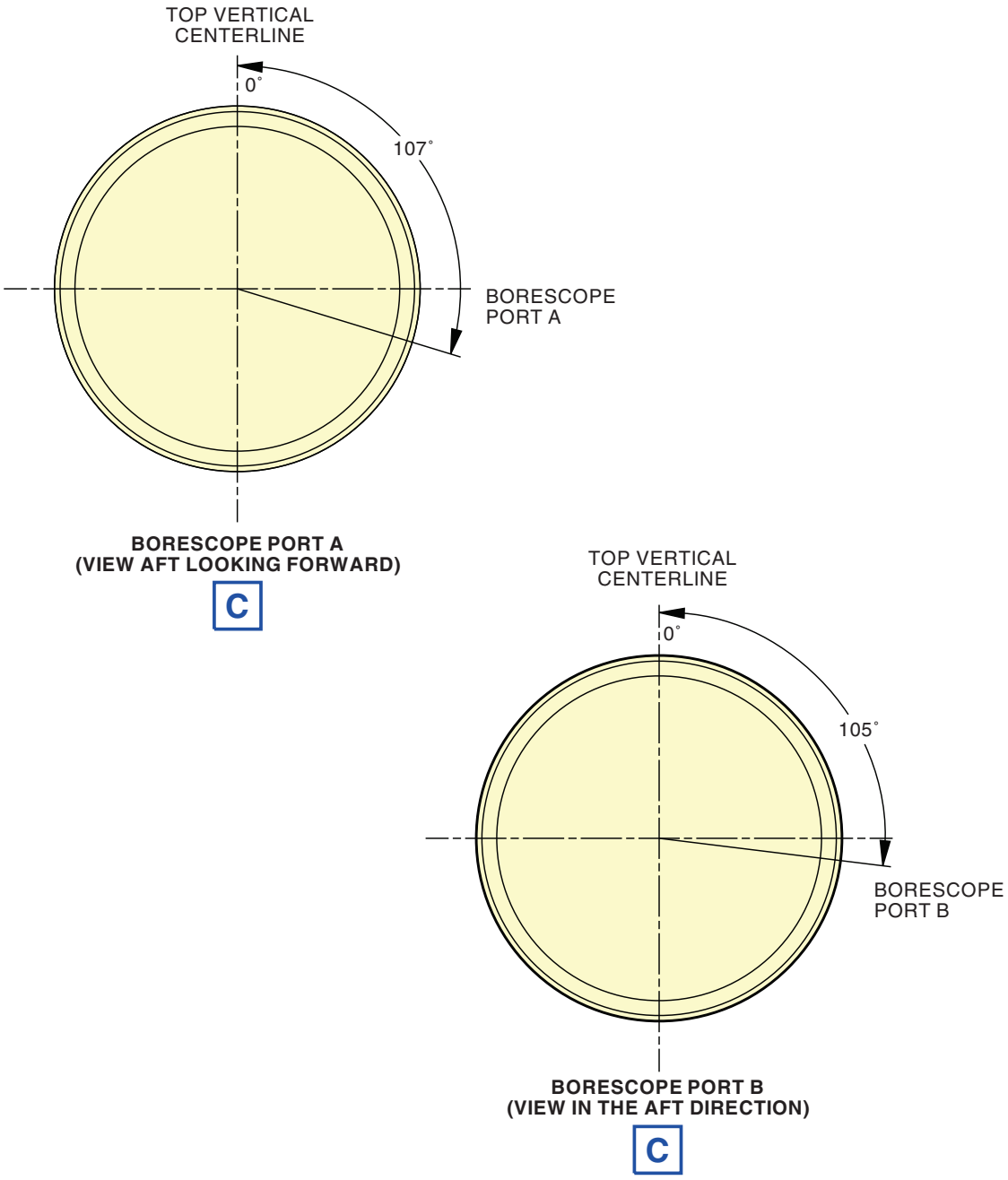
ECCN 9E991 BOEING PROPRIETARY - See title page for details



5021146-00
2433207 S0000562753_V2

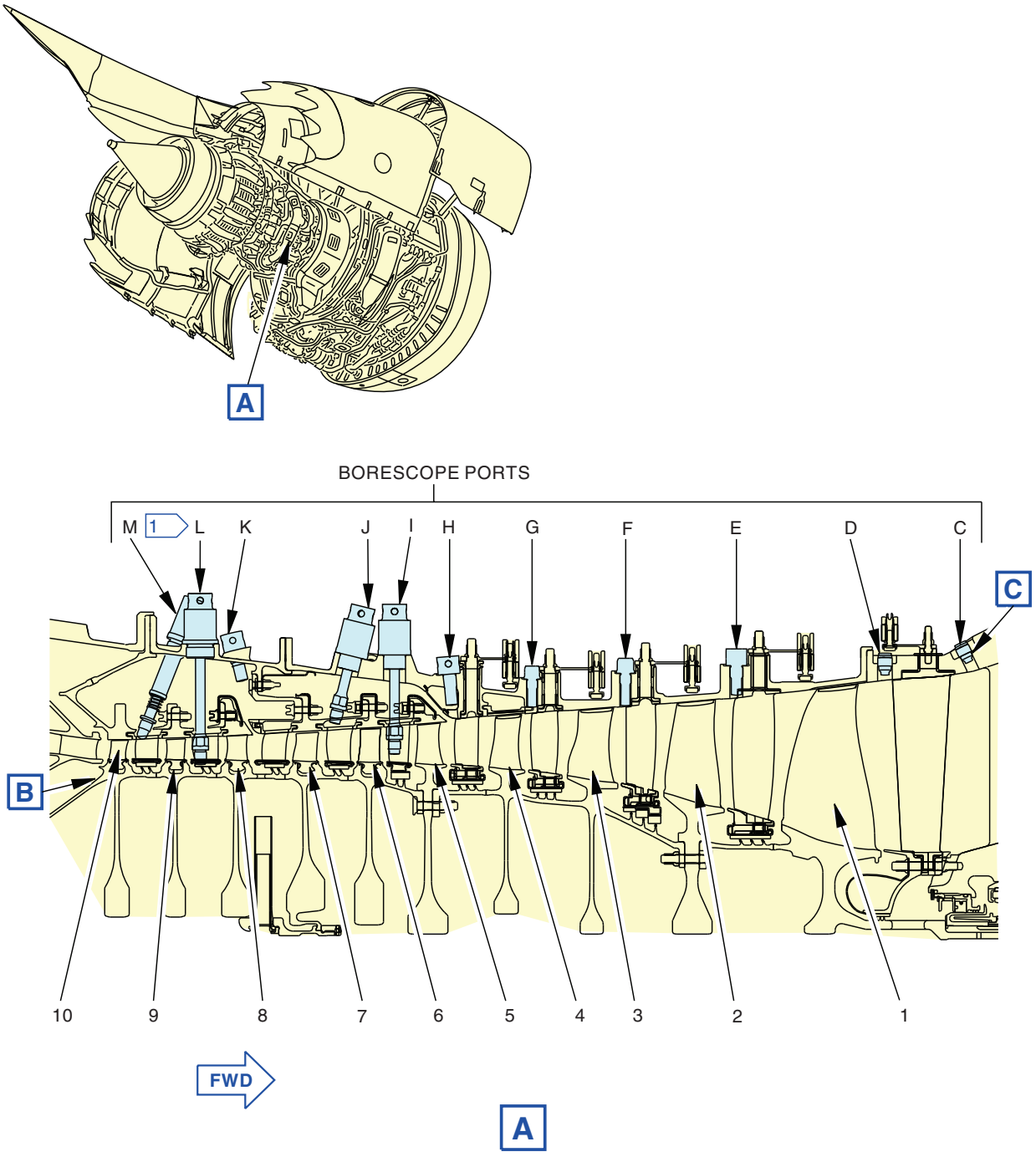
Booster Borescope Port Locations
Figure 603/72-00-00-990-840-G00 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Booster Borescope Port Locations
Figure 603/72-00-00-990-840-G00 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



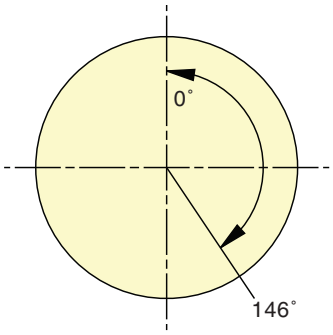
1 PRE LEAP-1B SERVICE BULLETIN 72-0161

5020998-00

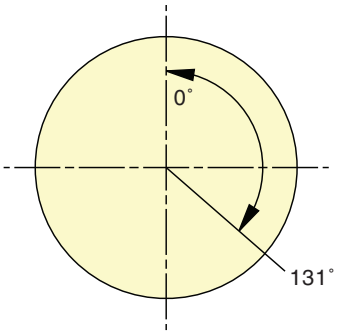
2433208 S0000562755_V3

High Pressure Compressor Section Borescope Port Locations
Figure 604/72-00-00-990-841-G00 (Sheet 1 of 3)

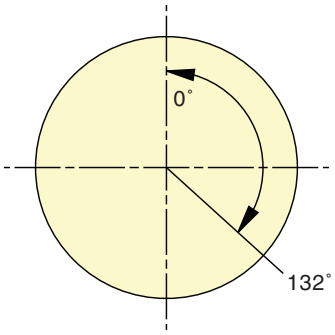
ECCN 9E991 BOEING PROPRIETARY - See title page for details



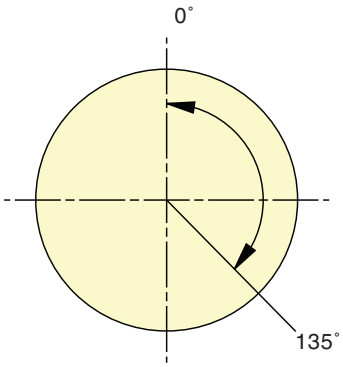
IGV AND STAGE 1 LE
FAN HUB FRAME
BORESCOPE PORT C
FORWARD OF IGV RING



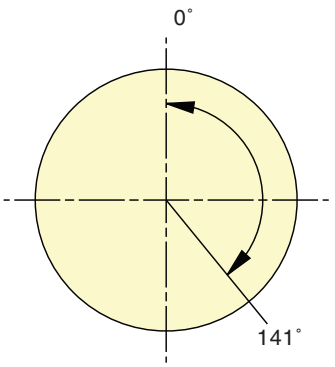
STAGE 1 LE (BLENDING)
FAN HUB FRAME
BORESCOPE PORT D
BEHIND IGV RING BRIDGE



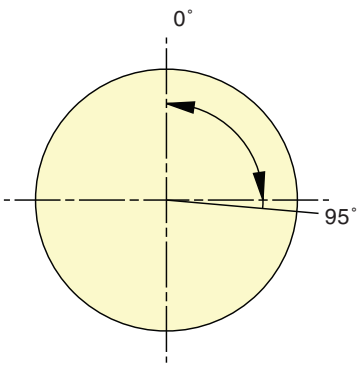
STAGE 2 LE
FORWARD HPC CASE
BORESCOPE PORT E



STAGE 3 LE
FORWARD HPC CASE
BORESCOPE PORT F



STAGE 4 LE
FORWARD HPC CASE
BORESCOPE PORT G



STAGE 5 LE
FORWARD HPC CASE
BORESCOPE PORT H

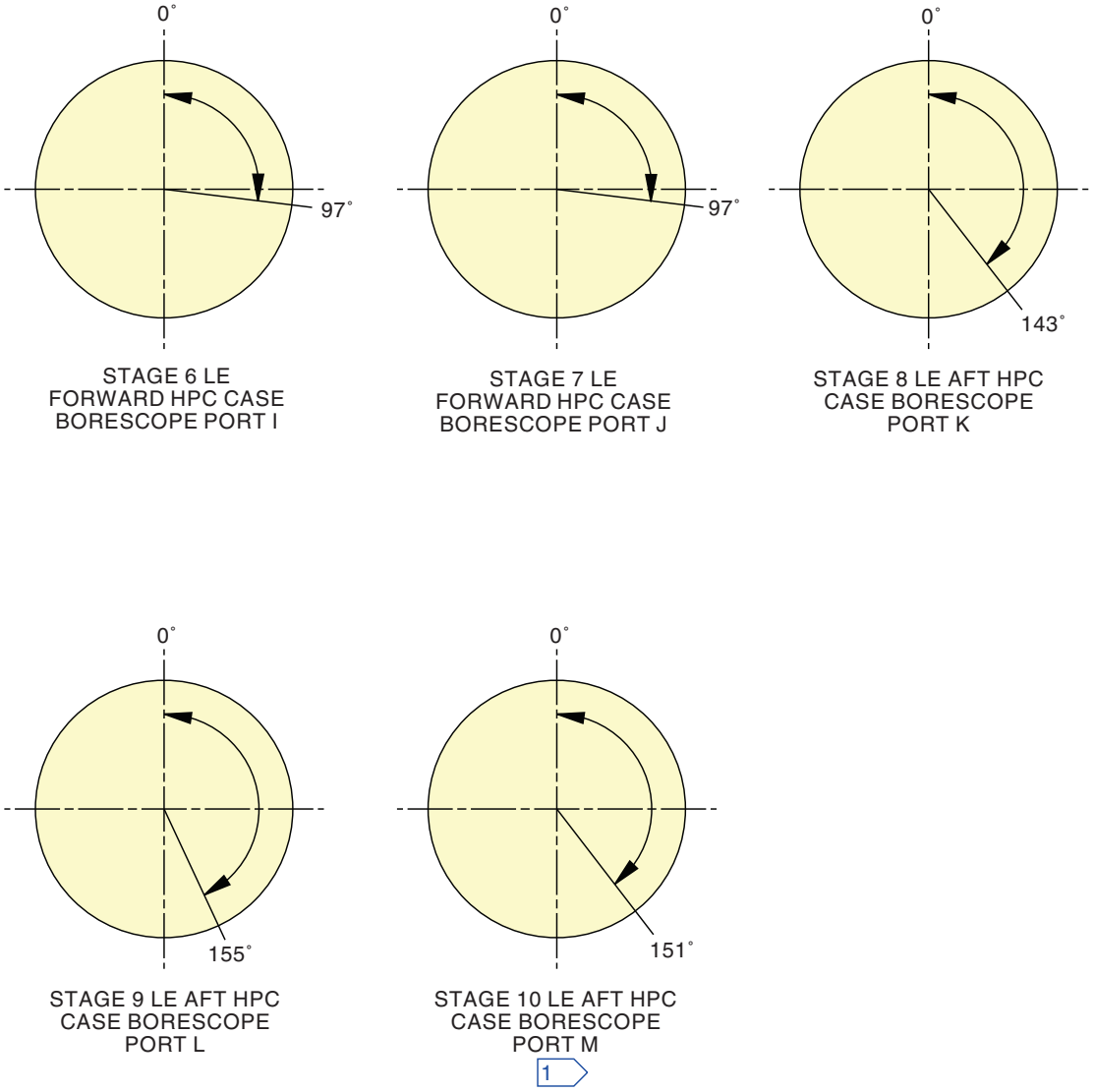
(VIEW IN THE AFT DIRECTION)

C

5021005-00
2433209 S0000562756_V3

High Pressure Compressor Section Borescope Port Locations
Figure 604/72-00-00-990-841-G00 (Sheet 2 of 3)

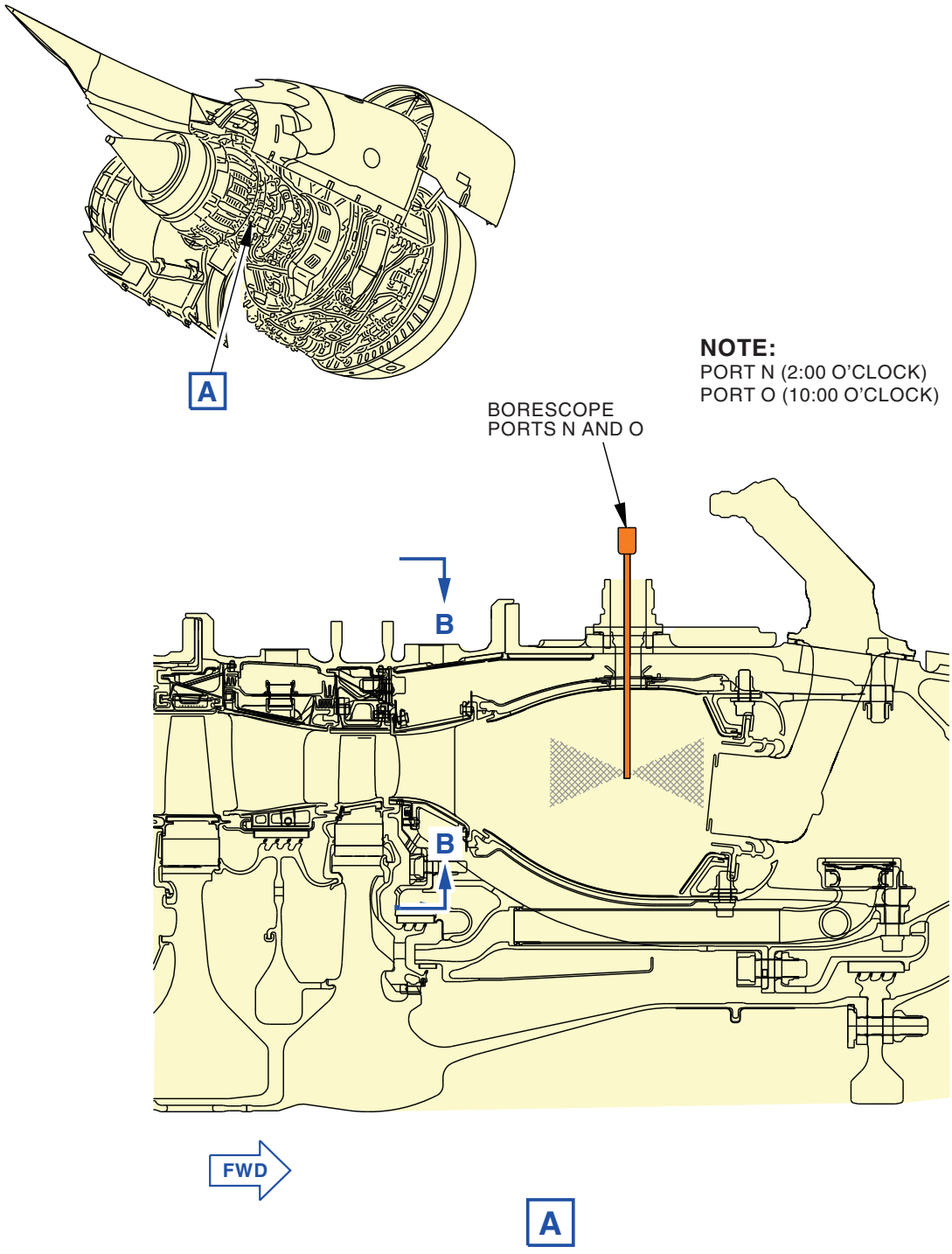
ECCN 9E991 BOEING PROPRIETARY - See title page for details



5021006-00
2433210 S0000562757_V3

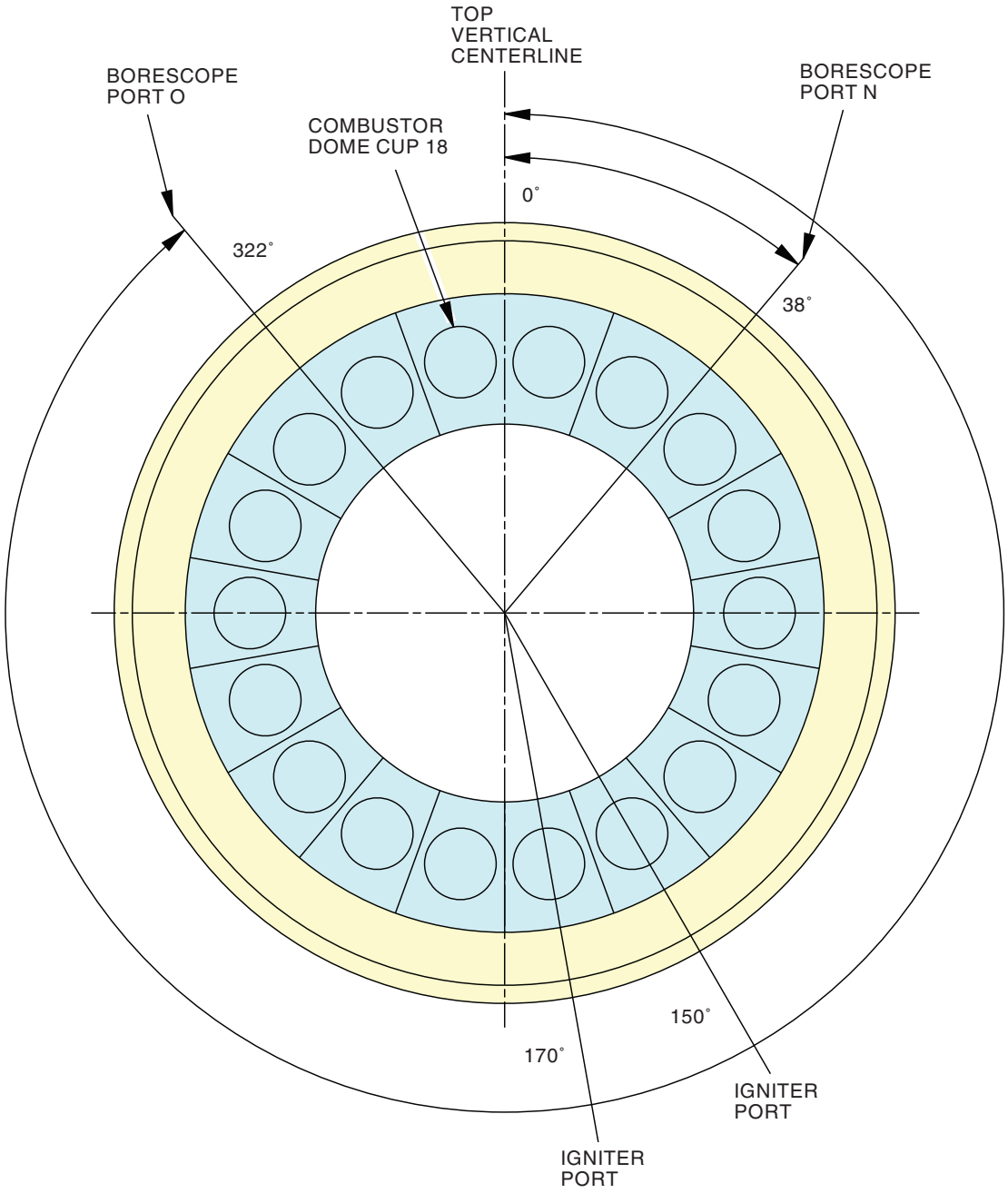
High Pressure Compressor Section Borescope Port Locations
Figure 604/72-00-00-990-841-G00 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Combustion Section Borescope Port Locations
Figure 605/72-00-00-990-842-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



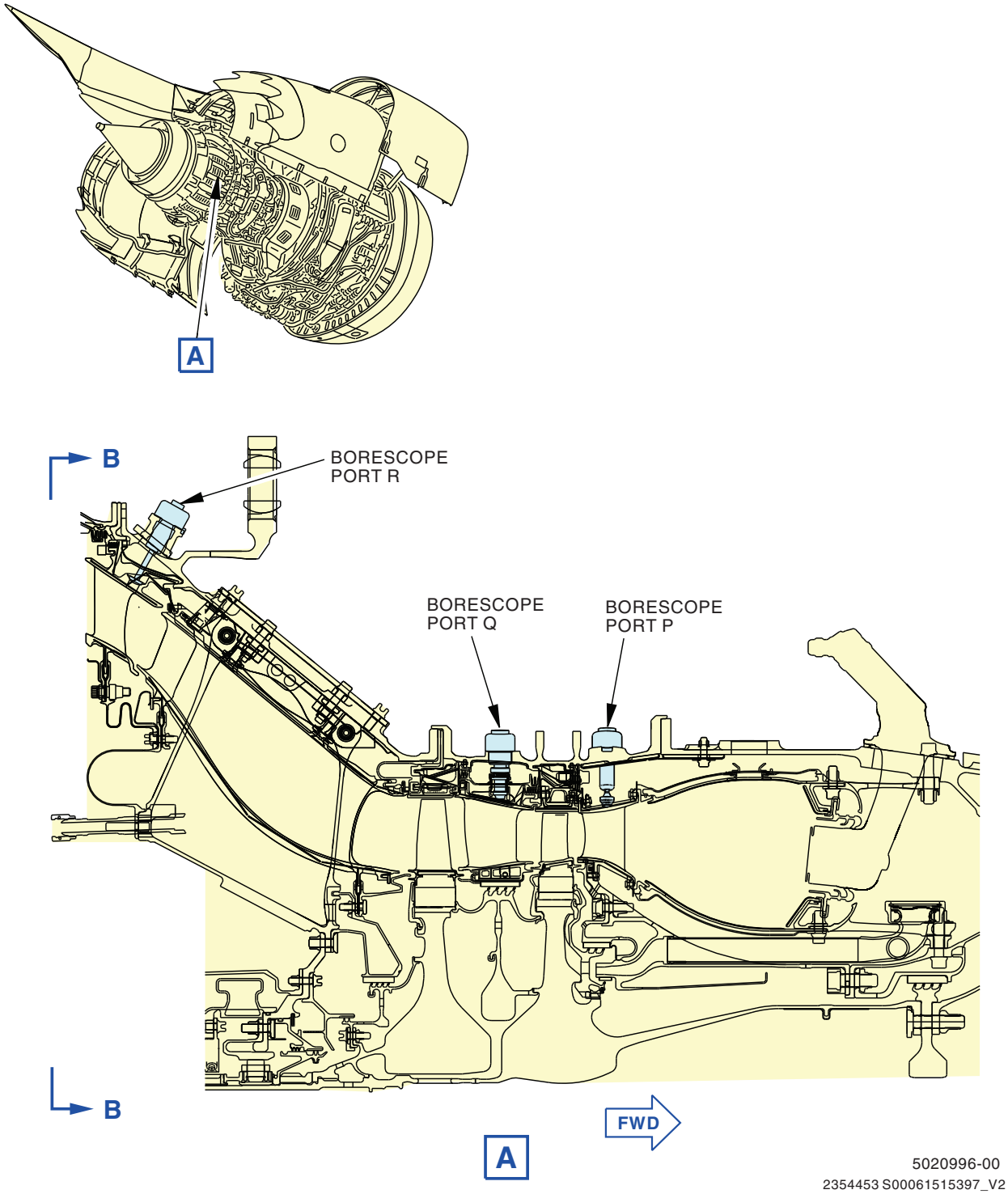
BORESCOPE PORTS N, O, AND DOME LOCATIONS
(AFT LOOKING FORWARD)

B-B

5029318-00
2354452 S00061515396_V5

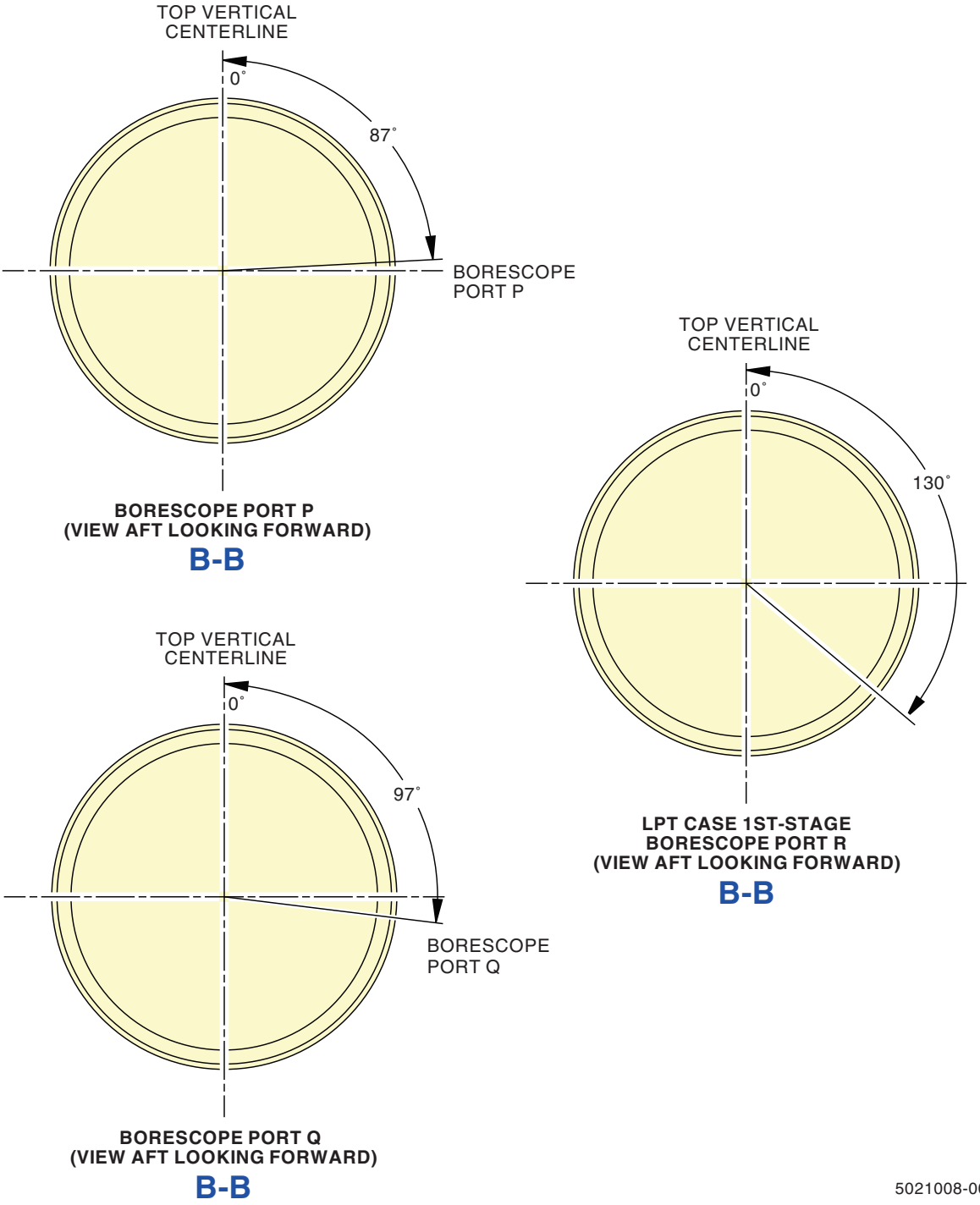
Combustion Section Borescope Port Locations
Figure 605/72-00-00-990-842-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



High Pressure Turbine Section Borescope Port Locations
Figure 606/72-00-00-990-843-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

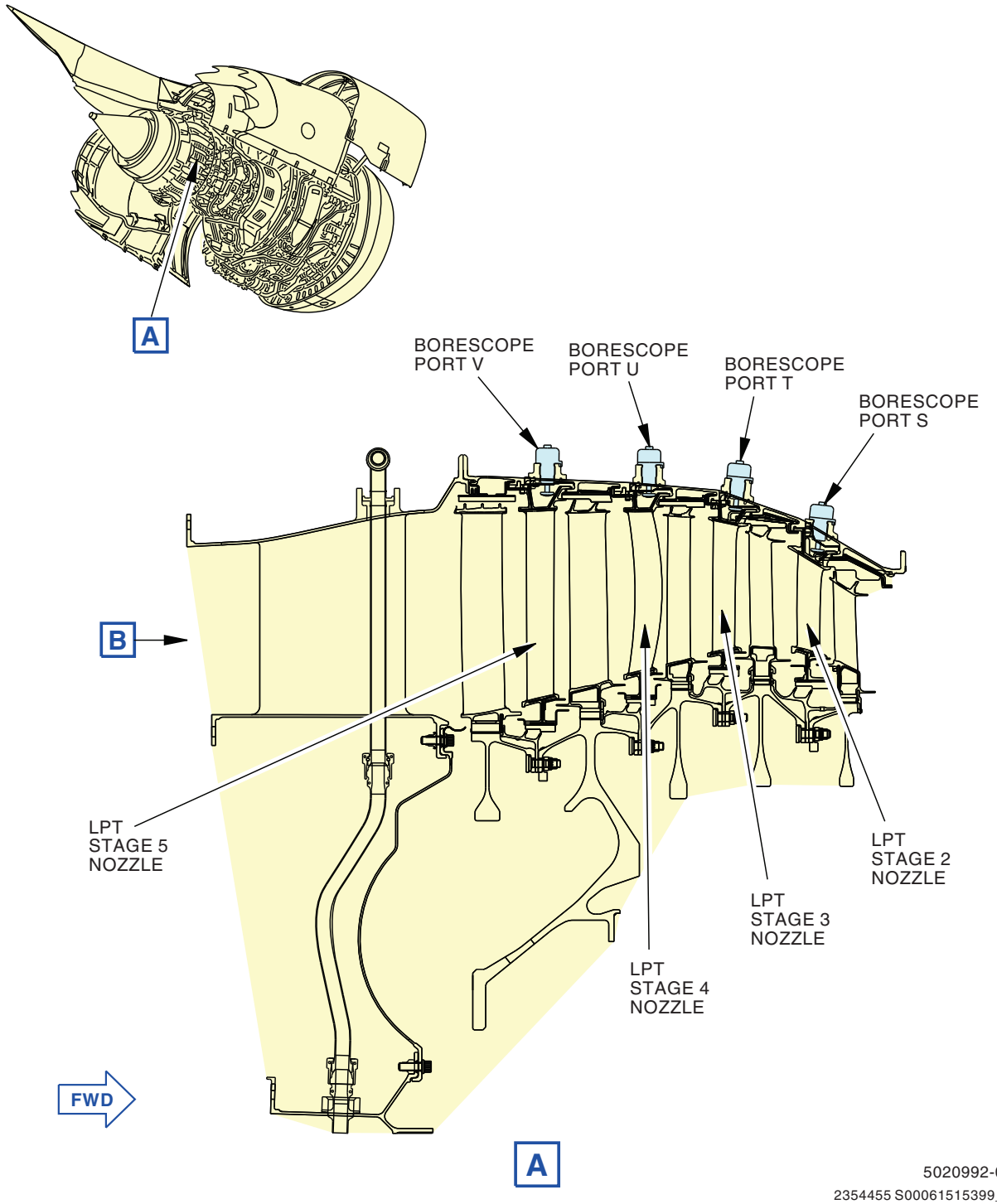


5021008-00

2354454 S00061515398_V2

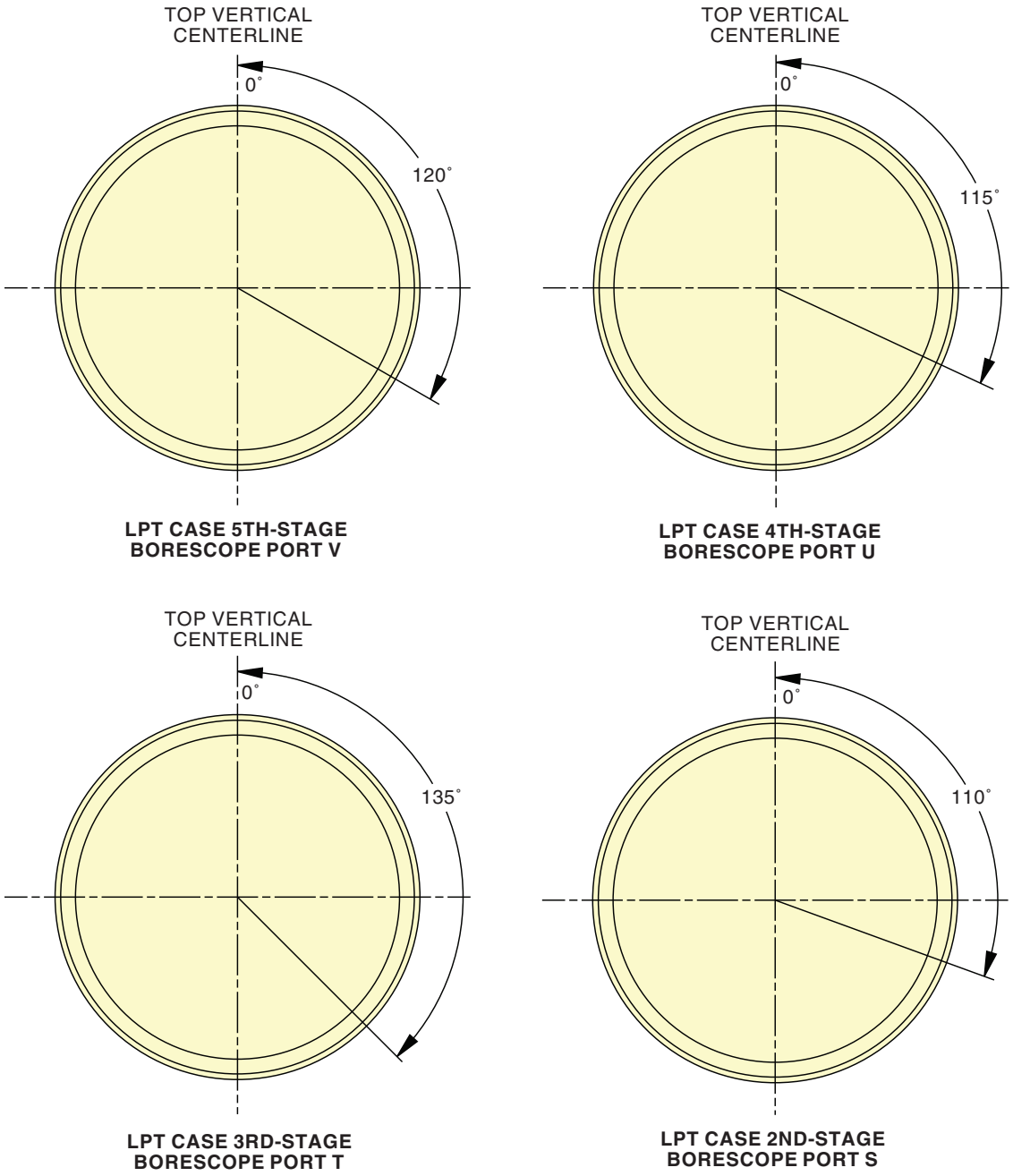
High Pressure Turbine Section Borescope Port Locations
Figure 606/72-00-00-990-843-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Low Pressure Turbine Rotor Section Borescope Port Locations
Figure 607/72-00-00-990-844-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Low Pressure Turbine Rotor Section Borescope Port Locations
Figure 607/72-00-00-990-844-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 72-00-00-290-803-G00

4. Booster Blades and Vanes Borescope Inspection

(Figure 608)

A. General

(1) This task gives the instructions to do the booster blades and vanes borescope inspection.

B. References

Reference	Title
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)
27-81-00-860-804	Leading Edge Flaps and Slats Retraction (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-4302	Set - Borescope, Rigid, Wolf Part #: 856A1320P05 Supplier: 58828
COM-4316	Fiberscope - Flexible Borescope Part #: USEF 2-2.4-1000 Supplier: 7CGG1 Part #: USEF 2-6-1000 Supplier: 7CGG1 Opt Part #: 7110561 Supplier: 32212 Opt Part #: IF2D5 Supplier: 32212 Opt Part #: IF6C5X1-8 Supplier: 32212
SPL-2207	Guide - Borescope Inspection - Booster Part #: 956A8078G01 Supplier: 58828 Part #: 956A8080G02 Supplier: 58828
SPL-4306	Light Source - Borescope, Wolf (300W) Part #: 856A1322P04 Supplier: 58828
STD-858	Tag - DO NOT OPERATE
STD-10718	Fiberscope (Meets General Electric or CFM Standard Practices Manual Specification)

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

F. Prepare for the Borescope Inspection

SUBTASK 72-00-00-860-016-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-00-00-860-017-G00

(2) Do this task: Leading Edge Flaps and Slats Retraction, TASK 27-81-00-860-804.

SUBTASK 72-00-00-860-018-G00

(3) Do this task: Leading Edge Flaps and Slats - Deactivation, TASK 27-81-00-040-801.

SUBTASK 72-00-00-860-019-G00

(4) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Booster Blades and Vanes Borescope Inspection

SUBTASK 72-00-00-211-001-G00

(1) Examine the booster rotor blades [1] and the booster stator vanes [2] as follows:

NOTE: Use non-halogenated solvents for titanium parts.

(a) Location of the borescope ports and borescope view.

1) The borescope ports A and B are located at the 3:30 o'clock position around the booster section.

a) You can see the borescope view of the stages 2 and 3 booster rotor blades [1] and stage 4 booster rotor blades [1] leading edge, through the borescope ports A and B and booster inlet.

b) You can see the borescope view of the stages 2 and 3 booster stator vanes [2] and stage 4 booster stator vanes [2] leading edge, next to the borescope ports A and B.

2) The borescope port C is located between at 4:45 o'clock position around the High Pressure Compressor (HPC) section.

a) You can see the borescope view of the stage 4 booster rotor blades [1] trailing edge and the stage 4 booster stator vanes [2] trailing edge through the HPC borescope port C.

(b) To examine the stage 2, 3 booster rotor blades [1] and the stage 4 booster rotor blades [1] leading edge, do these steps:

1) Use the rigid borescope, COM-4302, with a long right angle extension.

2) Use the light source, SPL-4306.

NOTE: If the fan blades are removed, you can see the leading edges of the stage 2 booster rotor blades without the borescope equipment.

3) Make sure that the input voltage of the light source, SPL-4306, agrees to that of the power supply.

4) Make sure that the light intensity is set to minimum before you connect the light source, SPL-4306, to the power supply.

5) Make sure that the light source and the power supply are correctly grounded.

6) Connect the light source, SPL-4306, to a grounded power supply.

7) Connect the fiber light cable to the rigid borescope, COM-4302, and to the port on the light source, SPL-4306.

8) Turn the light source on.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

9) Engage the borescope and the long right-angle extension as necessary for the inspection.

(c) To examine the stage 2 and 3 booster stator vanes [2] and stage 4 booster stator vanes [2] leading edge.

1) Use a fiberscope, COM-4316, with video probe.

a) If it is necessary, use the borescope guide, SPL-2207.

2) Engage the fiberscope, COM-4316, through the booster inlet and the inspection holes, port A and port B.

(d) To examine the stage 4 booster rotor blades [1] trailing edge and the stage 4 booster stator vanes [2] trailing edge.

1) Use a fiberscope, STD-10718, or a fiberscope, COM-4316, with video probe.

2) Insert through the HPC inspection hole, port C.

3) Engage the fiberscope, STD-10718, or the fiberscope, COM-4316, through the intermediate housing until the stage 4 booster stator vanes [2] trailing edge, and the stage 4 booster rotor blades [1] trailing edge.

WARNING BE CAREFUL WHEN YOU TURN THE FAN ROTOR. MAKE SURE THAT YOU ARE IN A STABLE POSITION. ALSO, YOU MUST WEAR HEAVY PROTECTIVE GLOVES TO PREVENT PERSONAL INJURY.

(e) Do a borescope inspection of the booster rotor blades [1] stages when you turn the fan rotor clockwise (aft looking forward):

1) If you find Foreign Object Damage (FOD) or missing pieces, examine these parts:

a) The downstream booster rotor blades [1].

b) The booster stator vanes [2].

c) The blades 1 and vanes 2 of the high pressure compressor.

2) Cracks and tears on the booster rotor blades [1].

a) Not permitted.

3) Nicks and dents in Area A1 on the booster rotor blades [1].

a) For stage 2: Any amount up to 0.008 in. (0.20 mm) in depth.

b) For stage 3: Any amount up to 0.002 in. (0.05 mm) in depth.

c) For stage 4: Any amount up to 0.01 in. (0.25 mm) in depth.

d) Protrusion is permitted.

e) The indication must not go through the metal.

4) Nicks and dents in Area A2 on the booster rotor blades [1].

a) For stage 2: Any amount up to 0.006 in. (0.15 mm) in depth.

b) For stage 3: Any amount up to 0.002 in. (0.05 mm) in depth.

c) For stage 4: Any amount up to 0.010 in. (0.25 mm) in depth.

d) Protrusion is permitted.

5) Nicks and dents in Area A3 on the booster rotor blades [1].

a) For stage 2: Any amount up to 0.010 in. (0.25 mm) in depth.

b) For stage 3: Any amount up to 0.002 in. (0.05 mm) in depth.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

c) For stage 4: Any amount up to 0.01 in. (0.25 mm) in depth.

d) Protrusion is permitted.

6) Nicks and dents in Area B1 on the booster rotor blades [1].

a) For stage 2: Any amount up to 0.002 in. (0.05 mm) in depth.

b) For stage 3: Any amount up to 0.002 in. (0.05 mm) in depth.

c) For stage 4: Any amount up to 0.0197 in. (0.50 mm) in depth.

d) Protrusion is permitted.

7) Nicks and dents in Area B2 on the booster rotor blades [1].

a) For stage 2: Any amount up to 0.002 in. (0.05 mm) in depth.

b) For stage 3: Any amount up to 0.0157 in. (0.40 mm) in depth.

c) For stage 4: Any amount up to 0.0197 in. (0.50 mm) in depth.

d) Protrusion is permitted.

8) Nicks and dents in Area B3 on the booster rotor blades [1].

a) For stage 2: Any amount up to 0.006 in. (0.15 mm) in depth.

b) For stage 3: Any amount up to 0.0157 in. (0.40 mm) in depth.

c) For stage 4: Any amount up to 0.0197 in. (0.50 mm) in depth.

d) Protrusion is permitted.

9) Nicks and dents in Area B4 on the booster rotor blades [1].

a) For stage 2: Any amount up to 0.002 in. (0.05 mm) in depth.

b) For stage 3: Any amount up to 0.0197 in. (0.50 mm) in depth.

c) For stage 4: Any amount up to 0.0197 in. (0.50 mm) in depth.

d) Protrusion is permitted.

10) Nicks and dents in Area D on the booster rotor blades [1].

a) For stage 2: Any amount up to 0.002 in. (0.05 mm) in depth.

b) For stage 3: Any amount up to 0.01 in. (0.25 mm) in depth.

c) For stage 4: Any amount up to 0.01 in. (0.25 mm) in depth.

d) Protrusion is permitted.

11) Nicks and dents in Area C on the booster rotor blades [1].

a) For stage 2: Any amount up to 0.008 in. (0.20 mm) in depth.

b) For stage 3: Any amount up to 0.01 in. (0.25 mm) in depth.

c) For stage 4: Any amount up to 0.01 in. (0.25 mm) in depth.

d) Protrusion is permitted.

e) The indication must not go through the metal.

12) Missing pieces at the blade tip corner on the booster rotor blades [1].

a) A maximum of 3 blades is permitted with these conditions:

<1> The axial length is less than 0.236 in. (6.0 mm).

<2> The radial length is less than 0.236 in. (6.0 mm).

b) A maximum of 2 blades is permitted with these conditions:

<1> The axial length is less than 0.157 in. (4.0 mm).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

<2> The radial length is less than 0.157 in. (4.0 mm).

13) Wear or local distortions at the blade tip corner on the booster rotor blades [1].
   a) A maximum of 3 blades is permitted with these conditions:
      <1> The axial length is less than 0.236 in. (6.0 mm).
      <2> The radial length is less than 0.236 in. (6.0 mm).
   b) A maximum of 2 blades is permitted with these conditions:
      <1> The axial length is less than 0.157 in. (4.0 mm).
      <2> The radial length is less than 0.157 in. (4.0 mm).
14) Local distortions in Areas A2 and A3 on the booster rotor blades [1].
   a) Not permitted.
15) Local distortions in Areas B2 and B4 on the booster rotor blades [1].
   a) Permitted if the circumferential displacement is less than 0.0197 in. (0.50 mm) from original contour.
   b) A maximum of five parts with distortion per stage is allowed .
16) Local distortions in Areas B1 and B3 on the booster rotor blades [1].
   a) Not permitted.
17) Deposit on the booster rotor blades [1].
   a) All amounts permitted.

(f) When FOD or missing pieces is found on the booster rotor blades [1], do a borescope inspection of the booster stator vanes [2] stage 2, 3 and 4.
1) Missing pieces of the booster stator vanes [2].
   a) Not permitted.
2) Broken vane of the booster stator vanes [2].
   a) Not permitted.
3) Crack and tears in all the areas of the booster stator vanes [2].
   a) Not permitted.
4) Nicks and dents in Area A on the leading edges of the booster stator vanes [2].
   a) For stage 2, 3 and 4: Any amount up to 0.010 in. (0.25 mm) in depth.
   b) Protrusion is permitted.
5) Nicks and dents in Area A on the booster stator vanes [2].
   a) For stage 2, 3 and 4: Any amount up to 0.010 in. (0.25 mm) in depth.
   b) Protrusion is permitted.
   c) The indication must not go through the metal.
6) Nicks and dents in Area A on the trailing edges of the booster stator vanes [2].
   a) For stage 2 and 3: Any amount up to 0.010 in. (0.25 mm) in depth.
   b) For stage 4: Any amount up to 0.002 in. (0.05 mm) in depth.
   c) Protrusion is permitted.
7) Nicks and dents in Area B on the leading edges of the booster stator vanes [2].
   a) For stage 2, 3, and 4: Any amount up to 0.010 in. (0.25 mm) in depth.
   b) Protrusion is permitted.

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8) Nicks and dents in Area B on the trailing edges of the booster stator vanes [2].
   a) For stage 2 and 3: Any amount up to 0.0197 in. (0.50 mm) in depth.
   b) For stage 4: Any amount up to 0.004 in. (0.10 mm) in depth.
   c) Protrusion is permitted.
9) Nicks and dents in Area E of the booster stator vanes [2].
   a) For stage 2, 3, and 4: Any amount up to 0.0197 in. (0.50 mm) in depth.
   b) Protrusion is permitted.
   c) The indication must not go through the metal.
10) Vane looseness of the booster stator vanes [2].
   a) Not permitted.
11) Local distortion in Area E of the booster stator vanes [2].
   a) Not permitted.
12) Local distortion on the leading edge and trailing edge (outside of Area F) of the booster stator vanes [2].
   a) Permitted with these conditions:
      <1> All local distortions are outside area F: Dimension H is more than 0.787 in. (20.0 mm).
      <2> The length L along the leading or trailing edge is not more than 0.295 in. (7.5 mm).
      <3> The maximum dimension X along the circumferential direction is not more than 0.029 in. (0.74 mm).
      <4> A maximum of six vanes affected per stage is allowed.
13) Erosion in Area B on the booster stator vanes [2].
   a) Permitted.
14) Deposit on the booster stator vanes [2].
   a) Permitted.
15) Erosion of abradable lining on outer shroud of the booster stator vanes [2].
   a) Permitted.
16) Separation, flaking, missing of abradable lining on the outer shroud of the booster stator vanes [2].
   a) For stage 2: Any amount, if the maximum surface affected is less than or equal to 15.50 in^2 (10,000 mm^2).
   b) For stage 3: Any amount, if the maximum surface affected is less than or equal to 12.40 in^2 (8000 mm^2).
   c) For stage 4: Any amount, if the maximum surface affected is less than or equal to 10.85 in^2 (7000 mm^2).
   d) A continue-in-service limit of 100 hours is permitted, if the finding is more than the maximum serviceable limit on one of the stage.
17) Pitting of abradable lining on the outer shroud of the booster stator vanes [2].
   a) All amounts permitted.
18) Swelling of abradable lining on the outer shroud of the booster stator vanes [2].

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a) All amounts permitted.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 72-00-00-211-002-G00

(1) Turn the light source to OFF position.

SUBTASK 72-00-00-211-003-G00

(2) Make sure that the main-light-source blower is on till the lamp and the housing are cool.

SUBTASK 72-00-00-860-020-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(3) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 72-00-00-440-008-G00

(4) Do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801.

SUBTASK 72-00-00-860-022-G00

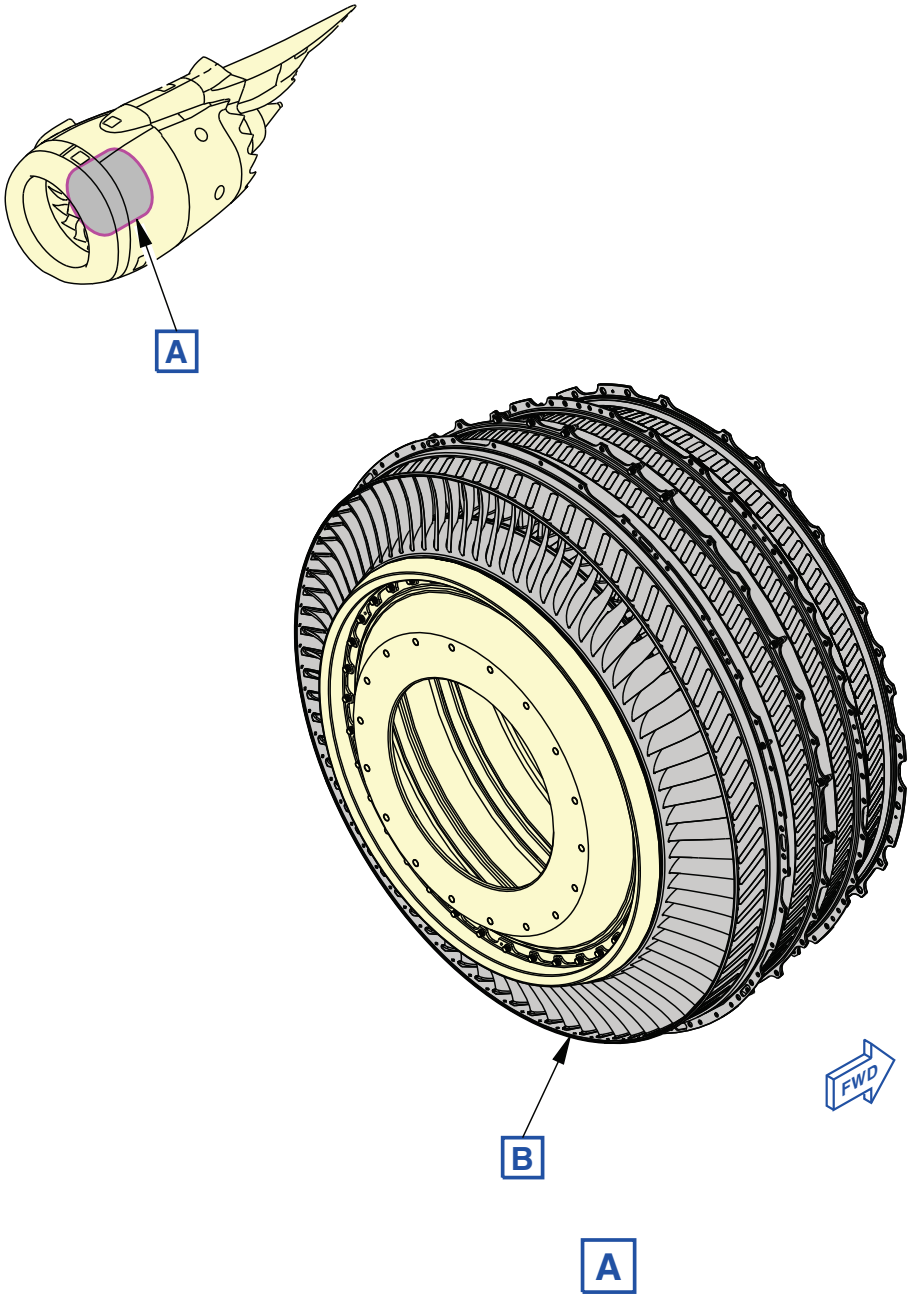
(5) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

END OF TASK

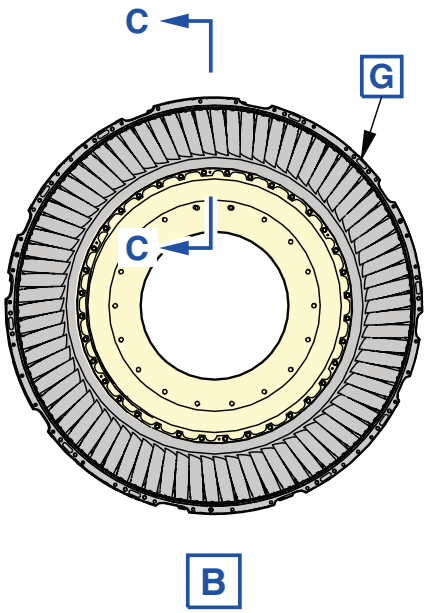
ECCN 9E991 BOEING PROPRIETARY - See title page for details



ICNLEAP1B720000DF030100001A00101
2439158 S0000565491_V3

Booster Blades and Vanes Borescope Inspection
Figure 608/72-00-00-990-849-G00 (Sheet 1 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



ROTOR BLADES STAGE	NUMBER OF BLADES	LEADING EDGE VIEW THRU:	TRAILING EDGE VIEW THRU:
1		--	--
2	58	BOOSTER INLET	PORT A
3	77	PORT A	PORT B
4	80	PORT B	PORT C / PORT B
STATOR VANES STAGE	NUMBER OF VANES	LEADING EDGE VIEW THRU:	TRAILING EDGE VIEW THRU:
1	78	BOOSTER INLET	BOOSTER INLET
2	120	BOOSTER INLET/ PORT A	PORT A
3	120	PORT B	PORT B
4	104	PORT C / PORT B	PORT C

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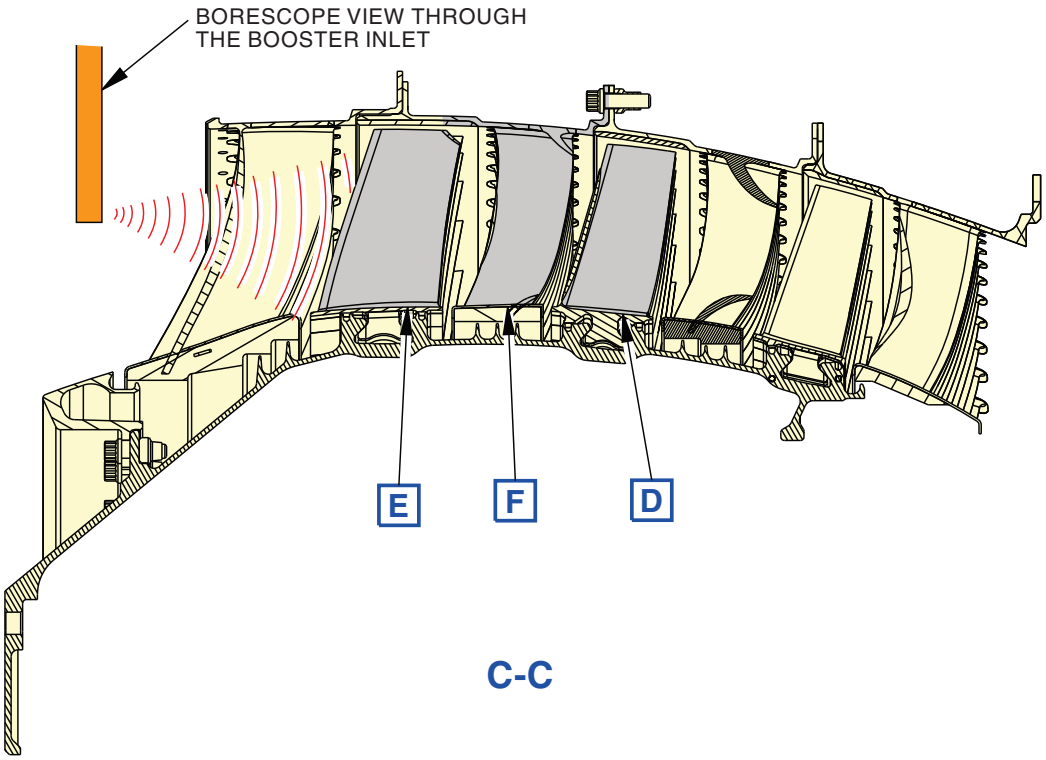
2572774 S0000617840_V2

Booster Blades and Vanes Borescope Inspection
Figure 608/72-00-00-990-849-G00 (Sheet 2 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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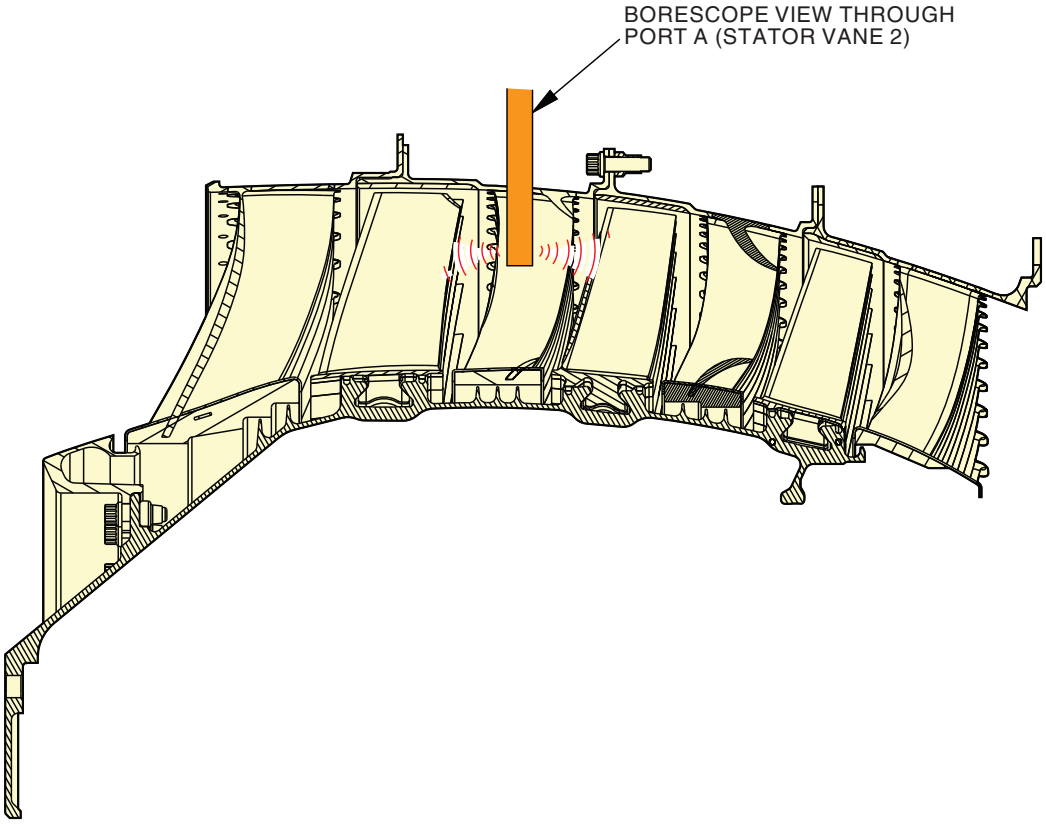
2439159 S0000565492_V2

Booster Blades and Vanes Borescope Inspection
Figure 608/72-00-00-990-849-G00 (Sheet 3 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



THE BORESCOPE PORT "A" IS AT 3:30 O'CLOCK POSITION, AFT LOOKING FORWARD

C-C

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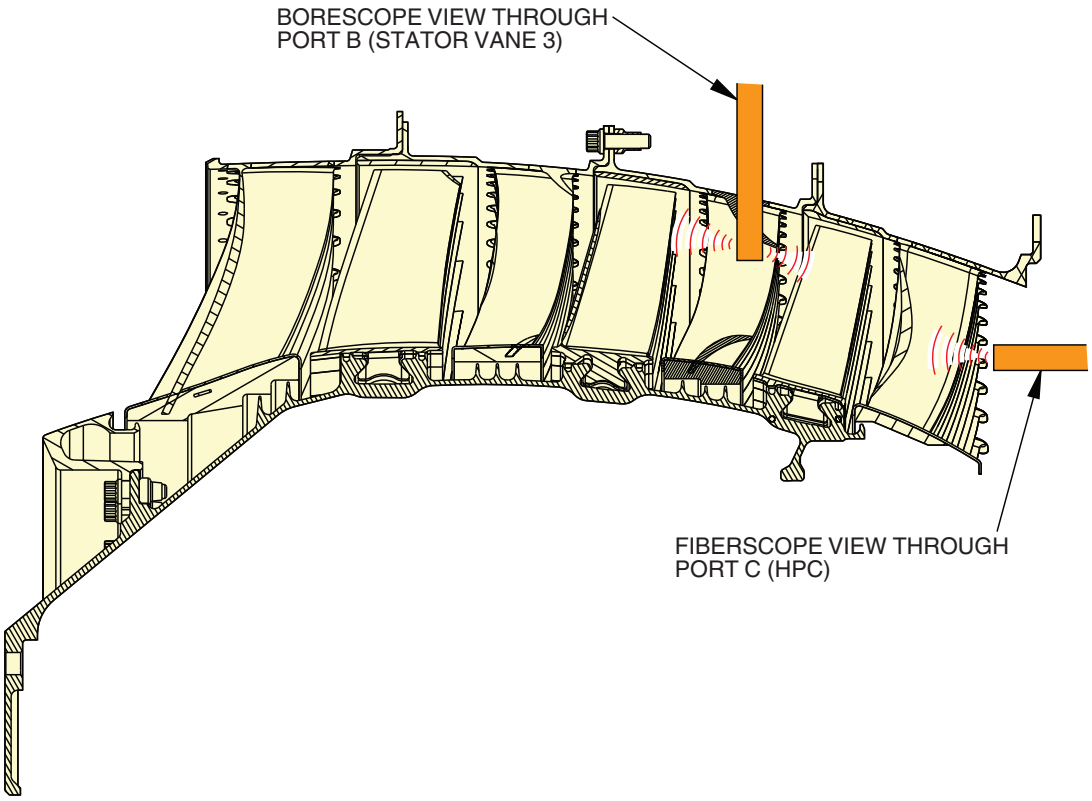
2439219 S0000565493_V3

Booster Blades and Vanes Borescope Inspection
Figure 608/72-00-00-990-849-G00 (Sheet 4 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



THE BORESCOPE PORT "B" IS AT 3:30 O'CLOCK POSITION, AFT LOOKING FORWARD

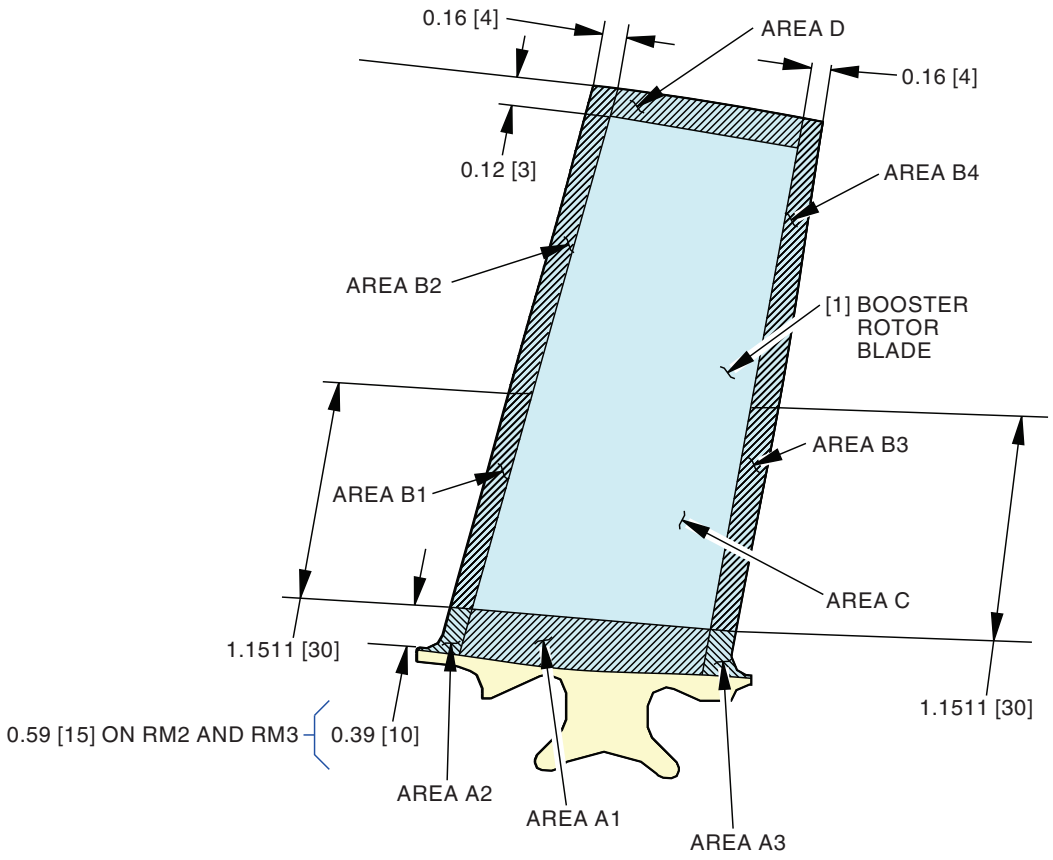
C-C

ICNLEAP1B720000DF030100004A00101

2439225 S0000565494_V4

Booster Blades and Vanes Borescope Inspection
Figure 608/72-00-00-990-849-G00 (Sheet 5 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



D

ZONING FOR PRESSURE AND SUCTION SIDE	AREAS
LEADING EDGES	AREA A2/B1/B2
TRAILING EDGES	AREA A3/B3/B4
BLADE TIP CORNER	NO SPECIFIC AREA DEFINED
FULL BLADE	AREA A1/C/D

NOTE:

DIMENSIONS ARE IN INCHES WITH MILLIMETERS IN BRACKETS.

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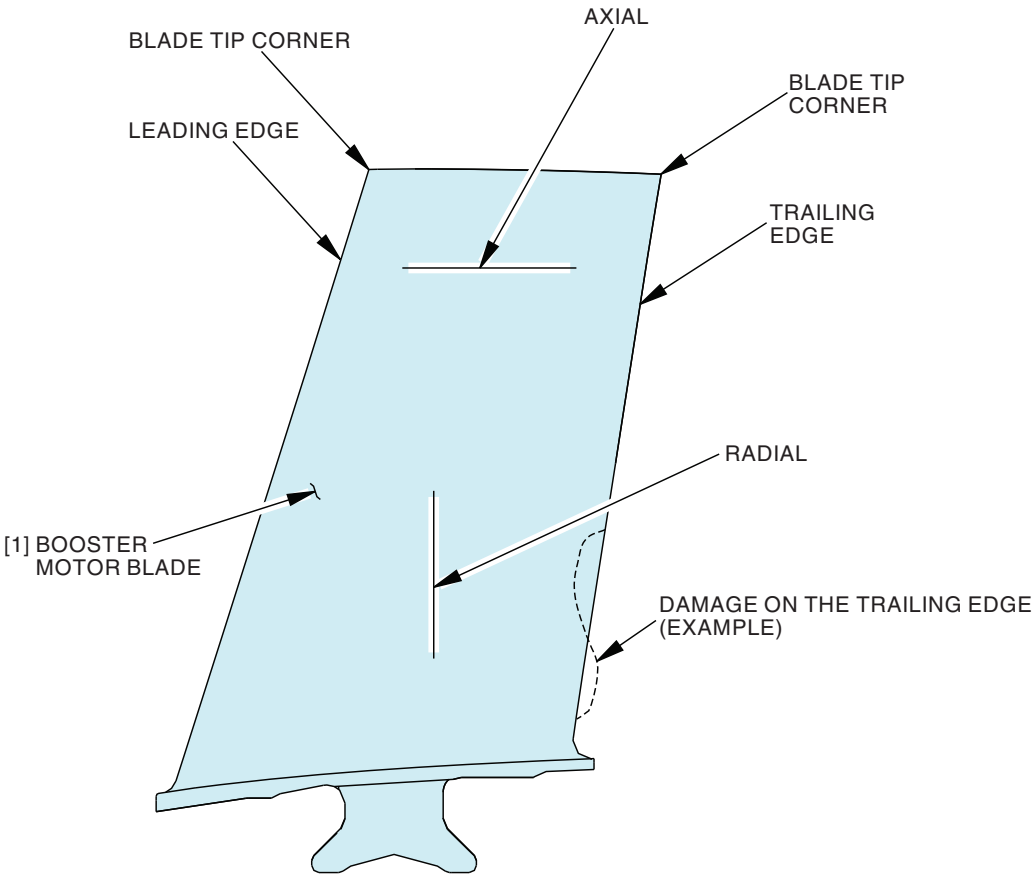
2439262 S0000565495_V4

Booster Blades and Vanes Borescope Inspection
Figure 608/72-00-00-990-849-G00 (Sheet 6 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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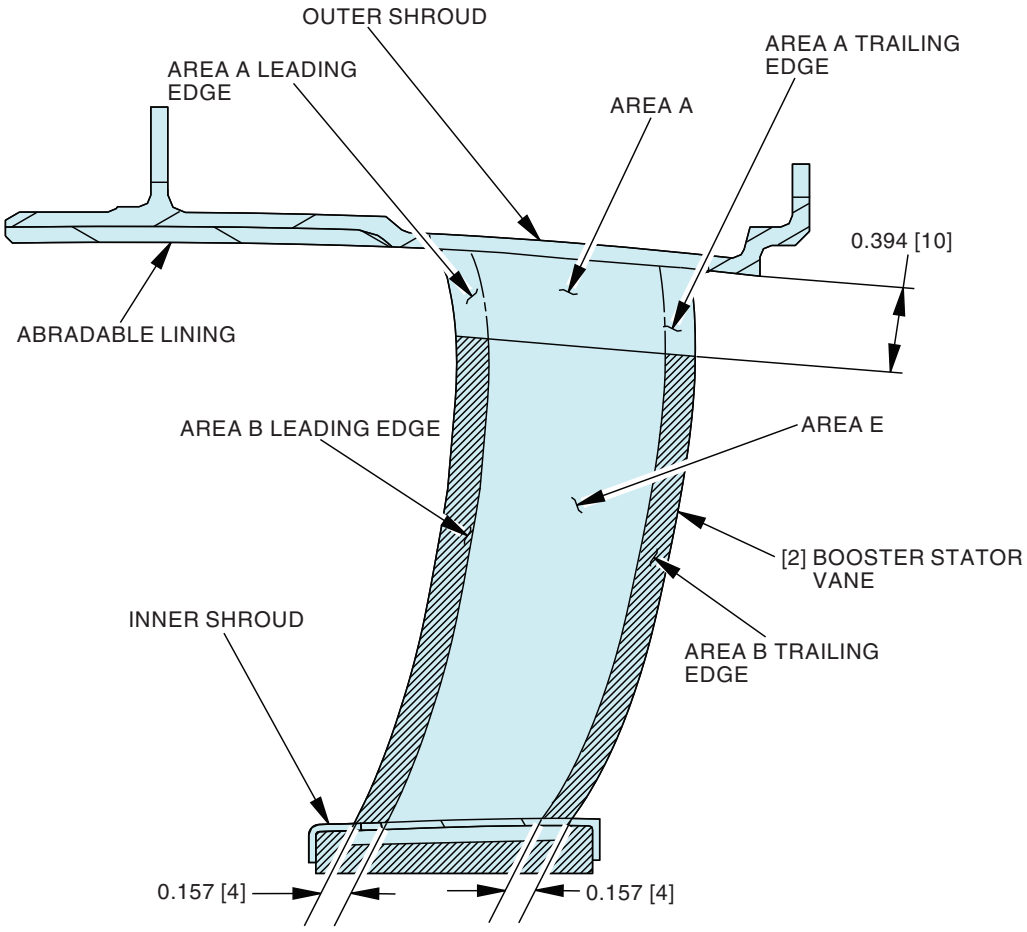
2439267 S0000565496_V2

Booster Blades and Vanes Borescope Inspection
Figure 608/72-00-00-990-849-G00 (Sheet 7 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



F

NOTE:

ALL DIMENSIONS ARE IN INCHES WITH MILLIMETERS IN BRACKETS.

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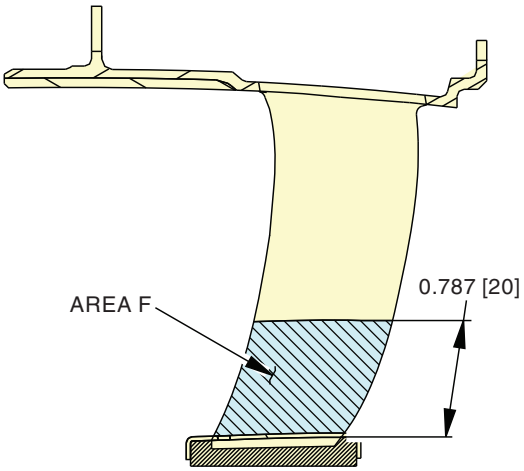
2439273 S0000565497_V4

Booster Blades and Vanes Borescope Inspection
Figure 608/72-00-00-990-849-G00 (Sheet 8 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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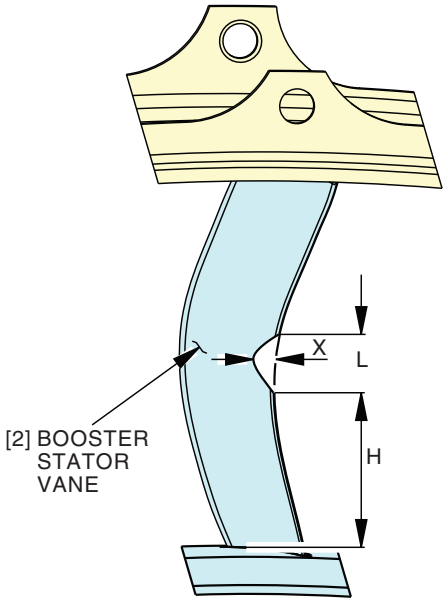

G

NOTE:

DIMENSIONS ARE IN INCHES WITH MILLIMETERS IN BRACKETS.

ICNLEAP1B720000DF030100008A00101

2439280 S0000565892_V3

Booster Blades and Vanes Borescope Inspection
Figure 608/72-00-00-990-849-G00 (Sheet 9 of 9)

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TASK 72-00-00-290-804-G00

5. High Pressure Compressor Section Borescope Detailed Inspection

(Figure 609, Figure 610, Figure 611, Figure 612, and Figure 613)

A. General

(1) This task gives the instructions to do the borescope inspection of the high pressure compressor section.

(2) You need a borescope equipment to do this task.

(3) The high pressure compressor is referred to as the HPC.

B. References

Reference	Title
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)
27-81-00-860-804	Leading Edge Flaps and Slats Retraction (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
72-00-00-000-801-G00	BSI Plug D Removal (P/B 401)
72-00-00-400-801-G00	BSI Plug D Installation (P/B 401)
72-00-00-822-801-G00	Zero-Index of the N2 Rotor Alignment (P/B 201)
72-00-00-910-801-G00	High Pressure Rotor (N2) Turning for Borescope (P/B 201)
72-23-04-000-801-G00	Fan Frame Module Segment Shrouds Removal (P/B 401)
72-23-04-400-801-G00	Fan Frame Module Segment Shrouds Installation (P/B 401)
72-30-00-350-801-G00	High Pressure Compressor (HPC) Rotor Blades/Blisks Borescope Blending Repair (P/B 801)
72-63-04-000-801-G00	Accessory Gearbox Manual Control Cover Removal (P/B 401)
75-31-00-910-801-G00	Variable Stator Vane Actuator Manual Operation (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE
STD-1432	Fiberscope - Flexible Borescope
STD-10717	Light Source - Borescope, (Meets General Electric or CFM Standard Practices Manual Specification)
STD-10718	Fiberscope (Meets General Electric or CFM Standard Practices Manual Specification)
STD-10719	Borescope - Rigid (Meets General Electric or CFM Standard Practices Manual Specification such as: model R857ELSV60)

D. Consumable Materials

Reference	Description	Specification
D00599 [CP2442]	Oil - Engine	
D50355 [CP2961]	Lubricant - Anti-Seize Compound	D6Y31C2

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E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Prepare for the Inspection

SUBTASK 72-00-00-860-023-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-00-00-860-024-G00

(2) Do this task: Leading Edge Flaps and Slats Retraction, TASK 27-81-00-860-804.

SUBTASK 72-00-00-040-001-G00

(3) Do this task: Leading Edge Flaps and Slats - Deactivation, TASK 27-81-00-040-801.

SUBTASK 72-00-00-860-026-G00

(4) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

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(Continued)

Number	Name/Location
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

SUBTASK 72-00-00-040-002-G00

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(5) Do the thrust reverser deactivation for ground maintenance (TASK 78-31-00-040-801-G00).

SUBTASK 72-00-00-941-001-G00

CAUTION MAKE SURE THE TEMPERATURE OF THE ENGINE AND THE ADJACENT AREAS NEAR THE BORESCOPE INSPECTION PORTS ARE LESS THAN 150°F (65.6°C). DO NOT DO A BORESCOPE INSPECTION IF THE ADJACENT AREAS ARE TOO HOT TO TOUCH. DO THE BORESCOPE INSPECTION WHEN THE ENGINE TEMPERATURE IS LESS THAN 150°F (65.6°C). THIS WILL PREVENT DAMAGE TO THE BORESCOPE EQUIPMENT.

(6) Do this task: Preparation for Borescope Inspection, TASK 72-00-00-290-801-G00.

SUBTASK 72-00-00-010-031-G00

(7) Remove the AGB manual control cover (TASK 72-63-04-000-801-G00).

SUBTASK 72-00-00-980-002-G00

(8) Install the electrical motoring tool or hand operated drive tool (TASK 72-00-00-910-801-G00).

SUBTASK 72-00-00-010-007-G00

(9) Remove the segment shroud 3 (TASK 72-23-04-000-801-G00).

SUBTASK 72-00-00-010-009-G00

(10) Remove the borescope plugs from ports C to M as follows (Figure 609):

NOTE: Do not remove the borescope plug D at this time unless the inspection of the stage 1 blisk airfoils shows damage to the stage 1 airfoils that you can repair by borescope blending.

NOTE: Borescope port M is only installed on engines with pre Service Bulletin 72-0161. However, it is not required to do the inspection of the stage 10 blades.

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(a) Remove each borescope plug and attach an identification tag onto the plug.

NOTE: Make sure that you attach an identification tag on each borescope plug to make sure that you re-install the plug in the correct borescope port after you complete the inspection.

(b) To remove the borescope plug D, do this task: BSI Plug D Removal, TASK 72-00-00-000-801-G00.

SUBTASK 72-00-00-980-003-G00

(11) Do this task: Zero-Index of the N2 Rotor Alignment, TASK 72-00-00-822-801-G00.

SUBTASK 72-00-00-480-003-G00

(12) Turn on the light source, STD-10717, and insert the rigid borescope, STD-10719, or fiberscope, STD-10718, or fiberscope, STD-1432, into position for the inspection.

(a) It can be necessary to manually shift the VSV system to permit inspection of stages 1 through 4 blisks for 6mm borescope probe (TASK 75-31-00-910-801-G00).

(b) As an alternative, to make less access restrictions, use a 4mm borescope probe.

SUBTASK 72-00-00-980-004-G00

(13) Manually or electrically motor the HPC rotor (TASK 72-00-00-910-801-G00).

SUBTASK 72-00-00-290-004-G00

(14) Make sure that you follow the inspection instructions and limits closely in the inspection task.

NOTE: Table 601 gives the HPC blade count by stage.

Table 601/72-00-00-993-812-G00 High Pressure Compressor Stage Blades

Stage Number	Number of Blades	Notes
PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0182		
1	25	Stage 1 of the HPC is a blisk
PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0182		
1	21	Stage 1 of the HPC is a blisk
PTF ALL		
2	34	Stage 2 of the HPC is a blisk
3	39	Stage 3 of the HPC is a blisk
4	49	Stage 4 of the HPC is a blisk
5	56	Stage 5 of the HPC is a blisk
6	66	
7	61	
8	67	
9	64	
10	68	

H. High Pressure Compressor Section Borescope Detailed Inspection

SUBTASK 72-00-00-211-004-G00

(1) Examine the HPC stage 1 blisk airfoils as follows (View B, Figure 612 (Sheet 2)):

(a) Cracks

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1) Not permitted.

(b) Particle buildup on airfoil

1) Permitted.

(c) Nicks, dents, pits, and scratches on the center of the airfoil (View H, Area E, Figure 612 (Sheet 5)):

1) Permitted with these conditions:

a) The damage is more than 0.100 in. (2.54 mm) from the leading edge, trailing edge, or tip.

NOTE: Root fillet is not included.

b) The nick or dent does not make a hole in the blade.

c) The depth is not over 0.003 in. (0.08 mm).

d) The defects are not on the same location on the opposite sides of the airfoil.

(d) Nicks, dents, pits, and scratches in the airfoil root radius and flowpath surface (View H, Area A, and flowpath, Figure 612 (Sheet 5)):

1) Permitted up to 0.002 in. (0.05 mm) in depth.

(e) Nicks, dents, pits, and material that is missing in the leading and trailing edges, and tip corners, but excluding the airfoil root radius (View H, Figure 612 (Sheet 5)):

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0182

1) Permitted up to 0.014 in. (0.36 mm) in depth.

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0182

2) Permitted up to 0.013 in. (0.33 mm) in depth.

PTF ALL

3) From 40% of DIM TH/TL up to and including the tip (Areas C and D).

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0182

a) Permitted up to 0.030 in. (0.76 mm) in depth.

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0182

b) Permitted up to 0.020 in. (0.51 mm) in depth.

PTF ALL

(f) Tears in the leading and trailing edges (View H, Areas B, C, and D, Figure 612 (Sheet 5))

1) Not permitted.

a) Refer to High Pressure Compressor (HPC) Rotor Blades/Blisks Borescope Blending Repair, TASK 72-30-00-350-801-G00 for the repair limits.

(g) Scratches in the in the leading and trailing edges (View H, Areas B, C, and D, Figure 612 (Sheet 5)).

1) Permitted up to 0.010 in. (0.25 mm) in depth.

(h) Airfoil tip corners that are missing

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1) Tip corners 0.085 in. (2.16 mm) missing in chord-wise and span-wise direction up to a maximum of 6 airfoils per stage is permitted.

NOTE: Some airfoil tip corners can be chamfered at both leading and trailing edges. The chamfers are used to identify blades with tip notches. This condition is fully serviceable and is not the same as missing corners (View D, Figure 612 (Sheet 3)).

(i) Curled blade tips

1) Total max allowable blockage is 0.500 in^2 (3.23 cm^2).

NOTE: There is no limit on the number of airfoils affected as long as it does not exceed the max allowable blockage.

2) Tears or cracks as a result of the curled blade tip are not permitted.

3) Measure the forward tip corner and axial displacements and the maximum deflection from the initial contour of a tip curl on the forward tip corner of the stage 1 HPC blisk airfoils as follows (View G, Figure 612 (Sheet 4)):

a) Measure and record the radial displacement (Dim. L) down the leading edge.

b) Measure and record the axial displacement (Dim. Y) along the blade tip chord.

c) Measure and record the maximum deflection (Angle A) from the initial contour.

4) Find the total blockage area corrective factor as follows:

a) If the maximum deflection (Angle A) from an initial contour is not more than 25 degrees, use the Corrective Factor (CF) of 0.22.

b) If the maximum deflection (Angle A) from an initial contour is more than 25 degrees and not more than 45 degrees, use the CF of 0.36.

c) If the maximum deflection (Angle A) from an initial contour is more than 45 degrees, use the CF of 0.50.

5) Calculate the Blockage Area (BA) as follows:

a) Multiply the radial displacement (L) by the axial displacement (Y) and by the appropriate Corrective Factor.

NOTE: BA = DIM. L x DIM. Y x CF.

6) If you find more than one tip curl, add the Blockage Area (BA) corresponding to each tip curl together and then consider the Total Blockage Area (TBA).

NOTE: TBA = BA1 + BA2 + BA3.

7) If the TBA is more than 0.500 in^2 (3.23 cm^2), remove the engine before the next flight.

(j) Distortion or bulging of the leading edge

1) Maximum bulge depth (T) of 0.040 in. (1.02 mm) on a maximum of 3 airfoils per stage is permitted (View C, Figure 612 (Sheet 3)).

(k) Erosion on the tip and tip corners

1) Maximum chord loss of 0.062 in. (1.57 mm) is permitted.

(l) Burrs and high metal on the blade tips due to rub

1) Permitted.

SUBTASK 72-00-00-220-003-G00

(2) Examine the HPC stages 2-5 blisk airfoils as follows (View B, Figure 612 (Sheet 2)):

(a) Particle buildup on airfoil

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1) Permitted.

(b) Cracks

1) Not permitted.

(c) Nicks, dents, pits, and scratches on the center of the airfoil (View H, Area E, Figure 612 (Sheet 5)):

1) Permitted with these conditions:

a) The damage is more than 0.10 in. (2.5 mm) from the leading edge, trailing edge, or tip.

NOTE: Root fillet is not included.

b) The nick or dent does not make a hole in the blade.

c) Permitted up to 0.003 in. (0.08 mm) in depth.

d) The defects are not on the same location on the opposite sides of the airfoil.

(d) Nicks, dents, pits, and scratches in the airfoil root radius and platform flowpath surface (View H, Area A and flowpath, Figure 612 (Sheet 5)):

1) Permitted up to 0.002 in. (0.05 mm) in depth.

(e) Nicks, dents, pits, and material that is missing in the leading and trailing edges, tip corners, but excluding the airfoil root radius (View H, Figure 612 (Sheet 5)):

1) Stage 1

a) See previous stage 1 limits.

2) Stage 2

a) From the top of the root radius to 40% of TH/TL, Area B

<1> Permitted up to 0.020 in. (0.51 mm) in depth.

b) From 40% of TH/TL up to and including the tip (Areas C and D)

<1> Permitted up to 0.022 in. (0.56 mm) in depth.

3) Stage 3

a) From the top of the root radius to 40% of TH/TL, Area B

<1> Permitted up to 0.030 in. (0.76 mm) in depth.

b) From 40% of TH/TL up to and including the tip (Areas C and D)

<1> Permitted up to 0.027 in. (0.69 mm) in depth.

4) Stage 4

a) From the top of the root radius to 40% of TH/TL, Area B

<1> Permitted up to 0.030 in. (0.76 mm) in depth.

b) From 40% of TH/TL up to and including the tip (Areas C and D)

<1> Permitted up to 0.012 in. (0.30 mm) in depth.

5) Stage 5

a) From the top of the root radius to 40% of TH/TL, Area B

<1> Permitted up to 0.030 in. (0.76 mm) in depth.

b) From 40% of TH/TL up to and including the tip (Areas C and D)

<1> Permitted up to 0.030 in. (0.76 mm) in depth.

(f) Tears in the leading and trailing edges (View H, Areas B, C, and D, Figure 612 (Sheet 5)):

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1) Not serviceable.

a) Refer to High Pressure Compressor (HPC) Rotor Blades/Blisks Borescope Blending Repair, TASK 72-30-00-350-801-G00 for the repair limits.

(g) Scratches in the in the leading and trailing edges (View H, Areas B, and C, Figure 612 (Sheet 5)):

1) Permitted up to 0.005 in. (0.13 mm) in depth.

(h) Distortion or bulging of the leading edge

1) Maximum bulge deflection (T) of 0.040 in. (1.02 mm) is permitted up to a maximum of 3 airfoils per stage (View C, Figure 612 (Sheet 3)).

(i) Curled blade tips

1) Curled blade tips less than 0.200 in. (5.08 mm) in length (DIM. Y) along the airfoil tip chord are permitted (View G, Figure 612 (Sheet 4)).

NOTE: Tears or cracks as a result of the curled blade tip are not permitted.

2) The maximum number of airfoils permitted with tip curls are as follows:

a) Stage 2: 3 airfoils

b) Stage 3: 4 airfoils

c) Stage 4: 5 airfoils

d) Stage 5: 5 airfoils

(j) Airfoil tip corners that are missing

1) Tip corners 0.085 in. (2.16 mm) missing in chord-wise and span wise direction up to a maximum of 6 airfoils per stage is permitted.

NOTE: Some airfoil tip corners can be chamfered at both leading and trailing edges. The chamfers are used to identify blades with tip notches. This condition is fully serviceable and is not the same as missing corners (View D, Figure 612 (Sheet 3)).

(k) Erosion of the tip and tip corners

1) Maximum chord loss of 0.066 in. (1.68 mm) is permitted.

(l) Burrs and high metal on the blade tips due to rub

1) Permitted.

SUBTASK 72-00-00-220-004-G00

(3) Examine the stage 6 - 9 HPC blades as follows (View E, Figure 612 (Sheet 2)):

NOTE: When you do a borescope inspection, gaps between blade platforms can be visible. The gaps are normal and there is no limit to the gap size.

NOTE: Borescope port M is only installed on engines with pre Service Bulletin 72-0161. However, it is not required to do the inspection of the stage 10 blades. If you do an inspection of the HPC stage 10 blades, all inspection criteria and limits from steps (a) to (m) apply.

(a) Particles on the blade surface

1) Permitted.

(b) Cracks

1) Not permitted.

(c) Scratches in all areas of the airfoil

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1) Scratches that are less than 0.003 in. (0.08 mm) in depth are permitted.

(d) Nicks, dents, pits, and scratches on the center of the airfoil (View J, Area E, Figure 612 (Sheet 6)):

1) Permitted with these conditions:

a) The nick or dent is more than 0.040 in. (1.02 mm) from the leading edge, trailing edge, and the tip.

b) The nick or dent does not make a hole in the blade.

c) The nick or dent is less than 0.005 in. (0.13 mm) in depth.

(e) Nicks, and dents in the airfoil root radius (View J, Area A, Figure 612 (Sheet 6)):

1) Nicks and dents that are less than 0.003 in. (0.08 mm) in depth are permitted.

(f) Nicks, dents, and scratches on the platform flowpath surface (View E, Figure 612 (Sheet 2)):

1) Any amount and length, 0.003 in. (0.08 mm) in depth, and not more than two defects with a maximum depth of 0.005 in. (0.13 mm).

(g) Nicks, dents, and scratches on the platform flowpath edges (View E, Figure 612 (Sheet 2)):

1) Any amount and length, 0.005 in. (0.13 mm) in depth.

(h) Nicks, dents, and missing material in the leading and trailing edges, tip corners, but excluding the airfoil root radius (View J, Figure 612 (Sheet 6)):

1) Stage 6

a) From the top of the root radius to 40% of TH/TL, Area B

<1> Permitted up to 0.027 in. (0.69 mm) in depth.

b) From 40% of TH/TL up to and including the tip (Areas C and D)

<1> Permitted up to 0.027 in. (0.69 mm) in depth.

2) Stage 7

a) From the top of the root radius to 40% of TH/TL, Area B

<1> Permitted up to 0.030 in. (0.76 mm) in depth.

b) From 40% of TH/TL up to and including the tip (Areas C and D)

<1> Permitted up to 0.035 in. (0.89 mm) in depth.

3) Stage 8

a) From the top of the root radius to 40% of TH/TL, Area B

<1> Permitted up to 0.030 in. (0.76 mm) in depth.

b) From 40% of TH/TL up to and including the tip (Areas C and D)

<1> Permitted up to 0.035 in. (0.89 mm) in depth.

4) Stage 9

a) From the top of the root radius to 40% of TH/TL, Area B

<1> Permitted up to 0.024 in. (0.61 mm) in depth.

b) From 40% of TH/TL up to and including the tip (Areas C and D)

<1> Permitted up to 0.026 in. (0.66 mm) in depth.

5) Stage 10

a) From the top of the root radius to 40% of TH/TL, Area B

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<1> Permitted up to 0.030 in. (0.76 mm) in depth.

b) From 40% of TH/TL up to and including the tip (Areas C and D)

<1> Permitted up to 0.030 in. (0.76 mm) in depth.

(i) Tears in the leading and trailing edges not including root radius (View J, Areas B, C, and D, Figure 612 (Sheet 6)):

1) Not permitted

a) Refer to High Pressure Compressor (HPC) Rotor Blades/Blisks Borescope Blending Repair, TASK 72-30-00-350-801-G00 for the repair limits.

(j) Airfoil tip corners that are missing

1) Tip corners 0.085 in. (2.16 mm) missing in chord-wise and span-wise direction up to a maximum of 6 airfoils per stage is permitted (View D, Figure 612 (Sheet 3)):

NOTE: Some airfoil tip corners can be chamfered at both leading and trailing edges. The chamfers are used to identify blades with tip notches. This condition is fully serviceable and is not the same as missing corners.

(k) Curled blade tips (View G, Figure 612 (Sheet 4)):

1) A maximum number of 6 blades can have curled blade tips with these conditions:

a) There are no tears or cracks as a result of the curled blade tip.

b) Stage 6

<1> The curled blade tip is not more than 0.148 in. (3.76 mm) in length along the airfoil tip chord (DIM. Y).

c) Stage 7

<1> The curled blade tip is not more than 0.163 in. (4.14 mm) in length along the airfoil tip chord (DIM. Y).

d) Stage 8

<1> The curled blade tip is not more than 0.151 in. (3.84 mm) in length along the airfoil tip chord (DIM. Y).

e) Stage 10

<1> The curled blade tip is not more than 0.131 in. (3.33 mm) in length along the airfoil tip chord (DIM. Y).

2) A maximum number of 7 HPC Stage 9 blades can have curled blade tips with these conditions:

a) There are no tears or cracks as a result of the curled blade tip.

b) The curled S9 blade tip is not more than 0.145 in. (3.68 mm) in length along the airfoil tip chord (DIM. Y).

(l) Distortion or bulging of the leading edge (View C, Figure 612 (Sheet 3))

1) Maximum bulge deflection (T) of 0.010 in. (0.25 mm) is permitted up to a maximum of 6 airfoils per stage.

(m) Burrs and high metal on the blade tips due to rub

1) Permitted.

(n) Radial tip cracks

1) Not permitted.

(o) Erosion of the tip and tip corners

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1) Not more than 0.030 in. (0.76 mm) is permitted in the chord wise direction.

(p) Blade platforms shingling or overlapping (Figure 612 (Sheet 7))

1) Not permitted.

SUBTASK 72-00-00-220-045-G00

(4) Examine the stage 6 - 10 blade locking lugs for cracks.

(a) One crack in circumferential direction on one lock lug is permitted for each stage. The other lock lug must be unaffected.

NOTE: The limit for circumferential cracking is applicable only in highlighted areas (lug's top outer face and sides) (Figure 613).

I. Put the Airplane Back to Its Usual Condition

SUBTASK 72-00-00-080-005-G00

(1) When the inspection is complete, do these steps:

(a) Remove the rigid borescope, STD-10719, or fiberscope, STD-10718, or fiberscope, STD-1432, from the borescope port.

(b) Turn the light source, STD-10717, to OFF.

(c) Let the light source blower stay ON until the lamp and case become cool.

SUBTASK 72-00-00-010-010-G00

(2) Install the borescope plug C as follows (Figure 609):

(a) Do a visual inspection of the borescope port (TASK 72-00-00-290-801-G00).

(b) Remove the identification tag from borescope plugs C.

(c) Apply engine oil, D00599 [CP2442], to the borescope plug threads and the mating surfaces.

(d) Install the plug in the borescope port.

(e) Tighten the plug to 161.0 in-lb (18.2 N·m) - 189.0 in-lb (21.4 N·m).

SUBTASK 72-00-00-420-006-G00

(3) If removed, install the borescope plug D as follows (Figure 609):

(a) Remove the identification tag from borescope plug D.

(b) Do this task: BSI Plug D Installation, TASK 72-00-00-400-801-G00.

SUBTASK 72-00-00-010-011-G00

(4) Install the borescope plugs E to H as follows (Figure 609):

(a) Do a visual inspection of the borescope plugs (TASK 72-00-00-290-801-G00).

(b) Remove the identification tags from borescope plugs E to H.

(c) Put Tiodize T8E-H grease, D50355 [CP2961], on the borescope plug threads.

NOTE: Do not apply anti-seize compound on the mating surfaces.

(d) Install the plugs in the correct borescope ports.

(e) Tighten each borescope plug to 189.0 in-lb (21.4 N·m) - 221.0 in-lb (25.0 N·m).

(f) Break the torque on the borescope plugs one plug at a time.

(g) Rotate the plugs a minimum of 180° to loosen the plugs.

(h) Tighten each borescope plug again to 189.0 in-lb (21.4 N·m) - 221.0 in-lb (25.0 N·m).

SUBTASK 72-00-00-010-012-G00

(5) Install the borescope plugs I, J, L, and M as follows (Figure 610):

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(a) Do a visual inspection of the borescope plugs (TASK 72-00-00-290-801-G00).

(b) Remove the identification tags from borescope plugs I, J, L, and M.

(c) Put Tiodize T8E-H grease, D50355 [CP2961], on the borescope plug threads and mating surfaces.

(d) Install the plugs in the correct borescope port as follows:

1) Put the plug into the correct port and hand tighten the inner thread.

NOTE: Make sure that the inner thread is correctly mated.

NOTE: While you install the inner thread, the outer thread is pushed up by the spring. Do not install both inner and outer threads at one time.

2) Tighten the inner thread to 76.0 in-lb (8.6 N·m) - 84.0 in-lb (9.5 N·m).

3) Loosen the inner borescope plug one quarter turn.

4) Tighten the inner borescope plug to 15.0 in-lb (1.7 N·m) - 20.0 in-lb (2.3 N·m) above the run-on torque.

5) Compress the spring and install the outer thread and hand-tighten the borescope plug.

6) Tighten the outer thread to 195.0 in-lb (22.0 N·m) - 215.0 in-lb (24.3 N·m).

SUBTASK 72-00-00-010-013-G00

(6) Install the borescope plug K as follows (Figure 609):

(a) Do a visual inspection of the borescope port (TASK 72-00-00-290-801-G00).

(b) Remove the identification tag from borescope plug K.

(c) Put Tiodize T8E-H grease, D50355 [CP2961], on the borescope plug threads.

NOTE: Do not apply anti-seize compound on the mating surfaces.

(d) Install the plug in the correct borescope port.

(e) Tighten the borescope plug to 189.0 in-lb (21.4 N·m) - 221.0 in-lb (25.0 N·m).

(f) Break the torque on the borescope plugs.

(g) Rotate the plug a minimum of 180° to loosen the plug.

(h) Tighten the borescope plug again to 189.0 in-lb (21.4 N·m) - 221.0 in-lb (25.0 N·m).

SUBTASK 72-00-00-410-008-G00

(7) Install the segment shroud 3 (TASK 72-23-04-400-801-G00).

SUBTASK 72-00-00-080-006-G00

(8) Remove the electrical motoring tool and install the Accessory Gear Box (AGB) cover plate (TASK 72-00-00-910-801-G00).

SUBTASK 72-00-00-860-027-G00

WARNING

OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(9) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 72-00-00-440-001-G00

(10) Do the activation procedure for the thrust reverser (TASK 78-31-00-440-801-G00).

SUBTASK 72-00-00-860-028-G00

(11) Do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801.

SUBTASK 72-00-00-860-029-G00

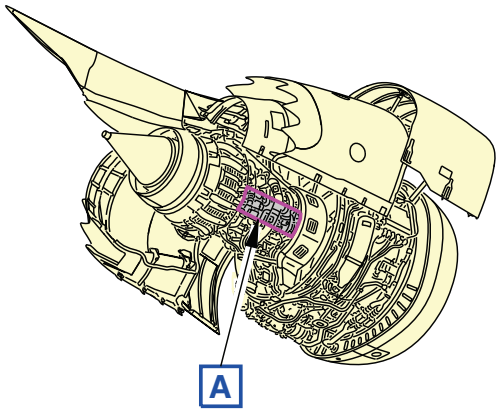
(12) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

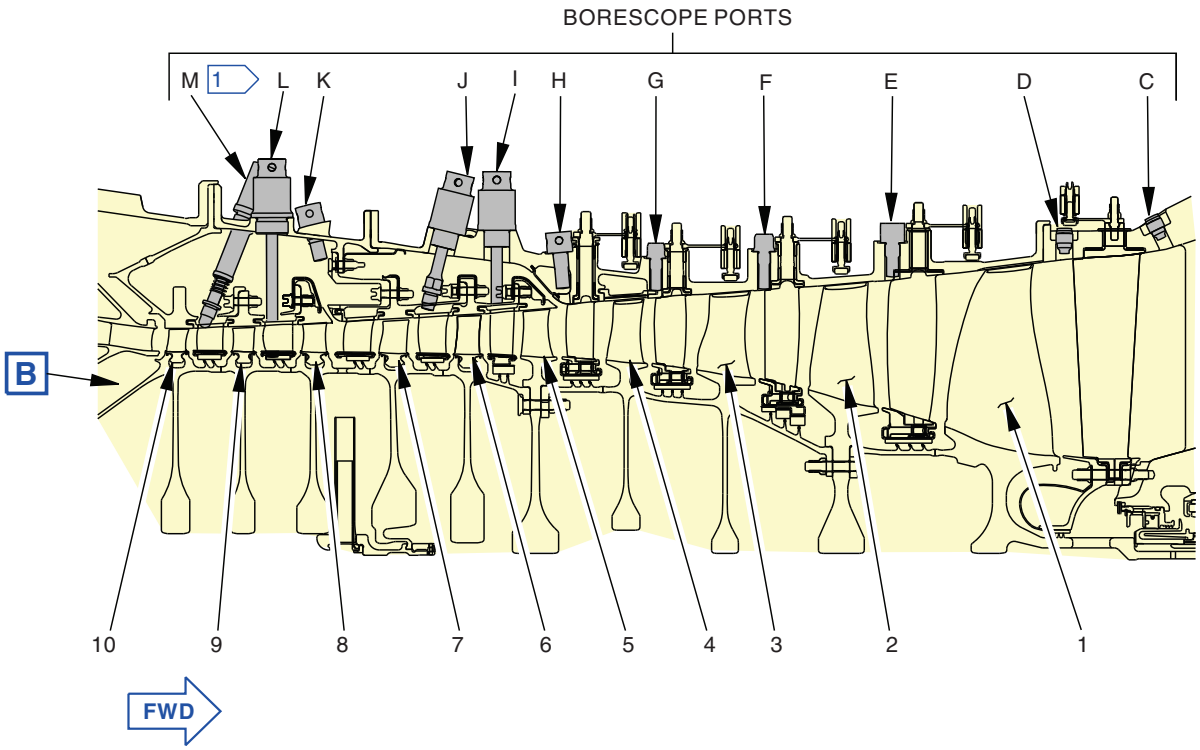
(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details





HIGH PRESSURE COMPRESSOR SECTION BORESCOPE PORT LOCATIONS

A

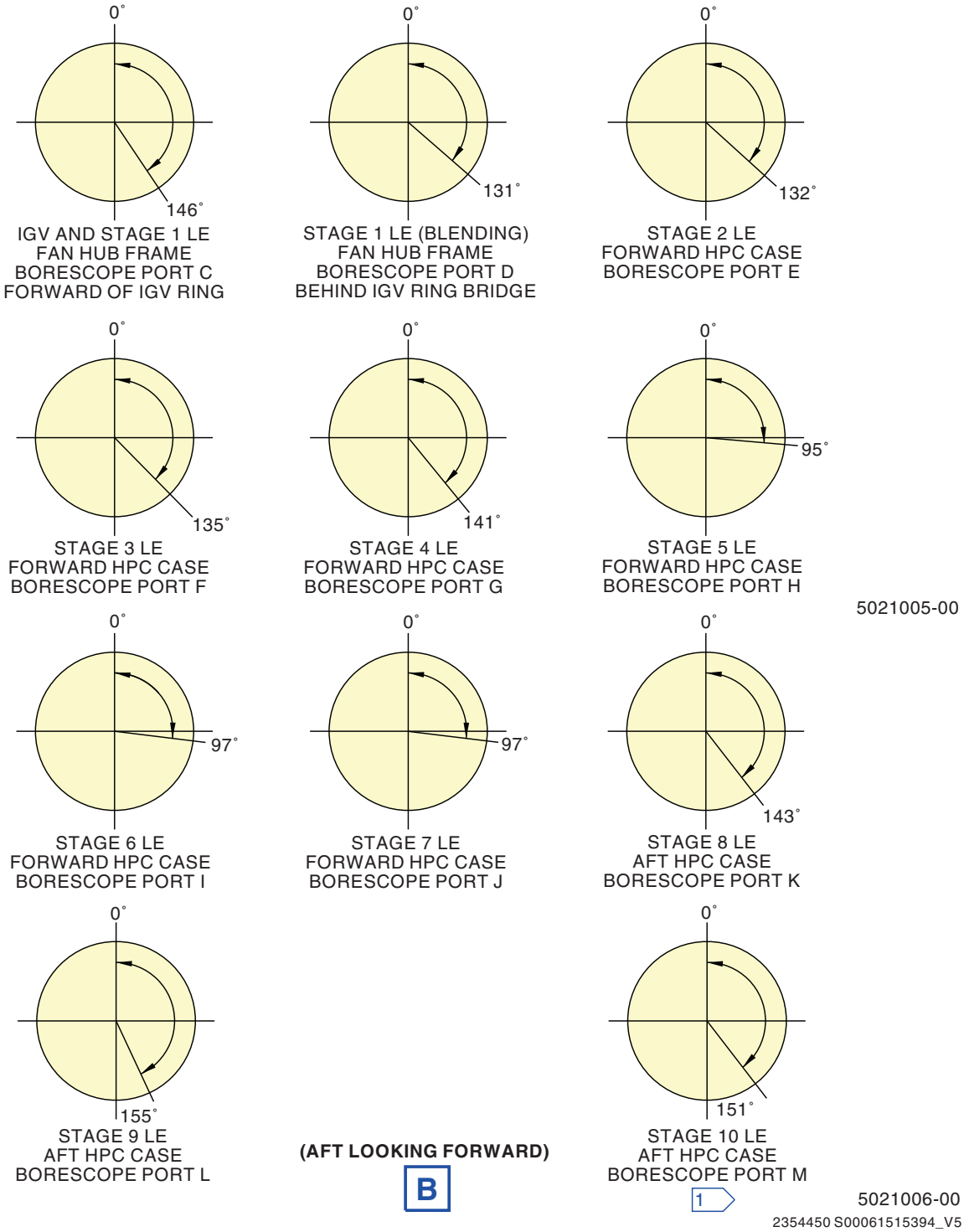
5036149-00

1 PRE LEAP-1B SERVICE BULLETIN 72-0161

2354449 S00061515393_V4

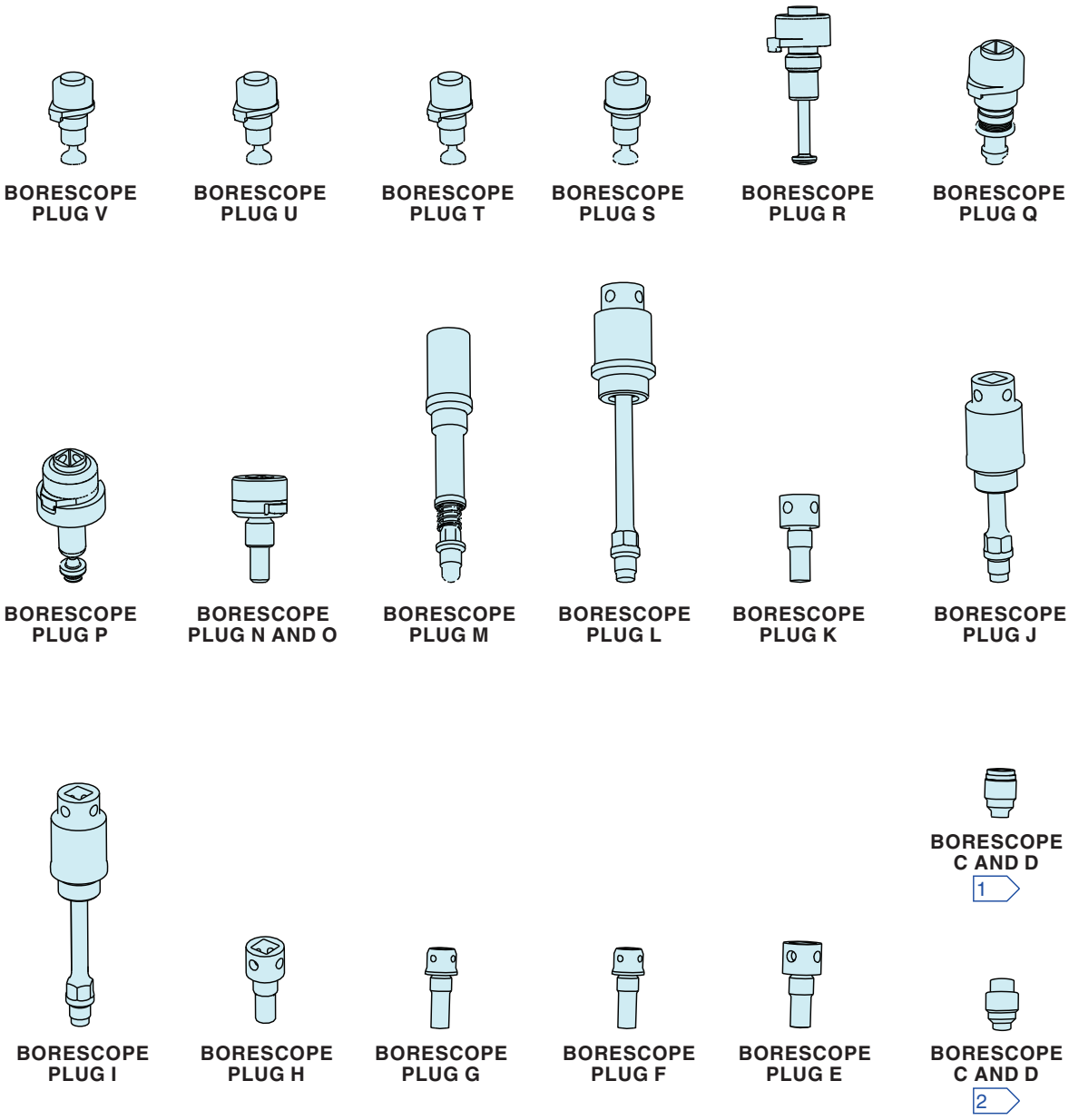
High Pressure Compressor Section Borescope Port Locations
Figure 609/72-00-00-990-812-G00 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



High Pressure Compressor Section Borescope Port Locations
Figure 609/72-00-00-990-812-G00 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

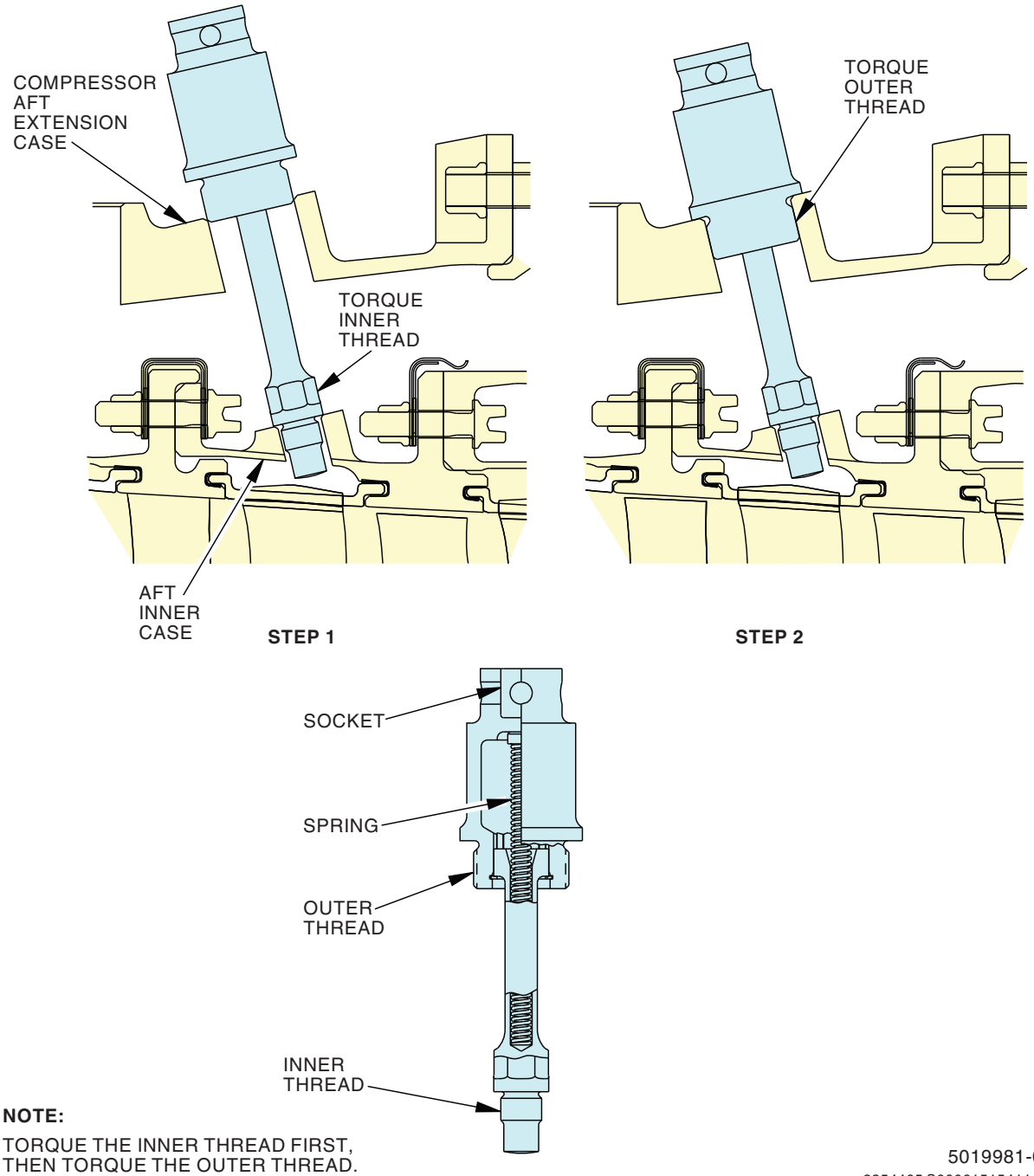


1 PRE LEAP-1B SERVICE BULLETIN 72-0179
2 POST LEAP-1B SERVICE BULLETIN 72-0179

2354463 S00061515409_V6

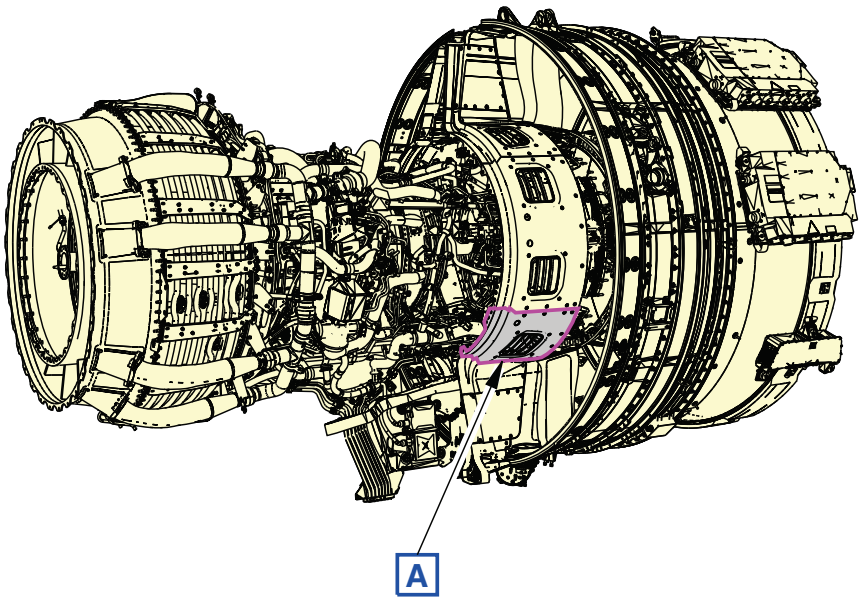
High Pressure Compressor Section Borescope Port Locations
Figure 609/72-00-00-990-812-G00 (Sheet 3 of 3)

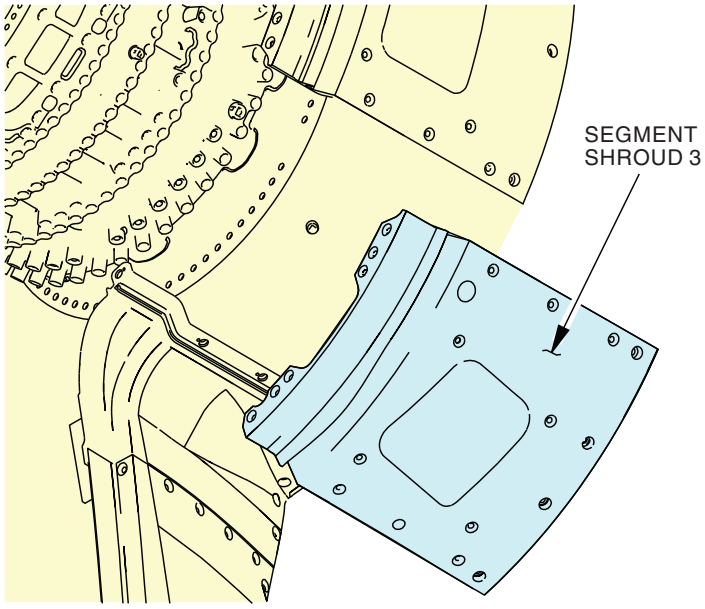
ECCN 9E991 BOEING PROPRIETARY - See title page for details



High Pressure Compressor for Ports I, J, L, and M Plug Installation
Figure 610/72-00-00-990-813-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details





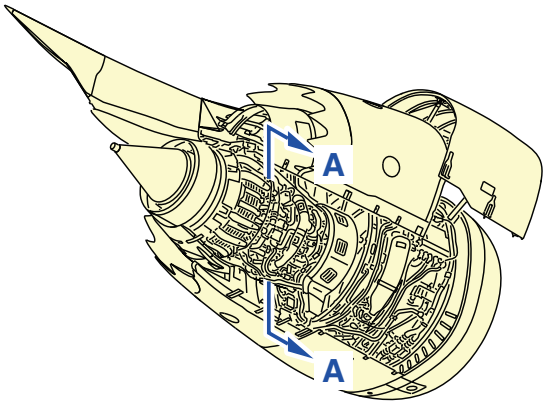
(TYPICAL)

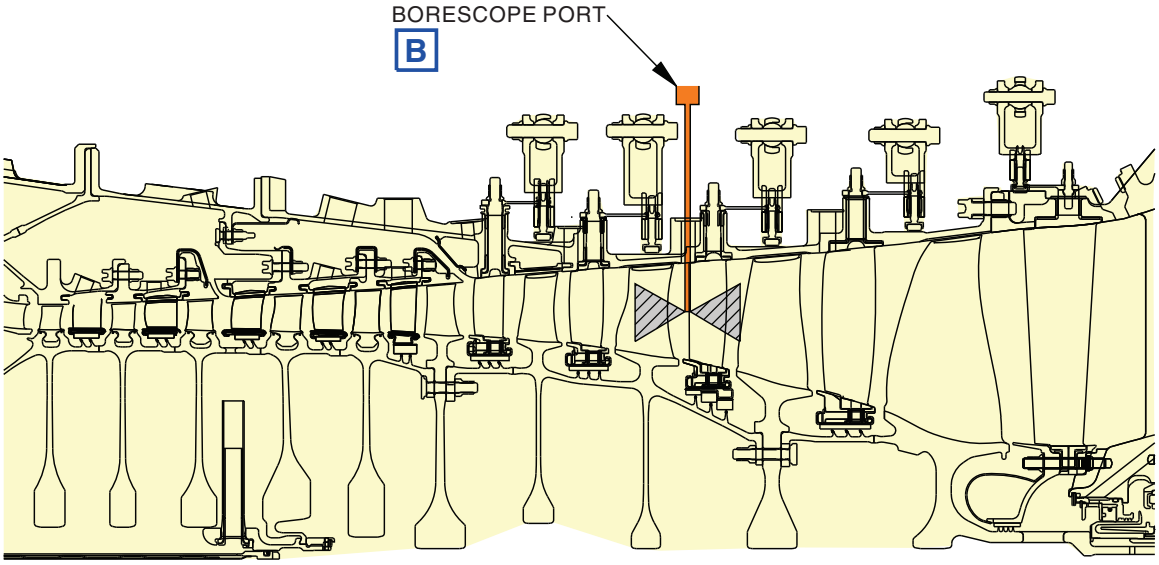
A

2851179 S0000667400_V1

High Pressure Compressor Borescope Plug Access
Figure 611/72-00-00-990-814-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details



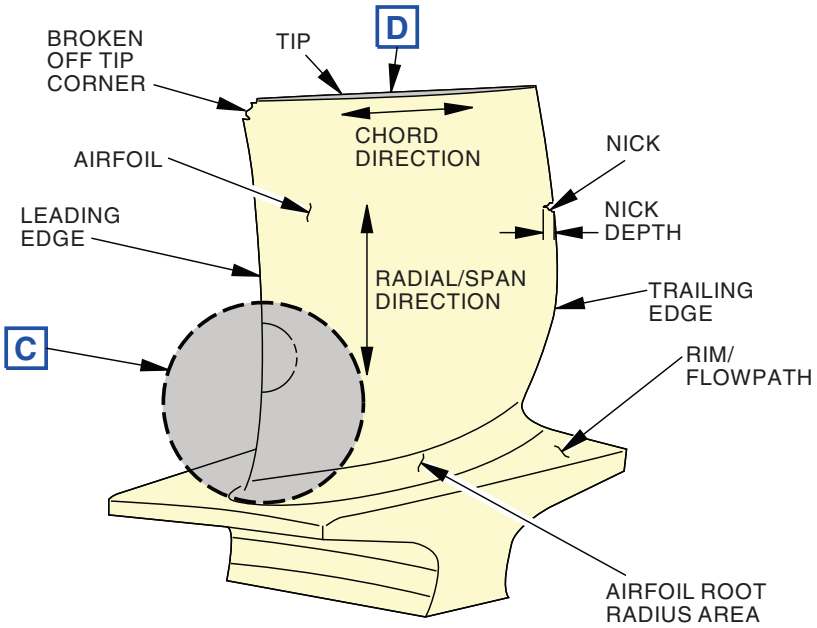


HIGH PRESSURE COMPRESSOR SECTION INSPECTION
A-A

5036152-00
2569204 S0000615782_V1

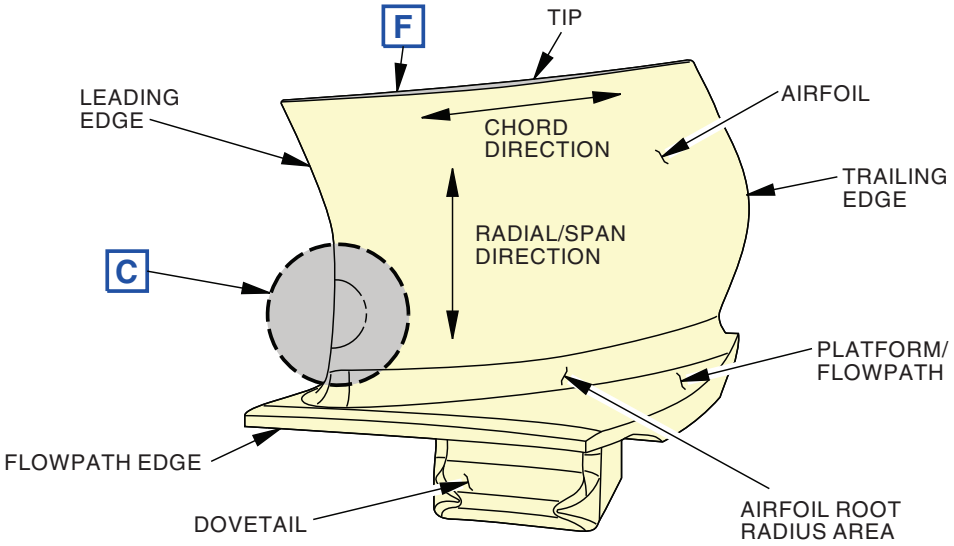
High Pressure Compressor Section Inspection
Figure 612/72-00-00-990-815-G00 (Sheet 1 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



EXAMPLE OF STAGE 1-5 BLISKS WITH TYPICAL DAMAGE

B



EXAMPLE OF STAGE 6-10 INSERTED BLADE

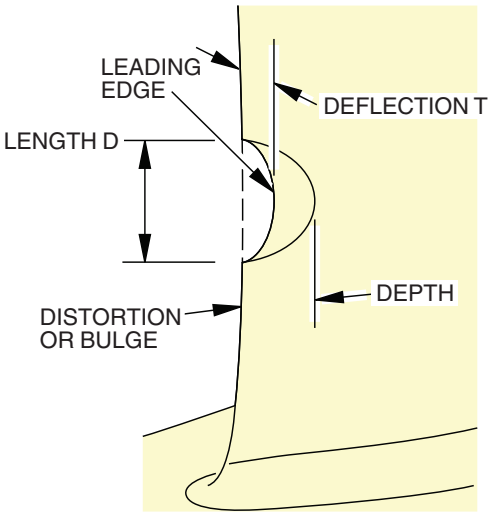
E

5036153-00

2569207 S0000615787_V3

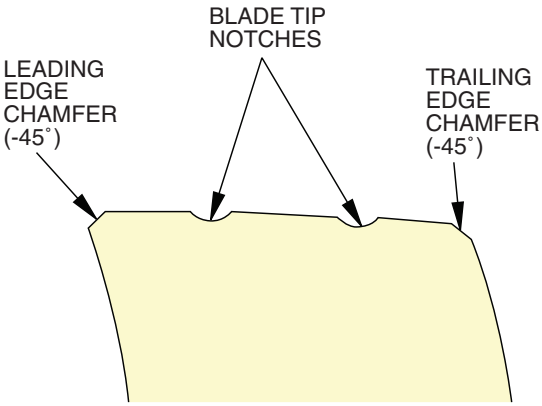
High Pressure Compressor Section Inspection
Figure 612/72-00-00-990-815-G00 (Sheet 2 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



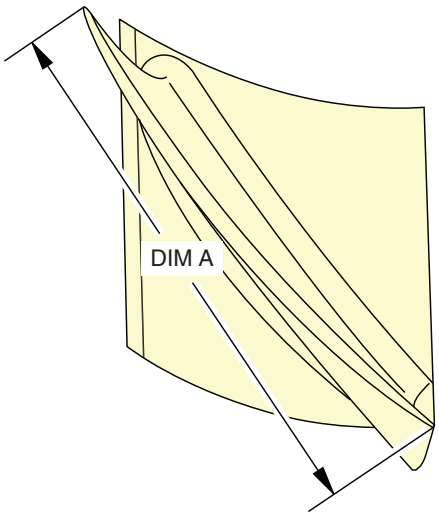
ALL STAGES: DISTORTION/BULGE

C



ALL STAGES: TYPICAL BLADE TIP NOTCHES AND CHAMFERS (SHOULD NOT BE INTERPRETED AS DAMAGE)

D



ALL STAGES: CHORD DIMENSION A

F

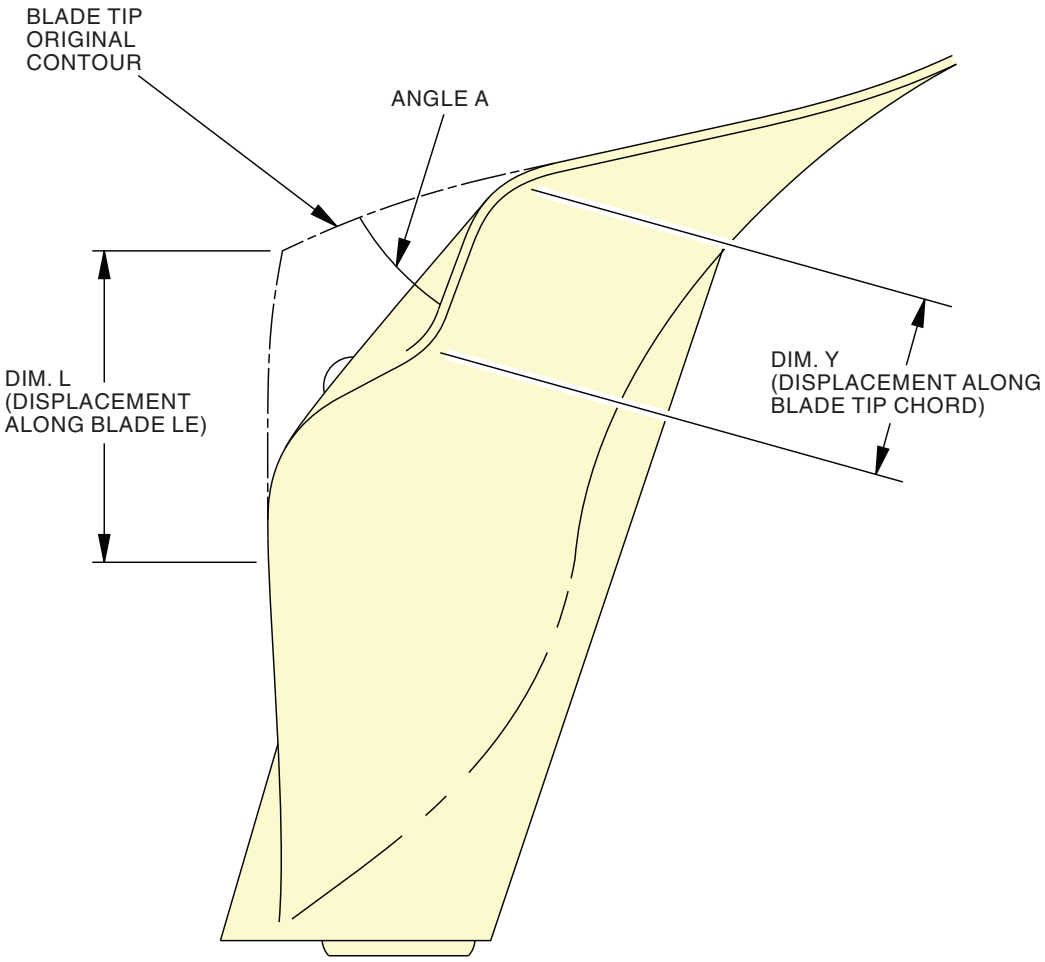
Stage	DIM A IN. (mm)	30% of DIM A IN. (mm)
1	3.221 (81.81)	0.966 (24.54)
1	3.327 (84.51)	0.998 (25.35)
2	2.190 (55.63)	0.657 (16.69)
3	1.670 (42.42)	0.501 (12.73)
4	1.262 (32.05)	0.379 (9.62)
5	1.100 (27.94)	0.330 (8.38)
6	0.986 (25.04)	0.296 (7.51)
7	1.090 (27.69)	0.327 (8.31)
8	1.007 (25.58)	0.302 (7.67)
9	0.971 (24.66)	0.291 (7.40)
10	0.878 (22.30)	0.263 (6.69)

1 PRE LEAP-1B SERVICE BULLETIN 72-0182
2 POST LEAP-1B SERVICE BULLETIN 72-0182

5036155-01
2569208 S0000615789_V4

High Pressure Compressor Section Inspection
Figure 612/72-00-00-990-815-G00 (Sheet 3 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



ALL STAGES: EXAMPLE OF TIP CURL DAMAGE

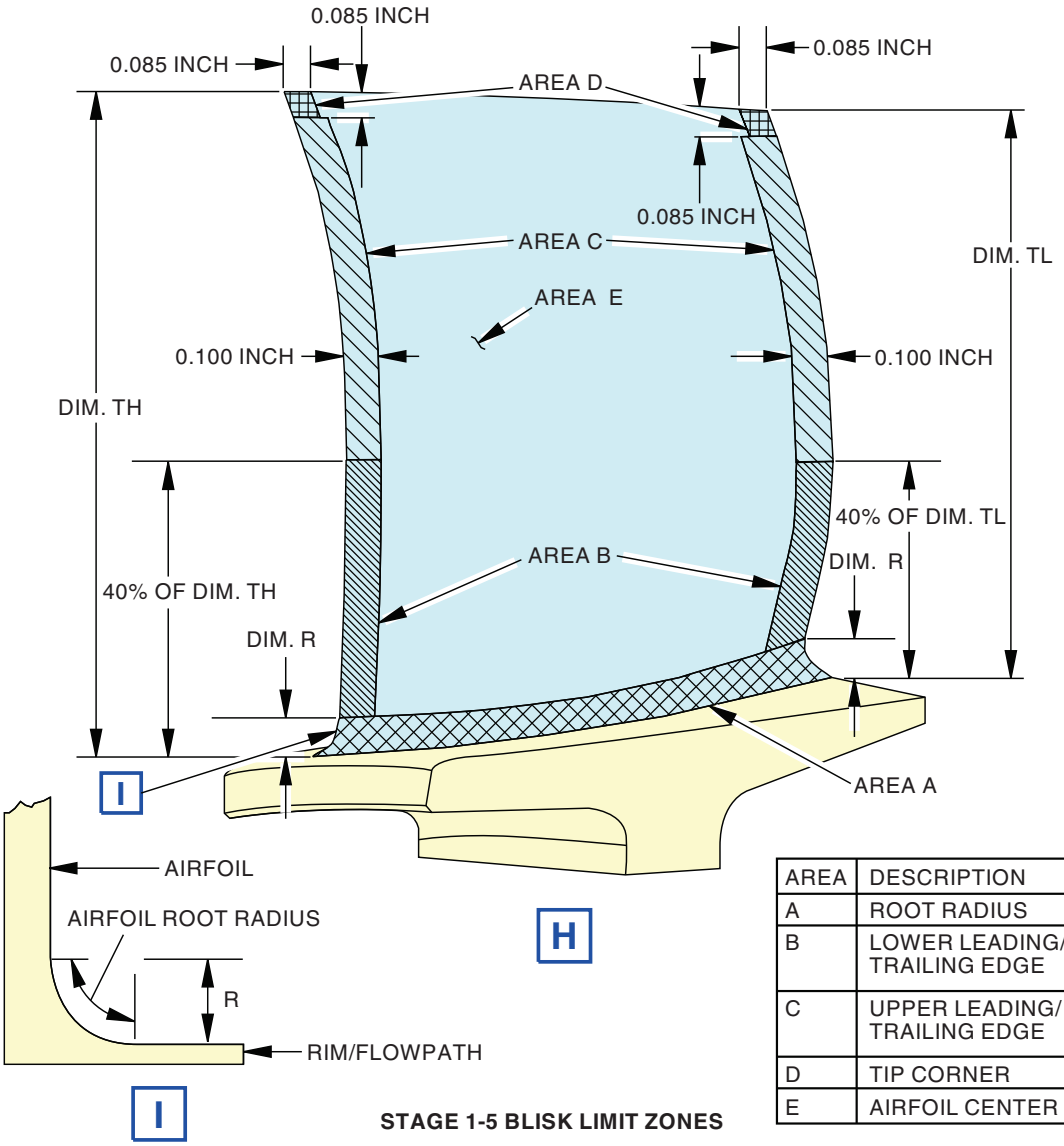
G

5036156-00

2569212 S0000615790_V1

High Pressure Compressor Section Inspection
Figure 612/72-00-00-990-815-G00 (Sheet 4 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



AREA	DESCRIPTION
A	ROOT RADIUS
B	LOWER LEADING/ TRAILING EDGE
C	UPPER LEADING/ TRAILING EDGE
D	TIP CORNER
E	AIRFOIL CENTER PANEL

STAGE 1-5 BLISK LIMIT ZONES

Stage	DIM TH IN. (mm)	40% OF DIM TH IN. (mm)	DIM TL IN. (mm)	40% OF DIM TL IN. (mm)	DIM R IN. (mm)
1	5.196 (131.98)	2.078 (52.79)	4.202 (106.73)	1.681 (42.69)	0.223 (5.66)
2	3.393 (86.18)	1.357 (34.47)	2.777 (70.54)	1.111 (28.21)	0.142 (3.61)
3	2.368 (60.15)	0.947 (24.06)	2.007 (50.98)	0.803 (20.39)	0.125 (3.18)
4	1.751 (44.48)	0.700 (17.79)	1.570 (39.88)	0.628 (15.95)	0.110 (2.79)
5	1.282 (32.56)	0.513 (13.03)	1.177 (29.90)	0.471 (11.96)	0.070 (1.78)

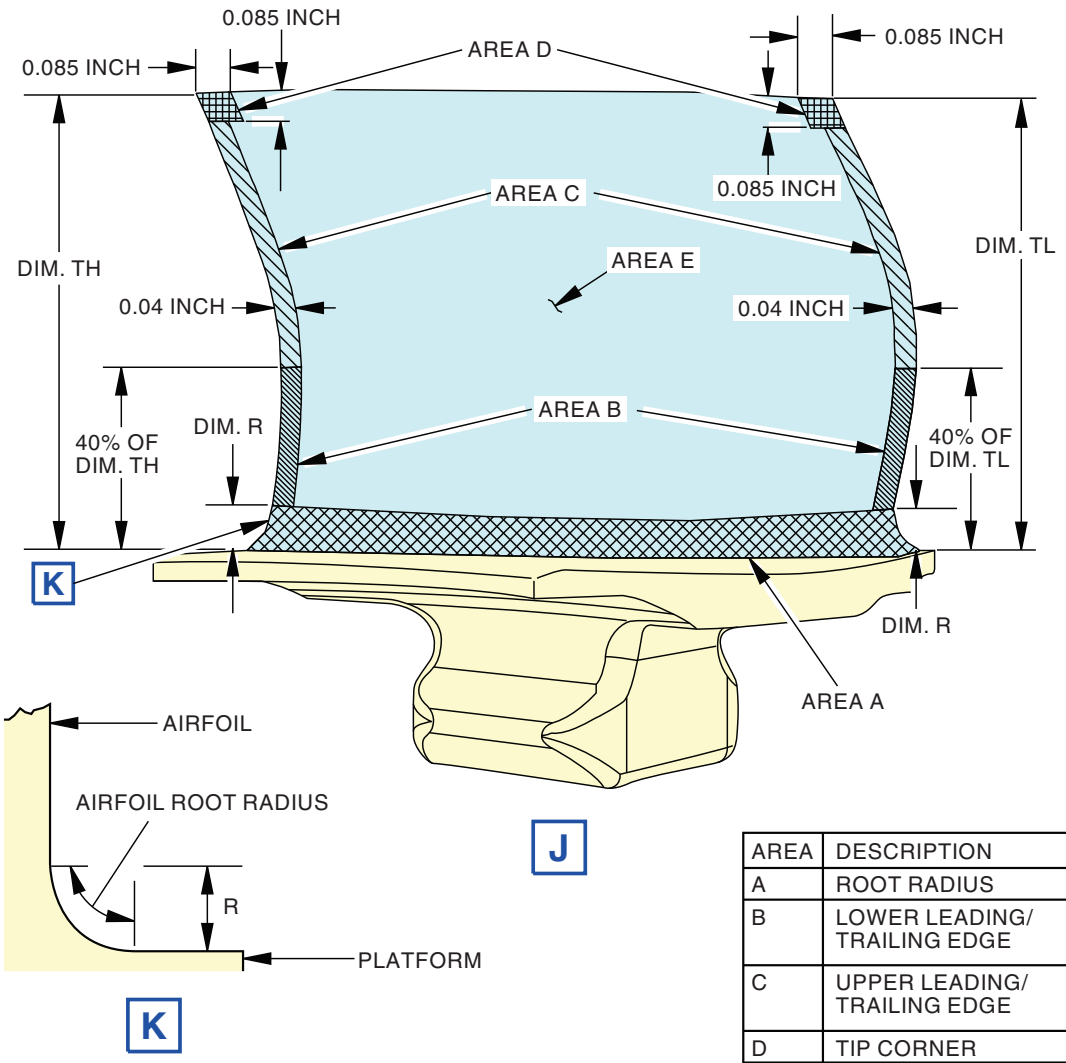
DAMAGE ZONE REFERENCE DIMENSIONS

5036157-00

2569231 S0000615791_V2

High Pressure Compressor Section Inspection
Figure 612/72-00-00-990-815-G00 (Sheet 5 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



AREA	DESCRIPTION
A	ROOT RADIUS
B	LOWER LEADING/ TRAILING EDGE
C	UPPER LEADING/ TRAILING EDGE
D	TIP CORNER
E	AIRFOIL CENTER PANEL

STAGE 6-10 INSERTED BLADE LIMIT ZONE

Stage	DIM TH IN. (mm)	40% OF DIM TH IN. (mm)	DIM TL IN. (mm)	40% OF DIM TL IN. (mm)	DIM R IN. (mm)
6	1.045 (26.54)	0.418 (10.62)	1.006 (25.55)	0.402 (10.22)	0.085 (2.15)
7	0.904 (22.96)	0.362 (9.18)	0.854 (21.69)	0.342 (8.68)	0.080 (2.03)
8	0.738 (18.75)	0.295 (7.50)	0.690 (17.53)	0.276 (7.01)	0.080 (2.03)
9	0.657 (16.69)	0.263 (6.68)	0.619 (15.72)	0.248 (6.29)	0.080 (2.03)
10	0.570 (14.48)	0.228 (5.79)	0.577 (14.66)	0.231 (5.86)	0.080 (2.03)

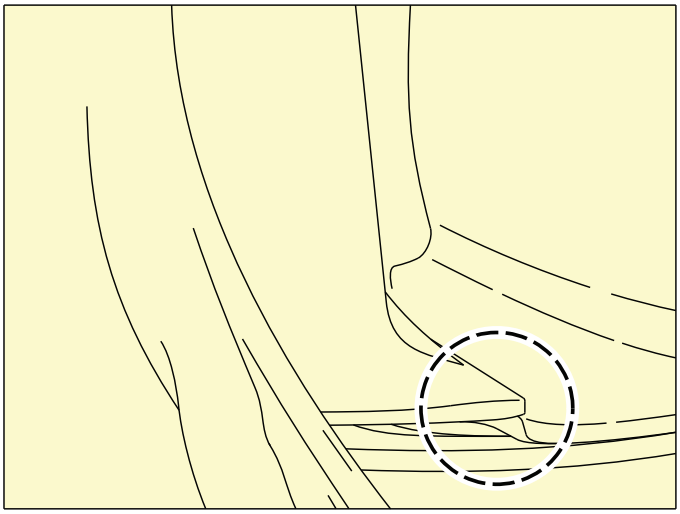
DAMAGE ZONE REFERENCE DIMENSIONS

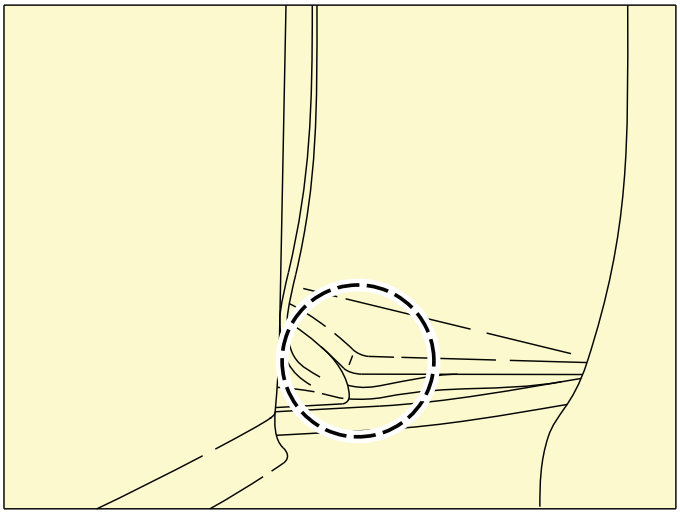
5036158-00

2569242 S0000615792_V4

High Pressure Compressor Section Inspection
Figure 612/72-00-00-990-815-G00 (Sheet 6 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details





BLADE PLATFORMS OVERLAPPING OR SHINGLING

NOTE:

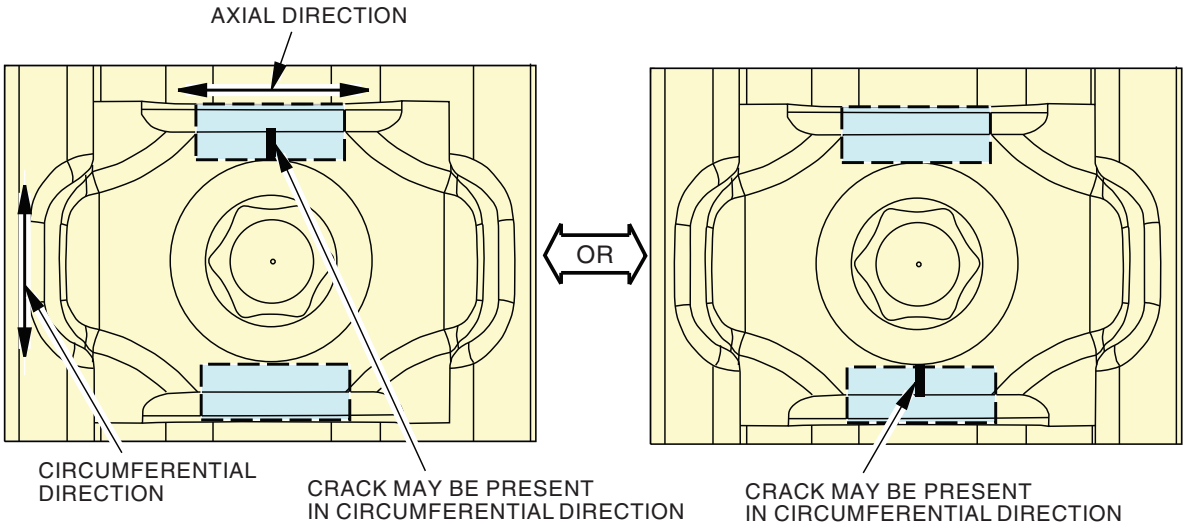
APPLICABLE FOR 6-10 BLADES

5051955-00

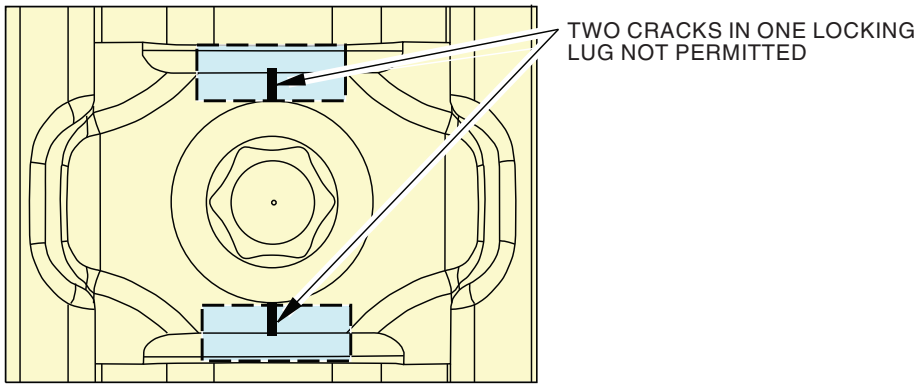
2823809 S0000651563_V2

High Pressure Compressor Section Inspection
Figure 612/72-00-00-990-815-G00 (Sheet 7 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



STAGE 6-10 CRACKING LIMIT ZONES (ACCEPTABLE)



STAGE 6-10 CRACKING LIMIT ZONES (NOT ACCEPTABLE)

LEGEND:

THE LIMIT FOR CIRCUMFERENTIAL CRACKING IS APPLICABLE ONLY FOR HIGHLIGHTED AREAS.

6018853-00

2956074 S0000733657_V1

High Pressure Compressor Section Inspection
Figure 613/72-00-00-990-861-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 72-00-00-290-805-G00

6. Combustion Section and High Pressure Turbine Section Stage One Nozzle Leading Edge Detailed Inspection

(Figure 614, Figure 615, Figure 616, Figure 617, Figure 618, Figure 619, Figure 620, and Figure 621)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task gives the instructions to examine the combustion section and high pressure turbine section stage one nozzle leading edge with a borescope.

B. References

Reference	Title
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
73-11-05-000-801-G00	Fuel Nozzle Removal (P/B 401)
73-11-05-400-801-G00	Fuel Nozzle Installation (P/B 401)
73-11-05-400-802-G00	Fuel Nozzle Set Installation (P/B 401)
74-21-02-000-801-G00	Left Igniter Plug Removal (P/B 401)
74-21-02-000-802-G00	Right Igniter Plug Removal (P/B 401)
74-21-02-220-801-G00	Igniter Inspection (P/B 601)
74-21-02-400-801-G00	Left Igniter Plug Installation (P/B 401)
74-21-02-400-802-G00	Right Igniter Plug Installation (P/B 401)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE
STD-10717	Light Source - Borescope, (Meets General Electric or CFM Standard Practices Manual Specification)
STD-10718	Fiberscope (Meets General Electric or CFM Standard Practices Manual Specification)
STD-10719	Borescope - Rigid (Meets General Electric or CFM Standard Practices Manual Specification such as: model R857ELSV60)

D. Consumable Materials

Reference	Description	Specification
D50017 [C02-071]	Compound - Antiseize And Lubricating - Pure Nickel Special Grade	GE Spec D6Y28C1 or A50TF198
D50034 [CP2619]	Compound - Antiseize - Pure Nickel Special, Never - Seez	
D50355 [CP2961]	Lubricant - Anti-Seize Compound	D6Y31C2

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Prepare for the Inspection

SUBTASK 72-00-00-941-002-G00

(1) Do this task: Preparation for Borescope Inspection, TASK 72-00-00-290-801-G00.

SUBTASK 72-00-00-010-024-G00

(2) If it is necessary, remove the igniter plugs.

(a) Do this task: Left Igniter Plug Removal, TASK 74-21-02-000-801-G00.

(b) Put an identification tag with a number 1 on the igniter plug.

(c) Do this task: Right Igniter Plug Removal, TASK 74-21-02-000-802-G00.

(d) Put an identification tag with a number 2 on the igniter plug.

SUBTASK 72-00-00-010-015-G00

(3) Remove the borescope plugs N and O.

NOTE: If using a rigid borescope, it is necessary to perform the Borescope Inspection (BSI) through both borescope ports (N and O) and both igniter ports. If using a flexible borescope, any borescope or igniter port may be used, depending on the technician's accessibility."

NOTE: In the event that damage is found during a BSI when using a rigid borescope, the expectation would then be to do a full 360 degree BSI with a flexible borescope.

(a) Apply a spray of penetrating oil to the ratcheting mechanism and let it soak.

(b) Apply break torque with a wrench to unseat the plug from the case.

(c) If it is necessary, apply another spray of penetrating oil to lubricate the plug threads and do step (b) again.

(d) Remove the plug.

(e) Put an identification tag that identifies the location for each borescope plug.

SUBTASK 72-00-00-210-029-G00

(4) For any plugs N or O that are not removed, do a check as follows:

(a) Make sure that the anti-rotation tab of the locking ring is fully engaged with the one of two slots on the borescope boss on the BSI port, and the plug is fully seated against the case.

(b) Use the borescope to check the condition in case there is no direct visual access.

(c) Apply hand pressure to the plug in the removal direction to check for loose plugs.

(d) If a plug is loose, tighten it to 100 in-lb (11.3 N·m) - 110 in-lb (12.4 N·m).

SUBTASK 72-00-00-480-004-G00

(5) Turn the light source, STD-10717, to ON.

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SUBTASK 72-00-00-480-016-G00

(6) Insert the rigid borescope, STD-10719, or the fiberscope, STD-10718, into position for the examination.

SUBTASK 72-00-00-290-005-G00

(7) Make sure that you follow the inspection instructions and limits closely in the inspection task.

NOTE: Table 602 provides the total number of fuel nozzles, High Pressure Turbine (HPT) stage one nozzle airfoils, deflector plates for the combustion section, and HPT stage one nozzle.

Table 602/72-00-00-993-813-G00 Fuel Nozzles, HPT Stage One Airfoil, and Combustion Section Deflector Plates

Component	Quantity
Fuel Nozzles	18
Stage 1 HPT Nozzle Airfoils	36
Combustion Section Deflector Plates	18

H. Combustor Mixer Retainer Hooks and Weld Pins Inspection

SUBTASK 72-00-00-211-025-G00

(1) If exposed, examine the combustor mixer retainer hooks and weld pins (Figure 614 (Sheet 1)) that keep the combustor radial mixers in place in the cold side of the combustor.

(a) For cups 8 and 9:

1) Broken mixer retainer hook(s) and/or broken or damaged weld pin(s).

a) Not permitted.

(b) For cups 1 - 7 and 10 - 18:

1) Broken mixer retainer hook(s) and/or broken or damaged weld pin(s) in six (6) or more cup positions.

a) Not permitted.

(c) For cups 1 - 7 and 10 - 18:

1) Broken mixer retainer hook(s) and/or broken or damaged weld pin(s) in five (5) or less cup positions.

a) Both mixer retainer hooks can be broken at each cup location.

NOTE: Broken mixer retainer hooks can be in consecutive cup positions.

b) Both retainer weld pins can be damaged and/or broken at each cup location.

NOTE: Broken or damaged weld pins can be in consecutive cup positions.

2) Do these steps for Continue-In-Service (CIS):

NOTE: The CIS limit is 600 flight cycles if the steps below are completed and hardware conditions are met.

a) Do cold side inspection of retainer hooks and weld pins at all 18 fuel nozzle cup positions.

<1> Refer to the section for Combustion Chamber Cold Side (FWD) Inspection below.

b) Make a record of all broken mixer retainer hook(s) and weld pin(s) by fuel nozzle cup position.

c) Put the mixer at the distressed cup location(s) in the correct position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

d) Make sure that the mixer is not pushed against the retainer hook that is not broken if one of the two retainer hooks in a cup position is broken.

e) Make sure that the mixer is not on top of the weld pin(s) (Figure 614 (Sheet 2) and Figure 614 (Sheet 3)).

f) Make sure that anti-rotation arms of adjacent mixers are in the correct position (engaged) (Figure 614 (Sheet 2) and Figure 614 (Sheet 3)).

g) Install fuel nozzles in the cup position(s) with broken mixer retainer hook(s) and/or damaged/broken weld pin(s), and at least two adjacent cup positions (TASK 73-11-05-400-801-G00 or TASK 73-11-05-400-802-G00).

<1> Fully bolt these fuel nozzles to the combustion case with new fuel nozzle bolts.

h) Do a borescope inspection of the cup position(s) with broken mixer retainer hook(s) and/or broken/damaged weld pin(s) after fuel nozzles installation.

<1> Make sure that the mixer at each distressed cup location is put in the correct position.

<2> Make sure that the mixer is not pushed against the retainer hook that is not broken if one of the two retainer hooks in a cup position is broken.

<3> Make sure that the mixer is not on top of the weld pin(s) (Figure 614 (Sheet 2) and Figure 614 (Sheet 3)).

<4> Make sure that anti-rotation arms of adjacent mixers are in the correct position (engaged) (Figure 614 (Sheet 2) and Figure 614 (Sheet 3)).

i) Do a borescope inspection HPT S1N flow-path surfaces and make sure that there is no evidence of oxidation/burning.

j) Create a report similar to Figure 614 (Sheet 2) and make sure that all the requirements are met.

<1> Keep this report in engine maintenance record for reference.

k) At the end of first CIS limit of 600 flight cycles, do an inspection of the hardware condition again (refer to the section for Combustion Chamber Cold Side (FWD) Inspection below).

<1> Create a new report similar to Figure 614 (Sheet 2) and make sure that requirements in steps h) and i) above are met.

NOTE: Use original report for reference.

<2> If all requirements are met, the time to do the CIS limit is 600 flight cycles.

NOTE: It is acceptable to do inspection steps again. You can make a report of creation with requirements validation and do it again for additional CIS limit of 600 flight cycles.

I. Combustion Section and High Pressure Turbine Section Stage One Nozzle Leading Edge Inspection

SUBTASK 72-00-00-211-006-G00

(1) Examine the surfaces of the combustor, fuel nozzle aft heat shield, and stage one HPT nozzle leading edge (Figure 616):

(a) Discoloration

1) Permitted.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Carbon accumulation

1) Permitted.

(c) Missing, chipped or spalled areas of the Thermal Barrier Coating (TBC)

1) Permitted.

SUBTASK 72-00-00-211-007-G00

(2) Examine the deflectors, and the dome plate (if it is visible):

(a) Distortion of the deflector

1) Permitted.

(b) Axial or radial cracks in the deflector are permitted as follows:

1) In area A (middle):

a) Cracks with a distance from crack tip to nearest deflector outer edge more than 0.112 in. (2.845 mm), are permitted.

b) All cracks that do not meet the above condition, but are less than or equal to 0.200 in. (5.08 mm) in length, are permitted.

c) The cracks are not connected.

2) In area B (wings):

a) All cracks less than or equal to 0.200 in. (5.08 mm) in length, are permitted.

b) The cracks are not connected.

(c) CIS limits for axial or radial cracks in the deflector (View E, Figure 616).

NOTE: This over-serviceable limit extensions below are for a reduced interval, repeat inspection.

1) In area A (middle):

a) Cracks with a distance from crack tip to nearest deflector outer edge less than or equal to 0.112 in. (2.845 mm), are permitted for 1000 cycles.

b) Crack with lifted deflector material is permitted for 250 cycles.

c) Single crack connecting eyelet and deflector outer edge is permitted for 500 cycles.

2) In area B (wings):

a) Cracks with a distance from crack tip to nearest deflector outer edge more than 0.400 in. (10.160 mm), are permitted for 250 cycles.

b) Cracks with a distance from crack tip to nearest deflector outer edge less than or equal to 0.400 in. (10.160 mm), are permitted for 200 cycles.

c) Single crack connecting eyelet and deflector outer edge is permitted for 200 cycles.

(d) Circumferential cracks in the deflector

1) Not permitted.

PTF ALL; AIRPLANES PRE SB-CFM-LEAP-1B-72-0145

(e) Examine the deflector-retainer braze joint and retainer eyelet area (Figure 616 (Sheet 5) and Figure 616 (Sheet 6)):

1) Retainer axial cracks in eyelet area are permitted.

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PTF ALL; AIRPLANES PRE SB-CFM-LEAP-1B-72-0145 (Continued)

2) Circumferential braze cracks up to 50 percent of circumference (4 in. (101.6 mm) in length) are permitted.

3) CIS limits for circumferential cracks in the deflector-retainer braze joint.

a) Circumferential cracks between 50-60 percent of circumference (4 in. (101.6 mm) - 4.8 in. (121.9 mm) in length) are permitted for 250 cycles.

NOTE: This over-serviceable limit extensions is for a reduced interval, repeat inspection.

b) Circumferential cracks more than 60 percent (4.8 in. (121.9 mm) in length) are permitted for 25 cycles.

NOTE: This over serviceable limit extension is for a one-time CIS application.

4) Retainer missing material is not permitted.

5) Retainer separation is not permitted.

PTF ALL

(f) Missing deflector material from the outer edge (View E, Figure 616)

1) Missing material is permitted on the deflector with these conditions:

NOTE: There are a total of 18 deflectors.

a) The maximum depth of missing material does not exceed 0.10 in. (2.54 mm) from the edge (first row of cooling holes fully visible).

2) CIS Limits

a) Missing material from the outer edge more than the serviceable limits, see the steps below for liberated deflector inspection.

(g) Missing material in the eyelet or wing area (Figure 620)

1) Condition is not serviceable.

2) CIS Limits

a) For cups 7, 8, 9 and 10:

<1> Missing material is permitted for 1 cycle.

NOTE: This over-serviceable limit extension is for a one-time CIS application.

b) For cups 1-6 and 11-18:

<1> Missing material at the eyelet or wing location no more than 0.5 in^2 (322.6 mm^2) for up to three deflectors is permitted for 500 cycles, provided there is no missing material on the deflector edge, there is no deflector lifted material, and cold (FWD) side inspection (paragraph Combustion Chamber Cold Side (FWD) Inspection) of the dome plate is performed to the limits specified in steps for radial dome plate cracks and circumferential dome plate cracks below.

In case of deflector lifted material, CIS limit is 250 cycles for this condition, provided all other criteria are met.

NOTE: This over-serviceable limit extension is for a reduced interval, repeat inspection.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

<2> Missing material more than 0.5 in^2 (322.6 mm^2), see steps below for liberated deflector inspection and perform cold (FWD) side inspection (paragraph Combustion Chamber Cold Side (FWD) Inspection) of the dome plate, performed to the limits specified in steps for radial dome plate cracks and circumferential dome plate cracks below.

(h) Liberated deflector (including partially liberated deflector)

1) For cups 7-10:

a) Liberated deflector is permitted for 1 cycle only on one of the cups.

NOTE: This over-serviceable limit extension is for a one-time CIS application.

2) For cups 1-6 and 11-18:

a) Liberated deflector with no crack indications or radial cracks less than 0.20 in. (5.08 mm) in length on the exposed dome plate, is permitted for 167 cycles.

NOTE: This over-serviceable limit extension is for a reduced interval, repeat inspection.

b) Liberated deflector with one or more cracks on the exposed dome plate, where the distance between cracks is more than 0.35 in. (8.89 mm), is permitted for a specific number of cycles, based on the number of dome plate cooling hole rows between the end of the crack and the flange, as shown in Table 603, Figure 619, and step for CIS Limits for exposed dome plate below.

NOTE: This over-serviceable limit extension is for a reduced interval, repeat inspection.

Table 603/72-00-00-993-817-G00 CIS Limits of the Cracks on the Dome Plate

Cooling Hole Rows Between End of Crack and Dome Plate Bend Radii	CIS Disposition (Cycles)
0	1
1	33
2	69
3	116
4+	167

c) Liberated deflector with a circumferential crack on the exposed dome plate, is not permitted.

d) Liberated deflector with more than one crack on the exposed dome plate, where the distance between cracks is less than 0.35 in. (8.89 mm), is permitted for 1 cycle.

NOTE: This over-serviceable limit extension is for a one-time CIS application.

(i) Radial dome plate cracks (View E, Figure 616)

1) There is no limit to the number of radial cracks permitted with these conditions:

a) All radial cracks are less than or equal to 0.20 in. (5.08 mm) in length.

b) The cracks are not connected.

c) There are no cracks on the bend radii of the dome plate.

2) CIS Limits for not exposed dome plate

a) Radial cracks between 0.20 in. (5.08 mm) and 0.40 in. (10.16 mm) in length.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

b) The cracks are not connected.

c) The cracks that have a minimum of 2.0 in. (50.8 mm) separation are permitted for 25 cycles.

NOTE: This over-serviceable limit extension is for a one-time CIS application.

3) CIS Limits for exposed dome plate, or if required (Table 603, Figure 619)

a) Radial cracks greater than 0.20 in. (5.08 mm) in length, propagating from the dome plate eyelet with less than 0.35 in. (8.89 mm) separation, are permitted for 1 cycle.

NOTE: This over-serviceable limit extension is for a one-time CIS application.

b) Radial cracking less than or equal to 0.20 in. (5.08 mm), is permitted for 167 cycles.

NOTE: This over-serviceable limit extension is for a reduced interval, repeat inspection.

c) All radial cracks greater than 0.20 in. (5.08 mm) in length, with 4 or more rows of cooling holes between crack tip and dome plate bend radii, are permitted for 167 cycles.

NOTE: This over-serviceable limit extension is for a reduced interval, repeat inspection.

d) All radial cracks greater than 0.20 in. (5.08 mm) in length, with 3 rows of cooling holes between crack tip and dome plate bend radii, are permitted for 116 cycles.

NOTE: This over-serviceable limit extension is for a reduced interval, repeat inspection.

e) All radial cracks greater than 0.20 in. (5.08 mm) in length, with 2 rows of cooling holes between crack tip and dome plate bend radii, are permitted for 69 cycles.

NOTE: This over-serviceable limit extension is for a reduced interval, repeat inspection.

f) All radial cracks greater than 0.20 in. (5.08 mm) in length, with 1 row of cooling holes between crack tip and dome plate bend radii, are permitted for 33 cycles.

NOTE: This over-serviceable limit extension is for a reduced interval, repeat inspection.

g) All radial cracks greater than 0.20 in. (5.08 mm) in length, with 0 row of cooling holes between crack tip and dome plate bend radii, are permitted for 1 cycle.

NOTE: This over-serviceable limit extension is for a one-time CIS application.

(j) Circumferential dome plate cracks (View E, Figure 616)

1) Not permitted.

2) CIS Limits

a) Circumferential cracks up to 0.40 in. (10.16 mm) that are not connected with 2.0 in. (50.8 mm) minimum separation are permitted for 25 cycles.

NOTE: This over serviceable limit extension is for a one-time CIS application.

SUBTASK 72-00-00-211-008-G00

(3) Examine the combustion chamber mixer Trailing Edge (TE) (View G, Section B-B, Figure 616):

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(a) Oxidation

1) Any amount permitted.

(b) Missing material

1) Up to 0.210 in. (5.33 mm) from the TE, limited to 270 degrees circumferentially around the mixer.

2) CIS Limit

a) Between 0.210 in. (5.33 mm) and 0.260 in. (6.60 mm) from the TE, limited to 270 degrees circumferentially around the mixer, is permitted for 25 cycles.

NOTE: This over-serviceable limit extension is for a one-time CIS application.

SUBTASK 72-00-00-211-009-G00

(4) Examine the fuel nozzles (View G, Figure 616):

(a) Heat shield (aft surface)

1) Discoloration or carbon deposits

a) Permitted.

2) Radial cracking

a) All amounts of radial cracks are permitted up to 0.10 in. (2.54 mm) long.

b) Up to eight radial cracks are permitted for the full length of the aft surface.

3) Circumferential cracking

a) Not permitted.

4) Oxidation or missing material (View I and View H, Figure 616):

a) Full exposure of the entire second row of the cooling holes is permitted.

NOTE: These over-serviceable limit extensions below are for a reduced interval, repeat the inspection.

b) Full exposure of any third row cooling hole is permitted for 250 cycles.

c) Full exposure of two adjacent cooling holes in the fourth row is permitted for 100 cycles. Multiple locations permitted.

d) Full exposure of three or more adjacent cooling holes in the fourth row permitted for 25 cycles if the conditions that follow are not there:

<1> Coke lip oxidation

<2> Cooling hole surface material loss, or

<3> Cracks connecting cooling holes.

(b) Heat shield coke lip for the following:

1) Oxidation or missing material for all fuel nozzles

a) Up to two areas of oxidation or missing material permitted for up to 0.30 in. (7.62 mm) of circumferential length each (any amount of depth). Distressed areas must be located at least 0.30 in. (7.62 mm) from each other.

2) CIS Limit

NOTE: This over serviceable limit extension is for a one-time CIS application.

a) Oxidation or missing material 0.30 in. (7.62 mm) of circumferential length in up to three locations is permitted for 25 cycles.

3) Discoloration or carbon deposits

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a) Permitted.

(c) Heat shield conic surface and splitter for the following:

1) Discoloration or carbon deposits

a) Permitted.

2) Axial cracking (conic surface)

a) All axial cracks are permitted up to 0.228 in. (5.79 mm) in length (entire length of the heat shield component).

3) Axial cracking (splitter)

a) Permitted

4) Circumferential cracking

a) Not permitted.

5) Oxidation or missing material (conic surface)

a) Not permitted.

6) Oxidation or missing material (splitter)

a) 0.050 in. (1.27 mm) of oxidation or missing material for 360 degrees permitted on conic surface trailing edge.

7) CIS Limit

NOTE: This over serviceable limit extension is for a one-time CIS application.

a) Oxidation or missing material between 0.050 in. (1.27 mm) and 0.075 in. (1.90 mm) for 360 degrees is permitted for 25 cycles.

SUBTASK 72-00-00-211-010-G00

(5) Examine the inner combustion liners (Figure 616):

(a) Circumferential cracks

1) All cracks permitted with this condition:

a) They are a maximum of 0.50 in. (12.7 mm) in length.

2) CIS Limit

NOTE: This over serviceable limit extension is for a one-time CIS application.

a) Cracks between 0.50 in. (12.7 mm) and 1.0 in. (25.4 mm) in length are permitted for 25 cycles.

(b) Axial or diagonal cracks that are not connected

1) All cracks up to 0.5 in. (12.7 mm) in length are permitted.

2) CIS Limits

NOTE: The over-serviceable limit extensions below are for a reduced interval, repeat the inspection.

a) Cracks between 0.5 in. (12.7 mm) and 1.0 in. (25.4 mm) in length are permitted for 250 cycles.

b) Cracks between 1.0 in. (25.4 mm) and 1.5 in. (38.1 mm) in length are permitted for 100 cycles.

c) Cracks between 1.5 in. (38.1 mm) and 2.0 in. (50.8 mm) in length are permitted for 50 cycles.

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d) Cracks between 2.0 in. (50.8 mm) and 2.25 in. (57.2 mm) in length are permitted for 25 cycles.

NOTE: This over serviceable limit extension is for a one-time CIS application.

(c) Axial, diagonal and circumferential connected cracks

1) Connected cracks are permitted with these conditions:

a) Connected cracks are permitted if the combined crack length is less than the limits listed below:

NOTE: The CIS limits listed below apply also for length of connected cracks.

NOTE: The total length of the connected cracks is the length measured from the most forward crack edge to the most aft crack edge.

<1> Circumferential cracks that are a maximum of 0.50 in. (12.7 mm) in length.

<2> Axial or diagonal cracks are up to 0.50 in. (12.7 mm).

b) Connected cracks in the same cooling hole row are not added.

c) Areas fully enclosed by connecting cracks are 0.385 in^2 (248.4 mm^2) or less.

2) CIS Limits

NOTE: This CIS limit is for a reduced interval, repeat inspection.

a) Connected cracks area that are more than the serviceable limit is permitted for 25 cycles.

b) Combined crack length is permitted for the 25 cycles CIS limit, if it is less than the values shown below:

NOTE: The 25 cycles CIS limit means that total length of axial-diagonal-circumferential cracks is not bigger than 2.25 in. (57.15 mm). Also, if these connected cracks are enclosed in the area not exceeding missing material area limit from burn through or missing metal that is more than the serviceable limit.

<1> For circumferential cracks:

<a> Cracks between 0.50 in. (12.7 mm) and 1.0 in. (25.4 mm) in length are permitted for 25 cycles.

<2> For axial or diagonal cracks that are not connected:

<a> Cracks between 0.5 in. (12.7 mm) and 1.0 in. (25.4 mm) in length are permitted for 250 cycles.

<b> Cracks between 1.0 in. (25.4 mm) and 1.5 in. (38.1 mm) in length are permitted for 100 cycles.

<c> Cracks between 1.5 in. (38.1 mm) and 2.0 in. (50.8 mm) in length are permitted for 50 cycles.

<d> Cracks between 2.0 in. (50.8 mm) and 2.25 in. (57.2 mm) in length are permitted for 25 cycles.

(d) Burn through or missing metal

1) Burn through or missing metal areas are permitted with these conditions:

a) A maximum permitted area for the multihole panel is not more than 0.01 in^2 (6.45 mm^2).

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b) Total permitted burn through area on the inner liner is less than 0.385 in^2 (248.4 mm^2).

2) CIS Limit

NOTE: This over serviceable limit extension is for a one-time CIS application.

a) Burn through or missing metal that is more than the serviceable limit is permitted for 25 cycles.

(e) Distortion

1) There is no limit to the amount permitted within 0.50 in. (12.7 mm) of the initial liner contour.

2) CIS Limit

a) Distortion between 0.50 in. (12.7 mm) and 0.75 in. (19.0 mm) is permitted for 25 cycles.

NOTE: This over serviceable limit extension is for a one-time CIS application.

SUBTASK 72-00-00-211-011-G00

(6) Examine the outer combustion liners (Figure 616):

(a) Circumferential cracks

1) Permitted with this condition:

a) The cracks have a maximum of 0.50 in. (12.7 mm) in length.

2) CIS Limit

NOTE: This over serviceable limit extension is for a one-time CIS application.

a) Cracks between 0.50 in. (12.7 mm) and 1.0 in. (25.4 mm) in length are permitted for 25 cycles.

(b) Axial or diagonal cracks that are not connected

1) Permitted with this condition:

a) The cracks have a maximum of 0.5 in. (12.7 mm) in length.

2) CIS Limits

NOTE: The over serviceable limit extensions below are for a reduced interval, repeat the inspection.

a) Cracks between 0.5 in. (12.7 mm) and 1.0 in. (25.4 mm) in length are permitted for 250 cycles.

b) Cracks between 1.0 in. (25.4 mm) and 1.5 in. (38.1 mm) in length are permitted for 100 cycles.

c) Cracks between 1.5 in. (38.1 mm) and 2.0 in. (50.8 mm) in length are permitted for 50 cycles.

d) Cracks between 2.0 in. (50.8 mm) and 2.25 in. (57.2 mm) in length are permitted for 25 cycles.

NOTE: This over serviceable limit extension is for a one-time CIS application.

(c) Axial or diagonal connected cracks

1) Connected cracks are permitted with these conditions:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

a) The total length of the connected cracks is less than the limit for axial or diagonal cracks (above) that are not connected.

NOTE: The total length of the connected cracks is the length measured from the most forward crack edge to the most aft crack edge.

b) Connected cracks in the same cooling hole row are not added.

c) Areas fully enclosed by connecting cracks are 0.385 in^2 (248.4 mm^2) or less.

(d) Burn through or missing metal

1) Burn through or missing metal is permitted with these conditions:

a) A maximum permitted area for the multihole panel is not more than 0.01 in^2 (6.45 mm^2).

b) Total permitted burn through area on the inner liner is less than 0.385 in^2 (248.4 mm^2).

2) CIS Limit

NOTE: This over serviceable limit extension is for a one-time CIS application.

a) Burn through or missing metal that is more than the serviceable limit is permitted for 25 cycles.

(e) Distortion

1) There is no limit to the amount permitted within 0.50 in. (12.7 mm) of the initial liner contour.

2) CIS Limit

NOTE: This over serviceable limit extension is for a one-time CIS application.

a) Distortion between 0.50 in. (12.7 mm) and 0.75 in. (19.0 mm) is permitted for 25 cycles.

SUBTASK 72-00-00-211-014-G00

(7) Examine the leading edge of the HPT stage one nozzle airfoil (Figure 617):

(a) Missing TBC

1) Permitted.

(b) Cracks that connect nose holes

1) Permitted with these conditions:

a) Not more than 0.175 in. (4.445 mm) in length.

b) Cracks are not connected.

c) No metal is missing.

(c) Cracks that extend from a nose hole(s) but do not connect with another nose hole(s)

1) Permitted if no metal is missing.

(d) Burns or erosion are permitted as follows:

NOTE: Burns that are on the surface and no holes are through on the parent metal.

1) The total damage is less than 0.175 in. (4.445 mm) in diameter for each vane.

2) There are no more than four burned vanes in a 90 degree arc.

(e) HPT stage one nozzle leading edge CIS limits

NOTE: This over serviceable limit extension is for a one-time CIS application.

1) Leading edge metal that is missing because of cracks or burn through:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

a) As much as 0.14 in^2 (0.903 cm^2) for each airfoil with a maximum of two vanes in a 90 degree arc is permitted for up to 25 cycles.

SUBTASK 72-00-00-211-016-G00

(8) Examine the inner and outer platforms of the HPT stage one nozzle (Figure 617):

NOTE: This inspection is only for HPT stage 1 nozzle assembly that is adjacent to the borescope port.

(a) Missing TBC

1) Permitted.

(b) Cracks in the parent material

1) Permitted.

(c) Nicks, scores, scratches and dents on the platform surface

1) Permitted.

(d) Burns

1) Permitted if no material is missing and the burns are not fully through.

(e) Bulges or bows

1) Not permitted.

(f) HPT stage one nozzle inner platform protruding or missing spline seal

1) Permitted.

(g) CIS Limits

NOTE: This over serviceable limit extension is for a one-time CIS application.

1) Bulges or bows are permitted for a maximum of 25 cycles.

2) A maximum area of 0.07 in^2 (0.4516 cm^2) of missing material, on each airfoil, because of cracks or burn through is permitted for a maximum of 25 cycles.

J. Combustion Chamber Cold Side (FWD) Inspection (If Required)

SUBTASK 72-00-00-010-046-G00

(1) Prepare to the inspection as follows:

(a) Identify the fuel nozzle on which the bolts or fuel nozzle itself will be removed for the purpose of cold side inspection.

NOTE: The fuel nozzle should be chosen based on the most convenient location for inspecting the defect.

NOTE: For a retainer broken tab condition, the fuel nozzle should not be an affected cup or one of the two adjacent cups.

(b) Remove a maximum of two bolts that are used to retain the fuel nozzle on the combustion case (TASK 73-11-05-000-801-G00 and Figure 621).

NOTE: TASK 73-11-05-000-801-G00 should be used only for bolt removal, do not proceed with actions unrelated to the bolts used to retain the fuel nozzle to the combustion case.

(c) If rigid borescope, STD-10719, with diameter 0.15 in. (3.8 mm) is not available, remove the fuel nozzle to be able to inspect required hardware with alternate, larger borescope (TASK 73-11-05-000-801-G00).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 72-00-00-290-009-G00

(2) Insert the rigid borescope, STD-10719, with diameter 0.15 in. (3.8 mm) through one of the bolt holes, or alternate borescope through the fuel nozzle hole, into the designated position for inspection.

SUBTASK 72-00-00-290-010-G00

(3) Do an inspection of required hardware.

(a) For maximum serviceable limits, refer to SUBTASK G72-00-00-211-006-G00.

SUBTASK 72-00-00-290-011-G00

(4) When the inspection is completed, remove the rigid borescope, STD-10719, with diameter 0.15 in. (3.8 mm), or alternate.

SUBTASK 72-00-00-410-037-G00

(5) Install the fuel nozzle or fuel nozzle bolts (TASK 73-11-05-400-801-G00).

K. Put the Airplane Back to Its Usual Condition

SUBTASK 72-00-00-090-001-G00

(1) When the inspection is complete, do these steps:

(a) Turn the light source, STD-10717, to OFF.

(b) Let the light source blower stay ON until the lamp and case become cool.

SUBTASK 72-00-00-010-016-G00

(2) Remove the rigid borescope, STD-10719, or the fiberscope, STD-10718.

SUBTASK 72-00-00-410-009-G00

CAUTION

OBEY ALL INSTRUCTIONS TO INSTALL THE BORESCOPE PLUG(S). IF YOU DO NOT OBEY THE INSTRUCTIONS, DAMAGE TO EXTERNAL EQUIPMENT CAN OCCUR.

(3) Install the borescope plugs N and O (Figure 615):

NOTE: When you install the borescope plug, make sure that the anti-rotation sleeve or sleeves on the plug are correctly engaged with the slots on the borescope boss before you fully tighten the plug (Figure 618).

(a) Do a visual inspection of the borescope plugs and the borescope ports N and O (TASK 72-00-00-290-801-G00).

(b) Remove the identification tags from the borescope plugs N and O before the installation of each plug.

(c) Apply a layer of Tiodize T8E-H grease, D50355 [CP2961], to the threads, mating surfaces, and ratcheting mechanism.

NOTE: You can use pure nickel special compound, D50017 [C02-071], or pure nickel special compound, D50034 [CP2619], as alternative.

(d) Install the borescope plugs in the combustor borescope ports N and O.

NOTE: You can spray penetrating oil into the locking mechanism to free the locking mechanism. This must be done before you replace the plug.

(e) Tighten the borescope plugs to 100 in-lb (11.3 N·m) - 110 in-lb (12.4 N·m).

(f) After plug has been tightened, do a visual check of the plug and make sure that the anti-rotation sleeve or sleeves are fully engaged with the slots on the borescope boss, and the head of the borescope plug is fully seated against the borescope boss.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(g) If either or both of the above conditions are not met, the ratcheting feature of the plug can be seized or locked, do the following steps:

1) Apply penetrating oil directly in between the locking ring and the head of the plug.

2) Apply torque to the plug again until it begins to turn freely.

3) Tighten the borescope plug to 100 in-lb (11.3 N·m) - 110 in-lb (12.4 N·m).

(h) Do a visual check again before you continue.

(i) If above conditions cannot be met, replace the plug.

(j) Use the borescope to check the condition in case there is no direct visual access.

SUBTASK 72-00-00-211-020-G00

(4) If you removed the igniter plugs, do the following:

(a) Do a visual inspection of the igniter plugs (TASK 74-21-02-220-801-G00).

SUBTASK 72-00-00-410-023-G00

(5) If you removed the igniter plugs, do the steps that follow:

(a) Remove the identification tag from igniter plug 1.

(b) Do this task: Left Igniter Plug Installation, TASK 74-21-02-400-801-G00.

(c) Remove the identification tag from igniter plug 2.

(d) Do this task: Right Igniter Plug Installation, TASK 74-21-02-400-802-G00.

SUBTASK 72-00-00-860-034-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(6) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 72-00-00-440-009-G00

(7) Do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 72-00-00-860-036-G00

(8) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

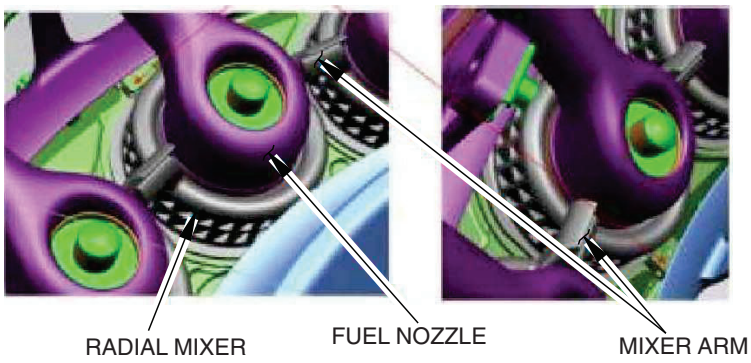
(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TWO RETAINER HOOKS AND TWO WELD PINS
PER FUEL NOZZLE CUP POSITION.



COMBUSTOR MIXER RETAINER HOOKS AND WELD PINS IN THE COLD SIDE OF THE COMBUSTOR WITH AND WITHOUT FUEL NOZZLE

6056783-00
3120578 S0000855571_V1

Combustor Mixer Retainer Hooks and Weld Pins Inspection
Figure 614/72-00-00-990-877-G00 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CONFIRMS PROPER MIXER PLACEMENT

CONFIRMS WELD PIN CONDITION

CONFIRMS NO OXIDATION ON LE HPTN S1

CONFIRMS SHEARED WELD PIN AND GOOD

MIXER ANTI-ROTATION ARMS PROPERLY ENGAGED

CONFIRMS USUAL MIXER/ COMBUSTOR GAP

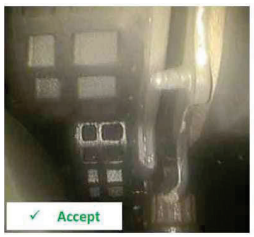
COMBUSTOR MIXER RETAINER HOOKS AND WELD PINS IN THE COLD SIDE OF THE COMBUSTOR

6056784-00
3120750 S0000855574_V1

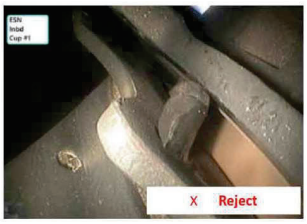
Combustor Mixer Retainer Hooks and Weld Pins Inspection
Figure 614/72-00-00-990-877-G00 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

MIXER SEATING

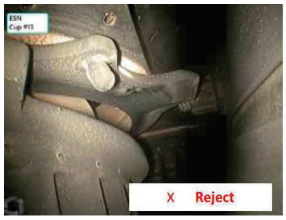


MIXER TAB IN CORRECT POSITION ALIGNED WITH BROKEN RETAINER HOOK - ACCEPT CONDITION

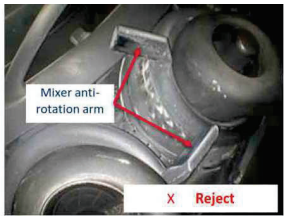


MIXER SEATED ON A WELD PIN - REJECT CONDITION

MIXER SEATING

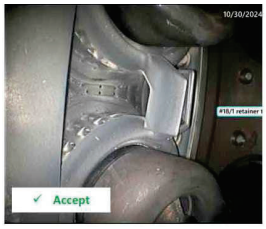


MIXER TAB OUTSIDE OF THE RETAINER HOOK LOCATION - REJECT CONDITION



MIXER ANTI-ROTATION ARM ENGAGED WITH SUPPORT COWL - REJECT CONDITION

ANTI-ROTATION ARMS ENGAGEMENT



MIXER ANTI-ROTATION ARMS ARE ENGAGED AND ALIGNED - ACCEPT CONDITION

ANTI-ROTATION ARMS ENGAGEMENT

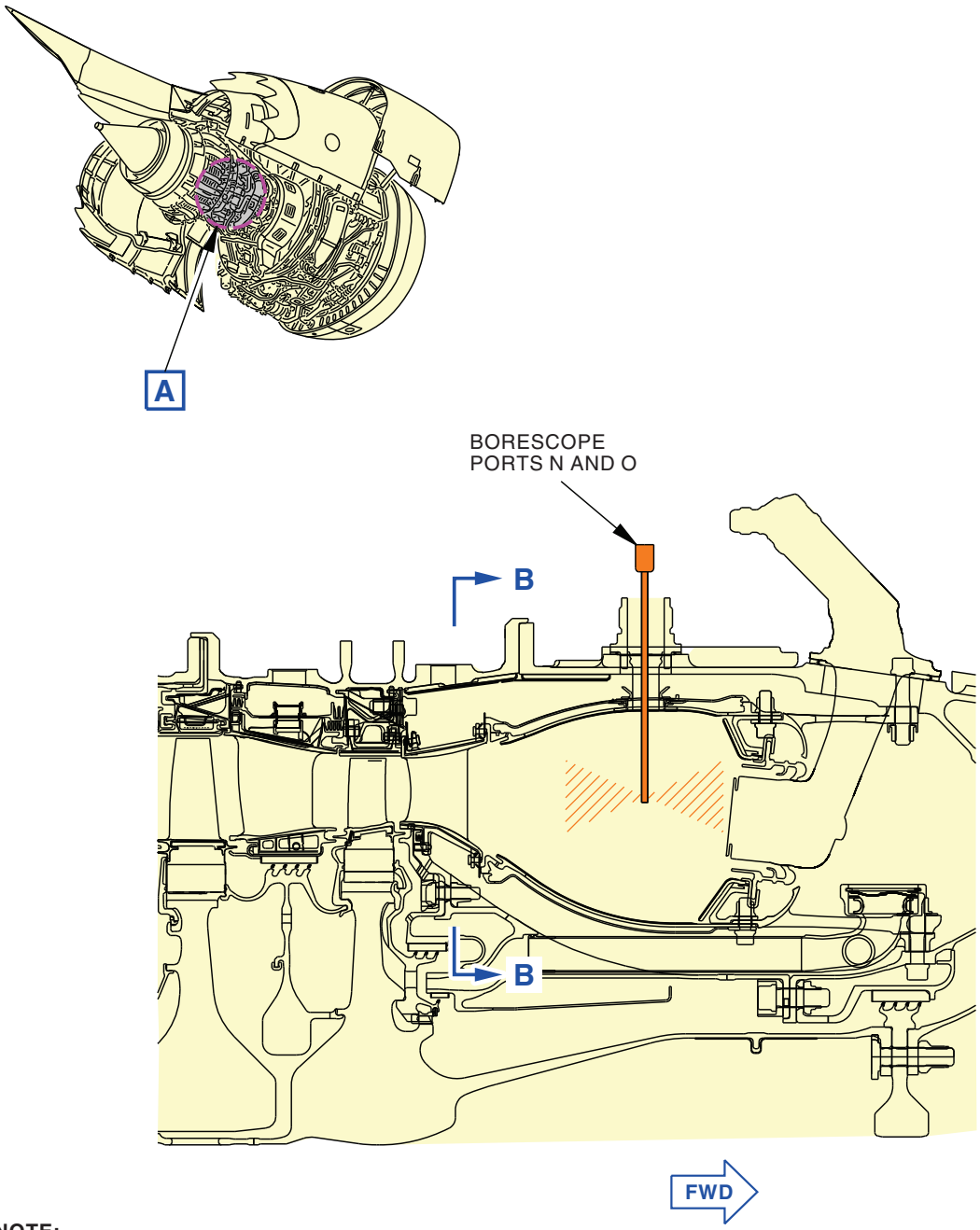
MIXER ANTI-ROTATION ARMS NOT ENGAGED/ALIGNED - REJECT CONDITION

EXAMPLES OF ACCEPTABLE AND REJECT CONDITIONS AFTER MIXER PLACEMENT AND FUEL NOZZLES INSTALLED

6056785-00
3120755 S0000855575_V1

Combustor Mixer Retainer Hooks and Weld Pins Inspection
Figure 614/72-00-00-990-877-G00 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOTE:

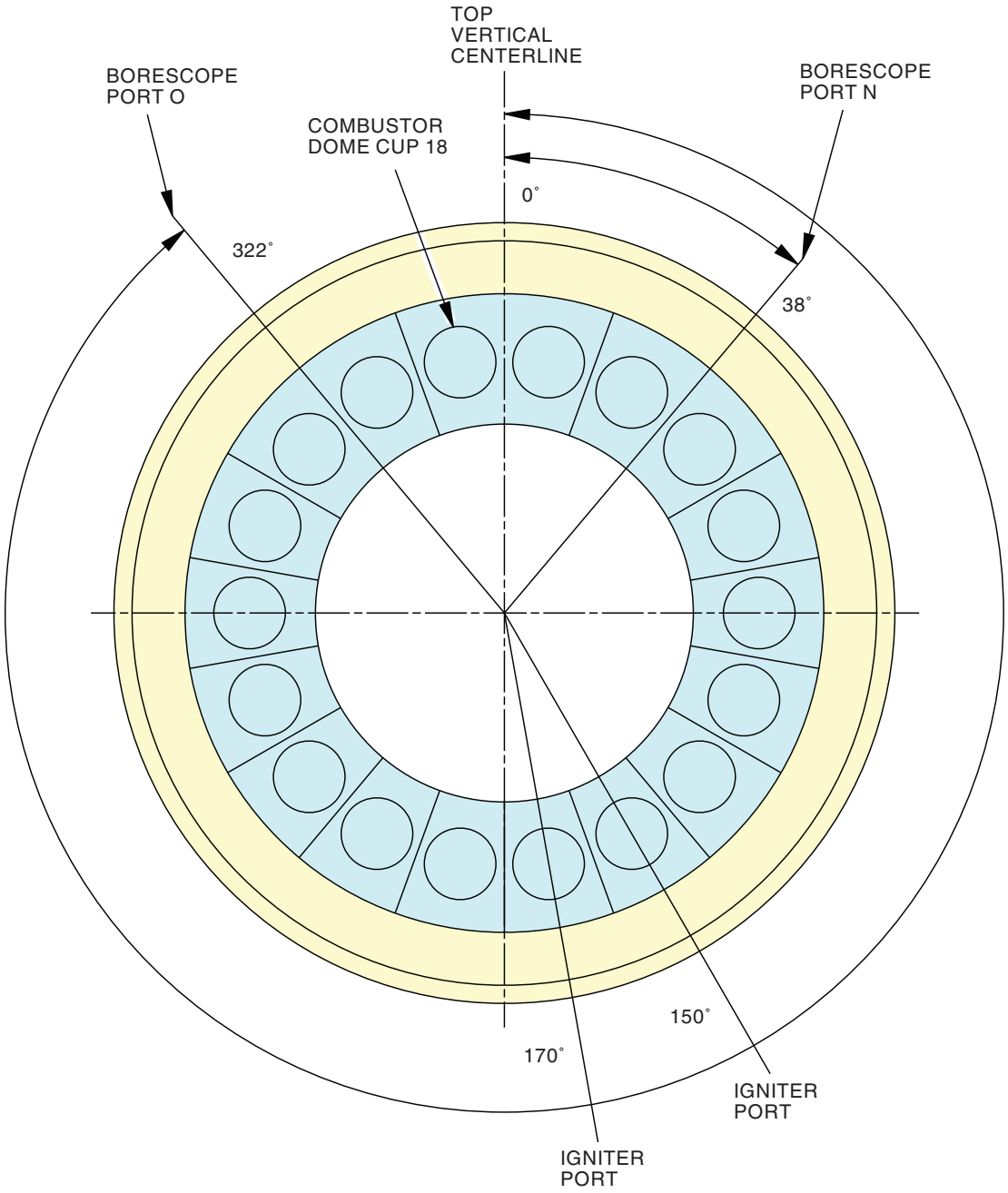
PORT N (2:00 OCLOCK)
PORT O (10:00 OCLOCK)

5020997-00

2354451 S00061515395_V3

Combustion Section Borescope Port Locations
Figure 615/72-00-00-990-816-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

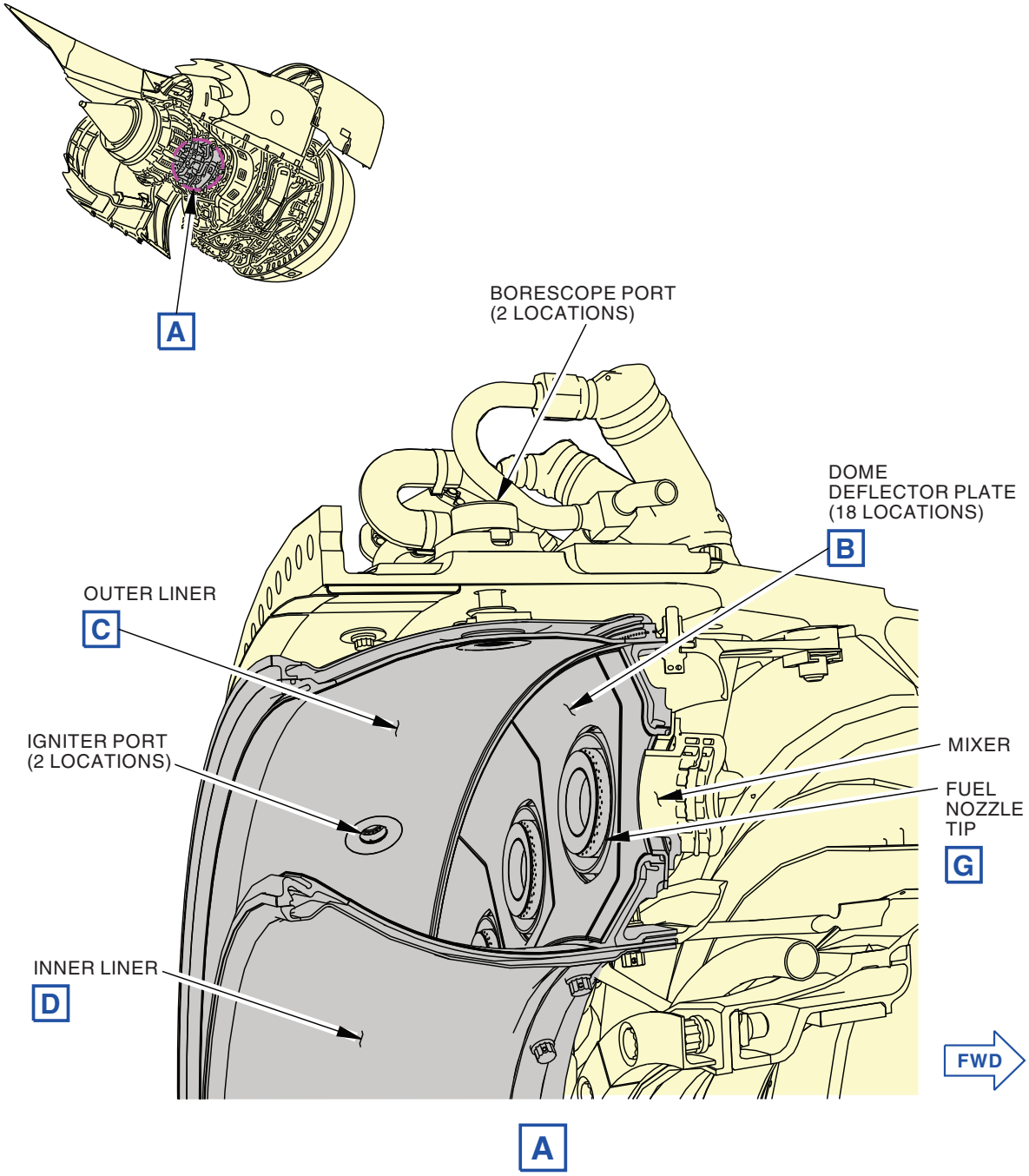


BORESCOPE PORTS N, O, AND DOME LOCATIONS
(AFT LOOKING FORWARD)
B-B

5029318-00
2354452 S00061515396_V5

Combustion Section Borescope Port Locations
Figure 615/72-00-00-990-816-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

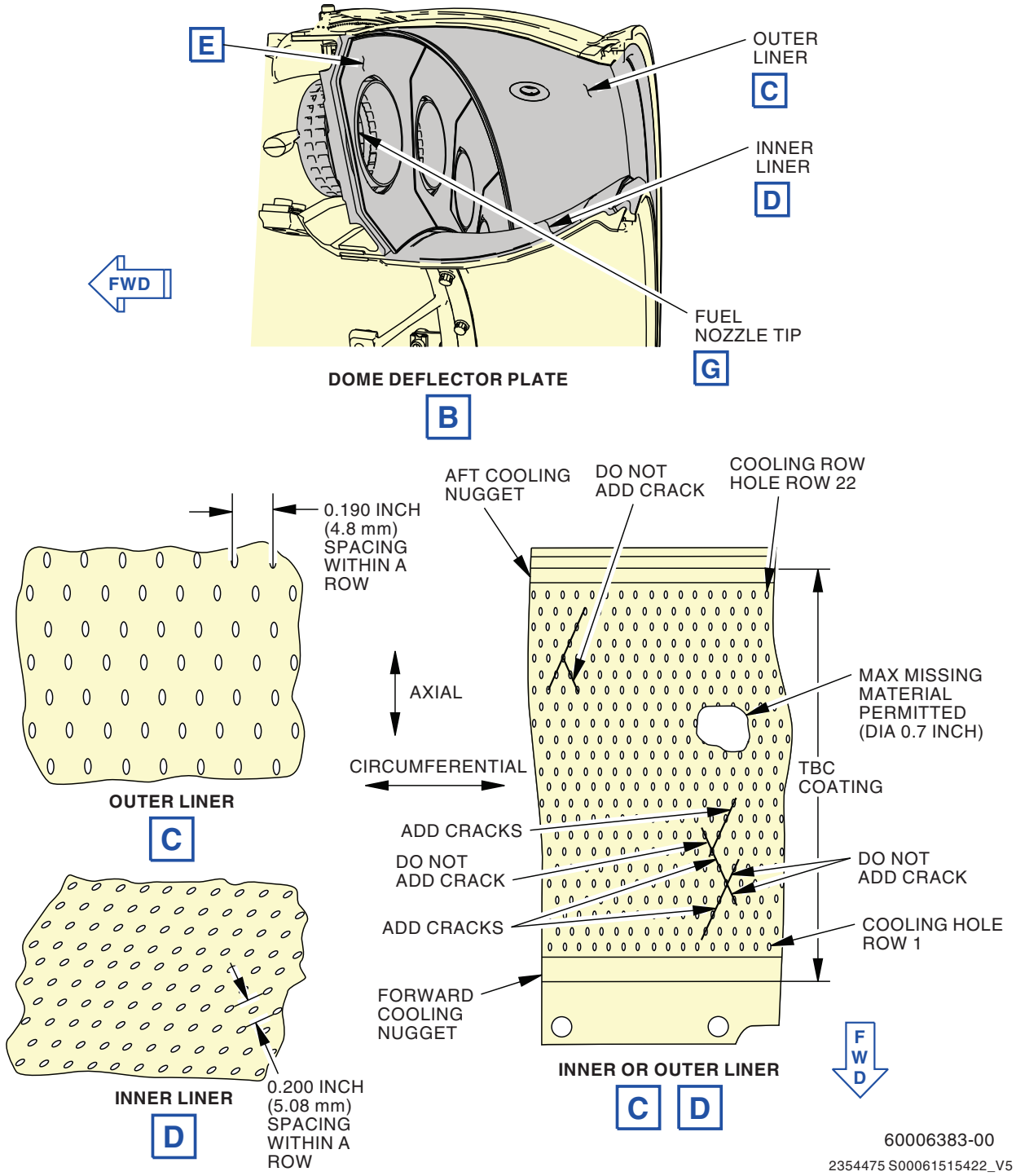


NOTE:
ITEMS REMOVED FOR CLARITY.

5021227-00
2354473 S00061515420_V5

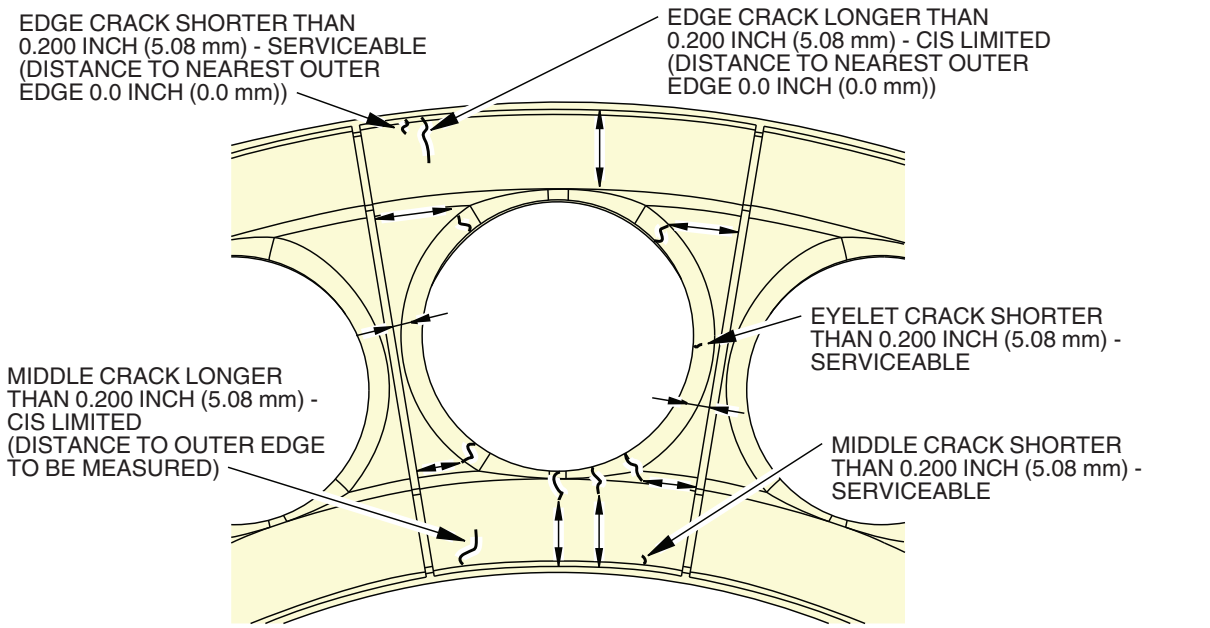
Combustion Section Inspection
Figure 616/72-00-00-990-817-G00 (Sheet 1 of 10)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



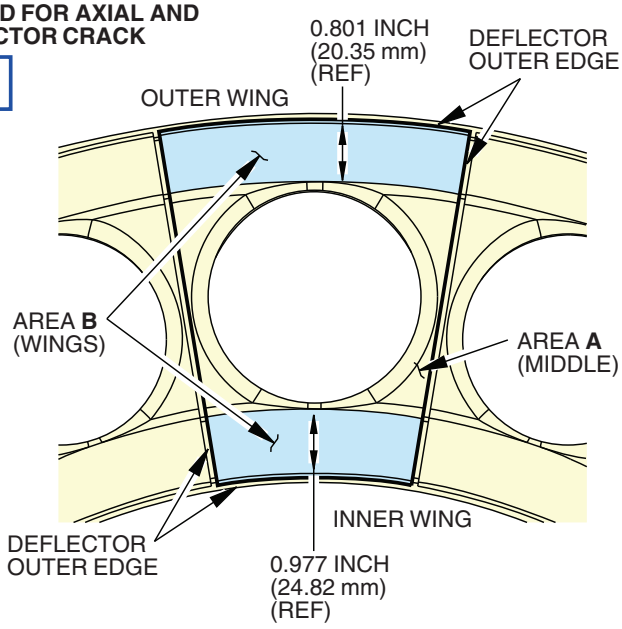
Combustion Section Inspection
Figure 616/72-00-00-990-817-G00 (Sheet 2 of 10)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



MEASURING METHOD FOR AXIAL AND RADIAL DEFLECTOR CRACK

E



LEGEND:

- NEAREST DEFLECTOR EDGE DISTANCE
- INDICATION
- DEFLECTOR OUTER EDGE

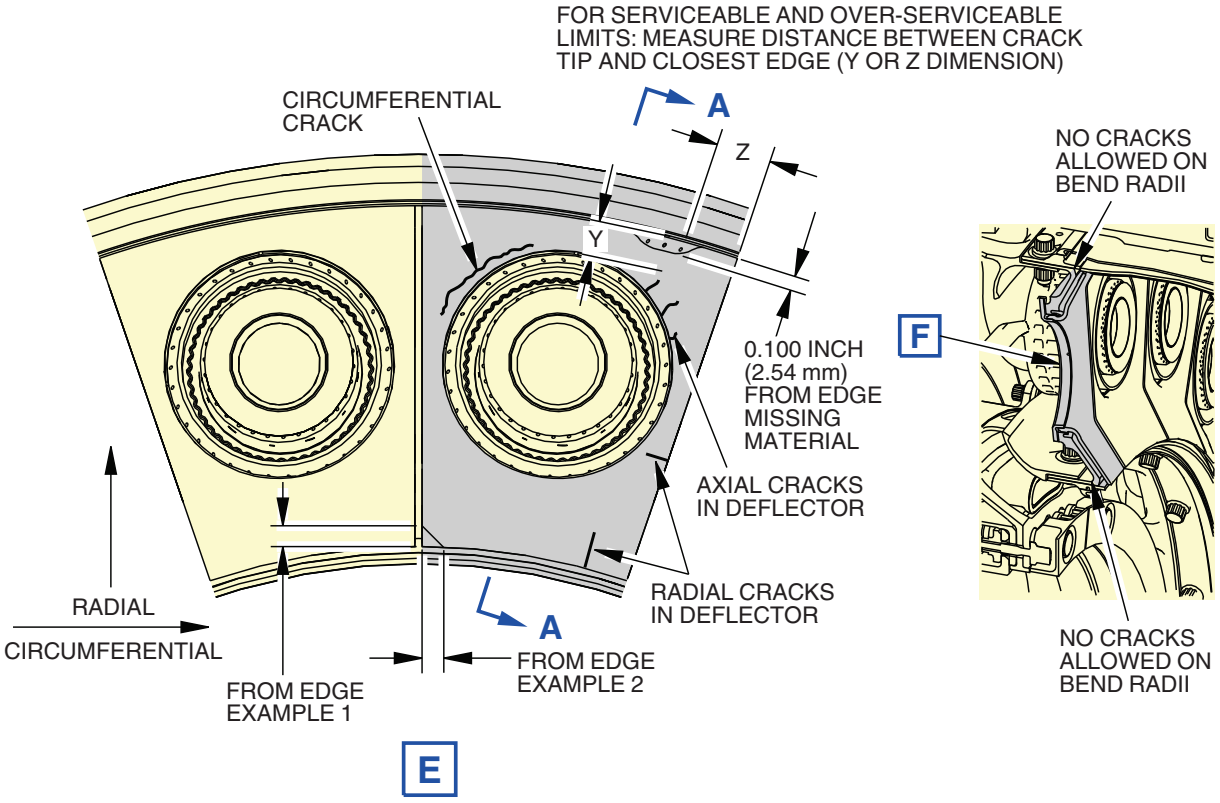
NOTE:

1. THE NEAREST EDGE OF THE DEFLECTOR DOES NOT HAVE TO BE THE TOP OR BOTTOM EDGE. THE NEAREST EDGE CAN ALSO BE THE SIDE EDGE.
2. THE CRACK LIMITS BELOW ARE FOR THE SHORTEST (PERPENDICULAR) DISTANCE TO THE NEAREST DEFLECTOR EDGE.
3. THE POINT TO LINE FUNCTION ON THE BORESCOPE WILL PROVIDE THE SHORTEST (PERPENDICULAR) DISTANCE TO THE NEAREST DEFLECTOR EDGE.

6035955-01
3036146 S0000805512_V2

Combustion Section Inspection
Figure 616/72-00-00-990-817-G00 (Sheet 3 of 10)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOTE:

IN EXAMPLE 1, THE DOME PLATE CANNOT BE SEEN BECAUSE THERE IS NO DEFLECTOR MISSING MATERIAL. THIS CONDITION IS SERVICEABLE.

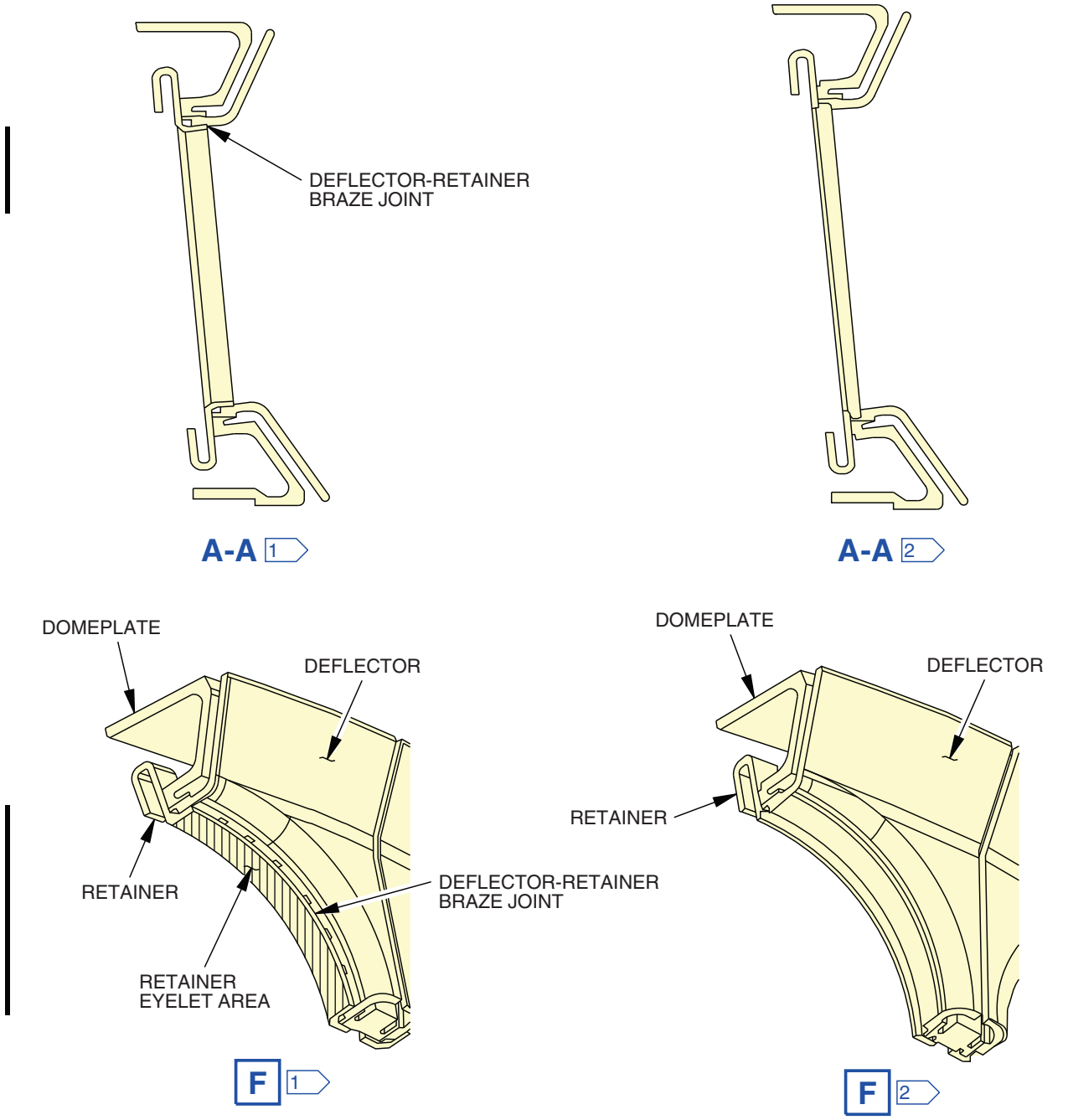
IN EXAMPLE 2, COOLING HOLES ON THE DOME PLATE CAN BE SEEN BECAUSE THERE IS DEFLECTOR MISSING MATERIAL.

IN EXAMPLE 2, NUMEROUS CRACKS AND MISSING MATERIAL IS SEEN. DEFLECTOR MISSING MATERIAL IS SERVICEABLE IF MISSING MATERIAL IS LESS THAN 0.100 INCH (2.54 mm) FROM THE EDGE. DEFLECTOR CRACKING IS SERVICEABLE IF THE CRACKS ARE AXIAL OR RADIAL AND LESS THAN 0.200 INCH (5.08 mm) IN LENGTH, THE CONDITION IS NOT SERVICEABLE IF CIRCUMFERENTIAL CRACKS EXIST.

5021259-01
2354476 S00061515423_V6

Combustion Section Inspection
Figure 616/72-00-00-990-817-G00 (Sheet 4 of 10)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

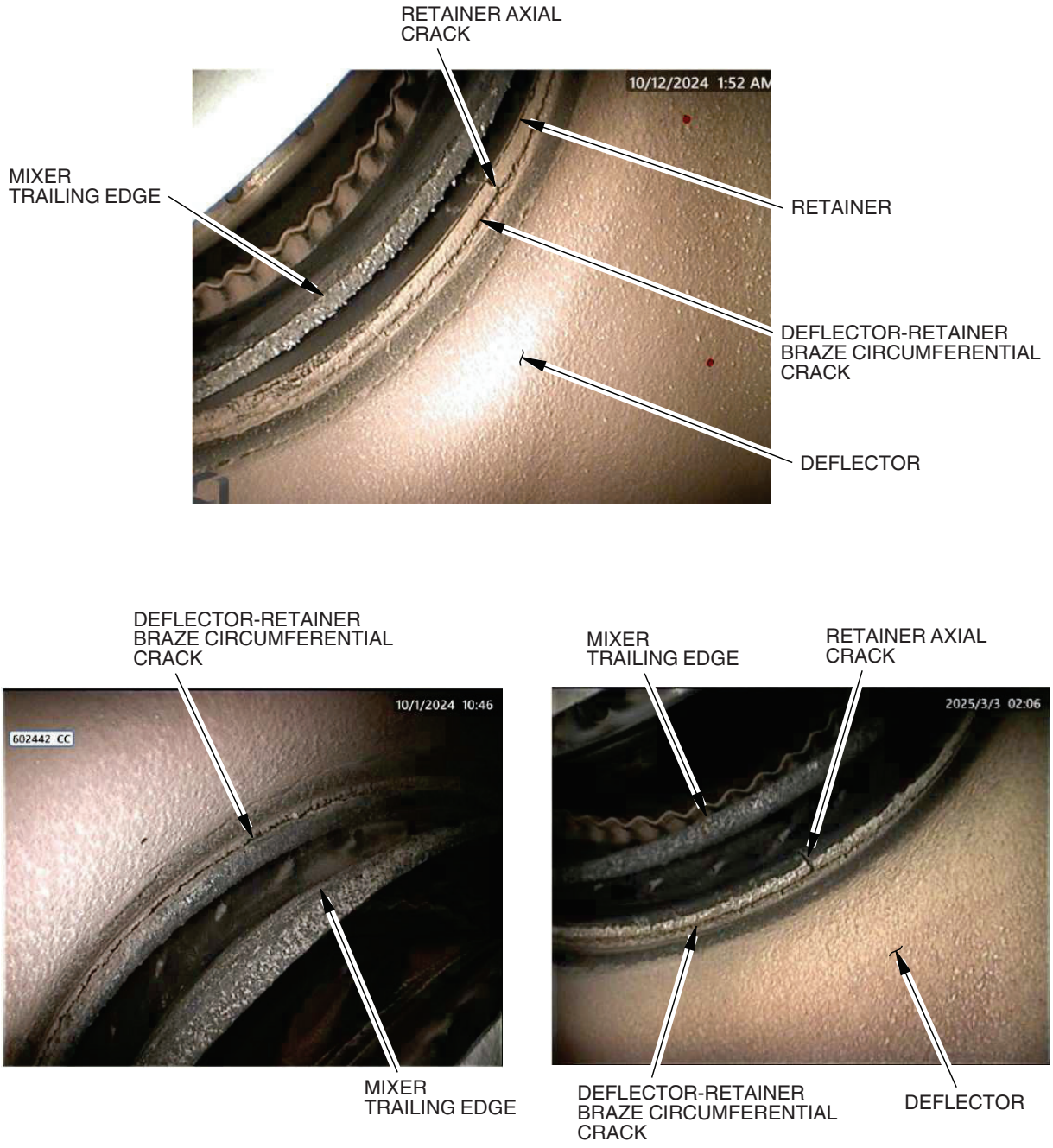


1 AIRPLANES WITH PRE LEAP - 1B SERVICE BULLETIN 72-0145
2 AIRPLANES WITH POST LEAP - 1B SERVICE BULLETIN 72-0145

2963704 S0000741769_V2

Combustion Section Inspection
Figure 616/72-00-00-990-817-G00 (Sheet 5 of 10)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

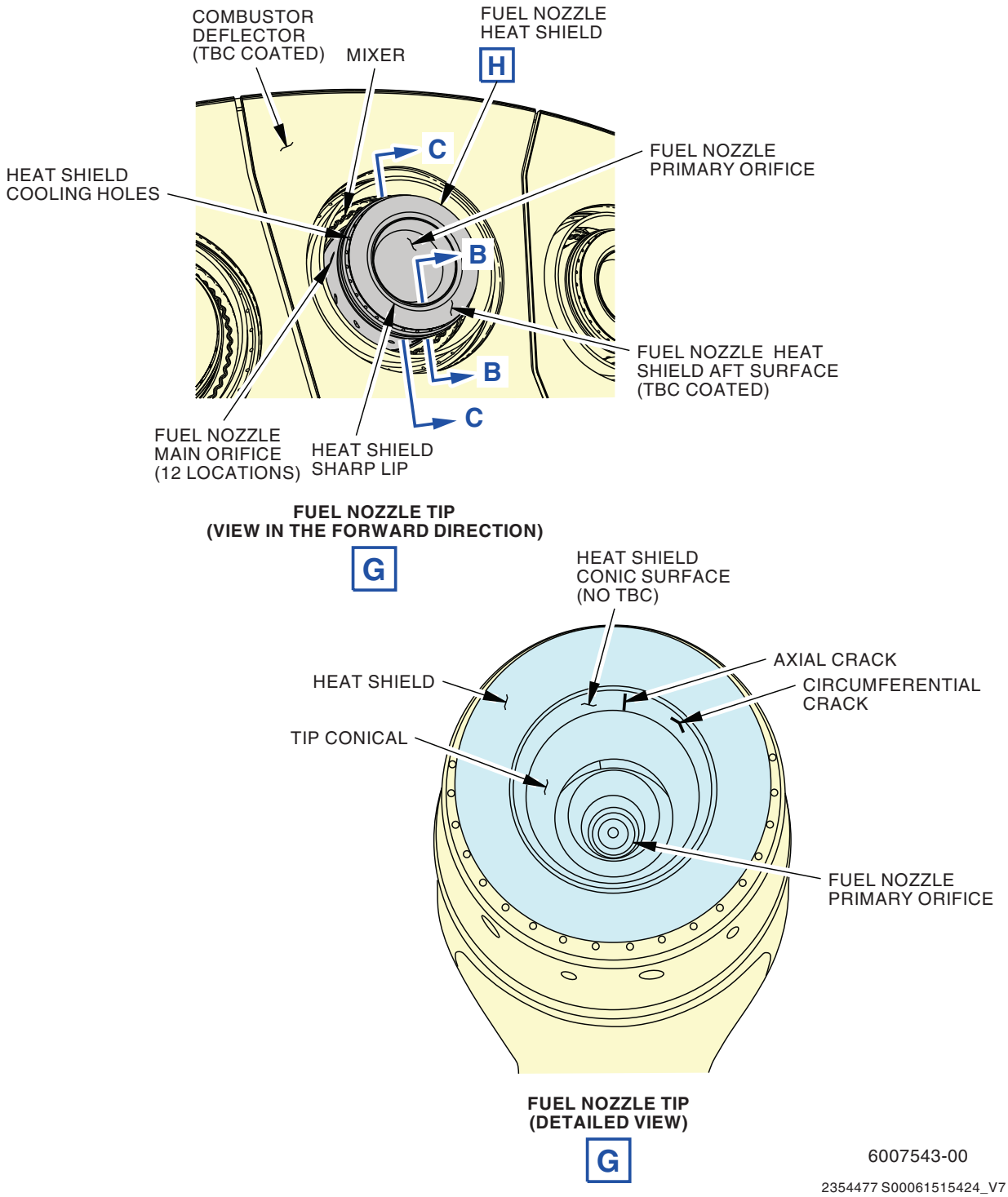


EXAMPLES OF DEFLECTOR-RETAINER CIRCUMFERENTIAL BRAZE CRACK AND RETAINER AXIAL CRACK

3123266 S0000857409_V1

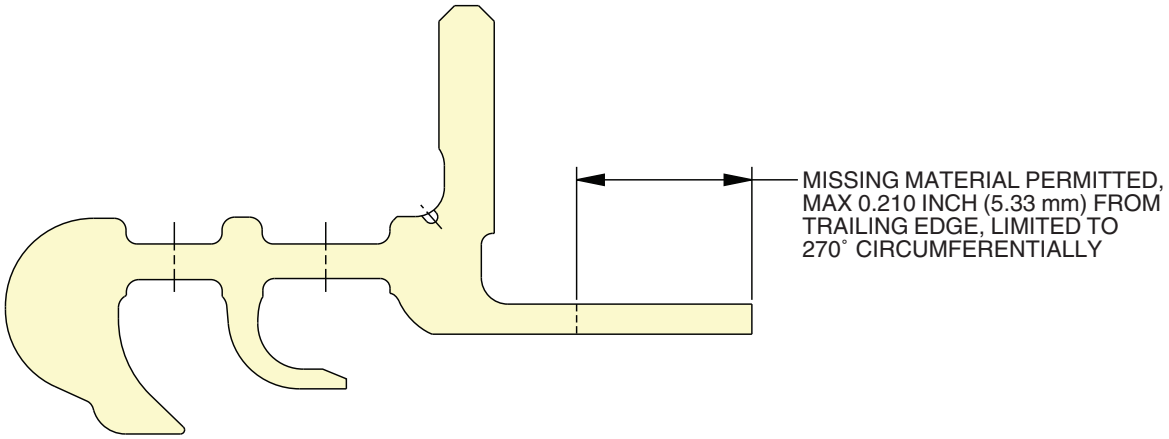
Combustion Section Inspection
Figure 616/72-00-00-990-817-G00 (Sheet 6 of 10)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

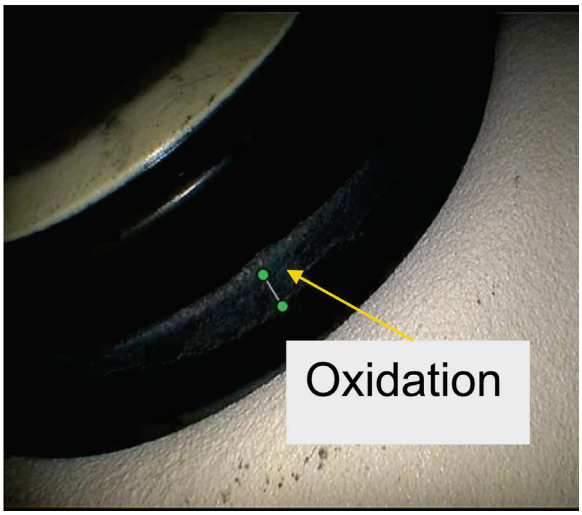


Combustion Section Inspection
Figure 616/72-00-00-990-817-G00 (Sheet 7 of 10)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



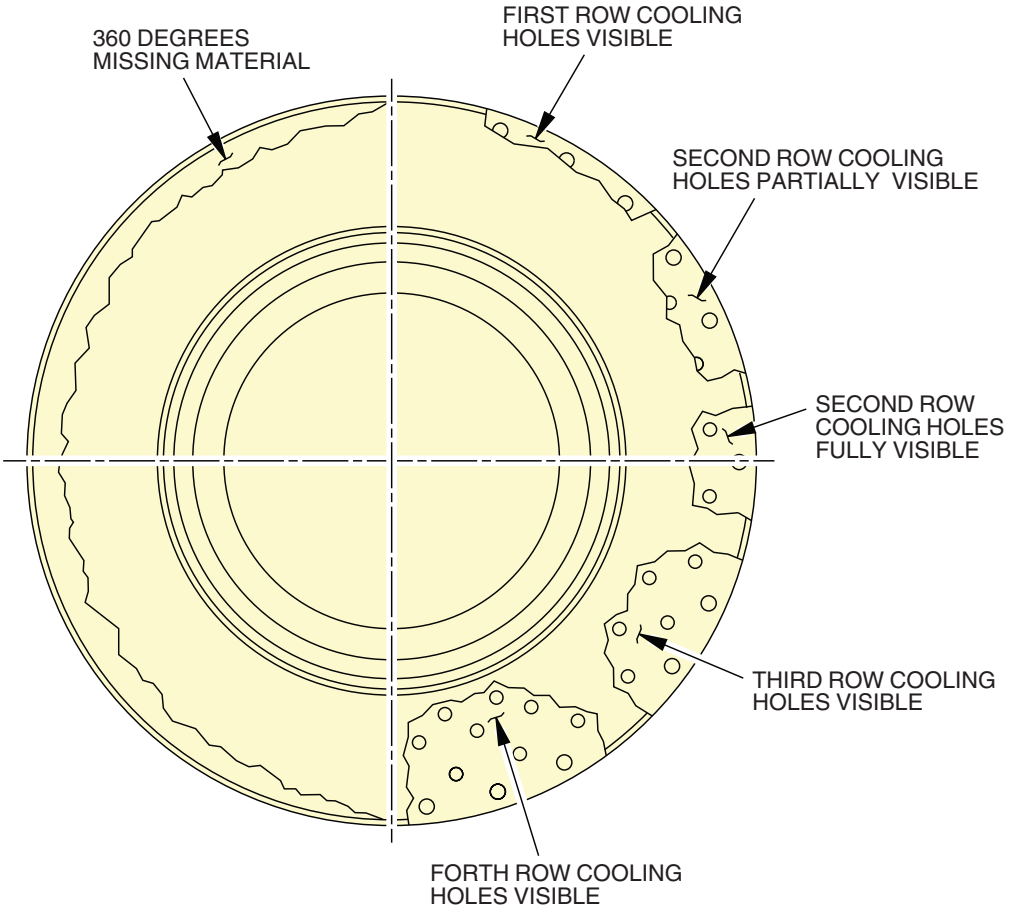
MIXER CONTOUR
B-B



5021274-00
2354478 S00061515425_V5

Combustion Section Inspection
Figure 616/72-00-00-990-817-G00 (Sheet 8 of 10)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



FUEL NOZZLE HEAT SHIELD
(VIEW IN THE FORWARD DIRECTION)

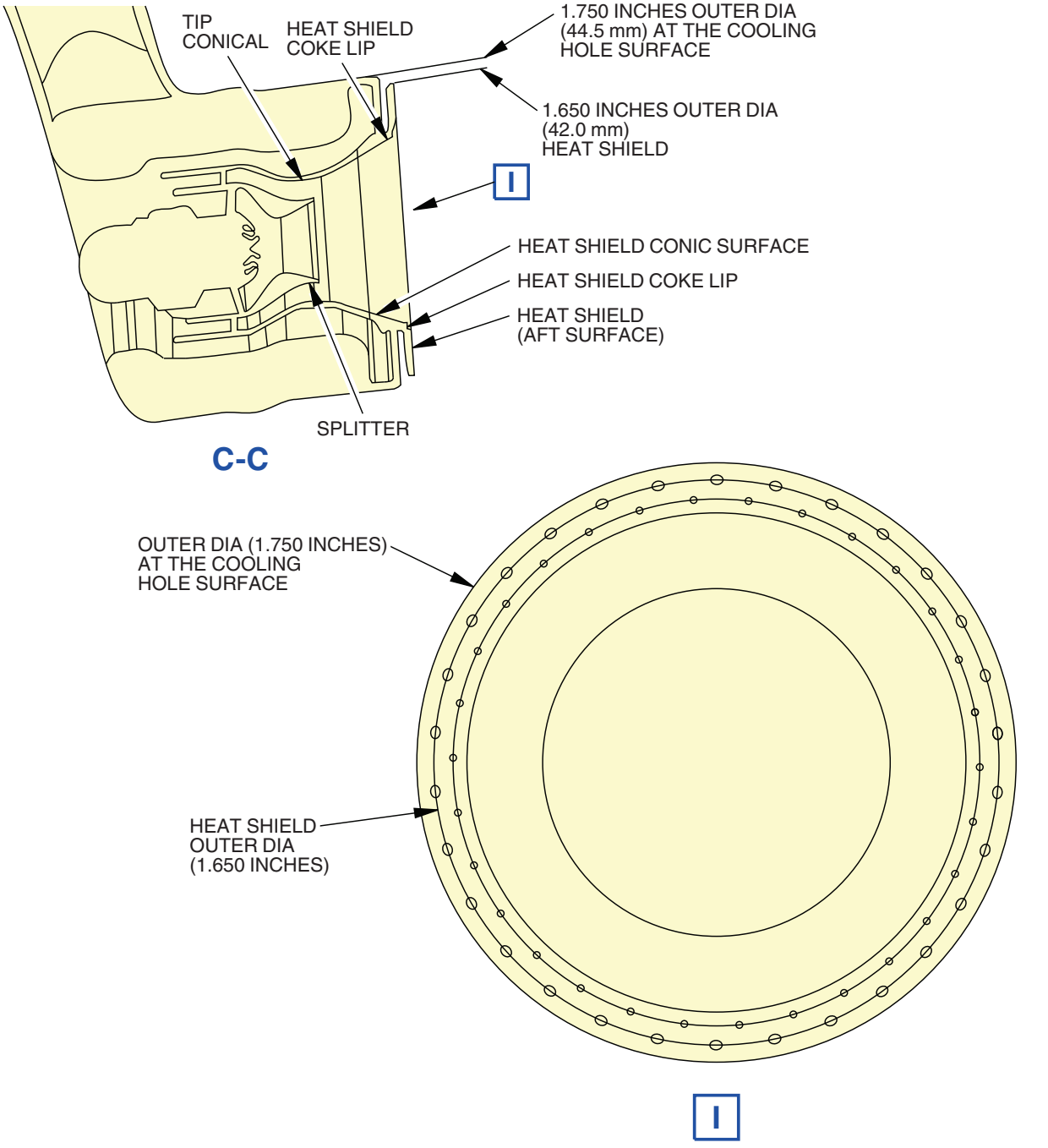
H

6018117-00

2354479 S00061515426_V10

Combustion Section Inspection
Figure 616/72-00-00-990-817-G00 (Sheet 9 of 10)

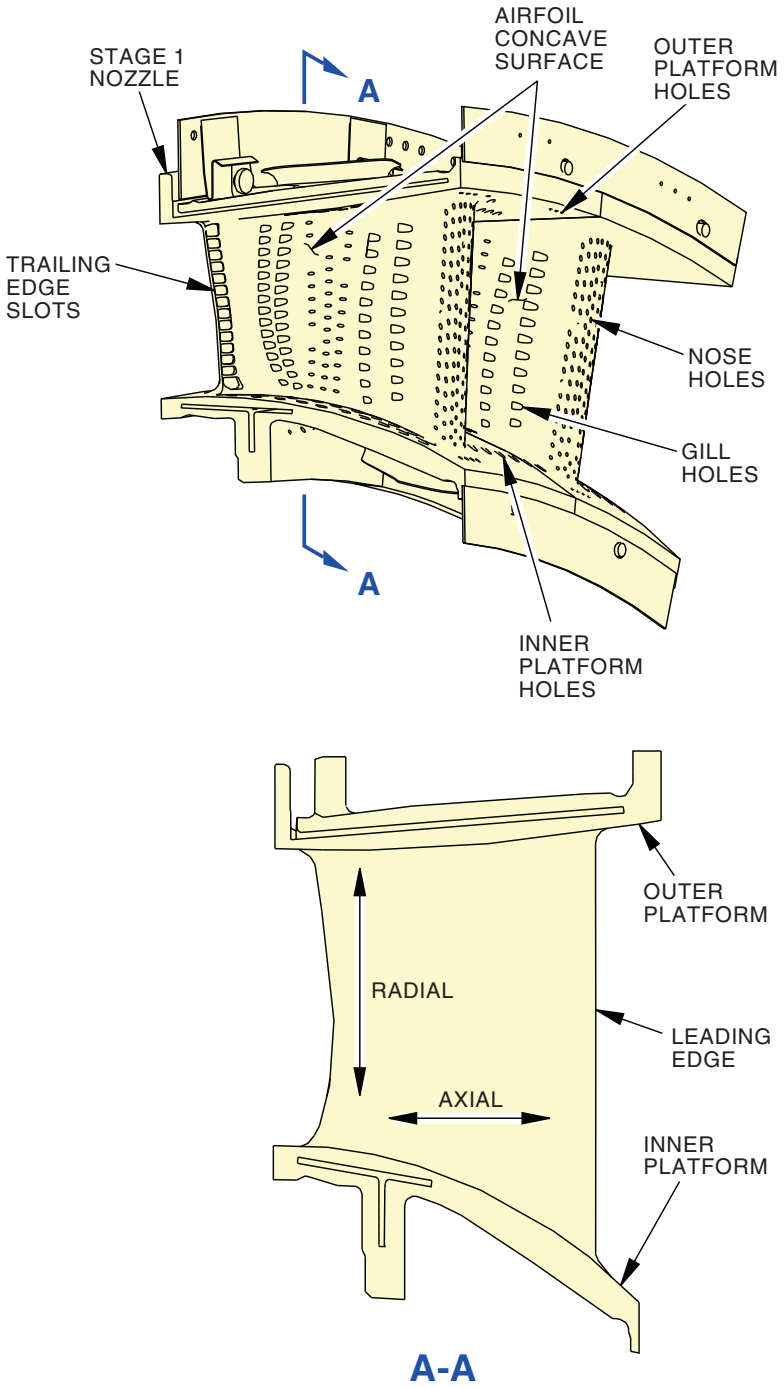
ECCN 9E991 BOEING PROPRIETARY - See title page for details



6018118-00
2947141 S0000723021_V5

Combustion Section Inspection
Figure 616/72-00-00-990-817-G00 (Sheet 10 of 10)

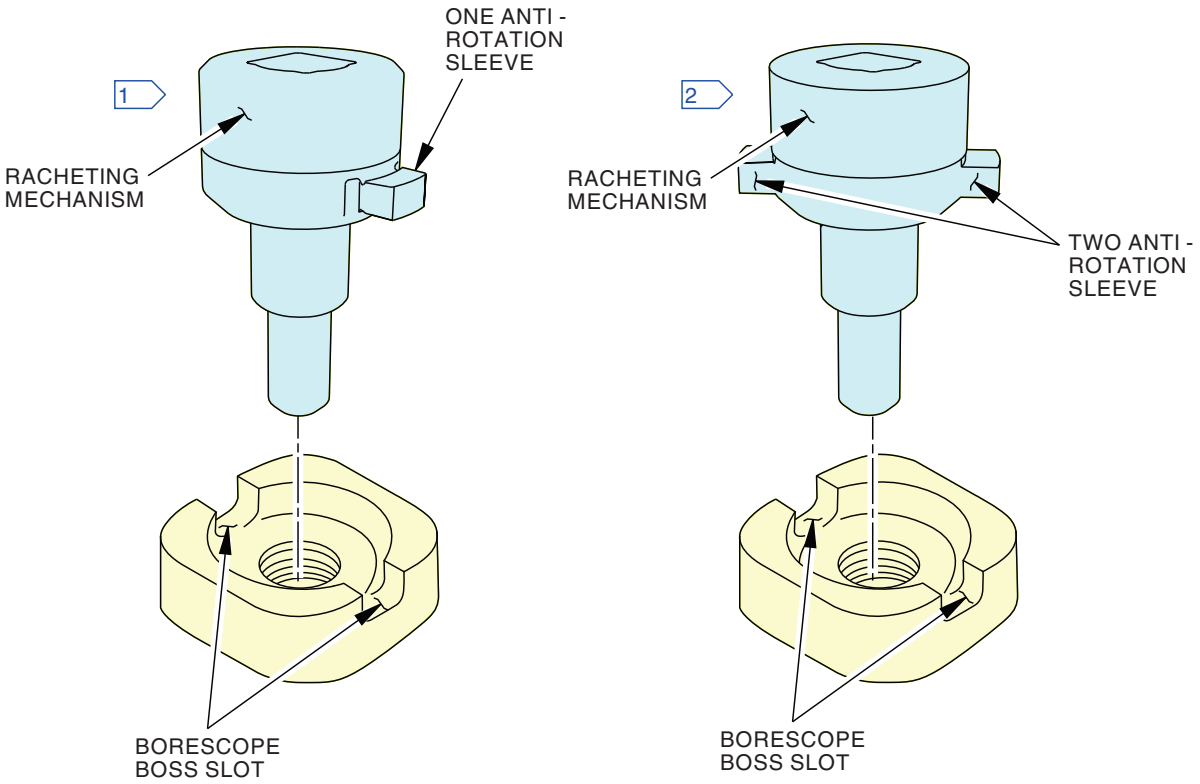
ECCN 9E991 BOEING PROPRIETARY - See title page for details



5021275-00
2354480 S00061515427_V3

HPT Stage One Nozzle Airfoil Inspection
Figure 617/72-00-00-990-818-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOTE:

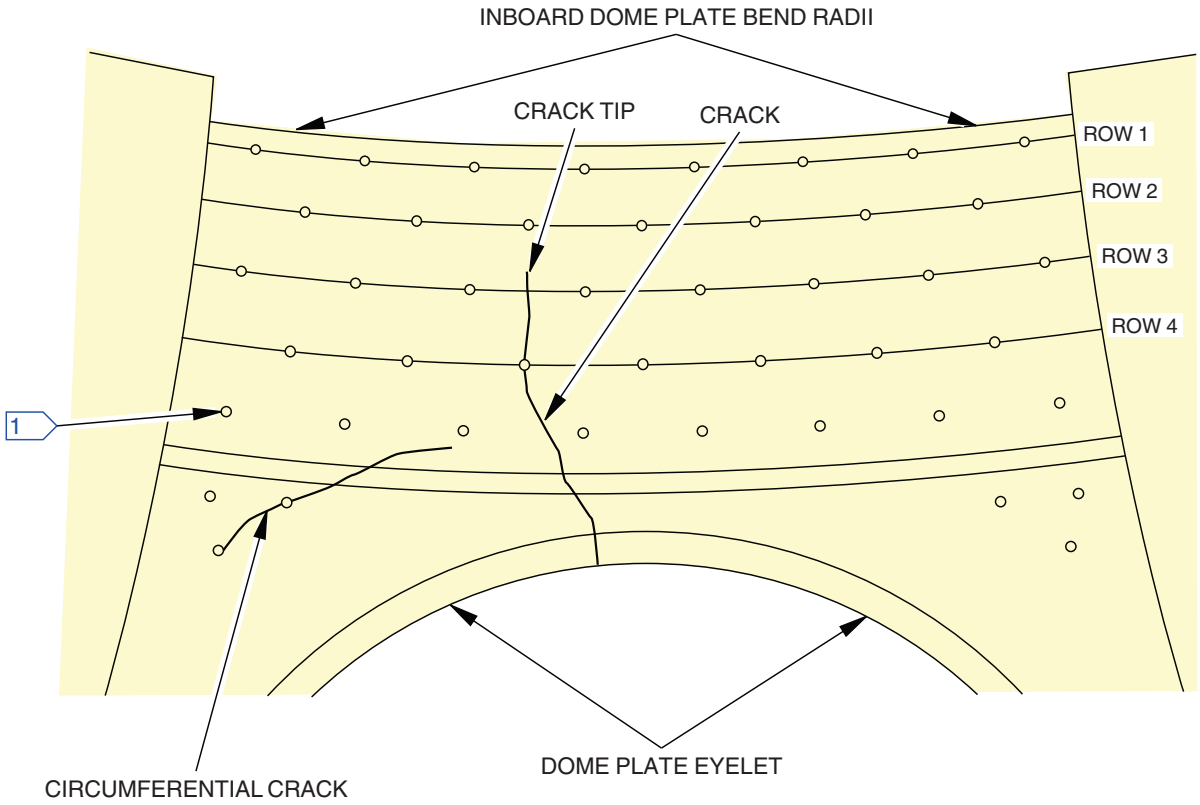
1 PRE LEAP - 1B SERVICE BULLETIN 72-0157

2 POST LEAP - 1B SERVICE BULLETIN 72-0157

5058859-00
2846322 S0000665059_V1

CDN Borescope Plug Installation
Figure 618/72-00-00-990-860-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

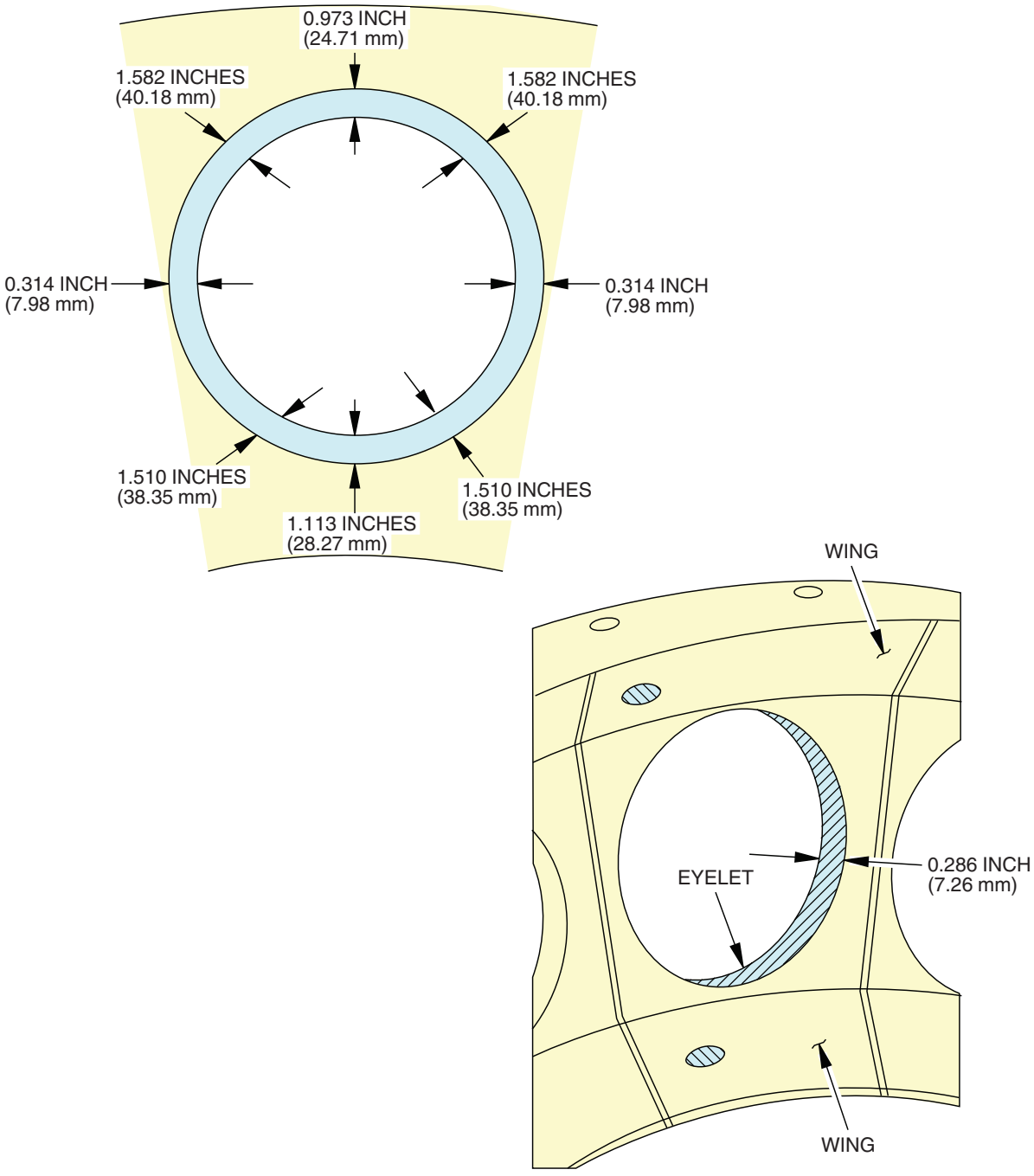


1 COOLING HOLE

6054786-00
3099153 S0000845361_V1

Combustion Chamber Dome Plate Bend Radii Inspection
Figure 619/72-00-00-990-874-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details



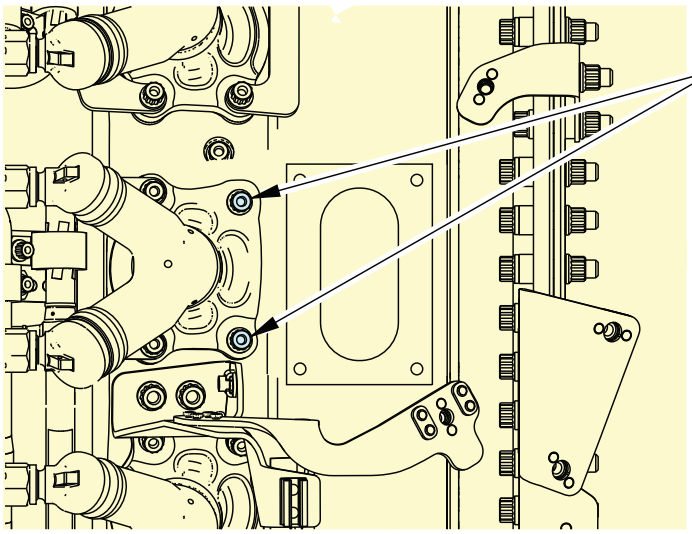
NOTE:

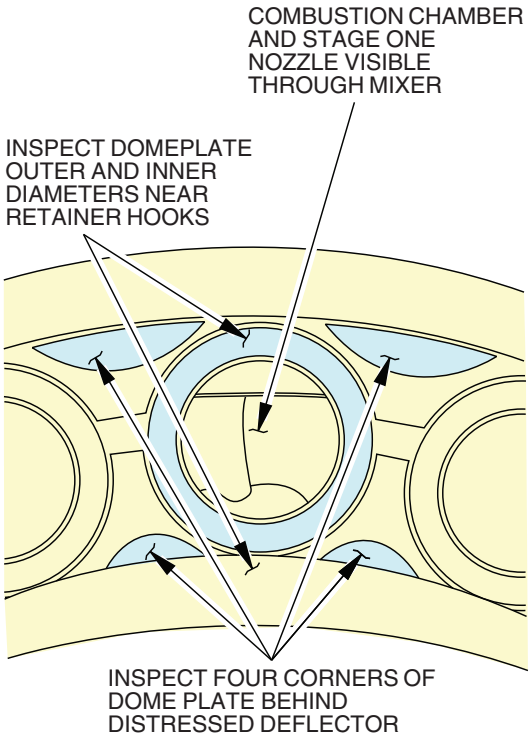
DIMENSIONAL DEFINITION OF THE EYELET MEASURED FROM THE OUTER EDGES OF THE DEFLECTOR.

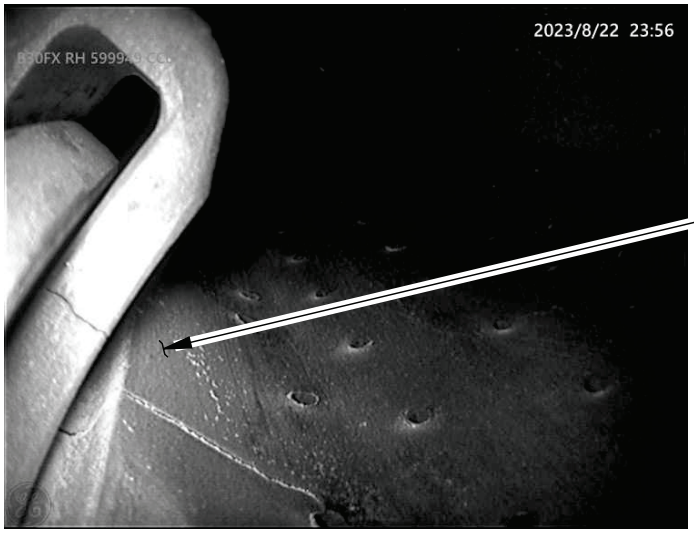
6054786-00
3099165 S0000845362_V1

Deflector Eyelet and Wing Definition Inspection
Figure 620/72-00-00-990-875-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details







6054783-00
6054784-00
6054785-00
3099147 S0000845368_V1

Combustion Chamber Cold Side (FWD) Inspection
Figure 621/72-00-00-990-876-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 72-00-00-290-806-G00

7. High Pressure Turbine Section Detailed Inspection

(Figure 622)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task gives the instructions to do the borescope inspection of the High Pressure Turbine (HPT) section.

B. References

Reference	Title
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)
27-81-00-860-804	Leading Edge Flaps and Slats Retraction (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
72-00-00-822-801-G00	Zero-Index of the N2 Rotor Alignment (P/B 201)
72-00-00-910-801-G00	High Pressure Rotor (N2) Turning for Borescope (P/B 201)
72-63-04-000-801-G00	Accessory Gearbox Manual Control Cover Removal (P/B 401)
72-63-04-400-801-G00	Accessory Gearbox Manual Control Cover Installation (P/B 401)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-14866	Turning Tool - Electronic Part #: 9481M92P05 Supplier: 58828 Part #: 9481M92P06 Supplier: 58828 Opt Part #: 9481M92P03 Supplier: 58828
STD-858	Tag - DO NOT OPERATE
STD-10717	Light Source - Borescope, (Meets General Electric or CFM Standard Practices Manual Specification)
STD-10718	Fiberscope (Meets General Electric or CFM Standard Practices Manual Specification)
STD-10719	Borescope - Rigid (Meets General Electric or CFM Standard Practices Manual Specification such as: model R857ELSV60)

D. Consumable Materials

Reference	Description	Specification
D00601 [CP2101]	Vaseline - Graphite Mineral	
D50355 [CP2961]	Lubricant - Anti-Seize Compound	D6Y31C2

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E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Prepare for the Inspection

SUBTASK 72-00-00-860-037-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-00-00-860-038-G00

(2) Do this task: Leading Edge Flaps and Slats Retraction, TASK 27-81-00-860-804.

SUBTASK 72-00-00-860-039-G00

(3) Do this task: Leading Edge Flaps and Slats - Deactivation, TASK 27-81-00-040-801.

SUBTASK 72-00-00-860-040-G00

(4) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

SUBTASK 72-00-00-941-003-G00

CAUTION MAKE SURE THE TEMPERATURE OF THE ENGINE AND THE ADJACENT AREAS NEAR THE BORESCOPE INSPECTION PORTS ARE LESS THAN 150°F (65.6°C). DO NOT DO A BORESCOPE INSPECTION IF THE ADJACENT AREAS ARE TOO HOT TO TOUCH. DO THE BORESCOPE INSPECTION WHEN THE ENGINE TEMPERATURE IS LESS THAN 150°F (65.6°C). THIS WILL PREVENT DAMAGE TO THE BORESCOPE EQUIPMENT.

(5) Do this task: Preparation for Borescope Inspection, TASK 72-00-00-290-801-G00.

SUBTASK 72-00-00-010-026-G00

(6) Do this task: Accessory Gearbox Manual Control Cover Removal, TASK 72-63-04-000-801-G00.

SUBTASK 72-00-00-490-001-G00

(7) Attach a 3/4 inch square drive hand tool or the electronic turning tool, SPL-14866, to the Accessory Gear Box (AGB) manual control pad (TASK 72-00-00-910-801-G00).

SUBTASK 72-00-00-010-017-G00

(8) Remove the borescope plugs from borescope ports P, Q, and R.

SUBTASK 72-00-00-480-023-G00

(9) Put an identification tag on each borescope plug.

SUBTASK 72-00-00-980-005-G00

(10) Do this task: Zero-Index of the N2 Rotor Alignment, TASK 72-00-00-822-801-G00.

SUBTASK 72-00-00-480-005-G00

(11) Turn the light source, STD-10717, to on and insert the rigid borescope, STD-10719, or fiberscope, STD-10718, into its position for the inspection.

NOTE: This inspection is limited to what can be seen with the rigid borescope.

SUBTASK 72-00-00-941-004-G00

(12) Manually or electrically motor the High Pressure Compressor (HPC) rotor (TASK 72-00-00-910-801-G00).

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SUBTASK 72-00-00-290-006-G00

(13) Make sure that the instructions and limits are followed closely in the inspection task.

NOTE: Table 604 provides the HPT blades, the nozzles, and the shroud segment count.

Table 604/72-00-00-993-811-G00

Component	Quantity of segments, blades, or airfoils	Notes
Stage 1 HPT Nozzle	18 Segments, 36 Airfoils	Each nozzle segment contains 2 airfoils
Stage 1 HPT Shroud	18 Segments	
Stage 1 HPT Blades	60 Blades	
Stage 2 HPT Nozzle	36 Segments, 36 Airfoils	Each nozzle segment contains 1 airfoil
Stage 2 HPT Shroud	36 Segments	
Stage 2 HPT Blades	68 Blades	

H. High Pressure Turbine Section Inspection

SUBTASK 72-00-00-220-010-G00

(1) Examine the concave surfaces of the HPT stage 1 nozzle as follows (Figure 622 (Sheet 1) and Figure 622 (Sheet 2)):

NOTE: The inspection is only of the HPT stage 1 nozzle assembly that is adjacent to the borescope port.

(a) Radial cracks that extend from the cooling holes

1) Permitted if there is no measurable through-wall hole.

(b) Other cracks that are not in the cooling hole rows

1) Permitted if there is no measurable through-wall hole.

(c) Burns (Surface oxidation resulting in erosion of base material)

1) Permitted if the burns are not fully through the wall.

(d) Cracks in the fillet area adjacent to the inner and outer platforms

1) Permitted if there is no measurable through-wall hole.

(e) Continue-In-Service (CIS) limit for the concave surfaces of the HPT stage 1 nozzle.

NOTE: This CIS limit is for a reduced interval repeat inspection.

1) A maximum measured hole area of 0.1750 in^2 (1.129 cm^2) on each airfoil, because of cracks or burn-through, is permitted for a maximum of 25 cycles.

SUBTASK 72-00-00-220-011-G00

(2) Examine the convex surfaces of the HPT stage 1 nozzle as follows (Figure 622 (Sheet 1) and Figure 622 (Sheet 2)):

NOTE: The inspection is only of the HPT stage 1 nozzle assembly that is adjacent to the borescope port.

(a) Radial cracks

1) Permitted if there is no measurable through-wall hole.

(b) Axial cracks

1) Permitted if there is no measurable through-wall hole.

(c) Burns (Surface oxidation resulting in erosion of base material)

1) Permitted if the burns are not fully through the wall.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(d) Cracks in the fillet area adjacent to the inner and outer platforms

1) Permitted if there is no measurable through-wall hole.

(e) CIS limit for the convex surfaces of the HPT stage 1 nozzle.

NOTE: This CIS limit is for a reduced interval repeat inspection.

1) A maximum measured holed area of 0.035 in^2 (0.226 cm^2) on each airfoil, because of cracks or burn-through is permitted for a maximum of 100 cycles.

2) A maximum measured hole area of 0.1750 in^2 (1.129 cm^2) on each airfoil because of cracks or burn-through is permitted for a maximum of 25 cycles.

SUBTASK 72-00-00-220-012-G00

(3) Examine the trailing edges of the HPT stage 1 nozzle airfoils as follows (Figure 622 (Sheet 1) and Figure 622 (Sheet 2)):

NOTE: The inspection is only of the HPT stage 1 nozzle assembly that is adjacent to the borescope port.

(a) Axial cracks that start at the slots adjacent to the trailing edge

1) Permitted.

(b) Buckling or bowing

1) Permitted.

(c) Material that is missing

1) The area of missing material must not be more than 0.75 in^2 (4.8 cm^2) for each airfoil.

(d) CIS limits for the trailing edges of the HPT stage 1 nozzle.

NOTE: This CIS limit is for a reduced interval repeat inspection.

1) A maximum area of 0.8750 in^2 (5.645 cm^2) of missing material on each airfoil is permitted for a maximum of 25 cycles.

SUBTASK 72-00-00-220-013-G00

(4) Examine all areas of the HPT stage 1 nozzle airfoil as follows (Figure 622 (Sheet 1) and Figure 622 (Sheet 2)):

NOTE: The inspection is only of the HPT stage 1 nozzle assembly that is adjacent to the borescope port.

(a) Craze cracks

1) Permitted.

(b) Nicks, scores, scratches and dents

1) Permitted.

(c) Metal splatter

1) Permitted.

(d) Missing thermal barrier coating.

1) Permitted.

SUBTASK 72-00-00-220-014-G00

(5) Examine the inner and outer platforms of the HPT stage 1 nozzle as follows (Figure 622 (Sheet 1) and Figure 622 (Sheet 2)):

NOTE: The inspection is only of the HPT stage 1 nozzle assembly that is adjacent to the borescope port.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Missing thermal barrier coating

1) Permitted.

(b) Cracks in the parent material

1) Permitted.

(c) Nicks, scores, scratches and dents on the platform surface

1) Permitted.

(d) Burns

1) Permitted if no material is missing and the burns are not fully through.

(e) Bulges or bows

1) Not permitted.

(f) HPT stage 1 nozzle inner platform protruding or missing spline seal

1) Permitted.

(g) CIS limits for the inner and outer platforms of the HPT stage 1 nozzle.

NOTE: This CIS limit is for a reduced interval repeat inspection.

1) Bulges or bows are permitted for a maximum of 25 cycles.

2) A maximum area of 0.070 in^2 (0.45 cm^2) of missing material, on each airfoil, because of cracks or burn-through is permitted for a maximum of 25 cycles.

SUBTASK 72-00-00-220-015-G00

(6) Examine the HPT stator stage 1 shroud segment as follows (Figure 622 (Sheet 1) and Figure 622 (Sheet 3)):

NOTE: This inspection is limited to what can be seen with the rigid borescope.

NOTE: For the HPT stage 1 shroud segments, it is recommended to do an inspection with the 360 degree flexible borescope. Refer to TASK 72-00-00-290-808-G00.

NOTE: The flow path coating is both the top coating and bond coating. Exposed base material is area where both the top coating and bond coating are fully missing.

(a) Blade rub

1) Permitted.

(b) Cracks in the shroud flowpath surface

1) Cracks in the base material are permitted up to 0.60 in. (15.24 mm) in length.

2) Cracks in the flow path coating are permitted.

(c) Missing flow path coating

1) Any amount of missing coating permitted in or less than 0.10 in. (2.5 mm) from the leading and trailing edges.

2) For any shroud segment that is missing complete circumferential flow path coating in or less than 0.10 in. (2.5 mm) of the trailing edge, do an inspection of the leading edges of the three adjacent stage 2 HPT nozzle segments.

3) For the areas more than 0.10 in. (2.5 mm) from the leading edge and trailing edge, a total of 0.024 in^2 (15.48 mm^2) of missing coating is permitted.

(d) Chipping of the shroud segment base material along the leading and trailing edge

1) Up to two chips on each shroud segment are permitted if they are in or less than 0.050 in. (1.27 mm) from the edge.

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(e) Chipping of the shroud segment base material along the circumferential split line

1) A maximum axial length of 1 in. (25.4 mm) is permitted on up to 10 shroud segments on each end.

a) Any number of missing or partially missing discourager spline seals is permitted.

b) Any number of missing or partially missing primary spline seals is permitted.

(f) CIS limits for the shroud segment flowpath surface:

1) If a flex scope is necessary to do an inspection of the shroud with condition on reduced interval, refer to High Pressure Turbine (HPT) Stage 1 Shroud Borescope Inspection, TASK 72-00-00-290-808-G00.

2) A 1600 cycle inspection interval with a flex scope is permitted with this condition, if all shrouds are reviewed:

a) Any amount of missing flow path coating and exposed base material is permitted.

3) A 800 cycle inspection interval with a flex scope is permitted with this condition, if all shrouds are reviewed:

a) Chipping of the shroud base material up to 0.25 in. (6.4 mm) from the leading edge or up to 0.15 in. (3.8 mm) from the trailing edge.

4) A 200 cycle inspection interval with a flex scope is permitted with this condition, if all shrouds are reviewed:

a) A maximum axial length of 1.0 in. (25.4 mm) of missing base material along the circumferential split spline is permitted on all shroud segments.

b) Any amount of circumferential cracking that does not extend across the entire shroud segment length.

5) A 100 cycle inspection interval with a flex scope is permitted with these conditions, if all shrouds are reviewed:

a) Chipping of the shroud base material up to 0.35 in. (8.9 mm) from the leading edge or up to 0.25 in. (6.4 mm) from the trailing edge.

6) A 30 cycle inspection interval with a flex scope is permitted with these conditions, if all shrouds are reviewed:

a) Any amount of axial cracking that does not extend across the entire shroud segment length.

b) Any amount of circumferential cracking that does extend across the entire shroud segment length.

7) Remove the engine in 10 cycles or less with this condition:

a) Any amount of axial or circumferential cracking, where cracking has intersected.

b) Any amount of axial cracking that extends across the entire shroud segment length.

c) Any amount of shroud segment radial displacement into the flow path.

d) If you find a through hole in shroud segment or missing portion of the shroud segment.

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SUBTASK 72-00-00-211-017-G00

(7) Examine the surface coating of the HPT rotor stage 1 blades as follows (Figure 622 (Sheet 1) and Figure 622 (Sheet 4)):

(a) Missing surface coating with metal oxidation that shows as green or black discoloration of the material

1) Permitted, if there are no holes or cracks in the blade.

NOTE: The serviceability for holes or cracks in the blade must be found by paragraphs (8) through (16) that follow to permit the missing surface coating with metal oxidation.

SUBTASK 72-00-00-220-016-G00

(8) Examine the leading edges of the HPT rotor stage 1 blades as follows (Figure 622 (Sheet 1) and Figure 622 (Sheet 4)):

(a) Axial and radial cracks in Area A

1) All cracks are permitted with these conditions:

a) A crack does not connect more than three holes.

b) Axial cracks do not extend farther than the leading edge holes.

c) Axial cracks do not extend farther than the convex surface cooling holes.

(b) CIS limits for axial and radial cracks in Area A of the HPT rotor stage 1 blade leading edge

NOTE: This CIS limit is for a reduced interval repeat inspection.

1) A 650 cycle inspection interval is permitted with these conditions:

a) Cracks can connect between holes, if there is no missing material.

b) Remove the engine in 25 cycles or less if the axial crack extends farther than the leading edge holes or the convex surface cooling holes.

(c) Radial cracks that go out of the Area B leading edge holes

1) Permitted with these conditions:

a) A crack does not connect more than two holes.

b) There are no radial cracks that turn in an axial direction.

(d) CIS limits for radial cracks in Area B of the HPT stage 1 blade leading edge

NOTE: These CIS limits are for a reduced interval repeat inspection.

1) A 250 cycle inspection interval is permitted with these conditions:

a) Cracks can connect between holes, if there is no missing material.

b) Remove the engine in 25 cycles or less if the radial cracks turn in an axial direction and extend farther than the leading edge holes or convex surface cooling holes.

(e) Axial cracks in Area B

1) Not permitted.

(f) CIS limits for axial cracks in Area B of the HPT stage 1 blade leading edge

NOTE: This CIS limit is for a reduced interval repeat inspection.

1) A 250 cycle inspection interval is permitted with these conditions:

a) Cracks can connect between holes, if there is no missing material.

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b) Cracks do not extend farther than the leading edge holes.

c) Cracks do not extend farther than the convex surface cooling holes.

d) Remove the engine in 25 cycles or less if a crack extends farther than the leading edge holes or convex surface cooling holes.

(g) Nicks and dents in Area A

1) Permitted with these conditions:

a) The nick, dent and all related cracks do not extend farther than the leading edge holes.

b) The nick, dent and all related cracks do not extend farther than the convex surface cooling holes.

(h) Nicks and dents in Area B

1) Permitted with these conditions:

a) The nick, dent, and all related cracks do not extend farther than the leading edge holes.

b) The nick, dent, and all related cracks do not extend farther than the convex surface cooling holes.

c) There are no related tears or material that is missing.

(i) Material that is missing in Area A

NOTE: Squealer tip is not part of Area A.

1) Material that is missing is not permitted.

(j) CIS limits for material that is missing from Area A of the HPT stage 1 blade leading edge

NOTE: This CIS limit is for a reduced interval repeat inspection.

1) A 650 cycle inspection interval is permitted with these conditions:

a) All amounts are permitted if it does not extend farther than the first internal rib.

NOTE: The first internal rib is forward of Row 2 on the convex side and aligned with Row 6 on the concave side.

b) All related axial cracks do not extend farther than the leading edge holes.

c) All related axial cracks do not extend farther than the convex surface cooling holes.

d) Remove the engine in 25 cycles or less if the related axial cracks extend farther than the leading edge holes or convex surface cooling holes.

(k) CIS limits for material that is missing from Area A of the HPT stage 1 blade leading edge that extends farther than the first internal rib:

NOTE: The first internal rib is forward of Row 2 on the convex side and aligned with Row 6 on the concave side.

NOTE: This CIS limit is for a reduced interval repeat inspection.

1) A 125 cycle inspection interval is permitted with these conditions:

a) All amounts are permitted.

b) All related axial cracks do not extend farther than the leading edge holes.

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PTF ALL PRE SB 737-CFM-LEAP-1B-72-0454 (Continued)

c) All related axial cracks do not extend farther than the convex surface cooling holes.

d) Remove the engine in 25 cycles or less if all related axial cracks extend farther than the leading edge holes or convex surface cooling holes.

e) Remove the engine in 15 cycles or less if the metal that is missing extends farther than the second internal rib.

NOTE: Missing material into or beyond Row 1 on the convex side and farther aft of Row 6 on the concave side.

(l) Material that is missing in Area B

1) Not permitted.

(m) CIS limits for material that is missing from Area B of the HPT stage 1 blade leading edge

NOTE: This CIS limit is for a reduced interval repeat inspection.

1) A 250 cycle inspection interval is permitted with these conditions:

a) All amounts are permitted if it does not extend farther than the first internal rib.

b) All related axial cracks are not permitted.

c) All related radial cracks are not permitted.

d) Remove the engine in 5 cycles or less if missing material extends farther than the first rib.

e) Remove the engine in 25 cycles or less if related axial or radial cracks are present.

(n) Blocked leading edge holes

1) One blocked cooling hole (row 1- 6) permitted in area B.

2) One blocked cooling hole (row 1- 6) permitted in area A.

(o) Surface coating.

1) Permitted with metal oxidation (green of black discoloration of the material) if there are no holes or cracks in the blade.

SUBTASK 72-00-00-220-017-G00

(9) Examine the trailing edges of the HPT rotor stage 1 blade as follows (Figure 622 (Sheet 1) and Figure 622 (Sheet 4)):

(a) Axial cracks in Area A

1) All cracks that are less than 0.20 in. (5.1 mm) from the trailing edge are permitted.

(b) CIS limits for axial cracks in Area A of the HPT stage 1 blade trailing edge

NOTE: This CIS limit is for a reduced interval repeat inspection.

1) A 650 cycle inspection interval is permitted with these conditions:

a) The cracks are less than 0.275 in. (6.99 mm) in length.

b) Remove the engine in 25 cycles or less if the cracks are more than 0.275 in. (6.99 mm).

(c) Axial cracks in Area B, but not in the trailing edge root fillet radius

1) Not permitted.

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PTF ALL PRE SB 737-CFM-LEAP-1B-72-0454 (Continued)

(d) CIS limits for axial cracks in Area B of the HPT stage 1 blade trailing edge but not in the trailing edge root fillet radius

NOTE: This CIS limit is for a reduced interval repeat inspection.

1) A 250 cycle inspection interval is permitted with these conditions:

a) The cracks are less than 0.275 in. (6.99 mm) in length.

b) Remove the engine in 15 cycles or less if the cracks are more than 0.275 in. (6.99 mm) in length.

(e) Axial cracks in the trailing edge root radius

1) Cracks that are less than 0.135 in. (3.43 mm) in length and do not turn radially are permitted.

(f) CIS limits for axial cracks in the trailing edge root radius

NOTE: This CIS limit is for a reduced interval repeat inspection.

1) A 125 cycle inspection interval is permitted with these conditions:

a) The cracks are less than 0.275 in. (6.99 mm) in length and do not turn radially.

b) Remove the engine in 25 cycles or less if the cracks are longer than 0.275 in. (6.99 mm) in length, or turn in a radial direction.

(g) Nicks in Area A

1) Permitted if there is no related missing or torn material, or cracks.

(h) CIS limits for nicks in Area A with related missing or torn material, or cracks

NOTE: This CIS limits are for a reduced interval repeat inspection.

1) A 650 cycle inspection interval is permitted with these conditions:

a) No internal cavities can be seen.

b) There are no cracks more than 0.275 in. (6.99 mm).

c) Remove the engine in 25 cycles or less if cracks are longer than 0.275 in. (6.99 mm).

2) If internal cavities can be seen, a 125 cycle inspection interval is permitted with these conditions:

a) There are no cracks more than 0.275 in. (6.99 mm).

b) Remove the engine in 25 cycles or less if cracks are longer than 0.275 in. (6.99 mm).

(i) Nicks in Area B

1) Permitted with these conditions:

a) There are no related torn material or cracks.

b) There are no nicks in the root fillet radius.

(j) The CIS limits for nicks in Area B

NOTE: This CIS limit is for a reduced interval repeat inspection.

1) A 650 cycle inspection interval is permitted with these conditions:

a) Nicks in the root fillet radius are less than 0.275 in. (6.99 mm).

b) There are is no torn material or cracks.

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c) The internal cavity of the blade cannot be seen.

2) A 250 cycle inspection interval is permitted with these conditions:

a) Nicks in the root fillet radius are less than 0.275 in. (6.99 mm).

b) The torn metal or cracks are less than 0.275 in. (6.99 mm) in length.

c) The internal cavity of the blade cannot be seen.

3) A 125 cycle inspection interval is permitted with these conditions:

a) Nicks in the root fillet radius are less than 0.275 in. (6.99 mm).

b) The torn metal or cracks are less than 0.275 in. (6.99 mm) in length.

c) The internal cavity of the blade can be seen.

4) Remove the engine in 25 cycles or less if torn metal or cracks are more than 0.275 in. (6.99 mm).

(k) Dents in Area A

1) Permitted with these conditions:

a) The length or diameter of the dent is not more than the length of one trailing edge slot 0.11 in. (2.8 mm).

b) The depth of the dent is not more than 0.03 in. (0.8 mm).

c) No metal extrusion from the dent into the trailing edge slot is permitted.

(l) The CIS limits for dents in Area A that have extrusion into the trailing edge slot

NOTE: This CIS limit is for a reduced interval repeat inspection.

1) A 650 cycle inspection interval is permitted with this condition:

a) There are no related cracks.

(m) Dents in Area B

1) Permitted with these conditions:

a) The length or diameter of the dent is not more than the length of one trailing edge slot 0.11 in. (2.79 mm).

b) The depth of the dent is not more than 0.03 in. (0.76 mm).

c) No metal extrusion from the dent into the trailing edge slot is permitted.

(n) CIS limits for the stage 1 blade with trailing edge dents in Area B that have extrusion into the trailing edge slot

NOTE: This CIS limit is for a reduced interval repeat inspection.

1) A 250 cycle inspection interval is permitted with this condition:

a) There is no cracking.

(o) Tears in Area A

1) Not permitted.

(p) The CIS limits for tears in Area A

NOTE: This CIS limit is for a reduced interval repeat inspection.

1) A 650 cycle inspection interval is permitted with these conditions:

a) The tear that is further than 0.275 in. (6.99 mm) from the trailing edge.

b) All axial cracks from the tears are less than 0.07 in. (1.8 mm) in length.

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c) Radial cracks in the direction of the tip from tears are permitted.

(q) Tears in Area B

1) Not permitted.

(r) The CIS limits for stage 1 tears in Area B

NOTE: This CIS limit is for a reduced interval repeat inspection.

1) A 250 cycle inspection interval is permitted with these conditions:

a) The tear is further than 0.275 in. (6.99 mm) from the trailing edge.

b) All axial cracks from the tears are less than 0.07 in. (1.8 mm) in length.

c) Radial cracks in the direction of the tip from tears are permitted.

(s) Oxidation in the root radius

1) Not permitted.

(t) CIS limits for oxidation in the stage 1 blade trailing edge root radius

NOTE: This CIS limit is for a reduced interval repeat inspection.

1) A 125 cycle inspection interval is permitted with these conditions:

a) The internal cavities cannot be seen.

b) Remove the engine in 10 cycles or less if internal cavities can be seen.

(u) Metal that is missing from the stage 1 blade trailing edge in Areas A or B but not in the root fillet radius

1) Permitted if the internal cavities of the blade cannot be seen.

(v) CIS limits for metal that is missing from the stage 1 blade trailing edge in Areas A or B but not in the root fillet radius where internal cavities can be seen

NOTE: This CIS limit is for a reduced interval repeat inspection.

1) A 650 cycle inspection interval is permitted with these conditions:

a) Axial cracks are permitted if they are not more than 0.07 in. (1.8 mm) in length.

b) Radial cracks are permitted if they are not more than 0.14 in. (3.6 mm) in length.

c) Missing metal is not more than 0.07 in. (1.8 mm) in diameter or equivalent area.

2) A 250 cycle inspection interval is permitted with these conditions:

a) Axial cracks are permitted if they are not more than 0.07 in. (1.8 mm) in length.

b) Radial cracks are permitted if they are not more than 0.14 in. (3.6 mm) in length.

(w) Metal that is missing from the stage 1 blade trailing edge in the root fillet radius

1) Not permitted.

a) If metal is missing, see the CIS limits that follow.

(x) CIS limits for metal that is missing from the stage 1 blade trailing edge in the root fillet radius with no cracks

NOTE: This CIS limit is for a one time use.

1) Remove the engine in 10 cycles or less.

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(y) Metal that is missing from the stage 1 blade trailing edge in the root fillet radius with axial or radial cracks.

1) Not permitted.

2) Remove the engine before the next flight.

SUBTASK 72-00-00-220-018-G00

(10) Examine the HPT rotor stage 1 blade concave and convex surfaces as follows (Figure 622 (Sheet 1) and Figure 622 (Sheet 4)):

(a) Radial cracks at the cooling hole

1) Permitted above the second cooling hole from the top with this condition:

a) All radial cracks connect a maximum of two holes and do not turn in an axial direction.

2) No cracks are permitted below the cooling holes on the concave surface.

a) See the CIS limits that follow.

3) The CIS limits for stage 1 blade concave surface radial cracks at the cooling hole:

NOTE: This CIS limit is for a reduced interval repeat inspection.

a) A 10 cycle inspection interval is permitted with these conditions:

<1> There are no cracks on all cooling holes on the concave surface.

<2> The cracks connect a maximum of three holes and do not turn in an axial direction.

(b) Radial cracks in Area A other than at the cooling holes

1) Permitted with these conditions:

a) There is no more than one crack on the concave surface and is no longer than the space between four cooling holes 0.30 in. (7.6 mm).

b) The crack does not turn in an axial direction.

c) The internal cavity of the blade cannot be seen.

2) CIS limits for radial cracks in Area A other than at the cooling holes

NOTE: This CIS limit is for a reduced interval repeat inspection.

a) A 250 cycle inspection interval is permitted with these conditions:

<1> There is no more than one crack on the concave surface and is no longer than 0.69 in. (17.5 mm) and less than 0.14 in. (3.6 mm) in width.

<2> The internal cavity of the blade cannot be seen.

(c) Radial cracks in Area B other than at the cooling holes

1) Permitted with these conditions:

a) There is no more than one crack and the crack is no longer than the space between four cooling holes 0.30 in. (7.6 mm).

b) The crack does not turn in an axial direction.

c) The internal cavity of the blade cannot be seen.

2) CIS limits for radial cracks in Area B other than at the cooling holes:

NOTE: This CIS limit is for a reduced interval repeat inspection.

a) A 250 cycle inspection interval is permitted with these conditions:

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<1> The internal cavity of the blade cannot be seen.

<2> There is no more than one crack on the concave surface and is no more than 0.42 in. (10.7 mm) in length.

<a> Remove the engine in 25 cycles or less if the crack is more than 0.42 in. (10.7 mm) in length.

<3> The crack does not go in an axial direction.

NOTE: This CIS limit is for a one time use.

<a> Remove the engine in 10 cycles or less if the crack is in an axial direction.

(d) Radial cracks that go out of the tips of the cooling holes

1) Three cracks are permitted with these conditions:

a) The cracks do not go into the next cooling hole.

b) The cracks do not go in the direction of the blade platform from the bottom hole.

c) The cracks do not go in an axial direction.

2) CIS limits for radial cracks that go out of the tip of the cooling holes

NOTE: This CIS limit is for a reduced interval repeat inspection.

a) A 250 cycle inspection interval is permitted with these conditions:

<1> The cracks do not connect more than two cooling holes.

<2> The cracks do not go in the direction of the platform from the bottom hole.

<3> The cracks do not turn in an axial direction.

<4> Remove the engine in 10 cycles or less if the crack turns axial or if it goes in the direction of the platform from the bottom hole.

b) A 25 cycle inspection interval is permitted with these conditions:

NOTE: This CIS limit is for a reduced interval repeat inspection.

<1> The cracks connect a maximum of three cooling holes.

<2> The cracks do not go in the direction of the platform from the bottom hole.

<3> The cracks do not turn in an axial direction.

<4> Remove the engine in 10 cycles or less if the crack connects more than 3 cooling holes or turn in an axial direction.

(e) Nicks and dents in Area A concave and convex surfaces

1) Permitted with this condition:

a) There are no tears, cracks, or holes that go through the surface metal.

(f) CIS limits for nicks and dents in Area A of the HPT stage 1 blade forward of nose of row 2 cooling holes on the convex surface

NOTE: This CIS limit is for a reduced interval repeat inspection.

1) A 150 cycle inspection interval is permitted with these conditions:

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a) All amounts are permitted if damage starts forward of the nose of row 2 cooling holes.

b) All related missing material and axial cracks do not extend farther than the Leading Edge (LE) holes.

NOTE: The LE holes are defined by row 3 to row 6.

c) All related missing material or axial cracks do not extend farther than row 2 cooling holes.

d) Remove the engine in 25 cycles or less if the related axial cracks extend farther than the LE holes or row 2 cooling holes.

e) Remove the engine in 15 cycles or less if the metal that is missing extends farther than the first internal rib.

NOTE: Missing material into or beyond row 2 on the convex side and farther aft of row 6 on the concave side.

f) Remove the engine before next flight cycle if the nicks and dents in Area A convex surface with cracks or holes through the metal surface initiate at or aft of row 2 cooling holes.

(g) Nicks and dents in Area B concave and convex surfaces

1) Permitted in the Area B concave surface with this condition:

a) The diameter of the nick or dent is not more than the space between two cooling holes 0.07 in. (1.8 mm).

2) Permitted in the Area B convex surface with this condition:

a) There are no tears, cracks or holes through the surface metal.

3) No nicks or dents are permitted in the root radius.

(h) Twisted, bent, burnt or melted airfoil surfaces

1) Not permitted.

(i) Axial cracks in area A and B.

1) Not permitted.

a) See the CIS limits that follow.

2) The CIS limits for axial cracks on the concave and convex surface areas A and B:

NOTE: This CIS limit is for a reduced interval repeat inspection.

a) A 250 cycle inspection interval is permitted with these conditions:

<1> There is a maximum of one crack that is less than 0.07 in. (1.8 mm) in length.

<2> The crack does not turn in a radial direction.

3) The CIS limits for axial cracks on the concave and convex surface areas A and B:

NOTE: This CIS limit is for a reduced interval repeat inspection.

a) A 125 cycle inspection interval is permitted with these conditions:

<1> There is a maximum of one crack that is less than 0.14 in. (3.6 mm) in length.

<2> The crack does not turn in a radial direction.

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<3> Remove the engine in 10 cycles or less if the crack is more than 0.14 in. (3.6 mm) in length or if the crack turns in a radial direction.

SUBTASK 72-00-00-220-019-G00

(11) Examine the HPT rotor stage 1 blade for radial cracks on the pressure (concave) airfoil:

(Figure 622 (Sheet 1), Figure 622 (Sheet 4))

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(Figure 622 (Sheet 6))

PTF ALL PRE SB 737-CFM-LEAP-1B-72-0454; POST LEAP-1B SERVICE BULLETIN 72-0138

(Figure 622 (Sheet 7))

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(a) Radial cracks on the HPT rotor stage 1 blade on the pressure (concave) airfoil between, but not related to, row 7 and 8 cooling air holes

1) Not permitted.

(b) Engines with radial cracks on the HPT rotor stage 1 blade on the pressure (concave) airfoil between, but not related to, row 9 and 10 cooling air holes are permitted to continue in service as follows:

NOTE: The CIS limits are reduced intervals repeat inspections.

1) A 1300 cycle inspection interval is permitted for this condition:

a) There is no more than one crack in region A (region A is different than Area A) and it is not more than 0.19 in. (4.8 mm).

2) A 650 cycle inspection interval is permitted for these conditions:

a) There is no more than one radial crack that extends farther than Area A and into Regions B or C, and the crack is no more than 0.48 in. (12.2 mm).

b) There is no more than one radial crack that goes through the upper Region C to the tip that is no more than 0.48 in. (12.2 mm).

3) Remove the engine in 25 cycles or less if there is a radial crack on the pressure surface that extends farther than Regions B or C.

SUBTASK 72-00-00-220-020-G00

(12) Examine the HPT rotor stage 1 blade tip as follows (Figure 622 (Sheet 1) and Figure 622 (Sheet 4)):

NOTE: The blade tip is defined as the area within 0.16 in. (4.1 mm) of the end of the blade.

(a) Radial cracks that begin at the end of the blade (the crack could open up at the tip and make a 'V' notch)

1) Concave side of the blade:

a) It is permitted to have a crack in the 'V' notch if the internal cavity cannot be seen.

b) It is permitted for cracks to intersect the top cooling holes in a concave cooling row.

c) Cracks are permitted if they do not go further than 0.20 in. (5.08 mm) from the end of the blade.

<1> See the CIS limits that follow.

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2) CIS limits for radial cracks on the concave side of the blade

NOTE: This CIS limit is for a reduced interval repeat inspection.

a) A 800 cycle inspection interval is permitted with these conditions:

<1> The radial tip cracks are between 0.20 in. (5.08 mm) and 0.25 in. (6.35 mm) in length.

<2> The internal cavity cannot be seen.

b) A 400 cycle inspection interval is permitted with these conditions:

<1> The radial tip cracks are between 0.25 in. (6.35 mm) and 0.30 in. (7.62 mm) in length.

<2> The internal cavity cannot be seen.

c) Remove the engine in 10 cycles or less if the radial cracks are longer than 0.30 in. (7.62 mm) or if the internal cavity can be seen.

3) Convex side of the blade:

a) A crack in the 'V' notch is permitted if the internal cavity cannot be seen.

b) Cracks are permitted if they do not go further than 0.20 in. (5.08 mm) from the end of the blade.

<1> See the CIS limits that follow.

4) The CIS limits for radial cracks on the convex side of the blade.

NOTE: This CIS limit is for a reduced interval repeat inspection.

a) A 800 cycle inspection interval is permitted with these conditions:

<1> The radial tip cracks are between 0.20 in. (5.08 mm) and 0.25 in. (6.35 mm) in length.

<2> The internal cavity cannot be seen.

b) A 400 cycle inspection interval is permitted with these conditions:

<1> The radial tip cracks are between 0.25 in. (6.35 mm) and 0.30 in. (7.62 mm) in length.

<2> The internal cavity cannot be seen.

c) Remove the engine in 10 cycles or less if the radial cracks are longer than 0.30 in. (7.62 mm) or if the internal cavity can be seen.

(b) Axial cracks

1) Permitted.

(c) Bent or curled blade tips

1) Any amount permitted if the bent or curled tips do not go below the most outboard trail edge slot and the internal cavity cannot be seen.

(d) Metal that is missing from the blade squealer tip.

1) Permitted with these conditions:

a) Missing material does not go below the tip cap.

b) The internal cavity cannot be seen.

2) The CIS limits for metal that is missing from the blade squealer tip:

NOTE: This CIS limit is for a one time use.

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a) Remove the engine in 10 cycles or less if the internal cavity can be seen.

3) Metal that is missing from the tip on the trailing edge:

a) Metal that is missing from the tip on the trailing edge is permitted with these conditions:

<1> Damage does not go below the first trailing edge slot.

<2> The internal cavity cannot be seen.

4) The CIS limits for metal that is missing from the tip on the trailing edge:

NOTE: This CIS limit is for a reduced interval repeat inspection.

a) A 125 cycle inspection interval is permitted with these conditions:

<1> The damage goes below the first trailing edge slot, but does not go farther than 0.20 in. (5.1 mm).

<2> The internal cavity cannot be seen.

SUBTASK 72-00-00-220-021-G00

(13) Examine for cracks in Area B on the concave side of the airfoil surface that go out of row 12 cooling holes toward row 9 cooling holes:

(Figure 622 (Sheet 1), Figure 622 (Sheet 4))

PTF ALL PRE SB 737-CFM-LEAP-1B-72-0454; PRE LEAP-1B SERVICE BULLETIN 72-0138

(Figure 622 (Sheet 6))

PTF ALL PRE SB 737-CFM-LEAP-1B-72-0454; POST LEAP-1B SERVICE BULLETIN 72-0138

(Figure 622 (Sheet 7))

PTF ALL PRE SB 737-CFM-LEAP-1B-72-0454

(a) Cracks that go out of the tips of row 12 cooling holes toward row 9 cooling holes are not permitted.

(b) A CIS limit for cracks that go out of the tips of row 12 cooling holes toward row 9 cooling holes are as follows:

NOTE: This CIS limit is for a reduced interval repeat inspection.

1) 250 cycle inspection interval is permitted with this condition:

a) Cracks from the bottom hole do not reach the blade platform.

2) Cracks are not within 0.10 in. (2.5 mm) of row 9 cooling holes.

3) Remove the engine in 10 cycles or less if crack from the bottom hole reaches the blade platform, or is within 0.10 in. (2.5 mm) of row 9 cooling holes.

SUBTASK 72-00-00-220-037-G00

(14) Examine for cracks in Area A on the concave side of the airfoil surface that go out of row 12 cooling holes toward row 9 cooling holes :

(Figure 622 (Sheet 1), Figure 622 (Sheet 4))

PTF ALL PRE SB 737-CFM-LEAP-1B-72-0454; PRE LEAP-1B SERVICE BULLETIN 72-0138

(Figure 622 (Sheet 6))

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PTF ALL PRE SB 737-CFM-LEAP-1B-72-0454; POST LEAP-1B SERVICE BULLETIN 72-0138

(Figure 622 (Sheet 7))

PTF ALL PRE SB 737-CFM-LEAP-1B-72-0454

(a) Cracks that go out of the tips of row 12 cooling holes toward row 9 cooling holes are not permitted.

(b) The CIS limit for cracks that go out of the tips of row 12 cooling holes toward row 9 cooling holes are as follows:

NOTE: This CIS limit is for a reduced interval repeat inspection.

1) A 650 cycle inspection interval is permitted with these conditions:

a) Cracks from the bottom hole of row 12 do not touch the blade platform.

b) Cracks from row 12 cooling holes are in or less than 0.10 in. (2.5 mm) of row 9 cooling holes.

c) If any other cracks are in or less 0.10 in. (2.5 mm) of row 9 or row 12 cooling holes, remove the engine in 10 cycles or less.

SUBTASK 72-00-00-220-022-G00

(15) Examine the HPT stage 1 rotor blade tips for buildup of first stage shroud material (Figure 622 (Sheet 1) and Figure 622 (Sheet 4)):

(a) Shroud material build-up on the blade tips is permitted.

SUBTASK 72-00-00-220-023-G00

(16) Examine the HPT rotor stage 1 blade platform as follows (Figure 622 (Sheet 1) and Figure 622 (Sheet 4)):

NOTE: Lifted, raised, or peeling material is defined as the blade platform material which appears above the platform surface as a result of oxidation or cracks.

(a) Cracks on the pressure side (concave side) of the platform.

1) Any amount of cracks less than 0.200 in. (5.08 mm) is permitted.

a) The CIS limits for HPT pressure side platform cracking are as follows:

<1> A 600 cycle inspection interval is permitted for pressure side (concave) platform cracks more than 0.200 in. (5.08 mm) but less than 0.300 in. (7.62 mm).

<2> A 350 cycle inspection interval is permitted for pressure side (concave) platform cracks that are between 0.300 in. (7.62 mm) and 0.350 in. (8.89 mm) in length.

<3> 100 cycle inspection interval is permitted for pressure side (concave) platform cracks that are 0.350 in. (8.89 mm) to 0.400 in. (10.16 mm) in length.

<4> Remove the engine in 5 cycles or less if crack is more than 0.400 in. (10.16 mm) or if any raised material is observed on the concave platform can be seen.

PTF ALL PRE SB 737-CFM-LEAP-1B-72-0454; POST LEAP-1B SERVICE BULLETIN 72-0138

2) One crack permitted from Row 14 hole 1, any length permitted.

a) A crack from this location is not permitted to connect with any other cracks or extend beyond to the forward edge of the platform (platform lead edge).

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PTF ALL PRE SB 737-CFM-LEAP-1B-72-0454; POST LEAP-1B SERVICE BULLETIN 72-0138 (Continued)

b) Raised of material is not permitted.

PTF ALL PRE SB 737-CFM-LEAP-1B-72-0454

(b) Cracks on the trailing edge of the platform.

PTF ALL PRE SB 737-CFM-LEAP-1B-72-0454; PRE LEAP-1B SERVICE BULLETIN 72-0054

1) No more than three cracks are permitted from the platform edge.
2) Cracks are permitted to extend from the platform trailing edge to the platform cooling holes.
3) Cracks are not permitted between the platform cooling holes.
4) Cracks are not permitted between the platform cooling holes and the airfoil root radius.
5) Raised material between cracks is not permitted.

PTF ALL PRE SB 737-CFM-LEAP-1B-72-0454; POST LEAP-1B SERVICE BULLETIN 72-0054

6) No more than three cracks are permitted from the platform edge.
7) Cracks are not permitted at the airfoil root radius.
8) Raised material between cracks is not permitted.

PTF ALL PRE SB 737-CFM-LEAP-1B-72-0454

(c) The CIS limits for cracks on the trailing edge of the platform are as follows:

NOTE: The CIS limits are for a reduced interval repeat inspection.

PTF ALL PRE SB 737-CFM-LEAP-1B-72-0454; PRE LEAP-1B SERVICE BULLETIN 72-0054

1) A 650-cycle inspection interval is permitted in this condition:
   a) The blade platform cracks are between the platform cooling holes on the convex side of the platform.
2) A 250-cycle inspection interval is permitted in this condition:
   a) The blade platform cracks are outside of the platform cooling holes on the convex side of the platform but does not meet the convex airfoil root radius.
3) Remove the engine in 25 cycles or less if any blade platform crack meets the convex airfoil root radius.
4) Remove the engine in 25 cycles or less if there is raised material between cracks.

PTF ALL PRE SB 737-CFM-LEAP-1B-72-0454; POST LEAP-1B SERVICE BULLETIN 72-0054

5) A 250-cycle inspection interval is permitted in this condition:
   a) The blade platform cracks are between the platform and the convex airfoil root radius.
6) Remove the engine in 25 cycles or less if the blade platform crack is at the convex airfoil root radius.
7) Remove the engine in 25 cycles or less if there is raised material.

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PTF ALL POST SB 737-CFM-LEAP-1B-72-0454

SUBTASK 72-00-00-220-047-G00

(17) Examine the surface coating of the HPT rotor 1 blades as follows (Figure 622 (Sheet 5), Figure 622 (Sheet 8), Figure 622 (Sheet 11), and Figure 622 (Sheet 13)):

NOTE: The HPT rotor stage 1 blade has a surface coating over the parent metal of the blade.

(a) Missing coating.

1) Any amount permitted.

(b) Coating cracks.

1) Any amount permitted.

(c) Discoloration.

1) Any amount permitted.

SUBTASK 72-00-00-220-048-G00

(18) Examine the LE of the HPT rotor stage 1 blades as follows (Figure 622 (Sheet 1) and Figure 622 (Sheet 5)):

NOTE: The LE is defined as the area located at cooling hole rows 3 through 6.

(a) Axial and radial cracks in area A or area B.

NOTE: Area B is the area on the blade between the platform and LE row 5 hole 4.

1) Not permitted.

(b) The over-serviceable limit extension for axial and radial cracks in area A of the HPT rotor stage 1 blade LE .

1) A 650 cycle inspection interval is permitted in these conditions:

a) If there is no missing material, cracks can connect any number of holes.

b) If the axial crack extends past the cooling holes in row 3 or row 6, remove the engine in 10 cycles or less.

(c) The over-serviceable limit extension for radial cracks in area B of the HPT stage 1 blade LE.

1) A 250 cycle inspection interval is permitted in these conditions:

a) If there is no missing material, cracks can connect any number of holes.

b) If the radial cracks turn in an axial direction and extend past the cooling holes in row 3 or row 6, remove the engine in 10 cycles or less.

(d) The over-serviceable limit extension for axial cracks in area B of the HPT stage 1 blade LE.

1) A 250 cycle inspection interval is permitted in these conditions:

a) If there is no missing material, cracks can connect any number of cooling holes.

b) Cracks do not extend past the cooling holes in row 3 or row 6.

c) If a crack extends past the cooling holes in row 3 or row 6, remove the engine in 10 cycles or less.

(e) Nicks and dents in area A or area B.

1) Any number of nicks and dents permitted in this condition:

a) The nick, dent, or any related crack does not extend further than the LE cooling holes.

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PTF ALL POST SB 737-CFM-LEAP-1B-72-0454 (Continued)

2) Tears or material that is missing not permitted.

(f) The over-serviceable limit extension for nicks and dents in area A or area B.

1) If tears or material that is missing associated with a dent or nick, remove in 10 cycles or less.

(g) Material that is missing in area A.

1) Not permitted.

(h) The over-serviceable limit extensions for material that is missing from area A of the HPT stage 1 blade LE.

1) A 650 cycle inspection interval is permitted in these conditions:

a) If it does not extend further than row 3 or row 6, any amount is permitted.

b) Any related axial cracks do not extend further than the LE cooling holes.

c) If the related axial cracks extend further than the cooling holes in row 3 or row 6, remove the engine in 10 cycles or less.

(i) Material that is missing in area B.

1) Not permitted.

(j) The over-serviceable limit extensions for material that is missing from area B of the HPT stage 1 blade LE.

1) A 250 cycle inspection interval is permitted in these conditions:

a) If it does not extend further than row 3 or row 6, any amount is permitted.

b) Any related axial cracks are not permitted.

c) Any related radial cracks are not permitted.

d) If there are related axial or radial cracks, remove the engine in 10 cycles or less.

SUBTASK 72-00-00-220-049-G00

(19) Examine the Trailing Edge (TE) of the HPT rotor stage 1 blades as follows (Figure 622 (Sheet 1) and Figure 622 (Sheet 5)):

(a) Axial cracks in area A.

1) Not permitted.

(b) The over-serviceable limit extensions for axial cracks in area A.

1) A 600 cycle interval is permitted in this condition:

a) Any number of cracks that are up to 0.100 in. (2.54 mm) from TE permitted.

2) A 250 cycle interval is permitted in these conditions:

a) Any number of cracks that are up to 0.200 in. (5.08 mm) permitted.

b) If the cracks are more than 0.200 in. (5.08 mm), remove the engine in 10 cycles or less.

(c) Axial cracks in area B.

NOTE: Area B is the area on the blade between the platform and TE hole 18.

1) Not permitted.

(d) The over-serviceable limit extensions for axial cracks in area B.

1) A 600 cycle interval is permitted in this condition:

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PTF ALL POST SB 737-CFM-LEAP-1B-72-0454 (Continued)

a) Any number of cracks permitted that are up to 0.035 in. (0.89 mm) from TE.

2) A 250 cycle interval is permitted in these conditions:

a) Any number of cracks permitted that are up to 0.070 in. (1.78 mm).

b) If the tears are more than 0.070 in. (1.78 mm), remove the engine in 10 cycles or less.

(e) Nicks and Dents in area A or B.

1) Any number of nicks and dents less than 0.020 in. (0.51 mm) in diameter permitted.

(f) Tears in area A.

1) Not permitted.

(g) The over-serviceable limit extensions for tears in area A.

1) A 600 cycle interval is permitted in this condition:

a) Tears up to 0.100 in. (2.54 mm) from TE.

2) A 250 cycle interval is permitted in these conditions:

a) Tears up to 0.200 in. (5.08 mm).

b) If the tears are more than 0.200 in. (5.08 mm), remove the engine in 10 cycles or less.

(h) Tears in area B.

1) Not permitted.

(i) The over-serviceable limit extensions for tears in area B.

1) A 600 cycle interval is permitted in this condition:

a) Tears up to 0.035 in. (0.89 mm) from TE.

2) A 250 cycle interval is permitted in these conditions:

a) Tears up to 0.070 in. (1.78 mm) from TE.

b) If the tears are more than 0.070 in. (1.78 mm), remove the engine in 10 cycles or less.

(j) Metal that is missing from the stage 1 blade TE in area A.

1) Not permitted.

(k) The over-serviceable limit extensions for metal that is missing from the stage 1 blade TE in area A.

1) A 600 cycle interval is permitted in this condition:

a) Missing metal up to 0.100 in. (2.54 mm) from TE.

2) A 250 cycle interval is permitted in these conditions:

a) Missing metal up to 0.200 in. (5.08 mm) from TE.

b) If the tears are more than 0.200 in. (5.08 mm), remove the engine in 10 cycles or less.

(l) Metal that is missing from the stage 1 blade TE in area B.

1) Not permitted.

(m) The over-serviceable limit extensions for metal that is missing from the stage 1 blade TE in area B.

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PTF ALL POST SB 737-CFM-LEAP-1B-72-0454 (Continued)

1) A 600 cycle interval is permitted in this condition:
   a) Missing metal up to 0.035 in. (0.89 mm) from TE.
2) A 250 cycle interval is permitted in these conditions:
   a) Missing metal up to 0.070 in. (1.78 mm) from TE.
   b) If the missing metals are more than 0.070 in. (1.78 mm), remove the engine in 10 cycles or less.

SUBTASK 72-00-00-220-050-G00

(20) Examine the HPT rotor stage 1 blade concave and convex surfaces as follows (Figure 622 (Sheet 1) and Figure 622 (Sheet 5)):

NOTE: The inspection is only of the HPT rotor stage 1 blade assembly that is adjacent to the borescope port.

(a) Radial cracks at the cooling holes.
   1) Permitted above the second cooling hole from the top with this condition:
      a) All radial cracks connect a maximum of two holes and do not turn in an axial direction.
   2) No cracks are permitted below the cooling holes on the concave surface.
      a) Refer to the CIS limits that follow.
   3) The CIS limits for stage 1 blade concave surface radial cracks at the cooling hole:

      NOTE: This CIS limit is for a reduced interval repeat inspection.

      a) A 10 cycle inspection interval is permitted with these conditions:
         <1> There are no cracks on all cooling holes on the concave surface.
         <2> The cracks connect a maximum of three holes and do not turn in an axial direction.

(b) Radial cracks in area A other than at the cooling holes.
   1) Radial cracks in area A other than at the cooling holes are permitted in these conditions:
      a) There is a maximum of one crack on the concave surface and it is no longer than 0.300 in. (7.62 mm).
      b) The crack does not turn in an axial direction.
      c) You cannot see the internal cavity of the blade.

(c) The over-serviceable limit extensions for radial cracks in area A other than at the cooling holes.
   1) A 650 cycle inspection interval is permitted in these conditions:
      a) There is a maximum of one crack on the concave surface and is no longer than 0.560 in. (14.22 mm) and less than 0.07 in. (1.78 mm) in width.
      b) You cannot see the internal cavity of the blade.
   2) A 250 cycle inspection interval is permitted in these conditions:
      a) There is a maximum of one crack on the concave surface and it is no longer than 0.690 in. (17.53 mm) and less than 0.140 in. (3.56 mm) in width.
      b) You cannot see the internal cavity of the blade.

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(d) Radial cracks on the concave side of the airfoil between Row 7 and Row 11.

NOTE: The inspection is only of the HPT rotor stage 1 blade assembly that is adjacent to the borescope port.

1) Radial cracks on the HPT rotor stage 1 blade on the pressure (concave) airfoil between, but not related to, row 7 and 9 cooling air holes.

a) Not permitted.

2) Engines with radial cracks on the HPT rotor stage 1 blade on the pressure (concave) airfoil between, but not related to, row 10 and 11 cooling air holes are permitted to continue in service as follows:

NOTE: The CIS limits are reduced intervals repeat inspections.

a) An 1300 cycle inspection interval is permitted in these conditions:

<1> There is no more than one crack in region A (region A is different than Area A).

<2> The crack is not more than 0.19 in. (4.8 mm).

b) A 650 cycle inspection interval is permitted in these conditions:

<1> There is no more than one radial crack that extends further than Region A.

<2> The crack is not more than 0.48 in. (12.2 mm).

c) If there is a radial crack on the pressure surface that extends beyond Region A, remove the engine in 25 cycles or less.

(e) Radial cracks on convex side.

1) Not permitted.

(f) The over-serviceable limit extensions for radial cracks on convex side.

1) A 250 cycle inspection interval is permitted for cracks on the convex side in these conditions:

a) There is a maximum of one crack on the convex surface and is no longer than 0.35 in. (8.9 mm).

b) Crack is no closer than 0.120 in. (3.0 mm) from the tip end of the blade.

c) Crack is no further than 0.70 in. (17.8 mm) from TE and no closer than 0.55 in. (13.97 mm) from the TE.

d) Crack does not turn axial.

e) Crack is not closer than 0.25 in. (6.35 mm) from any other crack or missing material.

2) If the crack is closer than 0.175 in. (4.44 mm) from any other crack or missing material or if the crack turns in an axial direction, remove the engine in 10 cycles or less.

(g) Radial cracks in area B other than at the cooling holes.

NOTE: Area B is the area on the blade between the platform and row 11 hole 7 on the concave surface. Area B is the area on the blade between the platform and row 2 hole 4 on the convex surface.

1) Radial cracks in area B other than at the cooling holes are permitted in these conditions:

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a) There is a maximum of one crack and the crack is no longer than 0.300 in. (7.62 mm).

b) The crack does not turn in an axial direction.

c) You cannot see the internal cavity of the blade.

(h) The over-serviceable limit extensions for radial cracks in area B other than at the cooling holes.

1) A 250 cycle inspection interval is permitted in these conditions:

a) You cannot see the internal cavity of the blade.

b) There is a maximum of one crack on the concave surface and it is no more than 0.420 in. (10.67 mm) in length.

c) If the crack is more than 0.420 in. (10.67 mm) in length, remove the engine in 10 cycles or less.

d) The crack does not go in an axial direction.

e) If the crack is in an axial direction, remove the engine in 10 cycles or less.

(i) Radial cracks that go out of the tips of the cooling holes.

1) Three cracks are permitted in these conditions:

a) The cracks do not go into the next cooling hole.

b) The cracks do not go in the direction of the blade platform from the bottom hole.

c) The cracks do not go in an axial direction.

(j) The over-serviceable limit extensions for radial cracks that go out of the tip of the cooling holes.

1) A 250 cycle inspection interval is permitted in these conditions:

a) The cracks do not connect more than two cooling holes.

b) The cracks do not go in the direction of the platform from the bottom hole.

c) The cracks do not turn in an axial direction.

d) If the crack turns axial or if it goes in the direction of the platform from the bottom hole, remove the engine in 10 cycles or less.

2) A 25 cycle inspection interval is permitted in these conditions:

a) The cracks connect a maximum of three cooling holes.

b) The cracks do not go in the direction of the platform from the bottom hole.

c) The cracks do not turn in an axial direction.

d) If the crack connects more than three cooling holes or turn in an axial direction, remove the engine in 10 cycles or less.

(k) Nicks and dents in area A concave and convex surfaces.

1) Any number of nicks and dents in area A in this condition:

a) There are no tears, cracks or holes that go through the surface metal.

(l) Nicks and dents in area B concave and convex surfaces.

1) Any number of nicks and dents permitted in the area B concave surface in this condition:

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a) The diameter of the nick or dent is not more than 0.070 in. (1.78 mm).

2) Any number of nicks and dents permitted in the area B convex surface in this condition:

a) There are no tears, cracks or holes through the surface metal.

(m) Twisted or bent airfoil.

1) Not permitted.

(n) Axial cracks in areas A and B.

1) Not permitted.

(o) The over-serviceable limit extensions for axial cracks on the concave and convex surface areas A and B.

1) An 125 cycle inspection interval is permitted in these conditions:

a) There is a maximum of one crack that is less than 0.070 in. (1.78 mm) in length.

b) The crack does not turn in a radial direction.

c) If the crack is more than 0.070 in. (1.78 mm) in length or if the crack turns in a radial direction, remove the engine in 10 cycles or less.

SUBTASK 72-00-00-220-051-G00

(21) Examine the HPT rotor stage 1 blade-tip as follows (Figure 622 (Sheet 5), Figure 622 (Sheet 8), Figure 622 (Sheet 11), and Figure 622 (Sheet 13)):

NOTE: The blade tip is defined as the area in or less than 0.16 in. (4.06 mm) of the end of the blade. Approximately TE hole 35 and above.

(a) Radial cracks that begin at the end of the blade (the crack can open up at the tip and make a 'V' notch).

1) Concave side of the blade.

a) If you do not see the internal cavity, it is permitted to have a crack in the 'V' notch.

b) It is permitted for cracks to intersect the top cooling holes in a concave cooling row.

c) If the cracks do not go further than 0.200 in. (5.08 mm) from the end of the blade, any number of cracks are permitted.

2) The over-serviceable limit extensions for radial cracks on the concave side of the blade.

NOTE: This over-serviceable limit extension is for a reduced interval, repeat inspection.

a) An 800 cycle inspection interval is permitted in these conditions:

<1> The radial tip cracks are between 0.200 in. (5.08 mm) and 0.250 in. (6.35 mm).

<2> You cannot see the internal cavity of the blade.

b) A 400 cycle inspection interval is permitted in these conditions:

<1> The radial tip cracks are between 0.250 in. (6.35 mm) and 0.300 in. (7.62 mm).

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<2> You cannot see the internal cavity of the blade.

c) If the radial cracks are longer than 0.300 in. (7.62 mm) or if you can see the internal cavity, remove the engine in 10 cycles or less.

3) Convex side of the blade.

a) If you cannot see the internal cavity, a crack in the 'V' notch is permitted.

b) If the cracks do not go further than 0.200 in. (5.08 mm) from the end of the blade, any number of cracks are permitted.

4) The over-serviceable limit extensions for radial cracks on the convex side of the blade.

NOTE: This over-serviceable limit extension is for a reduced interval, repeat inspection.

a) An 800 cycle inspection interval is permitted in these conditions:

<1> The radial tip cracks are between 0.200 in. (5.08 mm) and 0.250 in. (6.35 mm).

<2> You cannot see the internal cavity of the blade.

b) A 400 cycle inspection interval is permitted in these conditions:

<1> The radial tip cracks are between 0.250 in. (6.35 mm) and 0.300 in. (7.62 mm).

<2> You cannot see the internal cavity of the blade.

c) If the radial cracks are longer than 0.300 in. (7.62 mm) or if you can see the internal cavity, remove the engine in 10 cycles or less.

(b) Axial cracks.

1) Any number of axial cracks permitted.

(c) Bent or curled blade tips.

1) Bent or curled blade tips are permitted in these conditions:

a) If the bent or curled tips do not go below pressure side tip row 13 and TE exit hole row 12 hole 37.

b) You cannot see the internal cavity.

(d) Metal that is missing from the blade tip.

1) Metal that is missing from the blade tip is permitted in these conditions:

a) If the missing material does not go below pressure side tip row 13 and TE exit hole row 12 hole 37.

b) You cannot see the internal cavity.

(e) The over-serviceable limit extensions for metal that is missing from the blade tip.

1) If you can see the internal cavity, remove the engine in 10 cycles or less.

(f) Metal that is missing from the tip on the TE.

1) Metal that is missing from the tip on the TE is permitted in these conditions:

a) Damage does not go below the TE exit hole row 12 hole 35.

b) You cannot see the internal cavity.

(g) The over-serviceable limit extensions for metal that is missing from the tip on the TE.

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1) A 125 cycle inspection interval is permitted in these conditions:
   a) The damage goes below the first TE exit hole row 12 hole 35, but does not go further than 0.200 in. (5.08 mm).
   b) You cannot see the internal cavity.

SUBTASK 72-00-00-220-052-G00

(22) Examine the HPT stage 1 rotor blade tips for buildup of first stage shroud material (Figure 622 (Sheet 1) and Figure 622 (Sheet 5)):
   (a) Shroud material build-up on the blade tips is permitted.

SUBTASK 72-00-00-220-053-G00

(23) Examine the HPT rotor stage 1 blade platform as follows (Figure 622 (Sheet 5), Figure 622 (Sheet 8), Figure 622 (Sheet 11), and Figure 622 (Sheet 13)):
   (a) Cracks on the concave side of the platform.
      1) Cracks less than 0.200 in. (5.08 mm) in length:
         a) Any amount permitted.
         b) Raised material is not permitted.
         c) Does not extend beyond to the forward edge of the platform (platform LE) or connects to other platform holes.
   (b) The over-serviceable limit extensions for cracks on the HPT rotor stage 1 blade concave-platform.

      NOTE: The over-serviceable limit extension is for a reduced interval, repeat inspection.
      1) A 600 cycle inspection interval is permitted in this condition:
         a) Cracks on the concave platform less than 0.300 in. (7.62 mm) but more than 0.200 in. (5.08 mm).
         b) Connecting cracks or cracks connecting with other platform holes are permitted as long as there is no raised or missing material.
      2) A 350 cycle inspection interval is permitted in this condition:
         a) Cracks on the concave platform less than 0.350 in. (8.89 mm) but more than 0.300 in. (7.62 mm) and no raised or missing material.
         b) Connecting cracks or cracks connecting with other platform holes are permitted as long as there is no raised or missing material.
      3) A 100 cycle inspection interval is permitted in this condition:
         a) Cracks on the concave platform less than 0.400 in. (10.16 mm) but more than 0.350 in. (8.89 mm) and no raised or missing material.
         b) Connecting cracks or cracks connecting with other platform holes are permitted as long as there is no raised or missing material.
      4) If there are cracks more than 0.400 in. (10.16 mm) in length and/or you see any raised or missing material on the concave platform, remove the engine in 5 cycles or less.
   (c) Cracks on the convex side of the platform.
      1) No more than three cracks are permitted from the platform edge.
      2) Cracks are not permitted at the airfoil and platform interface.

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PTF ALL POST SB 737-CFM-LEAP-1B-72-0454 (Continued)

3) Raised or missing material between cracks is not permitted.

(d) The over-serviceable limit extensions for cracks on the HPT rotor stage 1 blade convex-platform.

NOTE: This over-serviceable limit extension is for a reduced interval, repeat inspection.

1) A 250 cycle inspection interval is permitted in this condition:

a) The blade platform cracks are between the platform and the convex- airfoil root radius.

2) If the blade platform crack reaches the convex airfoil-root radius or if there is raised or missing material between cracks, remove the engine in 10 cycles or less.

PTF ALL

SUBTASK 72-00-00-220-024-G00

(24) Examine the HPT rotor second stage blade leading edge as follows (Figure 622 (Sheet 1), Figure 622 (Sheet 17) and Figure 622 (Sheet 18)):

(a) Cracks

1) Permitted in Area A if they are not related to a dent.

2) Cracks are not permitted in Area B.

(b) Nicks and dents in Area A

1) Permitted.

(c) Dents in Area B (includes root radius):

1) Three dents out of contour and 0.025 in. (0.64 mm) apart are permitted.

NOTE: Surface distortion other than a dent is not permitted

(d) Nicks in Area B:

1) Four nicks 0.020 in. (0.51 mm) in length and 0.125 in. (3.18 mm) apart are permitted.

2) One nick 0.040 in. (1.02 mm) in length on the leading edge only is permitted.

(e) Metal that is missing in Area A

1) Not permitted.

2) The CIS limits for HPT stage 2 blade leading edge metal that is missing in Area A:

NOTE: This CIS limit is for a reduced interval repeat inspection.

NOTE: The first internal rib is defined as 0.164 in. (4.166 mm) in length from the leading edge.

a) A 650 cycle inspection interval is permitted with these conditions:

<1> All amounts less than 0.008 in^2 (0.05 cm^2) (or 0.164 in. (4.17 mm) in length from the leading edge) does not go farther than the first internal rib.

<2> Axial cracks do not go farther than the first internal rib.

<3> Remove the engine in 25 cycles or less if the missing material is more than 0.008 in^2 (0.05 cm^2) or goes farther than the first rib.

<4> Remove the engine in 15 cycles or less if the cracks go farther than the first internal rib.

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(f) Metal that is missing in Area B

1) Not permitted.

a) The CIS limits for HPT stage 2 blade leading edge metal that is missing in Area B:

<1> Remove the engine before the next flight if the metal that is missing goes farther than the first internal rib or 0.164 in. (4.17 mm) in length from the leading edge.

<2> Remove the engine in 5 cycles if the metal that is missing does not go beyond the first rib.

(g) Metal splatter.

1) Permitted with this condition:

a) If the metal splatter is seen on the convex side (suction side) of the blade in Area C, then do an inspection of all stage two nozzle segments for heat distress and burns in the inner platforms (TASK 72-00-00-290-809-G00).

SUBTASK 72-00-00-220-025-G00

(25) Examine the HPT rotor second stage blade trailing edge as follows (Figure 622 (Sheet 1) and Figure 622 (Sheet 17)):

(a) Cracks

1) Not permitted.

(b) Nicks and dents

1) Permitted if they are not within 0.35 in. (8.9 mm) of the blade platform.

(c) Metal that is missing and tears.

1) Not permitted.

SUBTASK 72-00-00-220-026-G00

(26) Examine the HPT rotor second stage blade concave and convex surfaces as follows (Figure 622 (Sheet 1) and Figure 622 (Sheet 17)):

(a) Cracks

1) Not permitted.

(b) Nicks and dents

1) Permitted with these conditions:

a) The diameter is less than 0.18 in. (4.6 mm).

b) There is no torn metal.

c) They are not within 0.35 in. (8.9 mm) of the blade platforms.

(c) Positive braze material

1) Positive braze material is permitted in these conditions:

a) The material is in Area D of the concave and convex surfaces.

b) The material extends not more than 0.003 in. (0.08 mm) above a concave or convex surface.

(d) Blades surface coating.

1) There is no limit to the amount of missing surface coating, with metal oxidation (green or black discoloration of the material), if there are no holes or cracks in the blade.

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SUBTASK 72-00-00-220-027-G00

(27) Examine the HPT rotor second stage blade for pitting (Figure 622 (Sheet 1) and Figure 622 (Sheet 17)):

(a) Four pits are permitted on the concave or convex surface in these conditions:

1) Each pit is less than 0.02 in. (0.5 mm) in length.

2) Each pit is more than 0.088 in. (2.24 mm) from an adjacent pit.

(b) There is no limit on the number of pits that are permitted on the concave or convex surface in these conditions:

1) Each pit is less than 0.011 in. (0.28 mm) in length.

2) There is a clear separation between each pit and an adjacent pit.

(c) One pit is permitted on the concave and convex surfaces in these conditions:

1) The pit is not more than 0.035 in. (0.89 mm) in length.

SUBTASK 72-00-00-220-028-G00

(28) Examine the HPT rotor second stage blade tip area as follows (Figure 622 (Sheet 1) and Figure 622 (Sheet 17)):

(a) Radial cracks

1) Two radial cracks for concave or convex sides are permitted with these conditions:

a) The cracks extend less than 0.07 in. (1.8 mm) below the blade or are less than 0.07 in. (1.8 mm) in length.

b) The internal cavity of the blade cannot be seen.

(b) Axial cracks

1) Permitted if they are not more than 0.35 in. (8.9 mm).

(c) The CIS limits for stage 2 blade tip area

1) All lengths of axial cracks above the tip cap are permitted for no more than 10 cycles.

NOTE: This CIS limit is for a one time use.

(d) Bent or curled blade tip

1) Permitted.

(e) Metal that is missing and tears

1) Permitted with these conditions:

a) The tip cavity shelf cannot be seen because part of or all of the tip cap is missing.

b) The opposite cavity wall is not missing.

c) There is only one area on each cavity wall that has damage 0.21 in. (5.3 mm) in length.

(f) Build up of second stage shroud material.

1) Permitted.

SUBTASK 72-00-00-211-024-G00

(29) Examine the HPT rotor second stage blades for missing Thermal Barrier Coating (TBC) as follows (Figure 622 (Sheet 1) and Figure 622 (Sheet 17)):

(a) Thermal Barrier Coating (TBC).

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1) There is no limit to the amount of missing TBC if there are no holes or cracks in the blade.

SUBTASK 72-00-00-220-029-G00

(30) Examine the inner and outer platforms of the HPT stage 2 nozzle segment as follows (Figure 622 (Sheet 1) and Figure 622 (Sheet 14)):

NOTE: The inspection is limited to the HPT stage 2 nozzle assembly that is adjacent to the borescope port.

(a) Cracks

1) Cracks that are 0.35 in. (8.89 mm) or less in length are permitted.

(b) Nicks, scores, scratches, and dents in the inner and outer platform surfaces

1) Nicks, scores, scratches, and dents that are 0.04 in. (1.0 mm) or less in length are permitted.

(c) Craze cracks

1) Permitted.

(d) Cracks in the outer platform trailing edge

1) Cracks that are 0.30 in. (7.6 mm) or less in length are permitted if the cracks do not go into the vane fillet radius.

(e) Coating that is missing

1) Permitted if the adjacent HPT rotor stage 1 shroud segment is not missing complete circumferential Environmental Barrier Coating (EBC) in or less than 0.100 in. (2.54 mm) of the shroud trailing edge.

2) It is permitted to have a total area 0.5 in^2 (322.6 mm^2) or less in Area B if the adjacent HPT rotor stage 1 shroud EBC is missing complete circumferential EBC in or less than 0.100 in. (2.54 mm) of the shroud trailing edge.

a) Any amount of missing coating is permitted outside Area B.

(f) The CIS limits for stage 2 missing coating in Area B with missing EBC of the trailing edge of the adjacent HPT rotor stage 1 shroud is as follows:

1) An 800 cycle inspection interval is permitted for any amount of missing coating.

NOTE: The CIS limit is for a reduced interval repeat inspection.

(g) Heat distress or oxidation on the outer platform

1) All amounts permitted if the adjacent HPT rotor stage 1 shroud segment is not missing complete circumferential EBC in or less than 0.100 in. (2.54 mm) of the shroud trailing edge.

2) It is permitted to have a total area 0.1 in^2 (64.5 mm^2) or less in Area B if the adjacent HPT rotor stage 1 shroud EBC is missing complete circumferential EBC in or less than 0.100 in. (2.54 mm) of the shroud trailing edge.

a) Any amount of missing coating is permitted outside Area B.

3) It is not permitted to have holes that are through the parent metal, regardless of HPT rotor stage 1 shroud EBC condition.

(h) The CIS limits for the stage 2 nozzle outer platform oxidation is as follows:

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1) A 200 cycle inspection interval is permitted for any amount of heat distress or oxidation in Area B if the adjacent HPT rotor stage 1 shroud EBC is missing complete circumferential EBC in or less than 0.100 in. (2.54 mm) of the shroud trailing edge.

NOTE: The CIS limit is for a reduced interval repeat inspection.

2) Remove the engine in 10 cycles or less if the oxidation is through the parent metal, regardless of HPT rotor stage 1 shroud EBC condition.

NOTE: The CIS limit is for a one time use.

(i) Cracks from the borescope hole on flowpath side (Figure 622 (Sheet 14))

1) Any amount permitted for cracks as follows:

a) No more than two cracks.

b) Cracks are less than 0.175 in. (4.44 mm) in length.

c) Cracks are more than 0.175 in. (4.44 mm) apart.

d) Cracks are minimum of 0.35 in. (8.89 mm) from other outer platform cracks.

e) Cracks do not intersect airfoil.

(j) Oxidation or erosion around the borescope hole on flowpath side

1) Any amount permitted, if there is no through-hole and the erosion is not on the airfoil surface.

(k) Cracks and oxidation around the borescope hole on non flowpath side

1) Any amount permitted inside the borescope hole.

2) Up to one crack permitted on the borescope boss.

(l) The CIS limits for the stage 2 nozzle borescope hole

1) A 600 cycle inspection is permitted for more than two cracks, or cracks greater than 0.175 in. (4.44 mm) in length, or less than 0.175 in. (4.44 mm) apart as follows:

NOTE: The CIS limit is for a reduced interval inspection.

a) Cracks that intersect the airfoil so long as the crack is not axial.

b) Does not extend past the outer band root fillet radius.

NOTE: Root fillet radius is bounded by the location of cooling hole A (Figure 622 (Sheet 15)).

c) Cracks that extend beyond the root fillet radius, should be evaluated per concave airfoil crack paragraph 7.H.(24).(d).

2) A 300 cycle inspection is permitted for cracks on the nozzle insert weld joint as follows:

a) Cracks only present on one of the four weld joints.

3) Remove the engine in 10 cycles or less, if erosion or material recession results in a through-hole.

4) Remove the engine for any cracks that turn in an axial direction at the root fillet, or if any cracks intersect with an axial airfoil crack.

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SUBTASK 72-00-00-220-030-G00

(31) Examine the vane airfoils of the HPT stage 2 nozzle assembly as follows (Figure 622 (Sheet 1) and Figure 622 (Sheet 14)):

NOTE: The inspection is only of the HPT stage 2 nozzle assembly that is adjacent to the borescope port.

(a) Axial cracks in the vane trailing edges that are more than 0.25 in. (6.4 mm) from the inner platform

1) Axial cracks in the vane trailing edges are permitted if they are not more than 0.06 in. (1.5 mm) in length.

2) Two cracks are permitted as follows:

a) The cracks are not more than 0.35 in. (8.9 mm) in length.

b) The cracks are separated from each other by a minimum of 0.25 in. (6.4 mm).

(b) All types of cracks in the vane trailing edges that are less than 0.25 in. (6.4 mm) from the inner platform

1) Permitted if they do not extend more than 0.35 in. (8.9 mm) from the trailing edge.

(c) Axial cracks in the vane leading edges

1) One axial crack is permitted if it is 0.20 in. (5.1 mm) or less in length.

2) Not permitted in Area C if they are parallel to the outer platform.

3) Permitted if there are no separation cracks in the surface coating.

(d) Cracks in the vane concave surfaces

1) Two cracks for each vane concave surface are permitted if they are 0.50 in. (12.7 mm) or less in length.

(e) Cracks in the convex side of the vane to the outer platform fillet area

1) Not permitted.

(f) Nicks, scores, scratches and dents

1) Permitted if they are not more than 0.17 in. (4.3 mm) in diameter.

(g) Oxidation or missing material in the concave and convex surfaces

1) Permitted to an area that is less than 0.80 in. (20.3 mm) in diameter or equivalent area of 0.50 in^2 (3.23 cm^2).

2) Holes that are through the airfoil and expose the airfoil internal cavity are not permitted.

(h) Oxidation or missing material in the leading edge

1) An area that is 0.50 in. (12.7 mm) or less in length and 0.30 in. (7.6 mm) or less in width is permitted if the adjacent HPT rotor stage 1 shroud segment is not missing complete circumferential EBC in or less than 0.100 in. (2.54 mm) of the shroud trailing edge.

2) It is permitted to have a total area 0.1 in^2 (64.5 mm^2) or less in Area A if the adjacent HPT rotor stage 1 shroud EBC is missing complete circumferential EBC in or less than 0.100 in. (2.54 mm) of the shroud trailing edge.

a) Any amount of missing coating is permitted outside Area A.

3) It is not permitted to have holes that are through the parent metal, regardless of HPT rotor stage 1 shroud EBC condition.

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(i) The CIS limits for the stage 2 nozzle oxidation or missing material in the leading edge is as follows:

1) A 200 cycle inspection interval is permitted for any amount of oxidation or missing material in Area A if the adjacent HPT rotor stage 1 shroud EBC is missing complete circumferential EBC in or less than 0.100 in. (2.54 mm) of the shroud trailing edge.

NOTE: The CIS limit is for a reduced interval repeat inspection.

2) Remove the engine in 10 cycles or less if the oxidation is through the parent metal, regardless of HPT rotor stage 1 shroud EBC condition.

NOTE: The CIS limit is for a one time use.

(j) Coating that is missing from the airfoil surfaces

1) All amounts missing from a single airfoil are permitted if the adjacent HPT rotor stage 1 shroud segment is not missing complete circumferential EBC in or less than 0.100 in. (2.54 mm) of the shroud trailing edge.

2) It is permitted to have a total area 0.5 in^2 (322.6 mm^2) or less in Area A if the adjacent HPT rotor stage 1 shroud EBC is missing complete circumferential EBC in or less than 0.100 in. (2.54 mm) of the shroud trailing edge.

a) Any amount of missing coating is permitted outside Area A.

(k) The CIS limits for stage 2 missing coating in Area A with missing EBC of the trailing edge of the adjacent HPT rotor stage 1 shroud is as follows:

1) An 800 cycle inspection interval is permitted for any amount of missing coating.

NOTE: The CIS limit is for a reduced interval repeat inspection.

(l) Material that is missing from the trailing edge

1) One area of 0.18 in. (4.6 mm) or less in axial direction by 0.35 in. (8.9 mm) or less in radial direction is permitted.

(m) Cracks in the vane convex surfaces.

1) It is permitted to have a maximum of two cracks on each vane with these conditions:

a) The cracks are 0.35 in. (8.9 mm) or less in length.

b) The cracks are separated from the inner and outer platform by 0.35 in. (8.9 mm) or more.

SUBTASK 72-00-00-220-031-G00

(32) Examine the HPT rotor stage 2 shroud as follows (Figure 622 (Sheet 1) and Figure 622 (Sheet 16)):

NOTE: This inspection is limited to what you can see with the rigid borescope.

(a) Blade rub

1) Permitted.

(b) Cracks

1) Not permitted.

(c) Surface oxidation or corrosion

1) All amounts are permitted as long as there is no burn through.

(d) Holes through the shroud flowpath surface.

1) Not permitted.

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2) CIS limit is to remove the engine in 25 cycles or less if there is a hole through the flowpath surface.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 72-00-00-940-001-G00

(1) When the inspection is complete, do these steps:

(a) Remove the rigid borescope, STD-10719, or fiberscope, STD-10718, from the HPT port.

(b) Turn the light source, STD-10717, to OFF.

(c) Let the light source blower stay ON until the lamp and case become cool.

SUBTASK 72-00-00-410-010-G00

(2) Install the borescope plugs P, Q, and R as follows:

(a) Do a visual inspection of the borescope plugs and the borescope ports (TASK 72-00-00-290-801-G00).

(b) Remove the identification tag from borescope plugs P, Q, and R before the installation of each plug.

(c) Put the Tiodize T8E-H grease, D50355 [CP2961], on the borescope plug threads and the mating surfaces.

NOTE: Use the graphite mineral vaseline, D00601 [CP2101], as an alternative.

(d) Install the borescope plugs P, Q, and R on the HPT ports.

(e) Make sure that installation running torque of the anti-rotation device on the borescope plugs is less than 8.8 in-lb (1.0 N·m).

1) If the running torque is more than 8.8 in-lb (1.0 N·m), replace the borescope plug.

(f) Tighten the borescope plugs P and Q to 85.0 in-lb (9.6 N·m) - 95.0 in-lb (10.7 N·m).

(g) Tighten the borescope plug R to 50.4 in-lb (5.7 N·m) - 55.6 in-lb (6.3 N·m).

SUBTASK 72-00-00-080-018-G00

(3) Remove the 3/4 inch square drive hand tool or the electronic turning tool, SPL-14866, from the AGB manual control pad (TASK 72-00-00-910-801-G00).

SUBTASK 72-00-00-410-026-G00

CAUTION

MAKE SURE THAT YOU INSTALL THE COVER ON THE GEARBOX. IF YOU DO NOT INSTALL THE COVER, ENGINE OIL LEAKS CAN OCCUR DURING OPERATION AND ENGINE DAMAGE CAN OCCUR.

(4) Do this task: Accessory Gearbox Manual Control Cover Installation, TASK 72-63-04-400-801-G00.

SUBTASK 72-00-00-860-041-G00

WARNING

OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(5) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 72-00-00-440-006-G00

(6) Do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801

SUBTASK 72-00-00-860-043-G00

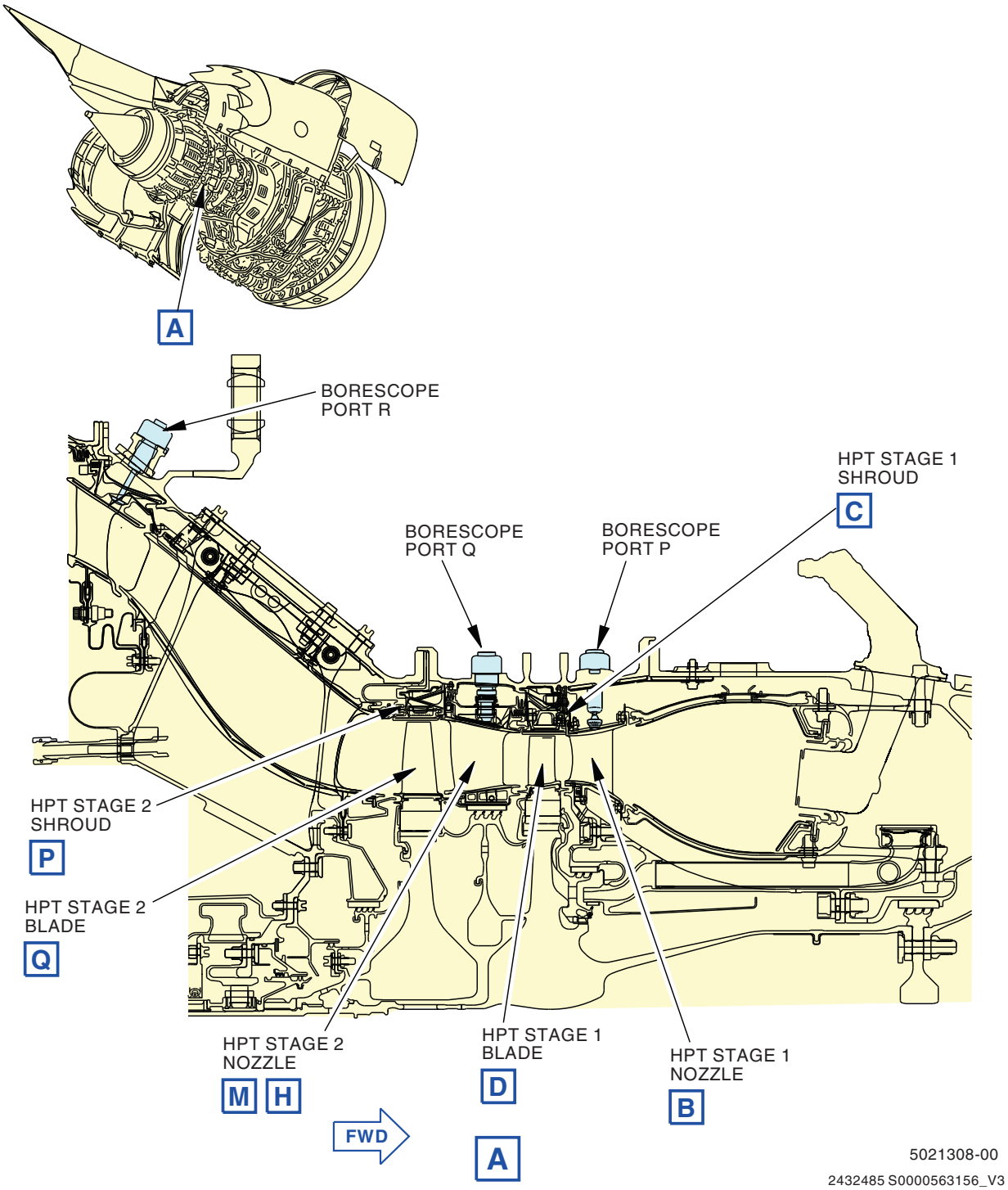
(7) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

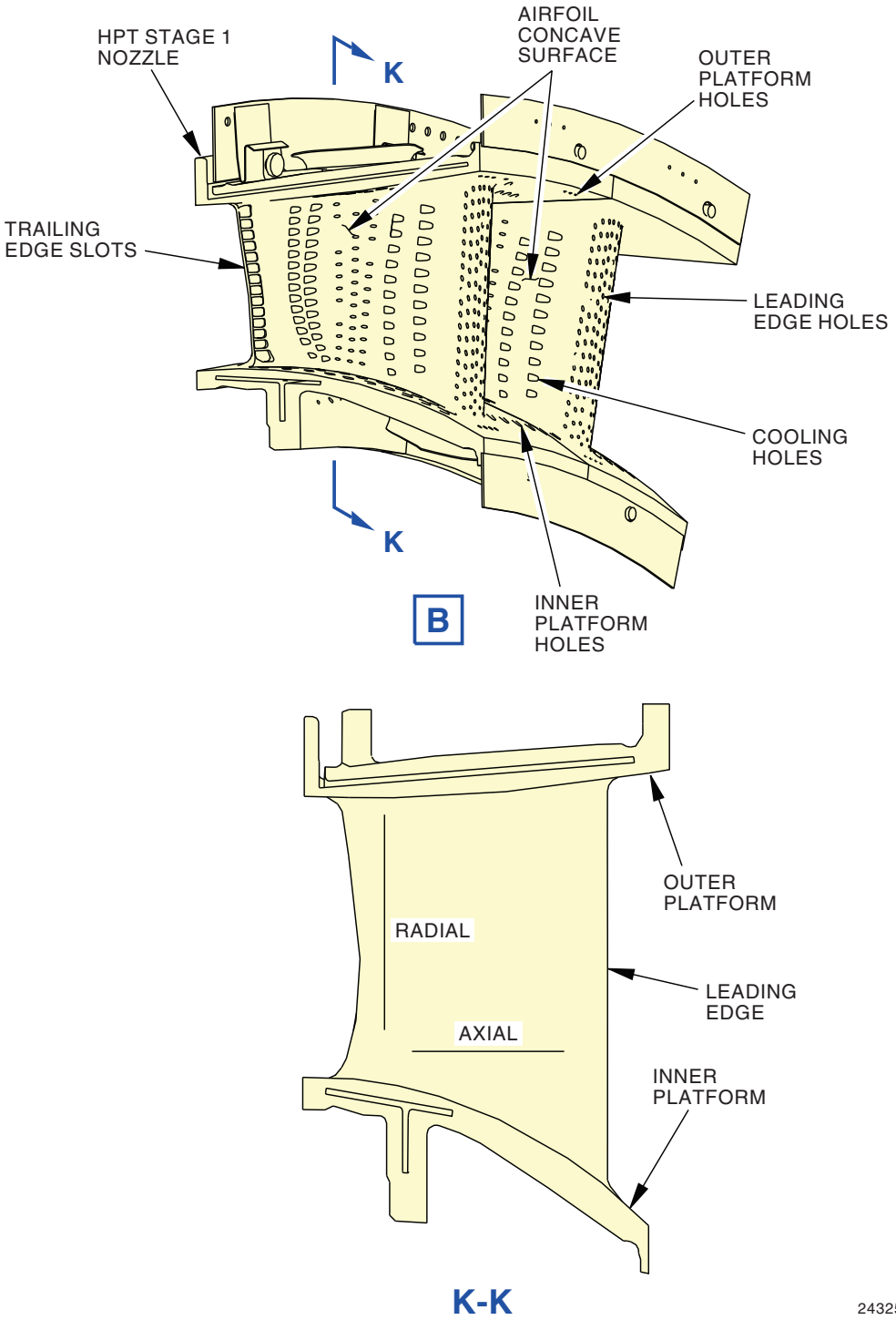
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



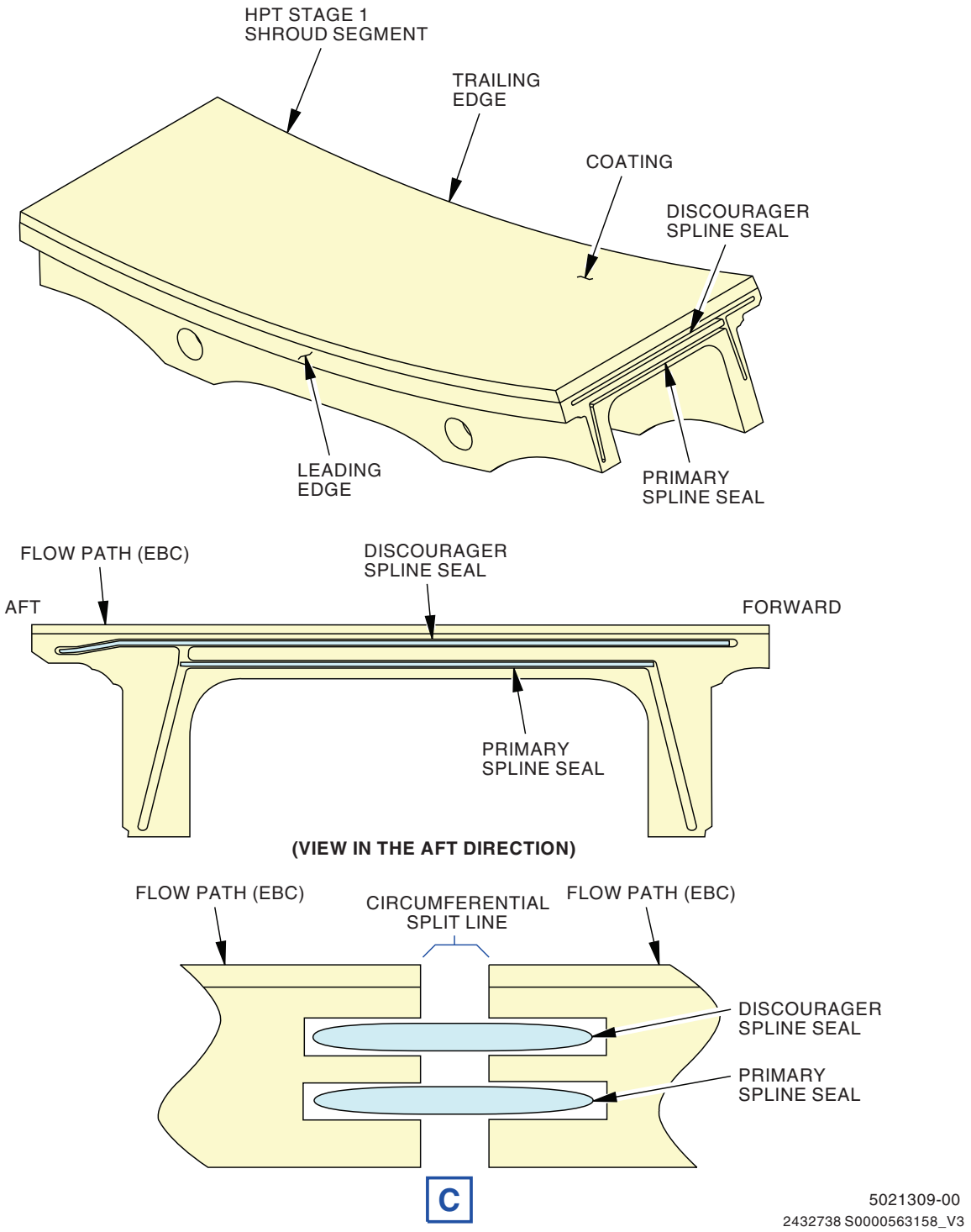
High Pressure Turbine Section Blade, Nozzle and Shroud Inspection
Figure 622/72-00-00-990-845-G00 (Sheet 1 of 18)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



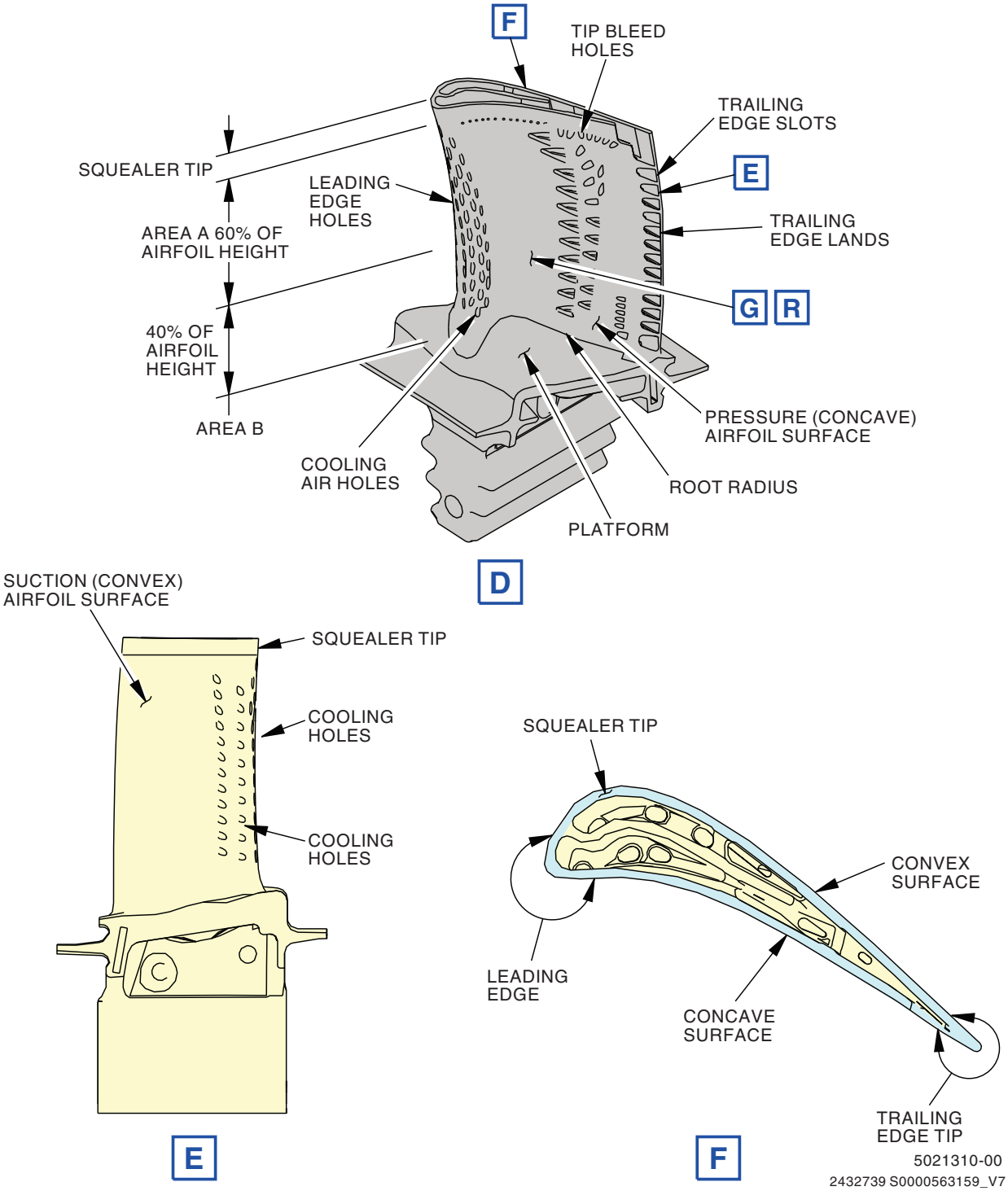
High Pressure Turbine Section Blade, Nozzle and Shroud Inspection
Figure 622/72-00-00-990-845-G00 (Sheet 2 of 18)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



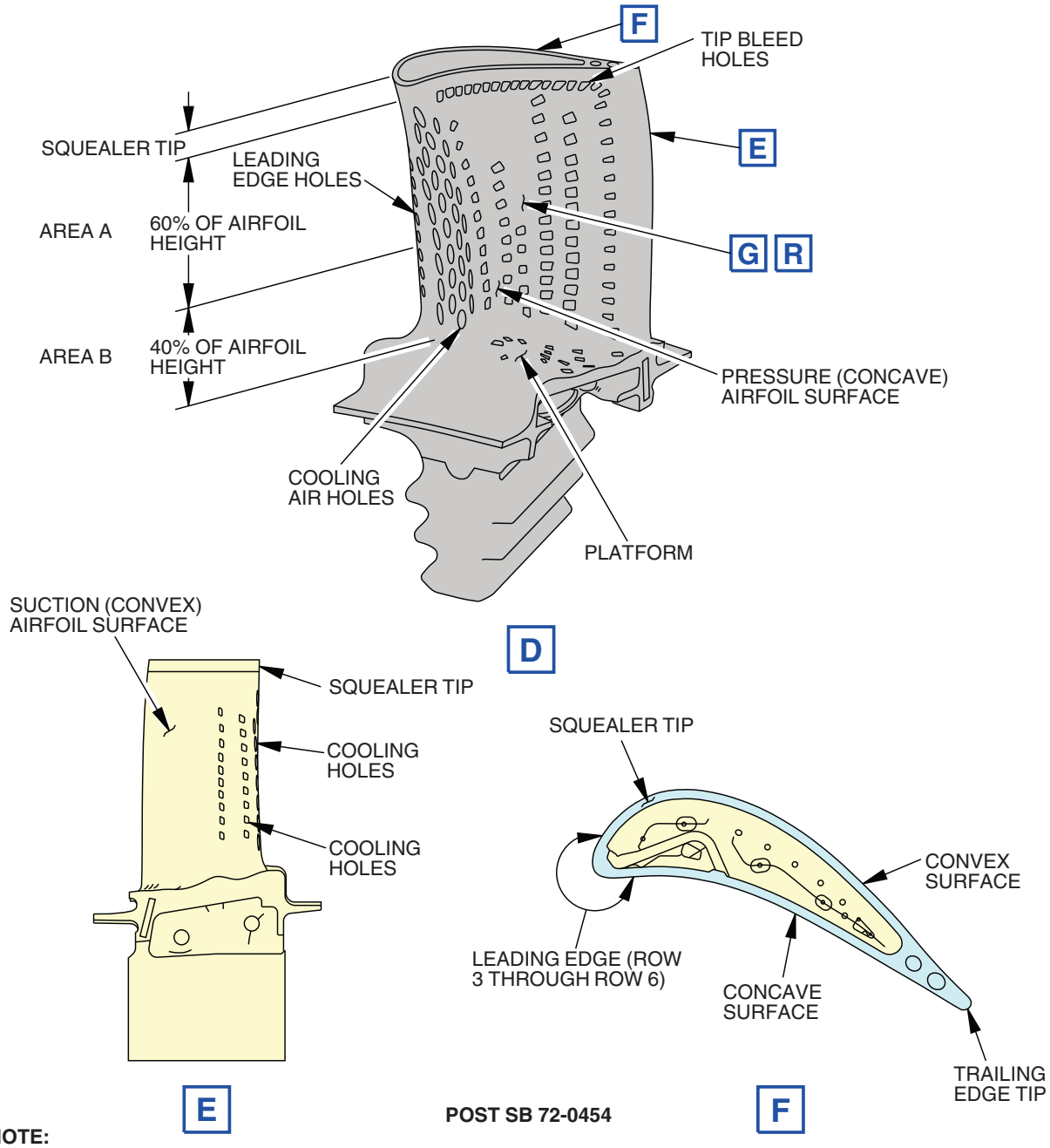
High Pressure Turbine Section Blade, Nozzle and Shroud Inspection
Figure 622/72-00-00-990-845-G00 (Sheet 3 of 18)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



High Pressure Turbine Section Blade, Nozzle and Shroud Inspection
Figure 622/72-00-00-990-845-G00 (Sheet 4 of 18)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOTE:

SQUEALER TIP AREA IS DEFINED FROM THE BOTTOM OF TIP BLEED HOLES TO TOP OF THE BLADE ALL AROUND THE AIRFOIL.

AREA A IS DEFINED AS THE AREA BETWEEN AREA B AND THE SQUEALER TIP ALL AROUND THE AIRFOIL.

AREA B IS DEFINED FROM THE TOP OF THE PLATFORM TO ROW 6 HOLE 3 AT THE LE AND ROW 12 HOLE 17 AT THE TE ALL AROUND THE AIRFOIL.

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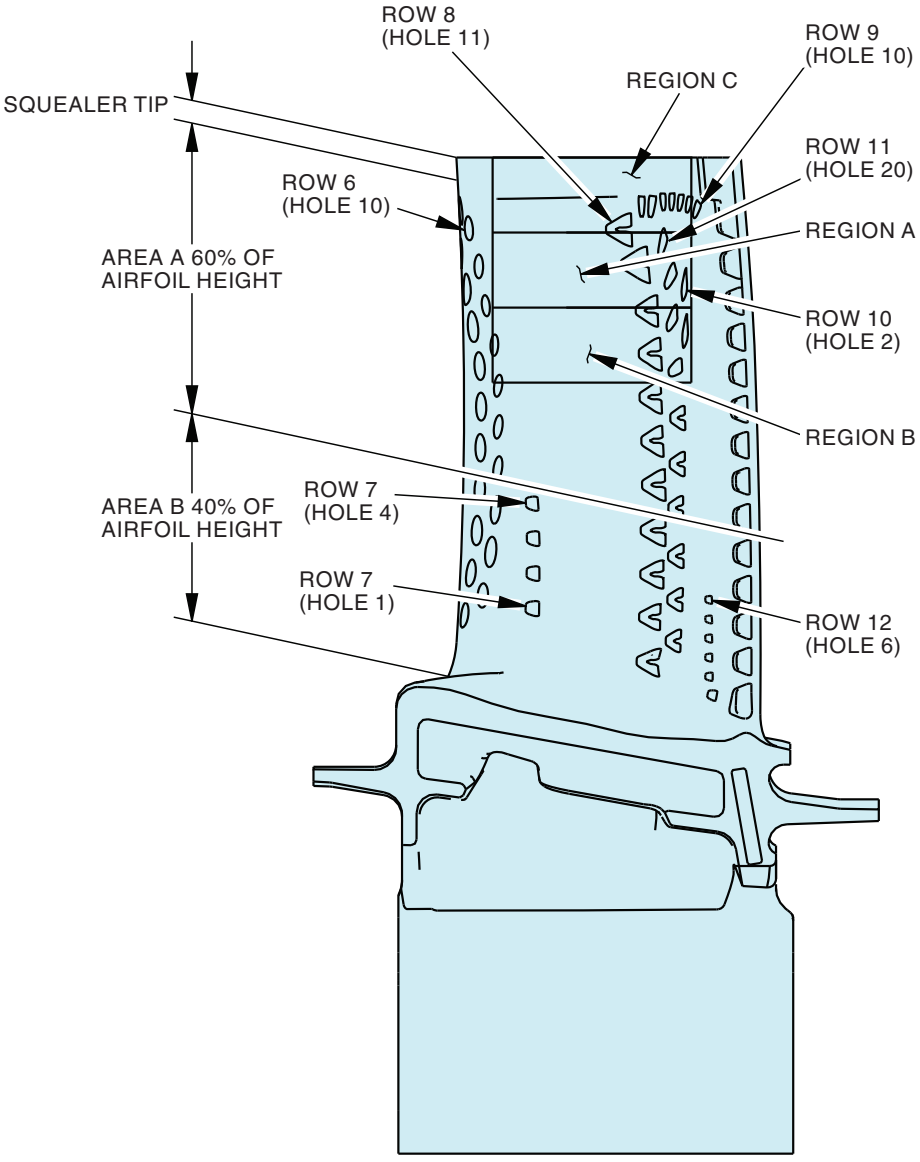
High Pressure Turbine Section Blade, Nozzle and Shroud Inspection
Figure 622/72-00-00-990-845-G00 (Sheet 5 of 18)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



PRE SB 72-0138

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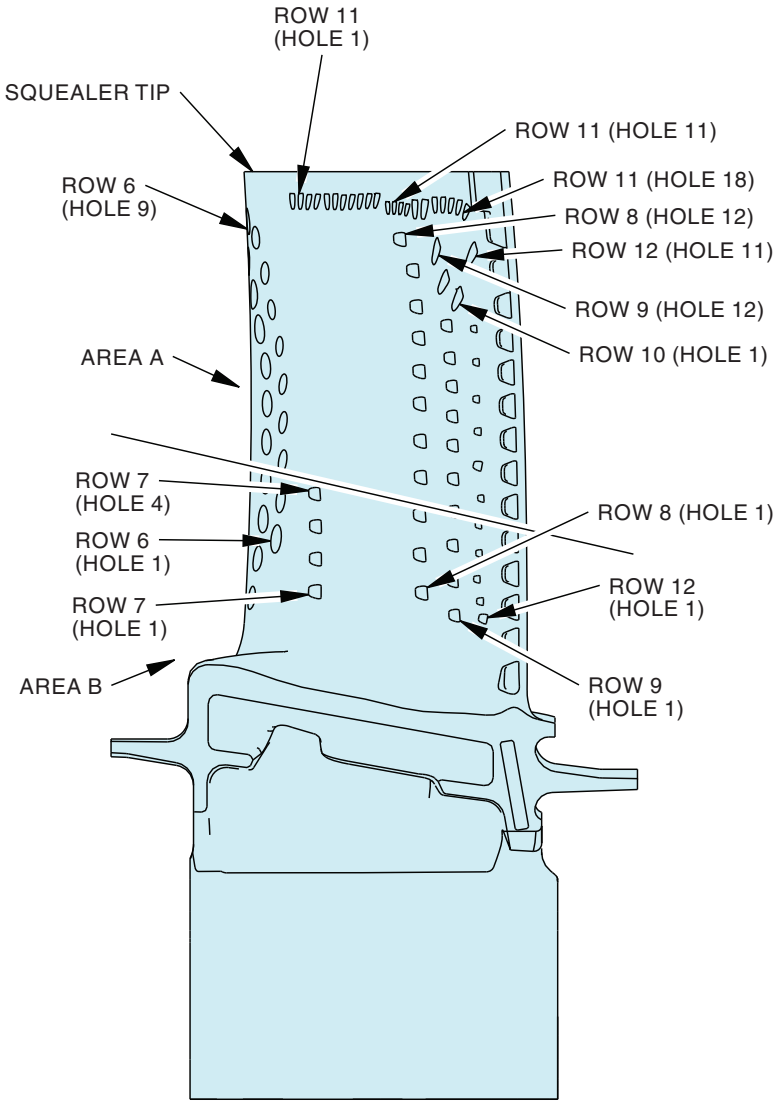
High Pressure Turbine Section Blade, Nozzle and Shroud Inspection
Figure 622/72-00-00-990-845-G00 (Sheet 6 of 18)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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POST SB 72-0138

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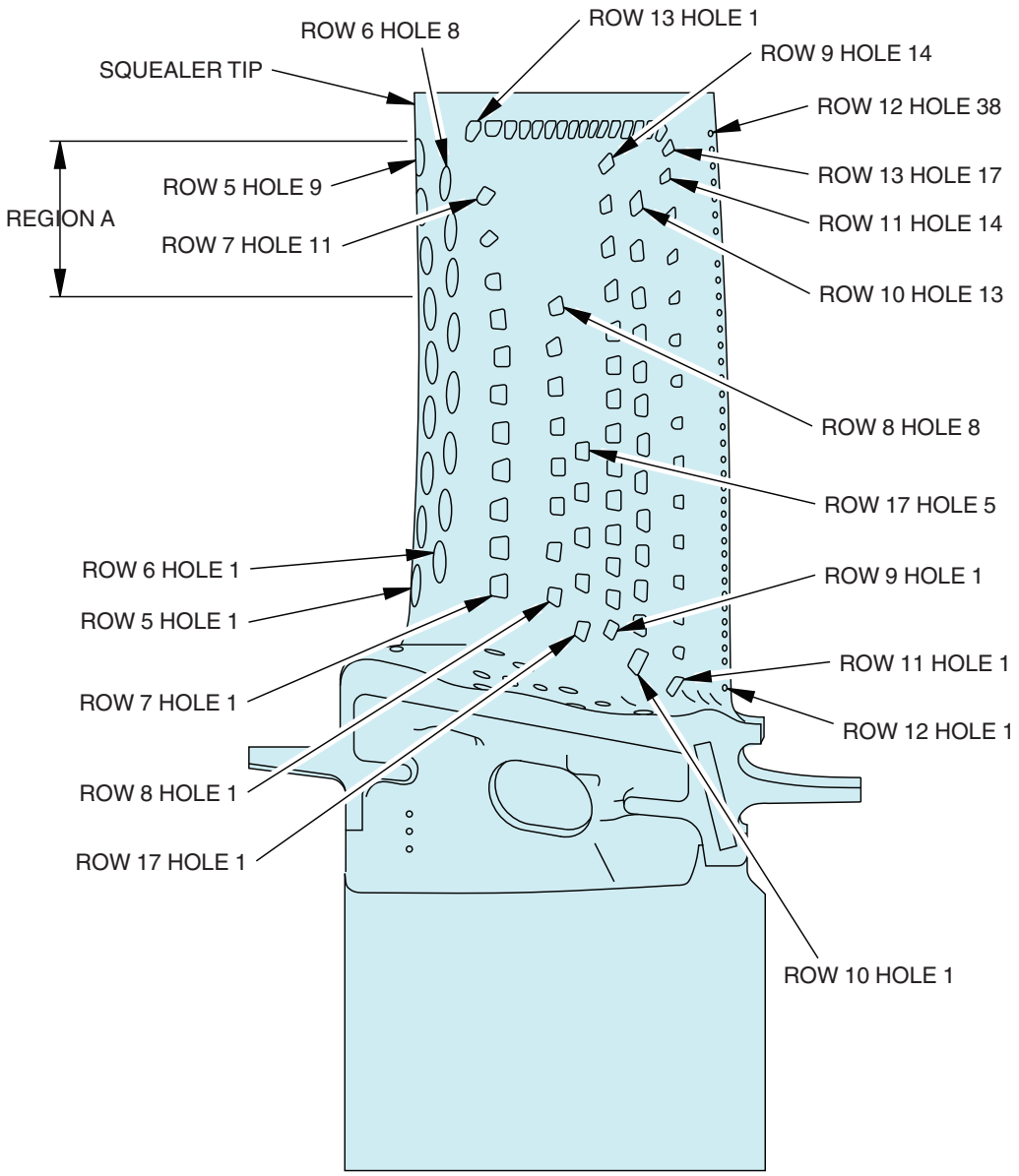
High Pressure Turbine Section Blade, Nozzle and Shroud Inspection
Figure 622/72-00-00-990-845-G00 (Sheet 7 of 18)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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POST SB 72-0454

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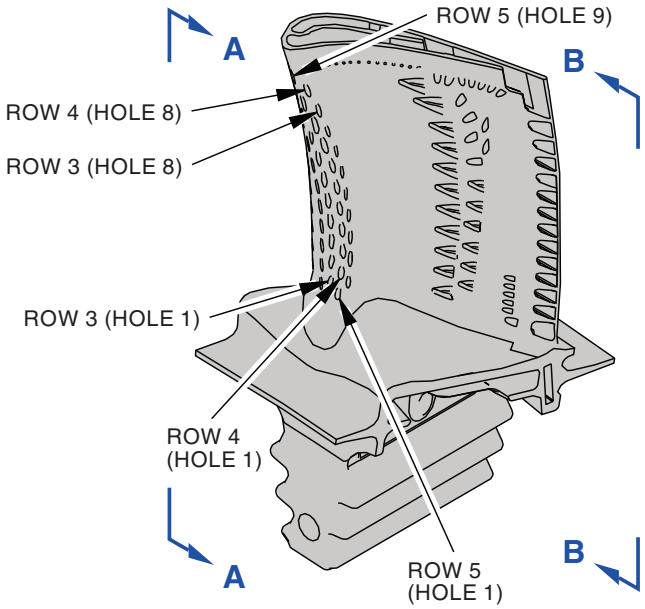
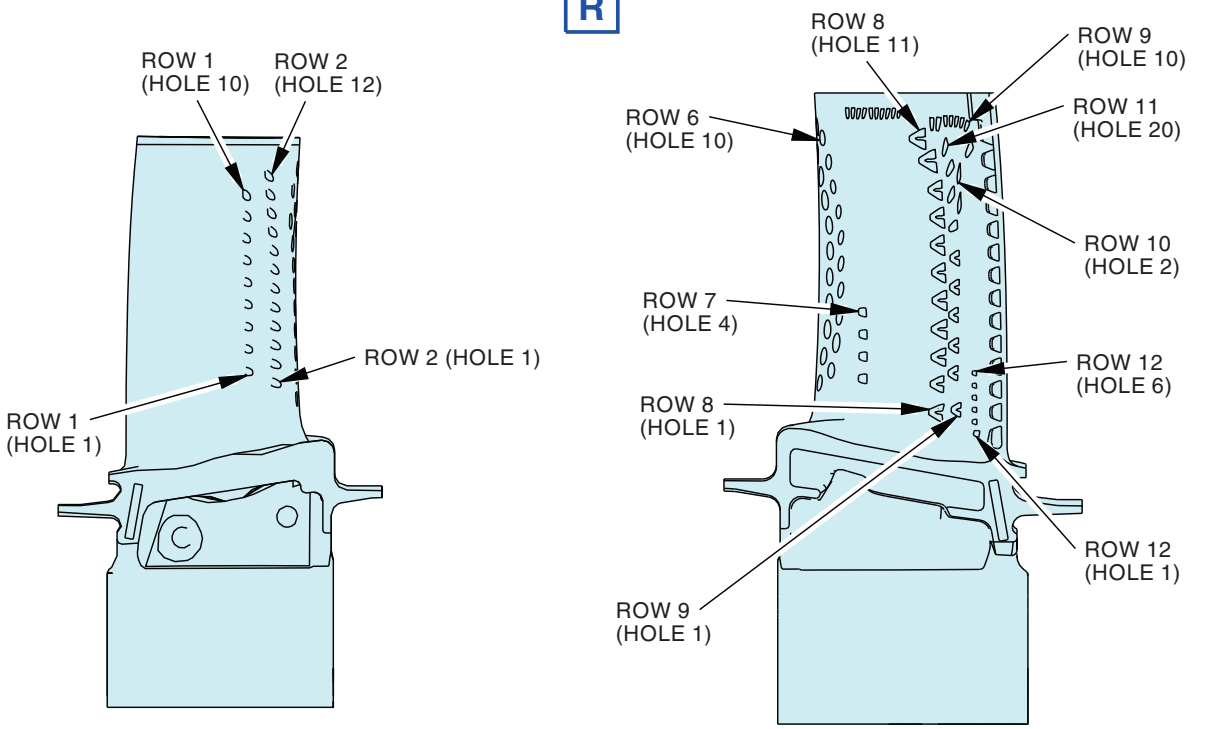
High Pressure Turbine Section Blade, Nozzle and Shroud Inspection
Figure 622/72-00-00-990-845-G00 (Sheet 8 of 18)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



HPT STAGE 1 TURBINE BLADE COOLING HOLE ROWS

R

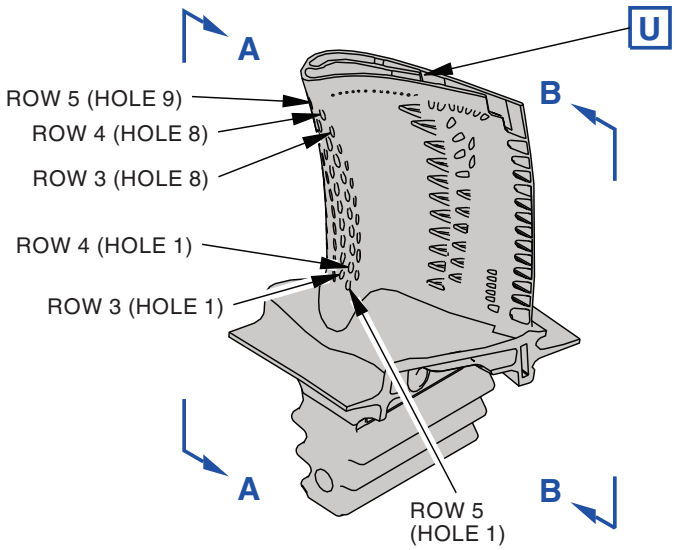


PRE SB 72-0138

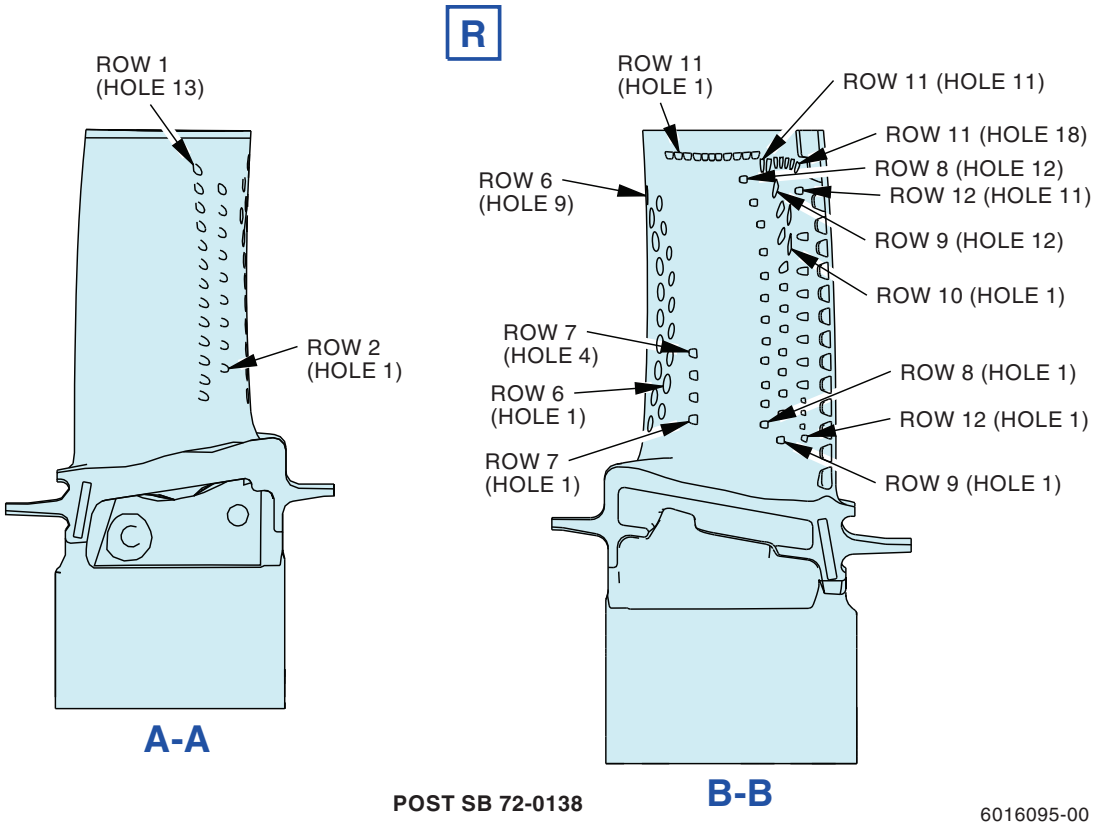
6016094-00
2915480 S0000700773_V3

High Pressure Turbine Section Blade, Nozzle and Shroud Inspection
Figure 622/72-00-00-990-845-G00 (Sheet 9 of 18)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



HPT STAGE 1 TURBINE BLADE COOLING HOLE ROWS



POST SB 72-0138

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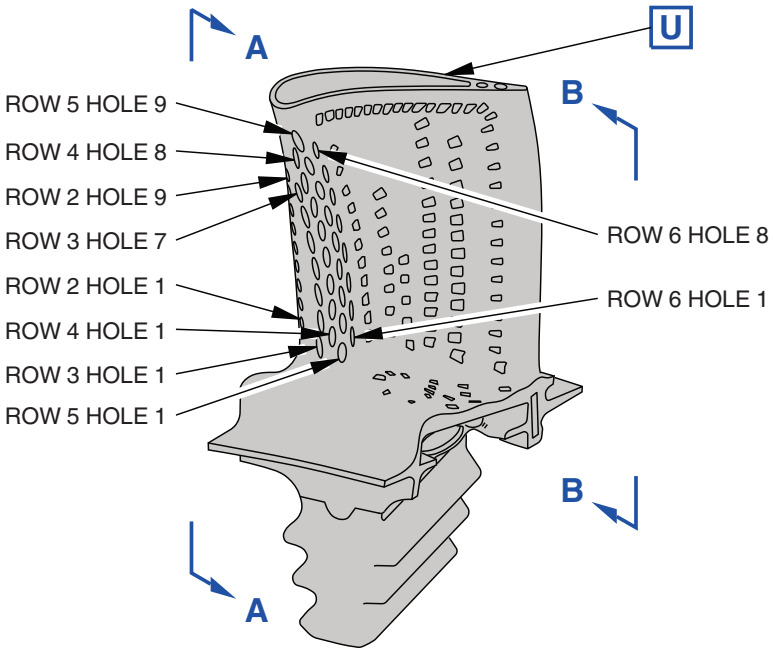
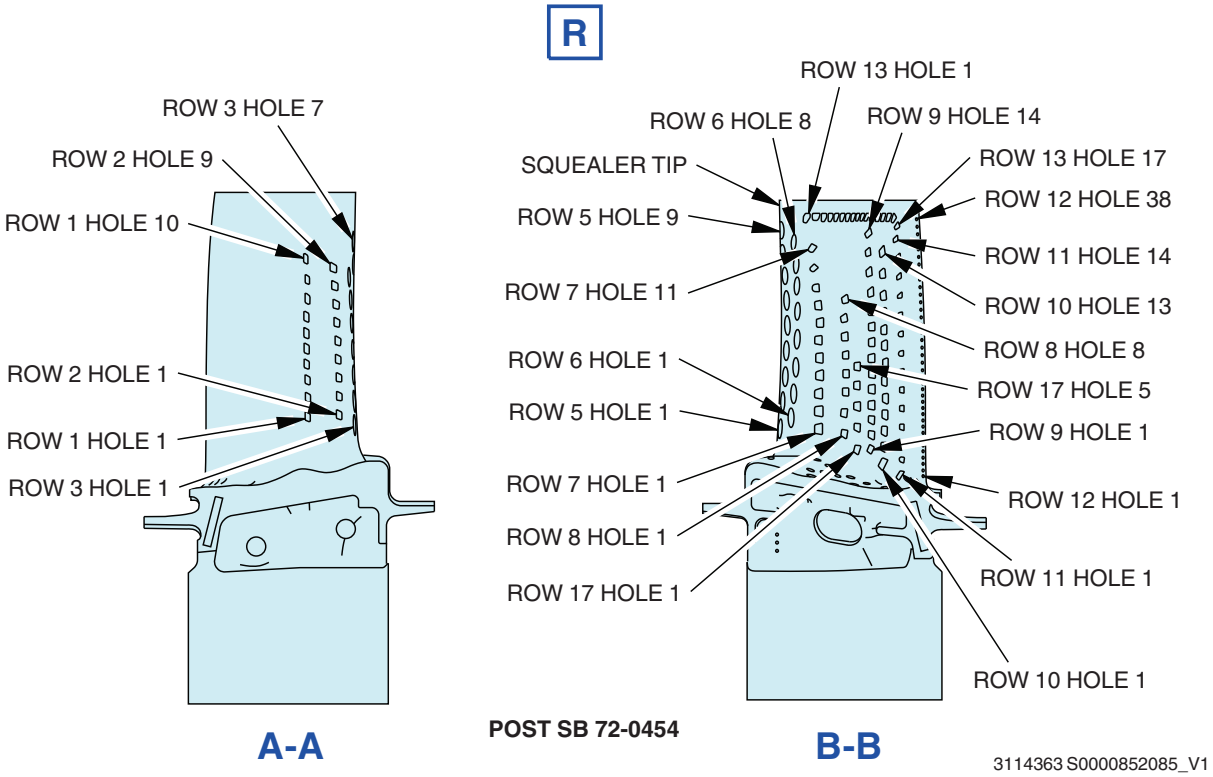
High Pressure Turbine Section Blade, Nozzle and Shroud Inspection
Figure 622/72-00-00-990-845-G00 (Sheet 10 of 18)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

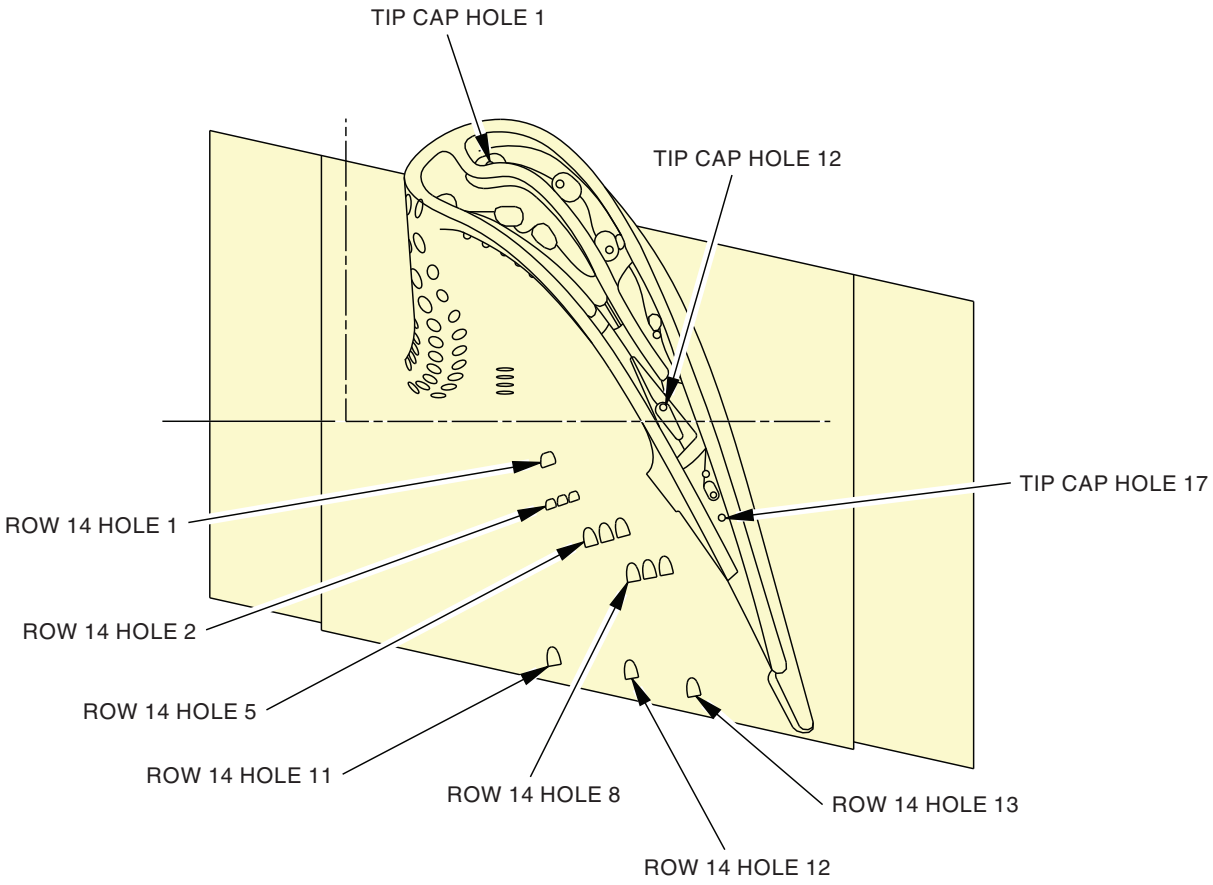


HPT STAGE 1 TURBINE BLADE COOLING HOLE ROWS



High Pressure Turbine Section Blade, Nozzle and Shroud Inspection
Figure 622/72-00-00-990-845-G00 (Sheet 11 of 18)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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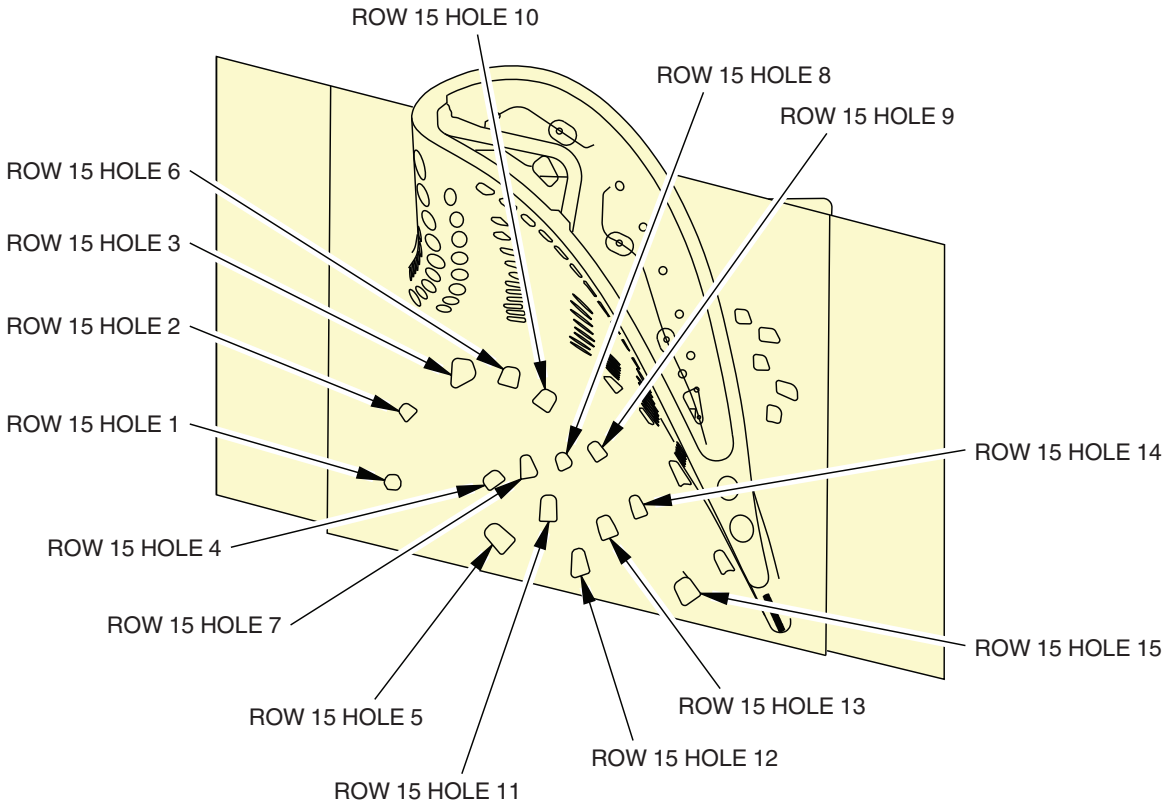
High Pressure Turbine Section Blade, Nozzle and Shroud Inspection
Figure 622/72-00-00-990-845-G00 (Sheet 12 of 18)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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POST SB 72-0454

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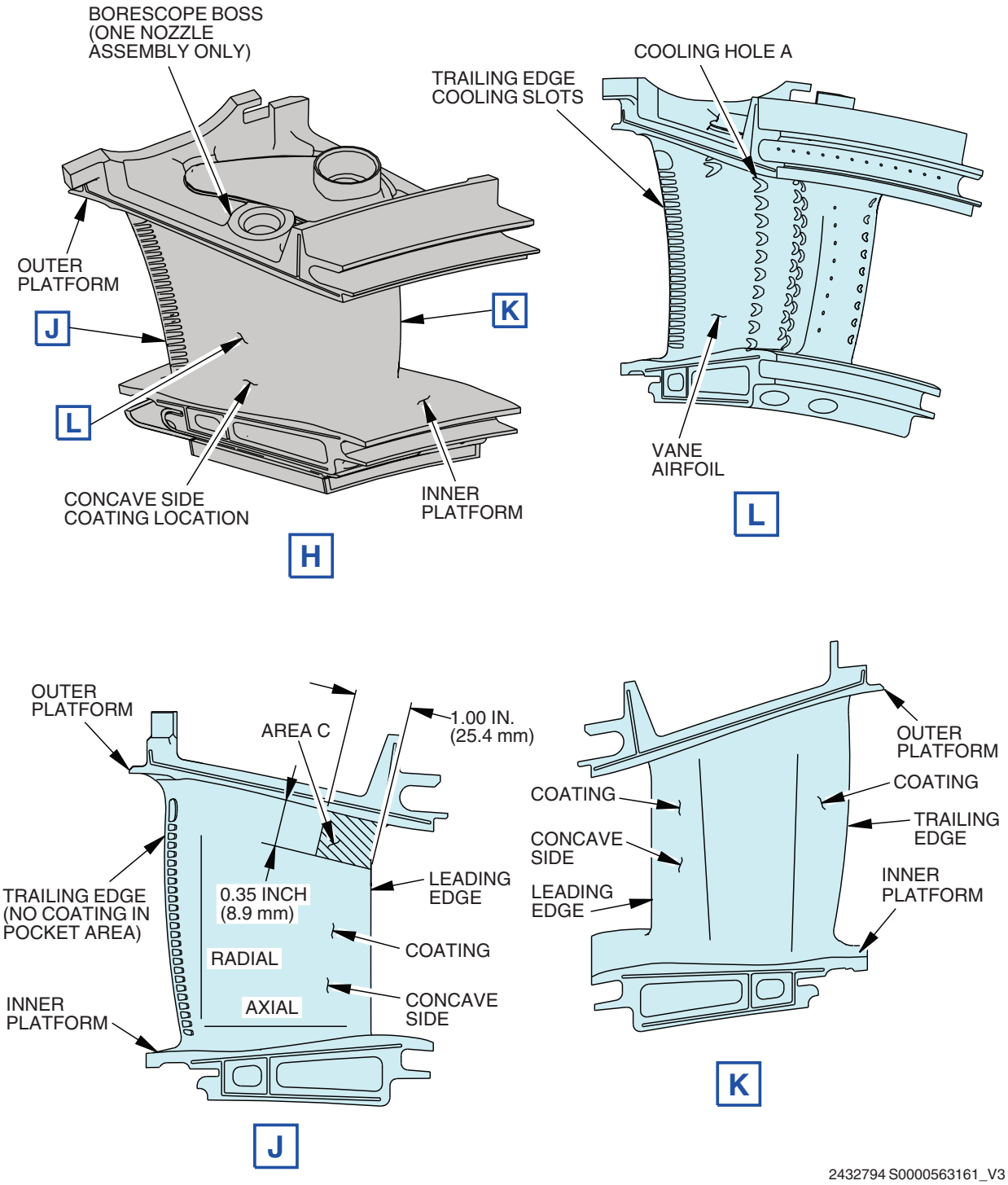
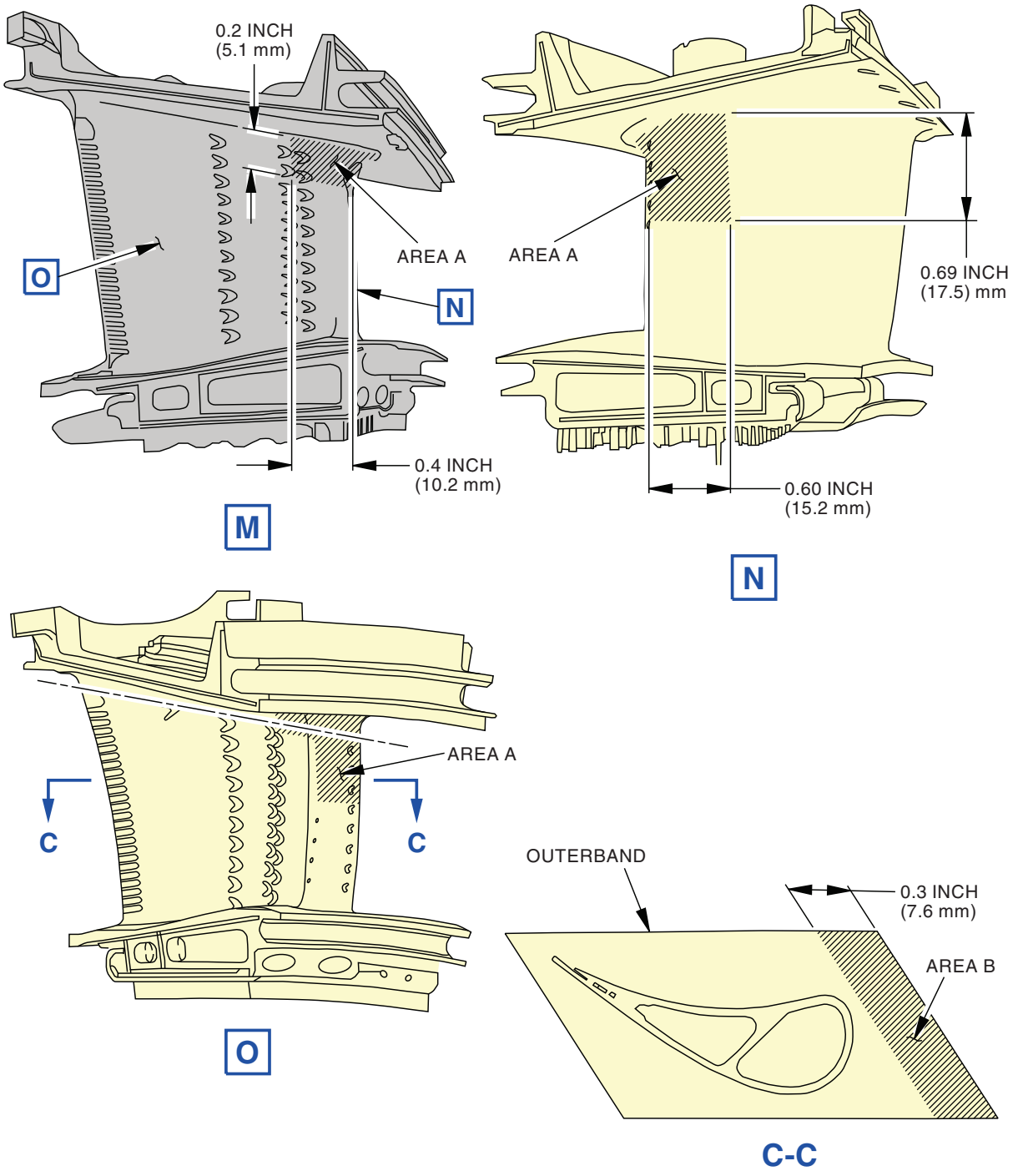
High Pressure Turbine Section Blade, Nozzle and Shroud Inspection
Figure 622/72-00-00-990-845-G00 (Sheet 13 of 18)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



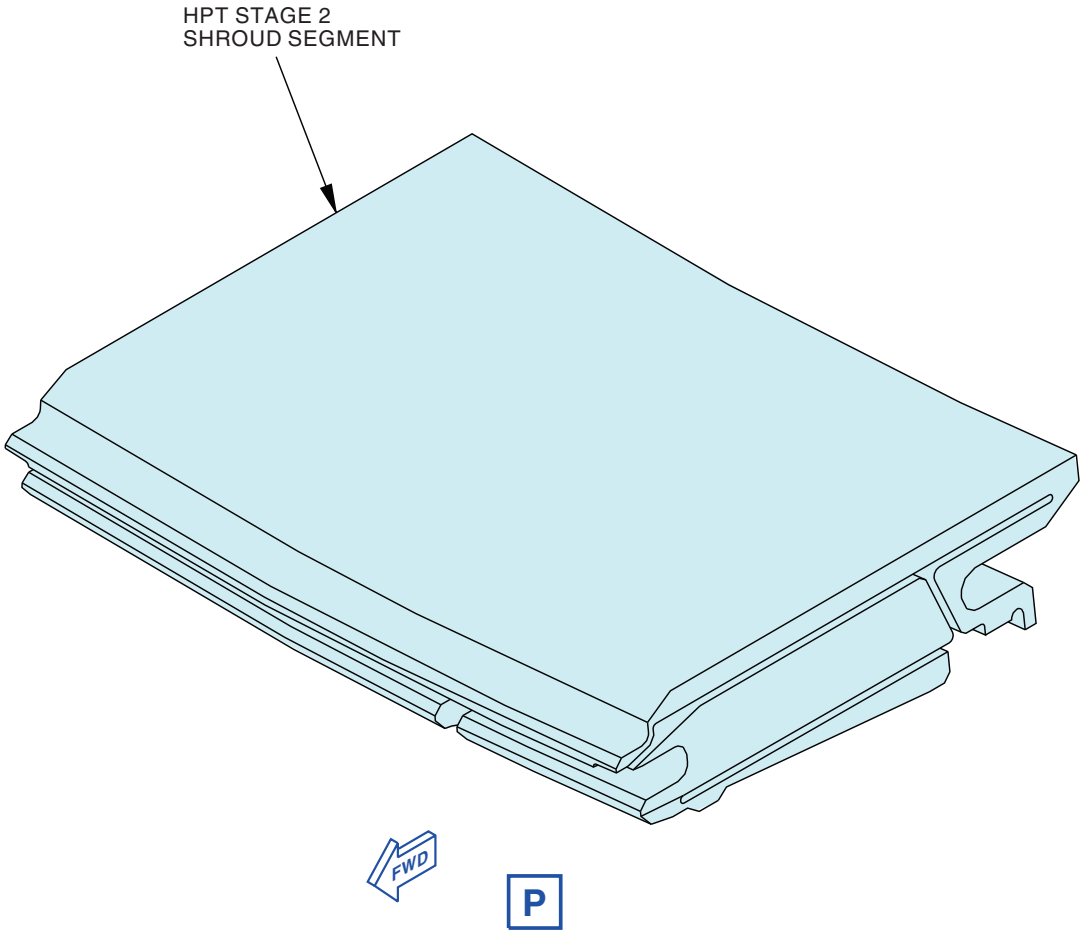
High Pressure Turbine Section Blade, Nozzle and Shroud Inspection
Figure 622/72-00-00-990-845-G00 (Sheet 14 of 18)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



High Pressure Turbine Section Blade, Nozzle and Shroud Inspection
Figure 622/72-00-00-990-845-G00 (Sheet 15 of 18)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



5021313-00

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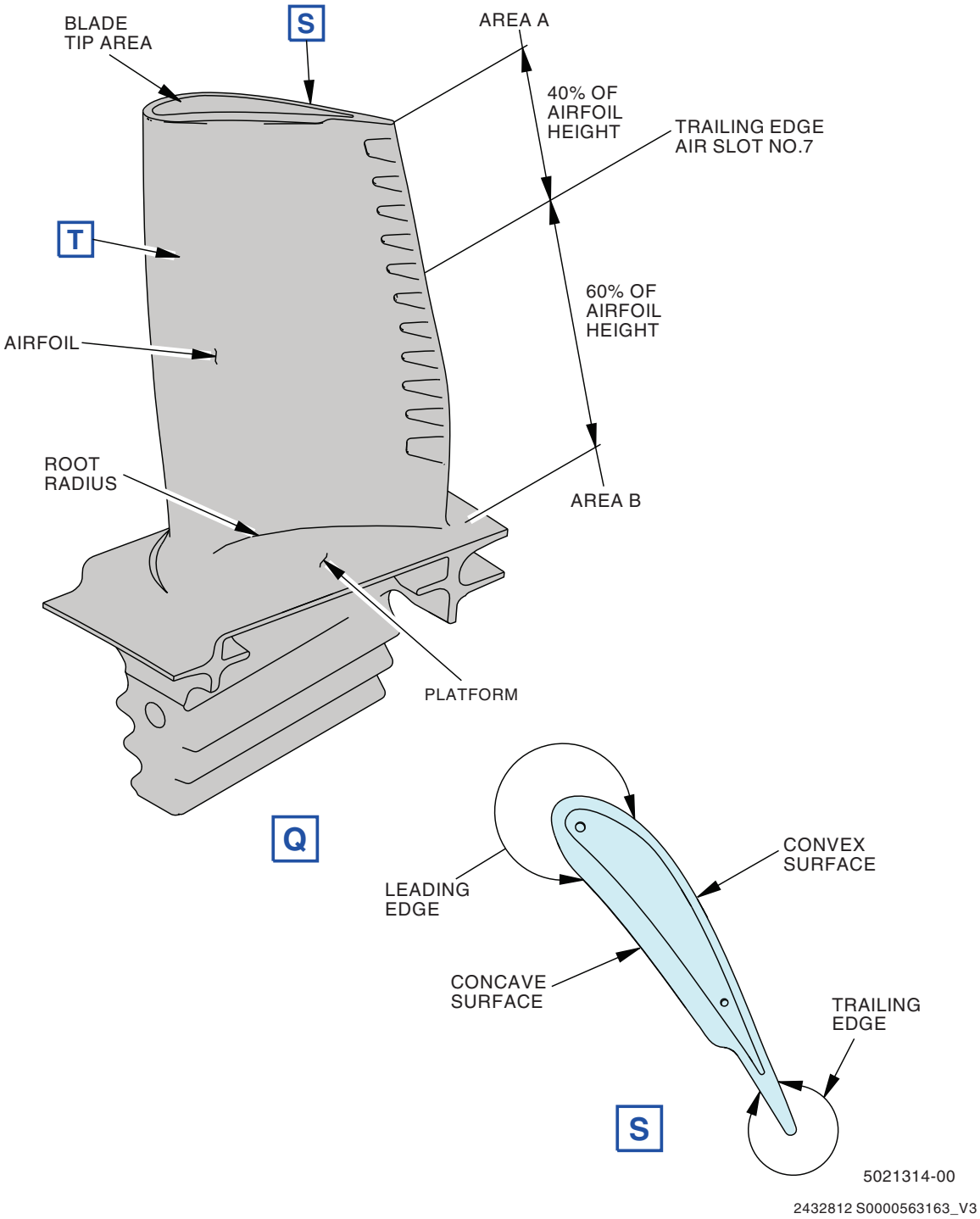
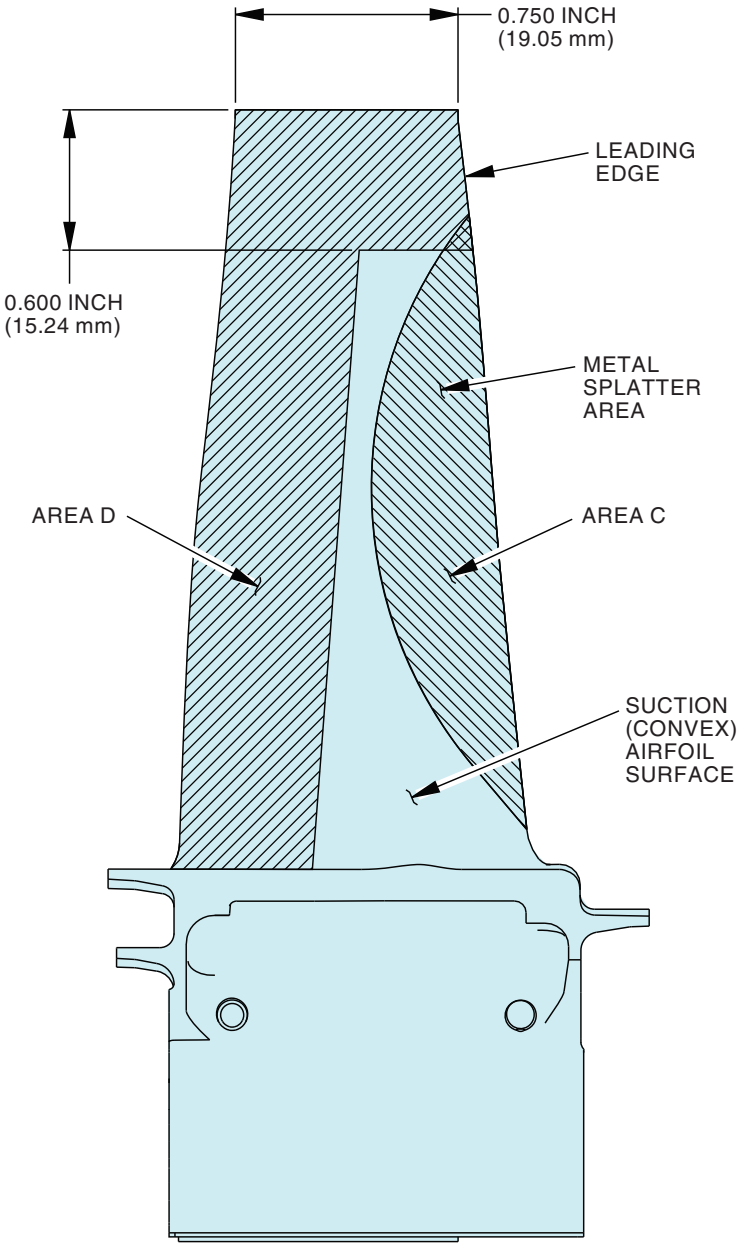
High Pressure Turbine Section Blade, Nozzle and Shroud Inspection
Figure 622/72-00-00-990-845-G00 (Sheet 16 of 18)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



High Pressure Turbine Section Blade, Nozzle and Shroud Inspection
Figure 622/72-00-00-990-845-G00 (Sheet 17 of 18)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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5021315-00

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High Pressure Turbine Section Blade, Nozzle and Shroud Inspection
Figure 622/72-00-00-990-845-G00 (Sheet 18 of 18)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 72-00-00-290-807-G00

8. High Pressure Turbine (HPT) Stage 1 Nozzle Borescope Inspection

(Figure 623)

A. General

(1) This task gives the instructions to do a 360 degree inspection of the stage 1 High Pressure Turbine (HPT) nozzle guide vane trailing edges with a flexible borescope.

(2) The high pressure turbine is referred to as the HPT.

(3) There are 18 segments and 36 foils in the stage 1 HPT nozzle.

(a) Each nozzle segment has 2 airfoils.

B. References

Reference	Title
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-14907	Borescope Guide Tubes Part #: 956A1603G01 Supplier: 58828
STD-858	Tag - DO NOT OPERATE
STD-1059	Platform - Engine and Strut Access
STD-1432	Fiberscope - Flexible Borescope
STD-10719	Borescope - Rigid (Meets General Electric or CFM Standard Practices Manual Specification such as: model R857ELSV60)

D. Consumable Materials

Reference	Description	Specification
D00601 [CP2101]	Vaseline - Graphite Mineral	
D50355 [CP2961]	Lubricant - Anti-Seize Compound	D6Y31C2

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Prepare for the Inspection

SUBTASK 72-00-00-480-006-G00

(1) Put the 3 ft (1 m) access platform, STD-1059, into position.

SUBTASK 72-00-00-010-018-G00

WARNING BE CAREFUL WHEN YOU WORK ON THE ENGINE COMPONENTS IMMEDIATELY AFTER THE ENGINE IS SHUTDOWN. THE ENGINE COMPONENTS CAN STAY HOT FOR UP TO ONE HOUR AND CAN CAUSE INJURY.

CAUTION BEFORE YOU USE THE BORESCOPE, MAKE SURE THAT THE TEMPERATURES OF THE INSPECTION AREAS ARE BELOW 150°F (66°C). IF THE ADJACENT CASES ARE TOO HOT TO KEEP YOUR HAND ON, THE INSPECTION AREAS ARE TOO HOT. IF THE INSPECTION AREA IS TOO HOT, DAMAGE TO THE BORESCOPE CAN OCCUR.

(2) Do this task: Preparation for Borescope Inspection, TASK 72-00-00-290-801-G00.

(a) Remove the borescope plug from the borescope port P.

NOTE: This port gives access to the trailing edges of the stage 1 HPT nozzle guide vanes for inspection.

(b) Put an identification tag on the borescope plug.

H. High Pressure Turbine (HPT) Stage 1 Nozzle Borescope Inspection

SUBTASK 72-00-00-211-015-G00

(1) Do a 360 degree borescope inspection of the trailing edges of the stage 1 HPT nozzle guide vanes:

NOTE: A 5 mm (or less) flexible borescope is necessary for this inspection if you use the guide tube. The borescope must be a minimum of 90.0 in. (2.29 m) in length. This is necessary to do the 360 degree borescope inspection of the trailing edges of the stage 1 HPT nozzle guide vanes.

NOTE: It is optional to use a 6 mm flexible borescope without the guide tubes.

CAUTION MAKE SURE THAT YOU DO NOT BEND THE FLEXIBLE BORESCOPE MORE THAN IS NECESSARY. IF YOU BEND THE BORESCOPE TOO MUCH, YOU WILL CAUSE DAMAGE.

CAUTION MAKE SURE THE FLEXIBLE BORESCOPE DOES NOT CATCH THE INTERNAL DETAILS OF THE ENGINE. IF YOU DO NOT, DAMAGE TO THE BORESCOPE COULD OCCUR. ALSO, DAMAGE TO THE POWERPLANT COULD OCCUR IF THE BORESCOPE BECOMES CAUGHT OR BROKEN INSIDE THE ENGINE.

(a) If you will use a guide tubes, SPL-14907, put it into the borescope port P.

(b) Insert the fiberscope, STD-1432, into the borescope port P.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Extend the fiberscope, STD-1432, into the flowpath between the stage 1 nozzle trailing edge and stage 1 blade leading edge.

(d) Continue to move the fiberscope, STD-1432, in the direction set by the stage 1 HPT nozzle vanes.

(e) Move the fiberscope, STD-1432, across the top of the engine and 360 degrees around until you can see the flexible borescope.

NOTE: It is important that you can see the point where the flexible borescope is between the two vanes of the stage 1 HPT nozzle. This necessary to make sure that you do a full 360 degree inspection.

(f) Examine the HPT stage one nozzle trailing edges using the serviceability limits in this task: High Pressure Turbine Section Detailed Inspection, TASK 72-00-00-290-806-G00.

NOTE: The Stage 1 HPT nozzle guide vane airfoil leading edges may be inspected using the fiberscope, STD-1432, or the rigid borescope, STD-10719.

(g) Examine the HPT stage one nozzle airfoils using the serviceability limits in this task: High Pressure Turbine Section Detailed Inspection, TASK 72-00-00-290-806-G00.

(h) Examine the HPT stage one nozzle inner and outer platforms using the serviceability limits in this task: High Pressure Turbine Section Detailed Inspection, TASK 72-00-00-290-806-G00.

(i) Examine all 36 stage 1 HPT nozzle airfoil trailing edges as you pull the flexible borescope back around the full 360 degree path.

SUBTASK 72-00-00-080-007-G00

(2) When the inspection is complete, remove the borescope from the engine.

SUBTASK 72-00-00-080-019-G00

(3) If it is necessary, remove the guide tubes, SPL-14907, from the borescope port P.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 72-00-00-410-011-G00

(1) Install the plug into the borescope port P as follows:

(a) Do a visual inspection of the borescope plug and the borescope port (TASK 72-00-00-290-801-G00).

(b) Remove the identification tag from the borescope plug.

(c) Put Tiodize T8E-H grease, D50355 [CP2961], on the borescope plug threads and the mating surfaces.

NOTE: You can use graphite mineral vaseline, D00601 [CP2101], as an alternative to the Tiodize T8E-H grease, D50355 [CP2961].

(d) Install the borescope plug into borescope port P.

(e) Tighten the borescope plug to 85.0 in-lb (9.6 N·m) - 95.0 in-lb (10.7 N·m).

SUBTASK 72-00-00-080-008-G00

(2) Remove the access platform, STD-1059, from the engine area.

SUBTASK 72-00-00-410-012-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(3) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 72-00-00-440-010-G00

(4) Do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801

SUBTASK 72-00-00-910-001-G00

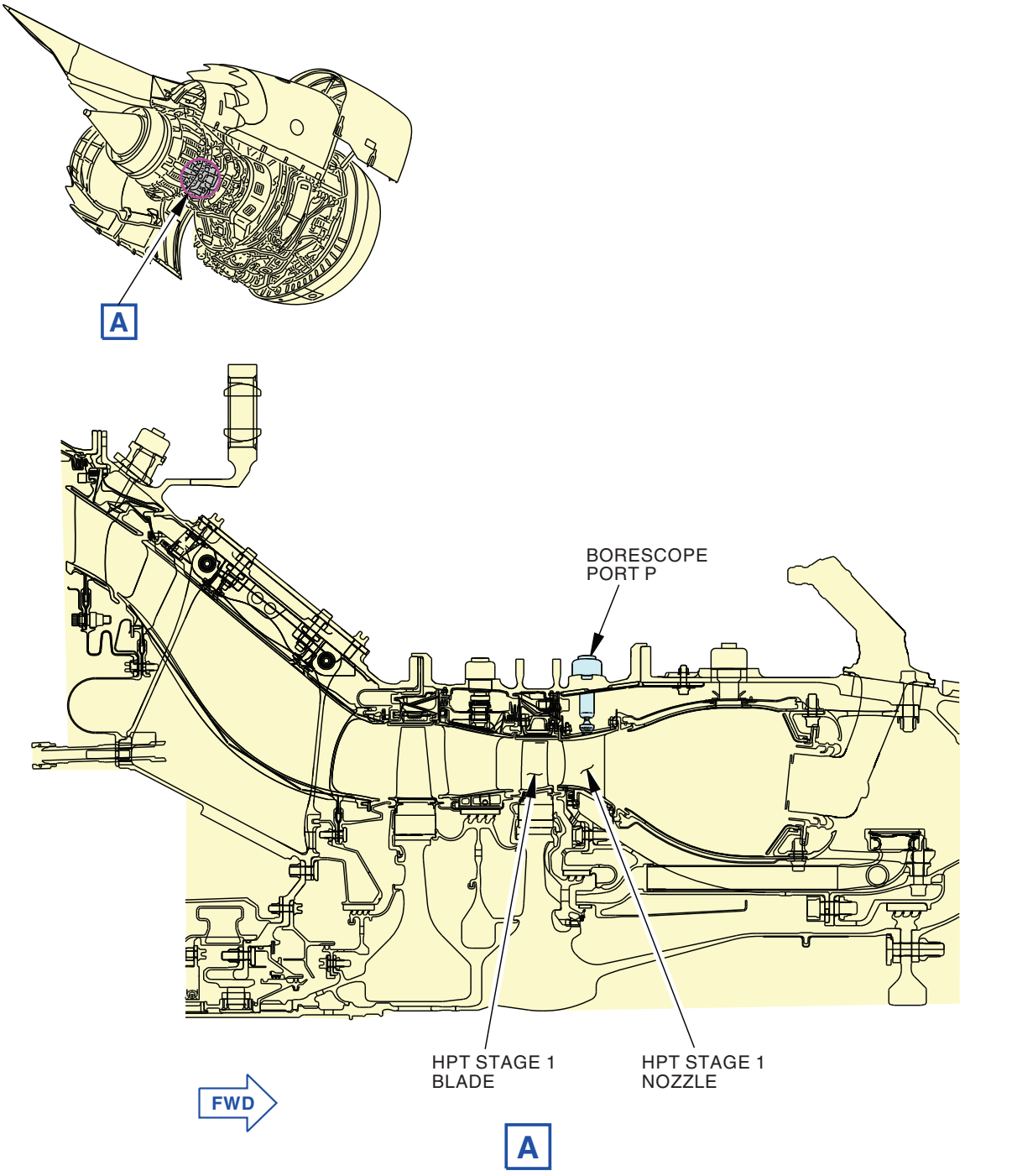
(5) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



High Pressure Turbine (HPT) Stage 1 Nozzle Borescope Inspection
Figure 623/72-00-00-990-847-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 72-00-00-290-808-G00

9. High Pressure Turbine (HPT) Stage 1 Shroud Borescope Inspection

(Figure 624)

A. General

(1) This task gives the instructions to do the borescope inspection of the stage 1 High Pressure Turbine (HPT) shroud segments.

(2) The stage 1 HPT shroud segments are inspected using a flexible borescope.

(3) There are 18 stage 1 HPT shroud segments.

B. References

Reference	Title
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-14907	Borescope Guide Tubes Part #: 956A1603G01 Supplier: 58828
STD-858	Tag - DO NOT OPERATE
STD-1059	Platform - Engine and Strut Access
STD-1432	Fiberscope - Flexible Borescope

D. Consumable Materials

Reference	Description	Specification
D00601 [CP2101]	Vaseline - Graphite Mineral	
D50355 [CP2961]	Lubricant - Anti-Seize Compound	D6Y31C2

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Prepare for the Inspection

SUBTASK 72-00-00-480-007-G00

WARNING BE CAREFUL WHEN YOU WORK ON THE ENGINE COMPONENTS IMMEDIATELY AFTER THE ENGINE IS SHUTDOWN. THE ENGINE COMPONENTS CAN STAY HOT FOR UP TO ONE HOUR AND CAN CAUSE INJURY.

(1) Put the 3 ft (1 m) access platform, STD-1059, into position.

SUBTASK 72-00-00-010-029-G00

CAUTION MAKE SURE THE TEMPERATURE OF THE ENGINE AND THE ADJACENT AREAS NEAR THE BORESCOPE INSPECTION PORTS ARE LESS THAN 150°F (65.6°C). DO NOT DO A BORESCOPE INSPECTION IF THE ADJACENT AREAS ARE TOO HOT TO TOUCH. DO THE BORESCOPE INSPECTION WHEN THE ENGINE TEMPERATURE IS LESS THAN 150°F (65.6°C). THIS WILL PREVENT DAMAGE TO THE BORESCOPE EQUIPMENT.

(2) Remove the applicable borescope plugs from the HPT case borescope ports.

NOTE: The borescope ports gives access for the inspection of the stage 1 HPT shrouds. The P port for the leading edge and the Q port for trailing edge of the shroud.

SUBTASK 72-00-00-930-003-G00

(3) Put an identification tag on each borescope plug.

SUBTASK 72-00-00-010-019-G00

(4) Do this task: Preparation for Borescope Inspection, TASK 72-00-00-290-801-G00.

H. High Pressure Turbine (HPT) Stage 1 Shroud Borescope Inspection

SUBTASK 72-00-00-480-008-G00

(1) Do a 360 degree borescope inspection of the stage 1 HPT shrouds as follows:

NOTE: A 5 mm (or less) flexible borescope is necessary for this inspection if you use the guide tube. The borescope must be a minimum of 90 in. (2.29 m) in length. This is necessary to do the 360 degree borescope inspection of the trailing edges of the stage 1 HPT shrouds.

NOTE: It is optional to use a 6 mm flexible borescope without the guide tubes.

CAUTION MAKE SURE THAT YOU DO NOT BEND THE FLEXIBLE BORESCOPE MORE THAN IS NECESSARY. IF YOU BEND THE BORESCOPE TOO MUCH, YOU WILL CAUSE DAMAGE.

CAUTION MAKE SURE THE FLEXIBLE BORESCOPE DOES NOT CATCH THE INTERNAL DETAILS OF THE ENGINE. IF YOU DO NOT, DAMAGE TO THE BORESCOPE COULD OCCUR. ALSO, DAMAGE TO THE POWERPLANT COULD OCCUR IF THE BORESCOPE BECOMES CAUGHT OR BROKEN INSIDE THE ENGINE.

(a) Port P:

If you will use a guide tubes, SPL-14907, put the guide tube into borescope port P as follows:

1) Move the end of the guide tube into the hole in the stage 1 HPT nozzle and into the flowpath.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) Put the fiberscope, STD-1432, into the guide tubes, SPL-14907, and extend it into the stage 1 HPT flowpath.

3) Continue to move the fiberscope, STD-1432, in the direction set by the stage 1 HPT nozzle vanes.

4) Move the fiberscope, STD-1432, on 360 degrees around until you can see the flexible borescope.

NOTE: Make sure that you can see the point where the flexible borescope is between the two vanes of the stage 1 HPT shroud. This is necessary to make sure that you do a full 360 degree inspection.

(b) Port Q:

If you will use a guide tubes, SPL-14907, put the guide tube into HPT case port Q as follows:

1) Move the end of the guide tube into the hole in the stage 2 HPT nozzle and into the flowpath.

2) Put the fiberscope, STD-1432, into the guide tubes, SPL-14907, and extend it into the stage 2 HPT flowpath.

3) Continue to move the fiberscope, STD-1432, in the direction set by the stage 2 HPT nozzle vanes.

4) Move the fiberscope, STD-1432, on 360 degrees around the engine until you can see the flexible borescope.

NOTE: Make sure that you can see the point where the flexible borescope is between the two vanes of the stage 2 HPT nozzle. This is necessary to make sure that you do a full 360 degree inspection.

(c) Examine all 18 stage 1 HPT shrouds as you pull the fiberscope, STD-1432, back around the full 360 degree path.

(d) Inspection limits for the stage 1 HPT shrouds:

NOTE: The flow path coating is both the top coating and bond coating. Exposed base material is area where both the top coating and bond coating are fully missing.

1) Blade rub

a) Permitted.

2) Cracks

a) Cracks in the base material are permitted up to 0.60 in. (15.24 mm) in length are permitted.

b) Cracks in the flow path coating are permitted.

3) Missing flow path coating

a) In or less than 0.10 in. (2.54 mm) from the leading and trailing edges, any amount of missing coating is permitted.

b) For any shroud segment that is missing complete circumferential flow path coating in or less than 0.10 in. (2.54 mm) of the trailing edge, do an inspection of the leading edges of the three adjacent stage 2 HPT nozzle segments.

c) For areas more than 0.10 in. (2.54 mm) from the leading and trailing edges, a total of 0.024 in^2 (15.48 mm^2) missing coating on each shroud is permitted.

4) Chipping of the shroud segment base material along the leading and trailing edge

ECCN 9E991 BOEING PROPRIETARY - See title page for details

a) Up to two chips on each shroud segment are permitted if they are in or less than 0.050 in. (1.270 mm) from the edge.

5) Chipping of the shroud segment base material along the circumferential split line

a) A maximum axial length of 1.0 in. (25.4 mm) is permitted on up to 10 shroud segments, each end.

6) Missing spline seals

a) Any number of missing or partially missing discourager spline seals is permitted.

b) Any number of missing or partially missing primary spline seals is permitted.

7) Continue-In-Service (CIS) limits for the shroud segment flowpath surface:

a) A 1600 cycle inspection interval is permitted with these conditions:

<1> Any amount of missing flow path coating and exposed base material is permitted.

b) A 800 cycle inspection interval is permitted with these conditions:

<1> Chipping of the shroud base material up to 0.25 in. (6.4 mm) from the leading edge or up to 0.15 in. (3.8 mm) from the trailing edge.

c) A 200 cycle inspection interval is permitted with this condition:

<1> A maximum axial length of 1.0 in. (25.4 mm) of missing base material along the circumferential split line is permitted on all shroud segments.

<2> Any amount of circumferential cracking that does not extend across the entire shroud segment length.

d) A 100 cycle inspection interval is permitted with these conditions:

<1> Chipping of the shroud base material up to 0.35 in. (8.9 mm) from the leading edge or up to 0.25 in. (6.4 mm) from the trailing edge.

e) A 30 cycle inspection interval is permitted with these conditions:

<1> Any amount of axial cracking that does not extend across the entire shroud segment length.

<2> Any amount of circumferential cracking that does extend across the entire shroud segment length.

f) Remove the engine in 10 cycles or less with these conditions:

<1> Any amount of axial or circumferential cracking, where cracking has intersected.

<2> Any amount of axial cracking that extends across the entire shroud segment length.

<3> Any amount of shroud segment radial displacement into the flow path.

<4> If you find a through hole in shroud segment or missing portion of the shroud segment

I. Put the Airplane back to Its Usual Condition

SUBTASK 72-00-00-080-009-G00

(1) Remove the fiberscope, STD-1432.

SUBTASK 72-00-00-410-013-G00

(2) Install the applicable borescope plugs P and Q into the borescope ports as follows:

(a) If it is necessary, remove the guide tubes, SPL-14907, from the borescope ports.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Remove the identification tags from the borescope plugs.

(c) Put high temperature grease graphite mineral vaseline, D00601 [CP2101], on the borescope plug threads and the mating surfaces.

NOTE: You can use Tiodize T8E-H grease, D50355 [CP2961], as an alternative.

(d) Install the applicable borescope plugs into the borescope ports.

(e) Torque the borescope plugs P and Q to 85.0 in-lb (9.6 N·m) - 95.0 in-lb (10.7 N·m).

SUBTASK 72-00-00-080-010-G00

(3) Remove the access platform, STD-1059, from the engine area.

SUBTASK 72-00-00-410-014-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(4) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 72-00-00-440-007-G00

(5) Do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801

SUBTASK 72-00-00-860-044-G00

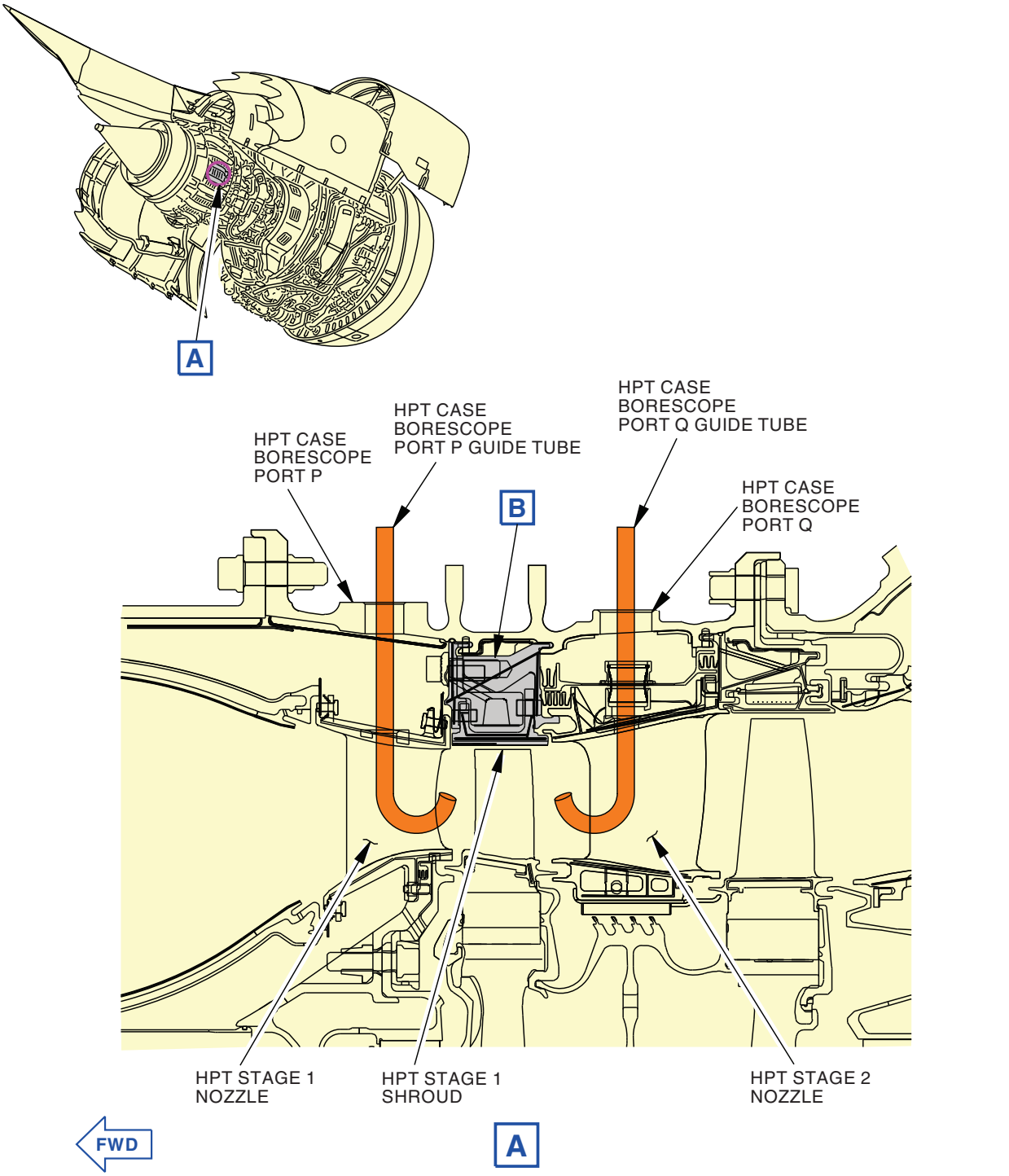
(6) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

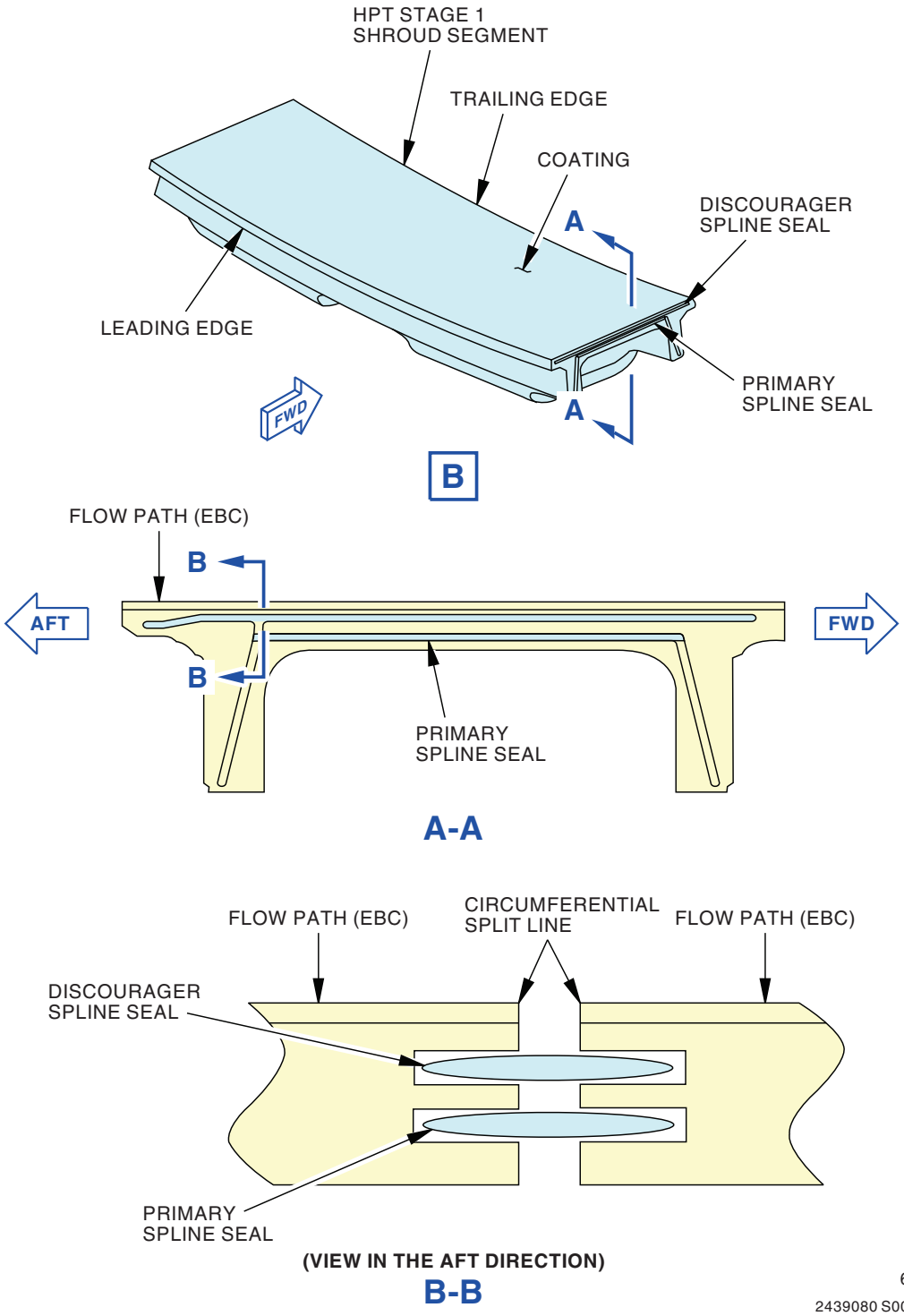
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



High Pressure Turbine (HPT) Stage 1 Shroud Borescope Inspection
Figure 624/72-00-00-990-846-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



High Pressure Turbine (HPT) Stage 1 Shroud Borescope Inspection
Figure 624/72-00-00-990-846-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 72-00-00-290-809-G00

10. High Pressure Turbine (HPT) Stage 2 Nozzle Borescope Inspection

(Figure 625)

A. General

(1) This task gives the instructions to do the 360 degree borescope inspection of the stage 2 High Pressure Turbine (HPT) nozzle guide vane leading and trailing edges and leading edges.

(2) There are 36 segments in the stage 2 HPT nozzle.

B. References

Reference	Title
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-14907	Borescope Guide Tubes Part #: 956A1603G01 Supplier: 58828
STD-858	Tag - DO NOT OPERATE
STD-1059	Platform - Engine and Strut Access
STD-1432	Fiberscope - Flexible Borescope

D. Consumable Materials

Reference	Description	Specification
D00601 [CP2101]	Vaseline - Graphite Mineral	
D50355 [CP2961]	Lubricant - Anti-Seize Compound	D6Y31C2

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Prepare for the Inspection

SUBTASK 72-00-00-480-009-G00

> **WARNING** BE CAREFUL WHEN YOU WORK ON THE ENGINE COMPONENTS IMMEDIATELY AFTER THE ENGINE IS SHUTDOWN. THE ENGINE COMPONENTS CAN STAY HOT FOR UP TO ONE HOUR AND CAN CAUSE INJURY.

(1) Put the 3 ft (1 m) access platform, STD-1059, into position.

SUBTASK 72-00-00-010-028-G00

> **CAUTION** BEFORE YOU USE THE BORESCOPE, MAKE SURE THAT THE TEMPERATURES OF THE INSPECTION AREAS ARE BELOW 150°F (66°C). IF THE ADJACENT CASES ARE TOO HOT TO KEEP YOUR HAND ON, THE INSPECTION AREAS ARE TOO HOT. IF THE INSPECTION AREA IS TOO HOT, DAMAGE TO THE BORESCOPE CAN OCCUR.

(2) Remove the HPT stage 2 borescope plug from the HPT case borescope port Q.

NOTE: This borescope port gives access for the inspection of the leading and trailing edges of the stage 2 HPT nozzle guide vanes.

SUBTASK 72-00-00-930-002-G00

(3) Put an identification tag on the borescope plug.

SUBTASK 72-00-00-010-020-G00

(4) Do this task: Preparation for Borescope Inspection, TASK 72-00-00-290-801-G00.

G. High Pressure Turbine (HPT) Stage 2 Nozzle Borescope Inspection

SUBTASK 72-00-00-290-007-G00

(1) Do a 360 degree borescope inspection of the stage 2 HPT nozzle guide vane leading and trailing edges as follows:

> **CAUTION** MAKE SURE THAT YOU DO NOT BEND THE FLEXIBLE BORESCOPE MORE THAN IS NECESSARY. IF YOU BEND THE BORESCOPE TOO MUCH, YOU WILL CAUSE DAMAGE.

> **CAUTION** MAKE SURE THE FLEXIBLE BORESCOPE DOES NOT CATCH THE INTERNAL DETAILS OF THE ENGINE. IF YOU DO NOT, DAMAGE TO THE BORESCOPE COULD OCCUR. ALSO, DAMAGE TO THE POWERPLANT COULD OCCUR IF THE BORESCOPE BECOMES CAUGHT OR BROKEN INSIDE THE ENGINE.

(a) If you will use a guide tubes, SPL-14907, put the guide tube into the HPT case borescope port Q as follows:

NOTE: A 5 mm (or less) flexible borescope is necessary for this inspection if you use guide tubes. A flexible borescope with a minimum length of 90 in. (2.29 m) is necessary to do the 360-degree borescope inspection of the leading and trailing edges of the stage 2 HPT nozzle guide vanes.

NOTE: It is optional to use a 6 mm flexible borescope without the guide tubes.

(b) Put the 5 mm or less fiberscope, STD-1432, into the guide tubes, SPL-14907, and extend it into the stage 2 HPT flowpath.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Continue to move the fiberscope, STD-1432, in the direction set by the stage 2 HPT nozzle vanes.

(d) Move the fiberscope, STD-1432, across the top of the engine and 360 around until you can see the flexible borescope.

NOTE: It is important that you can see the point where the flexible borescope is between the two vanes of the stage 2 HPT nozzle. This necessary to make sure that you do a full 360 degree inspection.

(e) Examine the stage 2 HPT nozzles guide vanes per the serviceability limits in the HPT section detailed inspection task (TASK 72-00-00-290-806-G00).

(f) Examine the stage 2 HPT nozzles guide vane inner and outer platforms per the serviceability limits in the HPT section detailed inspection task (TASK 72-00-00-290-806-G00).

(g) Examine the trailing edges of the stage 2 nozzles as follows:

1) Use the serviceability limits in this task: High Pressure Turbine Section Detailed Inspection, TASK 72-00-00-290-806-G00.

2) Put the fiberscope, STD-1432, into borescope port Q.

3) Extend the fiberscope, STD-1432, into the stage 2 HPT flowpath on the forward side of the stage 2 blade.

4) Continue to move the fiberscope, STD-1432, in the direction set by the stage 2 nozzle vanes.

NOTE: The flexible borescope will go into the flowpath in the opposite direction compared to when you examine the leading edge.

(h) Examine the leading edges of the stage 2 nozzles as follows:

1) Use the serviceability limits in this task: High Pressure Turbine Section Detailed Inspection, TASK 72-00-00-290-806-G00.

2) Put the fiberscope, STD-1432, into borescope port Q.

3) Extend the fiberscope, STD-1432, into the stage 2 HPT flowpath toward the stage 2 blade.

4) Guide the fiberscope, STD-1432, between the stage 2 nozzle airfoils toward the platform leading edge.

5) Move the fiberscope, STD-1432, in the direction set by the stage 2 nozzle vanes 360 degree until you can see the flexible borescope.

6) Slowly remove the fiberscope, STD-1432, to view the leading edges of all 36 stage 2 nozzles.

SUBTASK 72-00-00-080-011-G00

(2) When the inspection is complete, remove the fiberscope, STD-1432, from the engine.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 72-00-00-080-012-G00

(1) If it is necessary, remove the guide tubes, SPL-14907, from the borescope port.

SUBTASK 72-00-00-410-015-G00

(2) Install the borescope port Q plug as follows:

(a) Remove the identification tag from the borescope plug.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Put Tiodize T8E-H grease, D50355 [CP2961], on the borescope plug thread and the mating surface.

NOTE: You can use graphite mineral vaseline, D00601 [CP2101], as an alternative.

(c) Install the borescope plug into the borescope port Q.

(d) Tighten the borescope plug Q to 85.0 in-lb (9.6 N·m) - 95.0 in-lb (10.7 N·m).

SUBTASK 72-00-00-080-013-G00

(3) Remove the access platform, STD-1059, from the engine area.

SUBTASK 72-00-00-410-016-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(4) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 72-00-00-440-005-G00

(5) Do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801

SUBTASK 72-00-00-910-002-G00

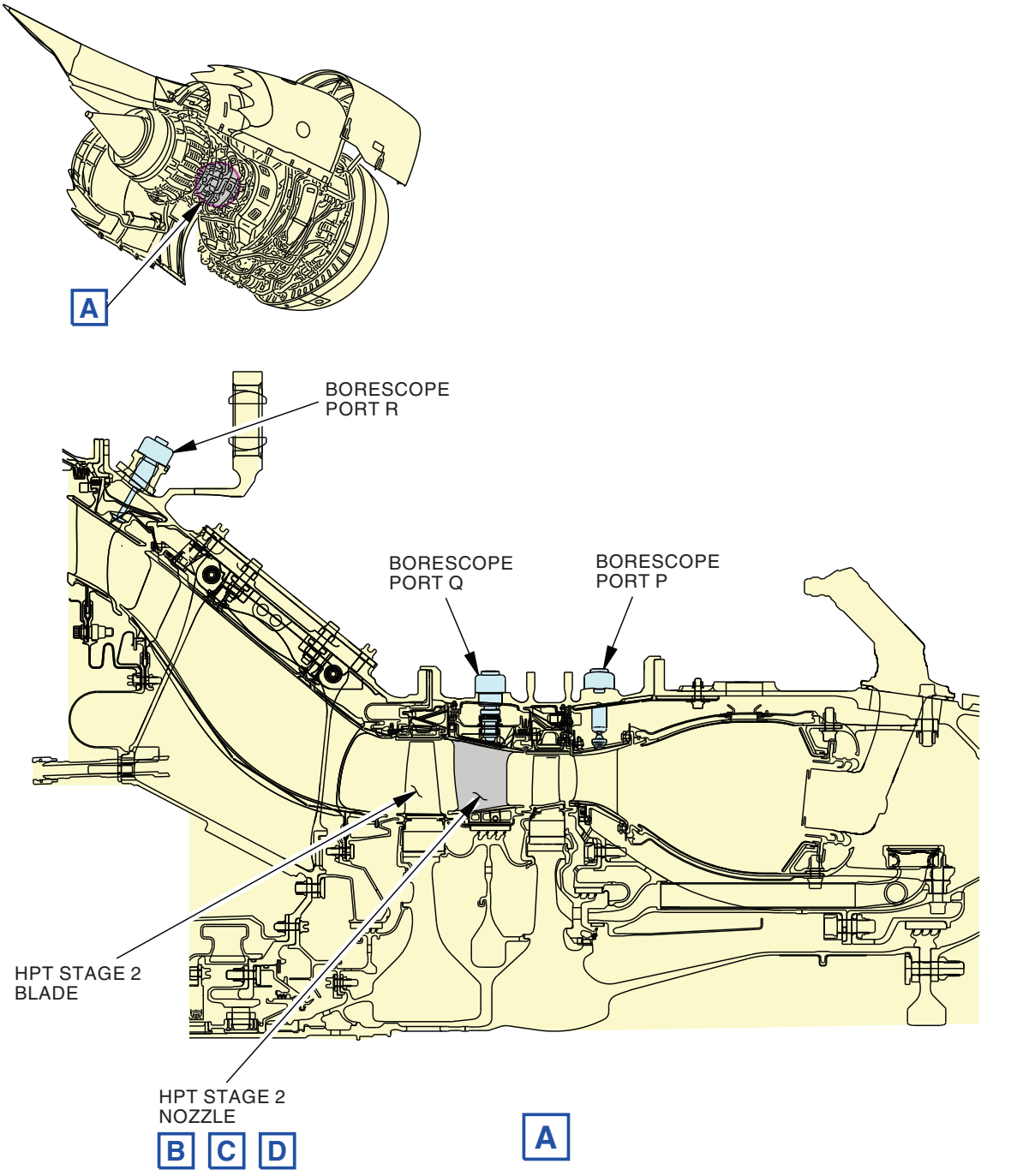
(6) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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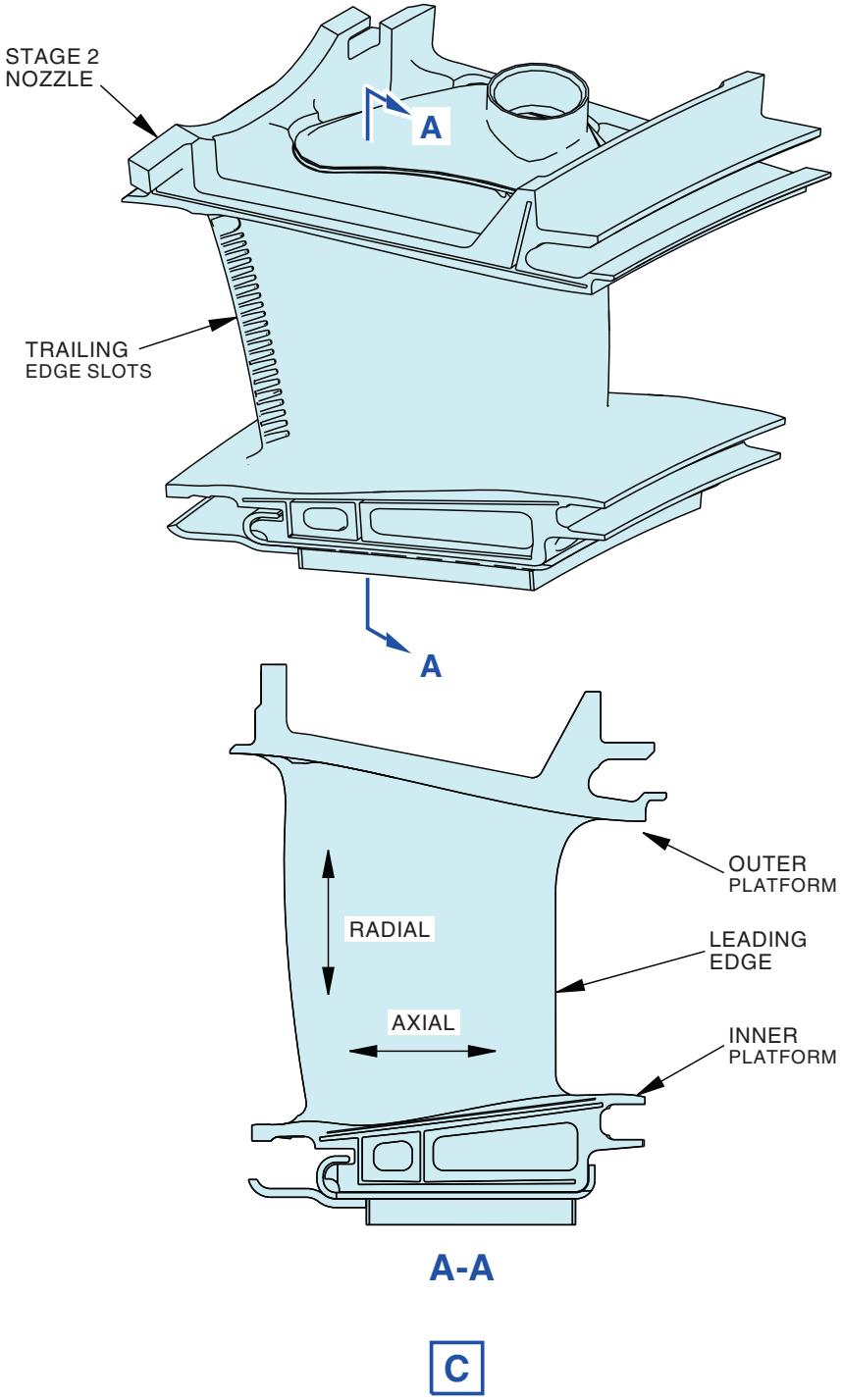
High Pressure Turbine (HPT) Stage 2 Nozzle Borescope Inspection
Figure 625/72-00-00-990-848-G00 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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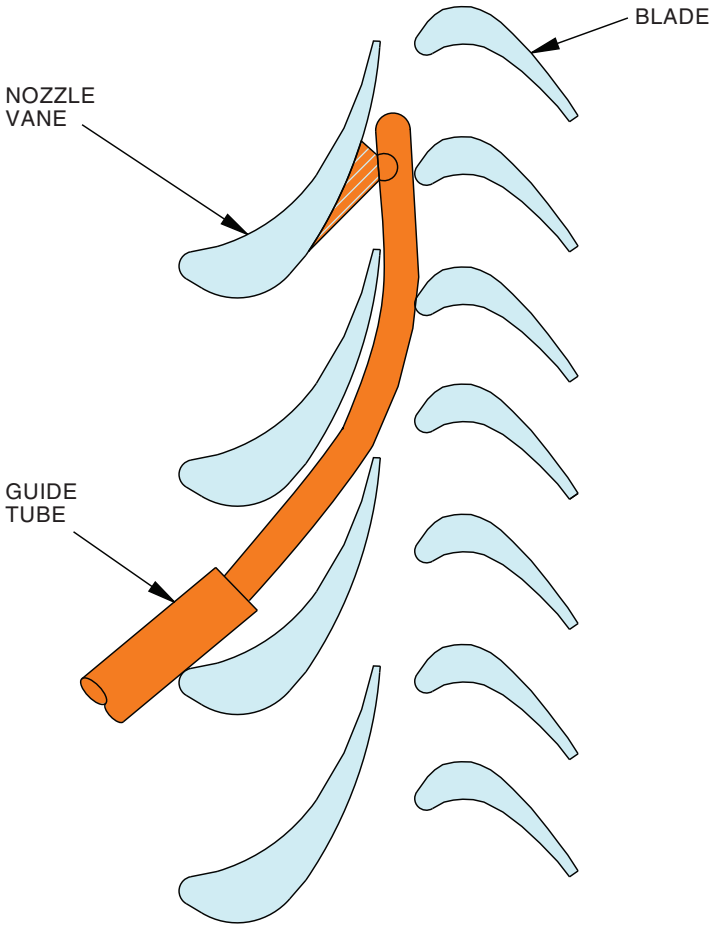
High Pressure Turbine (HPT) Stage 2 Nozzle Borescope Inspection
Figure 625/72-00-00-990-848-G00 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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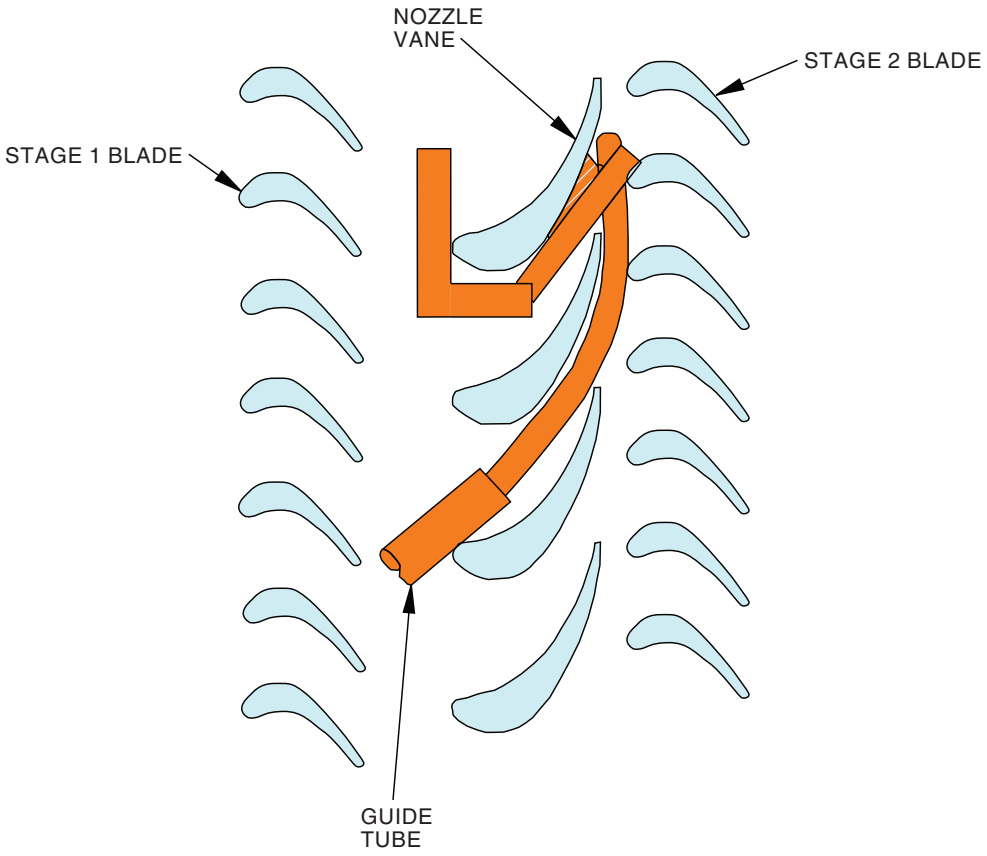
High Pressure Turbine (HPT) Stage 2 Nozzle Borescope Inspection
Figure 625/72-00-00-990-848-G00 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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High Pressure Turbine (HPT) Stage 2 Nozzle Borescope Inspection
Figure 625/72-00-00-990-848-G00 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 72-00-00-290-810-G00

11. High Pressure Turbine (HPT) Stage 2 Shroud Borescope Inspection

(Figure 626)

A. General

(1) This task provides the instructions for the borescope inspection of the High Pressure Turbine (HPT) stage 2 shroud.

(2) You will use a flexible borescope to do a 360 degree inspection of the stage 2 HPT shrouds.

(3) The stage 2 HPT shrouds are inspected using a rigid borescope (TASK 72-00-00-290-805-G00).

(4) There are 36 segments in the stage 2 HPT shroud.

B. References

Reference	Title
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-14907	Borescope Guide Tubes Part #: 956A1603G01 Supplier: 58828
STD-858	Tag - DO NOT OPERATE
STD-1059	Platform - Engine and Strut Access
STD-1432	Fiberscope - Flexible Borescope

D. Consumable Materials

Reference	Description	Specification
D50017 [C02-071]	Compound - Antiseize And Lubricating - Pure Nickel Special Grade	GE Spec D6Y28C1 or A50TF198
D50043 [C02-058]	Compound - Antiseize, Acheson GP460 (For Threaded Fasteners 0.250 Inches Diameter Or Larger, C02-079 Is An Alternative)	GE A50TF201 Class A

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1

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(Continued)

Number	Name/Location
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Prepare for the Inspection

SUBTASK 72-00-00-480-010-G00

WARNING

BE CAREFUL WHEN YOU WORK ON THE ENGINE COMPONENTS IMMEDIATELY AFTER THE ENGINE IS SHUTDOWN. THE ENGINE COMPONENTS CAN STAY HOT FOR UP TO ONE HOUR AND CAN CAUSE INJURY.

(1) Put the 3.0 ft (0.91 m) access platform, STD-1059, into position.

SUBTASK 72-00-00-010-027-G00

CAUTION

BEFORE YOU USE THE BORESCOPE, MAKE SURE THAT THE TEMPERATURES OF THE INSPECTION AREAS ARE BELOW 150°F (66°C). IF THE ADJACENT CASES ARE TOO HOT TO KEEP YOUR HAND ON, THE INSPECTION AREAS ARE TOO HOT. IF THE INSPECTION AREA IS TOO HOT, DAMAGE TO THE BORESCOPE CAN OCCUR.

(2) Remove the borescope plug from the HPT case borescope port Q.

NOTE: This borescope port gives access for the inspection of the stage 2 HPT shroud.

SUBTASK 72-00-00-930-001-G00

(3) Put an identification tag on the borescope plug.

SUBTASK 72-00-00-010-021-G00

(4) Do this task: Preparation for Borescope Inspection, TASK 72-00-00-290-801-G00.

H. High Pressure Turbine (HPT) Stage 2 Shroud Borescope Inspection

SUBTASK 72-00-00-290-008-G00

(1) Do a 360 degree borescope inspection of the stage 2 HPT shrouds as follows:

CAUTION

MAKE SURE THAT YOU DO NOT BEND THE FLEXIBLE BORESCOPE MORE THAN IS NECESSARY. IF YOU BEND THE BORESCOPE TOO MUCH, YOU WILL CAUSE DAMAGE.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(CAUTION PRECEDES)

CAUTION

MAKE SURE THE FLEXIBLE BORESCOPE DOES NOT CATCH THE INTERNAL DETAILS OF THE ENGINE. IF YOU DO NOT, DAMAGE TO THE BORESCOPE COULD OCCUR. ALSO, DAMAGE TO THE POWERPLANT COULD OCCUR IF THE BORESCOPE BECOMES CAUGHT OR BROKEN INSIDE THE ENGINE.

(a) If you will use a guide tubes, SPL-14907, put the guide tube into the HPT case borescope port Q as follows:

NOTE: A 5 mm (or less) flexible borescope is necessary for this inspection if you use guide tubes. A flexible borescope with a minimum length of 90.0 in. (2.29 m) is necessary to do the 360-degree borescope inspection of the stage 2 HPT shrouds.

NOTE: It is optional to use a 6 mm flexible borescope without the guide tubes.

1) Move the end of the guide tube into the hole in the stage 2 HPT nozzle and into the flowpath.

(b) Put the fiberscope, STD-1432, into the guide tubes, SPL-14907, and extend it into the stage 2 HPT flowpath.

(c) Continue to move the fiberscope, STD-1432, in the direction set by the stage 2 HPT nozzle vanes.

(d) Move the fiberscope, STD-1432, across the top of the engine and 360 around until you can see the flexible borescope.

NOTE: It is important that you can see the point where the flexible borescope is inserted through the stage 2 HPT nozzle outer platform. This is necessary to make sure that you do a full 360 degree inspection.

(e) Examine the stage 2 HPT shrouds using the serviceability limits in the HPT section detailed inspection task (TASK 72-00-00-290-806-G00).

(f) Examine for missing spline seals.

1) Up to 8 missing spline seals are permitted.

(g) Continue-In-Service (CIS) limit for the missing spline seals.

NOTE: This CIS limit is for a reduced interval repeat inspection.

1) More than 8 missing spline seals are permitted for a maximum of 500 cycles.

(h) Examine for missing material at the split line.

1) Up to 0.075 in. (1.90 mm) from the end face in the split line area is permitted.

(i) CIS limit for the missing split line material.

NOTE: This CIS limit is for a reduced interval repeat inspection.

1) Up to 0.110 in. (2.79 mm) from the end face in the split line area is permitted for a maximum of 500 cycles.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 72-00-00-080-014-G00

(1) When the inspection is complete, remove the fiberscope, STD-1432, from the engine.

SUBTASK 72-00-00-410-017-G00

(2) Install the borescope plug into port Q follows:

(a) If used, remove the guide tubes, SPL-14907, from the borescope port Q.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Remove the identification tag from the borescope plug.

(c) Put pure nickel special compound, D50017 [C02-071], on the borescope plug threads and the mating surfaces.

NOTE: You can use Acheson GP460 compound, D50043 [C02-058], as an alternative.

(d) Install the borescope plug into borescope port Q.

(e) Tighten the borescope plug to 85.0 in-lb (9.6 N·m) - 95.0 in-lb (10.7 N·m).

SUBTASK 72-00-00-080-015-G00

(3) Remove the access platform, STD-1059, from the engine area.

SUBTASK 72-00-00-410-018-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(4) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 72-00-00-440-003-G00

(5) Do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801.

SUBTASK 72-00-00-910-003-G00

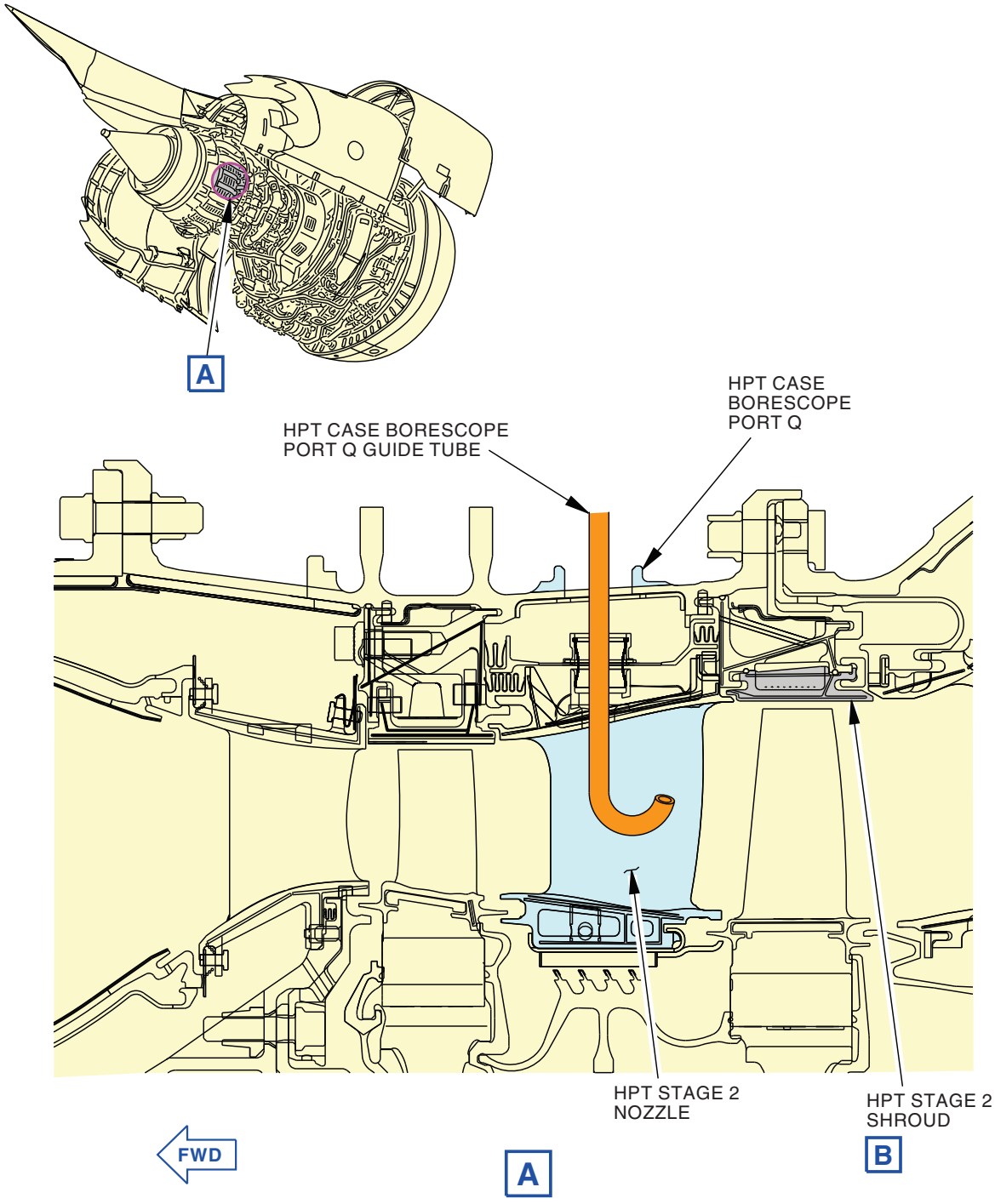
(6) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

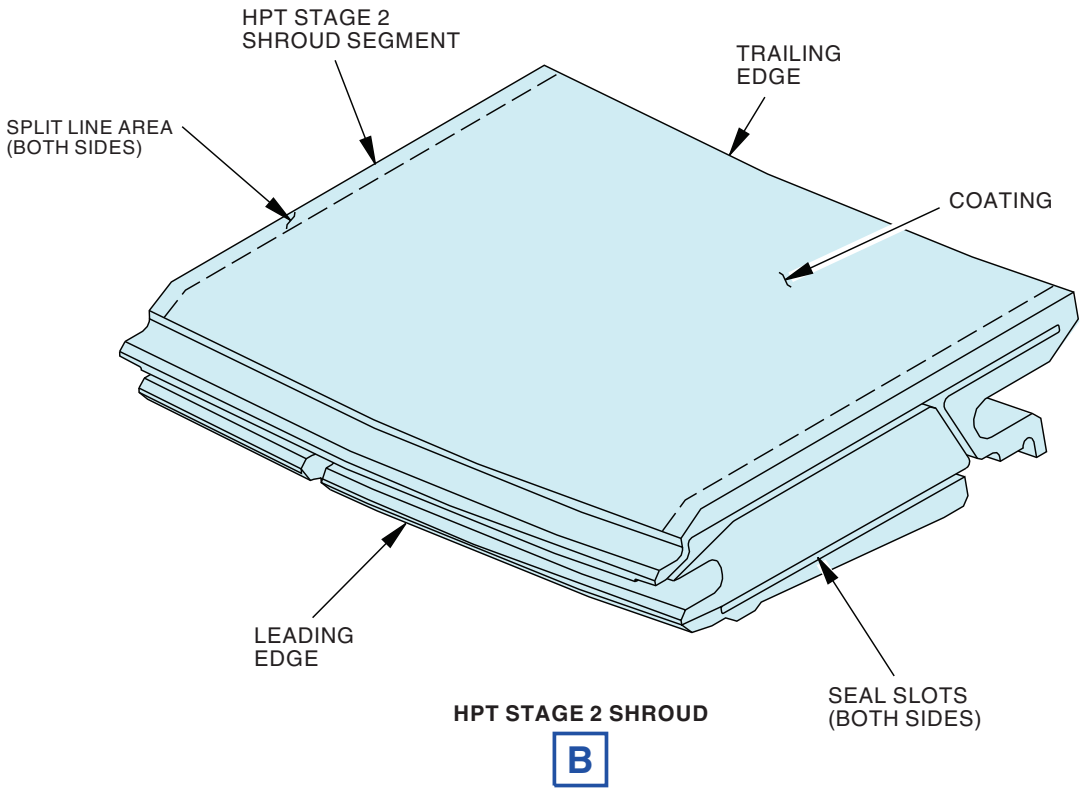
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



High Pressure Turbine (HPT) Stage 2 Shroud Borescope Inspection
Figure 626/72-00-00-990-850-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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High Pressure Turbine (HPT) Stage 2 Shroud Borescope Inspection
Figure 626/72-00-00-990-850-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 72-00-00-290-812-G00

12. High Pressure Turbine (HPT) Blade Tip Material Loss using Tip Notches Inspection

(Figure 627)

A. General

(1) This task gives the instructions to estimate the High Pressure Turbine (HPT) blade tip material loss using the blade tip notches.

NOTE: Notches are machined into 2 HPT stage 1 blade tips and 2 HPT stage 2 blade tips at assembly.

B. References

Reference	Title
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)
27-81-00-860-804	Leading Edge Flaps and Slats Retraction (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
72-00-00-822-801-G00	Zero-Index of the N2 Rotor Alignment (P/B 201)
72-00-00-910-801-G00	High Pressure Rotor (N2) Turning for Borescope (P/B 201)
72-63-04-000-801-G00	Accessory Gearbox Manual Control Cover Removal (P/B 401)
72-63-04-400-801-G00	Accessory Gearbox Manual Control Cover Installation (P/B 401)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-14866	Turning Tool - Electronic Part #: 9481M92P05 Supplier: 58828 Part #: 9481M92P06 Supplier: 58828 Opt Part #: 9481M92P03 Supplier: 58828
STD-858	Tag - DO NOT OPERATE
STD-10718	Fiberscope (Meets General Electric or CFM Standard Practices Manual Specification)
STD-10719	Borescope - Rigid (Meets General Electric or CFM Standard Practices Manual Specification such as: model R857ELSV60)

D. Consumable Materials

Reference	Description	Specification
D50017 [C02-071]	Compound - Antiseize And Lubricating - Pure Nickel Special Grade	GE Spec D6Y28C1 or A50TF198
D50043 [C02-058]	Compound - Antiseize, Acheson GP460 (For Threaded Fasteners 0.250 Inches Diameter Or Larger, C02-079 Is An Alternative)	GE A50TF201 Class A

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E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Prepare for the Inspection

SUBTASK 72-00-00-860-068-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-00-00-860-069-G00

(2) Do this task: Leading Edge Flaps and Slats Retraction, TASK 27-81-00-860-804.

SUBTASK 72-00-00-040-003-G00

(3) Do this task: Leading Edge Flaps and Slats - Deactivation, TASK 27-81-00-040-801.

SUBTASK 72-00-00-860-070-G00

(4) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

SUBTASK 72-00-00-040-004-G00

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(5) Do the thrust reverser deactivation for ground maintenance (TASK 78-31-00-040-801-G00).

SUBTASK 72-00-00-940-006-G00

CAUTION MAKE SURE THE TEMPERATURE OF THE ENGINE AND THE ADJACENT AREAS NEAR THE BORESCOPE INSPECTION PORTS ARE LESS THAN 150°F (65.6°C). DO NOT DO A BORESCOPE INSPECTION IF THE ADJACENT AREAS ARE TOO HOT TO TOUCH. DO THE BORESCOPE INSPECTION WHEN THE ENGINE TEMPERATURE IS LESS THAN 150°F (65.6°C). THIS WILL PREVENT DAMAGE TO THE BORESCOPE EQUIPMENT.

(6) Do this task: Preparation for Borescope Inspection, TASK 72-00-00-290-801-G00.

SUBTASK 72-00-00-010-032-G00

(7) Do this task: Accessory Gearbox Manual Control Cover Removal, TASK 72-63-04-000-801-G00.

SUBTASK 72-00-00-980-008-G00

(8) Attach a 3/4 inch square drive hand tool or the electronic turning tool, SPL-14866, to the AGB (Accessory Gearbox) manual control pad (TASK 72-00-00-910-801-G00).

SUBTASK 72-00-00-010-033-G00

(9) Remove the borescope plugs from borescope ports P, Q, and R.

(a) Put an identification tag on each borescope plug.

NOTE: Make sure that you attach an identification tag on each borescope plug to make sure that you re-install the plug in the correct borescope port after you complete the inspection.

SUBTASK 72-00-00-980-009-G00

(10) Do this task: Zero-Index of the N2 Rotor Alignment, TASK 72-00-00-822-801-G00.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 72-00-00-980-010-G00

(11) Manually or electrically motor the High Pressure Compressor (HPC) rotor (TASK 72-00-00-910-801-G00).

H. High Pressure Turbine (HPT) Blade Tip Material Loss using Tip Notches Inspection

SUBTASK 72-00-00-210-025-G00

(1) Identify the HPT stage 1 blade position 1 by the trailing edge corner chamfer.

NOTE: The HPT stage 1 and stage 2 blade tip notches are visible through borescope port Q.

SUBTASK 72-00-00-210-026-G00

(2) Identify the HPT stage 1 tip notched blades.

NOTE: There are 60 HPT stage 1 blades. Blade position 1 and blade position 31 are tip notched during assembly on the convex surface (Sheet 1).

SUBTASK 72-00-00-211-022-G00

(3) Use Table 1 to find the amount of material loss on the stage 1 blade position 1 and stage 1 blade position 31.

Table 605/72-00-00-993-814-G00 HPT STG1 Blade Tip Notch Condition

Number of visible tip notches	Condition
2	Material loss on the blade tip less than 0.010 in. (0.254 mm)
1	Material loss on the blade tip between 0.010 in. (0.254 mm) - 0.020 in. (0.508 mm)
0	Material loss on the blade tip greater than 0.020 in. (0.508 mm)

(a) 0.010 in. (0.254 mm) tip notch 0.39 in. (9.906 mm) from trailing edge at assembly

(b) 0.020 in. (0.508 mm) tip notch 0.54 in. (13.716 mm) from trailing edge at assembly

SUBTASK 72-00-00-210-027-G00

(4) Identify the HPT stage 2 tip notched blades.

NOTE: There are 68 HPT stage 2 blades. Blade position 1 and blade position 35 are tip notched during assembly on the concave surface (Sheet 2).

SUBTASK 72-00-00-211-023-G00

(5) Use Table 2 to find the amount of material loss on the stage 2 blade position 1 and stage 1 blade position 35

Table 606/72-00-00-993-815-G00 HPT STG2 Blade Tip Notch Condition

Number of visible tip notches	Condition
2	Material loss on the blade tip less than 0.010 in. (0.254 mm)
1	Material loss on the blade tip between 0.010 in. (0.254 mm) - 0.020 in. (0.508 mm)
0	Material loss on the blade tip greater than 0.020 in. (0.508 mm)

(a) 0.010 in. (0.254 mm) tip notch 0.80 in. (20.320 mm) from trailing edge at assembly

(b) 0.020 in. (0.508 mm) tip notch 0.94 in. (23.876 mm) from trailing edge at assembly

I. Put the Airplane Back to Its Usual Condition

SUBTASK 72-00-00-080-021-G00

(1) When the inspection is complete, do these steps:

(a) Remove the rigid borescope, STD-10719, or fiberscope, STD-10718, from the borescope port.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 72-00-00-410-027-G00

(2) Install the borescope plugs into the borescope ports P, Q, and R as follows:

(a) Do a visual inspection of the borescope plugs and the borescope ports (TASK 72-00-00-290-801-G00).

(b) Remove the identification tag from borescope plugs.

(c) Put the pure nickel special compound, D50017 [C02-071], on the borescope plugs threads and the mating surfaces.

NOTE: You can use the Acheson GP460 compound, D50043 [C02-058], as an alternative.

(d) Install the borescope plugs in the borescope ports.

(e) Make sure that installation running torque of the anti-rotation device on the borescope plugs is less than 8.8 in-lb (1.0 N·m).

(f) Tighten the borescope plugs P and Q to 85.0 in-lb (9.6 N·m) - 95.0 in-lb (10.7 N·m).

(g) Tighten the borescope plug R to 50.4 in-lb (5.7 N·m) - 55.6 in-lb (6.3 N·m).

SUBTASK 72-00-00-080-022-G00

(3) Remove the 3/4 inch square drive hand tool or the electronic turning tool, SPL-14866, from the AGB manual control pad (TASK 72-00-00-910-801-G00).

SUBTASK 72-00-00-860-071-G00

CAUTION

MAKE SURE THAT YOU INSTALL THE COVER ON THE GEARBOX. IF YOU DO NOT INSTALL THE COVER, ENGINE OIL LEAKS CAN OCCUR DURING OPERATION AND ENGINE DAMAGE CAN OCCUR.

(4) Do this task: Accessory Gearbox Manual Control Cover Installation, TASK 72-63-04-400-801-G00.

SUBTASK 72-00-00-860-072-G00

WARNING

OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(5) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

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(Continued)

Number	Name/Location
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 72-00-00-440-002-G00

(6) Do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801.

SUBTASK 72-00-00-860-073-G00

(7) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

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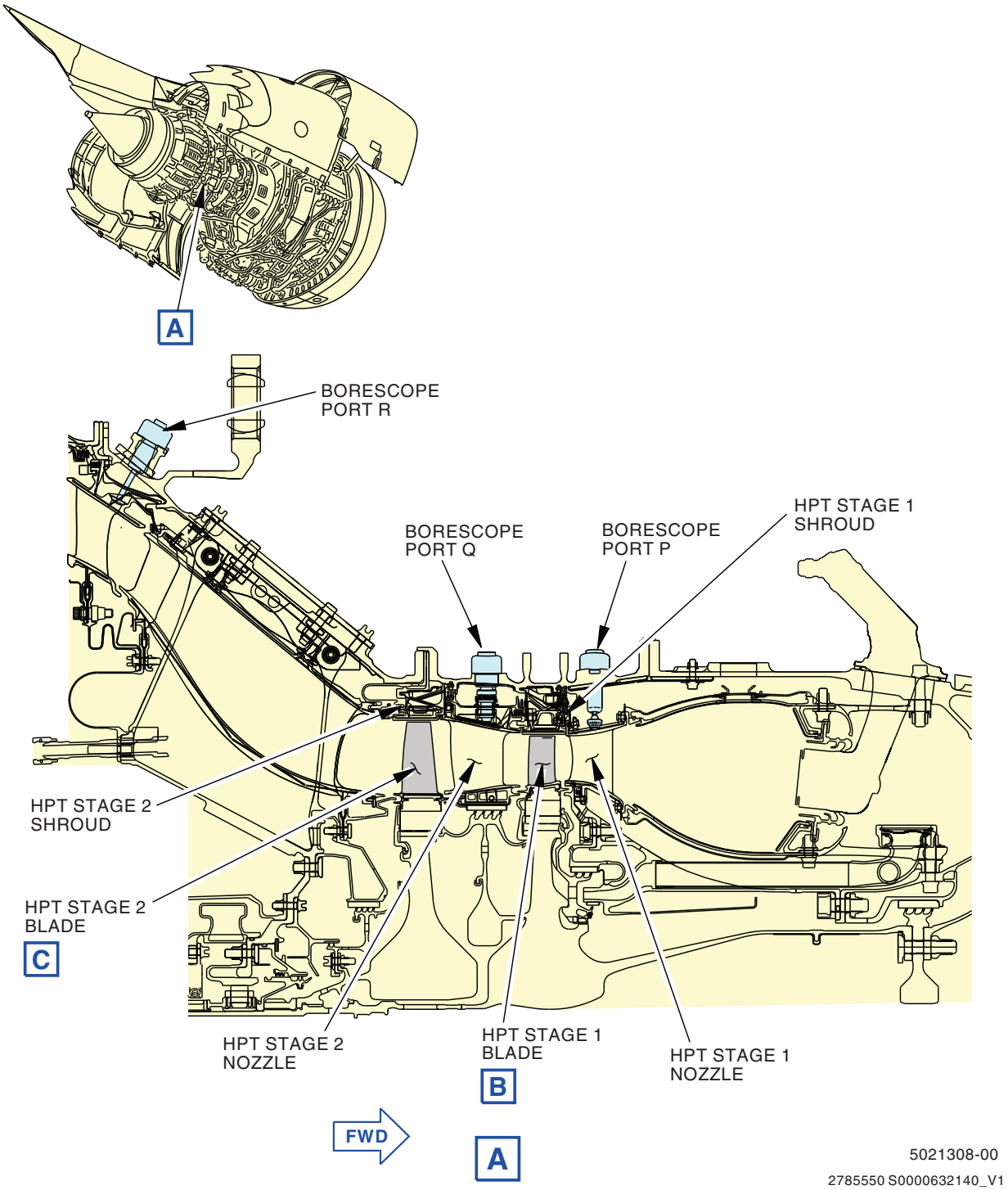
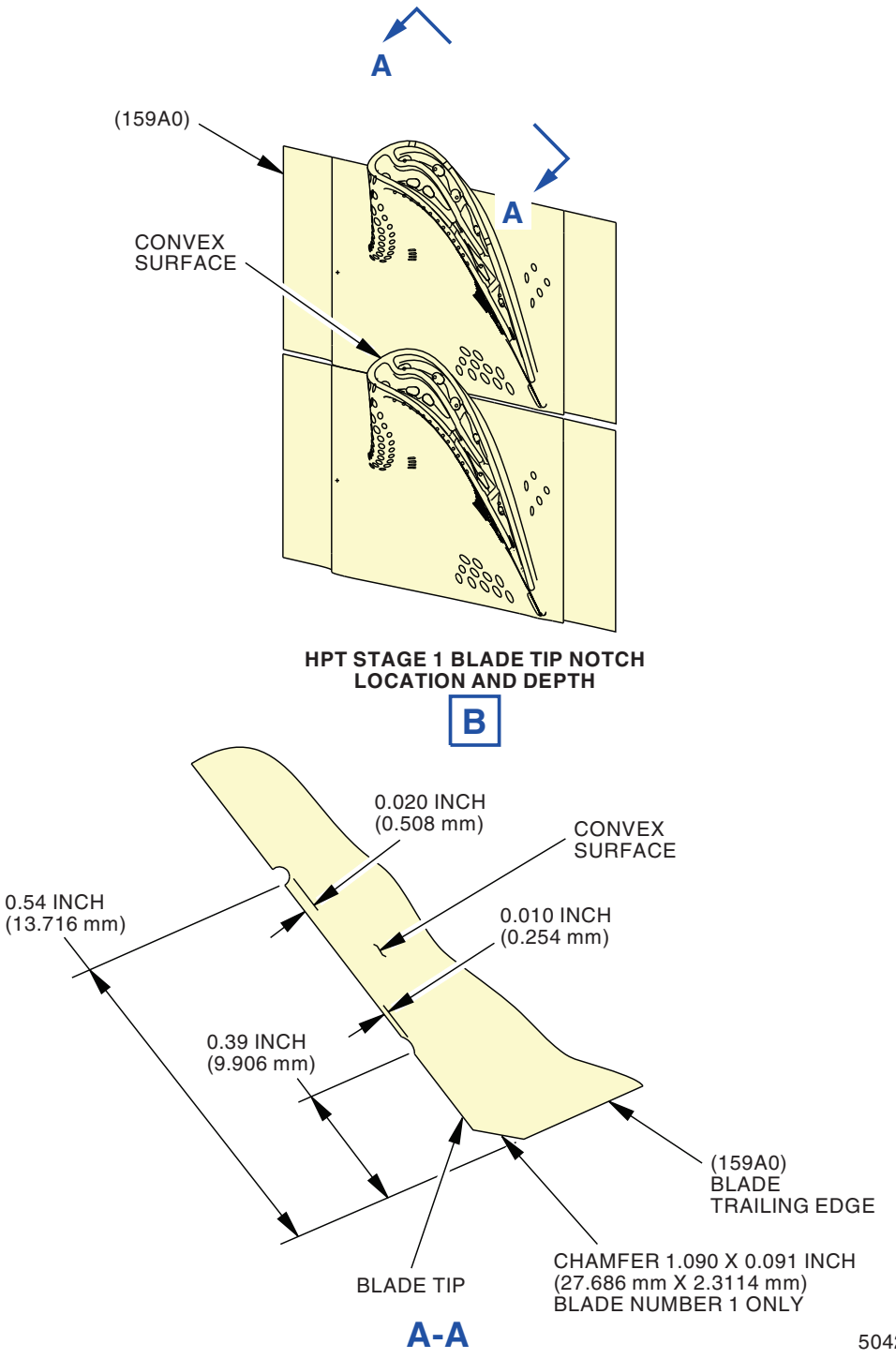


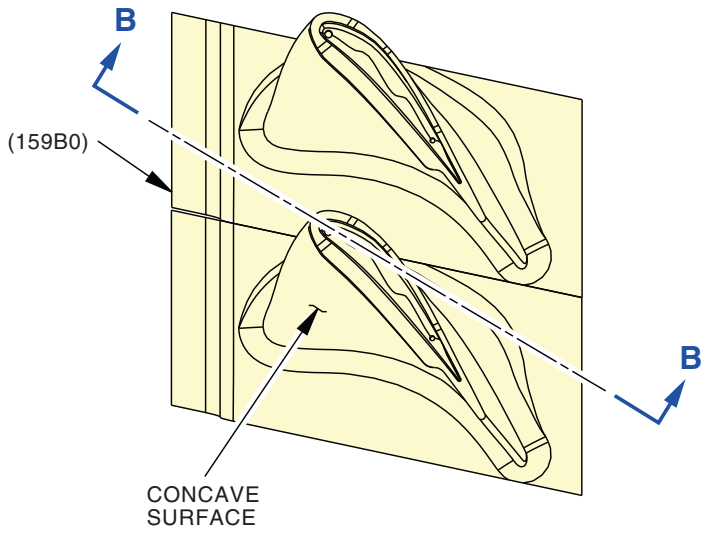
High Pressure Turbine (HPT) Blade Tip Material Loss using Tip Notches Inspection
Figure 627/72-00-00-990-859-G00 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



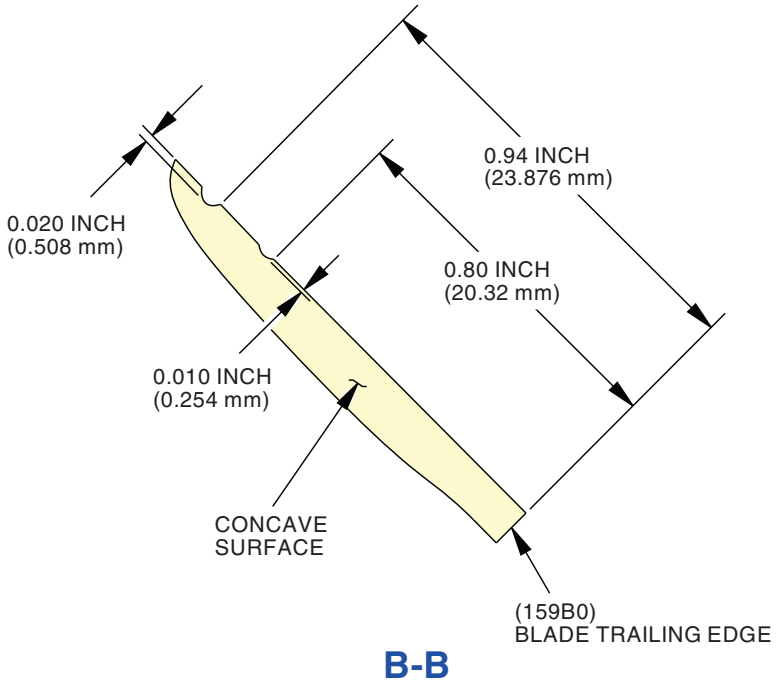
High Pressure Turbine (HPT) Blade Tip Material Loss using Tip Notches Inspection
Figure 627/72-00-00-990-859-G00 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



**HPT STAGE 2 BLADE TIP NOTCH
LOCATION AND DEPTH**

C



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**High Pressure Turbine (HPT) Blade Tip Material Loss using Tip Notches Inspection
Figure 627/72-00-00-990-859-G00 (Sheet 3 of 3)**

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TASK 72-00-00-290-811-G00

13. Low Pressure Turbine (LPT) Section Borescope Inspection

(Figure 628)

A. General

(1) This task gives the instructions to do the borescope inspection of the Low Pressure Turbine (LPT) section.

B. References

Reference	Title
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
72-00-00-350-801-G00	Low Pressure Turbine (LPT) Blades Boro - Blending Repair (P/B 801)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE
STD-1059	Platform - Engine and Strut Access
STD-10717	Light Source - Borescope, (Meets General Electric or CFM Standard Practices Manual Specification)
STD-10718	Fiberscope (Meets General Electric or CFM Standard Practices Manual Specification)
STD-10719	Borescope - Rigid (Meets General Electric or CFM Standard Practices Manual Specification such as: model R857ELSV60)

D. Consumable Materials

Reference	Description	Specification
D50068 [CP2544]	Lubricant - Molykote P37, Paste	

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2

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(Continued)

Number	Name/Location
426	Right Thrust Reverser, Engine 2

G. Prepare for the Inspection

SUBTASK 72-00-00-010-022-G00

(1) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

SUBTASK 72-00-00-480-018-G00

WARNING BE CAREFUL WHEN YOU WORK ON THE ENGINE COMPONENTS IMMEDIATELY AFTER THE ENGINE IS SHUTDOWN. THE ENGINE COMPONENTS CAN STAY HOT FOR UP TO ONE HOUR AND CAN CAUSE INJURY.

(2) Put the 3 ft (1 m) access platform, STD-1059, into position.

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SUBTASK 72-00-00-010-030-G00

CAUTION

MAKE SURE THE TEMPERATURE OF THE ENGINE AND THE ADJACENT AREAS NEAR THE BORESCOPE INSPECTION PORTS ARE LESS THAN 150°F (65.6°C). DO NOT DO A BORESCOPE INSPECTION IF THE ADJACENT AREAS ARE TOO HOT TO TOUCH. DO THE BORESCOPE INSPECTION WHEN THE ENGINE TEMPERATURE IS LESS THAN 150°F (65.6°C). THIS WILL PREVENT DAMAGE TO THE BORESCOPE EQUIPMENT.

(3) Remove the borescope plugs R, S, T, U, and V from the LPT section to examine the turbine stages 1 to 5.

SUBTASK 72-00-00-930-004-G00

(4) Put an identification tag on each borescope plug.

SUBTASK 72-00-00-941-005-G00

(5) Do this task: TASK 72-00-00-290-801-G00.

SUBTASK 72-00-00-480-011-G00

(6) Turn the light source, STD-10717, to on.

SUBTASK 72-00-00-480-012-G00

(7) Insert the rigid borescope, STD-10719, or fiberscope, STD-10718, into position for the inspection.

SUBTASK 72-00-00-980-006-G00

CAUTION

DO NOT USE THE LPT BLADES TO TURN THE N1 ROTOR. THE BLADES ARE BRITTLE AT ROOM TEMPERATURE, AND CAN EASILY BREAK. DAMAGE TO THE ENGINE CAN OCCUR.

(8) Manually turn the fan rotor in a Counter-Clock-Wise (CCW) direction (forward of the engine, looking aft) to turn the LPT rotor.

H. Low Pressure Turbine Section Borescope Inspection

SUBTASK 72-00-00-220-032-G00

(1) Examine the LPT stage 1 blades [2] as follows:

(a) Examine all areas for these conditions (Figure 628 (Sheet 1), Figure 628 (Sheet 3), Figure 628 (Sheet 2), Figure 628 (Sheet 9), Figure 628 (Sheet 10), and Figure 628 (Sheet 12)):

1) Peeling of particles

a) Permitted in the airfoil and platform flowpath if they are not bowed or melted.

NOTE: Do not confuse the peeling particles with metal deposits.

2) Metal deposits

a) Permitted to a maximum of 0.000129 in^3 (2.13 mm^3) in volume.

3) Casting marks

a) Permitted.

4) Corrosion

a) Permitted for 10,000 flight cycles if no missing material.

NOTE: This C-I-S limit is a repeat inspection.

5) Porosity

ECCN 9E991 BOEING PROPRIETARY - See title page for details

a) Permitted.

(b) Examine the airfoil leading edge for these conditions (Figure 628 (Sheet 2)):

1) Cracks

a) Not permitted.

2) Nicks and dents

a) Permitted to a maximum of 0.019 in. (0.5 mm) in depth.

3) Distortion

a) Not permitted.

(c) Examine the airfoil trailing edge for these conditions (Figure 628 (Sheet 2)):

1) Cracks

a) Not permitted.

2) Nicks and dents

a) Permitted to a maximum of 0.019 in. (0.5 mm) in depth.

3) Bent

a) Permitted to a maximum depth (D) of 0.00984 in. (0.25 mm) with L/D more than 20, for any number of blades with smooth and no sharp edges.

b) Permitted to a maximum depth (D) of 0.0149 in. (0.38 mm) with L/D more than 10, for no more than 5 blades per stage, one bent max per blade, with smooth, no sharp edges, no coating finding.

(d) Examine the concave or convex airfoil surfaces for these conditions (Figure 628 (Sheet 2)):

1) Cracks

a) Not permitted.

2) Nicks and dents

a) Permitted to a maximum of 0.0047 in. (0.12 mm) in depth in the Area E.

b) Permitted to a maximum of 0.0098 in. (0.25 mm) in depth in areas not in Area E.

(e) Examine the blade root platform flowpath for these conditions (Figure 628 (Sheet 2)):

1) Cracks

a) Not permitted.

2) Out of flush

a) Permitted to a maximum of 0.047 in. (1.2 mm) out of flush.

3) Gaps between two platforms

a) Permitted if there is no indication of wear.

4) Pitting, Corrosion

a) Permitted if no missing material.

(f) Examine the tip shroud platform flowpath for these conditions (Figure 628 (Sheet 2), Figure 628 (Sheet 9), Figure 628 (Sheet 10), Figure 628 (Sheet 11), Figure 628 (Sheet 12), Figure 628 (Sheet 14), Figure 628 (Sheet 32) and Figure 628 (Sheet 33)):

NOTE: You can see misalignment of the forward tip shrouds (Figure 628 (Sheet 13)).

1) Cracks

ECCN 9E991 BOEING PROPRIETARY - See title page for details

a) Not permitted.

2) Interlock gap

PTF ALL; PRE LEAP-1B SB 72-0293

a) Permitted if:

<1> No missing material detectable on hard face at the interlock loss.

<2> No discontinuity between hard face and base material.

b) Perform a 360 degree Borescope Inspection of the LPT stage 1 blades retainer ring [16].

<1> If there is an indication on the LPT stage 1 blades retainer ring [16], refer to the maximum serviceable limits of this part.

<2> If there is no indication on the LPT stage 1 blades retainer ring [16], examine again the tip shroud platform flowpath of the LPT stage 1 blades [2] in 1,600 cycles or less.

PTF ALL; POST LEAP-1B SB 72-0293

c) Permitted if:

<1> Gaps between two forward tip shrouds (leading edge side) are less than 0.1000 in. (2.54 mm).

<2> No more than three interlock gaps.

<3> Each interlock gap has a separation of a minimum of nine blades ± one blade.

<4> No missing material is detectable on hard face at the interlock loss.

<5> No discontinuity between hard face and base material.

PTF ALL

3) Gaps between tip shrouds

a) Permitted if there is interlock gap serviceable.

4) Shingled or unlatched

a) Not permitted.

5) Out of flush

a) Permitted to a maximum of 0.051 in. (1.3 mm) out of flush.

6) Nicks and dents on the platform flowpath

a) Permitted to a maximum of 0.0098 in. (0.25 mm) in depth.

7) Wear on seal teeth

a) Any amount if not associated with molten seal teeth.

8) Flaking on hardened seal teeth

a) Any amount.

9) Notch in corner radius (if any)

a) Any amount.

SUBTASK 72-00-00-220-033-G00

(2) Examine the LPT stage 2 blades [4] as follows:

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(a) Examine all areas for these conditions (Figure 628 (Sheet 1), Figure 628 (Sheet 3), Figure 628 (Sheet 4), Figure 628 (Sheet 9), Figure 628 (Sheet 10), and Figure 628 (Sheet 12)):

1) Peeling of particles

a) Permitted in the airfoil and platform flowpath if they are not bowed or melted.

NOTE: Do not confuse the peeling particles with metal deposits.

2) Metal deposits

a) Permitted to a maximum of 0.000129 in^3 (2.13 mm^3) in volume.

3) Casting marks

a) Permitted.

4) Corrosion

a) Permitted for 10,000 flight cycles if no missing material.

NOTE: This C-I-S limit is a repeat inspection.

5) Porosity

a) Permitted.

6) Temperature discoloration of the airfoil

a) Permitted.

(b) Examine the airfoil leading edge for these conditions (Figure 628 (Sheet 4)):

1) Cracks

a) Not permitted.

b) Maximum repairable limit: in the limits of the applicable repair.

c) Repair method: Low Pressure Turbine (LPT) Blades Boro - Blending Repair, TASK 72-00-00-350-801-G00.

2) Nicks and dents

a) Permitted to a maximum of 0.019 in. (0.5 mm) in depth.

b) Maximum repairable limit: in the limits of the applicable repair.

c) Repair method: Low Pressure Turbine (LPT) Blades Boro - Blending Repair, TASK 72-00-00-350-801-G00.

3) Distortion

a) Not permitted.

(c) Examine the airfoil trailing edge for these conditions (Figure 628 (Sheet 4)):

1) Cracks

a) Not permitted.

b) Maximum repairable limit: in the limits of the applicable repair.

c) Repair method: Low Pressure Turbine (LPT) Blades Boro - Blending Repair, TASK 72-00-00-350-801-G00.

2) Nicks and dents

a) Permitted to a maximum of 0.019 in. (0.5 mm) in depth.

b) Maximum repairable limit: in the limits of the applicable repair.

c) Repair method: Low Pressure Turbine (LPT) Blades Boro - Blending Repair, TASK 72-00-00-350-801-G00.

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3) Bent

a) Permitted to a maximum depth (D) of 0.0098 in. (0.25 mm) with L/D more than 20, for any number of blades with smooth and no sharp edges.

b) Permitted to a maximum depth (D) of 0.0149 in. (0.38 mm) with L/D more than 10, for no more than 5 blades per stage, one bent max per blade, with smooth, no sharp edges.

(d) Examine the concave or convex airfoil surfaces for these conditions (Figure 628 (Sheet 4)):

1) Cracks

a) Not permitted.

2) Nicks and dents

a) Permitted to a maximum of 0.0098 in. (0.25 mm) in depth.

(e) Examine the blade root platform flowpath for these conditions (Figure 628 (Sheet 4)):

1) Cracks

a) Not permitted.

2) Out of flush

a) Permitted to a maximum of 0.047 in. (1.2 mm) out of flush.

3) Gaps between two platforms

a) Permitted if there is no indication of wear.

4) Rub marks (Figure 628 (Sheet 5))

a) Any number up to 0.011 in. (0.3 mm) in depth, in the area W.

5) Nicks and dents on the platform flowpath

a) Permitted to a maximum of 0.0098 in. (0.25 mm) in depth.

(f) Examine the tip shroud platform flowpath for these conditions (Figure 628 (Sheet 4), Figure 628 (Sheet 9), Figure 628 (Sheet 10), Figure 628 (Sheet 12), Figure 628 (Sheet 14), and Figure 628 (Sheet 31)):

1) Cracks

a) Not permitted.

2) Interlock gap

a) Not permitted.

3) Gaps between tip shrouds

a) Permitted if there is interlock gap serviceable.

4) Shingled or unlatched

a) Not permitted.

5) Out of flush

a) Permitted to a maximum of 0.051 in. (1.3 mm) out of flush.

6) Nicks and dents on the platform flowpath

a) Permitted to a maximum of 0.0098 in. (0.25 mm) in depth.

7) Notch in corner radius (if any)

a) Permitted.

8) Wear on seal teeth

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a) Any amount if not associated with molten seal teeth.

9) Flaking on hardened seal teeth

a) Any amount.

SUBTASK 72-00-00-220-034-G00

(3) Examine the LPT stage 3 blades [6] as follows:

(a) Examine all areas for these conditions (Figure 628 (Sheet 1), Figure 628 (Sheet 3), Figure 628 (Sheet 4), Figure 628 (Sheet 9), Figure 628 (Sheet 10), and Figure 628 (Sheet 12)):

1) Peeling of particles

a) Permitted in the airfoil and platform flowpath if they are not bowed or melted.

NOTE: Do not confuse the peeling particles with metal deposits.

2) Metal deposits

a) Permitted to a maximum of 0.000129 in^3 (2.13 mm^3) in volume.

3) Casting marks

a) Permitted.

4) Corrosion

a) Permitted for 10,000 flight cycles if no missing material.

NOTE: This C-I-S limit if a repeat inspection.

5) Porosity

a) Permitted.

6) Temperature discoloration of the airfoil

a) Permitted.

(b) Examine the airfoil leading edge for these conditions (Figure 628 (Sheet 4)):

1) Cracks

a) Not permitted.

b) Maximum repairable limit: in the limits of the applicable repair.

c) Repair method: Low Pressure Turbine (LPT) Blades Boro - Blending Repair, TASK 72-00-00-350-801-G00.

2) Nicks and dents

a) Permitted to a maximum of 0.019 in. (0.5 mm) in depth.

b) Maximum repairable limit: in the limits of the applicable repair.

c) Repair method: Low Pressure Turbine (LPT) Blades Boro - Blending Repair, TASK 72-00-00-350-801-G00.

3) Distortion

a) Not permitted.

(c) Examine the airfoil trailing edge for these conditions (Figure 628 (Sheet 4)):

1) Cracks

a) Not permitted.

b) Maximum repairable limit: in the limits of the applicable repair.

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c) Repair method: Low Pressure Turbine (LPT) Blades Boro - Blending Repair, TASK 72-00-00-350-801-G00.

2) Nicks and dents

a) Permitted to a maximum of 0.019 in. (0.5 mm) in depth.

b) Maximum repairable limit: in the limits of the applicable repair.

c) Repair method: Low Pressure Turbine (LPT) Blades Boro - Blending Repair, TASK 72-00-00-350-801-G00.

3) Bent

a) Permitted to a maximum depth (D) of 0.0098 in. (0.25 mm) with L/D more than 20, for any number of blades with smooth and no sharp edges.

b) Permitted to a maximum depth (D) of 0.019 in. (0.5 mm) with L/D more than 10, for no more than 5 blades per stage, one bent max per blade, with smooth, no sharp edges.

(d) Examine the concave or convex airfoil surfaces for these conditions (Figure 628 (Sheet 4)):

1) Cracks

a) Not permitted.

2) Nicks and dents

a) Permitted to a maximum of 0.0098 in. (0.25 mm) in depth.

(e) Examine the blade root platform flowpath for these conditions (Figure 628 (Sheet 4)):

1) Cracks

a) Not permitted.

2) Out of flush

a) Permitted to a maximum of 0.047 in. (1.2 mm) out of flush.

3) Gaps between two platforms

a) Permitted if there is no indication of wear.

4) Nicks and dents on the platform flowpath

a) Permitted to a maximum of 0.0098 in. (0.25 mm) in depth.

5) Rub marks (Figure 628 (Sheet 6))

a) Any number up to 0.011 in. (0.3 mm) in depth, in the area W.

(f) Examine the tip shroud platform flowpath for these conditions (Figure 628 (Sheet 2), Figure 628 (Sheet 9), Figure 628 (Sheet 10), Figure 628 (Sheet 12), Figure 628 (Sheet 14), and Figure 628 (Sheet 31)):

NOTE: You can see misalignment of the forward tip shrouds (Figure 628 (Sheet 13)).

1) Cracks

a) Not permitted.

2) Interlock gap

a) Not permitted.

3) Gaps between two tip shrouds

a) Permitted if there is interlock gap serviceable.

4) Shingled or unlatched

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a) Not permitted.

5) Out of flush

a) Permitted to a maximum of 0.059 in. (1.5 mm) out of flush.

6) Nicks and dents on the platform flowpath

a) Permitted to a maximum of 0.0098 in. (0.25 mm) in depth.

7) Notch in corner radius (if any)

a) Permitted.

8) Wear on seal teeth

a) Any amount if not associated with molten seal teeth.

9) Flaking on hardened seal teeth

a) Any amount.

SUBTASK 72-00-00-220-035-G00

(4) Examine the LPT stage 4 blades [8] as follows:

(a) Examine all areas for these conditions (Figure 628 (Sheet 1), Figure 628 (Sheet 3), Figure 628 (Sheet 4), Figure 628 (Sheet 9), Figure 628 (Sheet 10), and Figure 628 (Sheet 12)):

1) Peeling of particles

a) Permitted in the airfoil and platform flowpath if they are not bowed or melted.

NOTE: Do not confuse the peeling particles with metal deposits.

2) Metal deposits

a) Permitted to a maximum of 0.000129 in^3 (2.13 mm^3) in volume.

3) Casting marks

a) Permitted.

4) Corrosion

a) Permitted for 10,000 flight cycles if no missing material.

NOTE: This C-I-S limit is a repeat inspection.

5) Porosity

a) Permitted.

6) Temperature discoloration of the airfoil

a) Permitted.

(b) Examine the airfoil leading edge for these conditions (Figure 628 (Sheet 4)):

1) Cracks

a) Not permitted.

b) Maximum repairable limit: in the limits of the applicable repair.

c) Repair method: Low Pressure Turbine (LPT) Blades Boro - Blending Repair, TASK 72-00-00-350-801-G00.

2) Nicks and dents

a) Permitted to a maximum of 0.019 in. (0.5 mm) in depth.

b) Maximum repairable limit: in the limits of the applicable repair.

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c) Repair method: Low Pressure Turbine (LPT) Blades Boro - Blending Repair, TASK 72-00-00-350-801-G00.

3) Distortion

a) Not permitted.

(c) Examine the airfoil trailing edge for these conditions (Figure 628 (Sheet 4)):

1) Cracks

a) Not permitted.

2) Nicks and dents

a) Permitted to a maximum of 0.019 in. (0.5 mm) in depth.

3) Bent

a) Permitted to a maximum depth (D) of 0.0098 in. (0.25 mm) with L/D more than 20, for any number of blades with smooth and no sharp edges.

b) Permitted to a maximum depth (D) of 0.019 in. (0.5 mm) with L/D more than 10, for no more than 5 blades per stage, one bent max per blade, with smooth, no sharp edges.

(d) Examine the concave or convex airfoil surfaces for these conditions (Figure 628 (Sheet 4)):

1) Cracks

a) Not permitted.

2) Nicks and dents

a) Permitted to a maximum of 0.0098 in. (0.25 mm) in depth.

(e) Examine the blade root platform flowpath for these conditions (Figure 628 (Sheet 4)):

1) Cracks

a) Not permitted.

2) Out of flush

a) Permitted to a maximum of 0.047 in. (1.2 mm) out of flush.

3) Gaps between two platforms

a) Permitted if there is no indication of wear.

4) Rub marks (Figure 628 (Sheet 8))

a) Any number up to 0.0019 in. (0.05 mm) in depth, in the areas W.

b) Any number up to 0.011 in. (0.3 mm) in depth, in the area S.

5) Nicks and dents on the platform flowpath

a) Permitted to a maximum of 0.0098 in. (0.25 mm) in depth.

(f) Examine the tip shroud platform flowpath for these conditions (Figure 628 (Sheet 4), Figure 628 (Sheet 9), Figure 628 (Sheet 10), Figure 628 (Sheet 12), Figure 628 (Sheet 14), and Figure 628 (Sheet 31)):

1) Cracks

a) Not permitted.

2) Interlock gap

a) Not permitted.

3) Gaps between two tip shrouds

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a) Permitted if there is interlock gap serviceable.

4) Shingled or unlatched

a) Not permitted.

5) Out of flush

a) Permitted to a maximum of 0.0610 in. (1.55 mm) out of flush.

6) Nicks and dents on the platform flowpath

a) Permitted to a maximum of 0.0098 in. (0.25 mm) in depth.

7) Lack of welded deposit of the tip shroud

a) Permitted if dimension A is more than 0.063 in. (1.6 mm).

8) Notch in corner radius (if any)

a) Permitted.

9) Wear on seal teeth

a) Any amount if not associated with molten seal teeth.

10) Flaking on hardened seal teeth

a) Any amount.

SUBTASK 72-00-00-220-036-G00

(5) Examine the LPT stage 5 blades [10] as follows:

(a) Examine all areas for these conditions (Figure 628 (Sheet 1), Figure 628 (Sheet 4), Figure 628 (Sheet 9), Figure 628 (Sheet 10), and Figure 628 (Sheet 12)):

1) Temperature discoloration of the airfoil

NOTE: Local discoloration can be seen on the airfoil.

a) Permitted.

2) Metal deposit

a) Permitted.

3) Casting marks

a) Permitted.

4) Pull outs

a) Permitted.

(b) Examine the airfoil leading edge for these conditions (Figure 628 (Sheet 4)):

1) Cracks

a) Not permitted.

b) Maximum repairable limit: in the limits of the applicable repair.

c) Repair method: Low Pressure Turbine (LPT) Blades Boro - Blending Repair, TASK 72-00-00-350-801-G00.

2) Nicks and dents

a) Permitted to a maximum of 0.0177 in. (0.45 mm) in depth.

b) Maximum repairable limit: in the limits of the applicable repair.

c) Repair method: Low Pressure Turbine (LPT) Blades Boro - Blending Repair, TASK 72-00-00-350-801-G00.

3) Corrosion pitting or sulfidation blistering

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a) Not permitted.

4) Distortion

a) Not permitted.

(c) Examine the airfoil trailing edge for these conditions (Figure 628 (Sheet 4)):

1) Cracks

a) Not permitted.

b) Maximum repairable limit: in the limits of the applicable repair.

c) Repair method: Low Pressure Turbine (LPT) Blades Boro - Blending Repair, TASK 72-00-00-350-801-G00.

2) Nicks and dents

a) Permitted to a maximum of 0.0177 in. (0.45 mm) in depth.

b) Maximum repairable limit: in the limits of the applicable repair.

c) Repair method: Low Pressure Turbine (LPT) Blades Boro - Blending Repair, TASK 72-00-00-350-801-G00.

3) Corrosion pitting or sulfidation blistering

a) Not permitted.

4) Distortion

a) Not permitted.

(d) Examine the concave or convex airfoil surfaces for these conditions (Figure 628 (Sheet 4)):

1) Cracks

a) Not permitted.

2) Nicks and dents

a) Permitted to a maximum of 0.0177 in. (0.45 mm) in depth.

3) Corrosion pitting or sulfidation blistering

a) Not permitted.

(e) Examine the blade root platform flowpath for these conditions (Figure 628 (Sheet 2)):

1) Cracks

a) Not permitted.

2) Out of flush

a) Permitted to a maximum of 0.031 in. (0.8 mm) out of flush.

3) Gaps between two platforms

a) Permitted if there is no indication of wear.

4) Nicks and dents on the platform flowpath

a) Permitted to a maximum of 0.0177 in. (0.45 mm) in depth.

(f) Examine the tip shroud platform flowpath for these conditions (Figure 628 (Sheet 4), Figure 628 (Sheet 9), Figure 628 (Sheet 10), and Figure 628 (Sheet 12)):

1) Cracks

a) Not permitted.

2) Interlock gap

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a) Not permitted.

3) Gaps between two tip shrouds

a) Permitted if there is interlock gap serviceable.

4) Shingled or unlatched

a) Not permitted.

5) Out of flush

a) Permitted to a maximum of 0.043 in. (1.1 mm) out of flush.

6) Nicks and dents on the platform flowpath

a) Permitted to a maximum of 0.0177 in. (0.45 mm) in depth.

7) Wear in interlocks

a) Permitted with these conditions:

<1> The misalignment of aft face on the tip shroud is less or equal than 0.039 in. (1.0 mm).

<2> The axial preload is serviceable.

8) Flaking on the hardened seal teeth

a) Any amount.

SUBTASK 72-00-00-220-038-G00

(6) Examine the LPT stage 1 nozzle segments [1].

NOTE: You can see misalignment of the aft upper platforms (Figure 628 (Sheet 18)).

(a) Examine all areas.

1) Castings marks

a) Any amount.

2) Presence of oxides

a) Not permitted.

(b) Examine all airfoils areas (Leading Edge, Trailing Edge, Concave and Convex surfaces and airfoil inlets).

1) Nicks, dents, scratches

a) Any number if not more than 0.0031 in. (0.08 mm) in depth.

SUBTASK 72-00-00-220-039-G00

(7) Examine the LPT stage 2 nozzle segments [3].

(a) Examine all areas for these conditions (Figure 628 (Sheet 16)):

1) Casting marks

a) Any amount.

2) Corrosion

a) Permitted if there is no missing material.

(b) Examine the airfoil leading edges (Area D) for these conditions (Figure 628 (Sheet 16)):

1) Discoloration and heat scaling

a) Permitted if there is no distortion and there is no missing metal.

2) Cracks

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a) Permitted up to 0.059 in. (1.5 mm) in length if the cumulative length with a crack in the trailing edge section is not more than 0.157 in. (4.0 mm) on 4 airfoils maximum per segment.

3) Indications of burning, missing metal, leading edge distortion

a) Not permitted.

(c) Examine the inner platform concave side hard face area and inner platform convex side hard face area (Figure 628 (Sheet 28) and Figure 628 (Sheet 29)):

1) Missing material

a) A maximum permitted area of missing metal on the inner platform concave side hard face is not more than 0.0124 in^2 (8 mm^2).

<1> Missing material must not cross the inner platform from internal to external surfaces.

b) A maximum permitted area of missing metal on the inner platform convex side hard face is not more than 0.0124 in^2 (8 mm^2).

<1> Missing material must not cross the inner platform from internal to external surfaces.

(d) Examine the airfoil trailing edges (Area E) for these conditions (Figure 628 (Sheet 16) and Figure 628 (Sheet 17)):

1) Discoloration and heat scaling

a) Permitted if there is no distortion and there is no missing metal.

2) Cracks

a) Permitted, all cracks up to 0.118 in. (3.0 mm) in length if the cumulative length with a crack in the leading edge section is not more than 0.157 in. (4.0 mm) on 4 airfoils maximum per segment.

3) Indications of burning or missing metal

a) Not permitted.

(e) Examine the flow path area (Area W) for this condition (Figure 628 (Sheet 28)):

1) Coating bubbles

a) Any amount.

SUBTASK 72-00-00-220-040-G00

(8) Examine the LPT stage 3 nozzle segments [5] for these conditions (Figure 628 (Sheet 16)):

(a) Burns in all areas

1) Not permitted.

(b) Cracks in the leading and trailing edges

1) Not permitted.

(c) Corrosion in all areas

1) Permitted if no missing material.

SUBTASK 72-00-00-220-041-G00

(9) Examine the LPT stage 4 nozzle segments [7] (Figure 628 (Sheet 16) and Figure 628 (Sheet 17)).

(a) Examine all areas for burns (Figure 628 (Sheet 16)).

1) Burns are not permitted.

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(b) Examine the trailing edges for local deformation (Figure 628 (Sheet 16) and Figure 628 (Sheet 17)).

1) Permitted in area F (airfoil 1 only), dimension P up to 0.0098 in. (0.25 mm) and dimension L more than 10 times dimension P.

2) Permitted in all other areas, dimension P up to 0.031 in. (0.8 mm) and dimension L more than 3 times dimension P.

(c) Corrosion in all areas

1) Permitted if no missing material.

SUBTASK 72-00-00-220-042-G00

(10) Examine the LPT stage 5 nozzle segments [9] for these conditions:

(a) Burns in all areas (Figure 628 (Sheet 16))

1) Not permitted.

(b) Cracks in the leading and trailing edge

1) Not permitted.

(c) Corrosion in all areas

1) Permitted if no missing material.

SUBTASK 72-00-00-220-043-G00

(11) Examine the LPT stage 1 shroud segments [11] honeycomb for distortion (Figure 628 (Sheet 19) and Figure 628 (Sheet 20)).

(a) Distortion is permitted if the honeycomb does not extend into the air stream.

(b) Wear cells are permitted with no limit.

(c) Impact marks, missing material on cells are permitted (Figure 628 (Sheet 27)).

1) If located in the teeth rubmarks area: Up to 6 neighbor cells and not more than 4 cells of the width per Shroud segment.

2) If located outside of the teeth rubmarks area: Any amount.

SUBTASK 72-00-00-220-044-G00

(12) Examine the LPT stage 2 shroud segments [12] honeycomb for distortion (Figure 628 (Sheet 19) and Figure 628 (Sheet 20)).

(a) Distortion is permitted if the honeycomb does not extend into the air stream.

(b) Wear cells are permitted with no limit.

(c) Impact marks, missing material on cells are permitted (Figure 628 (Sheet 27)).

1) If located in the teeth rubmarks area: Up to 6 neighbor cells and not more than 4 cells of the width per Shroud segment.

2) If located outside of the teeth rubmarks area: Any amount.

SUBTASK 72-00-00-210-020-G00

(13) Examine the LPT stage 3 shroud segments [13] honeycomb for distortion (Figure 628 (Sheet 19) and Figure 628 (Sheet 20)).

(a) Distortion is permitted if the honeycomb does not extend into the air stream.

(b) Wear cells are permitted with no limit.

(c) Impact marks, missing material on cells are permitted (Figure 628 (Sheet 27)).

1) If located in the teeth rubmarks area: Up to 6 neighbor cells and not more than 4 cells of the width per Shroud segment.

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2) If located outside of the teeth rubmarks area: Any amount.

SUBTASK 72-00-00-210-021-G00

(14) Examine the LPT stage 4 shroud segments [14] honeycomb for distortion (Figure 628 (Sheet 19) and Figure 628 (Sheet 20)).

(a) Distortion is permitted if the honeycomb does not extend into the air stream.

(b) Wear cells are permitted with no limit.

(c) Impact marks, missing material on cells are permitted (Figure 628 (Sheet 27)).

1) If located in the teeth rubmarks area: Up to 6 neighbor cells and not more than 4 cells of the width per Shroud segment.

2) If located outside of the teeth rubmarks area: Any amount.

SUBTASK 72-00-00-210-022-G00

(15) Examine the LPT stage 5 shroud segments [15] honeycomb for distortion (Figure 628 (Sheet 19) and Figure 628 (Sheet 20)).

(a) Distortion is permitted if the honeycomb does not extend into the air stream.

(b) Wear cells are permitted with no limit.

(c) Impact marks, missing material on cells are permitted (Figure 628 (Sheet 27)).

1) If located in the teeth rubmarks area: Up to 6 neighbor cells and not more than 4 cells of the width per Shroud segment.

2) If located outside of the teeth rubmarks area: Any amount.

SUBTASK 72-00-00-210-031-G00

(16) Examine the LPT Stage 2 to 5 shims for these conditions (Figure 628 (Sheet 30)):

(a) Visible

1) Permitted.

(b) Not visible

1) Permitted.

SUBTASK 72-00-00-210-024-G00

(17) Examine the LPT borescope plugs for these conditions (Figure 628 (Sheet 21)):

(a) Cracks

1) Not permitted.

(b) Threads torn, worn, bent, missing fully or not

1) Not permitted.

(c) Worn locking pin

1) Not permitted.

(d) Discoloration and heat scaling

1) Permitted if no functional issue.

(e) Self-locking spring with no click when you rotate the head of the borescope plug

1) Not permitted.

(f) Compression spring with no automatic return of stem on the highest position

1) Not permitted.

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SUBTASK 72-00-00-210-030-G00

(18) Examine the LPT stage 1 blades retainer ring [16] for these conditions (Figure 628 (Sheet 22), Figure 628 (Sheet 23), Figure 628 (Sheet 24), Figure 628 (Sheet 25), and Figure 628 (Sheet 26):

(a) Radial cracks

1) One crack maximum permitted.

2) Re-inspect LPT stage 1 blades retainer ring [16] within 500 cycles if the cumulated opening is less than 0.118 in. (3.0 mm).

NOTE: Cumulated is for crack and junction.

(b) Anti-rotation pin bent (only for PRE SB 72-0122):

1) Bent length less than 17% (0.078 in. (2.0 mm)):

a) Re-inspect LPT stage 1 blades retainer ring [16] within 500 cycles.

2) Bent length more than 17% (0.078 in. (2.0 mm)):

a) Re-inspect LPT stage 1 blades retainer ring [16] within 50 cycles.

(c) Anti-rotation pin missing (only for PRE SB 72-0122):

1) Do an inspection of the LPT flowpath (stage 1 - stage 5) per Low Pressure Turbine (LPT) Section Borescope Inspection, TASK 72-00-00-290-811-G00.

2) If the LPT flowpath is serviceable per Low Pressure Turbine (LPT) Section Borescope Inspection, TASK 72-00-00-290-811-G00, re-inspect within 1,600 cycles.

(d) LPT Stage 1 Blade Hook missing (only for PRE SB 72-0122):

1) One blade hook missing maximum permitted.

a) Do an inspection of the LPT flowpath (stage 1 - stage 5) per Low Pressure Turbine (LPT) Section Borescope Inspection, TASK 72-00-00-290-811-G00.

b) If the LPT flowpath is serviceable per Low Pressure Turbine (LPT) Section Borescope Inspection, TASK 72-00-00-290-811-G00, re-inspect within 1,600 cycles.

(e) Other indications:

1) Not permitted.

2) A continue-In-Service limit of 10 cycles is permitted before CFM disposition except if the cumulated opening of the retainer ring is greater than one blade slot, 0.6300 in. (16.00 mm).

PTF ALL

I. Put the Airplane Back to Its Usual Condition

SUBTASK 72-00-00-080-016-G00

(1) Turn the light source, STD-10717, to off.

NOTE: Let the light source blower stay on until the lamp and case become cool.

SUBTASK 72-00-00-080-017-G00

(2) Remove the rigid borescope, STD-10719, or fiberscope, STD-10718, from the LPT ports.

SUBTASK 72-00-00-410-019-G00

(3) Install the borescope plugs R, S, T, U, and V as follows:

(a) Remove the identification tag from the borescope plugs R, S, T, V, and U.

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(b) Put lubricant, D50068 [CP2544], on the borescope plug threads and the mating surfaces.

(c) Install the borescope plugs R, S, T, U, and V.

(d) Tighten the borescope plugs to 53 in-lb (6 N·m) (TASK 72-00-00-290-801-G00).

SUBTASK 72-00-00-080-020-G00

(4) Remove the access platform, STD-1059, from the engine.

SUBTASK 72-00-00-410-020-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(5) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 72-00-00-440-004-G00

(6) Do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801

SUBTASK 72-00-00-860-045-G00

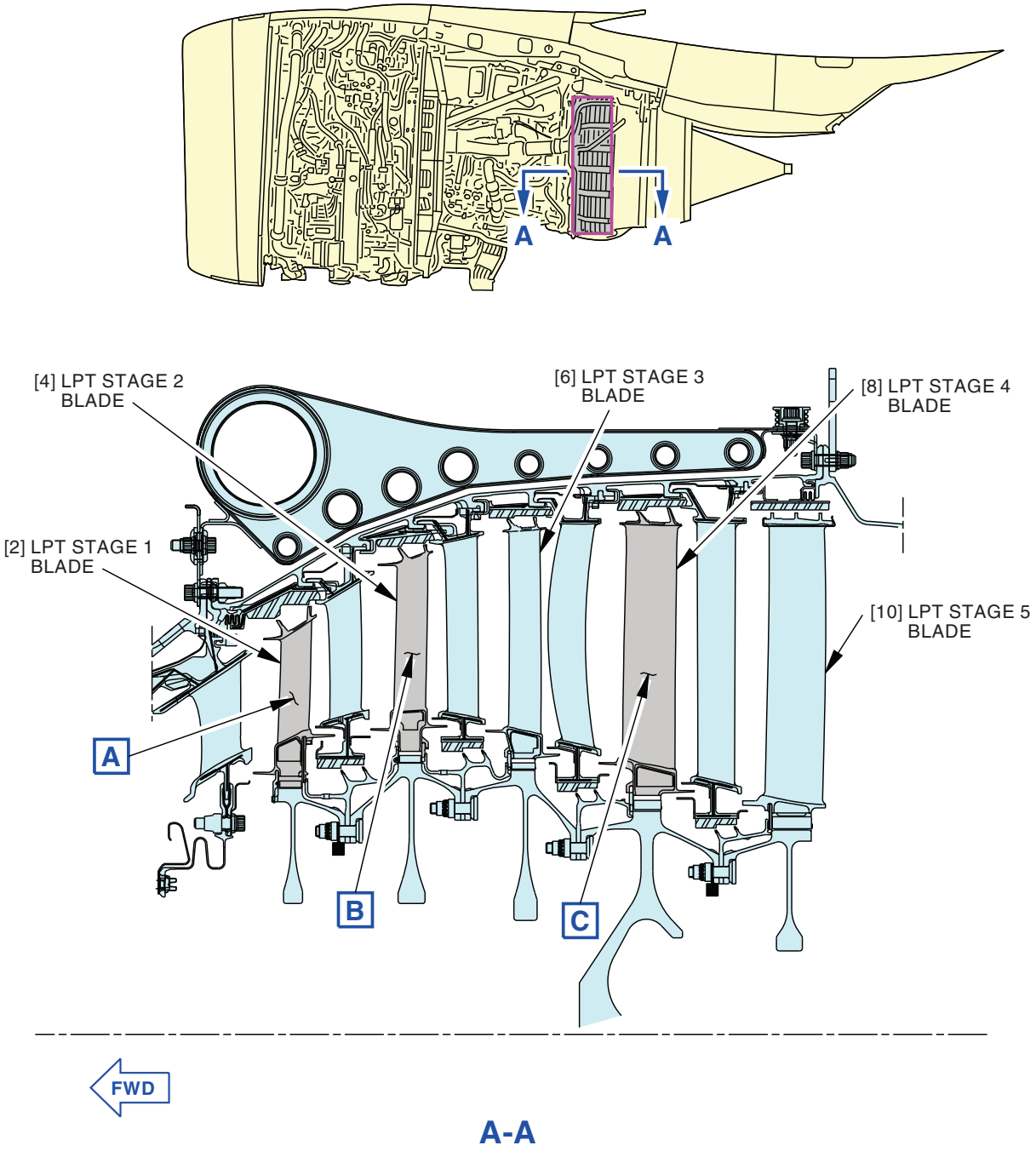
(7) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

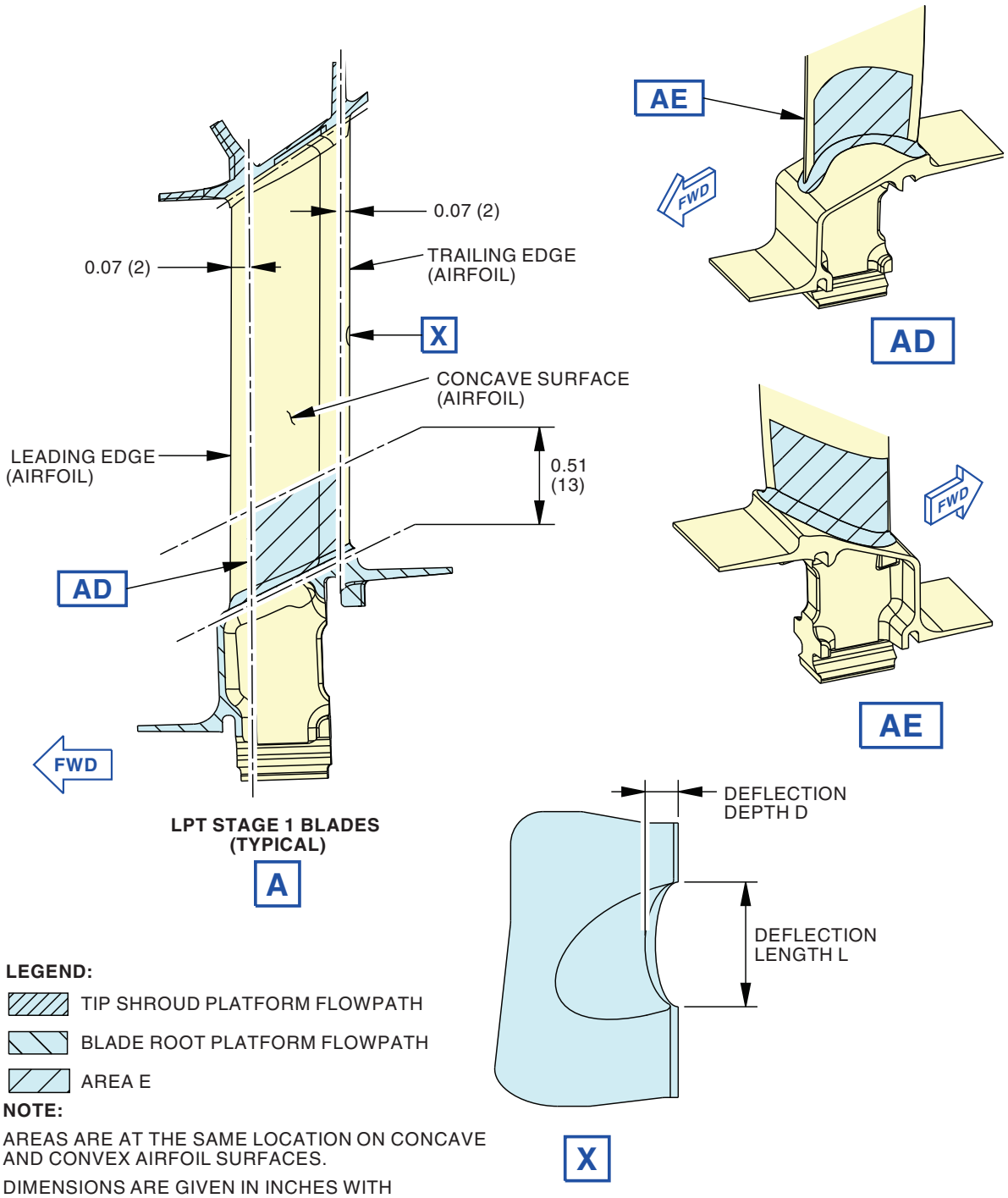


ICNLEAP1B720000D5882800001A00101

2539233 S0000602300_V2

Low Pressure Turbine (LPT) Section Borescope Inspection
Figure 628/72-00-00-990-858-G00 (Sheet 1 of 33)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



ICNLEAP1B720000D5882800002A00201

2539334 S0000602307_V4

Low Pressure Turbine (LPT) Section Borescope Inspection
Figure 628/72-00-00-990-858-G00 (Sheet 2 of 33)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

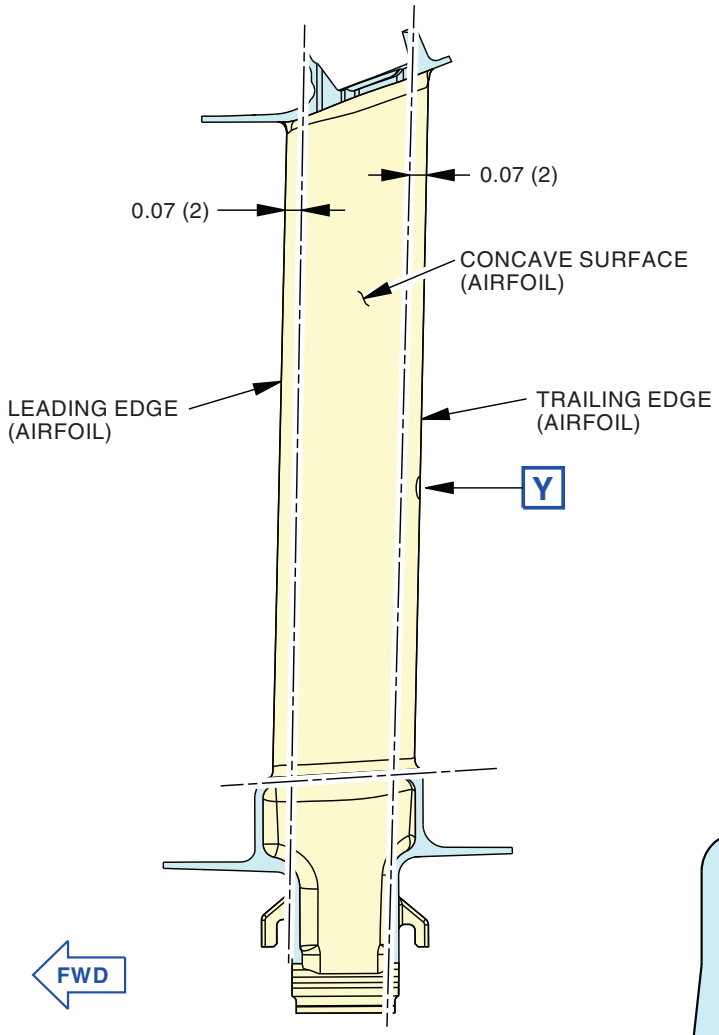
POROSITIES ARE MANUFACTURING-RELATED INDICATIONS THAT DO NOT HAVE SHARP CORNERS

ICNLEAP1B720000D5882800047A00101

3012743 S0000786048_V2

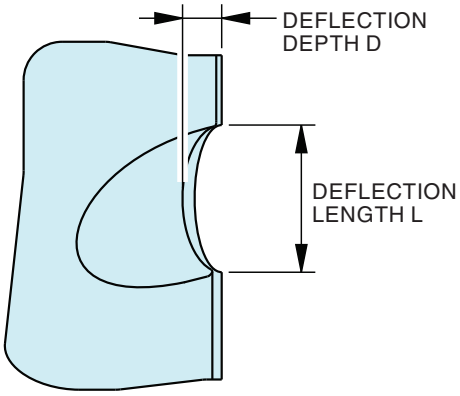
Low Pressure Turbine (LPT) Section Borescope Inspection
Figure 628/72-00-00-990-858-G00 (Sheet 3 of 33)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LPT STAGE 2 TO 5 BLADES (TYPICAL)

B



Y

LEGEND:

TIP SHROUD PLATFORM FLOWPATH

BLADE ROOT PLATFORM FLOWPATH

NOTE:

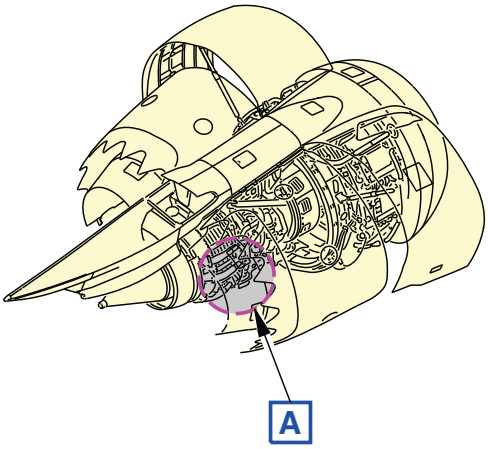
DIMENSIONS ARE GIVEN IN INCHES WITH MILLIMETERS IN PARENTHESES.

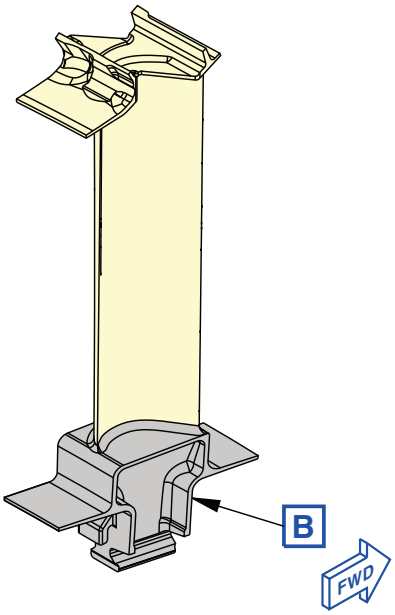
ICN-LEAP-1B-720000-D-58828-00003-A-002-01

2539367 S0000602308_V3

Low Pressure Turbine (LPT) Section Borescope Inspection
Figure 628/72-00-00-990-858-G00 (Sheet 4 of 33)

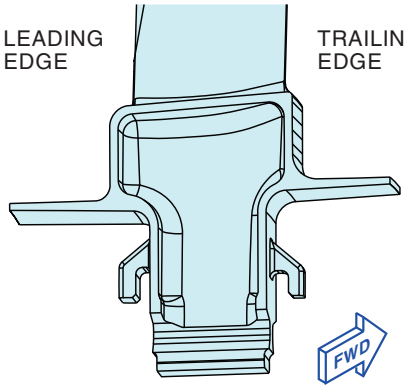
ECCN 9E991 BOEING PROPRIETARY - See title page for details





TYPICAL VIEW FOR LPT STAGE 2 BLADES

A



B

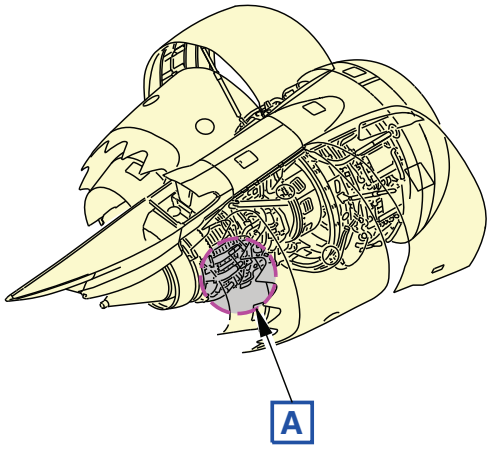
LEGEND:

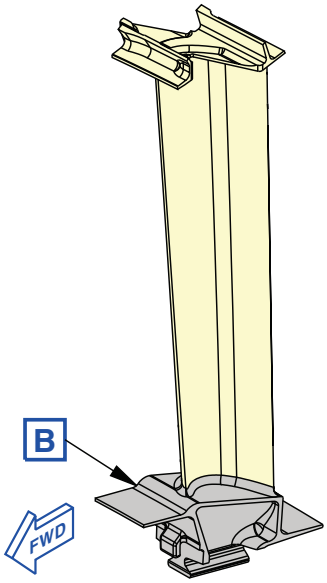
AREA **W**

ICNLEAP1B704351D5882800014A00101
2966255 S0000745245_V1

Low Pressure Turbine (LPT) Section Borescope Inspection
Figure 628/72-00-00-990-858-G00 (Sheet 5 of 33)

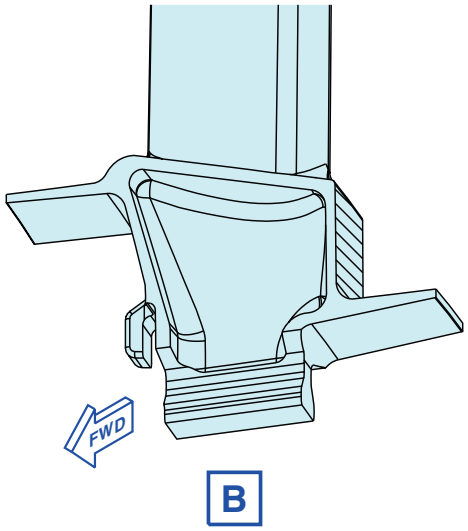
ECCN 9E991 BOEING PROPRIETARY - See title page for details





TYPICAL VIEW FOR LPT STAGE 3 BLADES

A



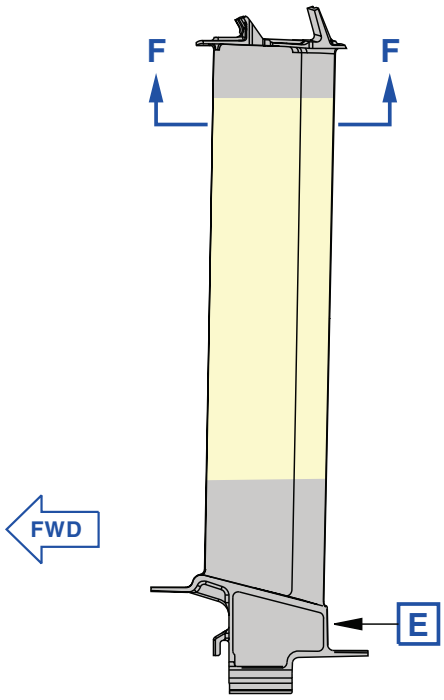
LEGEND:

AREA **W**

ICNLEAP1B704351D5882800015A00101
2966256 S0000745252_V1

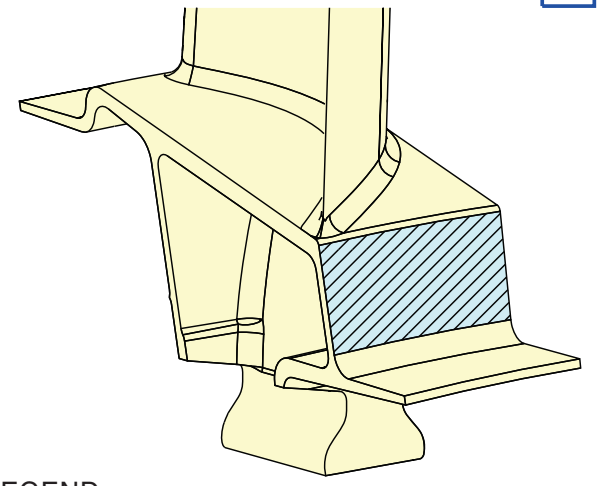
Low Pressure Turbine (LPT) Section Borescope Inspection
Figure 628/72-00-00-990-858-G00 (Sheet 6 of 33)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



TYPICAL VIEW OF LPT STAGE 4 BLADES

D

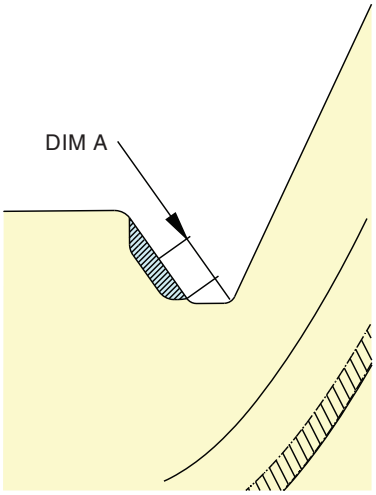


LEGEND:

AREA S

STELLITE DEPOSIT

E

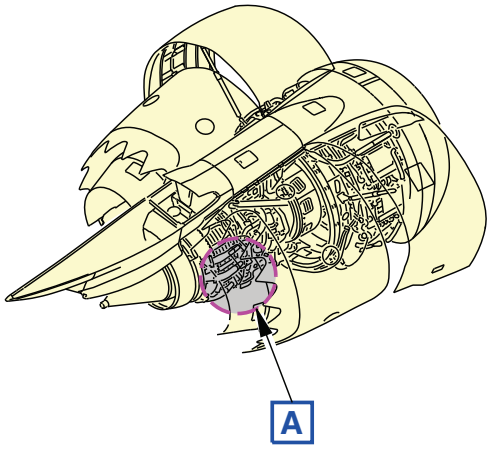


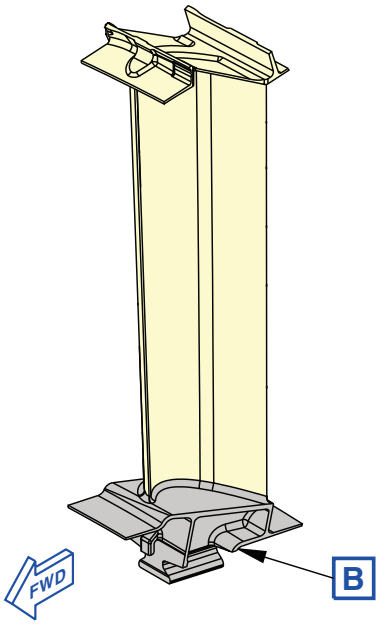
F-F

ICNLEAP1B720000D5882800012A00101
2563386 S0000612465_V2

Low Pressure Turbine (LPT) Section Borescope Inspection
Figure 628/72-00-00-990-858-G00 (Sheet 7 of 33)

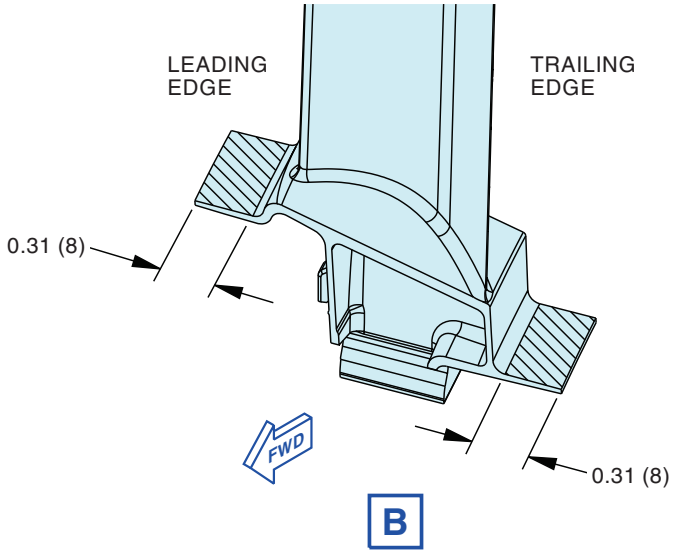
ECCN 9E991 BOEING PROPRIETARY - See title page for details





TYPICAL VIEW FOR LPT STAGE 4 BLADES

A



B

LEGEND:

AREA W

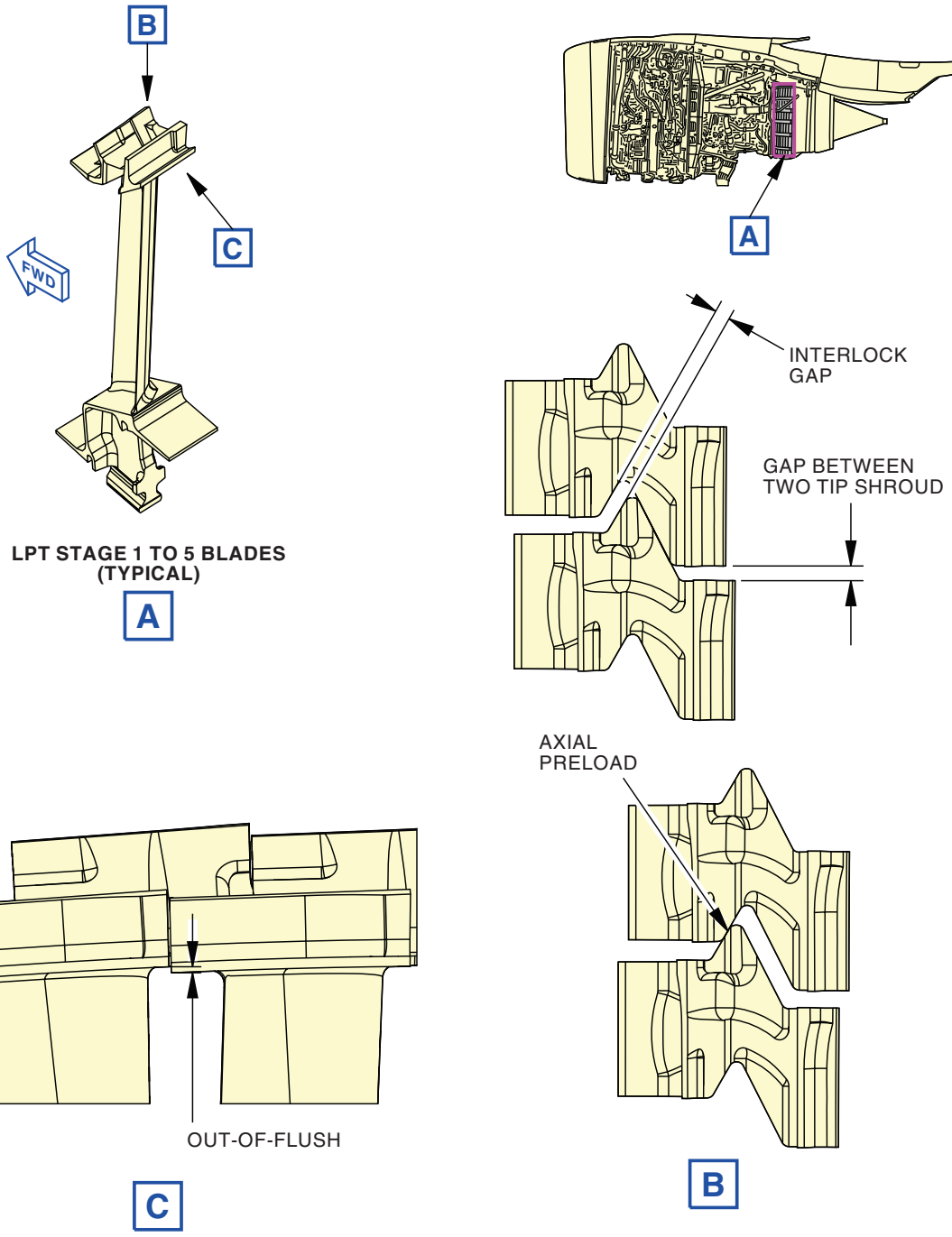
NOTE:

ALL DIMENSIONS ARE IN INCHES (mm).

ICNLEAP1B704351D5882800016A00101
2966257 S0000745253_V1

Low Pressure Turbine (LPT) Section Borescope Inspection
Figure 628/72-00-00-990-858-G00 (Sheet 8 of 33)

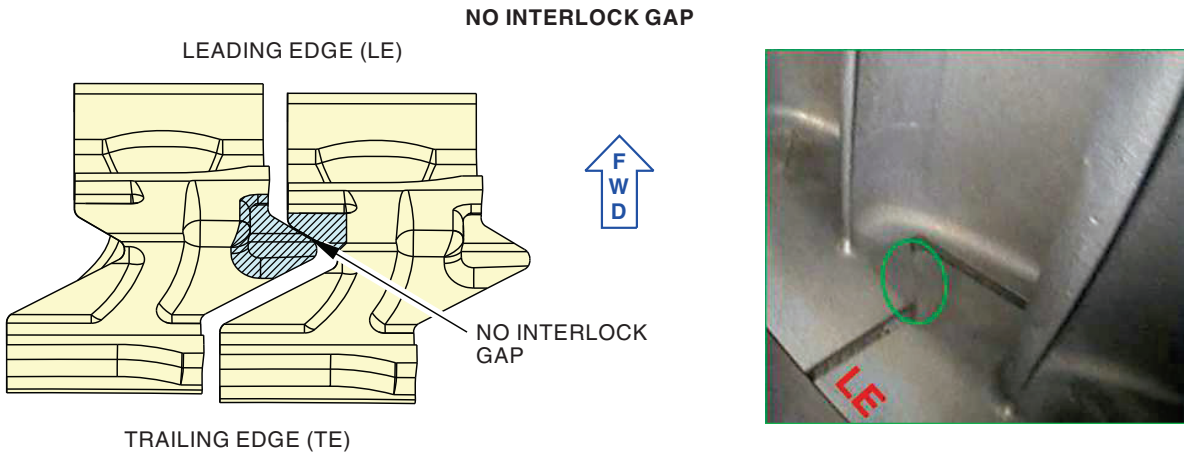
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2539369 S0000602309_V3

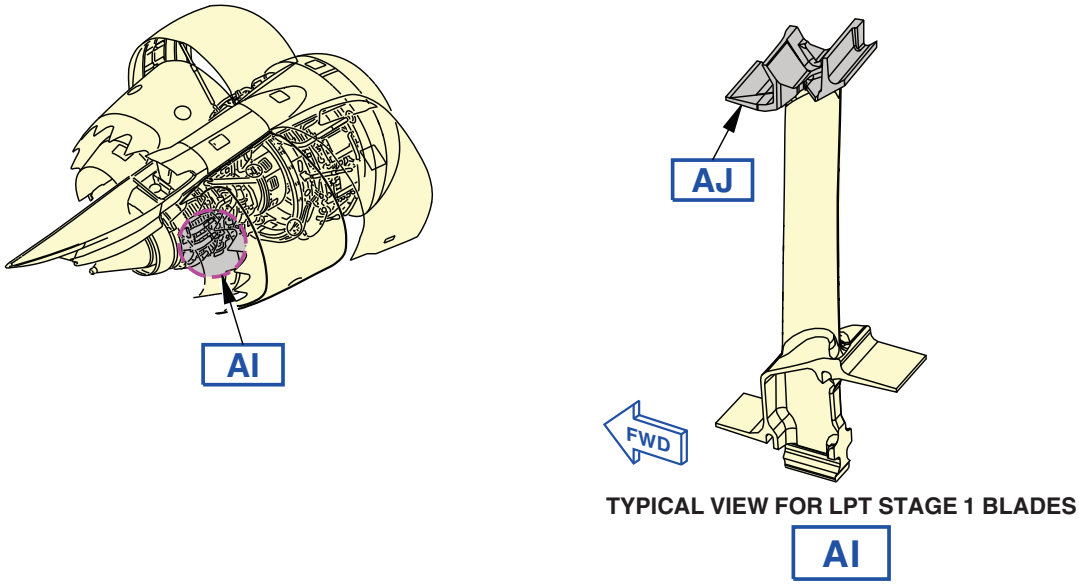
Low Pressure Turbine (LPT) Section Borescope Inspection
Figure 628/72-00-00-990-858-G00 (Sheet 9 of 33)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

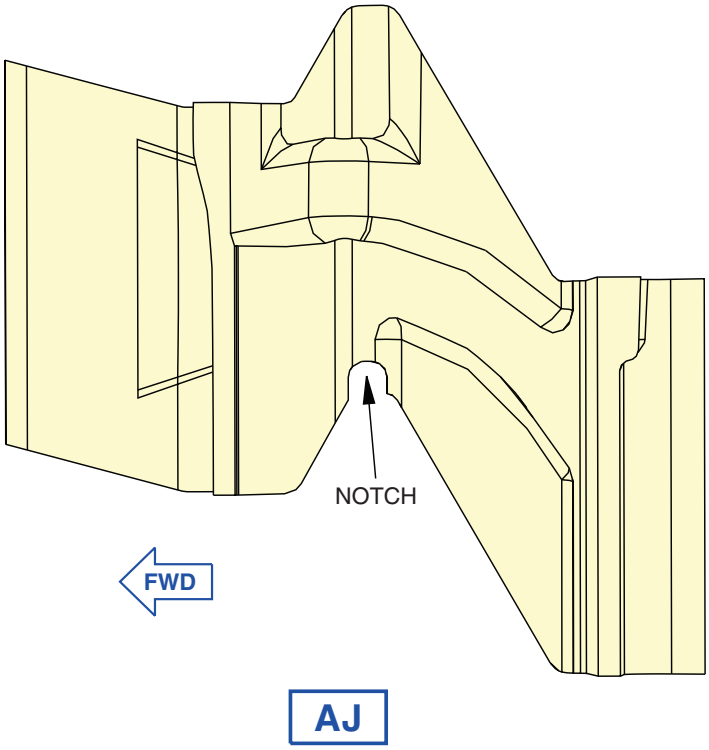


Low Pressure Turbine (LPT) Section Borescope Inspection
Figure 628/72-00-00-990-858-G00 (Sheet 10 of 33)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



TYPICAL VIEW FOR LPT STAGE 1 BLADES



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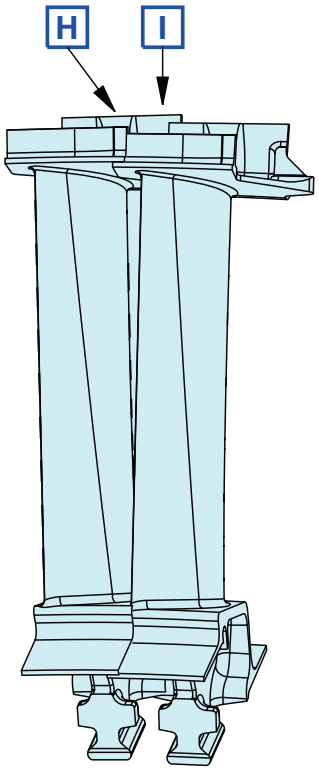
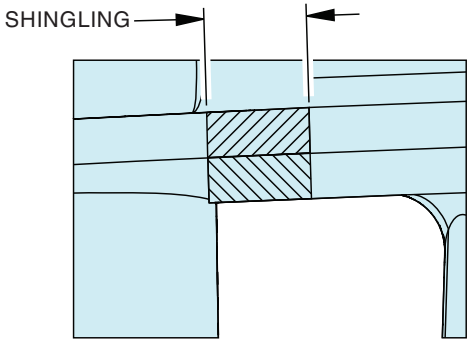
3083035 S0000838266_V1

Low Pressure Turbine (LPT) Section Borescope Inspection
Figure 628/72-00-00-990-858-G00 (Sheet 11 of 33)

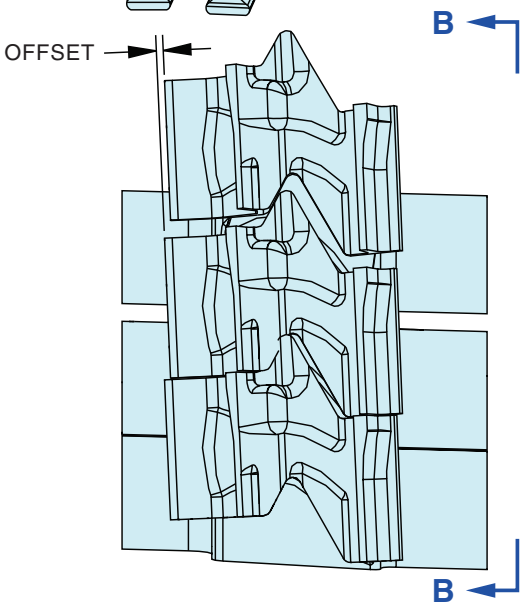
ECCN 9E991 BOEING PROPRIETARY - See title page for details





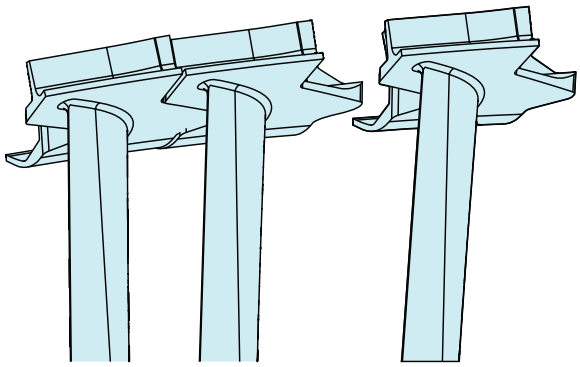
(VIEW IN THE DOWN DIRECTION)
(SHINGLED)

H



(VIEW IN THE DOWN DIRECTION)

I



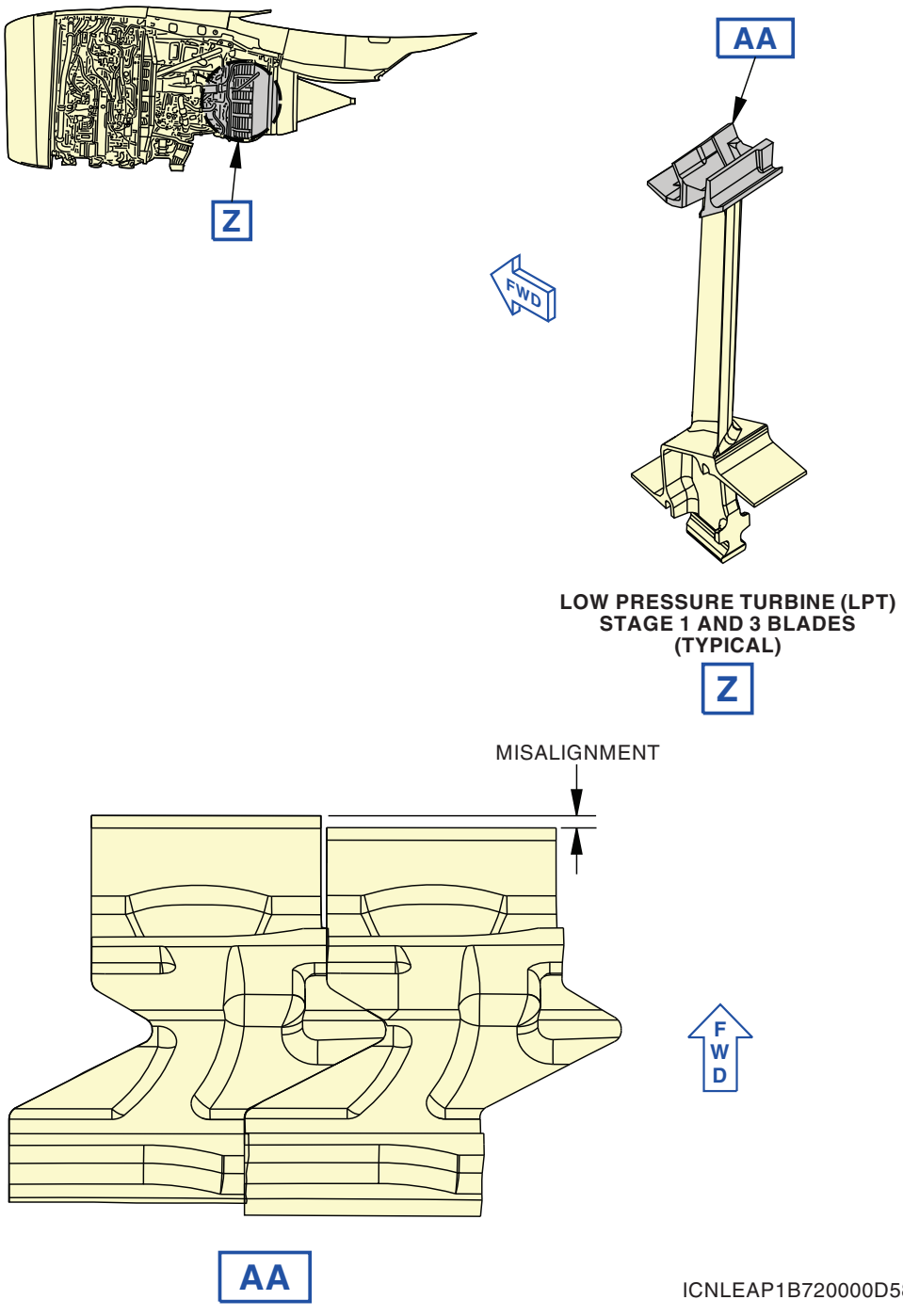
(VIEW ROTATED 90 DEGREES CW)
(UNLATCHED)

B-B

ICNLEAP1B720000D5882800005A00101
2539375 S0000602310_V3

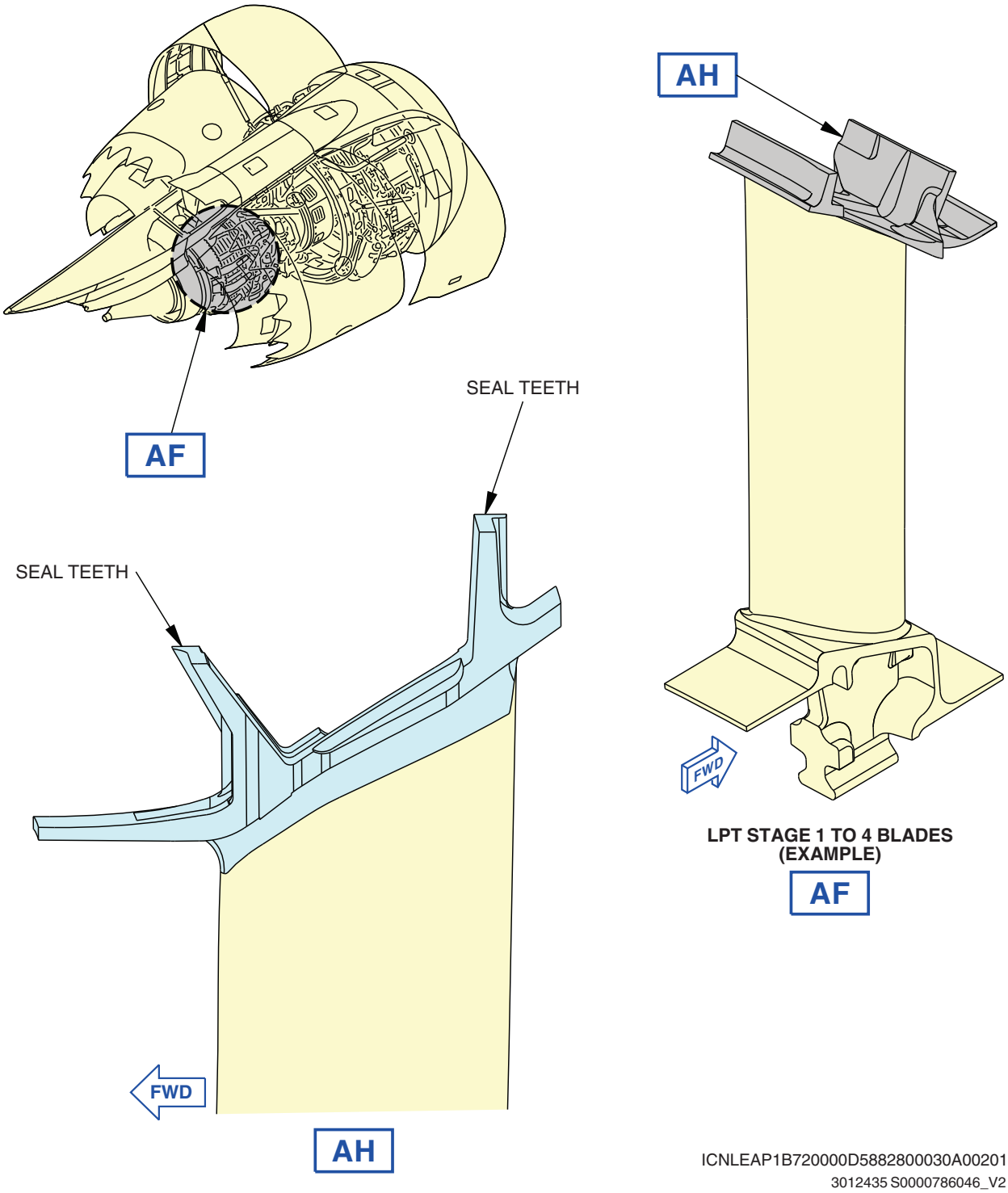
Low Pressure Turbine (LPT) Section Borescope Inspection
Figure 628/72-00-00-990-858-G00 (Sheet 12 of 33)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Low Pressure Turbine (LPT) Section Borescope Inspection
Figure 628/72-00-00-990-858-G00 (Sheet 13 of 33)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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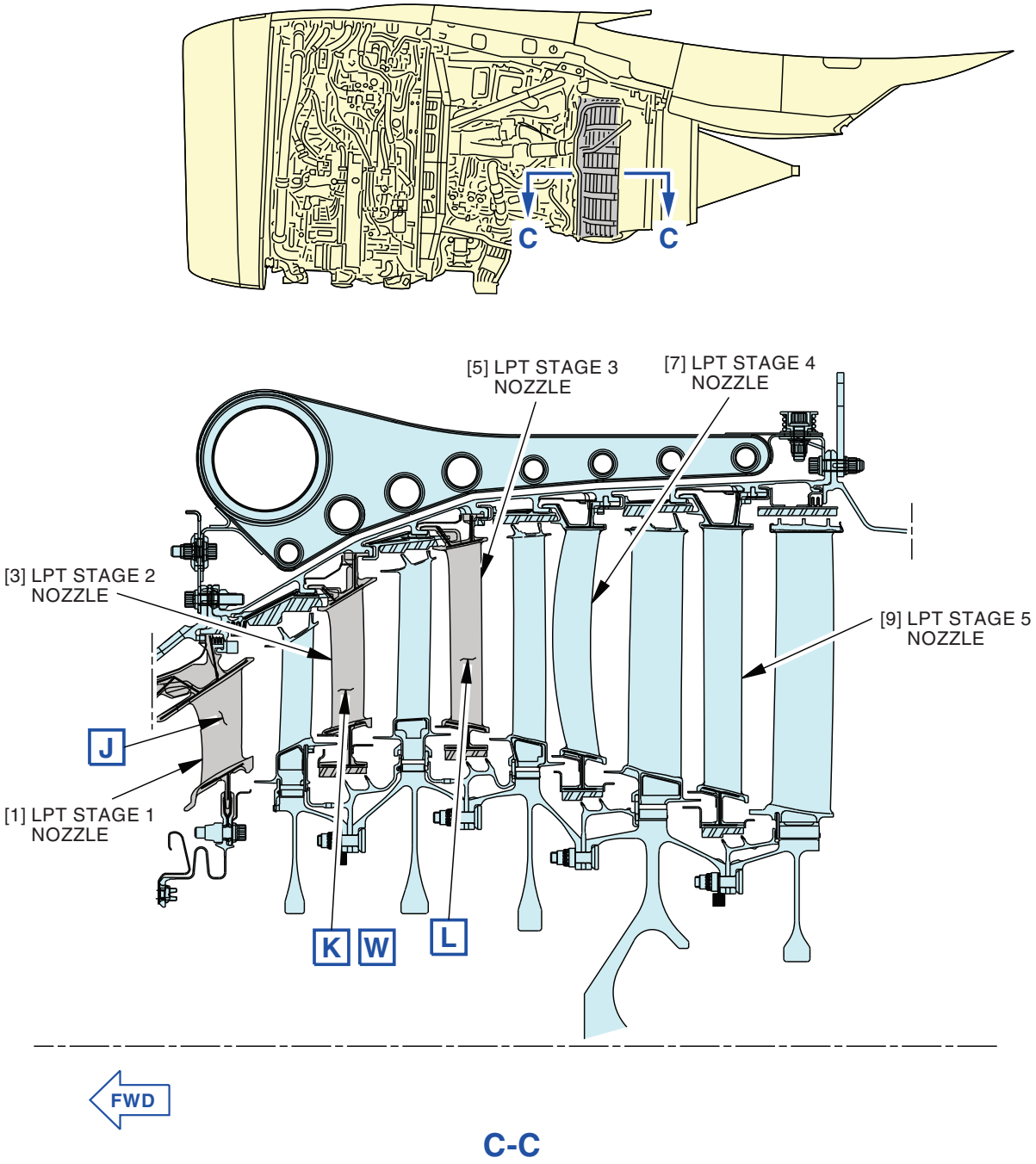
Low Pressure Turbine (LPT) Section Borescope Inspection
Figure 628/72-00-00-990-858-G00 (Sheet 14 of 33)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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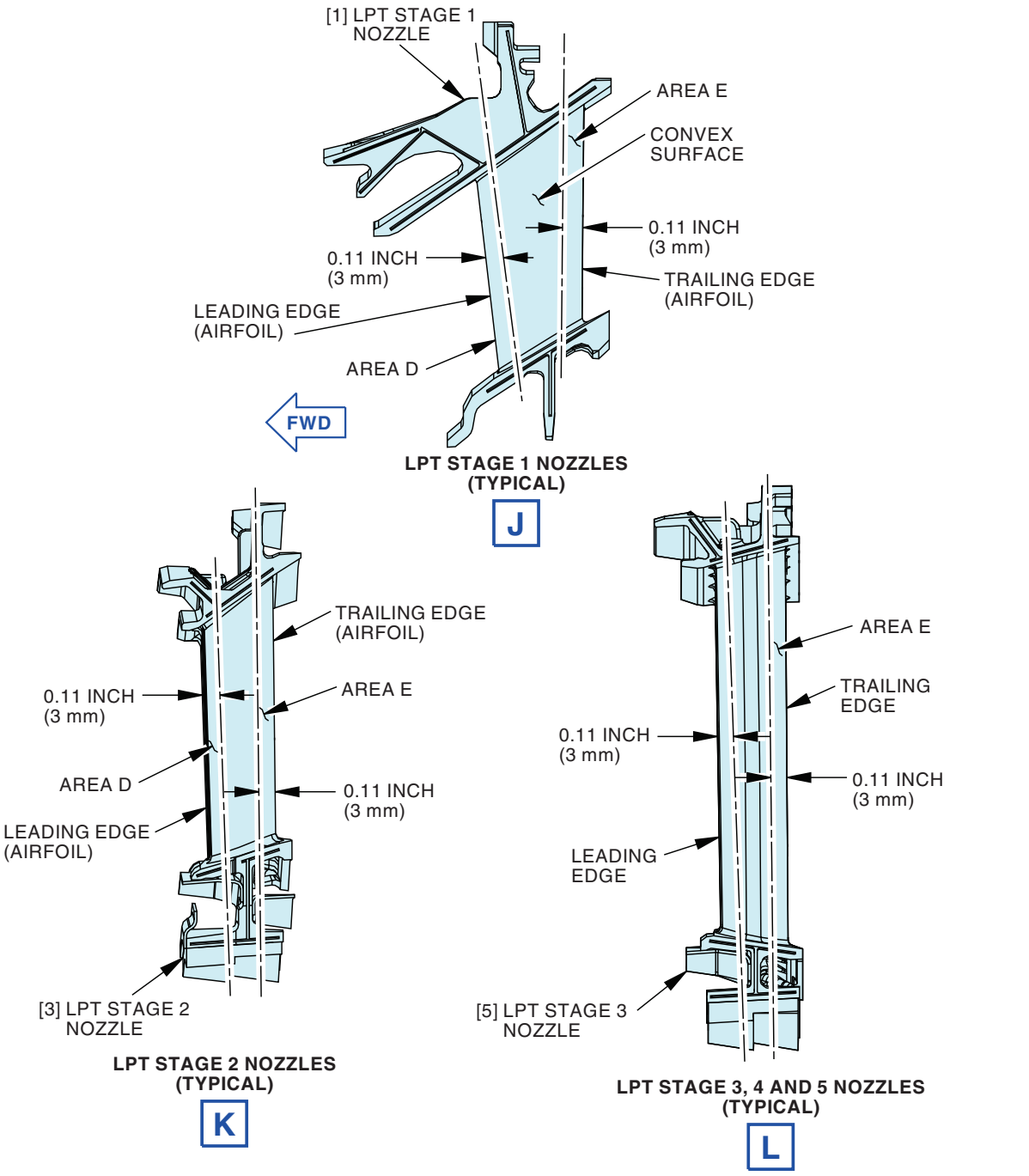
Low Pressure Turbine (LPT) Section Borescope Inspection
Figure 628/72-00-00-990-858-G00 (Sheet 15 of 33)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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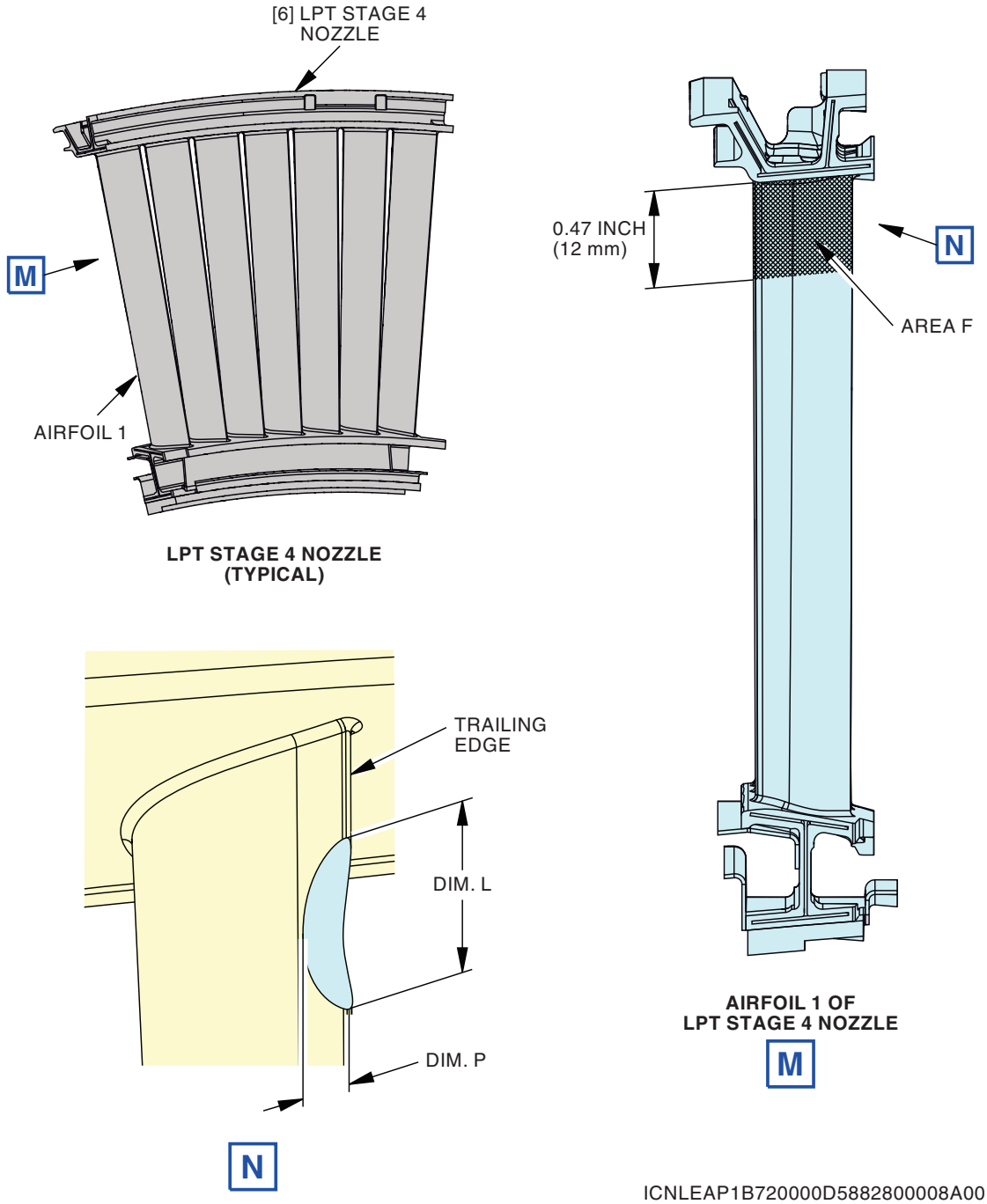
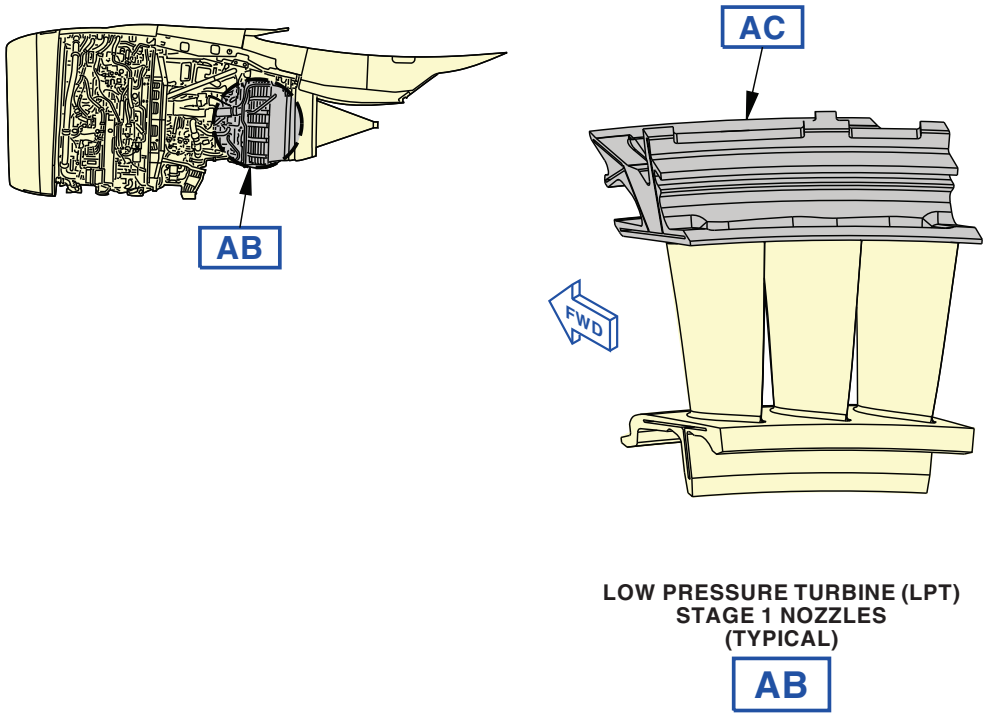
Low Pressure Turbine (LPT) Section Borescope Inspection
Figure 628/72-00-00-990-858-G00 (Sheet 16 of 33)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Low Pressure Turbine (LPT) Section Borescope Inspection
Figure 628/72-00-00-990-858-G00 (Sheet 17 of 33)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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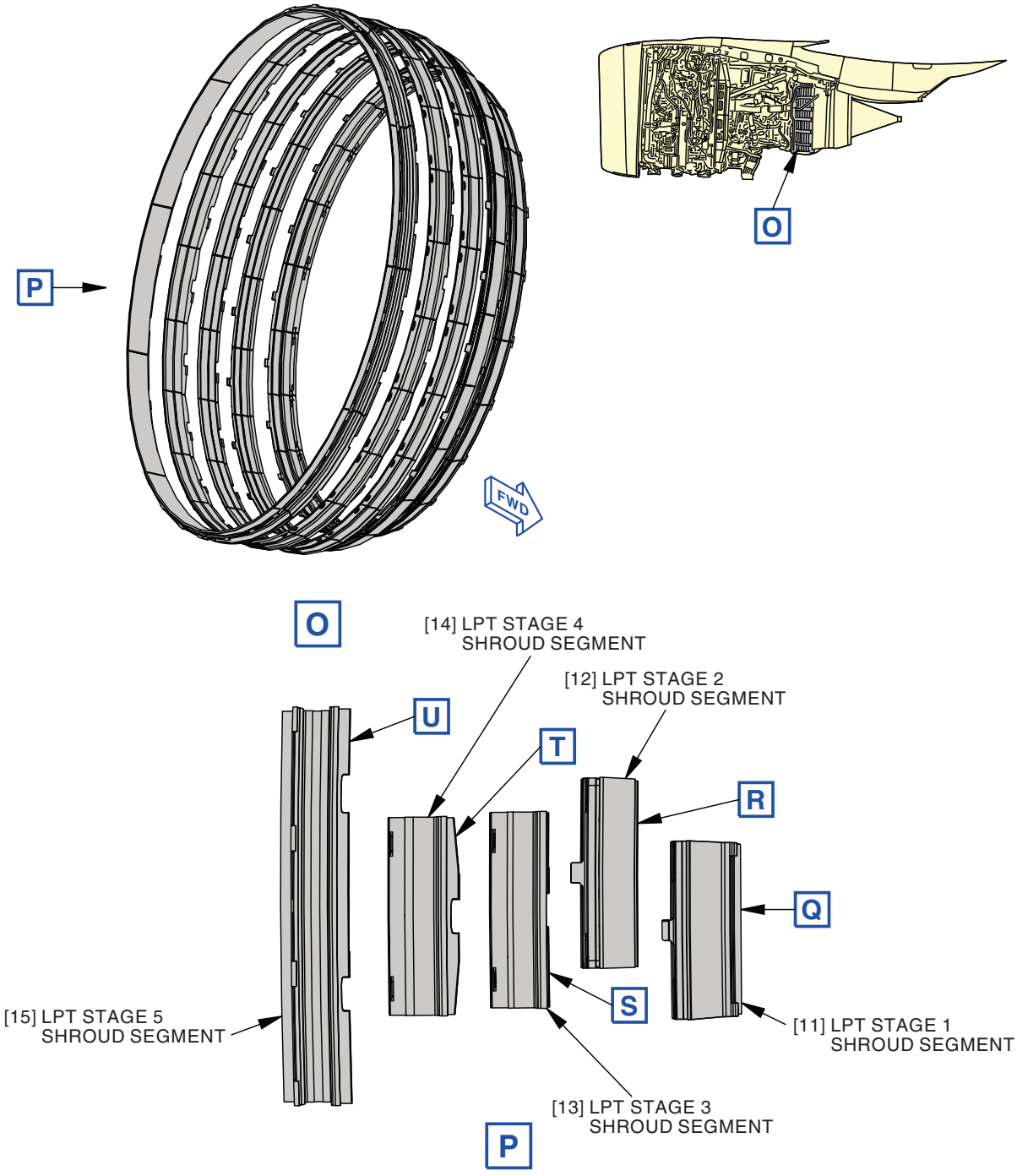
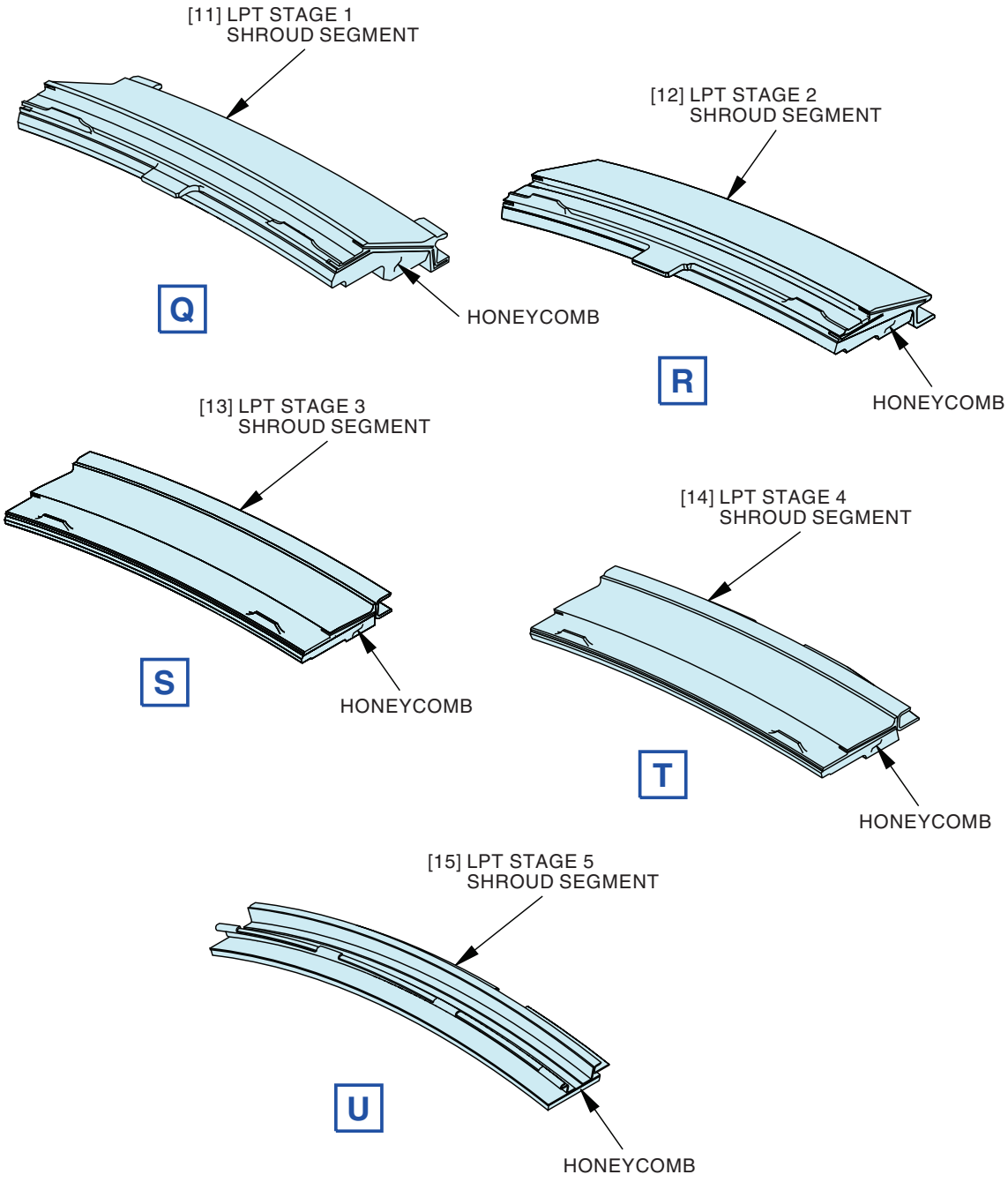
Low Pressure Turbine (LPT) Section Borescope Inspection
Figure 628/72-00-00-990-858-G00 (Sheet 18 of 33)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Low Pressure Turbine (LPT) Section Borescope Inspection
Figure 628/72-00-00-990-858-G00 (Sheet 19 of 33)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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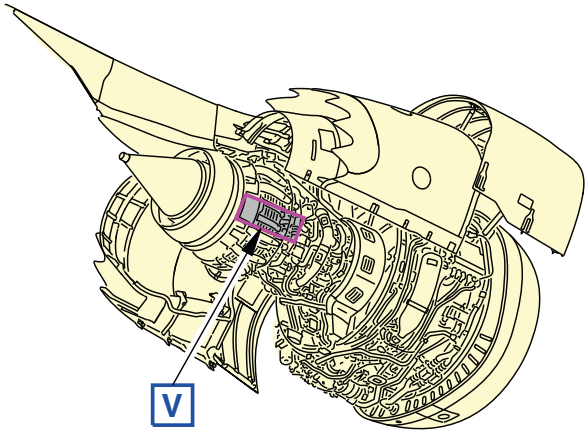
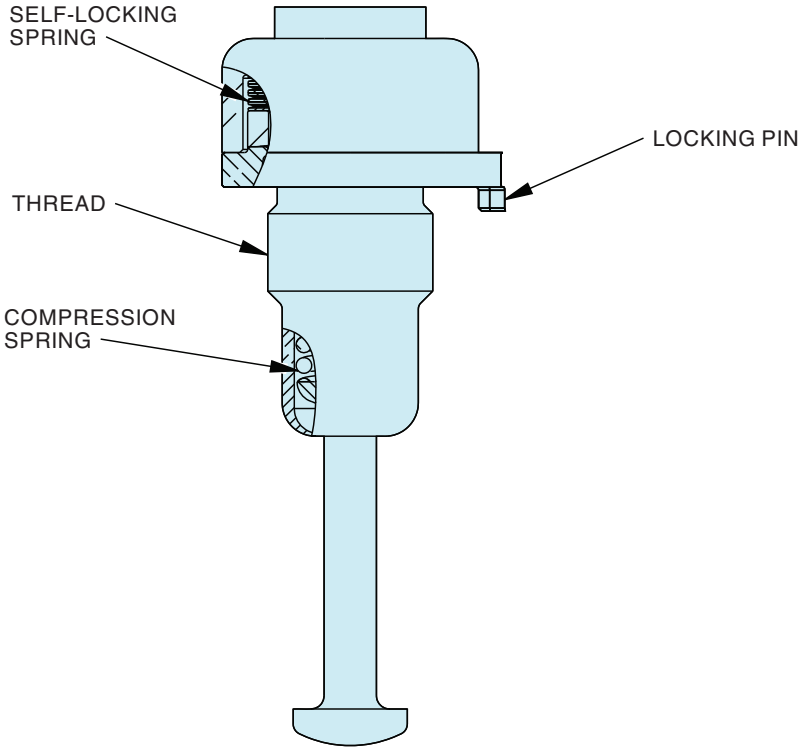
2539866 S0000602316_V2

Low Pressure Turbine (LPT) Section Borescope Inspection
Figure 628/72-00-00-990-858-G00 (Sheet 20 of 33)

ECCN 9E991 BOEING PROPRIETARY - See title page for details





LPT BORESCOPE PLUGS (TYPICAL)

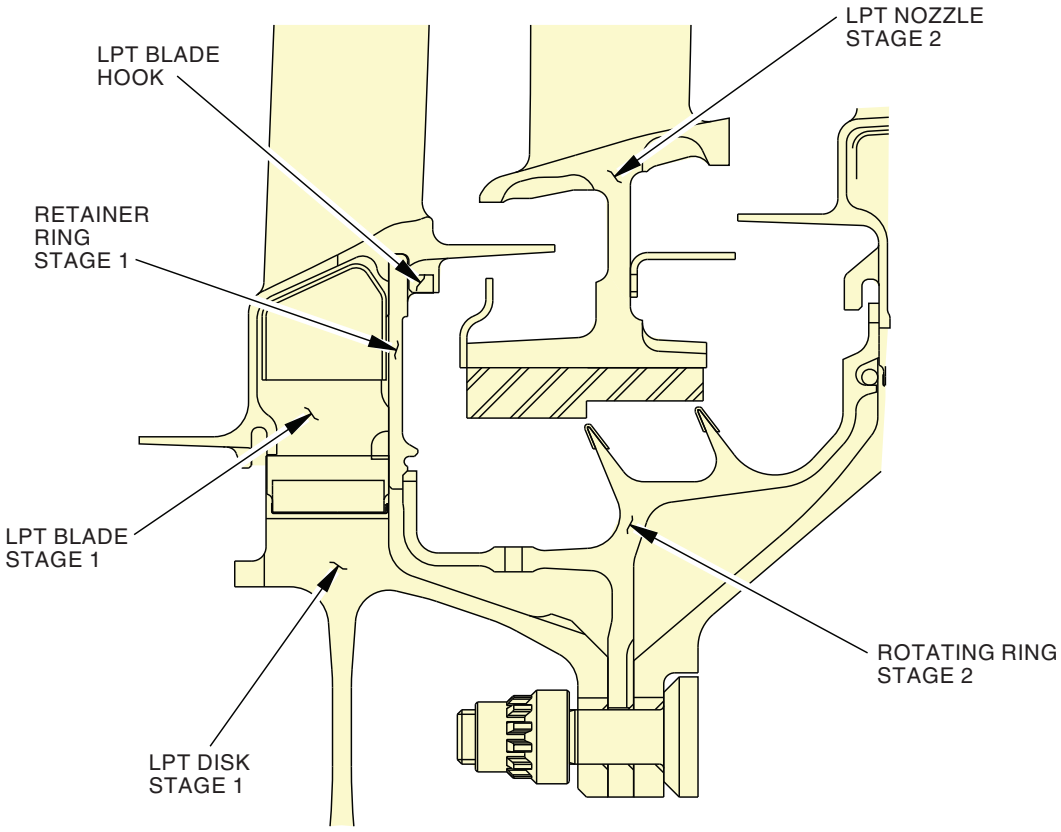
V

ICNLEAP1B720000D5882800011A00101

2539886 S0000602317_V3

Low Pressure Turbine (LPT) Section Borescope Inspection
Figure 628/72-00-00-990-858-G00 (Sheet 21 of 33)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



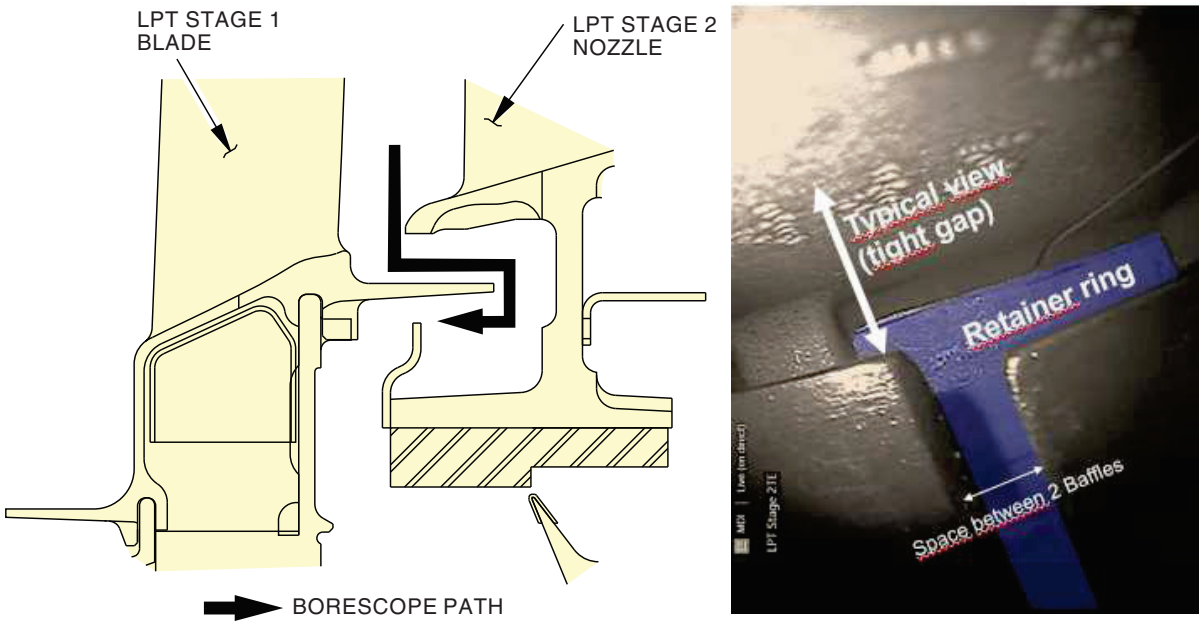
ICNLEAP1B720000D5882800023A00101

2946544 S0000723240_V1

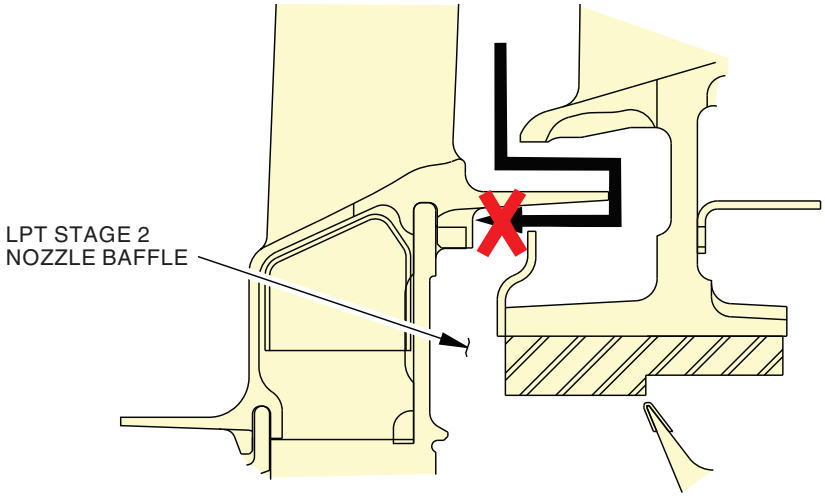
Low Pressure Turbine (LPT) Section Borescope Inspection
Figure 628/72-00-00-990-858-G00 (Sheet 22 of 33)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

BORESCOPE POSITION TO INSPECT THE RING FROM BORESCOPE PORTS



DO NOT PASS THROUGH THE NOZZLE FWD BAFFLE TO AVOID BORESCOPE SEIZURE



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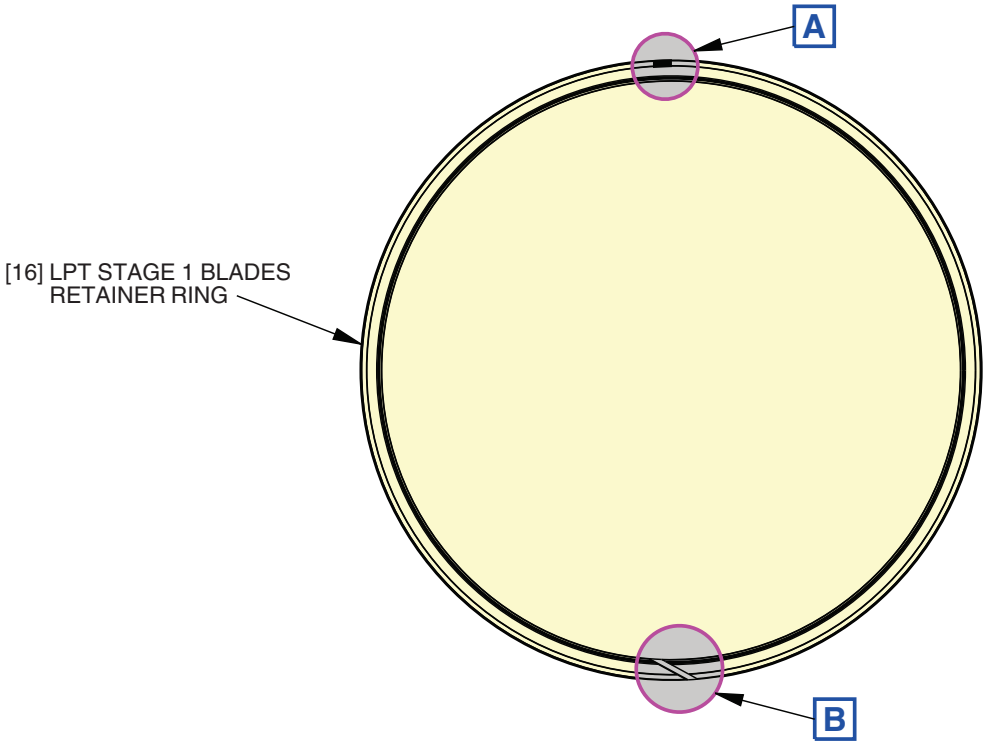
2946633 S0000723241_V1

Low Pressure Turbine (LPT) Section Borescope Inspection
Figure 628/72-00-00-990-858-G00 (Sheet 23 of 33)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



RETAINER RING ANTI-ROTATING PIN
(PRE SB LEAP-1B-72-00-0122-01A-930A-C)

A

RETAINER RING JUNCTION

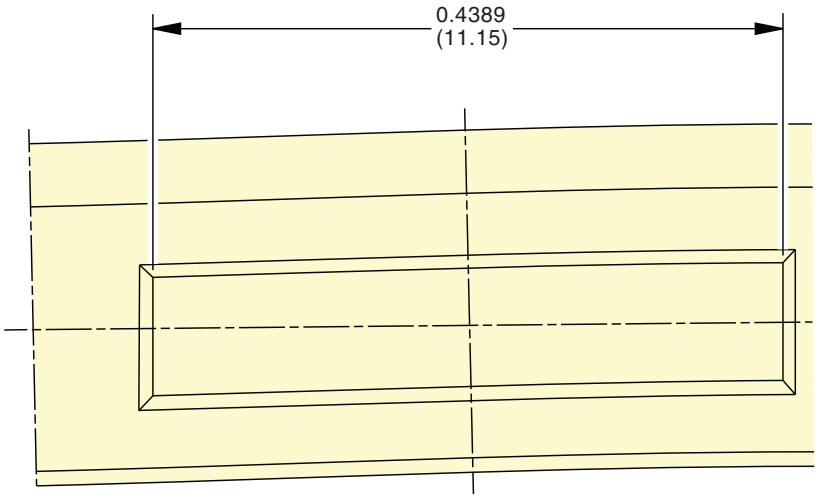
B

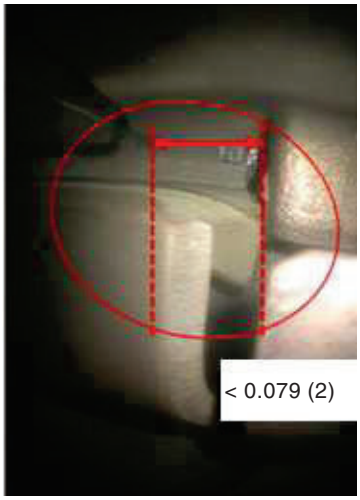
ICNLEAP1B720000D5882800025A00201

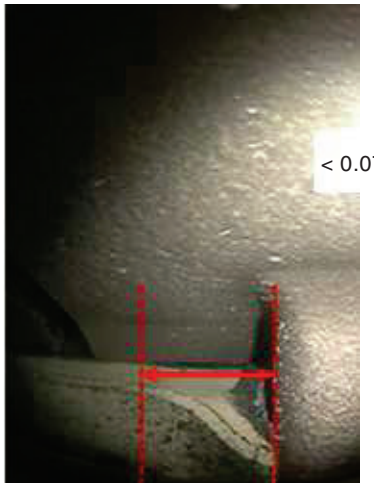
2946634 S0000723242_V2

Low Pressure Turbine (LPT) Section Borescope Inspection
Figure 628/72-00-00-990-858-G00 (Sheet 24 of 33)

ECCN 9E991 BOEING PROPRIETARY - See title page for details







NOTE:

17% OF ANTI-ROTATION PIN IS EQUIVALENT TO 0.079 (2).

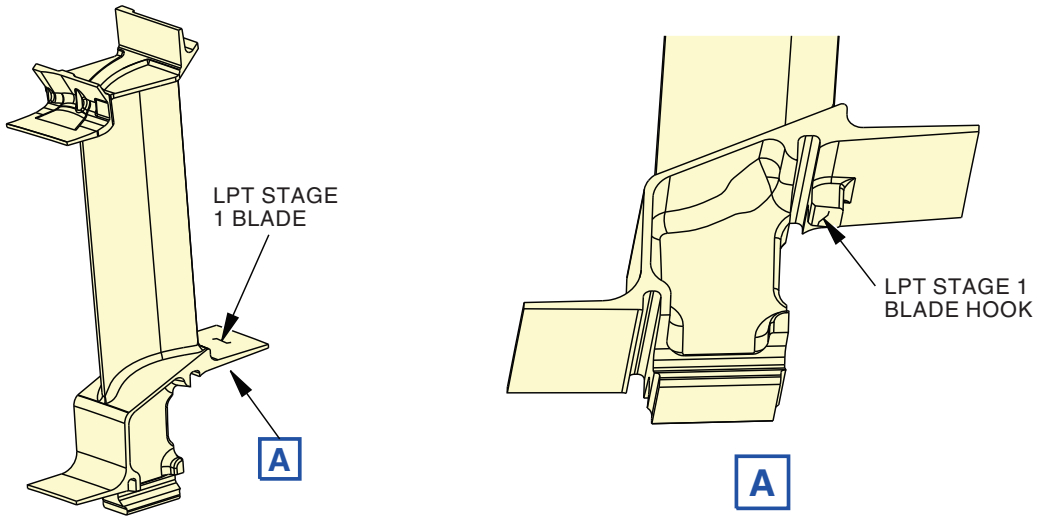
ALL DIMENSIONS ARE IN INCHES (MILLIMETERS ARE IN PARENTHESES).

ICNLEAP1B720000D5882800026A00101

2946635 S0000723243_V1

Low Pressure Turbine (LPT) Section Borescope Inspection
Figure 628/72-00-00-990-858-G00 (Sheet 25 of 33)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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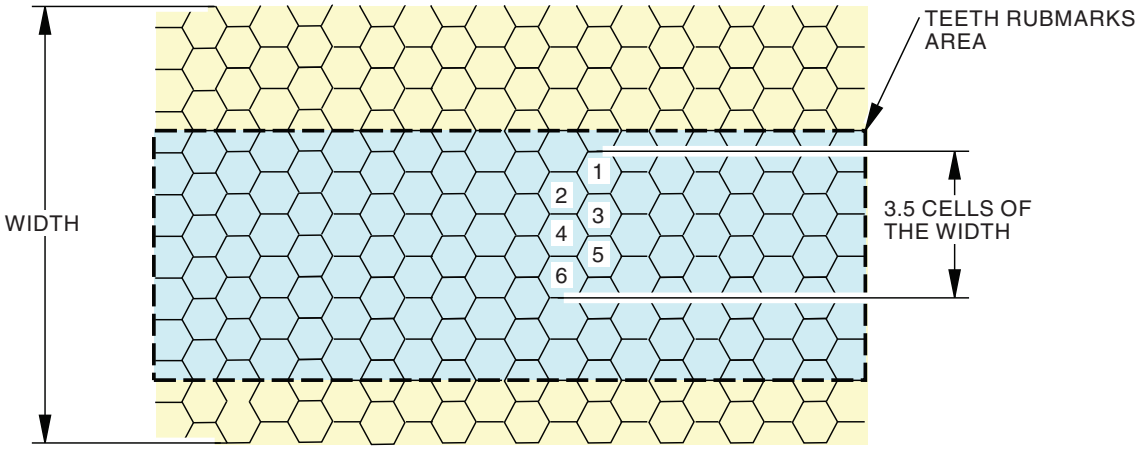
Low Pressure Turbine (LPT) Section Borescope Inspection
Figure 628/72-00-00-990-858-G00 (Sheet 26 of 33)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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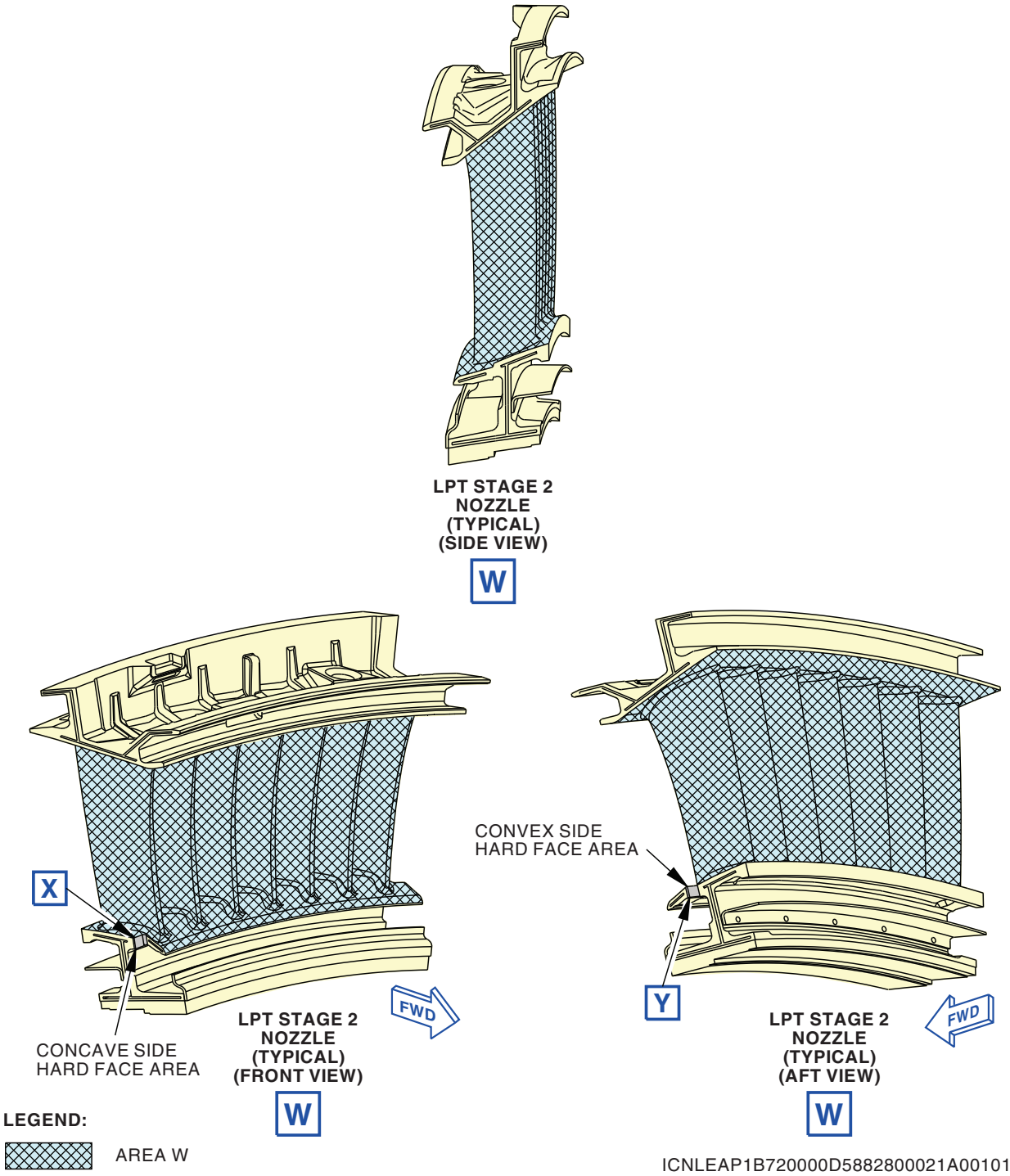
2920971 S0000703357_V1

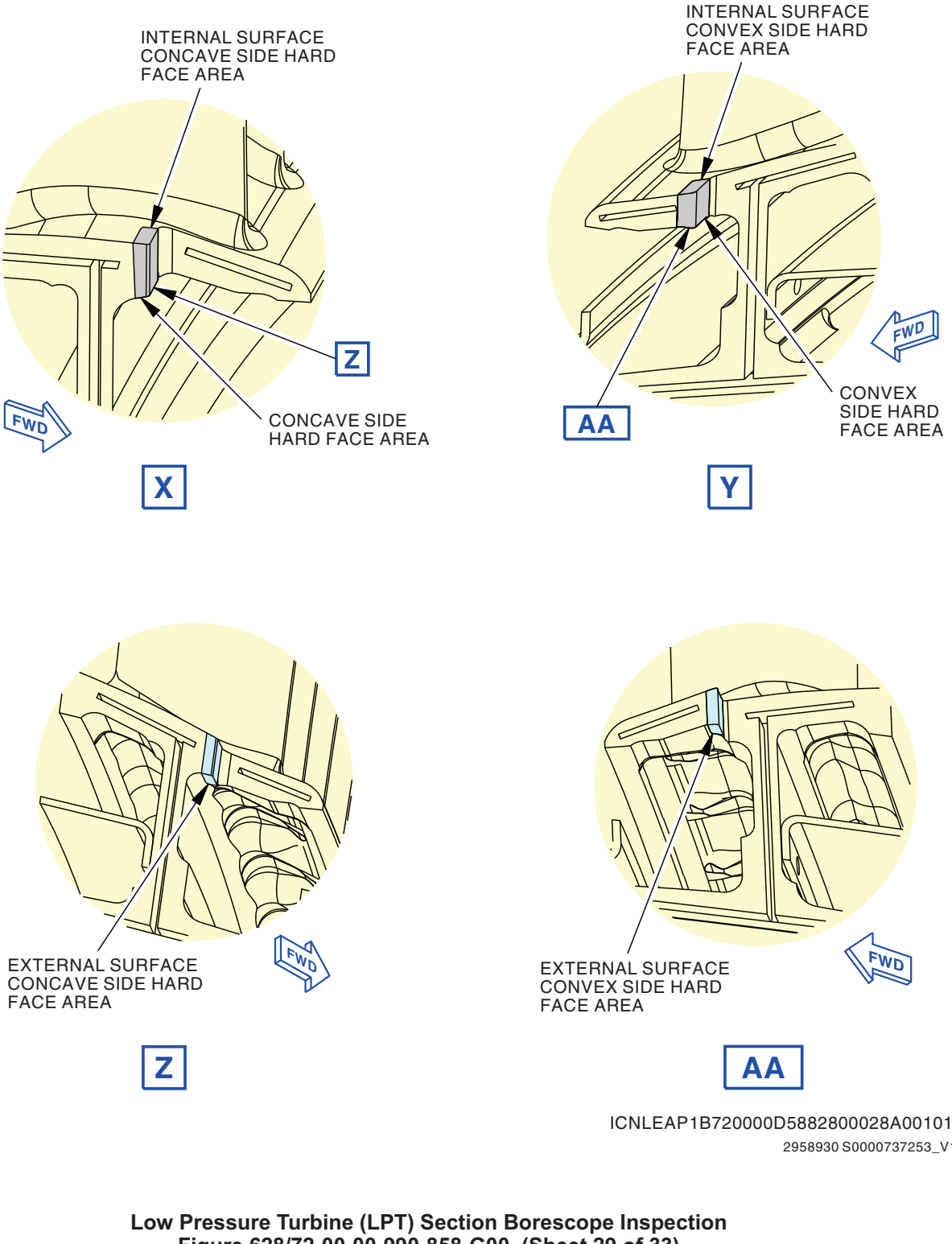
Low Pressure Turbine (LPT) Section Borescope Inspection
Figure 628/72-00-00-990-858-G00 (Sheet 27 of 33)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Low Pressure Turbine (LPT) Section Borescope Inspection
Figure 628/72-00-00-990-858-G00 (Sheet 28 of 33)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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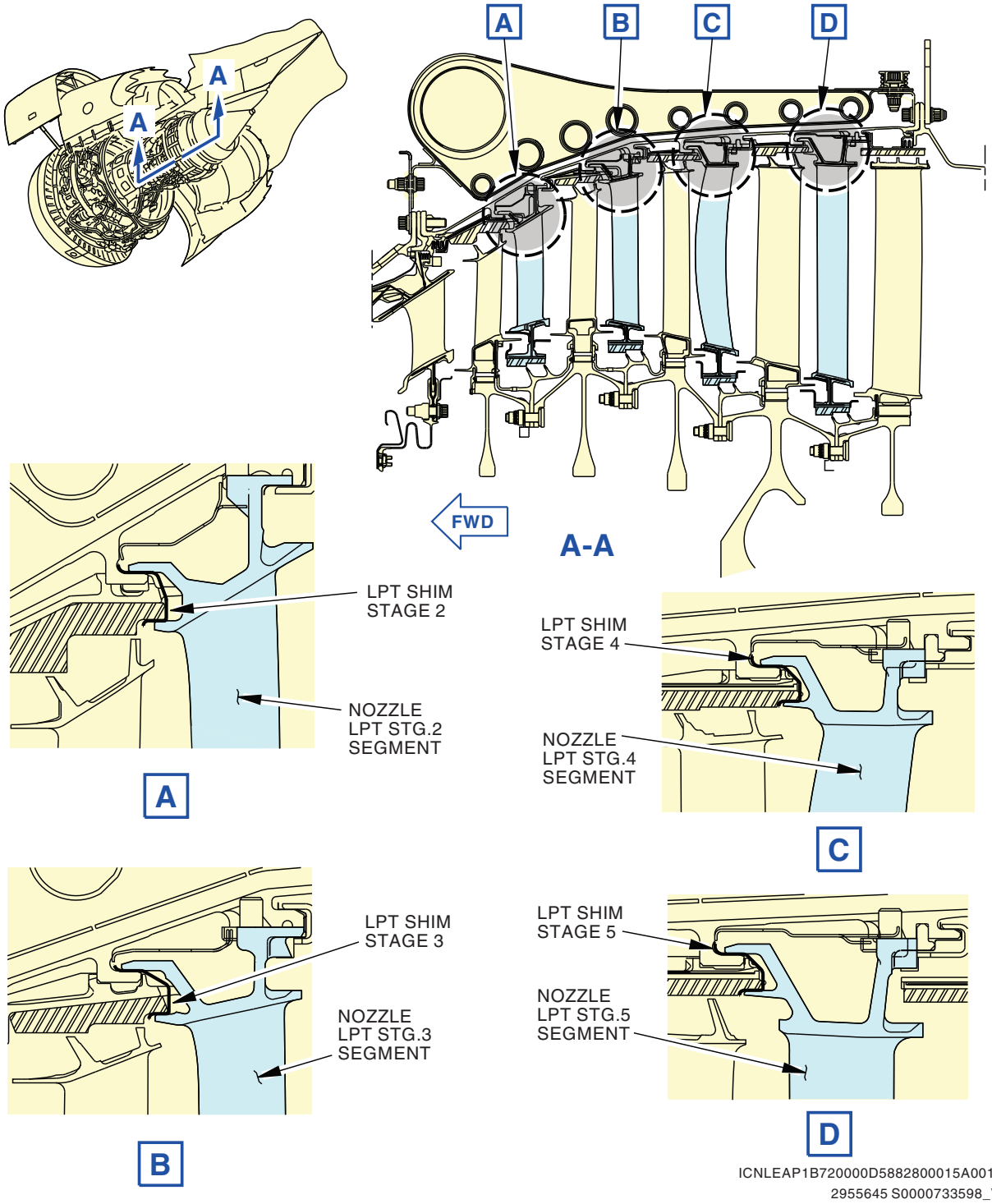
2958930 S0000737253_V1

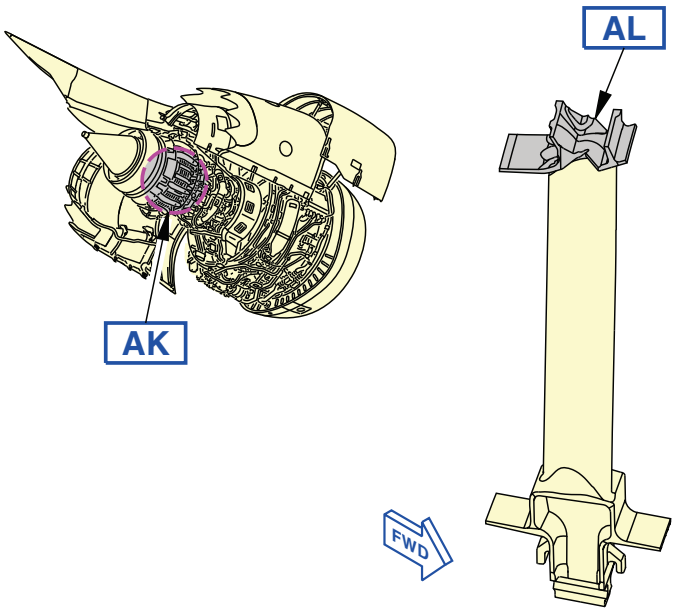
Low Pressure Turbine (LPT) Section Borescope Inspection
Figure 628/72-00-00-990-858-G00 (Sheet 29 of 33)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



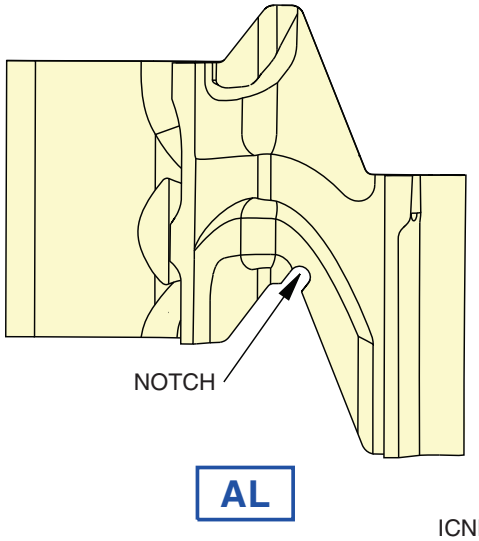
Low Pressure Turbine (LPT) Section Borescope Inspection
Figure 628/72-00-00-990-858-G00 (Sheet 30 of 33)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



TYPICAL VIEW FOR LPT STAGE 2, 3, 4 BLADES

AK

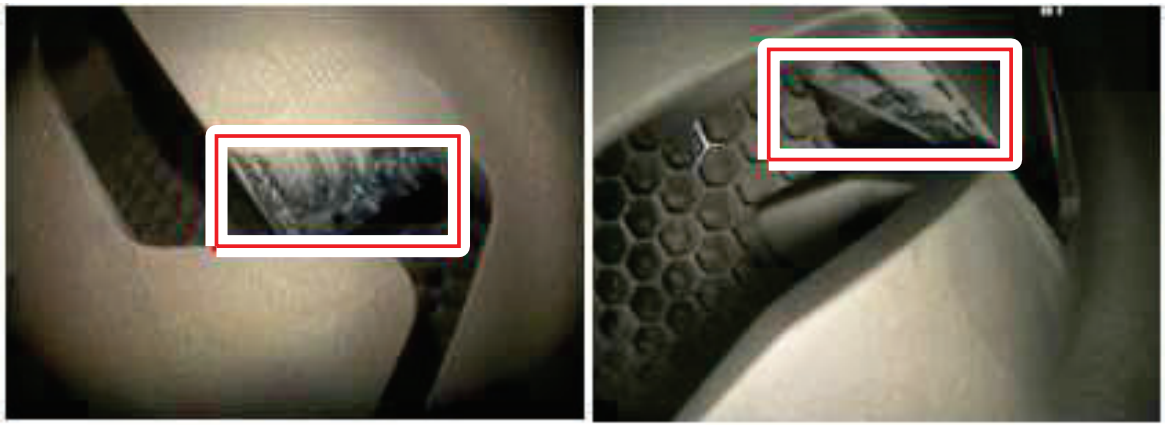


ICNLEAP1B720000D5882800056A00101

3083165 S0000838299_V1

Low Pressure Turbine (LPT) Section Borescope Inspection
Figure 628/72-00-00-990-858-G00 (Sheet 31 of 33)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



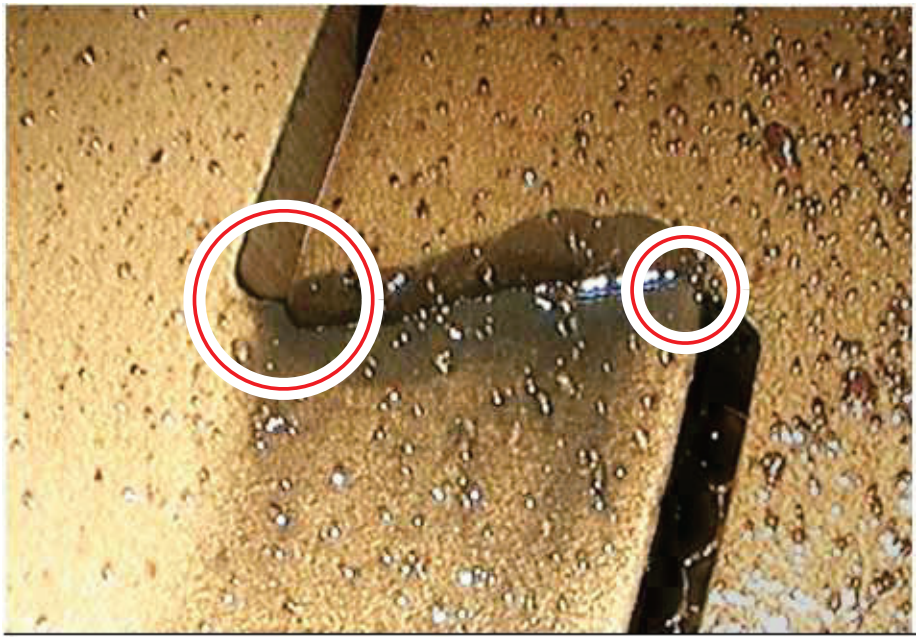
NORMAL STATE OF THE HARD FACE AFTER THE OPERATION OF THE ENGINE

ICNLEAP1B720000D5882800057A00101

3113027 S0000851473_V1

Low Pressure Turbine (LPT) Section Borescope Inspection
Figure 628/72-00-00-990-858-G00 (Sheet 32 of 33)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



DISCONTINUITY BETWEEN HARD FACE AND BASE MATERIAL

ICNLEAP1B720000D5882800058A00101

3113029 S0000851474_V1

Low Pressure Turbine (LPT) Section Borescope Inspection
Figure 628/72-00-00-990-858-G00 (Sheet 33 of 33)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL POST SB 737-CFM-LEAP-1B-75-0003

TASK 72-00-00-210-810-G00

14. High Pressure Turbine (HPT) Active Clearance Control (ACC) Air Manifolds Inspection

(Figure 629)

A. General

(1) This task provides the instructions for the inspection of the High Pressure Turbine (HPT) Active Clearance Control (ACC) air manifolds.

B. References

Reference	Title
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)
27-81-00-860-804	Leading Edge Flaps and Slats Retraction (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

F. Prepare for the Inspection

SUBTASK 72-00-00-860-094-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL POST SB 737-CFM-LEAP-1B-75-0003 (Continued)

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-00-00-860-096-G00

(2) Do this task: Leading Edge Flaps and Slats Retraction, TASK 27-81-00-860-804.

SUBTASK 72-00-00-860-097-G00

WARNING DO THE DEACTIVATION PROCEDURE FOR THE LEADING EDGE FLAPS. THE ACCIDENTAL MOVEMENT OF THE FLAPS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(3) Do this task: Leading Edge Flaps and Slats - Deactivation, TASK 27-81-00-040-801.

SUBTASK 72-00-00-010-045-G00

(4) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL POST SB 737-CFM-LEAP-1B-75-0003 (Continued)

SUBTASK 72-00-00-040-005-G00

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(5) Do the thrust reverser deactivation for ground maintenance (TASK 78-31-00-040-801-G00).

G. High Pressure Turbine (HPT) Active Clearance Control (ACC) Air Manifolds Inspection

SUBTASK 72-00-00-210-032-G00

WARNING BE CAREFUL WHEN YOU WORK ON THE ENGINE COMPONENTS IMMEDIATELY AFTER THE ENGINE IS SHUTDOWN. THE ENGINE COMPONENTS CAN STAY HOT FOR UP TO ONE HOUR AND CAN CAUSE INJURY.

(1) Do a visual inspection of the High Pressure Turbine Active Clearance Control (HPTACC) air manifolds 1, 2, 3, and 4 as follows (View A, Figure 629):

(a) Cracks in the manifolds 1, 2, 3, and 4 forward center finger bracket.

1) Permitted for 500 flight cycles if the crack is in the area of the bracket adjacent to the brazing with the integral support bracket (View C, Figure 629).

NOTE: This Continue-In-Service (CIS) limit is an inspection that occurs again and again.

2) If crack is outside the permitted area, replace the HPTACC air manifold.

(b) Cracks in the manifolds 1, 2, 3, and 4 on all other brackets (excluding forward center finger bracket) are not permitted.

1) Cracks on the forward center finger bracket are not permitted outside the permitted area.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 72-00-00-410-035-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL POST SB 737-CFM-LEAP-1B-75-0003 (Continued)

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 72-00-00-440-013-G00

(2) Do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801.

SUBTASK 72-00-00-860-095-G00

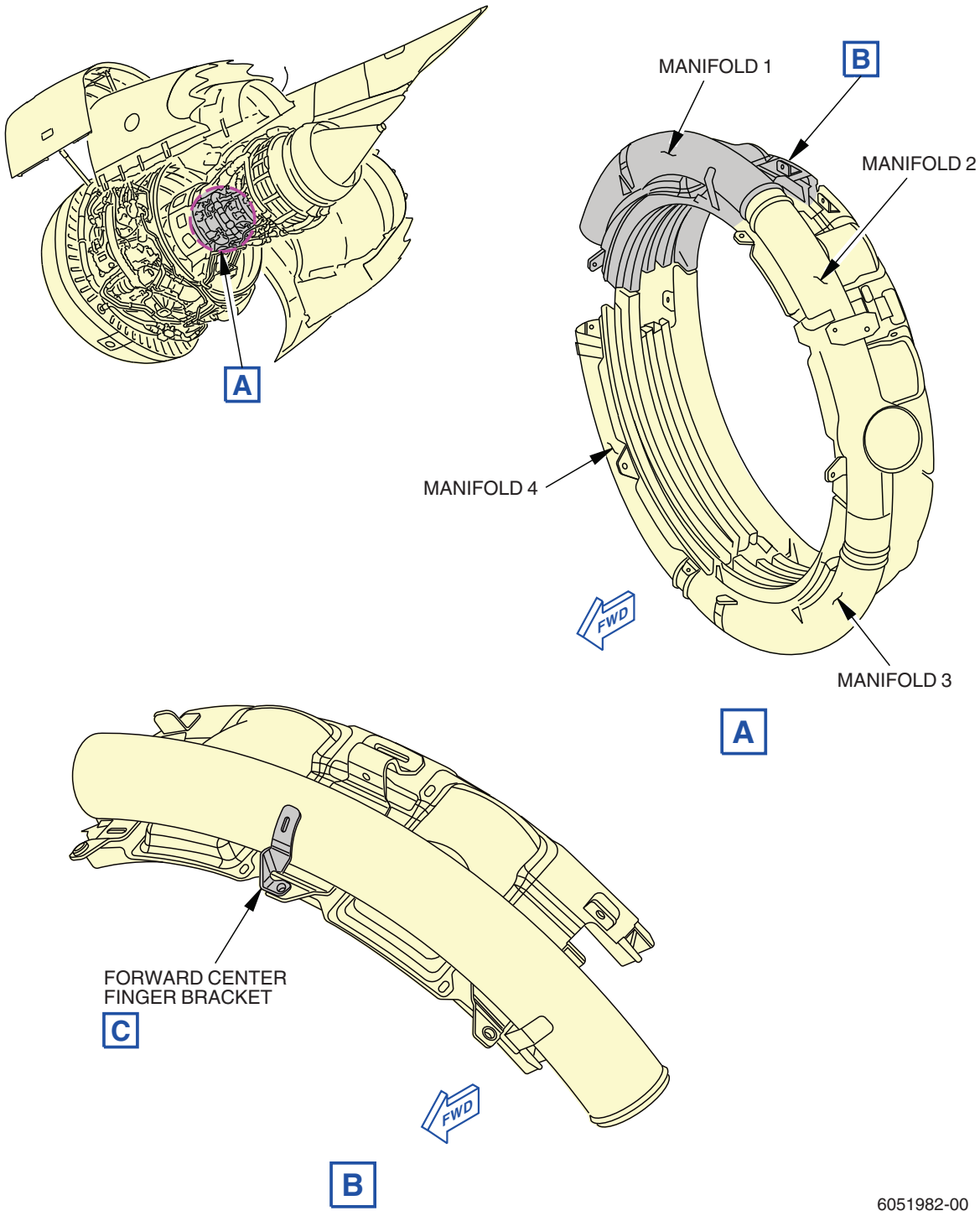
(3) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

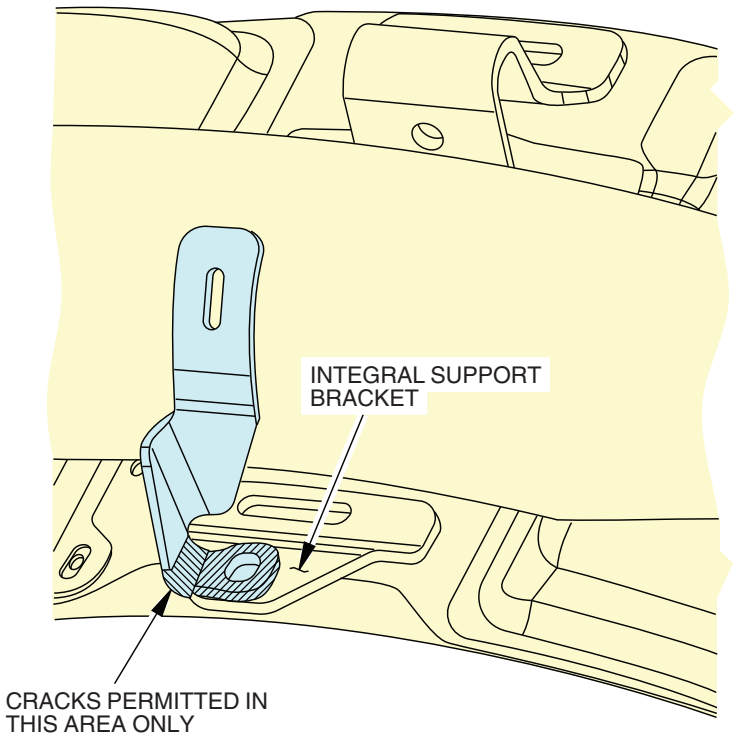
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



HPTACC Air Manifolds Inspection
Figure 629/72-00-00-990-873-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



FORWARD CENTER FINGER BRACKET

C

NOTE:
TYPICAL FORWARD CENTER BRACKET VIEW

6051983-00
3077351 S0000834755_V2

HPTACC Air Manifolds Inspection
Figure 629/72-00-00-990-873-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ENGINE - CLEANING/PAINTING

1. General

A. This procedure has these tasks:

(1) Engine cleaning specifications

(2) Engine gas path cleaning with pure water

(3) Engine gas path cleaning with cleaning solution

(4) Engine external surfaces cleaning

(5) Exhaust tube support cleaning

(6) Injection nozzle cleaning

(7) Exhaust air tube cleaning

(8) Outer exhaust air tube cleaning

(9) Rear ventilation tube cleaning.

TASK 72-00-00-100-801-G00

2. Engine Cleaning Specifications

A. General

(1) This task gives the specifications for engine cleaning.

B. References

Reference	Title
71-00-00-910-803-G00	Start the Engine (Normal Start) (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-14318	TDS - Digital Meter, Handheld, Range 0-9990 PPM, +/-2% Accuracy Part #: TDS-3 Supplier: 6VU32

D. Consumable Materials

Reference	Description	Specification
B00676 [CP1041]	Alcohol - Isopropyl	
B01024 [CP5060]	Cleaner - Gas Path	
B50378 [CP5060]	Cleaner - Gas Path - ZOK 27	
B50379 [CP5060]	Cleaner - Gas Path - R-MC G21	
B50380 [CP5060]	Cleaner - Gas Path - Ardrox 6367	

E. Engine Cleaning Specifications

SUBTASK 72-00-00-100-004-G00

CAUTION

DO NOT USE METHANOL SOLVENTS TO CLEAN THE ENGINE. DAMAGE TO TITANIUM PARTS CAN OCCUR.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(CAUTION PRECEDES)

CAUTION DO NOT CLEAN THE ENGINE WHEN THE EGT INDICATION IS HOTTER THAN THE SPECIFIED LIMIT. A FAST TEMPERATURE CHANGE CAN CAUSE DAMAGE TO THE ENGINE COMPONENTS.

CAUTION DO NOT CLEAN AN ENGINE AFTER EXPOSURE TO DRY CHEMICAL FIRE EXTINGUISHING AGENTS. DAMAGE TO THE ENGINE CAN OCCUR.

CAUTION DO NOT CLEAN AN ENGINE WHEN THE OUTSIDE AIR TEMPERATURE IS BELOW 14°F (-10°C). ICE WILL FORM ON THE ENGINE.

(1) You can wash the engine with pure water only or with a solution of ZOK 27 cleaner, B50378 [CP5060], R-MC G21 cleaner, B50379 [CP5060], or Ardrox 6367 cleaner, B50380 [CP5060].

(a) For best results, it is recommended to use gas path cleaner, B01024 [CP5060], heated to 158 ±25°F (70 ±14°C) to do the wash of the engine gas path.

1) If gas path cleaner, B01024 [CP5060], is not used, you can use water, heated to 158 ±25°F (70 ±14°C), as an alternative.

(b) An engine gas-path water cleaning with detergent provides the most effective method to clean the engine compressors.

(c) When a detergent is used, it is necessary to follow strict application of the engine rinsing and bleed system cycling procedures to prevent fumes in the flight compartment and cabin when the airplane goes back into service.

1) It may be necessary to clean the engine more than once if initial results are not satisfactory.

SUBTASK 72-00-00-940-004-G00

(2) If the air temperature is less than 41°F (5°C), use isopropyl alcohol, B00676 [CP1041], only as an anti-freeze agent with the water rinse or the cleaner solution.

SUBTASK 72-00-00-940-005-G00

CAUTION DO NOT CLEAN THE ENGINE WHEN THE EGT INDICATION IS HOTTER THAN THE SPECIFIED LIMIT. A FAST TEMPERATURE CHANGE CAN CAUSE DAMAGE TO THE ENGINE COMPONENTS.

(3) Make sure that the engine Exhaust Gas Temperature (EGT) is below 150°F (66°C).

SUBTASK 72-00-00-840-001-G00

(4) Water

(a) Water properties

1) Use water with less than 500 parts per million (ppm) Total Dissolved Solids (TDS).

2) Use water with less than 50 ppm of the total of chlorine and chloride and with a pH of 6.8 - 8.0.

NOTE: Potable water usually meets these requirements.

(b) Water flow

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CAUTION

DO NOT APPLY WATER FLOW MORE THAN 5 GPM (19 L/MINUTE). TOO MUCH CAN CAUSE DAMAGE TO THE ENGINE.

1) Apply a water flow between 4.5 gpm (17.0 l/min) and 5 gpm (19 l/min).

NOTE: If the water cart does not have a flowmeter, you can calculate the flow rate using a container of known quantity and a stop watch.

(c) Water Temperature

CAUTION

DO NOT INCREASE THE WATER TEMPERATURE MORE THAN 183°F (84°C). IF YOU DO NOT OBEY THIS INSTRUCTION, YOU CAN CAUSE DAMAGE TO THE ENGINE.

1) For best results, increase the temperature of the water to 158 ±25°F (70 ±14°C) to wash and clean the gas path.

SUBTASK 72-00-00-840-002-G00

(5) Anti-freeze

CAUTION

DO NOT CLEAN AN ENGINE WHEN THE OUTSIDE AIR TEMPERATURE IS BELOW 14°F (-10°C). ICE WILL FORM ON THE ENGINE.

(a) If the air temperature is below 41°F (5°C), prepare the anti-freeze mixture for the water wash or the cleaning solution as follows:

1) For air temperatures between 41°F (5°C) and 23°F (-5°C), use a mixture of 25% of alcohol, B00676 [CP1041], and 75% water.

2) For air temperatures between 23°F (-5°C) and 14°F (-10°C), use a mixture of 35% of alcohol, B00676 [CP1041], and 65% water.

3) For air temperatures below 14°F (-10°C), do not clean the engine.

SUBTASK 72-00-00-840-003-G00

(6) Cleaning solutions

(a) Pure water only

(b) 20-25% ZOK 27 cleaner, B50378 [CP5060], with 75-80% pure water

(c) 20-25% R-MC G21 cleaner, B50379 [CP5060], with 75-80% pure water

(d) 20-25% Ardrox 6367 cleaner, B50380 [CP5060], with 75-80% pure water.

SUBTASK 72-00-00-750-002-G00

(7) Use a TDS meter, COM-14318, or equivalent, to measure the TDS near the exhaust nozzle.

SUBTASK 72-00-00-110-001-G00

(8) Dry the engine after the engine gas path cleaning procedure is done.

(a) If there is no icing condition, start the engine not more than 2 hours after you have cleaned and serviced the engine (TASK 71-00-00-910-803-G00).

(b) If there is icing condition, start the engine not more than 30 minutes after you have cleaned and serviced the engine (TASK 71-00-00-910-803-G00).

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 72-00-00-100-802-G00

3. Engine Gas-Path Cleaning With Pure Water

(Figure 701)

A. General

(1) This task gives the instructions to clean the engine gas path with pure water.

B. References

Reference	Title
12-13-11-600-801	Engine Servicing (Oil Replenishing) (P/B 301)
21-00-01-100-801	Oil Contamination Removal from Air Conditioning and Pneumatic Systems (P/B 201)
71-00-00-700-805-G00	Test No. 29 - Water Wash Special Functions Test (P/B 501)
71-00-00-790-801-G00	Test No. 1 - Pneumatic Leak Test (P/B 501)
71-00-00-800-802-G00	Engine Operation Limits (P/B 201)
71-00-00-910-803-G00	Start the Engine (Normal Start) (P/B 201)
71-00-00-910-806-G00	Stop the Engine (Usual Engine Stop) (P/B 201)
71-00-00-910-808-G00	Dry Motor Procedure (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
73-21-11-000-801-G00	Pressure Sub-System Removal (P/B 401)
73-21-11-400-801-G00	Pressure Sub-System Installation (P/B 401)
74-11-01-400-801-G00	Left Ignition Exciter (Upper) Installation (P/B 401)
74-11-01-400-802-G00	Right Ignition Exciter (Lower) Installation (P/B 401)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-2288	Cart - Water Wash, Engine Part #: 31658 Supplier: 6C9S2 Part #: 35030 Supplier: 6C9S2 Part #: JMP/CFM56/D/4777/C200 Supplier: U2066 Part #: JMP/GE90/D/4038/C200 Supplier: U2066 Part #: PWP2100 Supplier: 6C9S2 Part #: PWP3600 Supplier: 6C9S2 Part #: PWP3700 Supplier: 6C9S2
COM-11191	Probe - Long, Water Wash Part #: JMP/LEAP/D/7185/B Supplier: U2066
COM-14318	TDS - Digital Meter, Handheld, Range 0-9990 PPM, +/-2% Accuracy Part #: TDS-3 Supplier: 6VU32

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(Continued)

Reference	Description
COM-14887	Collection System - Engine Wash Part #: JMP/KYOTO/D/6777/C200 Supplier: U2066
COM-15151	Probe - Short, Water Wash, LEAP-1B Engine Part #: JMP/LEAP/D/7028 Supplier: U2066
STD-858	Tag - DO NOT OPERATE
STD-3906	Mallet - Rubber
STD-3940	Air Source - Regulated, Dry Filtered, 0-150 psig

D. Consumable Materials

Reference	Description	Specification
G00018	Nitrogen - Gaseous, Pressurizing, 99.5 Percent Pure	A-A-59503 Type I Grade B, MIL-PRF-27401 Type I Grade A

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
11	E-seal	36-11-01-01-220	PTF ALL

F. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

G. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

H. Prepare for the Cleaning

SUBTASK 72-00-00-860-059-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-00-00-010-023-G00

(2) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

SUBTASK 72-00-00-940-002-G00

CAUTION DO NOT CLEAN AN ENGINE WHEN THE OUTSIDE AIR TEMPERATURE IS BELOW 14°F (-10°C). ICE WILL FORM ON THE ENGINE.

(3) Make sure that the outside air temperature is above 14°F (-10°C).

SUBTASK 72-00-00-940-003-G00

CAUTION DO NOT CLEAN THE ENGINE WHEN THE EGT INDICATION IS HOTTER THAN THE SPECIFIED LIMIT. A FAST TEMPERATURE CHANGE CAN CAUSE DAMAGE TO THE ENGINE COMPONENTS.

(4) Make sure that the engine Exhaust Gas Temperature (EGT) is below 150°F (66°C).

SUBTASK 72-00-00-100-001-G00

(5) Make sure that the water properties, water flow, water temperature, cleaning solutions and anti-ice mixtures are specific for this procedure (TASK 72-00-00-100-801-G00).

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SUBTASK 72-00-00-020-003-G00

CAUTION

DO NOT USE METHANOL SOLVENTS TO CLEAN THE ENGINE. DAMAGE TO TITANIUM PARTS CAN OCCUR.

CAUTION

DO NOT CLEAN AN ENGINE AFTER EXPOSURE TO DRY CHEMICAL FIRE EXTINGUISHING AGENTS. DAMAGE TO THE ENGINE CAN OCCUR.

(6) Prepare the engine for washing.

(a) Disconnect the PS3 line [2] and P3B line [3] from the PSS box [1] (TASK 73-21-11-000-801-G00).

1) Install the protective caps on the PS3 line [2] and P3B line [3].

2) Install the protective covers on the PSS box [1].

(b) Remove the E-seal [11] and coupling [10] installed between the upper CDP duct assembly [8] and lower CDP duct assembly [9].

1) Remove the coupling [10].

2) Slightly separate the upper CDP duct assembly [8] and lower CDP duct assembly [9] to remove the E-seal [11].

a) Examine the E-seal [11] for cracks, dents, or other damage.

b) Replace the E-seal [11] if necessary.

CAUTION

YOU MUST COVER OR REMOVE IGNITION EXCITERS BEFORE THE WATER WASH. IGNITION EXCITERS CAN GET MOISTURE THROUGH THE WELDS OF THE EXCITER BOX DURING WATER WASH. IF YOU DO NOT OBEY, DAMAGE TO THE IGNITION EXCITERS CAN OCCUR.

(c) Cover the ignition exciters, wiring, and tubes at the 6:00 o'clock location to protect from excessive water.

(d) If it is necessary, prepare the anti-freeze mixture (TASK 72-00-00-100-801-G00).

(e) Do this task: Engine Servicing (Oil Replenishing), TASK 12-13-11-600-801.

SUBTASK 72-00-00-860-065-G00

(7) On the P5 overhead panel, make sure that the BLEED 1 and BLEED 2 switches are in the OFF position.

SUBTASK 72-00-00-860-066-G00

(8) On the P5 overhead panel, make sure that the L PACK and R PACK switches are in the OFF position.

SUBTASK 72-00-00-860-067-G00

(9) On the P5 overhead panel, make sure that the WING ANTI-ICE, ENG 1 ANTI-ICE and ENG 2 ANTI-ICE switches are in the OFF position.

SUBTASK 72-00-00-480-019-G00

(10) Install the core wash probes per manufacturer instructions.

NOTE: For the probes installed between the booster and the Fan Blades, the left and right wands must be installed such that it injects water into the booster and is fed from behind the fan outlet guide vanes.

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SUBTASK 72-00-00-480-021-G00

(11) Secure wands to the fan bypass.

SUBTASK 72-00-00-480-024-G00

(12) Install an engine wash collection system, COM-14887, below the engine.

NOTE: An engine wash collection system, COM-14887, collects the dirty water that falls from the engine during gas-path cleaning.

NOTE: It is optional to use an engine wash collection system, COM-14887, only if local environmental restrictions allow this. Consult the regulatory guidelines before you make a decision not to use this tool.

I. Engine Gas-Path General Cleaning With Pure Water

SUBTASK 72-00-00-170-001-G00

WARNING USE TWO PERSONS TO DO THIS PROCEDURE. ONE PERSON OPERATES THE ENGINE CONTROLS. ONE PERSON TO MAKE SURE THAT PERSONNEL ARE CLEAR OF SURFACES THAT MOVE AND ALSO MONITORS THE ENGINE. INJURY TO PERSONS OR DAMAGE TO EQUIPMENT COULD OCCUR.

(1) Clean the engine gas path with pure water as follows:

(a) Collect a sample of the wash supply water.

1) Measure the total dissolved solid with the TDS meter, COM-14318, or equivalent.

(b) An engine water wash cart, COM-2288, with immersion heater.

NOTE: If you use anti-freeze, do not use the immersion heater.

NOTE: You can use a short wash probe, COM-15151, with the thrust reverser in the open position, and the long wash probe, COM-11191, with the thrust reverser in the closed position.

CAUTION DO NOT DRY-MOTOR THE ENGINE UNLESS THE INLET FUEL PRESSURE IS POSITIVE. THE FUEL PRESSURE IS NECESSARY TO LUBRICATE THE ENGINE FUEL PUMP AND FUEL METERING UNIT TO PREVENT DAMAGE.

CAUTION DO NOT OPERATE THE ENGINE STARTER MORE THAN THE SPECIFIED LIMITS IN THE DUTY CYCLE. IF YOU OPERATE THE STARTER MORE THAN THE SPECIFIED LIMITS, YOU CAN CAUSE DAMAGE TO THE STARTER OR THE GEARBOX.

CAUTION MAKE SURE THAT THE N1 ROTOR TURNS FREELY DURING THE FULL CLEAN PROCEDURE. IF YOU DO NOT OBEY THIS INSTRUCTION, OIL SYSTEM CONTAMINATION COULD OCCUR.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(CAUTION PRECEDES)

CAUTION

DO NOT APPLY WATER FLOW MORE THAN 5 GPM (19 L/MINUTE). TOO MUCH CAN CAUSE DAMAGE TO THE ENGINE.

(c) Start the water wash special function on the Multi Function Display (MFD) (TASK 71-00-00-700-805-G00).

NOTE: For the engine starter operation limits, refer to this task: Engine Operation Limits, TASK 71-00-00-800-802-G00.

NOTE: This function is conducted as part of the overall water wash procedure. When this function runs, the Variable Stator Vane (VSV) and Variable Bleed Valve (VBV) systems move to facilitate an engine water wash. The function opens the VSVs and closes the VBVs so that the water wash mixture can enter the compressor. This is a dry motoring test.

1) Follow the instructions on the MFD.

(d) Once the bowed rotor motoring has been completed and the “motoring” light has extinguished, wait 10 seconds and then do this step:

1) Inject the water 360 degrees around the Low Pressure Compressor (LPC) inlet through the fan blades.

NOTE: If you use an engine water wash cart, COM-2288 and short wash probe, COM-15151, you can inject the water directly at the LPC inlet during motoring.

(e) Stop the water injection before the end of the engine motoring.

(f) Do these steps:

1) During the engine coast-down, collect a sample of the wash water that drains from the exhaust area of the engine.

2) Use a TDS meter, COM-14318, or equivalent to measure and make a record of the total dissolved solid.

NOTE: Calibration of the Total Dissolved Solids (TDS) meter is not necessary if the same TDS meter is used to collect the sample of the wash supply water.

3) Subtract the initial TDS value measured from the second TDS value in the above step.

4) If the value is more than 150 Parts Per Million (PPM), do this step:

a) Do additional wash/soak cycles until the TDS delta falls below 150 PPM.

NOTE: Up to a maximum of four wash cycles.

(g) Let the water soak for 10-15 minutes between cycles.

NOTE: The 10-15 minutes between cycles will maximize the cleaning of the engine.

(h) Do the steps above again as necessary up to a maximum of four times to dry motor, inject water and soak the engine.

NOTE: You may do this step up to maximum of four times for engines with heavy dirt deposits.

(i) Start to dry motor the engine again (TASK 71-00-00-910-808-G00).

(j) When the engine has stopped, select “STOP TEST” on the MFD to stop the water wash special function.

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J. Put the Airplane Back to Its Usual Condition

SUBTASK 72-00-00-080-023-G00

(1) Remove the engine wash collection system, COM-14887.

SUBTASK 72-00-00-860-092-G00

(2) In case ignition exciter is covered, remove the cover from the ignition exciter and wiring.

SUBTASK 72-00-00-860-093-G00

(3) In case ignition exciter (left/right) is removed, re-install the ignition exciter (TASK 74-11-01-400-801-G00, TASK 74-11-01-400-802-G00).

SUBTASK 72-00-00-160-002-G00

(4) Clean the PS3 line [2] and P3B line [3] as follows:

(a) Remove the protective caps from the PS3 line [2] and P3B line [3].

(b) Remove the protective covers from the PSS box [1].

(c) Use the regulated air source, STD-3940, or nitrogen, G00018, to blow into the PS3 line [2] and clear it of water.

1) Use approximately 100 psig (689 kPa) (not to exceed 200 psig (1379 kPa)) to dry the air tubes.

a) Blow air into the air tubes for a minimum of 3 minutes.

NOTE: For the best results, compressed nitrogen is recommended.

(d) Use the regulated air source, STD-3940, or nitrogen, G00018, to blow into the P3B line [3] and clear it of water.

1) Use approximately 100 psig (689 kPa) (not to exceed 200 psig (1379 kPa)) to dry the air tubes.

a) Blow air into the air tubes for a minimum of 3 minutes.

NOTE: For the best results, compressed nitrogen is recommended.

(e) Do a visual inspection of the weep holes on the PS3 line [2] and P3B line [3] to make sure that they are clear of all blockage.

CAUTION

USE TWO WRENCHES TO TIGHTEN THE COUPLING NUT. USE ONE TO HOLD THE FITTING, AND THE OTHER TO TURN THE COUPLING NUT. IF YOU DO NOT USE TWO WRENCHES, DAMAGE TO THE EQUIPMENT CAN OCCUR.

(f) Connect the PS3 line [2] and P3B line [3] to the PSS box [1] (TASK 73-21-11-400-801-G00).

(g) Install the E-seal [11] and coupling [10]:

1) Examine the E-seal [11] for cracks, dents, or other damage.

2) Replace the E-seal [11] if necessary.

3) Install the E-seal [11] and coupling [10] between the upper CDP duct assembly [8] and lower CDP duct assembly [9].

a) Turn the coupling [10] up, to give the maximum clearance with the ignition lead.

b) Tighten the coupling [10] to 170 in-lb (19.2 N·m) - 180 in-lb (20.3 N·m).

c) Lightly tap the outer surface of the coupling [10] with a rubber mallet, STD-3906, or equivalent.

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d) Tighten the coupling [10] to 170 in-lb (19.2 N·m) - 180 in-lb (20.3 N·m) again.

SUBTASK 72-00-00-010-043-G00

(5) Do these steps to remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

SUBTASK 72-00-00-790-003-G00

(6) To check for leakage from the E-seal [11] between the upper CDP duct assembly [8] and lower CDP duct assembly [9], do this task: Test No. 1 - Pneumatic Leak Test, TASK 71-00-00-790-801-G00, for the affected duct only.

SUBTASK 72-00-00-010-044-G00

(7) Do these steps to remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch:

(a) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

SUBTASK 72-00-00-410-021-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(8) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

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K. Flush Water From the Lube, Sumps, and Bleed Systems

SUBTASK 72-00-00-860-061-G00

WARNING MAKE SURE THAT ALL PERSONNEL AND EQUIPMENT ARE CLEAR OF THE ENGINE DANGER AREAS. THE ENGINE INTAKE AND EXHAUST CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT IN THE DANGER AREAS.

(1) Remove the water ingested into the engine systems during cleaning as follows:

(a) If there is no icing condition, start the engine not more than 2 hours after you cleaned and serviced the engine (TASK 71-00-00-910-803-G00).

(b) If there is icing condition, start the engine not more than 30 minutes after you cleaned and serviced the engine (TASK 71-00-00-910-803-G00).

SUBTASK 72-00-00-710-002-G00

(2) Operate the engine at idle for 5 minutes.

SUBTASK 72-00-00-170-007-G00

(3) Flush the inlet cowl anti-ice system as follows:

(a) Put the APU BLEED switch, on the P5-10 overhead panel, to the OFF position.

1) Make sure that the DUAL BLEED light goes out.

(b) Make sure that the applicable BLEED switch, on the P5-10 overhead panel, is in the OFF position.

(c) Put the applicable ENG ANTI-ICE 1 or ENG ANTI-ICE 2 switch, on the P5-11 overhead panel, to the ON position.

1) Make sure that the applicable COWL VALVE light comes on for 1-3 seconds and then goes off.

CAUTION DO NOT KEEP THE COWL TAI VALVE OPEN FOR MORE THAN THE SPECIFIED TIME WHEN THE TEMPERATURE IS MORE THAN THE SPECIFIED LIMIT. DAMAGE TO THE INLET COWL CAN OCCUR.

(d) Within 30 seconds, put the applicable ENG ANTI-ICE 1 or ENG ANTI-ICE 2 switch, on the P5-11 overhead panel, to the OFF position.

1) Make sure that the TAI indication on the engine display N1 indication goes off.

SUBTASK 72-00-00-170-008-G00

(4) Flush the wing anti-ice system as follows:

(a) Put the applicable BLEED switch, on the P5-10 overhead panel, to the ON position.

(b) Do one of the following steps on the P5-10 overhead panel:

1) If one engine operates, put the ISOLATION VALVE switch to the OPEN position.

2) If both engines operate, put the ISOLATION VALVE switch to the CLOSE position.

(c) Make sure that each duct pressure indicator shows approximately 10 or more psi to indicate that the Pressure Regulating and Shutoff Valve (PRSOV) is opened.

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CAUTION

DO NOT OPERATE THE WING ANTI-ICE SYSTEM MORE THAN 30 SECONDS. DAMAGE TO THE AIRCRAFT CAN OCCUR.

(d) Put the WING ANTI-ICE switch, on the P5-11 overhead panel, to the ON position.

(e) Make sure that the L and R VALVE annunciator lights comes on 1 to 3 seconds and then goes off.

(f) After no more than 30 seconds, put the WING ANTI-ICE switch to the OFF position.

(g) Make sure that the L and R VALVE annunciator lights comes on 1 to 3 seconds and then goes off.

(h) Put the applicable engine 1 BLEED or 2 BLEED switch, on the P5-10 overhead panel, to the OFF position.

(i) Make sure that the duct pressure indicator goes to zero.

SUBTASK 72-00-00-910-004-G00

(5) Operate the engine at idle for 15 minutes.

(a) Make sure that there are no odors or fumes in the flight compartment and cabin as follows:

1) Put the L and R PACK switches to AUTO.

2) Put the 1 BLEED and 2 BLEED switches to ON.

3) Operate the engine at idle for 15 minutes and do a check for fumes or odors in the flight compartment and cabin.

4) Do a check for smoke or anti-freeze mixture odor in the flight compartment and cabin.

a) If there is no smoke or anti-freeze mixture odor, stop the engine (TASK 71-00-00-910-806-G00).

b) If you find smoke or anti-freeze odor, continue to operate the engine at idle until no odor/smoke remains in the flight compartment or cabin.

c) Put the L and R PACK switches to the OFF position.

d) If the odor/smoke remains in the flight compartment or cabin after 10 minutes engine running, do these steps:

<1> Do this task: Stop the Engine (Usual Engine Stop), TASK 71-00-00-910-806-G00.

<2> Decontaminate the environmental control system (TASK 21-00-01-100-801).

——— END OF TASK ———

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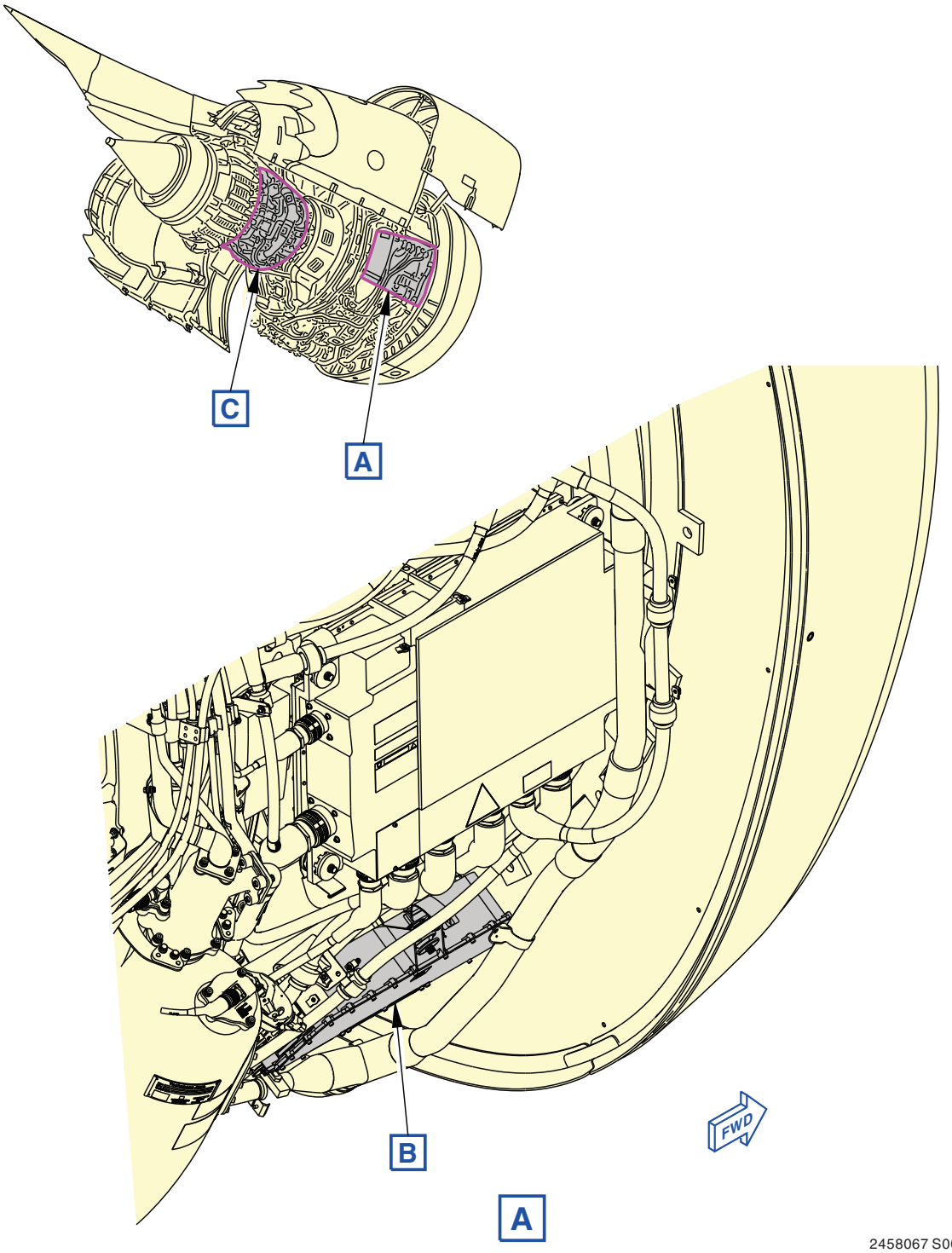

Engine Gas Path General Cleaning
Figure 701/72-00-00-990-853-G00 (Sheet 1 of 2)

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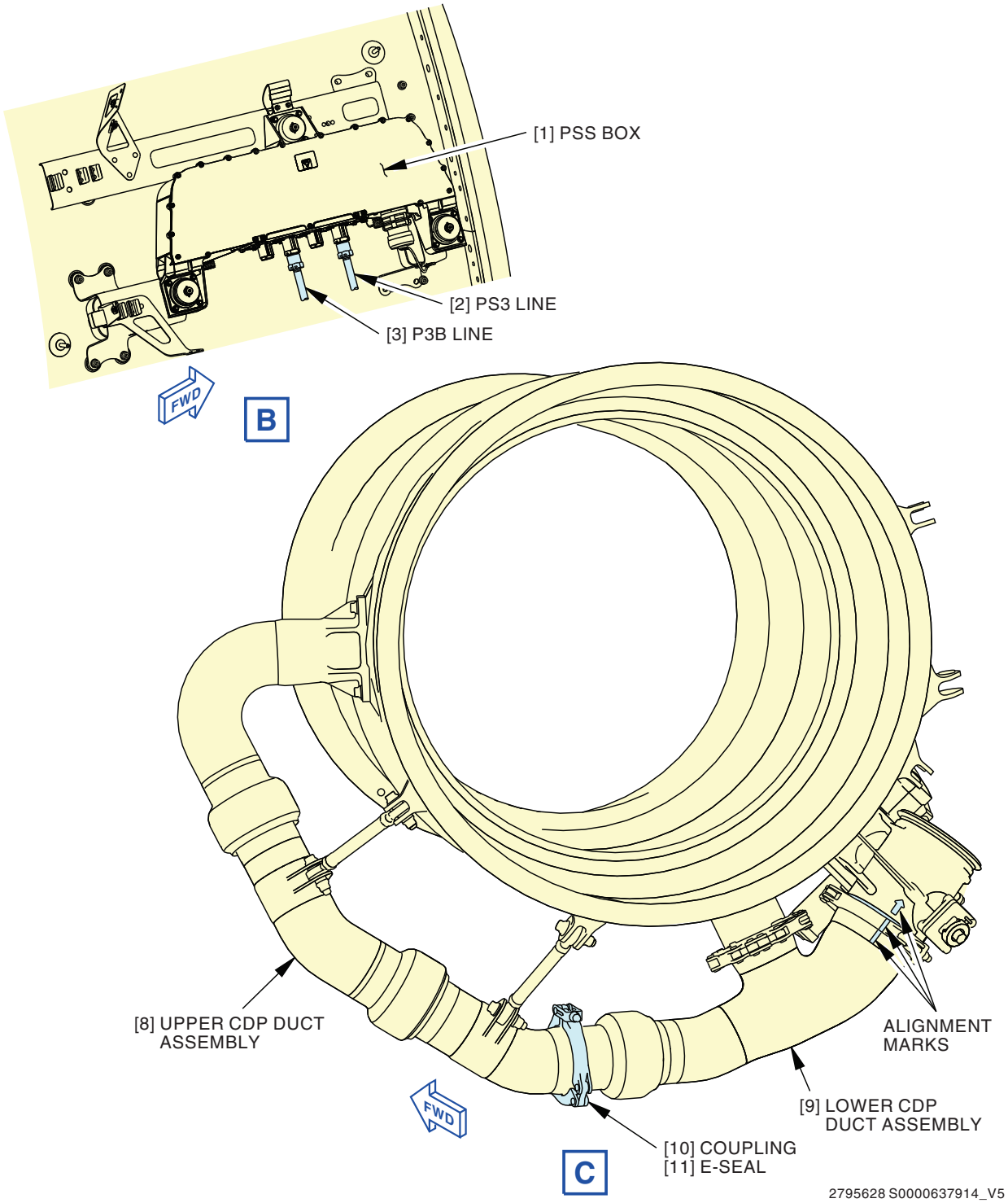

Engine Gas Path General Cleaning
Figure 701/72-00-00-990-853-G00 (Sheet 2 of 2)

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TASK 72-00-00-100-807-G00

4. Engine Gas-Path Cleaning With Cleaning Solution

(Figure 701)

A. General

(1) This task is for engine gas path cleaning with cleaning solution.

B. References

Reference	Title
12-13-11-600-801	Engine Servicing (Oil Replenishing) (P/B 301)
21-00-01-100-801	Oil Contamination Removal from Air Conditioning and Pneumatic Systems (P/B 201)
71-00-00-700-805-G00	Test No. 29 - Water Wash Special Functions Test (P/B 501)
71-00-00-790-801-G00	Test No. 1 - Pneumatic Leak Test (P/B 501)
71-00-00-800-801-G00	Engine Ground Safety Precautions (P/B 201)
71-00-00-800-802-G00	Engine Operation Limits (P/B 201)
71-00-00-910-803-G00	Start the Engine (Normal Start) (P/B 201)
71-00-00-910-806-G00	Stop the Engine (Usual Engine Stop) (P/B 201)
71-00-00-910-808-G00	Dry Motor Procedure (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
73-21-11-000-801-G00	Pressure Sub-System Removal (P/B 401)
73-21-11-400-801-G00	Pressure Sub-System Installation (P/B 401)
74-11-01-400-801-G00	Left Ignition Exciter (Upper) Installation (P/B 401)
74-11-01-400-802-G00	Right Ignition Exciter (Lower) Installation (P/B 401)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-2288	Cart - Water Wash, Engine Part #: 31658 Supplier: 6C9S2 Part #: 35030 Supplier: 6C9S2 Part #: JMP/CFM56/D/4777/C200 Supplier: U2066 Part #: JMP/GE90/D/4038/C200 Supplier: U2066 Part #: PWP2100 Supplier: 6C9S2 Part #: PWP3600 Supplier: 6C9S2 Part #: PWP3700 Supplier: 6C9S2
COM-11191	Probe - Long, Water Wash Part #: JMP/LEAP/D/7185/B Supplier: U2066
COM-14318	TDS - Digital Meter, Handheld, Range 0-9990 PPM, +/-2% Accuracy Part #: TDS-3 Supplier: 6VU32

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Reference	Description
COM-14887	Collection System - Engine Wash Part #: JMP/KYOTO/D/6777/C200 Supplier: U2066
COM-15151	Probe - Short, Water Wash, LEAP-1B Engine Part #: JMP/LEAP/D/7028 Supplier: U2066
SPL-2462	Deactivator - Check Valve Part #: C21006-7 Supplier: 81205 Opt Part #: C21006-1 Supplier: 81205 Opt Part #: PF80-012-500 Supplier: 053H3
SPL-15817	Adapter - Hose, Exhaust Part #: C21011-1 Supplier: 81205
STD-858	Tag - DO NOT OPERATE
STD-3906	Mallet - Rubber
STD-3940	Air Source - Regulated, Dry Filtered, 0-150 psig

D. Consumable Materials

Reference	Description	Specification
G00018	Nitrogen - Gaseous, Pressurizing, 99.5 Percent Pure	A-A-59503 Type I Grade B, MIL-PRF-27401 Type I Grade A

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
11	E-seal	36-11-01-01-220	PTF ALL

F. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

G. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

H. Prepare for the Cleaning

SUBTASK 72-00-00-860-074-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-00-00-010-034-G00

(2) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

SUBTASK 72-00-00-940-007-G00

CAUTION DO NOT CLEAN AN ENGINE WHEN THE OUTSIDE AIR TEMPERATURE IS BELOW 14°F (-10°C). ICE WILL FORM ON THE ENGINE.

(3) Make sure that the outside air temperature is above 14°F (-10°C).

SUBTASK 72-00-00-940-008-G00

CAUTION DO NOT CLEAN THE ENGINE WHEN THE EGT INDICATION IS HOTTER THAN THE SPECIFIED LIMIT. A FAST TEMPERATURE CHANGE CAN CAUSE DAMAGE TO THE ENGINE COMPONENTS.

(4) Make sure that the engine Exhaust Gas Temperature (EGT) is below 150°F (66°C).

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SUBTASK 72-00-00-100-005-G00

(5) Make sure that the water properties, water flow, water temperature, cleaning solutions and anti-ice mixtures are specific for this procedure (TASK 72-00-00-100-801-G00).

SUBTASK 72-00-00-020-005-G00

CAUTION

DO NOT USE METHANOL SOLVENTS TO CLEAN THE ENGINE. DAMAGE TO TITANIUM PARTS CAN OCCUR.

CAUTION

DO NOT CLEAN AN ENGINE AFTER EXPOSURE TO DRY CHEMICAL FIRE EXTINGUISHING AGENTS. DAMAGE TO THE ENGINE CAN OCCUR.

(6) Prepare the engine for washing.

(a) Disconnect the PS3 line [2] and P3B line [3] from the PSS box [1] (TASK 73-21-11-000-801-G00).

1) Install the protective caps on the PS3 line [2] and P3B line [3].

2) Install the protective covers on the PSS box [1].

(b) Remove the E-seal [11] and coupling [10] installed between the upper CDP duct assembly [8] and lower CDP duct assembly [9].

1) Remove the coupling [10].

2) Slightly separate the upper CDP duct assembly [8] and lower CDP duct assembly [9] to remove the E-seal [11].

a) Examine the E-seal [11] for cracks, dents, or other damage.

b) Replace the E-seal [11] if necessary.

CAUTION

YOU MUST COVER OR REMOVE IGNITION EXCITERS BEFORE THE WATER WASH. IGNITION EXCITERS CAN GET MOISTURE THROUGH THE WELDS OF THE EXCITER BOX DURING WATER WASH. IF YOU DO NOT OBEY, DAMAGE TO THE IGNITION EXCITERS CAN OCCUR.

(c) Cover the ignition exciters, wiring, and tubes at the 6:00 O'clock location to protect from excessive water.

(d) If it is necessary, prepare the anti-freeze mixture (TASK 72-00-00-100-801-G00).

(e) Do this task: Engine Servicing (Oil Replenishing), TASK 12-13-11-600-801.

SUBTASK 72-00-00-860-075-G00

(7) On the P5 overhead panel, make sure that the BLEED 1 and BLEED 2 switches are in the OFF position.

SUBTASK 72-00-00-860-076-G00

(8) On the P5 overhead panel, make sure that the L PACK and R PACK switches are in the OFF position.

SUBTASK 72-00-00-860-077-G00

(9) On the P5 overhead panel, make sure that the WING ANTI-ICE, ENG 1 ANTI-ICE and ENG 2 ANTI-ICE switches are in the OFF position.

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SUBTASK 72-00-00-480-020-G00

(10) Install the core wash probes per manufacturer instructions.

NOTE: For the probes installed between the booster and the Fan Blades, the left and right wands must be installed such that it injects the water solution into the booster and is fed from behind the fan outlet guide vanes.

SUBTASK 72-00-00-480-022-G00

(11) Secure wands to the fan bypass.

SUBTASK 72-00-00-480-025-G00

(12) Install an engine wash collection system, COM-14887, below the engine.

NOTE: An engine wash collection system, COM-14887, collects the dirty water that falls from the engine during gas-path cleaning.

NOTE: It is optional to use an engine wash collection system, COM-14887, only if local environmental restrictions allow this. Consult the regulatory guidelines before you make a decision not to use this tool.

I. Engine Gas-Path General Cleaning With Cleaning Solution

SUBTASK 72-00-00-170-012-G00

WARNING MAKE SURE THAT YOU USE ANTI-FREEZE SOLUTIONS CORRECTLY. ANTI-FREEZE SOLUTIONS WITH LOW FLASH POINTS CAN CAUSE A FIRE. INJURY TO PERSONNEL CAN OCCUR.

CAUTION DO NOT CLEAN THE ENGINE WHEN THE EGT INDICATION IS HOTTER THAN THE SPECIFIED LIMIT. A FAST TEMPERATURE CHANGE CAN CAUSE DAMAGE TO THE ENGINE COMPONENTS.

CAUTION DO NOT CLEAN AN ENGINE AFTER EXPOSURE TO DRY CHEMICAL FIRE EXTINGUISHING AGENTS. DAMAGE TO THE ENGINE CAN OCCUR.

(1) Clean the engine gas path with cleaning solution as follows:

NOTE: An engine gas-path water cleaning with detergent gives the most effective method to clean the engine compressors.

NOTE: If you use a cleaning agent, make sure that aircraft maintenance configuration is correct. Make sure that you complete the engine rinse and post engine wash test. This will prevent the possibility of fumes in the cabin when the aircraft returns to service.

NOTE: The engine bleed system must be operational prior to washing the engine with cleaning agent. Do a check to make sure that there is no failure message on the engine bleed system.

(a) Collect a sample of the wash supply water.

(b) Measure the total dissolved solid with the TDS meter, COM-14318, or equivalent.

1) An engine water wash cart, COM-2288, with immersion heater.

NOTE: If you use anti-freeze, do not use the immersion heater.

NOTE: You can use a short wash probe, COM-15151, with the thrust reverser in the open position, and long wash probe, COM-11191, with the thrust reverser in the closed position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CAUTION DO NOT DRY-MOTOR THE ENGINE UNLESS THE INLET FUEL PRESSURE IS POSITIVE. THE FUEL PRESSURE IS NECESSARY TO LUBRICATE THE ENGINE FUEL PUMP AND FUEL METERING UNIT TO PREVENT DAMAGE.

CAUTION DO NOT OPERATE THE ENGINE STARTER MORE THAN THE SPECIFIED LIMITS IN THE DUTY CYCLE. IF YOU OPERATE THE STARTER MORE THAN THE SPECIFIED LIMITS, YOU CAN CAUSE DAMAGE TO THE STARTER OR THE GEARBOX.

CAUTION MAKE SURE THAT THE N1 ROTOR TURNS FREELY DURING THE FULL CLEAN PROCEDURE. IF YOU DO NOT OBEY THIS INSTRUCTION, OIL SYSTEM CONTAMINATION COULD OCCUR.

CAUTION DO NOT APPLY WATER FLOW MORE THAN 5 GPM (19 L/MINUTE). TOO MUCH CAN CAUSE DAMAGE TO THE ENGINE.

(c) Start the Water Wash Special Function on the Multi Function Display (MFD) (TASK 71-00-00-700-805-G00).

NOTE: For the engine starter operation limits, refer to this task: Engine Operation Limits, TASK 71-00-00-800-802-G00.

NOTE: It is necessary for two persons to do this task. One person operates the engine controls. The second person makes sure that personnel are clear when surfaces move and also monitors the engine.

NOTE: This function is conducted as part of the overall water wash procedure. When this function runs, the Variable Stator Vane (VSV) and Variable Bleed Valve (VBV) systems move to facilitate an engine water wash. The function opens the VSVs and closes the VBVs so that the water wash mixture can enter the compressor. This is a dry motoring test.

(d) Follow the instructions on the MFD.

(e) After 10 seconds of motoring with a positive fan rotation, inject the water on the Low Pressure Compressor (LPC) inlet through the fan blades (or at the LPC inlet if you will use the short wash probe, COM-15151 and engine water wash cart, COM-2288).

(f) Stop the water injection before the end of the engine motoring.

(g) Do these steps:

1) During the engine coast-down, collect a sample of the wash water that drains from the exhaust area of the engine.

2) Use a TDS meter, COM-14318, or equivalent to measure and make a record of the total dissolved solid.

NOTE: Calibration of the Total Dissolved Solids (TDS) meter is not necessary if the same TDS meter is used to collect the sample of the wash supply water.

3) Subtract the initial TDS value measured from the second TDS value in the above step.

4) If the value is more than 150 Parts Per Million (PPM), do this step:

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a) Do additional wash/soak cycles until the TDS delta falls below 150 PPM.

NOTE: Up to a maximum of four wash cycles.

(h) Let the water soak for 10-15 minutes between cycles.

NOTE: The 10-15 minutes between cycles will maximize the cleaning of the engine.

(i) Do the steps above again as necessary up to a maximum of four times to dry motor, inject water and soak the engine.

NOTE: You may do this step up to maximum of four times for engines with heavy dirt deposits.

(j) Start to dry motor the engine again (TASK 71-00-00-910-808-G00).

SUBTASK 72-00-00-160-003-G00

(2) Rinse the cleaning solution from the engine gas path as follows:

NOTE: This will prevent fumes in the flight compartment and cabin when the airplane returns to service.

(a) Dry motor the engine for two minutes (TASK 71-00-00-910-808-G00).

(b) After 10 seconds of motoring with positive fan rotation, do this step:

1) Inject water 360 degrees around the LPC inlet through the fan blades.

NOTE: If you use an engine water wash cart, COM-2288 and short wash probe, COM-15151, you can inject the water directly at the LPC inlet during motoring.

(c) Let the engine soak for 5 minutes.

(d) Dry motor the engine again for two minutes (TASK 71-00-00-910-808-G00).

(e) After 10 seconds of motoring with positive fan rotation, do this step:

1) Inject water 360 degrees around the LPC inlet through the fan blades.

NOTE: If you use an engine water wash cart, COM-2288 and short wash probe, COM-15151, you can inject the water directly at the LPC inlet during motoring.

(f) Let the engine soak for 5 minutes.

(g) When the engine has stopped, select “STOP TEST” on the MFD to stop the water wash special function.

J. Put the Airplane Back to Its Usual Condition

SUBTASK 72-00-00-080-024-G00

(1) Remove an engine wash collection system, COM-14887.

SUBTASK 72-00-00-010-040-G00

(2) Remove the cover from the ignition exciter and wiring if it was covered.

SUBTASK 72-00-00-400-002-G00

(3) Reinstall the left or right ignition exciter if it was removed (TASK 74-11-01-400-801-G00 or TASK 74-11-01-400-802-G00).

SUBTASK 72-00-00-160-004-G00

(4) Clean the PS3 line [2] and P3B line [3] as follows:

(a) Remove the protective caps from the PS3 line [2] and P3B line [3].

(b) Remove the protective covers from the PSS box [1].

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(c) Use the regulated air source, STD-3940, or nitrogen, G00018, to blow into the PS3 line [2] and clear it of water.

1) Use approximately 100 psig (689 kPa) (not to exceed 200 psig (1379 kPa)) to dry the air tubes.

a) Blow air into the air tubes for a minimum of 3 minutes.

NOTE: For the best result, compressed nitrogen is recommended.

(d) Use the regulated air source, STD-3940, or nitrogen, G00018, to blow into the P3B line [3] and clear it of water.

1) Use approximately 100 psig (689 kPa) (not to exceed 200 psig (1379 kPa)) to dry the air tubes.

a) Blow air into the air tubes for a minimum of 3 minutes.

NOTE: For the best result, compressed nitrogen is recommended.

(e) Do a visual inspection of the weep holes on the PS3 line [2] and P3B line [3] to make sure that they are clear of all blockage.

CAUTION

USE TWO WRENCHES TO TIGHTEN THE COUPLING NUT. USE ONE TO HOLD THE FITTING, AND THE OTHER TO TURN THE COUPLING NUT. IF YOU DO NOT USE TWO WRENCHES, DAMAGE TO THE EQUIPMENT CAN OCCUR.

(f) Connect the PS3 line [2] and P3B line [3] to the PSS box [1] (TASK 73-21-11-400-801-G00).

(g) Install the E-seal [11] and coupling [10].

1) Turn the coupling [10] up, to give the maximum clearance with the ignition lead.

2) Examine the E-seal [11] for cracks, dents, or other damage.

3) Replace the E-seal [11] if necessary.

4) Install the E-seal [11] and coupling [10] between the upper CDP duct assembly [8] and lower CDP duct assembly [9].

a) Tighten the coupling [10] to 170 in-lb (19.2 N·m) - 180 in-lb (20.3 N·m).

b) Lightly tap the outer surface of the coupling [10] with a rubber mallet, STD-3906, or equivalent.

c) Tighten the coupling [10] to 170 in-lb (19.2 N·m) - 180 in-lb (20.3 N·m) again.

SUBTASK 72-00-00-010-042-G00

(5) Do these steps to remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

SUBTASK 72-00-00-790-001-G00

(6) To check for leakage from the E-seal [11] between the upper CDP duct assembly [8] and lower CDP duct assembly [9], do this task: Test No. 1 - Pneumatic Leak Test, TASK 71-00-00-790-801-G00, for the affected duct only.

SUBTASK 72-00-00-010-041-G00

(7) Do these steps to remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch:

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(a) On the P10 control standl, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

SUBTASK 72-00-00-410-028-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(8) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

K. Flush Cleaning Solution From the Lube, Sumps, and Bleed Systems.

SUBTASK 72-00-00-170-015-G00

(1) To prevent fumes in the cockpit and cabin when the airplane is put back into service, make sure that you do the flush procedures.

SUBTASK 72-00-00-160-005-G00

(2) Do one of these steps to make sure that there is an exit for the conditioned air:

(a) Make sure that the cabin pressure outflow valve is opened.

(b) Make sure that at least one passenger entry door is opened.

SUBTASK 72-00-00-860-101-G00

(3) Do this task: Engine Ground Safety Precautions, TASK 71-00-00-800-801-G00.

SUBTASK 72-00-00-860-079-G00

WARNING MAKE SURE THAT ALL PERSONNEL AND EQUIPMENT ARE CLEAR OF THE ENGINE DANGER AREAS. THE ENGINE INTAKE AND EXHAUST CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT IN THE DANGER AREAS.

(4) Remove the water ingested into the engine systems during cleaning as follows:

(a) If there is no icing condition, start the engine not more than 2 hours after you cleaned and serviced the engine (TASK 71-00-00-910-803-G00).

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(b) If there is icing condition, start the engine not more than 30 minutes after you cleaned and serviced the engine (TASK 71-00-00-910-803-G00).

SUBTASK 72-00-00-710-001-G00

(5) Operate the engine at idle for 5 minutes.

SUBTASK 72-00-00-170-013-G00

(6) Do these steps to flush the inlet cowl anti-ice system:

(a) Put the APU BLEED switch, on the P5-10 overhead panel, to the OFF position.

1) Make sure that the DUAL BLEED light goes out.

(b) Make sure that the applicable BLEED switch, on the P5-10 overhead panel, is in the OFF position.

(c) Put the ISOLATION VALVE switch, on the P5-10 overhead panel, to the CLOSE position.

(d) Move the applicable thrust lever to 60 percent N1.

NOTE: Do not increase the engine power to more than 80 percent N1.

(e) Put the applicable ENG ANTI-ICE 1 or ENG ANTI-ICE 2 switch, on the P5-11 overhead panel, to the ON position.

1) Make sure that the applicable COWL VALVE light comes on brightly for one to three seconds, and then goes dim.

CAUTION

DO NOT KEEP THE COWL TAI VALVE OPEN FOR MORE THAN THE SPECIFIED TIME WHEN THE TEMPERATURE IS MORE THAN THE SPECIFIED LIMIT. DAMAGE TO THE INLET COWL CAN OCCUR.

(f) After not more than 30 seconds, put the applicable ENG ANTI-ICE 1 or ENG ANTI-ICE 2 switch, on the P5-11 overhead panel, to the OFF position.

1) Make sure that the applicable COWL VALVE light comes on brightly for one to three seconds, and then goes off.

(g) Move the applicable thrust lever to the IDLE position.

1) Operate the engine at idle for five minutes.

SUBTASK 72-00-00-170-014-G00

(7) Flush the wing anti-ice system as follows:

(a) Put the applicable BLEED switch, on the P5-10 overhead panel, to the ON position.

(b) Do one of the following steps on the P5-10 overhead panel:

1) If one engine operates, put the ISOLATION VALVE switch to the OPEN position.

2) If both engines operate, put the ISOLATION VALVE switch to the CLOSE position.

(c) Make sure that each duct pressure indicator show approximately 10 or more psi to indicate that the Pressure Regulating and Shutoff Valve (PRSOV) is opened.

CAUTION

DO NOT OPERATE THE WING ANTI-ICE SYSTEM MORE THAN 30 SECONDS. DAMAGE TO THE AIRCRAFT CAN OCCUR.

(d) Put the WING ANTI-ICE switch, on the P5-11 overhead panel, to the ON position.

(e) Make sure that the L and R VALVE annunciator lights comes on 1 to 3 seconds and then goes off.

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(f) After no more than 30 seconds, put the WING ANTI-ICE switch to the OFF position.

(g) Make sure that the L and R VALVE annunciator lights comes on 1 to 3 seconds and then goes off.

(h) Put the applicable engine 1 BLEED or 2 BLEED switch, on the P5-10 Panel, to the OFF position.

(i) Make sure that the duct pressure indicator goes to zero.

SUBTASK 72-00-00-410-036-G00

WARNING MAKE SURE THAT ALL PERSONNEL AND EQUIPMENT ARE CLEAR OF THE ENGINE DANGER AREAS. THE ENGINE INTAKE AND EXHAUST CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT IN THE DANGER AREAS.

WARNING OBEY THE PRECAUTIONS FOR ENTRY/EXIT CORRIDORS AND ENGINE INLET HAZARD AREAS. FAILURE TO OBEY WARNINGS COULD RESULT IN INJURY OR DEATH.

WARNING PUT ON GLOVES FOR PROTECTION FROM HOT SURFACES WHEN YOU CONNECT OR DISCONNECT PNEUMATIC FITTINGS. PACK OPERATION CAN MAKE THE GROUND AIR CONNECTOR VERY HOT. A HOT GROUND AIR CONNECTOR WILL BURN YOUR BARE HANDS.

(8) Install the deactivator and exhaust hose adapter, SPL-15817, in the ground air connector as follows:

NOTE: This step is not mandatory. Its aim is to prevent fumes in the cockpit and cabin when the airplane is put back into service. Use of the deactivator helps decrease contamination of the bleed system during the engine wash.

(a) Put the deactivator and exhaust hose adapter, SPL-15817, in the ground air connector.

(b) Make sure that the check valve flappers are open.

(c) If it is necessary to use deactivator, SPL-2462, install the coupling on the supply hose from the ground air cart onto the nipple of the ground air connector check valve assembly.

(d) Make sure that the deactivator, SPL-2462, is in the correct position.

1) Make sure that you do not connect the supply hose to the ground cart.

2) Make sure that the exhaust from the hose flows to ambient.

SUBTASK 72-00-00-860-098-G00

(9) If the engine 1 was cleaned, do the steps that follow:

(a) Put the ISOLATION VALVE switch, on the P5-10 forward overhead panel, to the OPEN position.

(b) Put the engine 1 BLEED switch, on the P5-10 forward overhead panel, to the ON position.

(c) Operate the engine at idle for 5 minutes.

SUBTASK 72-00-00-860-099-G00

(10) If the engine 2 was cleaned, do the steps that follow:

(a) Put the ISOLATION VALVE switch, on the P5-10 forward overhead panel, to the CLOSED position.

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(b) Put the engine 2 BLEED switch, on the P5-10 forward overhead panel, to the ON position.

(c) Operate the engine at idle for 5 minutes.

SUBTASK 72-00-00-860-100-G00

(11) Put the applicable engine 1 BLEED or 2 BLEED switch, on the P5-10 forward overhead panel, to the OFF position.

SUBTASK 72-00-00-910-007-G00

WARNING MAKE SURE THAT ALL PERSONNEL AND EQUIPMENT ARE CLEAR OF THE ENGINE DANGER AREAS. THE ENGINE INTAKE AND EXHAUST CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT IN THE DANGER AREAS.

WARNING OBEY THE PRECAUTIONS FOR ENTRY/EXIT CORRIDORS AND ENGINE INLET HAZARD AREAS. FAILURE TO OBEY WARNINGS COULD RESULT IN INJURY OR DEATH.

WARNING PUT ON GLOVES FOR PROTECTION FROM HOT SURFACES WHEN YOU CONNECT OR DISCONNECT PNEUMATIC FITTINGS. PACK OPERATION CAN MAKE THE GROUND AIR CONNECTOR VERY HOT. A HOT GROUND AIR CONNECTOR WILL BURN YOUR BARE HANDS.

(12) If the exhaust hose adapter, SPL-15817, is installed, do the steps that follow:

(a) Disconnect the exhaust hose adapter, SPL-15817, from the ground air connector and remove the deactivator.

SUBTASK 72-00-00-020-009-G00

(13) If the deactivator, SPL-2462, is used, disconnect the coupling on the supply hose from the ground air connector and remove the deactivator.

SUBTASK 72-00-00-910-006-G00

(14) Operate the engine at idle for 15 minutes.

(a) Make sure that there are no odors or fumes in the flight compartment and cabin as follows:

1) Put the L and R PACK switches to AUTO.

2) Put the 1 BLEED and 2 BLEED switches to ON.

3) Operate the engine at idle for 15 minutes and do a check for fumes or odors in the flight compartment and cabin.

4) Do a check for smoke or anti-freeze mixture odor in the flight compartment and cabin.

a) If there is no smoke or anti-freeze mixture odor, stop the engine (TASK 71-00-00-910-806-G00).

b) If you find smoke or anti-freeze odor, continue to operate the engine at idle until no odor/smoke remains in the flight compartment or cabin.

c) Put the L and R PACK switches to the OFF position.

d) If the odor/smoke remains in the flight compartment or cabin after 10 minutes engine running, do these steps:

<1> Do this task: TASK 71-00-00-910-806-G00.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

<2> Decontaminate the environmental control system (TASK 21-00-01-100-801).

——— END OF TASK ———

TASK 72-00-00-100-806-G00

5. Engine External Surfaces Cleaning

A. General

(1) This task provides the instructions on how to clean the engine external surfaces.

B. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

C. Consumable Materials

Reference	Description	Specification
B00067	Alcohol - Denatured, Ethyl (Ethanol)	AMS 3002 (Supersedes O-E-760)
B00676 [CP1041]	Alcohol - Isopropyl	
B01058 [CP1039]	Solvent - Acetone, Reagent Grade	
B50190 [CP1040]	Solvent - Methyl Ethyl Ketone (MEK)	ASTM D740

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Prepare for the Cleaning

SUBTASK 72-00-00-860-062-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

F. Engine External Surfaces Cleaning

SUBTASK 72-00-00-100-002-G00

WARNING HYDROCARBONS ARE FLAMMABLE AND TOXIC. AVOID PROLONGED CONTACT WITH THE SKIN AND OBSERVE PRECAUTIONS AGAINST FIRE. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

CAUTION SOME PLASTICS AND RUBBER-BASE MATERIALS ARE DAMAGED BY HYDROCARBON SOLVENTS. IF YOU ARE NOT CAREFUL, DAMAGE TO THE ENGINE COULD OCCUR.

(1) Clean the external surfaces of the engine as follows:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Hydrocarbon method

1) Light quantities of grease, oil, gum and dirt can be removed as follows:

NOTE: This method is not good in the removal of baked-on oil deposits.

a) For metallic parts without coating or component made of non metallic material:

CAUTION

DO NOT USE ETHANOL ON TITANIUM PARTS OR DAMAGE TO PARTS WILL OCCUR.

<1> Clean manually with a cloth moist with only the acetone solvent, B01058 [CP1039] or the solvent, B50190 [CP1040] or the alcohol, B00676 [CP1041] or the alcohol, B00067.

b) For all the other parts:

<1> Clean manually with a cloth moist with only the alcohol, B00676 [CP1041] to permit handling and inspection.

<2> Remove the accumulation with a soft cloth.

<3> Wipe the cleaned area dry with a clean cloth or rinse with clean water.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 72-00-00-860-063-G00

(1) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

TASK 72-00-00-100-811-G00

6. Exhaust Tube Support - Cleaning

(Figure 702)

A. General

(1) This task gives instructions to clean the exhaust tube support.

B. References

Reference	Title
72-50-03-200-801-G00	Turbine Rear Vane Module Inspection (P/B 601)
72-59-00-000-801-G00	Injection Nozzle Removal (P/B 401)
72-59-00-400-801-G00	Injection Nozzle Installation (P/B 401)

C. Consumable Materials

Reference	Description	Specification
B00676 [CP1041]	Alcohol - Isopropyl	
B50415 [CP2829]	Degreaser - Cleaner, Aviation 2000	
B50416 [CP2828]	Cleaner - Solvent, SkyKleen 1000 Aviation Solvent	
B50417 [CP2827]	Cleaner - Solvent, Discovert DPLF	

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(Continued)

Reference	Description	Specification
B50434 [CP2841]	Solvent - Cleaning, SOCOSOLV A3582	
D50186 [CP2691]	Fluid - Penetrating - Aerokroil	
G51328 [CP2674]	Cloth - Lint-Free	

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Prepare for the Cleaning

WARNING REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THESE PRODUCTS.

WARNING BE CAREFUL WHEN YOU DO WORK ON THE ENGINE PARTS AFTER YOU STOP THE ENGINE. THE ENGINE PARTS CAN STAY HOT FOR ALMOST 1 HOUR. DO NOT TOUCH HOT PARTS WITHOUT APPLICABLE GLOVES. HOT PARTS CAN CAUSE INJURIES TO PERSONNEL.

WARNING USE PROTECTIVE GLOVES TO PREVENT INJURY FROM SHARP EDGES AND ROUGH SURFACES. IF YOU DO NOT OBEY, INJURY TO PERSONNEL CAN OCCUR.

CAUTION DO NOT CLEAN AN ENGINE IF DRY CHEMICAL FIRE EXTINGUISH AGENTS APPLIED. IF YOU DO NOT OBEY, DAMAGE TO ENGINE TITANIUM PARTS CAN OCCUR.

CAUTION DO NOT CLEAN AN ENGINE UNLESS THE EGT IS BELOW 150°F (66°C). IF YOU DO NOT OBEY, DAMAGE TO ENGINE TITANIUM PARTS CAN OCCUR.

CAUTION DO NOT CLEAN AN ENGINE WHEN THE AIR TEMPERATURE IS BELOW 14°F (-10°C). IF YOU DO NOT OBEY, DAMAGE TO ENGINE TITANIUM PARTS CAN OCCUR.

SUBTASK 72-00-00-000-001-G00

(1) Remove the injection nozzle, do this task: Injection Nozzle Removal, TASK 72-59-00-000-801-G00.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 72-00-00-840-004-G00

CAUTION

APPLY PROTECTION ON PART THAT YOU CLEAN AND ADJACENT PARTS. USE PROTECTION TO PREVENT SPRAYS AROUND THE AREA. IF SPRAYS APPLIED ON THESE PARTS, IMMEDIATELY FLUSH THE AREA WITH ISOPROPYL ALCOHOL AND CLEAN-LINT-FREE CLOTH. IF YOU DO NOT OBEY, DAMAGE TO THE PART CAN OCCUR.

(2) Protect adjacent parts of the exhaust tube support [21].

NOTE: It is recommended to use plastic tarpaulins or plastic sheets wrapped around the part. It is not recommended to use any kind of protection that can stay attached to the part.

(a) If you use penetrating fluid, D50186 [CP2691], do these steps:

1) Protect the 16 inserts of the LPT rear frame [22] (View C, Figure 702).

2) Protect the 2 inserts of the exhaust tube support [21] (View B, Figure 702).

F. Exhaust Tube Support Cleaning

SUBTASK 72-00-00-110-003-G00

(1) Do these steps to remove coking:

CAUTION

DO NOT TOUCH THE SURFACE OF THE PART WITH A MECHANICAL TOOL. IF YOU DO NOT OBEY, DAMAGE TO THE SURFACE OF THE PART CAN OCCUR.

(a) If it is necessary, remove coking with a mechanical tool.

NOTE: Use a mechanical tool to remove the bulk of coking. It is permitted to leave a thin layer of coking. The thin layer of coking will be removed later by a plastic tool.

CAUTION

THESE MATERIALS INCLUDE PETROLEUM. DO NOT APPLY THEM TO A HOT ENGINE TO PREVENT RISK OF FIRE. IF YOU DO NOT OBEY, FIRE ON THE ENGINE CAN OCCUR.

(b) Select only one of the following materials to clean the exhaust tube support [21]:

- SkyKleen 1000 solvent, B50416 [CP2828]
- Aviation 2000 degreaser, B50415 [CP2829]
- penetrating fluid, D50186 [CP2691]
- Discovert DPLF solvent, B50417 [CP2827]
- SOCOSOLV A3582 solvent, B50434 [CP2841].

(c) Make sure that the adjacent parts of the exhaust tube support [21] are protected.

1) If you use penetrating fluid, D50186 [CP2691], make sure that the inserts of the LPT rear frame [22] and inserts of the exhaust tube support [21] are protected.

CAUTION

DO NOT USE THE MATERIALS ON THE SURFACE FOR MORE THAN MAXIMUM PERMITTED TIME. MATERIALS MUST NOT BE MORE THAN 6 HOURS ON AREAS THAT YOU CLEAN. IF YOU DO NOT OBEY, DAMAGE TO THE PART CAN OCCUR.

(d) Apply one of the materials on the exhaust tube support [21].

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1) If penetrating fluid, D50186 [CP2691], contacts the inserts of the LPT rear frame [22] or inserts of the exhaust tube support [21], clean the inserts as soon as possible with alcohol, B00676 [CP1041].
   a) Make sure that penetrating fluid, D50186 [CP2691], is fully removed from the inserts.

CAUTION DO NOT LET PARTICLES FROM TOOLS MADE FROM PLASTIC STAY ON THE PART. PARTICLES FROM TOOLS MADE OF PLASTIC CAN CAUSE DAMAGE TO THE PART.

(e) Remove as much as possible of the remaining coking with a type plastic spatula or plastic brush.

SUBTASK 72-00-00-110-002-G00

CAUTION DO NOT USE ISOPROPYL ALCOHOL MORE THAN 6 HOURS ON THE PARTS THAT YOU CLEAN. IF YOU DO NOT OBEY, DAMAGE TO THE PART CAN OCCUR.

(2) Clean the exhaust tube support [21] with alcohol, B00676 [CP1041], and lint-free cloth, G51328 [CP2674].

SUBTASK 72-00-00-200-002-G00

(3) Inspect the exhaust tube support [21] (TASK 72-50-03-200-801-G00).
   (a) Make sure that there is no damage on the surface of the exhaust tube support [21].

G. Put the Airplane Back to Its Usual Condition

SUBTASK 72-00-00-840-005-G00

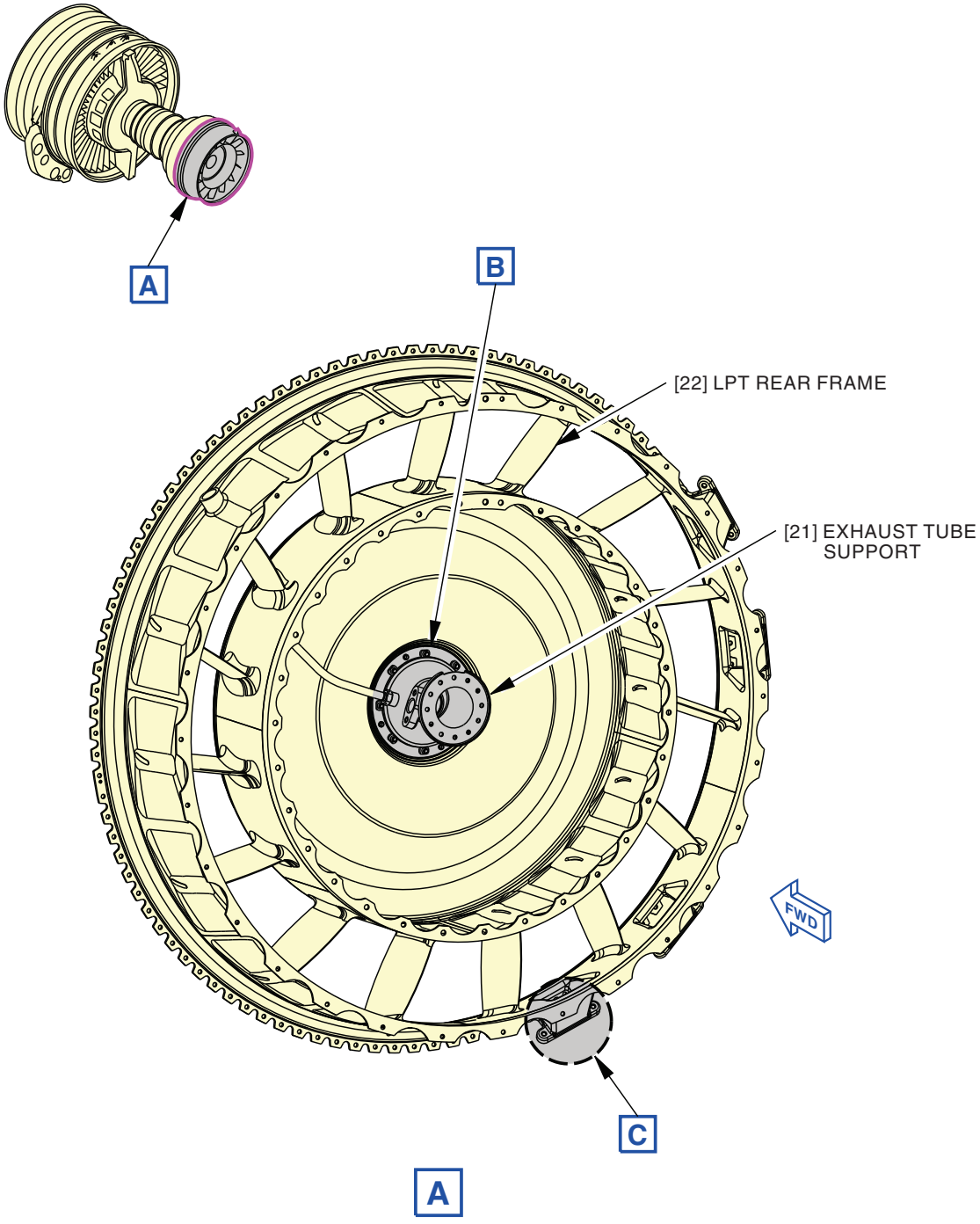
(1) Remove the protection from the adjacent parts of the exhaust tube support [21].
   (a) If it is installed, remove the protection from the inserts of the LPT rear frame [22] and inserts of the exhaust tube support [21].

SUBTASK 72-00-00-400-003-G00

(2) Install the injection nozzle, do this task: Injection Nozzle Installation, TASK 72-59-00-400-801-G00.

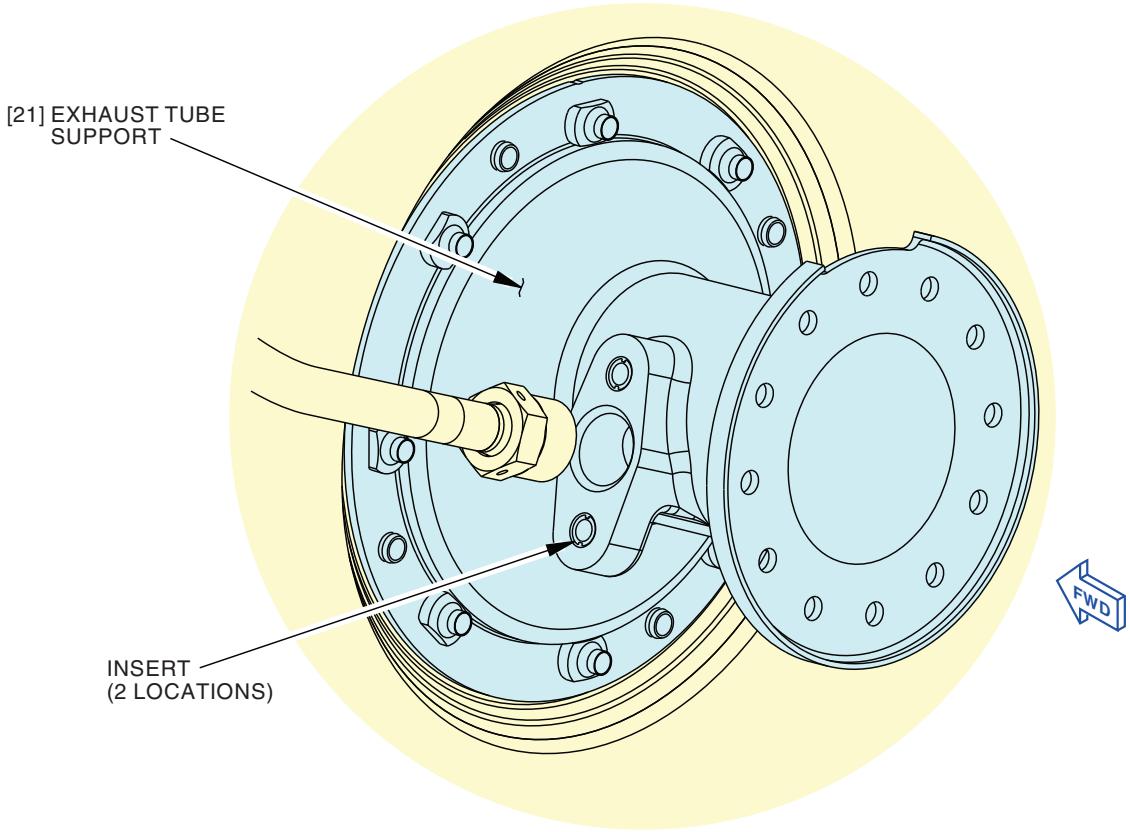
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Exhaust Tube Support Cleaning
Figure 702/72-00-00-990-868-G00 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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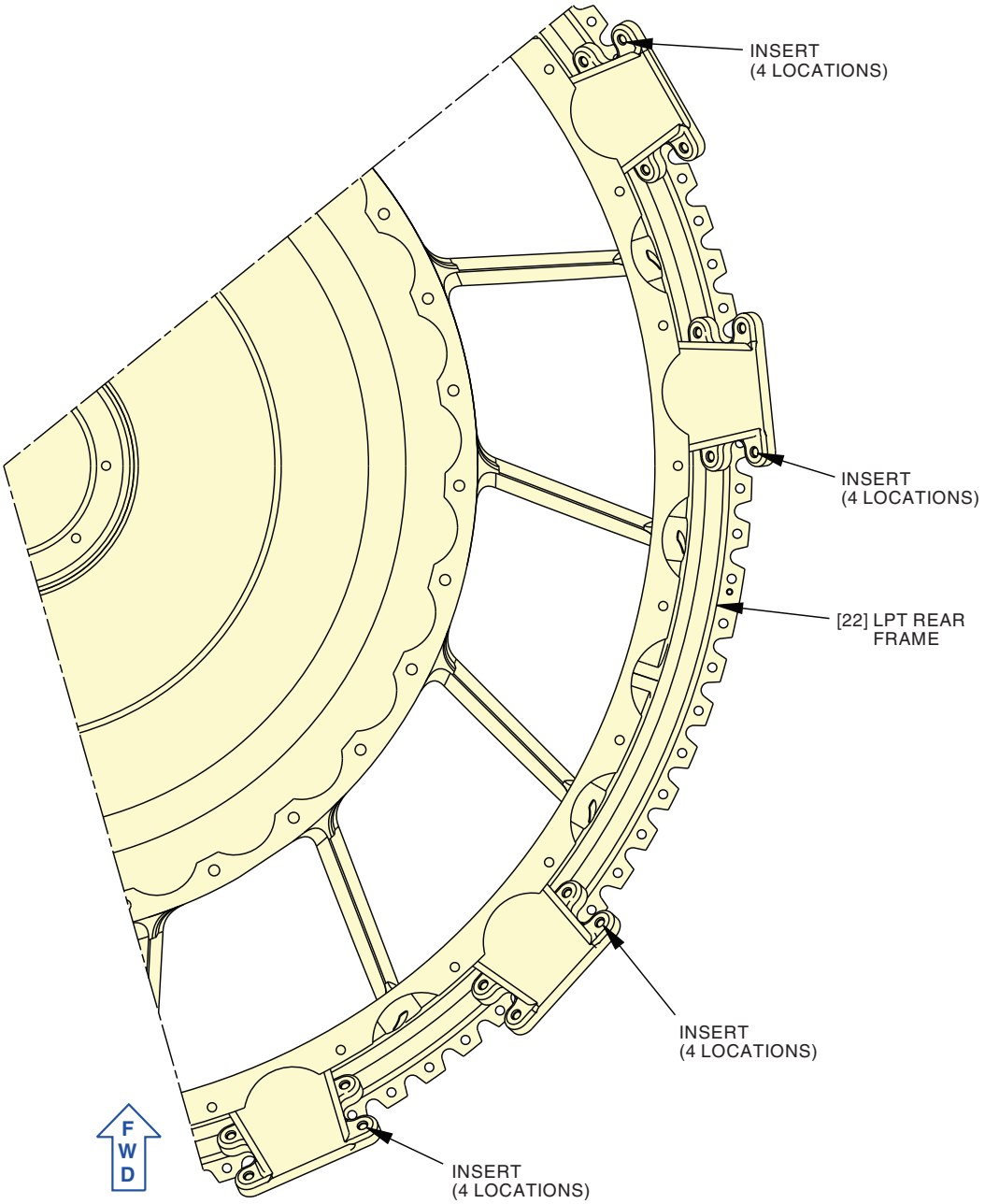
Exhaust Tube Support Cleaning
Figure 702/72-00-00-990-868-G00 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Exhaust Tube Support Cleaning
Figure 702/72-00-00-990-868-G00 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 72-00-00-100-812-G00

7. Injection Nozzle - Cleaning

A. General

(1) This task gives instructions to clean the injection nozzle.

B. References

Reference	Title
72-50-03-200-801-G00	Turbine Rear Vane Module Inspection (P/B 601)
72-59-00-000-801-G00	Injection Nozzle Removal (P/B 401)
72-59-00-400-801-G00	Injection Nozzle Installation (P/B 401)

C. Consumable Materials

Reference	Description	Specification
B00676 [CP1041]	Alcohol - Isopropyl	
B50415 [CP2829]	Degreaser - Cleaner, Aviation 2000	
B50416 [CP2828]	Cleaner - Solvent, SkyKleen 1000 Aviation Solvent	
B50417 [CP2827]	Cleaner - Solvent, Discovert DPLF	
B50434 [CP2841]	Solvent - Cleaning, SOCOSOLV A3582	
D50186 [CP2691]	Fluid - Penetrating - Aerokroil	
G51328 [CP2674]	Cloth - Lint-Free	

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Injection Nozzle Cleaning

WARNING REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THESE PRODUCTS.

WARNING BE CAREFUL WHEN YOU DO WORK ON THE ENGINE PARTS AFTER YOU STOP THE ENGINE. THE ENGINE PARTS CAN STAY HOT FOR ALMOST 1 HOUR. DO NOT TOUCH HOT PARTS WITHOUT APPLICABLE GLOVES. HOT PARTS CAN CAUSE INJURIES TO PERSONNEL.

WARNING USE PROTECTIVE GLOVES TO PREVENT INJURY FROM SHARP EDGES AND ROUGH SURFACES. IF YOU DO NOT OBEY, INJURY TO PERSONNEL CAN OCCUR.

CAUTION DO NOT CLEAN AN ENGINE IF DRY CHEMICAL FIRE EXTINGUISH AGENTS APPLIED. IF YOU DO NOT OBEY, DAMAGE TO ENGINE TITANIUM PARTS CAN OCCUR.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(CAUTION PRECEDES)

CAUTION

DO NOT CLEAN AN ENGINE UNLESS THE EGT IS BELOW 150°F (66°C). IF YOU DO NOT OBEY, DAMAGE TO ENGINE TITANIUM PARTS CAN OCCUR.

CAUTION

DO NOT CLEAN AN ENGINE WHEN THE AIR TEMPERATURE IS BELOW 14°F (-10°C). IF YOU DO NOT OBEY, DAMAGE TO ENGINE TITANIUM PARTS CAN OCCUR.

SUBTASK 72-00-00-000-004-G00

(1) Remove the injection nozzle, do this task: Injection Nozzle Removal, TASK 72-59-00-000-801-G00.

SUBTASK 72-00-00-100-006-G00

(2) Clean the injection nozzle, do these steps:

CAUTION

DO NOT TOUCH THE SURFACE OF THE PART WITH A MECHANICAL TOOL. IF YOU DO NOT OBEY, DAMAGE TO THE SURFACE OF THE PART CAN OCCUR.

(a) If it is necessary, remove coking with a mechanical tool.

NOTE: Use a mechanical tool to remove the bulk of coking. It is permitted to leave a thin layer of coking. The thin layer of coking will be removed later by a plastic tool.

CAUTION

THESE MATERIALS INCLUDE PETROLEUM. DO NOT APPLY THEM TO A HOT ENGINE TO PREVENT RISK OF FIRE. IF YOU DO NOT OBEY, FIRE ON THE ENGINE CAN OCCUR.

(b) Select only one of the following product to clean the injection nozzle:

- SkyKleen 1000 solvent, B50416 [CP2828]
- Aviation 2000 degreaser, B50415 [CP2829]
- penetrating fluid, D50186 [CP2691]
- Discovert DPLF solvent, B50417 [CP2827]
- SOCOSOLV A3582 solvent, B50434 [CP2841].

CAUTION

DO NOT USE THE MATERIALS ON THE SURFACE FOR MORE THAN MAXIMUM PERMITTED TIME. MATERIALS MUST NOT BE MORE THAN 6 HOURS ON AREAS THAT YOU CLEAN. IF YOU DO NOT OBEY, DAMAGE TO THE PART CAN OCCUR.

(c) Apply one of the cleaning products on the injection nozzle.

CAUTION

DO NOT LET PARTICLES FROM TOOLS MADE FROM PLASTIC STAY ON THE PART. PARTICLES FROM TOOLS MADE OF PLASTIC CAN CAUSE DAMAGE TO THE PART.

(d) Remove as much as possible the remaining coking with a type plastic spatula or plastic brush.

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CAUTION

DO NOT USE ISOPROPYL ALCOHOL MORE THAN 6 HOURS ON THE PARTS THAT YOU CLEAN. IF YOU DO NOT OBEY, DAMAGE TO THE PART CAN OCCUR.

(e) Clean the injection nozzle with lint-free cloth, G51328 [CP2674], that is moist with alcohol, B00676 [CP1041].

SUBTASK 72-00-00-200-005-G00

(3) Inspect the injection nozzle (TASK 72-50-03-200-801-G00).

(a) Make sure that there is no damage on the surface of the injection nozzle.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 72-00-00-400-006-G00

(1) Install the injection nozzle, do this task: Injection Nozzle Installation, TASK 72-59-00-400-801-G00.

——— END OF TASK ———

TASK 72-00-00-100-813-G00

8. Exhaust Air Tube - Cleaning

(Figure 703)

A. General

(1) This task gives instructions to clean the exhaust air tube.

B. References

Reference	Title
72-50-03-200-801-G00	Turbine Rear Vane Module Inspection (P/B 601)
72-59-00-000-801-G00	Injection Nozzle Removal (P/B 401)
72-59-00-400-801-G00	Injection Nozzle Installation (P/B 401)

C. Consumable Materials

Reference	Description	Specification
B00676 [CP1041]	Alcohol - Isopropyl	
B50415 [CP2829]	Degreaser - Cleaner, Aviation 2000	
B50416 [CP2828]	Cleaner - Solvent, SkyKleen 1000 Aviation Solvent	
B50417 [CP2827]	Cleaner - Solvent, Discovert DPLF	
B50434 [CP2841]	Solvent - Cleaning, SOCOSOLV A3582	
D50186 [CP2691]	Fluid - Penetrating - Aerokroil	
G51328 [CP2674]	Cloth - Lint-Free	

D. Location Zones

Zone	Area
411	Engine 1 - Engine
412	Engine 1 - Nose Inlet Cowl

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Prepare for the Cleaning

WARNING REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THESE PRODUCTS.

WARNING BE CAREFUL WHEN YOU DO WORK ON THE ENGINE PARTS AFTER YOU STOP THE ENGINE. THE ENGINE PARTS CAN STAY HOT FOR ALMOST 1 HOUR. DO NOT TOUCH HOT PARTS WITHOUT APPLICABLE GLOVES. HOT PARTS CAN CAUSE INJURIES TO PERSONNEL.

WARNING USE PROTECTIVE GLOVES TO PREVENT INJURY FROM SHARP EDGES AND ROUGH SURFACES. IF YOU DO NOT OBEY, INJURY TO PERSONNEL CAN OCCUR.

CAUTION DO NOT CLEAN AN ENGINE IF DRY CHEMICAL FIRE EXTINGUISH AGENTS APPLIED. IF YOU DO NOT OBEY, DAMAGE TO ENGINE TITANIUM PARTS CAN OCCUR.

CAUTION DO NOT CLEAN AN ENGINE UNLESS THE EGT IS BELOW 150°F (66°C). IF YOU DO NOT OBEY, DAMAGE TO ENGINE TITANIUM PARTS CAN OCCUR.

CAUTION DO NOT CLEAN AN ENGINE WHEN THE AIR TEMPERATURE IS BELOW 14°F (-10°C). IF YOU DO NOT OBEY, DAMAGE TO ENGINE TITANIUM PARTS CAN OCCUR.

SUBTASK 72-00-00-020-008-G00

(1) Remove the exhaust air tube [23] (TASK 72-59-00-000-801-G00).

SUBTASK 72-00-00-840-010-G00

CAUTION APPLY PROTECTION ON PART THAT YOU CLEAN AND ADJACENT PARTS. USE PROTECTION TO PREVENT SPRAYS AROUND THE AREA. IF SPRAYS APPLIED ON THESE PARTS, IMMEDIATELY FLUSH THE AREA WITH ISOPROPYL ALCOHOL AND CLEAN-LINT-FREE CLOTH. IF YOU DO NOT OBEY, DAMAGE TO THE PART CAN OCCUR.

(2) Protect the nut [24] and adjacent parts of the exhaust air tube [23].

NOTE: It is recommended to use plastic tarpaulins or plastic sheets wrapped around the part. It is not recommended to use any kind of protection that can stay attached to the part.

F. Exhaust Air Tube Cleaning

SUBTASK 72-00-00-110-008-G00

(1) Do these steps to remove coking:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CAUTION DO NOT TOUCH THE SURFACE OF THE PART WITH A MECHANICAL TOOL. IF YOU DO NOT OBEY, DAMAGE TO THE SURFACE OF THE PART CAN OCCUR.

(a) If it is necessary, remove coking with a mechanical tool.

NOTE: Use a mechanical tool to remove the bulk of coking. It is permitted to leave a thin layer of coking. The thin layer of coking will be removed later by a plastic tool.

CAUTION THESE MATERIALS INCLUDE PETROLEUM. DO NOT APPLY THEM TO A HOT ENGINE TO PREVENT RISK OF FIRE. IF YOU DO NOT OBEY, FIRE ON THE ENGINE CAN OCCUR.

(b) Select only one of the following product to clean the exhaust air tube [23]:

- SkyKleen 1000 solvent, B50416 [CP2828]
- Aviation 2000 degreaser, B50415 [CP2829]
- penetrating fluid, D50186 [CP2691]
- Discovert DPLF solvent, B50417 [CP2827]
- SOCOSOLV A3582 solvent, B50434 [CP2841].

(c) Make sure that the adjacent parts of the exhaust air tube [23] and nut [24] are protected.

CAUTION DO NOT USE THE MATERIALS ON THE SURFACE FOR MORE THAN MAXIMUM PERMITTED TIME. MATERIALS MUST NOT BE MORE THAN 6 HOURS ON AREAS THAT YOU CLEAN. IF YOU DO NOT OBEY, DAMAGE TO THE PART CAN OCCUR.

(d) Apply one of the product on the exhaust air tube [23].

1) If penetrating fluid, D50186 [CP2691], Aviation 2000 degreaser, B50415 [CP2829], Discovert DPLF solvent, B50417 [CP2827], SkyKleen 1000 solvent, B50416 [CP2828], or SOCOSOLV A3582 solvent, B50434 [CP2841], contacts the nut [24], clean the nut [24] as soon as possible with alcohol, B00676 [CP1041].

a) Make sure that the nut [24] is fully clean.

CAUTION DO NOT LET PARTICLES FROM TOOLS MADE FROM PLASTIC STAY ON THE PART. PARTICLES FROM TOOLS MADE OF PLASTIC CAN CAUSE DAMAGE TO THE PART.

(e) Remove as much as possible of the remaining coking with a type plastic spatula or plastic brush.

SUBTASK 72-00-00-110-009-G00

CAUTION DO NOT USE ISOPROPYL ALCOHOL MORE THAN 6 HOURS ON THE PARTS THAT YOU CLEAN. IF YOU DO NOT OBEY, DAMAGE TO THE PART CAN OCCUR.

(2) Clean the exhaust air tube [23] with alcohol, B00676 [CP1041], and lint-free cloth, G51328 [CP2674].

SUBTASK 72-00-00-200-006-G00

(3) Inspect the exhaust air tube [23] (TASK 72-50-03-200-801-G00).

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(a) Make sure that there is no damage on the surface of the exhaust air tube [23].

G. Put the Airplane Back to Its Usual Condition

SUBTASK 72-00-00-840-011-G00

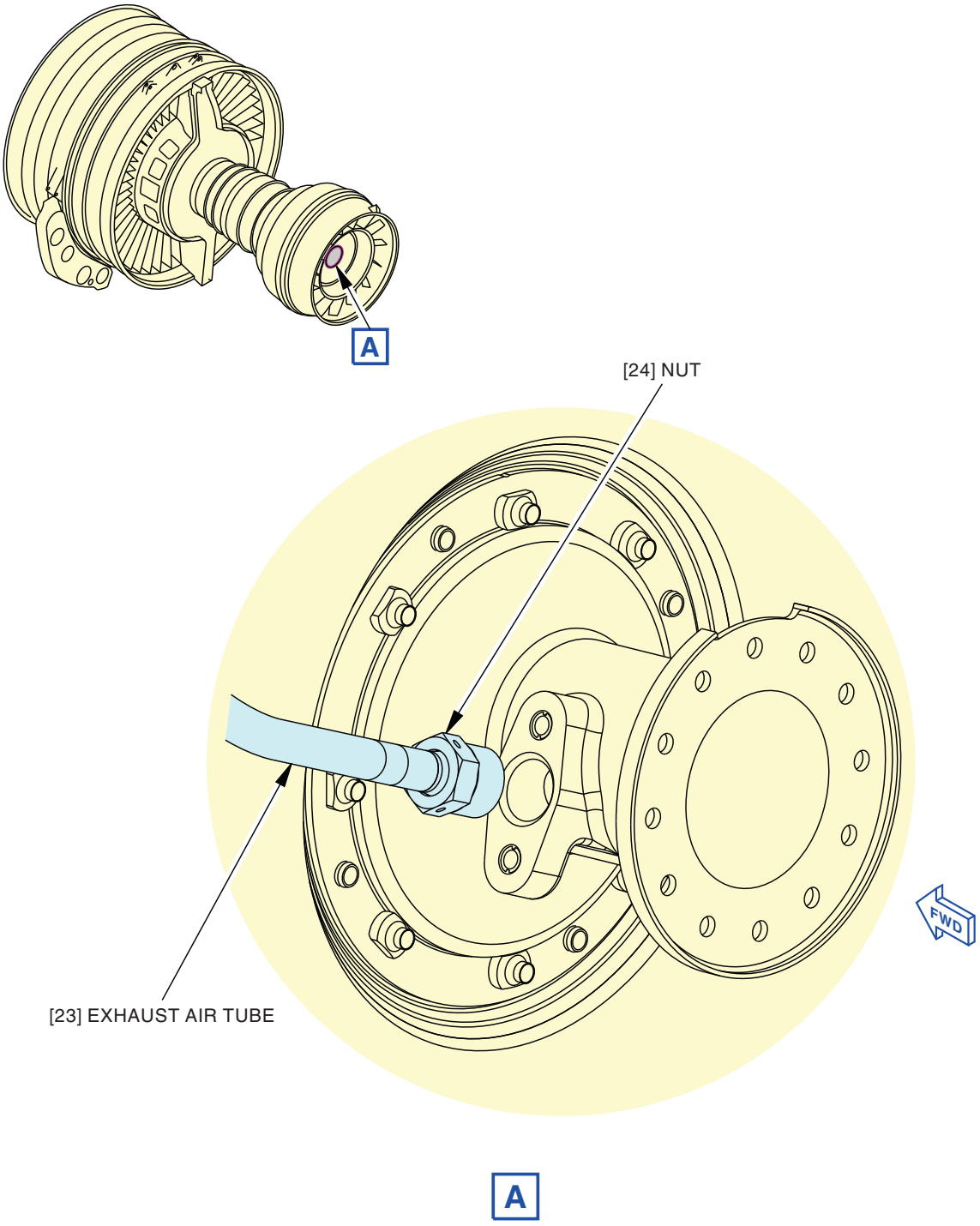
(1) Remove the protection from the nut [24] and adjacent parts of the exhaust air tube [23].

SUBTASK 72-00-00-420-008-G00

(2) Install the exhaust air tube [23] (TASK 72-59-00-400-801-G00).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Exhaust Air Tube - Cleaning
Figure 703/72-00-00-990-872-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 72-00-00-100-814-G00

9. Outer Exhaust Air Tube - Cleaning

(Figure 704)

A. General

(1) This task gives instructions to clean the outer exhaust air tube.

B. References

Reference	Title
72-50-03-200-801-G00	Turbine Rear Vane Module Inspection (P/B 601)
72-59-00-000-801-G00	Injection Nozzle Removal (P/B 401)
72-59-00-400-801-G00	Injection Nozzle Installation (P/B 401)

C. Consumable Materials

Reference	Description	Specification
B00676 [CP1041]	Alcohol - Isopropyl	
B50415 [CP2829]	Degreaser - Cleaner, Aviation 2000	
B50416 [CP2828]	Cleaner - Solvent, SkyKleen 1000 Aviation Solvent	
B50417 [CP2827]	Cleaner - Solvent, Discovert DPLF	
B50434 [CP2841]	Solvent - Cleaning, SOCOSOLV A3582	
D50186 [CP2691]	Fluid - Penetrating - Aerokroil	
G51328 [CP2674]	Cloth - Lint-Free	

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Prepare for the Cleaning

WARNING REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THESE PRODUCTS.

WARNING BE CAREFUL WHEN YOU DO WORK ON THE ENGINE PARTS AFTER YOU STOP THE ENGINE. THE ENGINE PARTS CAN STAY HOT FOR ALMOST 1 HOUR. DO NOT TOUCH HOT PARTS WITHOUT APPLICABLE GLOVES. HOT PARTS CAN CAUSE INJURIES TO PERSONNEL.

WARNING USE PROTECTIVE GLOVES TO PREVENT INJURY FROM SHARP EDGES AND ROUGH SURFACES. IF YOU DO NOT OBEY, INJURY TO PERSONNEL CAN OCCUR.

CAUTION DO NOT CLEAN AN ENGINE IF DRY CHEMICAL FIRE EXTINGUISH AGENTS APPLIED. IF YOU DO NOT OBEY, DAMAGE TO ENGINE TITANIUM PARTS CAN OCCUR.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(CAUTION PRECEDES)

CAUTION

DO NOT CLEAN AN ENGINE UNLESS THE EGT IS BELOW 150°F (66°C). IF YOU DO NOT OBEY, DAMAGE TO ENGINE TITANIUM PARTS CAN OCCUR.

CAUTION

DO NOT CLEAN AN ENGINE WHEN THE AIR TEMPERATURE IS BELOW 14°F (-10°C). IF YOU DO NOT OBEY, DAMAGE TO ENGINE TITANIUM PARTS CAN OCCUR.

SUBTASK 72-00-00-000-002-G00

(1) Remove the exhaust air tube [23] (TASK 72-59-00-000-801-G00).

SUBTASK 72-00-00-840-006-G00

CAUTION

APPLY PROTECTION ON PART THAT YOU CLEAN AND ADJACENT PARTS. USE PROTECTION TO PREVENT SPRAYS AROUND THE AREA. IF SPRAYS APPLIED ON THESE PARTS, IMMEDIATELY FLUSH THE AREA WITH ISOPROPYL ALCOHOL AND CLEAN-LINT-FREE CLOTH. IF YOU DO NOT OBEY, DAMAGE TO THE PART CAN OCCUR.

(2) Protect adjacent parts of the outer exhaust air tube [25] (Figure 704).

NOTE: It is recommended to use plastic tarpaulins or plastic sheets wrapped around the part. It is not recommended to use any kind of protection that can stay attached to the part.

(a) If you use penetrating fluid, D50186 [CP2691], do these steps:

1) Protect the 16 inserts of the LPT rear frame [22] (View C, Figure 704).

2) Protect the 2 inserts of the exhaust tube support [21] (View B, Figure 704).

F. Outer Exhaust Air Tube Cleaning

SUBTASK 72-00-00-110-004-G00

(1) Do these steps to remove coking:

CAUTION

DO NOT TOUCH THE SURFACE OF THE PART WITH A MECHANICAL TOOL. IF YOU DO NOT OBEY, DAMAGE TO THE SURFACE OF THE PART CAN OCCUR.

(a) If it is necessary, remove coking with a mechanical tool.

NOTE: Use a mechanical tool to remove the bulk of coking. It is permitted to leave a thin layer of coking. The thin layer of coking will be removed later by a plastic tool.

CAUTION

THESE MATERIALS INCLUDE PETROLEUM. DO NOT APPLY THEM TO A HOT ENGINE TO PREVENT RISK OF FIRE. IF YOU DO NOT OBEY, FIRE ON THE ENGINE CAN OCCUR.

(b) Select only one of the following materials to clean the outer exhaust air tube [25]:

- SkyKleen 1000 solvent, B50416 [CP2828]
- Aviation 2000 degreaser, B50415 [CP2829]
- penetrating fluid, D50186 [CP2691]
- Discovert DPLF solvent, B50417 [CP2827]

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- SOCOSOLV A3582 solvent, B50434 [CP2841].

(c) Make sure that the adjacent parts of the outer exhaust air tube [25] are protected.

1) If you use penetrating fluid, D50186 [CP2691], make sure that the inserts of the LPT rear frame [22] and inserts of the exhaust tube support [21] are protected.

CAUTION

DO NOT USE THE MATERIALS ON THE SURFACE FOR MORE THAN MAXIMUM PERMITTED TIME. MATERIALS MUST NOT BE MORE THAN 6 HOURS ON AREAS THAT YOU CLEAN. IF YOU DO NOT OBEY, DAMAGE TO THE PART CAN OCCUR.

(d) Apply one of the materials on the outer exhaust air tube [25].

1) If penetrating fluid, D50186 [CP2691], contacts the inserts of the LPT rear frame [22] or inserts of the exhaust tube support [21], clean the inserts as soon as possible with alcohol, B00676 [CP1041].

a) Make sure that penetrating fluid, D50186 [CP2691], is fully removed from the inserts.

CAUTION

DO NOT LET PARTICLES FROM TOOLS MADE FROM PLASTIC STAY ON THE PART. PARTICLES FROM TOOLS MADE OF PLASTIC CAN CAUSE DAMAGE TO THE PART.

(e) Remove as much as possible of the remaining coking with a type plastic spatula or plastic brush.

SUBTASK 72-00-00-110-005-G00

CAUTION

DO NOT USE ISOPROPYL ALCOHOL MORE THAN 6 HOURS ON THE PARTS THAT YOU CLEAN. IF YOU DO NOT OBEY, DAMAGE TO THE PART CAN OCCUR.

(2) Clean the outer exhaust air tube [25] with alcohol, B00676 [CP1041], and lint-free cloth, G51328 [CP2674].

SUBTASK 72-00-00-200-003-G00

(3) Inspect the outer exhaust air tube [25] (TASK 72-50-03-200-801-G00).

(a) Make sure that there is no damage on the surface of the outer exhaust air tube [25].

G. Put the Airplane Back to Its Usual Condition

SUBTASK 72-00-00-840-007-G00

(1) Remove the protection from the adjacent parts of the outer exhaust air tube [25].

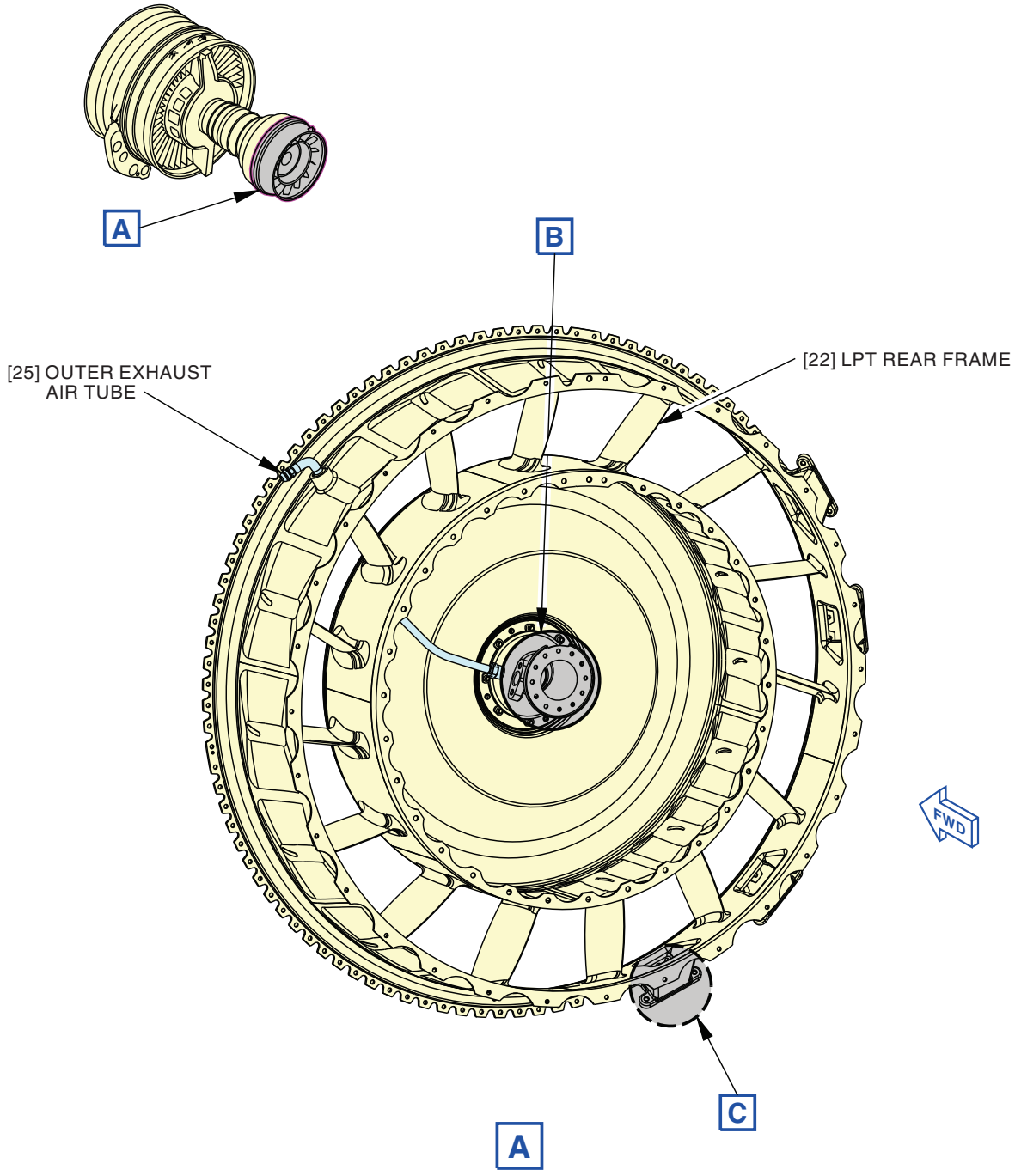
(a) If it is installed, remove the protection from the inserts of the LPT rear frame [22] and inserts of the exhaust tube support [21].

SUBTASK 72-00-00-400-004-G00

(2) Install the exhaust air tube [23] (TASK 72-59-00-400-801-G00).

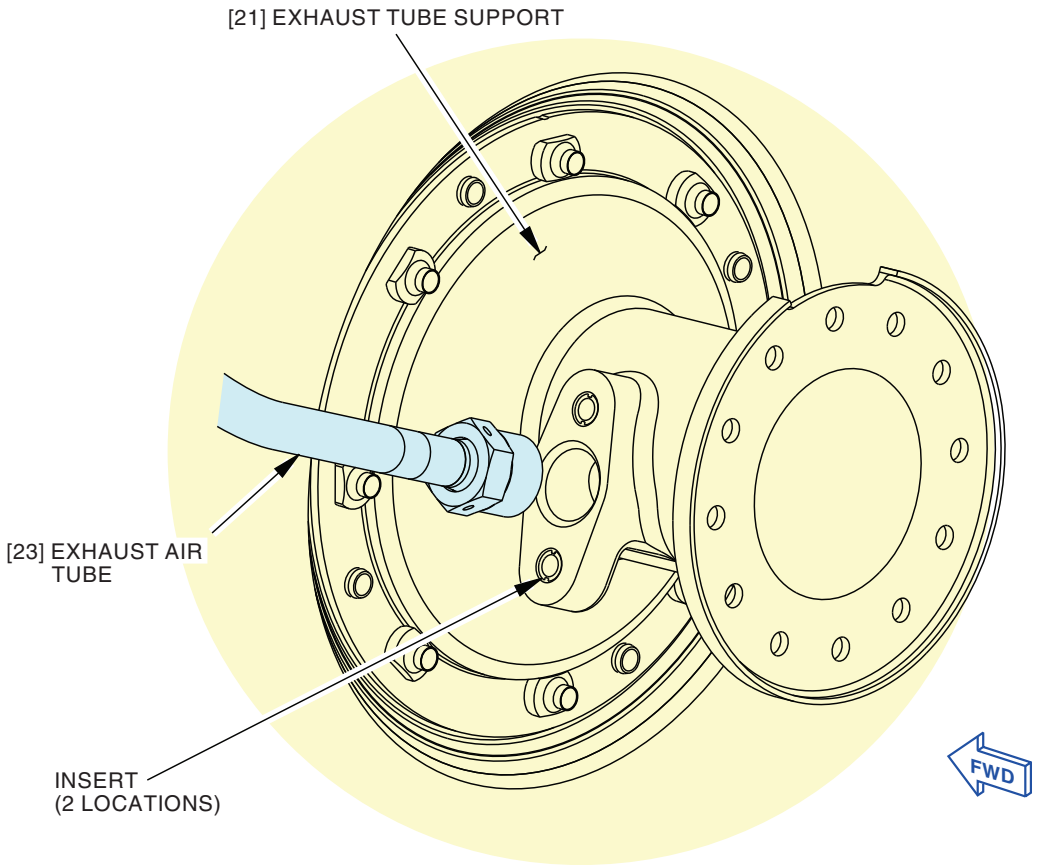
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Outer Exhaust Air Tube Cleaning
Figure 704/72-00-00-990-869-G00 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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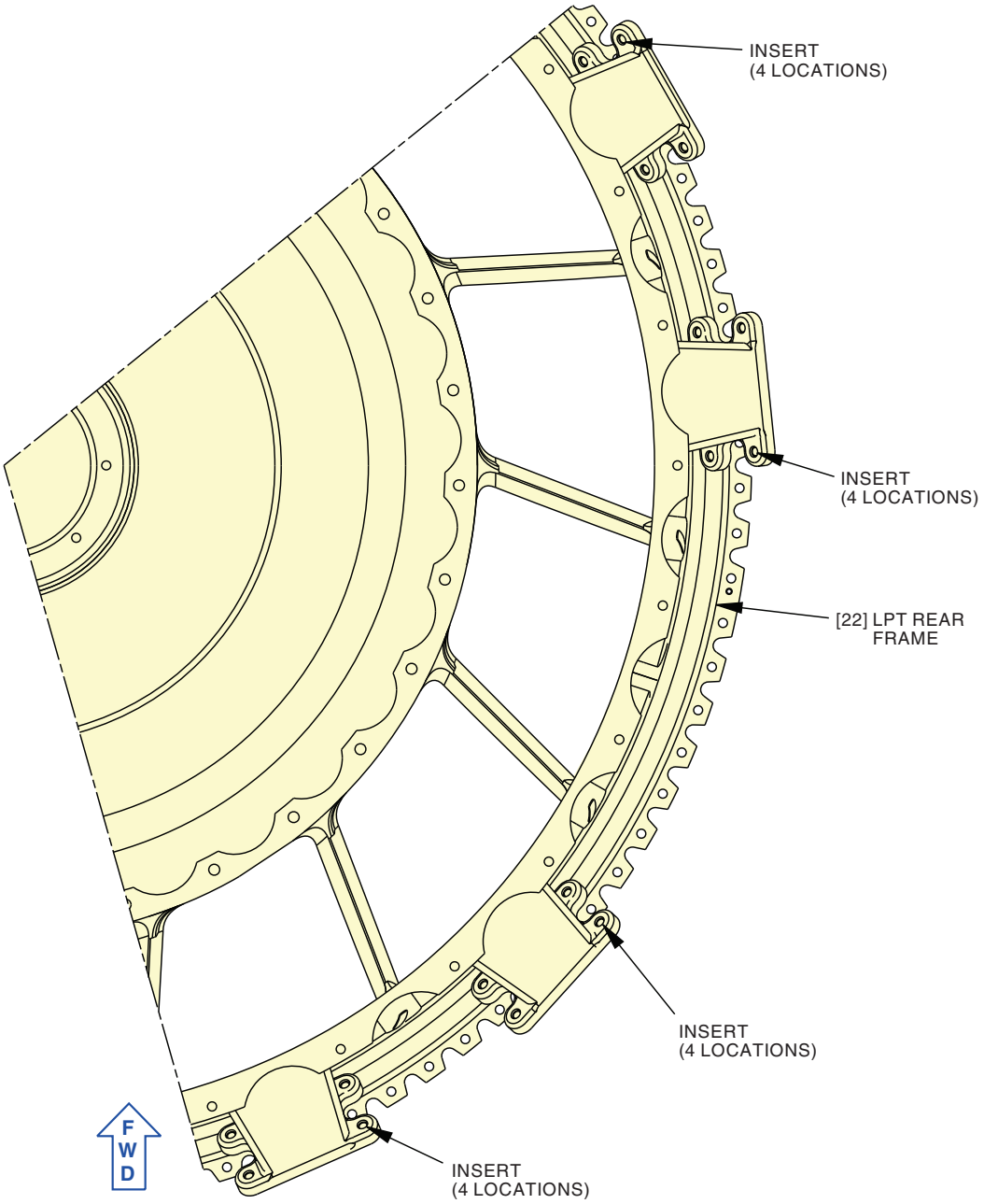
Outer Exhaust Air Tube Cleaning
Figure 704/72-00-00-990-869-G00 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Outer Exhaust Air Tube Cleaning
Figure 704/72-00-00-990-869-G00 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 72-00-00-100-815-G00

10. Rear Ventilation Tube - Cleaning

(Figure 705, Figure 706)

A. General

(1) This task gives instructions to clean the rear ventilation tube.

B. References

Reference	Title
72-50-03-200-801-G00	Turbine Rear Vane Module Inspection (P/B 601)
72-59-00-000-801-G00	Injection Nozzle Removal (P/B 401)
72-59-00-400-801-G00	Injection Nozzle Installation (P/B 401)

C. Consumable Materials

Reference	Description	Specification
B00676 [CP1041]	Alcohol - Isopropyl	
B50415 [CP2829]	Degreaser - Cleaner, Aviation 2000	
B50416 [CP2828]	Cleaner - Solvent, SkyKleen 1000 Aviation Solvent	
B50417 [CP2827]	Cleaner - Solvent, Discovert DPLF	
B50434 [CP2841]	Solvent - Cleaning, SOCOSOLV A3582	
D50186 [CP2691]	Fluid - Penetrating - Aerokroil	
G51328 [CP2674]	Cloth - Lint-Free	

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Prepare for the Cleaning

WARNING REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THESE PRODUCTS.

WARNING BE CAREFUL WHEN YOU DO WORK ON THE ENGINE PARTS AFTER YOU STOP THE ENGINE. THE ENGINE PARTS CAN STAY HOT FOR ALMOST 1 HOUR. DO NOT TOUCH HOT PARTS WITHOUT APPLICABLE GLOVES. HOT PARTS CAN CAUSE INJURIES TO PERSONNEL.

WARNING USE PROTECTIVE GLOVES TO PREVENT INJURY FROM SHARP EDGES AND ROUGH SURFACES. IF YOU DO NOT OBEY, INJURY TO PERSONNEL CAN OCCUR.

CAUTION DO NOT CLEAN AN ENGINE IF DRY CHEMICAL FIRE EXTINGUISH AGENTS APPLIED. IF YOU DO NOT OBEY, DAMAGE TO ENGINE TITANIUM PARTS CAN OCCUR.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(CAUTION PRECEDES)

CAUTION DO NOT CLEAN AN ENGINE UNLESS THE EGT IS BELOW 150°F (66°C). IF YOU DO NOT OBEY, DAMAGE TO ENGINE TITANIUM PARTS CAN OCCUR.

CAUTION DO NOT CLEAN AN ENGINE WHEN THE AIR TEMPERATURE IS BELOW 14°F (-10°C). IF YOU DO NOT OBEY, DAMAGE TO ENGINE TITANIUM PARTS CAN OCCUR.

SUBTASK 72-00-00-000-003-G00

(1) Remove the injection nozzle, do this task: Injection Nozzle Removal, TASK 72-59-00-000-801-G00.

SUBTASK 72-00-00-840-008-G00

CAUTION APPLY PROTECTION ON PART THAT YOU CLEAN AND ADJACENT PARTS. USE PROTECTION TO PREVENT SPRAYS AROUND THE AREA. IF SPRAYS APPLIED ON THESE PARTS, IMMEDIATELY FLUSH THE AREA WITH ISOPROPYL ALCOHOL AND CLEAN-LINT-FREE CLOTH. IF YOU DO NOT OBEY, DAMAGE TO THE PART CAN OCCUR.

(2) Protect adjacent parts of the rear ventilation tube [26].

NOTE: It is recommended to use plastic tarpaulins or plastic sheets wrapped around the part. It is not recommended to use any kind of protection that can stay attached to the part.

(a) If you use penetrating fluid, D50186 [CP2691], do these steps:

1) Protect the 16 inserts of the LPT rear frame [22] (View C, Figure 705).

2) Protect the 2 inserts of the exhaust tube support [21] (View B, Figure 705).

F. Rear Ventilation Tube Cleaning

SUBTASK 72-00-00-110-006-G00

(1) Do these steps to remove coking:

CAUTION DO NOT TOUCH THE SURFACE OF THE PART WITH A MECHANICAL TOOL. IF YOU DO NOT OBEY, DAMAGE TO THE SURFACE OF THE PART CAN OCCUR.

(a) If it is necessary, remove coking with a mechanical tool.

NOTE: Use a mechanical tool to remove the bulk of coking. It is permitted to leave a thin layer of coking. The thin layer of coking will be removed later by a plastic tool.

CAUTION THESE MATERIALS INCLUDE PETROLEUM. DO NOT APPLY THEM TO A HOT ENGINE TO PREVENT RISK OF FIRE. IF YOU DO NOT OBEY, FIRE ON THE ENGINE CAN OCCUR.

(b) Select only one of the following materials to clean the rear ventilation tube [26]:

- SkyKleen 1000 solvent, B50416 [CP2828]
- Aviation 2000 degreaser, B50415 [CP2829]
- penetrating fluid, D50186 [CP2691]

ECCN 9E991 BOEING PROPRIETARY - See title page for details

- Discovert DPLF solvent, B50417 [CP2827]
- SOCOSOLV A3582 solvent, B50434 [CP2841].

(c) Make sure that the adjacent parts of the rear ventilation tube [26] are protected.

1) If you use penetrating fluid, D50186 [CP2691], make sure that the inserts of the LPT rear frame [22] and inserts of the exhaust tube support [21] are protected.

CAUTION DO NOT USE THE MATERIALS ON THE SURFACE FOR MORE THAN MAXIMUM PERMITTED TIME. MATERIALS MUST NOT BE MORE THAN 6 HOURS ON AREAS THAT YOU CLEAN. IF YOU DO NOT OBEY, DAMAGE TO THE PART CAN OCCUR.

(d) Apply one of the materials on the rear ventilation tube [26].

1) If penetrating fluid, D50186 [CP2691], contacts the inserts of the LPT rear frame [22] and inserts of the exhaust tube support [21], clean the inserts as soon as possible with alcohol, B00676 [CP1041].

a) Make sure that penetrating fluid, D50186 [CP2691], is fully removed from the inserts.

CAUTION DO NOT LET PARTICLES FROM TOOLS MADE FROM PLASTIC STAY ON THE PART. PARTICLES FROM TOOLS MADE OF PLASTIC CAN CAUSE DAMAGE TO THE PART.

(e) Remove as much as possible of the remaining coking with a type plastic spatula or plastic brush.

SUBTASK 72-00-00-110-007-G00

CAUTION DO NOT USE ISOPROPYL ALCOHOL MORE THAN 6 HOURS ON THE PARTS THAT YOU CLEAN. IF YOU DO NOT OBEY, DAMAGE TO THE PART CAN OCCUR.

(2) Clean the rear ventilation tube [26] with alcohol, B00676 [CP1041], and lint-free cloth, G51328 [CP2674].

SUBTASK 72-00-00-200-004-G00

(3) Inspect the rear ventilation tube [26] (TASK 72-50-03-200-801-G00).

(a) Make sure that there is no damage on the surface of the rear ventilation tube [26].

G. Put the Airplane Back to Its Usual Condition

SUBTASK 72-00-00-840-009-G00

(1) Remove the protection from the adjacent parts of the rear ventilation tube [26].

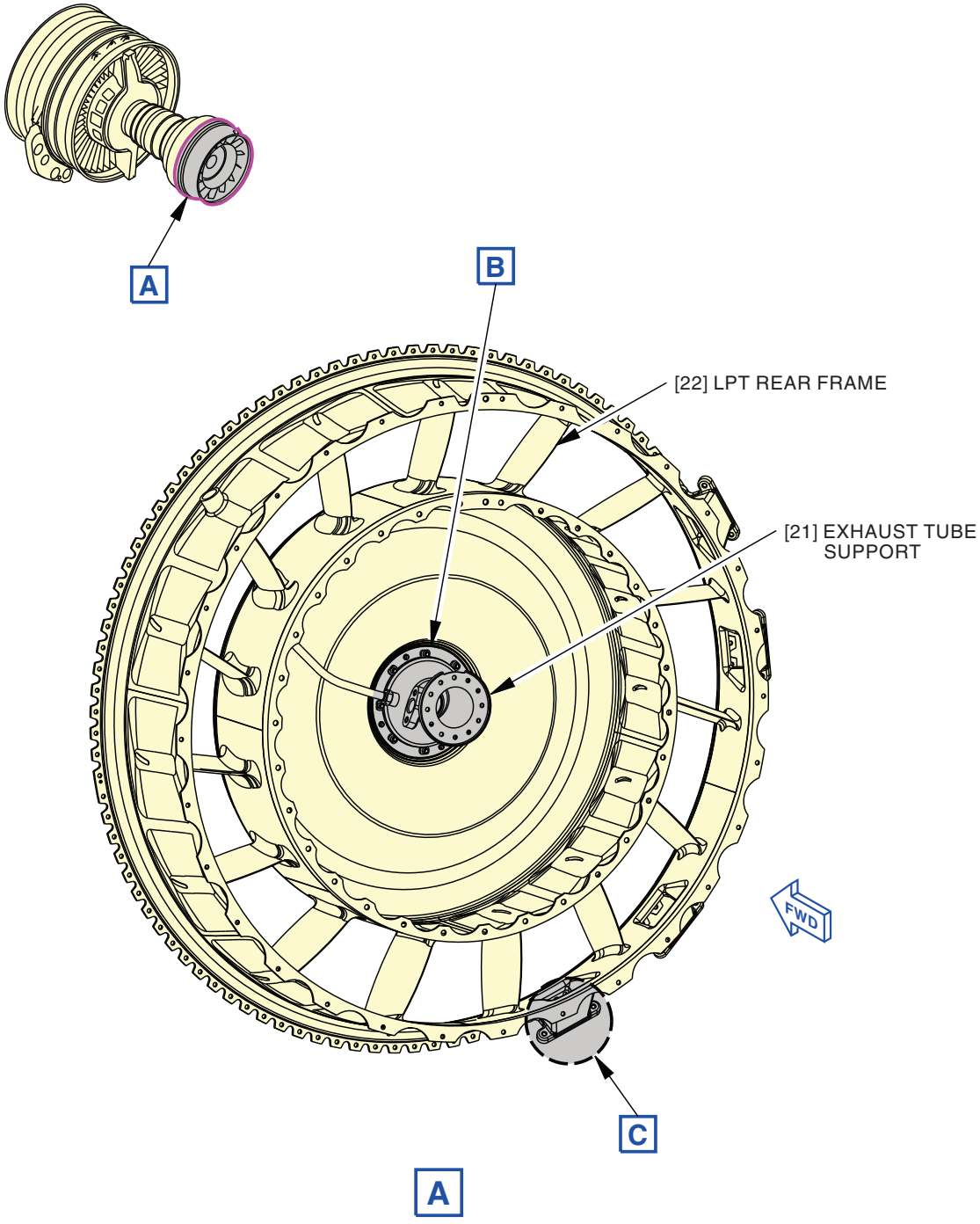
(a) If it is installed, remove the protection from the inserts of the LPT rear frame [22] and inserts of the exhaust tube support [21].

SUBTASK 72-00-00-400-005-G00

(2) Install the injection nozzle, do this task: Injection Nozzle Installation, TASK 72-59-00-400-801-G00.

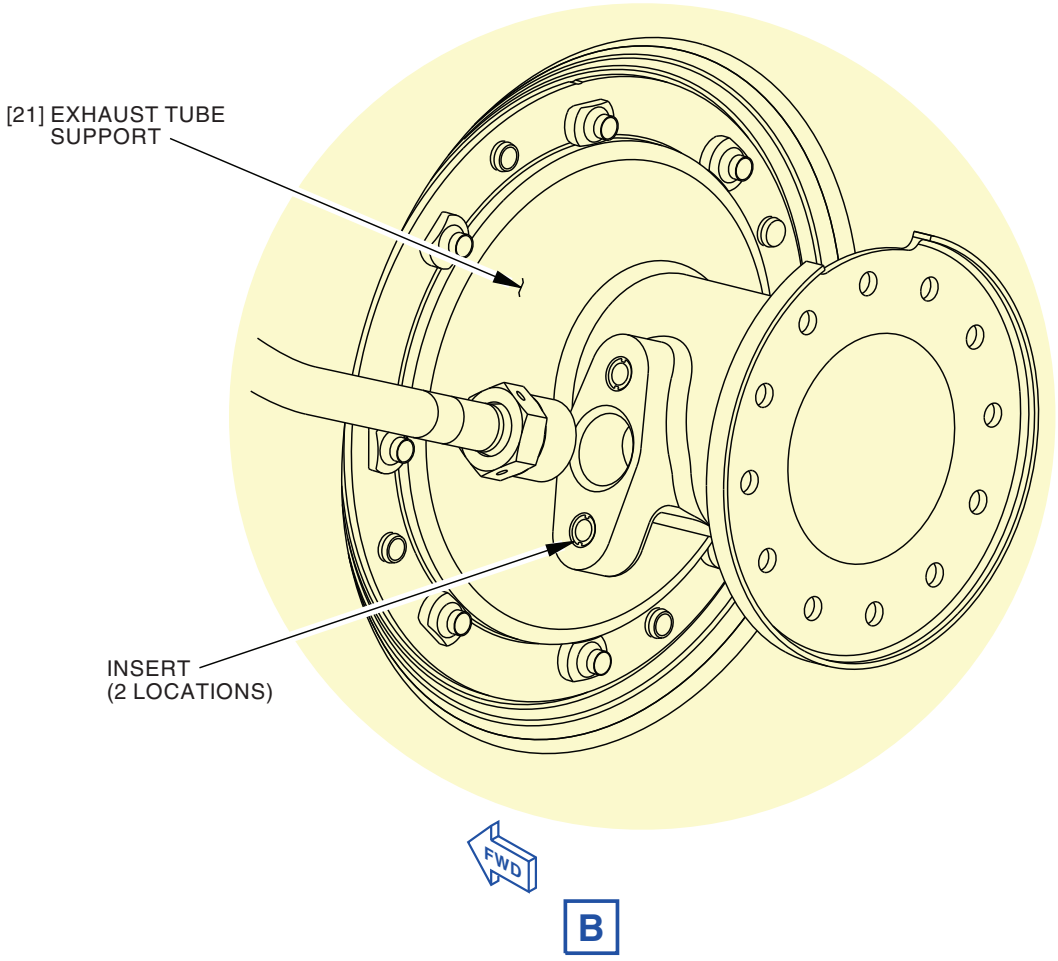
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Exhaust Tube Support and LPT Rear Frame Inserts Protection
Figure 705/72-00-00-990-870-G00 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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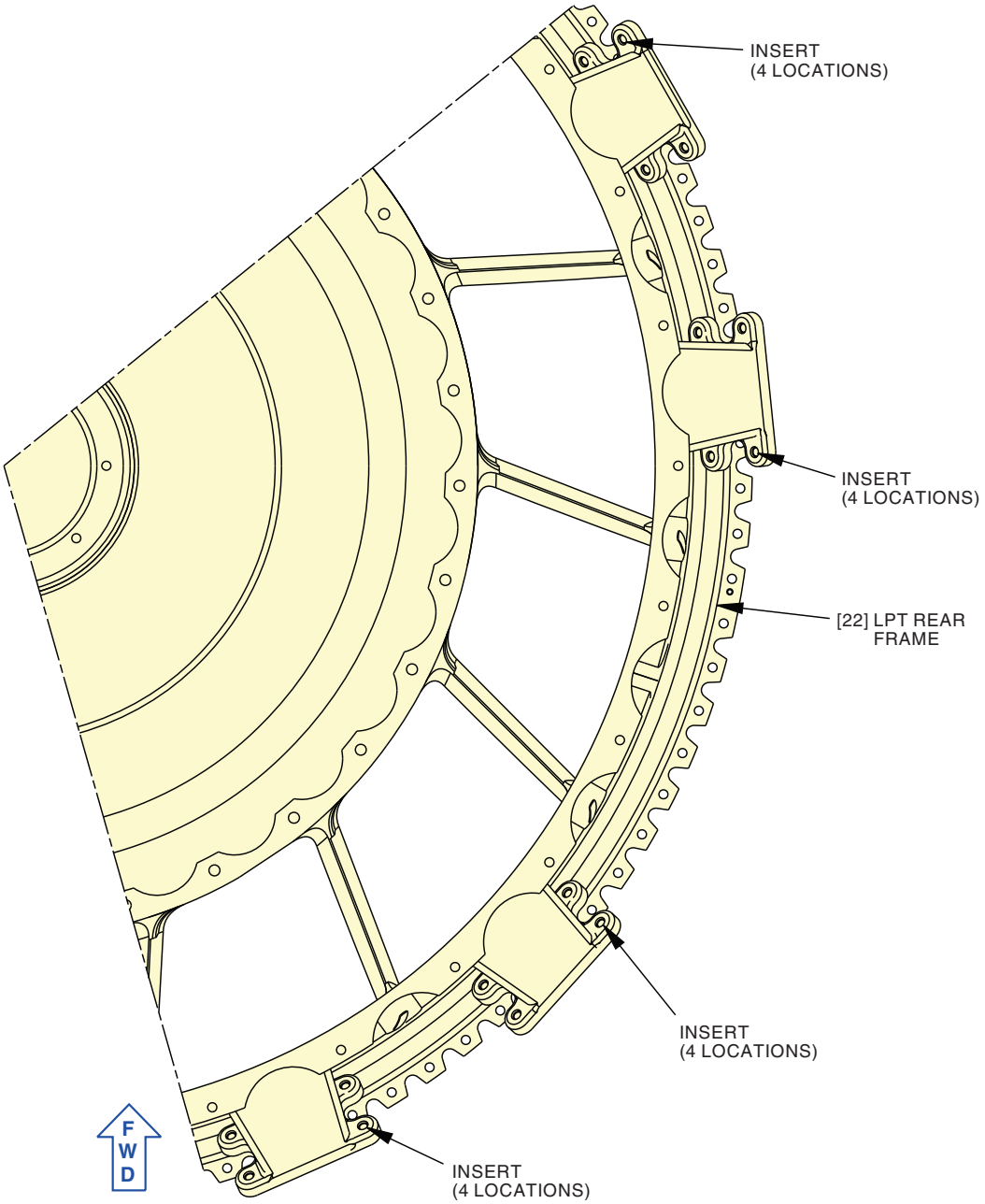
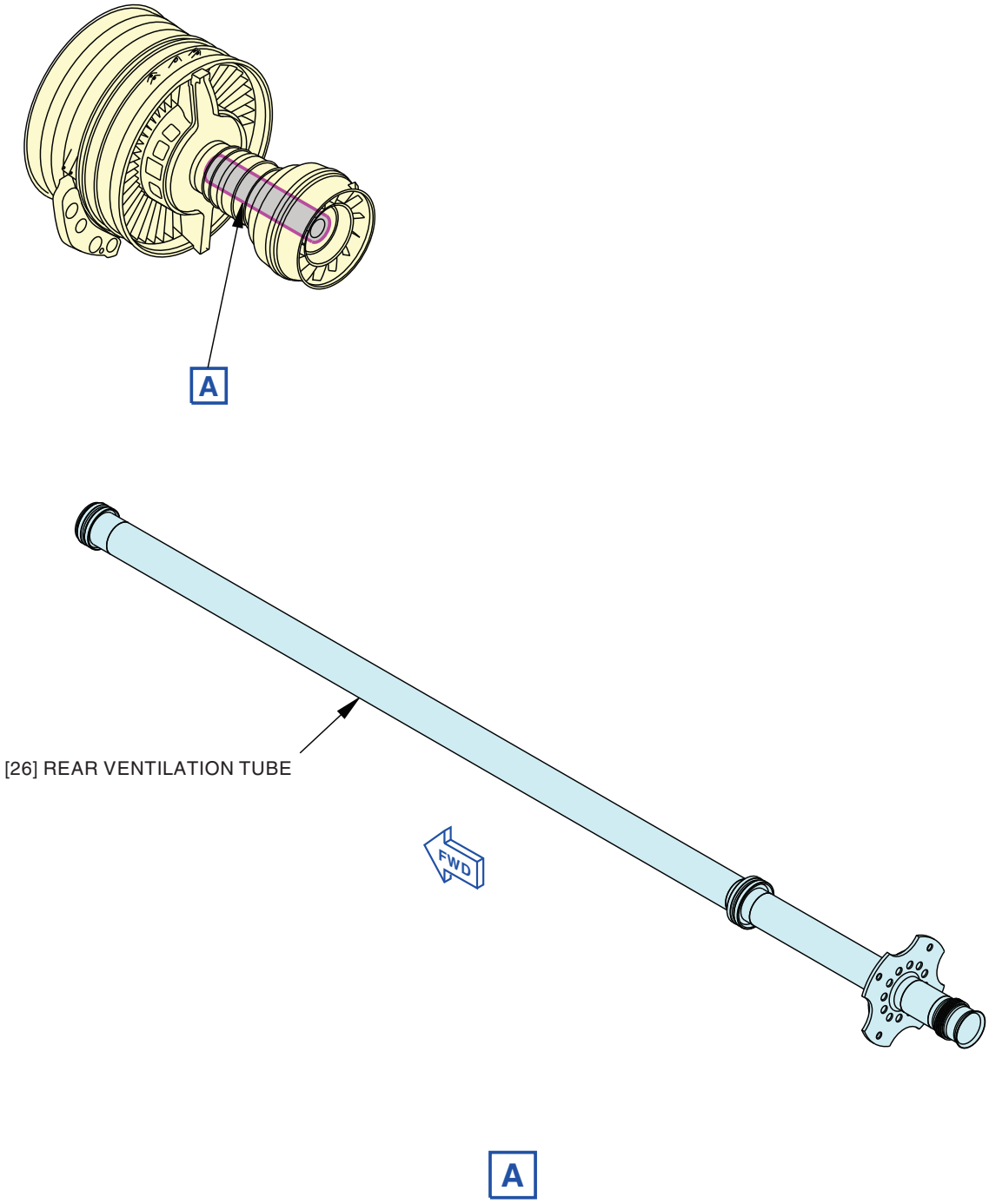
Exhaust Tube Support and LPT Rear Frame Inserts Protection
Figure 705/72-00-00-990-870-G00 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Exhaust Tube Support and LPT Rear Frame Inserts Protection
Figure 705/72-00-00-990-870-G00 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Rear Ventilation Tube Cleaning
Figure 706/72-00-00-990-871-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ENGINE - BSI PLUG D - REMOVAL/INSTALLATION

1. General

A. This procedure has two tasks:

(1) Borescope plug D Removal

(2) Borescope plug D Installation.

TASK 72-00-00-000-801-G00

2. BSI Plug D Removal

(Figure 401)

A. General

(1) This task gives instructions to remove the BSI (Borescope Inspection) plug D.

B. References

Reference	Title
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)
27-81-00-860-804	Leading Edge Flaps and Slats Retraction (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
72-23-04-000-801-G00	Fan Frame Module Segment Shrouds Removal (P/B 401)
75-31-00-910-801-G00	Variable Stator Vane Actuator Manual Operation (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
G50065 [CP8006]	Cable, Safety, Stainless Steel, 0.032 inch (0.8 mm) Diameter	M50 TF 9 CL-A

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

F. Prepare for the Borescope Plug D Removal

SUBTASK 72-00-00-860-086-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-00-00-860-083-G00

(2) Do this task: Leading Edge Flaps and Slats Retraction, TASK 27-81-00-860-804.

SUBTASK 72-00-00-860-084-G00

(3) Do this task: Leading Edge Flaps and Slats - Deactivation, TASK 27-81-00-040-801.

SUBTASK 72-00-00-860-087-G00

(4) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Borescope Plug D Removal

SUBTASK 72-00-00-010-036-G00

(1) Remove the shroud segment 3 (TASK 72-23-04-000-801-G00).

SUBTASK 72-00-00-010-037-G00

(2) If required remove the shroud segment 2 for access purpose (TASK 72-23-04-000-801-G00).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 72-00-00-010-038-G00

(3) Remove the borescope plug D [1] as follows:

(a) If necessary, to gain access, operate manually the Variable Stator Vane (VSV) system to make sure that the borescope plug D [1] access is aligned with Inlet Guide Vane (IGV) rings clearance, do this task: (Variable Stator Vane Actuator Manual Operation, TASK 75-31-00-910-801-G00).

NOTE: If you shift the VSVs to FULLY OPEN position, it could create adequate axial and circumferential clearance that enables removal of Plug D without the need to remove the IGV connecting link [2].

(b) If necessary, to gain access, remove the IGV connecting link [2] connecting the upper IGV ring [3] and lower IGV ring [4].

1) Remove the nut [7] and bolt [8] securing the IGV turnbuckle [6] to the clevis of the IGV connecting link [2].

2) Remove and discard the safety cable, G50065 [CP8006], from the four bolts [5] that attached the IGV connecting link [2] to the IGV rings.

3) Remove the IGV connecting link [2] as follow:

NOTE: Hold manually the IGV connecting link [2] link during the steps that follow.

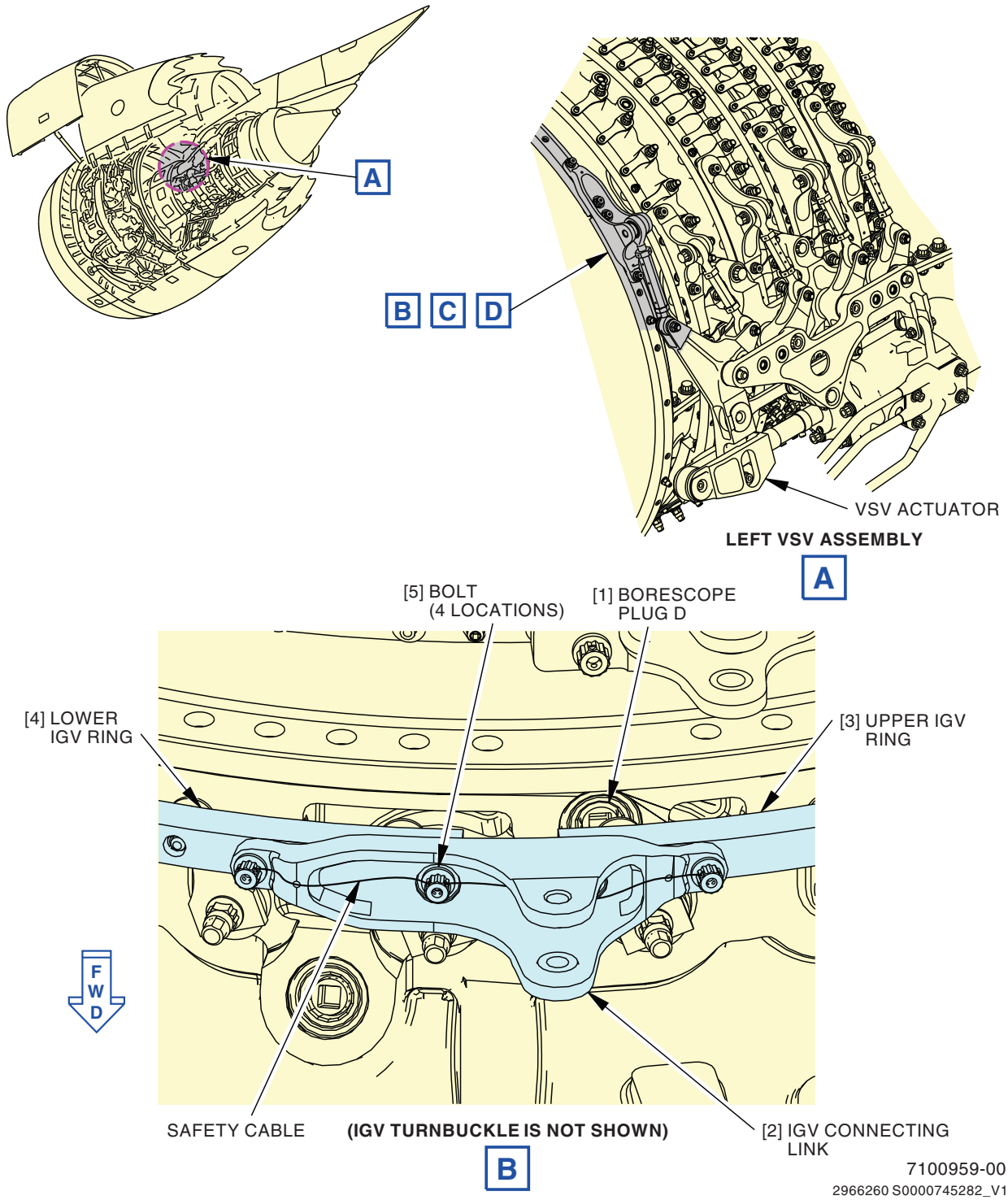
a) Remove the four bolts [5] attaching the IGV connecting link [2] to the lower IGV ring [4] and upper IGV ring [3].

b) Remove the IGV connecting link [2] from the lower IGV ring [4] and upper IGV ring [3].

4) Remove the borescope plug D [1] from the port D.

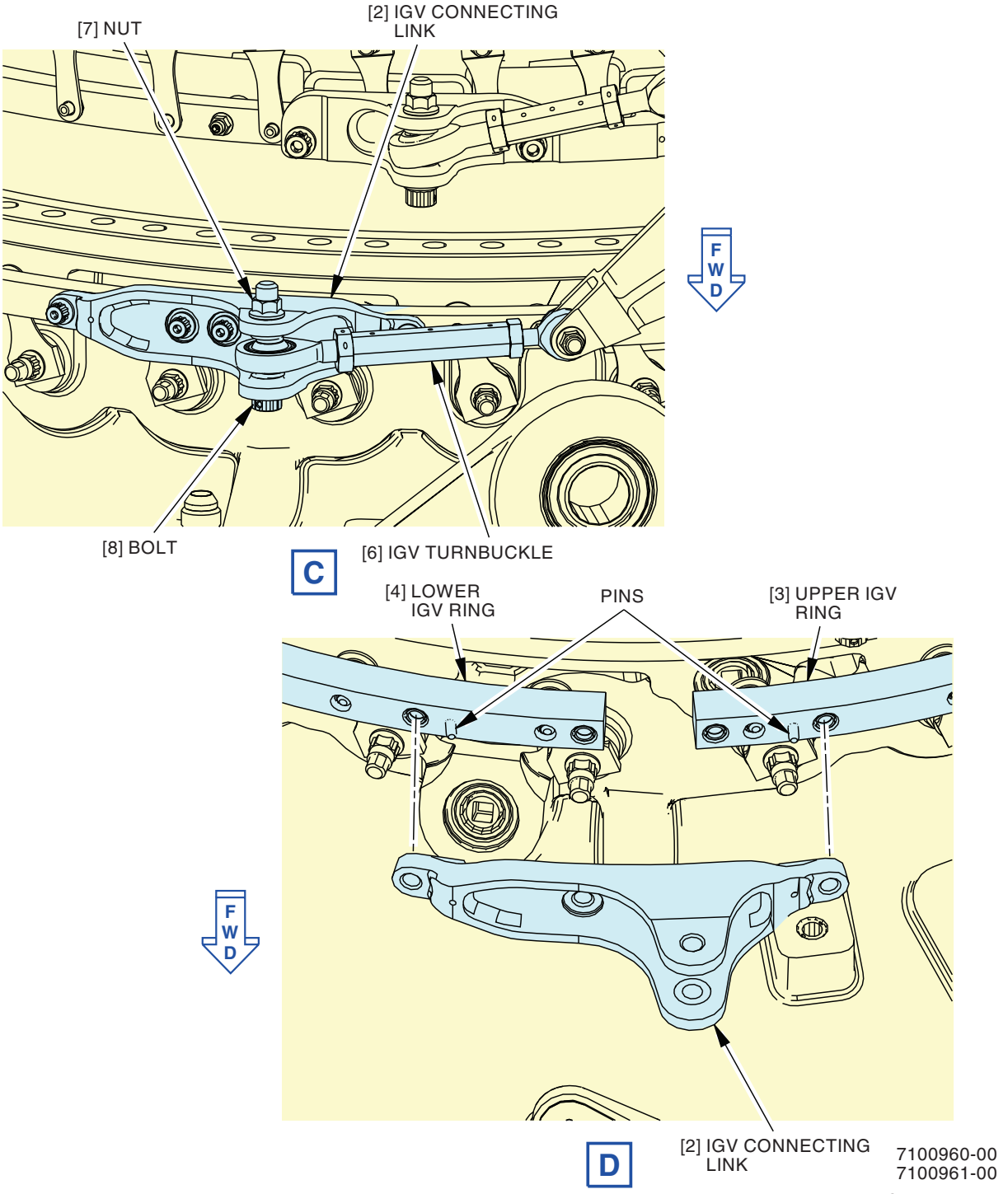
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



BSI Plug D Installation
Figure 401/72-00-00-990-862-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



BSI Plug D Installation
Figure 401/72-00-00-990-862-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 72-00-00-400-801-G00

3. BSI Plug D Installation

(Figure 401)

A. General

(1) This task gives instructions to install the BSI (Borescope Inspection) plug D.

B. References

Reference	Title
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
72-00-00-290-801-G00	Preparation for Borescope Inspection (P/B 601)
72-23-04-400-801-G00	Fan Frame Module Segment Shrouds Installation (P/B 401)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D00599 [CP2442]	Oil - Engine	
G50065 [CP8006]	Cable, Safety, Stainless Steel, 0.032 inch (0.8 mm) Diameter	M50 TF 9 CL-A

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

F. Borescope Plug D Installation

SUBTASK 72-00-00-400-001-G00

(1) Install the borescope plug D [1] in the port D as follows:

(a) Do a visual inspection of the borescope plug D [1] and the borescope port D (TASK 72-00-00-290-801-G00).

(b) Put the oil, D00599 [CP2442], on the borescope plug D [1] threads and the mating surfaces.

(c) Install borescope plug D [1] in the borescope port D.

(d) Torque the borescope plug D [1] to 161 in-lb (18.2 N·m) - 189 in-lb (21.4 N·m).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 72-00-00-410-030-G00

(2) If previously removed for access, reinstall the IGV connecting link [2] connecting the lower IGV ring [4] and upper IGV ring [3] as follows:

NOTE: It is necessary to hold manually the IGV connecting link [2] during the steps that follow.

(a) Lubricate the four bolts [5] with oil, D00599 [CP2442].

(b) Align the holes of the IGV connecting link [2] with the holes of the lower IGV ring [4] and the holes of the upper IGV ring [3].

(c) Put and hold the IGV connecting link [2] in position on the lower IGV ring [4] and on the upper IGV ring [3] at 4:30 o'clock location.

(d) Attach the IGV connecting link [2] to the lower IGV ring [4] and to upper IGV ring [3] with four bolts [5].

1) Install and slightly tighten the four bolts [5] with hand.

NOTE: This is to make sure that the IGV connecting link [2] is in the correct position.

2) Torque the four bolts [5] to 53 in-lb (6.0 N·m) - 57 in-lb (6.4 N·m).

3) Install the safety safety cable, G50065 [CP8006], through the four bolts [5].

(e) Attach the IGV turnbuckle [6] to the IGV connecting link [2] as follows:

1) Apply a layer of oil, D00599 [CP2442], to the threads of the nut [7].

2) Move the IGV turnbuckle [6] into the clevis of the IGV connecting link [2].

3) Attach the IGV turnbuckle [6] to the IGV connecting link [2] with the bolt [8] and nut [7].

a) Make sure that you install bolt [8] head FWD.

b) Torque the nut [7] to 106 in-lb (12.0 N·m) - 124 in-lb (14.0 N·m).

G. Put the Airplane to Its Usual Condition

SUBTASK 72-00-00-410-031-G00

(1) If necessary, install shroud segment 3 (TASK 72-23-04-400-801-G00).

SUBTASK 72-00-00-410-032-G00

(2) If necessary, install the shroud segment 2 (TASK 72-23-04-400-801-G00).

SUBTASK 72-00-00-860-088-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(3) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 72-00-00-440-011-G00

(4) Do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801

SUBTASK 72-00-00-860-089-G00

(5) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ENGINE - REPAIR

1. General

A. This procedure has this task:

(1) A repair of the Low Pressure Turbine (LPT) blades boro-blending.

TASK 72-00-00-350-801-G00

2. Low Pressure Turbine (LPT) Blades Boro - Blending Repair

(Figure 801, Figure 802, Figure 803)

A. General

(1) This task gives instructions to repair the leading and trailing edges of the Low Pressure Turbine (LPT) blades stage 2 to 4 and the leading edge of stage 5 with boro-blending, and to repair the trailing edge of stage 5 with blending.

B. References

Reference	Title
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)
70-11-06-230-802-G00	Fluorescent Penetrant Inspection (P/B 201)
70-11-06-230-803-G00	Fluorescent Penetrant Inspection (General) (P/B 201)
70-43-42-300-802-G00	Blending Procedures - Hand Blending (P/B 201)
70-43-42-300-803-G00	Blending Procedures - Power Blending (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
72-00-00-290-801-G00	Preparation for Borescope Inspection (P/B 601)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-2332	Borescope Blend Tool Part #: 9471M44P01 Supplier: 58828
SPL-14908	Borescope Blend Tool Part #: 956A1604P01 Supplier: 58828
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

F. Prepare for the Low Pressure Turbine (LPT) Blades Boro - Blending Repair

SUBTASK 72-00-00-860-090-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) On the P5 overhead panel, make sure that the applicable ENGINE START switch is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) On the P10 control stand, make sure that the applicable START LEVER switch is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-00-00-010-039-G00

(2) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

SUBTASK 72-00-00-350-001-G00

(3) Remove the borescope plugs (TASK 72-00-00-290-801-G00).

SUBTASK 72-00-00-350-002-G00

CAUTION

DO NOT INSTALL OR REMOVE A TOOL POINT FROM THE HEAD WITH THE BEARING IF IT IS SET IN THE 0 DEGREE POSITION. IF YOU DO NOT OBEY, THE DRIVE BELT WILL DISENGAGE. DAMAGE TO THE BORESCOPE CAN OCCUR.

(4) Install the borescope blend tool, SPL-14908, do these steps (Figure 803):

(a) Make sure that the belt is properly installed in the borescope blend tool, SPL-14908.

NOTE: The grinding tool must turn clockwise aft looking forward.

(b) Put the drive shaft head [2] at the 90 degrees position.

(c) Install the tool onto the drive shaft head [2].

(d) Tighten the lock screw [7].

(e) Set the light source switch to ON position.

(f) Put the tool at the 0 degree position.

1) Put the shaft [4] into the borescope port.

(g) Check the tool, and put the drive shaft head [2] at the 90 degrees position.

SUBTASK 72-00-00-350-003-G00

CAUTION

DO NOT INSTALL OR REMOVE A TOOL POINT FROM THE HEAD WITH THE BEARING IF IT IS SET IN THE 0 DEGREE POSITION. IF YOU DO NOT OBEY, THE DRIVE BELT WILL DISENGAGE. DAMAGE TO THE BORESCOPE CAN OCCUR.

(5) Remove the borescope blend tool, SPL-14908, do these steps:

(a) Put the drive shaft head [2] at the 0 degree position.

1) Remove the shaft [4] from the borescope port.

(b) Set the light source switch to OFF position.

(c) Put the drive drive shaft head [2] at the 90 degrees position.

(d) Loosen the lock screw [7] and remove the tool from the drive shaft head [2].

SUBTASK 72-00-00-350-004-G00

(6) Install the borescope plugs (TASK 72-00-00-290-801-G00).

SUBTASK 72-00-00-220-046-G00

(7) For dimensional information refer the table below:

NOTE: Dimensions are given in inches with millimeters in parentheses.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Table 801/72-00-00-993-816-G00

Description	Minimum In-Process Dimension	Maximum In-Process Dimension	Minimum Finish Dimension	Maximum Finish Dimension	Reference Dimension
Stage 2					
DIM. D	-	-	2 × the blending depth (DIM. Q or DIM. R)	-	-
DIM. G	-	-	7.5 × DIM. Q	-	-
DIM. H	-	-	7.5 × DIM. R	-	-
L1 + L2	-	-	-	0.39 (10)	-
L3 + L4	-	-	-	0.47 (12)	-
DIM. M	-	-	0.39 (10)	-	-
DIM. N	-	-	0.39 (10)	-	-
DIM. Q	-	-	-	0.03 (1)	-
DIM. R	-	-	-	0.03 (1)	-
RADIUS C	-	-	-	0.019 (0.5)	-
RADIUS B	-	-	0.08 (2)	-	-
Stage 3					
DIM. D	-	-	2 × the blending depth (DIM. Q or DIM. R)	-	-
DIM. G	-	-	5 × DIM. Q	-	-
DIM. H	-	-	5 × DIM. R	-	-
L1 + L2	-	-	-	0.39 (10)	-
L3 + L4	-	-	-	0.47 (12)	-
DIM. M	-	-	0.39 (10)	-	-
DIM. N	-	-	0.39 (10)	-	-
DIM. Q	-	-	-	0.0492 (1.25)	-
DIM. R	-	-	-	0.0492 (1.25)	-
DIM. S	-	-	1.88 (48)	-	-
DIM. T	-	-	1.96 (50)	-	-
RADIUS C	-	-	-	0.019 (0.5)	-
RADIUS B	-	-	0.08 (2)	-	-
Stage 4					
DIM. G	-	-	7.5 × DIM. Q	-	-
L1 + L2	-	-	-	0.39 (10)	-
DIM. M	-	-	0.39 (10)	-	-

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Table 801/72-00-00-993-816-G00 (Continued)

Description	Minimum In-Process Dimension	Maximum In-Process Dimension	Minimum Finish Dimension	Maximum Finish Dimension	Reference Dimension
DIM. N	-	-	0.39 (10)	-	-
DIM. Q	-	-	-	0.03 (1)	-
DIM. S	-	-	3.15 (80)	-	-
RADIUS C	-	-	-	0.019 (0.5)	-
RADIUS B	-	-	0.08 (2)	-	-
Stage 5					
DIM. D	-	-	2 × the blending depth (DIM. Q or DIM. R)	-	-
DIM. G	-	-	5 × DIM. Q	-	-
DIM. H	-	-	7.5 × DIM. R	-	-
L1 + L2	-	-	-	0.39 (10)	-
L3 + L4	-	-	-	0.47 (12)	-
DIM. M	-	-	0.39 (10)	-	-
DIM. N	-	-	0.39 (10)	-	-
DIM. Q	-	-	-	0.0492 (1.25)	-
DIM. R	-	-	-	0.03 (1)	-
RADIUS C	-	-	-	0.019 (0.5)	-
RADIUS B	-	-	0.08 (2)	-	-

G. Low Pressure Turbine (LPT) Blades Boro - Blending Repair

WARNING

REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THIS PRODUCT. IF NOT, IT CAN CAUSE INJURY.

SUBTASK 72-00-00-320-001-G00

(1) Do a boro - blending of the the LPT blades stage 2 to 4 and of the leading edge of the stage 5 (Figure 801 and Figure 802).

(a) Do a borescope inspection as follows:

1) Turn the fan rotor clockwise (aft looking forward).

2) Find the first blade you inspect.

NOTE: This is to make sure to inspect all the blades.

3) Find and measure the area to repair.

a) If it is necessary, do this task for the blades that you repair: Fluorescent Penetrant Inspection, TASK 70-11-06-230-802-G00.

b) Write down the inspection results: size and location for each blade.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

c) Make sure that the recorded inspection results are in the permitted blending limits (Table 801).

d) Calculate the minimum blend radius.

(b) Do the boro-blending as follows:

1) Install the grinding tool.

a) Use the shorter grinding tool tips for the forward stages and the longer grinding tool tips for the aft stages.

<1> Put the drive shaft head [2] at the 90 degrees position.

<2> Install the grinding tool onto the drive shaft head [2].

<3> Tighten the lock screw [7].

<4> Set the light source switch to ON position.

<5> Put the tool at the 0 degree position.

<a> Put the shaft [4] into the borescope port.

<6> Check the grinding tool, and put the drive shaft head [2] at the 90 degrees position.

b) Make sure that the blend tip is pointed away from the operator in the 90 degrees position.

c) Start the drive motor [5].

<1> Make sure that the grinding tip [1] rotates in the clockwise direction.

<a> If the direction of rotation is counterclockwise, do a check of the drive belt for possible twisted installation.

d) Make sure that the maximum number of LPT blades repairable per stage is 10%.

<1> Make sure that 16 blades rounded down for stage 2 and 3.

<2> Make sure that 10 blades rounded down for stage 4 and 5.

2) Put the grinding tip [1] near the area to repair.

3) Align the blending tool tip with the area to be blended.

a) Make sure that the movement is correct across the total length of the area.

b) Do not start the drive motor [5] at this time.

c) Check the blend limits (Figure 802).

4) Make sure that the drive motor [5] is off.

a) Apply pressure to the grinding tip [1] at the area to repair to practice the movement to use to blend the area to repair.

b) After sufficient practice with the grinding tip [1] on the blade leading edge surface, use the switch [3] to start the drive motor.

<1> Use the drive motor [5] to increase the speed.

CAUTION DO NOT APPLY TOO MUCH PRESSURE TO PREVENT VIBRATION. THE GRINDING TOOL WILL LOOSEN. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

c) Blend the area without too much pressure.

<1> If it is necessary, tighten the lock screw [7] again.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

<a> Check the blend limits (Figure 802).

d) Set the switch [3] to OFF position to stop the drive motor [5].

e) Remove the grinding tool.

<1> Put the drive shaft head [2] at the 0 degree position.

<a> Remove the shaft [4] from the borescope port.

<2> Set the light source switch to OFF position.

<3> Put the drive shaft head [2] at the 90 degrees position.

<4> Loosen the lock screw [7] and remove the tool from the drive shaft head [2].

<5> Check the blend limits (Figure 802).

(c) Do a polishing of the blade as follows:

1) Install the polishing tool.

a) Use the shorter polishing tool tips for the forward stages and the longer polishing tool tips for the aft stages.

<1> Put the drive shaft head [2] at the 90 degrees position.

<2> Install the polishing tool onto the drive shaft head [2].

<3> Tighten the lock screw [7].

<4> Set the light source switch to ON position.

<5> Put the polishing tool at the 0 degree position.

<a> Put the shaft [4] into the borescope port.

<6> Check the polishing tool, and put the drive shaft head [2] at the 90 degrees position.

2) Set the switch [3] to ON position to start the drive motor [5].

3) Polish the blended surface.

4) Set the switch [3] to OFF position to stop the drive motor [5].

5) Remove the polishing tool.

a) Put the drive shaft head [2] at the 0 degree position.

<1> Remove the shaft [4] from the borescope port.

b) Set the light source switch to OFF position.

c) Put the drive shaft head [2] at the 90 degrees position.

d) Loosen the lock screw [7] and remove the polishing tool from the drive shaft head [2].

(d) Do the borescope inspection of the blended areas.

1) Use the borescope blend tool, SPL-14908, or borescope blend tool, SPL-2332.

2) Make sure that all the areas to repair are blended.

(e) For the blade that you repair, do this task: Fluorescent Penetrant Inspection, TASK 70-11-06-230-802-G00.

NOTE: Cracks are not permitted.

SUBTASK 72-00-00-320-002-G00

(2) Do a blending of the trailing edges of LPT blades stage 5 (Figure 801 and Figure 802).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Do an inspection as follows:

1) Turn the fan rotor clockwise (aft looking forward).

2) Find the first blade you inspect.

NOTE: This is to make sure to inspect all the blades.

3) Find and measure the area to repair.

a) If it is necessary, do one of these tasks for the blades that you repair: Fluorescent Penetrant Inspection, TASK 70-11-06-230-802-G00, Fluorescent Penetrant Inspection (General), TASK 70-11-06-230-803-G00.

4) Write down the inspection results: size and location for each blade.

5) Make sure that the recorded inspection results are in the permitted blending limits (Table 801).

6) If the recorded inspection results are out of the permitted limits, find the length of the blend necessary to keep the minimum blend radius.

7) Calculate the minimum blend radius.

a) Make sure that the maximum number of repairable LPT blades per stage is 10% (10 blades rounded down for stage 5).

(b) For the blending, do one of these tasks for the blades that you repair: Blending Procedures - Hand Blending, TASK 70-43-42-300-802-G00, Blending Procedures - Power Blending, TASK 70-43-42-300-803-G00.

(c) Do an inspection of the blended areas.

1) Make sure that all the areas to repair are blended.

2) Do one of applicable tasks for the blades that you repair: Fluorescent Penetrant Inspection, TASK 70-11-06-230-802-G00, Fluorescent Penetrant Inspection (General), TASK 70-11-06-230-803-G00.

NOTE: Cracks are not permitted.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 72-00-00-350-005-G00

(1) Install the borescope plugs.

SUBTASK 72-00-00-410-034-G00

WARNING

OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS, BUT DO NOT DO THE THRUST REVERSER OR LEADING EDGE ACTIVATION. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 72-00-00-440-012-G00

(3) Do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801.

SUBTASK 72-00-00-860-091-G00

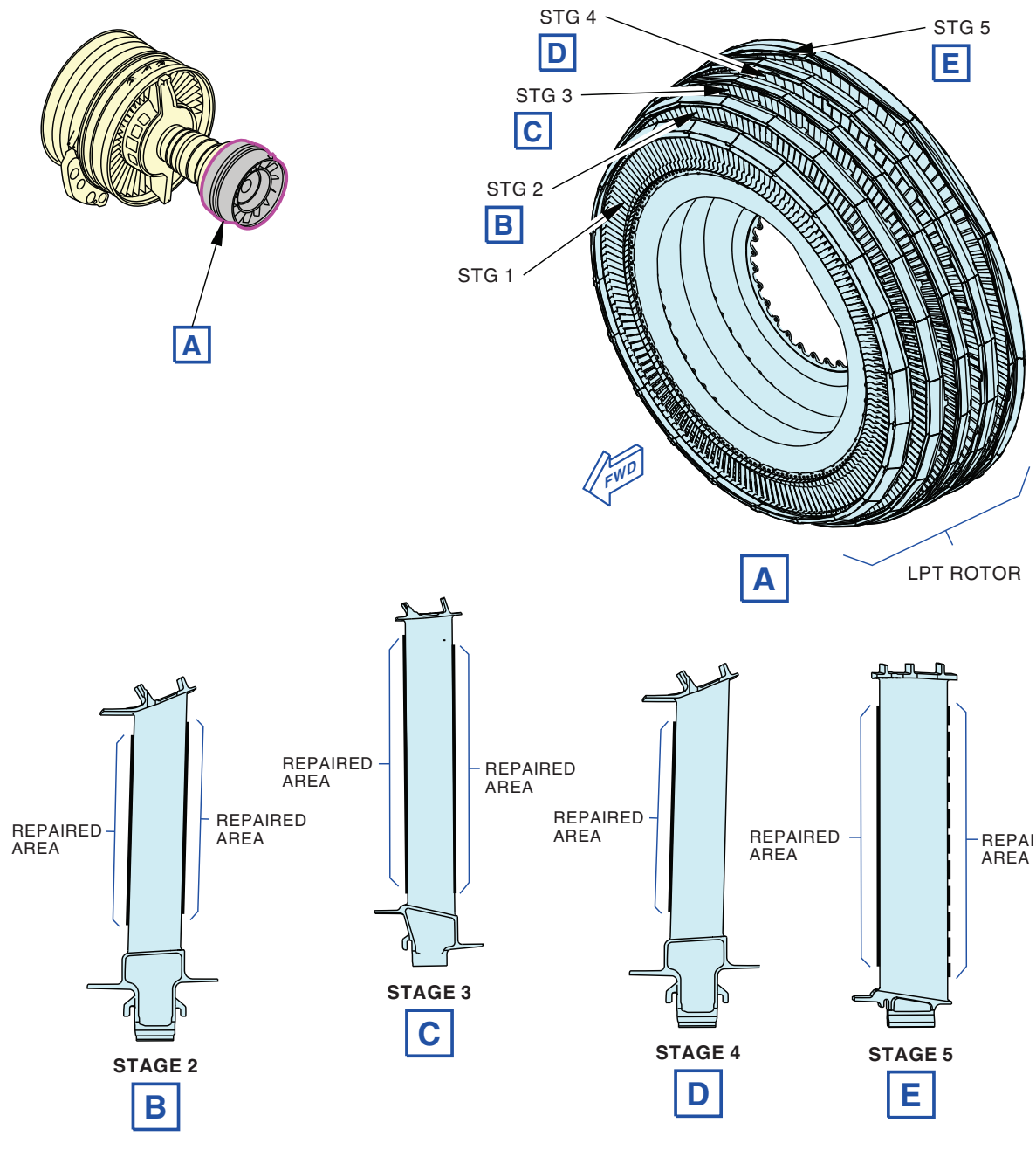
(4) Do these steps to remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LEGEND:

BLENDING

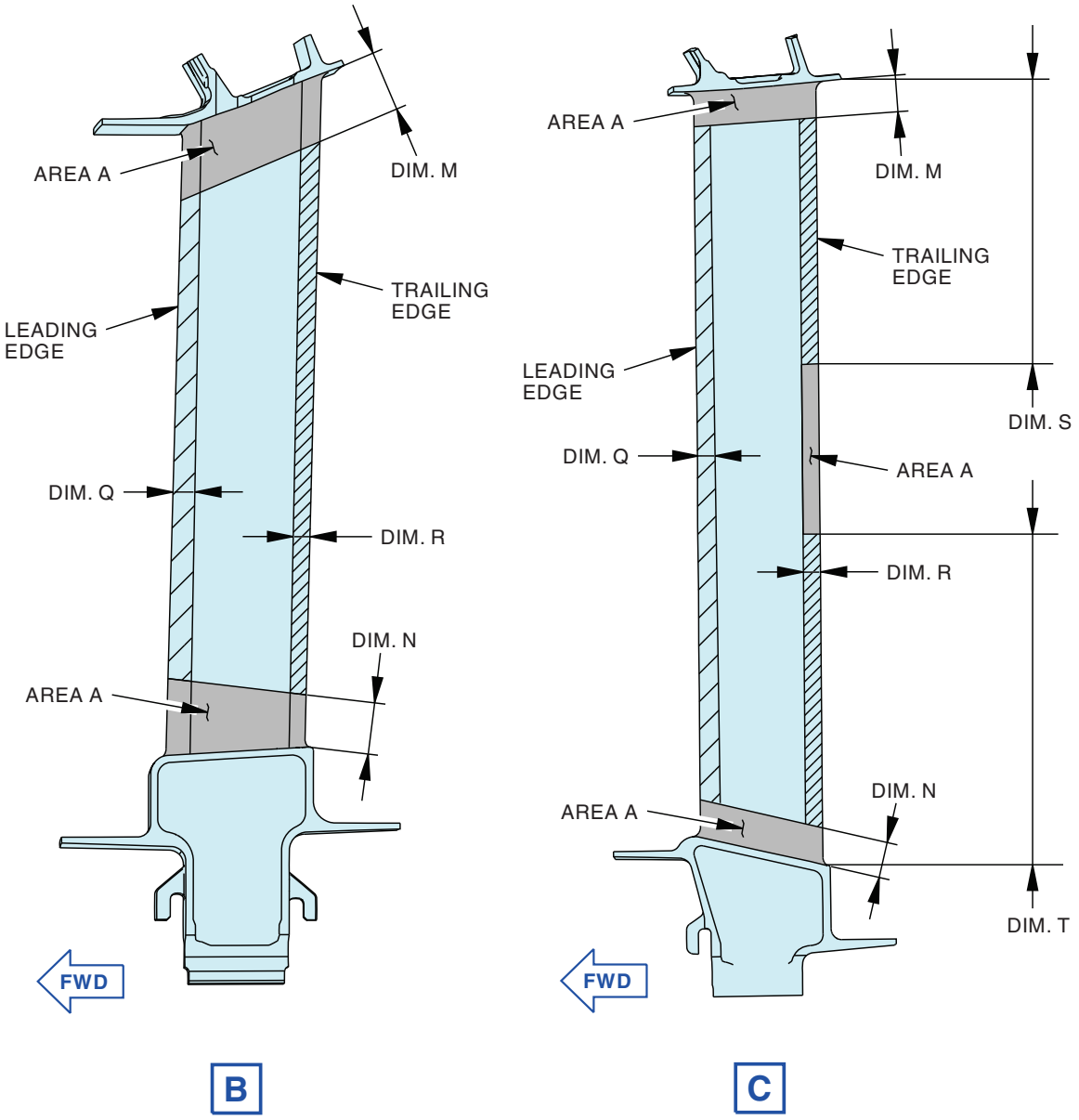
BORO-BLENDING

ICNLEAP1B720000CF030100101A00101

2991418 S0000768082_V1

Low Pressure Turbine (LPT) Blades Repair
Figure 801/72-00-00-990-863-G00 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LEGEND:

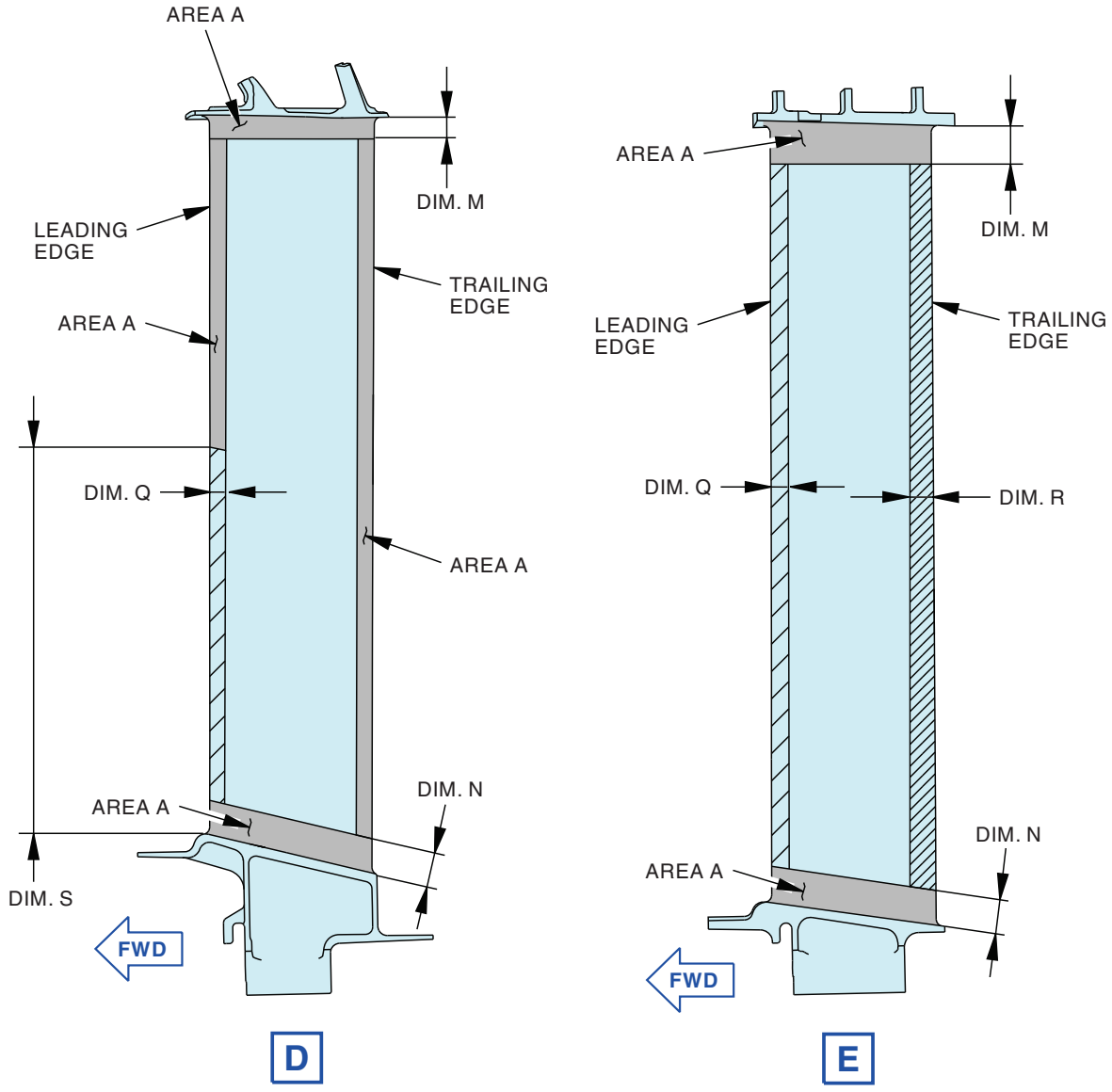
- TRAILING EDGE AREA: BLENDING PERMITTED
- LEADING EDGE AREA: BLENDING PERMITTED
- AREA A: NO BLENDING PERMITTED

ICNLEAP1B720000CF030100103A00101
ICNLEAP1B720000CF030100104A00101

2991431 S0000768086_V1

Low Pressure Turbine (LPT) Blades Repair
Figure 801/72-00-00-990-863-G00 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LEGEND:

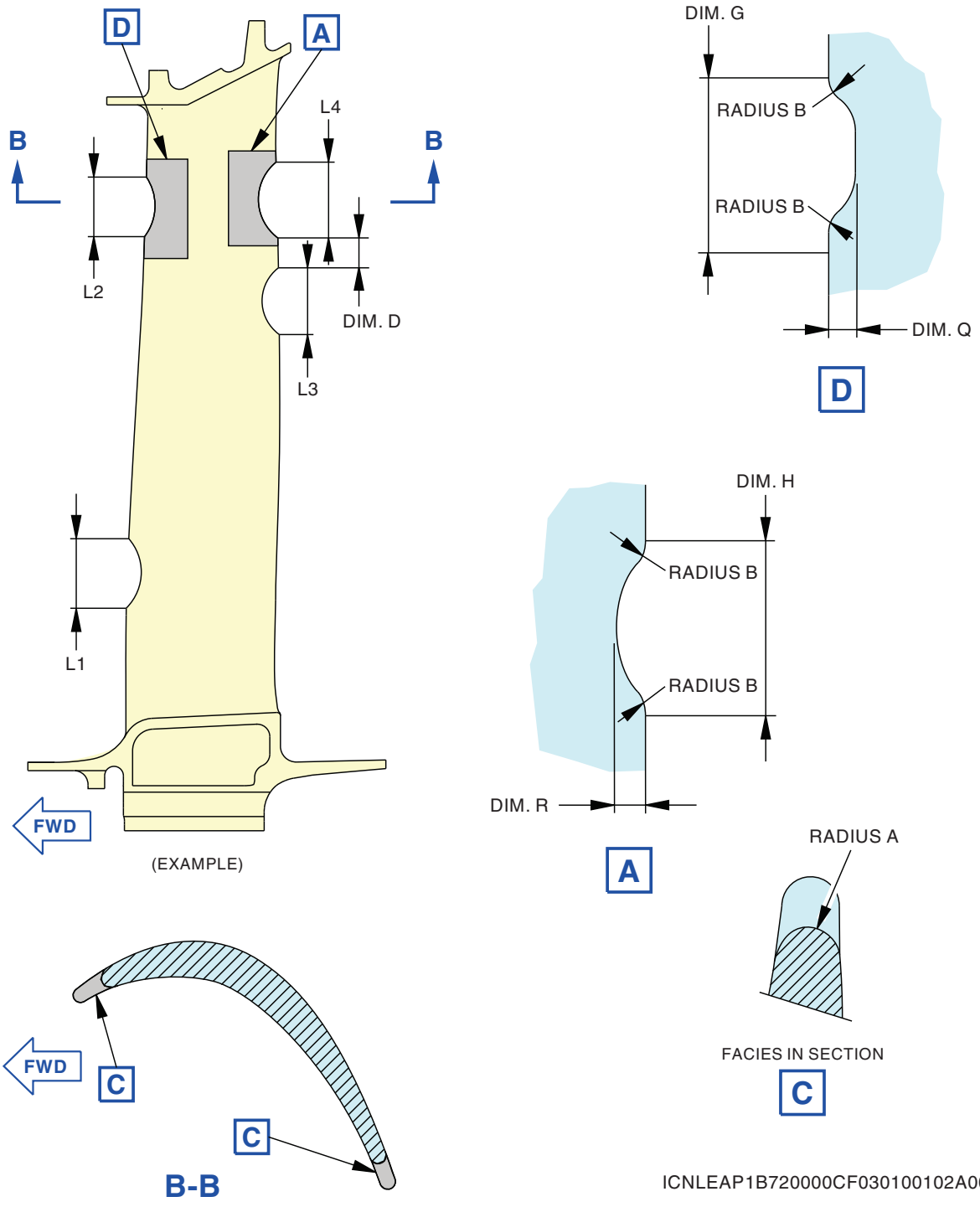
- TRAILING EDGE AREA: BLENDING PERMITTED
- LEADING EDGE AREA: BLENDING PERMITTED
- AREA A: NO BLENDING PERMITTED

ICNLEAP1B720000CF030100105A00101
ICNLEAP1B720000CF030100106A00101

2991436 S0000768096_V1

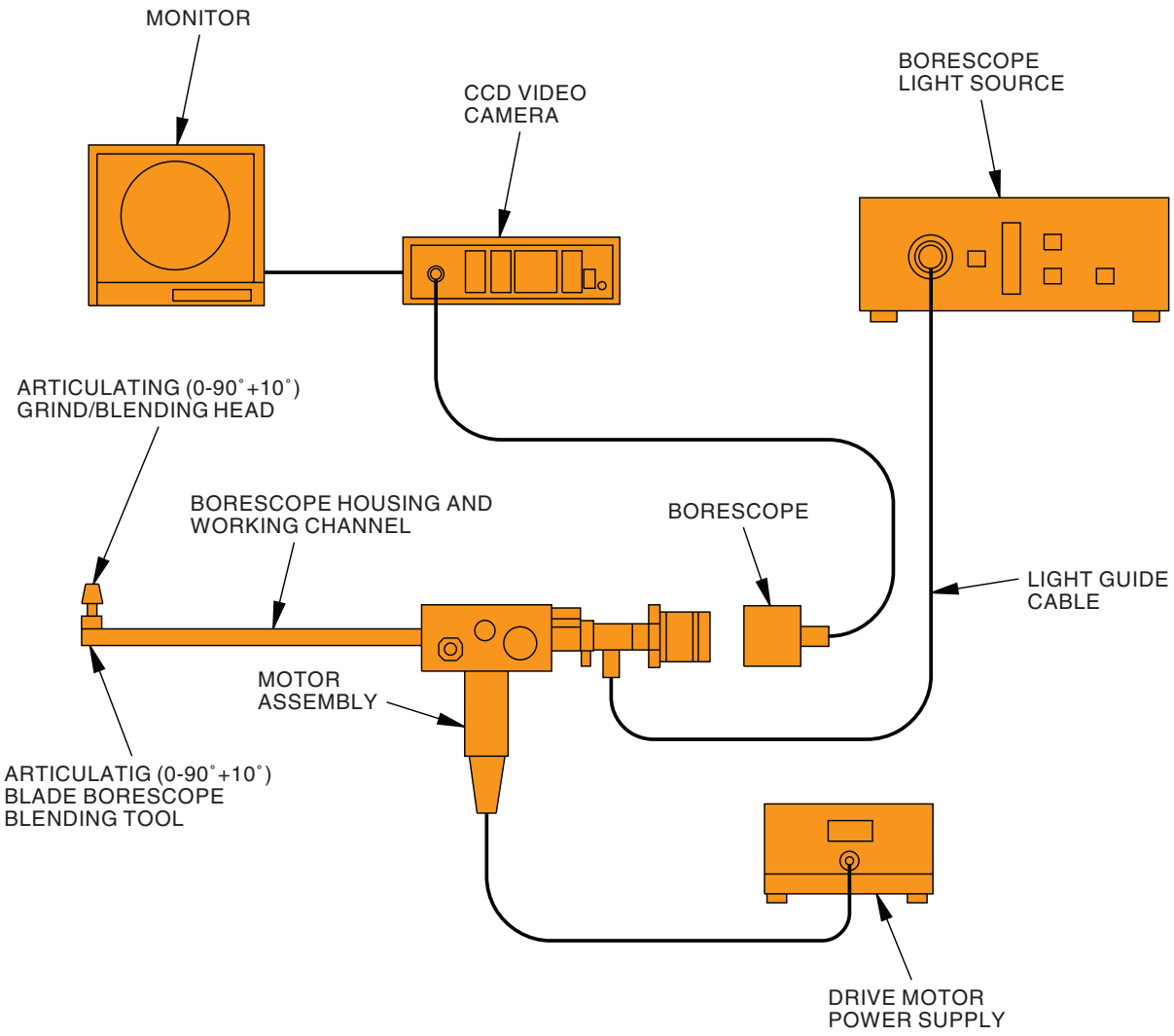
Low Pressure Turbine (LPT) Blades Repair
Figure 801/72-00-00-990-863-G00 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Low Pressure Turbine (LPT) Blades Blending Dimensions
Figure 802/72-00-00-990-864-G00

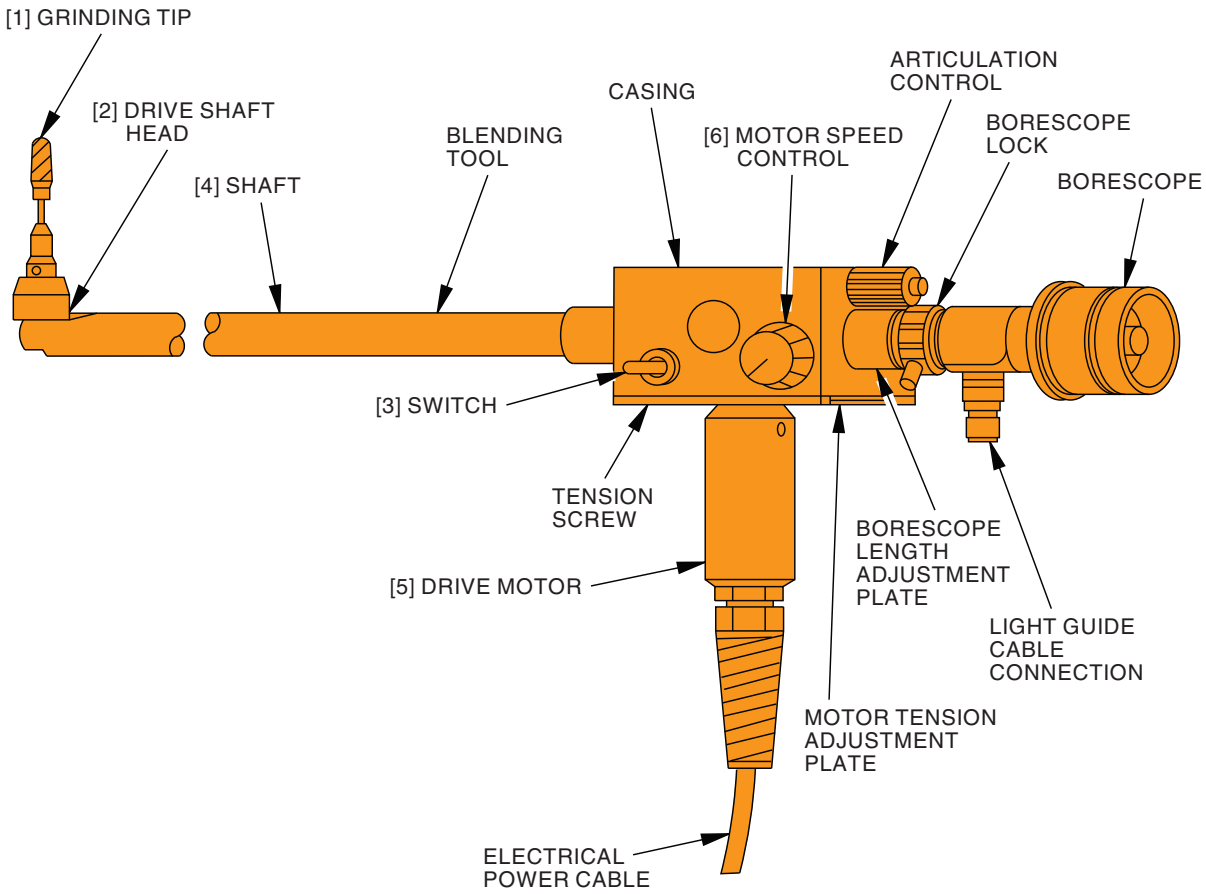
ECCN 9E991 BOEING PROPRIETARY - See title page for details



ICNLEAP1B720000CF030100107A00101
2991463 S0000768088_V1

Blade Borescope Blending Tool
Figure 803/72-00-00-990-866-G00 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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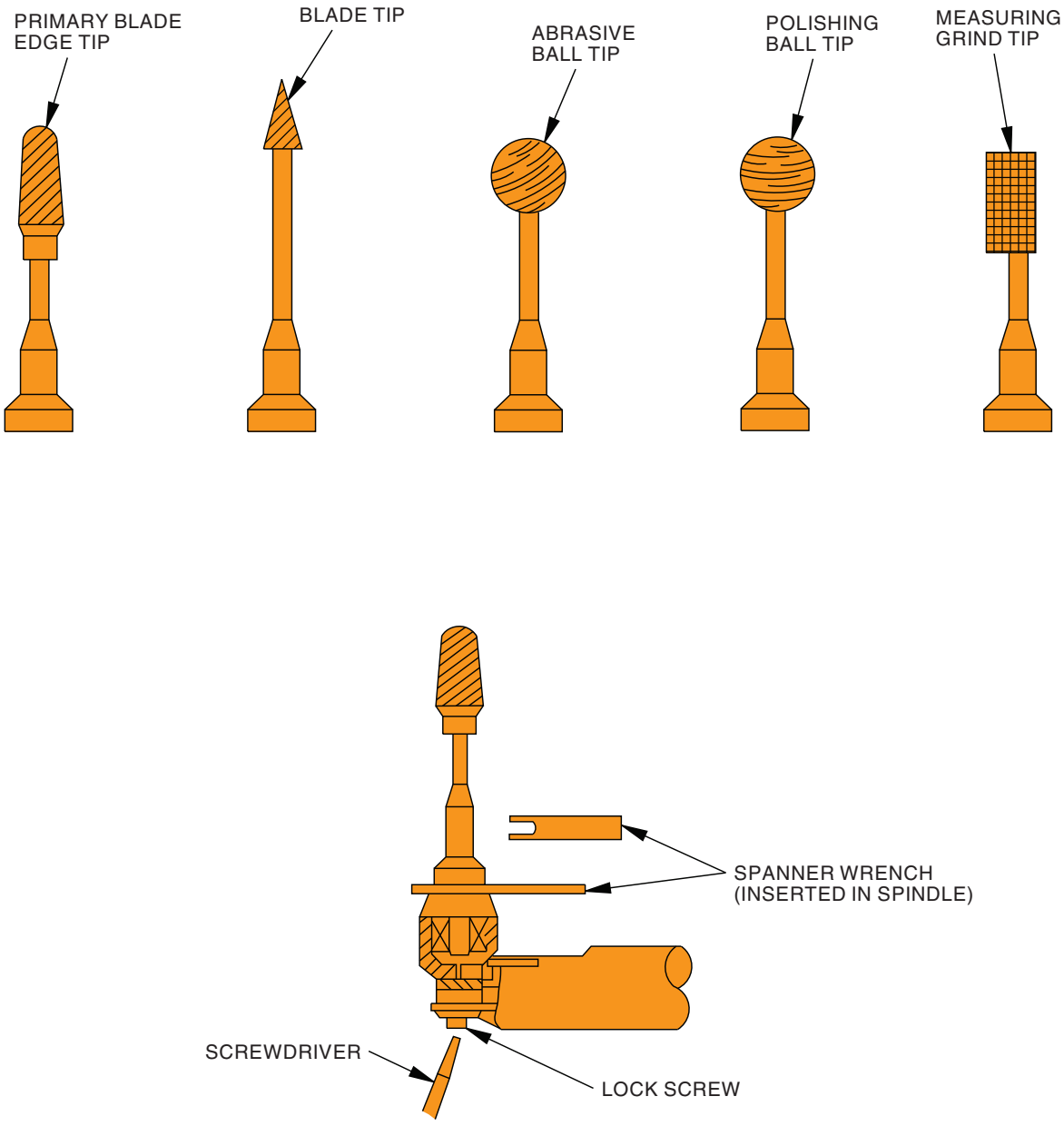
Blade Borescope Blending Tool
Figure 803/72-00-00-990-866-G00 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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2991566 S0000768098_V1

Blade Borescope Blending Tool
Figure 803/72-00-00-990-866-G00 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FAN STATOR ASSEMBLY - INSPECTION/CHECK

1. General

A. This procedure has these tasks:

(1) Fan Case Module Inspection

(2) Confirmed Air Leakage in the Fan Compartment Fan Case Color Inspection

(3) Low Pressure Compressor Stage 1 Inlet Guide Vane and Rotating Ring Spacer Inspection.

TASK 72-21-00-200-802-G00

2. Fan Case Module Inspection

(Figure 601)

A. General

(1) This task gives the instructions to examine the fan case module.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
72-21-05-300-801-G00	Acoustic Shroud Repair by Filling Up (P/B 801)
72-21-05-300-802-G00	Acoustic Shroud Seal Replacement (P/B 801)
72-21-08-300-801-G00	Acoustic Panel Impact Repair by Filling Up (P/B 801)
72-23-01-300-801-G00	Fan Case Abradable Coating Repair (P/B 801)
72-23-01-300-802-G00	Fan Case Fireproof Layer Local Repair (P/B 801)
72-23-01-300-803-G00	Fan Case Abradable Liner Cracks Local Repair (P/B 801)
72-23-01-960-801-G00	Fan Case Acoustic Shroud Bracket and/or Attachment Parts Replacement (P/B 801)
72-23-01-960-802-G00	Fan Case Counterplates Replacement (P/B 801)
72-23-01-960-803-G00	Fan Case Impact Repair on the Trench Filler (P/B 801)
72-23-01-960-804-G00	Fan Case Rivetless Nut Plate Assembly Replacement (P/B 801)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Prepare for the Inspection

SUBTASK 72-21-00-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-21-00-860-002-G00

(2) Open these applicable circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 72-21-00-860-003-G00

(3) Do this task: Open the Fan Cowl Panels (Selection), TASK 71-11-04-010-801-G00.

(a) On the applicable engine, open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

G. Fan Case Module Inspection

SUBTASK 72-21-00-211-001-G00

(1) Do a visual examination of the fan case:

(a) Examine all areas for these conditions:

1) Buckling of carbon tows (Figure 601 (Sheet 2)).

a) Permitted.

2) Resin reach area (Figure 601 (Sheet 3)).

a) Permitted.

3) Surface porosity except on abradable (Figure 601 (Sheet 4)).

a) Permitted.

4) Etched composite surface (Figure 601 (Sheet 5)).

a) Not permitted.

5) Coloration (Figure 601 (Sheet 6)).

a) Permitted.

(b) Examine the glass fiber coating areas for these conditions (Figure 601 (Sheet 1)):

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1) Cracks, tears in glass fiber protective plies, missing protective plies, missing or disbonded Room Temperature Vulcanization (RTV) on fibre glass reworked area (Figure 601 (Sheet 26) and Figure 601 (Sheet 27)).
   a) Permitted as follows:
      <1> Not more than 0.6200 in^2 (400 mm^2) each.
      <2> Distance between the cracks, tears, or missing protective plies area is more than 7.87 in. (200 mm).
      <3> Not more than 6.2000 in^2 (4000 mm^2) in cumulative surface.
   b) Maximum repairable limit: in the limit of the applicable repair.
   c) Repair method: Fan Case Fireproof Layer Local Repair, TASK 72-23-01-300-802-G00.
2) Delamination of glass fiber protective plies.
   a) Permitted as follows:
      <1> Not more than 0.6200 in^2 (400 mm^2) each.
      <2> Distance between delamination is more than 7.87 in. (200 mm).
      <3> Not more than 6.2000 in^2 (4000 mm^2) in cumulative surface.
   b) Maximum repairable limit: in the limit of the applicable repair.
   c) Repair method: Fan Case Fireproof Layer Local Repair, TASK 72-23-01-300-802-G00.
3) Fretting of glass fiber protective plies.
   a) Permitted as follows:
      <1> Not more than 0.003 in. (0.1 mm) in depth.
      <2> Not more than 0.6200 in^2 (400 mm^2) each.
      <3> Distance between fretting is more than 7.87 in. (200 mm).
   b) Maximum repairable limit: in the limit of the applicable repair.
   c) Repair method: Fan Case Fireproof Layer Local Repair, TASK 72-23-01-300-802-G00.
4) Crazing or scratches.
   a) Permitted up to 0.003 in. (0.1 mm) in depth.
   b) Maximum repairable limit: in the limit of the applicable repair.
   c) Repair method: Fan Case Fireproof Layer Local Repair, TASK 72-23-01-300-802-G00.
5) Nicks, dents, scratches on RTV in fiber glass reworked area (Figure 601 (Sheet 26) and Figure 601 (Sheet 27)).
   a) Permitted as follows:
      <1> No carbon visible.
      <2> One surface impacted: not more than 0.6200 in^2 (400 mm^2).
      <3> Minimal distances between two indications: 7.87 in. (200 mm).
      <4> Not more than 6.200 in^2 (4000 mm^2) in cumulative surface.
   b) Maximum repairable limit: in the limit of the applicable repair.

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c) Repair method: Fan Case Fireproof Layer Local Repair, TASK 72-23-01-300-802-G00.

(c) Examine the outer diameter of A1 flange for these conditions (Figure 601 (Sheet 7) and Figure 601 (Sheet 9)):

1) Nicks, dents, scratches with no carbon visible.

a) Permitted as follows:

<1> Not more than 0.011 in. (0.3 mm) in depth.

<2> Not more than 3.93 in. (100 mm) in length.

<3> Not more than 39.37 in. (1000 mm) in cumulative length.

b) Maximum repairable limit: in the limit of the applicable repair.

c) Repair method: Fan Case Fireproof Layer Local Repair, TASK 72-23-01-300-802-G00.

2) Nicks, dents, scratches with carbon visible.

a) Permitted as follows:

<1> Not more than 0.027 in. (0.7 mm) in depth with fiberglass.

<2> Not more than 0.019 in. (0.5 mm) in depth without fiberglass.

<3> Not more than 5.90 in. (150 mm) in length.

<4> Distance between the nicks, dents, scratches is more than 14.18 in. (360 mm).

<5> Not more than 11.81 in. (300 mm) in cumulative length.

b) Maximum repairable limit: in the limit of the applicable repair.

c) Repair method: Fan Case Fireproof Layer Local Repair, TASK 72-23-01-300-802-G00.

3) Missing protective plies area (from edge to edge).

a) Permitted as follows:

<1> Max cumulated surface up to 3.4645 in^2 (2235 mm^2).

<2> No carbon visible.

b) Maximum repairable limit: in the limit of the applicable repair.

c) Repair method: Fan Case Fireproof Layer Local Repair, TASK 72-23-01-300-802-G00.

(d) Examine the outer diameter of the B2 flange for these conditions (Figure 601 (Sheet 7) and Figure 601 (Sheet 9)):

1) Nicks, dents, scratches with no carbon visible.

a) Permitted as follows:

<1> Not more than 0.011 in. (0.3 mm) in depth.

<2> Not more than 3.93 in. (100 mm) in length.

<3> Not more than 39.37 in. (1000 mm) in cumulative length.

b) Maximum repairable limit: in the limit of the applicable repair.

c) Repair method: Fan Case Fireproof Layer Local Repair, TASK 72-23-01-300-802-G00.

2) Nicks, dents, scratches with carbon visible.

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a) Permitted as follows:

<1> Not more than 0.027 in. (0.7 mm) in depth with fiberglass.

<2> Not more than 0.019 in. (0.5 mm) in depth without fiberglass.

<3> Not more than 1.57 in. (40 mm) in length.

<4> Distance between the nicks, dents, scratches is more than 14.18 in. (360 mm).

<5> Not more than 11.81 in. (300 mm) in cumulative length.

b) Maximum repairable limit: in the limit of the applicable repair.

c) Repair method: Fan Case Fireproof Layer Local Repair, TASK 72-23-01-300-802-G00.

(e) Examine the outer diameter edges of A1 and B2 flanges (Figure 601 (Sheet 29)).

1) Nicks, dents, scratches.

a) Permitted as follows:

<1> Not more than 0.0787 in. (2 mm) in depth.

<2> Not more than 0.236 in. (6 mm) in length.

<3> Distance between two indications is more than 14.18 in. (360 mm).

<4> Not more than 10 indications per flange.

<5> No carbon visible.

b) Maximum repairable limit: in the limit of the applicable repair.

c) Repair method: Fan Case Fireproof Layer Local Repair, TASK 72-23-01-300-802-G00.

(f) Examine the forward face of A1 flange for these conditions (Figure 601 (Sheet 7), Figure 601 (Sheet 9), Figure 601 (Sheet 10), and Figure 601 (Sheet 11)):

NOTE: This surface is only visible when the inlet is removed from the engine. If the inlet is installed, do not remove it.

1) Nicks, dents, and scratches.

a) Permitted as follows:

<1> No carbon visible.

<2> Not more than 0.03 in. (1 mm) in depth.

<3> Not more than 3.93 in. (100 mm) in length.

<4> Not more than 39.37 in. (1000 mm) in cumulative length.

b) Maximum repairable limit: in the limit of the applicable repair.

c) Repair method: Fan Case Fireproof Layer Local Repair, TASK 72-23-01-300-802-G00.

2) Elongated fastening holes on inner diameter.

a) Permitted as follows:

<1> Not more than 0.610 in. (15.5 mm) on the diameter.

NOTE: Nominal diameter is 0.3980 in. (10.11 mm).

<2> Minimal distance of 15.74 in. (400 mm) between two elongated holes.

b) Maximum repairable limit: in the limit of the applicable repair.

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c) Repair method: Fan Case Fireproof Layer Local Repair, TASK 72-23-01-300-802-G00.

3) Elongated locating holes on the inner diameter.

NOTE: On the A1 flange, there are two locating holes. All other holes on inner diameter correspond to fastening holes.

a) Permitted if not more than 0.271 in. (6.9 mm) on the diameter.

NOTE: Nominal diameter is 0.2673 in. (6.79 mm).

b) Maximum repairable limit: in the limit of the applicable repair.

c) Repair method: Fan Case Fireproof Layer Local Repair, TASK 72-23-01-300-802-G00.

4) Elongated brackets attachment holes on the outer diameter.

a) Permitted if not more than 0.2803 in. (7.12 mm) on the diameter.

NOTE: Nominal diameter is 0.2760 in. (7.01 mm).

b) Maximum repairable limit: in the limit of the applicable repair.

c) Repair method: Fan Case Fireproof Layer Local Repair, TASK 72-23-01-300-802-G00.

5) Elongated brackets attachment counterbore on outer diameter.

a) Maximum repairable limit: in the limit of the applicable repair.

b) Repair method: Fan Case Fireproof Layer Local Repair, TASK 72-23-01-300-802-G00.

(g) Examine the forward face of A1 flange around the bolt holes for these conditions (Figure 601 (Sheet 7) and Figure 601 (Sheet 9)):

NOTE: The radius of area around bolt holes is 0.19 in. (5 mm).

1) Fraying at the bolt hole edges/resin missing.

a) Permitted up to 0.03 in. (1 mm) around the bolt hole on the whole hole inner face.

b) Maximum repairable limit: in the limit of the applicable repair.

c) Repair method: Fan Case Fireproof Layer Local Repair, TASK 72-23-01-300-802-G00.

(h) Examine the aft face of A1 flange around bolt holes for these conditions (Figure 601 (Sheet 7), Figure 601 (Sheet 9), Figure 601 (Sheet 10), and Figure 601 (Sheet 11)):

NOTE: The radius of area around bolt holes is 0.19 in. (5 mm).

1) Fretting.

a) Permitted up to 0.0098 in. (0.25 mm) in depth over 20 percent of area around each hole.

2) Nicks, dents, scratches.

a) Permitted up to 0.0019 in. (0.05 mm) in depth.

3) Elongated fastening inner diameter holes.

NOTE: The two locating holes on inner diameter are located at the same position as the ones on forward face of A1 flange.

a) Permitted if not more than 0.527 in. (13.4 mm) on the diameter.

NOTE: Nominal diameter is 0.3980 in. (10.11 mm).

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4) Elongated locating holes on inner diameter (Figure 601 (Sheet 10)).

NOTE: The two locating holes on inner diameter are located at the same position as the ones on forward face of A1 flange.

a) Permitted if not more than 0.2716 in. (6.90 mm) on the diameter.

NOTE: Nominal diameter is 0.2673 in. (6.79 mm).

5) Elongated support gear outer diameter holes.

a) Permitted if not more than 0.51 in. (13 mm) on the diameter.

NOTE: Nominal diameter is 0.2760 in. (7.01 mm).

(i) Examine the other areas of the aft face of A1 flange for these conditions (Figure 601 (Sheet 7) and Figure 601 (Sheet 9)):

1) Nicks, dents, scratches.

a) Permitted as follows:

<1> All amounts up to 0.011 in. (0.3 mm) in depth.

<2> All amounts up to 3.93 in. (100 mm) in length.

<3> Not more than 11.81 in. (300 mm) in cumulative length per stiffener plate.

b) Maximum repairable limit: any amount.

c) If it is necessary, do this task: Fan Case Counterplates Replacement, TASK 72-23-01-960-802-G00.

2) Stiffener plate rivet missing.

a) Permitted.

b) Maximum repairable limit: any amount.

c) If it is necessary, do this task: Fan Case Counterplates Replacement, TASK 72-23-01-960-802-G00.

(j) Examine the inner radius of A1 and B2 flange for these conditions (Figure 601 (Sheet 7) and Figure 601 (Sheet 9)):

1) Nicks, dents, scratches of A1 flange.

a) Permitted as follows:

<1> All amounts up to 0.011 in. (0.3 mm) in depth.

<2> All amounts up to 1.18 in. (30 mm) in length.

<3> Distance between the nicks, dents, scratches is more than 14.18 in. (360 mm).

<4> Not more than 11.81 in. (300 mm) in cumulative length.

b) Maximum repairable limit: in the limit of the applicable repair.

c) Repair method: Fan Case Fireproof Layer Local Repair, TASK 72-23-01-300-802-G00.

2) Nicks, dents, scratches of B2 flange.

a) Permitted as follows:

<1> All amounts up to 0.011 in. (0.3 mm) in depth.

<2> All amounts up to 1.18 in. (30 mm) in length.

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<3> Distance between the nicks, dents, scratches is more than 14.18 in. (360 mm).

<4> Not more than 11.81 in. (300 mm) in cumulative length.

b) Maximum repairable limit: in the limit of the applicable repair.

c) Repair method: Fan Case Fireproof Layer Local Repair, TASK 72-23-01-300-802-G00.

(k) Examine the outer skin of the fan case [1] for these conditions (Figure 601 (Sheet 1), Figure 601 (Sheet 2), and Figure 601 (Sheet 8)):

1) Worn thread or worn back bag on fixed insert.

a) Not permitted.

2) Loose or missing fixed insert.

a) Not permitted.

3) Twisted, torn or missing captive nut.

a) Not permitted.

4) Nicks, dents, scratches on glass fiber plies.

a) Permitted as follows:

<1> No carbon is visible.

<2> Not more than 0.011 in. (0.3 mm) in depth.

<3> Not more than 11.811 in. (300 mm) in length.

<4> Not more than 118.11 in. (3000 mm) in cumulative length.

b) Maximum repairable limit: in the limit of the applicable repair.

c) Repair method: Fan Case Fireproof Layer Local Repair, TASK 72-23-01-300-802-G00.

5) Loose, missing or broken stud.

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 75-0006

a) Any amount if the associated heat shield is still maintained either by Velcro without wear or by brackets to the flange of the fan frame shroud and if the associated washer and nut are missing.

NOTE: Removal of the heat shield per Service Bulletin LEAP-1B-72-00-0006-01A-930A-C is recommended at first opportunity.

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b) Any amount.

PTF ALL

6) Missing or worn Velcro.

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 75-0006

a) Any amount if the associated heat shield is still maintained by stud, washer, and nut or by brackets to the flange of the fan frame shroud.

NOTE: Removal of the heat shield per Service Bulletin LEAP-1B-72-00-0006-01A-930A-C is recommended at first opportunity.

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PTF ALL; POST LEAP-1B SERVICE BULLETIN 75-0006

b) Any amount.

PTF ALL

(l) Examine the flowpath for these conditions (Figure 601 (Sheet 12)):

1) Impact marks or craters (missing material) on abradable (Figure 601 (Sheet 31)).

a) Permitted as follows:

<1> Not more than 3.8750 in^2 (2500 mm^2) for a single impacted surface.

<2> Not more than 155 in^2 (100,000 mm^2) in cumulated impacted surface.

<3> The minimum distance between affected areas is equal to the maximum width of the nearest one affected area (Figure 601 (Sheet 31)).

<4> Depths: the carbon plies of the trench filler do not show through.

b) Maximum repairable limit: Any amount.

c) If it is necessary, do this task: Fan Case Abradable Coating Repair, TASK 72-23-01-300-801-G00.

NOTE: If the carbon plies show through, a Continue-In-Service (CIS) limit of 50 cycles or 125 hours is permitted. Carbon plies must remain within the carbon plies criteria under visual inspection. A daily inspection of the crater must be performed.

2) Circumferential groove on abradable, oriented within 80 degrees to 90 degrees from engine axis (Figure 601 (Sheet 32)) and (Figure 601 (Sheet 12)).

NOTE: For each AREA, the measure is equal to the angular location where the depth is the highest.

NOTE: Depth is measured from area of the abradable, which shows no sign of missing material.

a) Permitted as follows in AREA 1:

<1> Whether or not there is a rub.

<a> Any number and less than 0.59 in. (15 mm) in cumulative width.

<b> Depth: the carbon plies of the trench filler do not show through.

<c> The width of the unitary groove is less than or equal to 0.098 in. (2.5 mm).

NOTE: If the width of the groove cumulative is more than 0.59 in. (15 mm) and less than 0.78 in. (20 mm), a CIS limit of 50 cycles is permitted.

<2> If a rub is NOT reported in the same area.

<a> If the average depth (based on measurement at eight equally spaced locations) is up to 0.118 in. (3.0 mm): any number permitted up to 0.39 in. (10 mm) in width.

<b> If average depth (based on measurement at eight equally spaced locations) is between 0.118 in. (3.0 mm) and 0.196 in. (5.0 mm): any number permitted up to 0.59 in. (15 mm) in cumulated width.

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<c> Both criteria can be cumulated up to 1.29 in. (33 mm) in total width.

NOTE: Rub criteria is still applicable.

<3> Maximum repairable limit: Any amount.

b) Permitted as follows in AREA 2 and AREA 3:

<1> Any number and less than 1.96 in. (50 mm) in cumulative width (with respect of the grooves criteria in the AREA 1.

<2> The width of the unitary groove is less than or equal to 0.098 in. (2.5 mm).

<3> Maximum repairable limit: Any amount.

c) Depth: the carbon plies of the trench filler do not show through.

d) If it is necessary, do this task: Fan Case Abradable Coating Repair, TASK 72-23-01-300-801-G00.

NOTE: If the carbon plies show through, a CIS limit of 50 cycles or 125 hours is permitted. A daily inspection of the groove or rub must be performed.

3) Other groove on abradable (Figure 601 (Sheet 33)).

a) Permitted as follows:

<1> Up to 4 grooves, if not more than 1.10 in. (28 mm) in width.

<2> All depths permitted.

<3> Depth: the carbon plies of the trench filler do not show through.

b) If it is necessary, do this task: Fan Case Abradable Coating Repair, TASK 72-23-01-300-801-G00.

NOTE: If the carbon plies show through, a CIS limit of 50 cycles or 125 hours is permitted. A daily inspection of the groove must be performed.

4) Rub along airflow on abradable (Figure 601 (Sheet 32)) and (Figure 601 (Sheet 12)).

NOTE: Rub are defined as shown in (Figure 601 (Sheet 34)).

NOTE: Depth is measured from area of the abradable, which shows no sign of missing material.

NOTE: For each AREA, the measure is equal to the angular location where the depth is the highest.

a) Permitted as follows in AREA 1:

<1> Permitted up to 0.027 in. (0.7 mm) in average depth (based on measurement at eight equally spaced locations on abradable) (Figure 601 (Sheet 35)).

<2> Any number.

<3> Depth: the carbon plies of the trench filler do not show through.

b) Permitted as follows in AREA 2:

<1> Permitted up to 0.027 in. (0.7 mm) in average depth (based on measurement at eight equally spaced locations on abradable) (Figure 601 (Sheet 35)).

<2> Any number.

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<3> Depth: the carbon plies of the trench filler do not show through.

c) Permitted as follows in AREA 3:

<1> Permitted up to 0.118 in. (3.0 mm) in average depth (based on measurement at eight equally spaced locations on abradable) (Figure 601 (Sheet 35)).

<2> Any number.

<3> Depth: the carbon plies of the trench filler do not show through.

d) Maximum repairable limit: Any amount.

e) If it is necessary, do this task: Fan Case Abradable Coating Repair, TASK 72-23-01-300-801-G00.

NOTE: If the carbon plies show through, a CIS limit of 50 cycles or 125 hours is permitted. A daily inspection of the groove or rub must be performed.

5) Cracks on abradable (Figure 601 (Sheet 36)) and (Figure 601 (Sheet 37)).

a) Permitted as follows: Any amount.

b) Maximum repairable limit: in the limit of the applicable repair.

c) If it is necessary, do this task: Fan Case Abradable Liner Cracks Local Repair, TASK 72-23-01-300-803-G00.

6) Riddled with impacts aspect on abradable (Figure 601 (Sheet 12)) and (Figure 601 (Sheet 38)).

a) Permitted as follows:

<1> Aft of X plane: Any amount.

<2> Depth: the carbon plies of the trench filler do not show through.

7) Missing RTV between each acoustic panel.

a) Permitted.

8) Nicks, dents, scratches, impact marks on carbon plies (Figure 601 (Sheet 13)).

NOTE: Carbon plies are visible when abradable is missing. Do not remove abradable.

a) Permitted as follows:

<1> Not more than 0.011 in. (0.3 mm) in depth.

<2> Not more than 0.78 in. (20 mm) in length.

<3> Not more than 0.11 in. (3 mm) in width.

<4> Distance between the nicks, dents, scratches, impact marks is more than 7.88 in. (200 mm).

9) Tears, cracks, punctures on carbon plies (Figure 601 (Sheet 13)).

a) Not permitted.

b) Maximum repairable limit: in the limit of the applicable repair.

c) If it is necessary, do this task: Fan Case Impact Repair on the Trench Filler, TASK 72-23-01-960-803-G00.

10) Local collapse of inner surface on carbon plies (Figure 601 (Sheet 13)).

a) Permitted as follows:

<1> Not more than 5.

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<2> Distance between the local collapses is more than 9.85 in. (250 mm).

<3> Not more than 1.57 in. (40 mm) in diameter.

b) Maximum repairable limit: in the limit of the applicable repair.

c) If it is necessary, do this task: Fan Case Impact Repair on the Trench Filler, TASK 72-23-01-960-803-G00.

11) Surface porosity on abradable strip (Figure 601 (Sheet 15)).

a) Permitted any amount.

NOTE: Craters on abradable and riddled abradable must not be mistaken with porosities on abradable strip (Figure 601 (Sheet 31) and Figure 601 (Sheet 38)).

(m) Examine the inner skin of the fan case for these conditions (Figure 601 (Sheet 1)):

NOTE: The inner skin of the fan case is visible when the acoustic shroud or acoustic panels are removed. If they are installed, do not remove the parts.

1) Nicks, dents, scratches.

a) Permitted as follows:

<1> Not more than 0.011 in. (0.3 mm) in depth.

<2> Not more than 3.93 in. (100 mm) in length.

<3> Distance between the nicks, dents, scratches is more than 14.18 in. (360 mm).

<4> Not more than 39.37 in. (1000 mm) in cumulative length.

2) Porosities in the adhesive flash between the edge of trenchfiller and fan case (Figure 601 (Sheet 28)).

a) Maximum Serviceable Limit: Any amount.

(n) Examine the edges of the trench filler for these conditions (Figure 601 (Sheet 7)):

1) Missing RTV between trenchfiller and fan case.

a) Permitted except on the 19.68 in. (500 mm) circumferential length area centered at 6:00 o'clock on the forward edge.

2) Fretting.

a) Permitted up 0.003 in. (0.1 mm) in depth.

3) Nicks, dents, scratches.

a) Permitted as follows:

<1> Not more than 0.011 in. (0.3 mm) in depth.

<2> Not more than 0.78 in. (20 mm) in length.

<3> Not more than 0.11 in. (3 mm) in width.

<4> Distance between the nicks, dents, scratches is more than 7.87 in. (200 mm).

4) Tears, cracks, punctures on carbon plies (Figure 601 (Sheet 13)):

a) Not permitted.

(o) Examine the radius of the 13 brackets on the fan case for these conditions (Figure 601 (Sheet 14)):

NOTE: Brackets are visible when the acoustic shroud is removed. If it is installed, do not remove it.

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1) Nicks, dents, scratches, cracks.
   a) Not permitted.
   b) Maximum repairable limit: Any amount.
   c) If it is necessary, do this task: Fan Case Acoustic Shroud Bracket and/or Attachment Parts Replacement, TASK 72-23-01-960-801-G00.

(p) Examine the area around the holes on each of the 13 brackets on the fan case for these conditions (Figure 601 (Sheet 14)):

NOTE: The radius of area around bolt holes is 0.11 in. (3 mm).

1) Nicks, dents, scratches, cracks.
   a) Permitted up to 0.0019 in. (0.05 mm) in depth.
   b) Maximum repairable limit: Any amount.
   c) If it is necessary, do this task: Fan Case Acoustic Shroud Bracket and/or Attachment Parts Replacement, TASK 72-23-01-960-801-G00.

(q) Examine the other surfaces of the 13 brackets on the fan case for these conditions (Figure 601 (Sheet 14)):

1) Missing bolt or captive nut on the bracket.
   a) Not permitted.
   b) Maximum repairable limit: Any amount.
   c) If it is necessary, do this task: Fan Case Acoustic Shroud Bracket and/or Attachment Parts Replacement, TASK 72-23-01-960-801-G00.
2) Twisted or deformed bracket.
   a) Not permitted.
   b) Maximum repairable limit: Any amount.
   c) If it is necessary, do this task: Fan Case Acoustic Shroud Bracket and/or Attachment Parts Replacement, TASK 72-23-01-960-801-G00.
3) Torn, broken or unbonded bracket.
   a) Not permitted.
   b) Maximum repairable limit: Any amount.
   c) If it is necessary, do this task: Fan Case Acoustic Shroud Bracket and/or Attachment Parts Replacement, TASK 72-23-01-960-801-G00.
4) Nicks, dents, scratches, cracks.
   a) Permitted as follows:
      <1> Not more than 0.003 in. (0.1 mm) in depth.
      <2> Not more than 0.59 in. (15 mm) in length.
      <3> Not more than 0.007 in. (0.2 mm) in width.
   b) Maximum repairable limit: Any amount.
   c) If it is necessary, do this task: Fan Case Acoustic Shroud Bracket and/or Attachment Parts Replacement, TASK 72-23-01-960-801-G00.
5) Fretting.
   a) Permitted up to 0.070 in. (1.8 mm) minimum thickness for the bracket.
   b) Maximum repairable limit: Any amount.

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c) If it is necessary, do this task: Fan Case Acoustic Shroud Bracket and/or Attachment Parts Replacement, TASK 72-23-01-960-801-G00.

(r) Examine the rivetless nuts for these conditions:

1) Missing replaceable nut.

a) Not permitted.

b) Repairable limit: any amount.

c) If it is necessary, do this task: Fan Case Rivetless Nut Plate Assembly Replacement, TASK 72-23-01-960-804-G00.

2) Looseness.

a) Not permitted.

b) Repairable limit: any amount.

c) If it is necessary, do this task: Fan Case Rivetless Nut Plate Assembly Replacement, TASK 72-23-01-960-804-G00.

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(2) Do a visual examination of the acoustic shroud as follows (Figure 601 (Sheet 16)):

(a) Examine the inner surface of the acoustic shroud for these conditions (Figure 601 (Sheet 16) and Figure 601 (Sheet 21)):

PTF ALL; AIRPLANES WITHOUT SMOOTH ACOUSTIC SHROUD

1) Non-consecutive clogged hole or missing hole.

a) Permitted.

2) Consecutive clogged holes.

a) Not more than 2 affected areas of diameter 1.18 in. (30 mm).

3) Consecutive missing holes.

a) Not more than 2 holes.

NOTE: Missing hole is due to the manufacturing process.

PTF ALL

4) Nicks, dents, scratches, cracks, flaking on protective coating.

a) Permitted.

5) Tears, punctures, cracks through the skin of the protective coating.

a) Permitted.

6) Missing of protective coating.

a) Permitted.

7) Nicks, dents, scratches, impact marks on carbon plies.

a) Permitted as follows:

<1> Not more than 0.011 in. (0.3 mm) in depth.

<2> Not more than 0.78 in. (20 mm) in length.

<3> Not more than 0.11 in. (3 mm) in width.

<4> Distance between the nicks, dents, scratches, impact marks is more than 7.88 in. (200 mm).

8) Tears, cracks, punctures on carbon plies.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

a) Not permitted.

b) Maximum repairable limit: in the limit of the applicable repair.

c) If it is necessary, do this task: Acoustic Shroud Repair by Filling Up, TASK 72-21-05-300-801-G00.

9) Coloration aspect due to resin deposit.

a) Any amount.

NOTE: Coloration aspect is due to the manufacturing process.

PTF ALL; AIRPLANES WITHOUT SMOOTH ACOUSTIC SHROUD

10) Micro-shrinkage on inner face around the acoustic hole.

a) Any amount.

NOTE: Micro-shrinkage is due to the manufacturing process.

PTF ALL; AIRPLANES WITH SMOOTH ACOUSTIC SHROUD

11) Micro-shrinkage on inner face.

a) Any amount.

NOTE: Micro-shrinkage is due to the manufacturing process.

PTF ALL

12) Local collapse of inner surface.

a) Not permitted.

b) Maximum repairable limit: in the limit of the applicable repair.

c) If it is necessary, do this task: Acoustic Shroud Repair by Filling Up, TASK 72-21-05-300-801-G00.

13) Wrinkles or folds on protective coating, excluding radii between inner surface and panel edges.

a) Not more than 0.0118 in. (0.30 mm) in depth.

b) Not more than 5.90 in. (150 mm) cumulative length.

NOTE: Wrinkles and folds are due to the manufacturing process.

(b) Examine the outer surface of the acoustic shroud for these conditions (Figure 601 (Sheet 16) and Figure 601 (Sheet 17)):

NOTE: The outer surface is only visible if the acoustic shroud is removed. If it is installed, do not remove it.

1) Nicks, dents, scratches, impact marks.

a) Not more than 0.011 in. (0.3 mm) in depth.

b) Not more than 0.78 in. (20 mm) in length.

c) Not more than 0.11 in. (3 mm) in width.

d) Distance between the nicks, dents, scratches, impact marks is more than 7.88 in. (200 mm).

2) Tears, cracks, punctures on carbon plies.

a) Not permitted.

3) Local collapse.

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a) Any amount.

NOTE: The local collapse criterion is due to manufacturing.

4) Porosities.

a) Permitted if less than 0.0118 in. (0.30 mm) in depth.

NOTE: Porosities are due to the manufacturing process.

5) Coloration aspect.

a) Resin deposit.

<1> Permitted if the cumulative area is not more than 50% of outer surface.

b) Green-blue-white.

<1> Permitted if not more than 1.55 in^2 (10 cm^2) cumulative area.

NOTE: Coloration aspect is due to the manufacturing process.

(c) Examine the edges and radius of the acoustic shroud for these conditions (Figure 601 (Sheet 17) and Figure 601 (Sheet 19)):

NOTE: The forward edge is only visible if the inlet cowl is removed. The aftward edge is only visible if the acoustic shroud is removed. If parts are installed, do not remove them.

1) Porosities.

a) Any amount.

NOTE: Porosities are due to the manufacturing process.

2) Dry area.

a) Any amount.

NOTE: Dry areas are due to the manufacturing process.

3) Nicks, dents, scratches, impact marks on carbon plies.

a) Not more than 4 in. (101.6 mm) cumulative length.

b) Not more than 0.20 in. (5 mm) in width.

c) Not more than 0.040 in. (1 mm) in depth.

4) Fretting on forward edge.

a) Permitted if not more that 0.004 in. (0.1 mm) in depth.

5) Tears, cracks, punctures on carbon plies.

a) Not permitted.

6) Voids.

a) Permitted.

7) Coloration aspect (grey-blue-white) on forward edge and radius.

a) Permitted.

NOTE: Coloration aspect is due to the manufacturing process.

8) Coloration aspect on aftward edge.

a) Resin deposit.

<1> Permitted if the cumulative area is not more than 50% of edge surface.

b) Green-blue-white.

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<1> Permitted if not more than 0.775 in^2 (5 cm^2) cumulative area.

NOTE: Coloration aspect is due to the manufacturing process.

(d) Examine seal.

NOTE: If the inlet cowl and the acoustic shroud are installed, do not remove them.

1) Missing bonded seal.
   a) Permitted if not more than 39.37 in. (1000 mm) in cumulative length.
   b) Maximum repairable limit: Any amount.
   c) If it is necessary, do this task: Acoustic Shroud Seal Replacement, TASK 72-21-05-300-802-G00.
2) Cut, worn through and torn seal.

   NOTE: Cut the seal when it is visible in the flowpath.
   a) Permitted if not more than 39.37 in. (1000 mm) in cumulative length.
   b) Maximum repairable limit: Any amount.
   c) If it is necessary, do this task: Acoustic Shroud Seal Replacement, TASK 72-21-05-300-802-G00.
3) Crazing in seal.
   a) Permitted.
   b) Maximum repairable limit: Any amount.
   c) If it is necessary, do this task: Acoustic Shroud Seal Replacement, TASK 72-21-05-300-802-G00.

(e) Examine the radius of the brackets attached to the acoustic shroud for these conditions (Figure 601 (Sheet 18)):

NOTE: The brackets are only visible if the acoustic shroud is removed. If it is installed, do not remove it.

1) Nicks, dents, scratches, cracks.
   a) Not permitted.

(f) Examine the holes on the 13 brackets attached to the acoustic shroud for these conditions (Figure 601 (Sheet 18)):

NOTE: The brackets are only visible if the acoustic shroud is removed. If it is installed, do not remove it.

1) Fretting.
   a) Permitted up to 0.0098 in. (0.25 mm) in depth over 20% of the area around each hole.
2) Elongated screw holes.
   a) Permitted if not more than 0.2799 in. (7.11 mm).

      NOTE: The nominal diameter is 0.2756 in. (7 mm).
3) Elongated pin holes.
   a) Permitted if not more than 0.32116 in. (8.1575 mm).

      NOTE: The nominal diameter is 0.3209 in. (8.15 mm).

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(g) Examine the other surfaces of the brackets attached to the acoustic shroud for these conditions (Figure 601 (Sheet 18)):

NOTE: The brackets are only visible if the acoustic shroud is removed. If it is installed, do not remove it.

1) Fretting.
   a) Permitted up to 0.070 in. (1.8 mm) minimum thickness of the bracket.
2) Nicks, dents, scratches, cracks.
   a) Permitted as follows:
      <1> Not more than 0.003 in. (0.1 mm) in depth.
      <2> Not more than 0.59 in. (15 mm) in length.
      <3> Not more than 0.007 in. (0.2 mm) in width.
3) Torn, broken or unbounded bracket.
   a) Not permitted.
4) Twisted or deformed bracket.
   a) Not permitted.
5) Missing rivets on bracket.
   a) Not permitted.
6) Porosities in the sealing product under bracket.
   a) Any amount.
      NOTE: Porosities are due to the manufacturing process.

SUBTASK 72-21-00-211-014-G00

(3) Examine the acoustic panel as follows (Figure 601 (Sheet 22) and Figure 601 (Sheet 23)):

(a) Examine the inner and outer surfaces of the acoustic panel for these conditions:

NOTE: The outer surface is only visible if the acoustic panel is removed. If it is installed, do not remove it.

1) Local blind multi-perforations.
   a) Permitted.
2) Erosion.
   a) Permitted any amount on protective coating (inner skin).
   b) Permitted up to 0.007 in. (0.2 mm) in depth on other surfaces.
3) Nicks, dents, scratches, impact marks on the cups.
   a) Permitted up to 0.003 in. (0.1 mm) in depth.
4) Nicks, dents, scratches, flaking on protective coating (inner surface).
   a) Any amount permitted.
5) Tears, punctures, cracks on the protective coating (inner surface).
   a) Any amount permitted.
6) Missing of protective coating.
   a) Permitted.
7) Nicks, dents, scratches, impact marks on carbon plies on inner face.
   a) Permitted as follows:

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<1> Not more than 0.011 in. (0.3 mm) in depth.

<2> Not more than 0.78 in. (20 mm) in length.

<3> Not more than 0.11 in. (3 mm) in width.

<4> Distance between the nicks, dents, scratches, impact marks is more than 7.88 in. (200 mm).

b) Maximum repairable limit: in the limits of the applicable repair.

c) If it is necessary, do this task: Acoustic Panel Impact Repair by Filling Up, TASK 72-21-08-300-801-G00.

8) Tears, cracks, punctures on carbon plies of inner face.

a) Not permitted.

b) Maximum repairable limit: in the limits of the applicable repair.

c) If it is necessary, do this task: Acoustic Panel Impact Repair by Filling Up, TASK 72-21-08-300-801-G00.

9) Local collapse of surface on inner face.

a) Permitted as follows:

<1> Not more than 5.

<2> The distance between the local collapses is more than 9.85 in. (250 mm).

<3> The minimum distance with the fixation screw is 1.97 in. (50 mm).

b) Maximum repairable limit: in the limits of the applicable repair.

c) If it is necessary, do this task: Acoustic Panel Impact Repair by Filling Up, TASK 72-21-08-300-801-G00.

10) Nicks, dents, scratches, impact marks on carbon plies of outer face.

a) Permitted as follows:

<1> Not more than 0.011 in. (0.3 mm) in depth.

<2> Not more than 0.78 in. (20 mm) in length.

<3> Not more than 0.11 in. (3 mm) in width.

<4> Distance between the nicks, dents, scratches, impact marks is more than 7.88 in. (200 mm).

11) Tears, cracks, punctures on carbon plies of outer face.

a) Not permitted.

12) Local collapse of surface on outer face.

a) Permitted as follows:

<1> Not more than 5.

<2> The distance between the local collapses is more than 9.85 in. (250 mm).

<3> The minimum distance with the fixation screw is 1.97 in. (50 mm).

13) Unbonded or missing cup.

a) Not permitted.

14) Missing screw.

a) Not permitted.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

15) Surface oxidation on screw head (Figure 601 (Sheet 30)):
    a) Permitted.
16) Missing spacer.
    a) Not permitted.
17) Missing acoustic panel.
    a) Not permitted.
18) Shrinkage, folds on the inner surface (Figure 601 (Sheet 20)):
    a) Permitted as follows:
        <1> Up to a cumulated surface of 25 percent.
        <2> Up to 0.039 in. (1 mm) in depth.

(b) Examine the edges of the acoustic panel for these conditions (Figure 601 (Sheet 22) and Figure 601 (Sheet 23)):
1) Nicks, dents, scratches, impact marks on carbon plies.
    a) Permitted up to 0.007 in. (0.2 mm) in depth on all surfaces.
2) Tears, cracks, punctures on carbon plies.
    a) Not permitted.
3) Local collapse of the surface.
    a) Not permitted.
4) Dry composite areas (fibres with a lack of resin) on area A and area B.
    a) Any amount.

SUBTASK 72-21-00-211-015-G00

(4) Examine the stiffener plates as follows (Figure 601 (Sheet 24)):
(a) Examine the area around the bolt holes on the stiffener plates for these conditions:
1) Nicks, dents, scratches or cracks.
    a) Permitted up to 0.0019 in. (0.05 mm) in depth.
2) Fretting around the bolt holes.
    a) Permitted up to 0.0098 in. (0.25 mm) in depth over 20% of the area around each hole.

(b) Examine the other areas these conditions:
1) Nicks, dents, scratches, cracks.
    a) Permitted as follows:
        <1> Not more than 0.011 in. (0.3 mm) in depth.
        <2> Not more than 3.93 in. (100 mm) in length.
        <3> Not more than 11.81 in. (300 mm) in cumulative length per stiffener plate.

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 75-0006

SUBTASK 72-21-00-211-016-G00

(5) Examine the heat shield as follows (Figure 601 (Sheet 25)):

NOTE: Removal of the heat shield per Service Bulletin LEAP-1B-72-00-0006-01A-930A-C is recommended at first opportunity.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 75-0006 (Continued)

(a) Examine all areas for these conditions:

1) Missing material.

a) Any amount permitted if the microporous material is not exposed.

2) Torn grommet.

a) Any amount permitted if the heat shield is still maintained either by Velcro without wear, or by brackets to the flange of the fan frame shroud.

3) Tears of protection Min-K or thermal insulation.

a) Any amount permitted if the microporous material is not exposed.

4) Nicks, dents or scratches on silicone or scoring on the surface.

a) Any amount permitted if the microporous material is not exposed.

5) Local unsticking of the aluminum coating and/or missing aluminum coating.

a) Any amount permitted.

6) Missing wall.

a) Any amount permitted if the microporous material is not exposed.

7) Missing heat shield.

a) Any amount permitted.

(b) Examine the Velcro for this condition:

1) Missing or worn Velcro.

a) Any amount permitted if the heat shield is still maintained either by stud and nut, or by brackets to the flange of the fan frame shroud.

(c) Examine the brackets on the heat shield for these conditions:

1) Missing captive nut on bracket.

a) Any amount permitted if the heat shield is still maintained either by Velcro without wear, or by stud and nut.

2) Twisted or deformed bracket.

a) Permitted.

3) Broken bracket.

a) Any amount permitted if the heat shield is still maintained either by Velcro without wear, or by stud and nut.

4) Nicks, dents, scratches.

a) Any amount permitted if the heat shield is still maintained either by Velcro without wear, or by stud and nut.

5) Cracks.

a) Any amount permitted if the heat shield is still maintained either by Velcro without wear, or by stud and nut.

6) Fretting on the bracket.

a) Permitted.

7) Elongated hole.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 75-0006 (Continued)

a) Any amount permitted if the heat shield is still maintained either by Velcro without wear, or by stud and nut.

(d) Examine the washer for this condition:

1) Missing washer.

a) Any amount permitted if the associated stud is missing or broken and the associated nut is missing.

(e) Examine the nut for this condition:

1) Missing nut.

a) Any amount permitted if the associated stud is missing or broken and the associated washer is missing.

PTF ALL

H. Put the Airplane Back to Its Usual Condition

SUBTASK 72-21-00-860-004-G00

(1) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

(a) On the applicable engine, close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

SUBTASK 72-21-00-860-005-G00

(2) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 72-21-00-860-006-G00

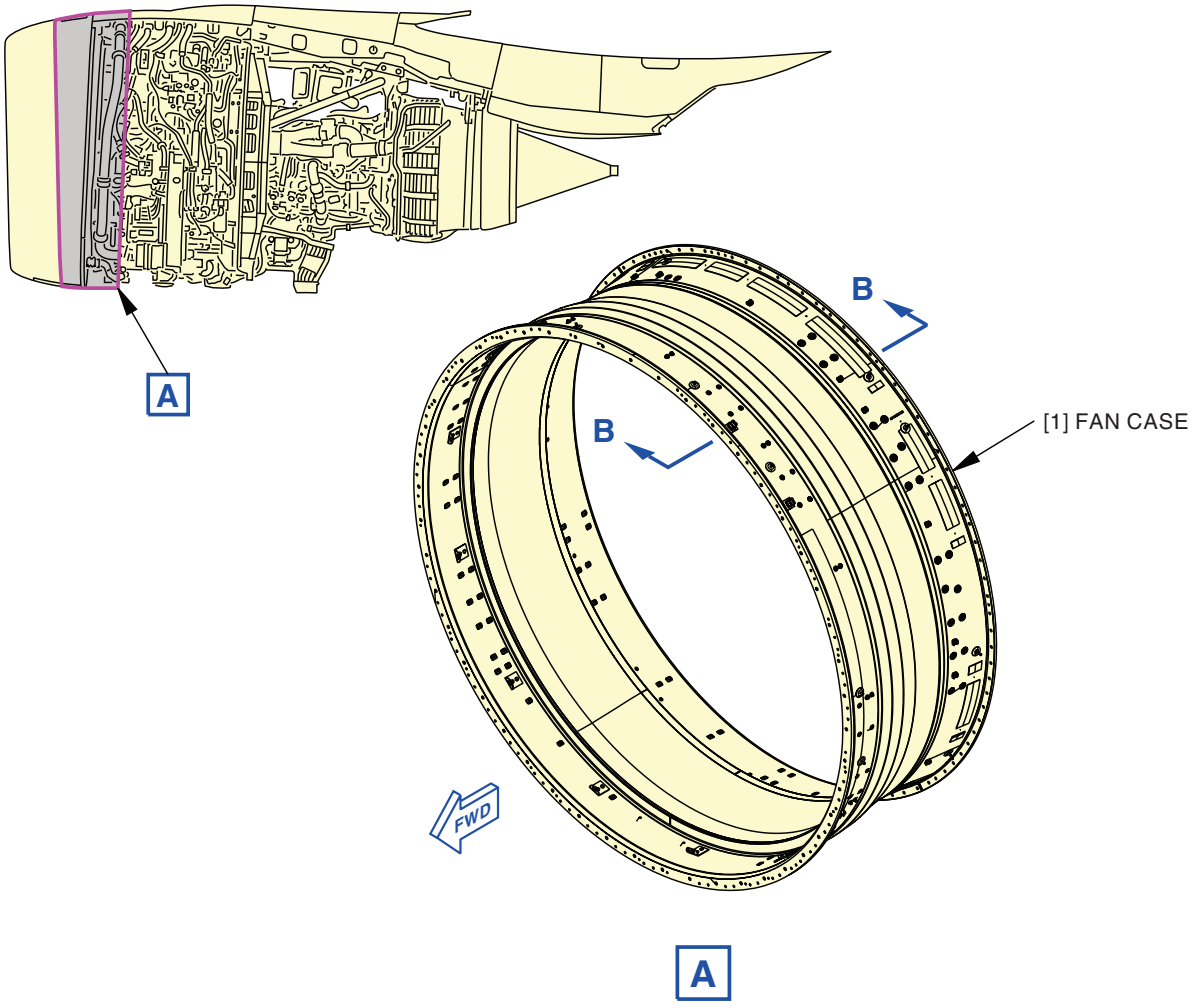
(3) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fan Case Module Inspection
Figure 601/72-21-00-990-802-G00 (Sheet 1 of 38)

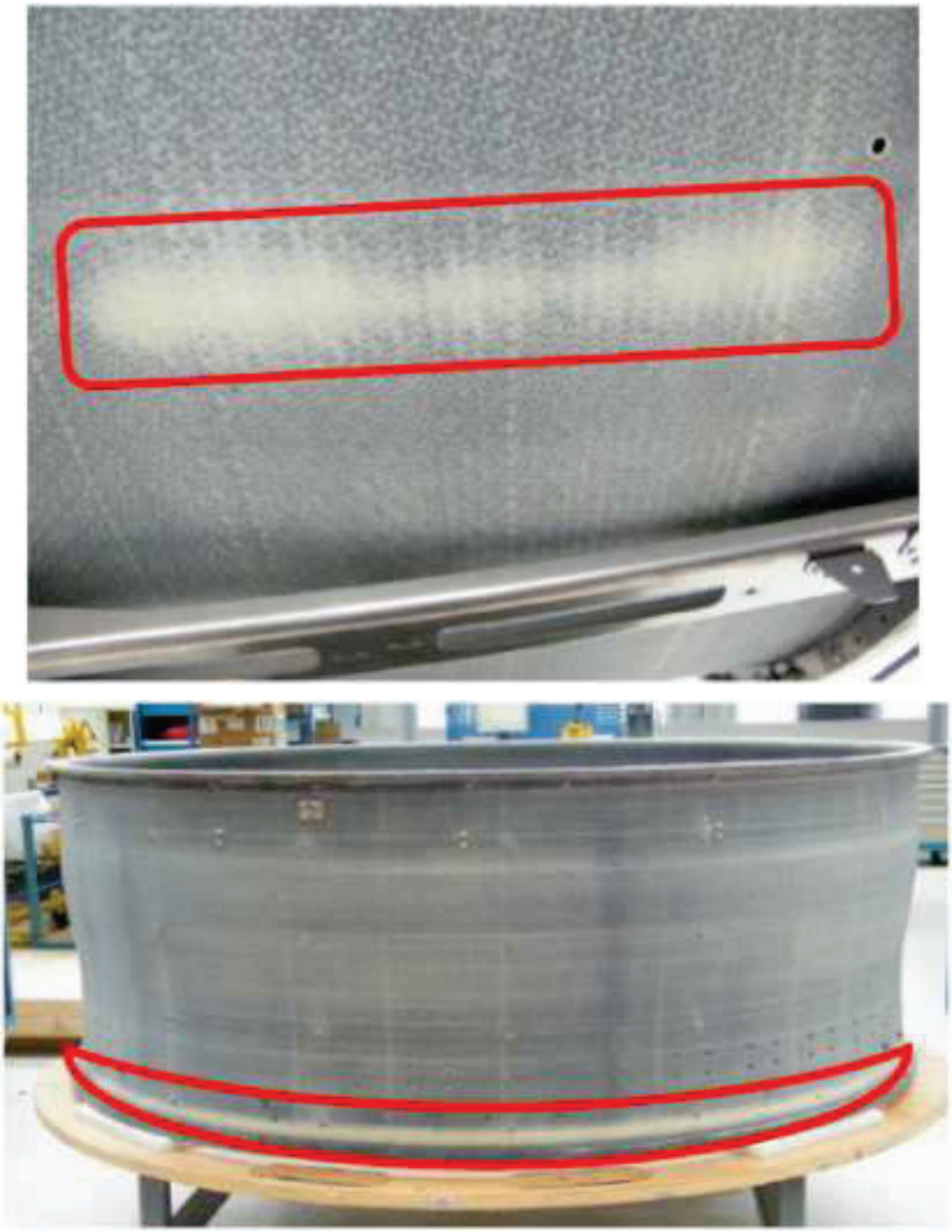
ECCN 9E991 BOEING PROPRIETARY - See title page for details

BUCKLING OF CARBON TOWS

2354518 S00061515483_V1

Fan Case Module Inspection
Figure 601/72-21-00-990-802-G00 (Sheet 2 of 38)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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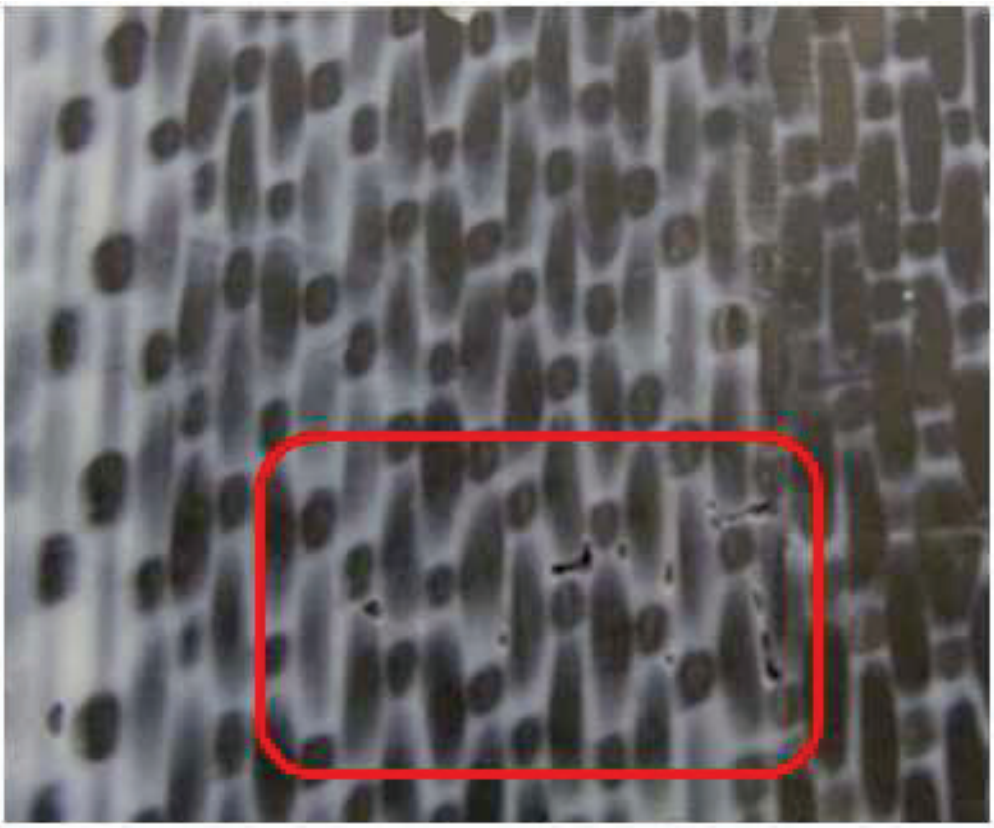
Fan Case Module Inspection
Figure 601/72-21-00-990-802-G00 (Sheet 3 of 38)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



SURFACE POROSITY EXCEPT ON ABRADABLE

ICNLEAP1B722100D5882800004A00201
2480943 S0000582902_V2

Fan Case Module Inspection
Figure 601/72-21-00-990-802-G00 (Sheet 4 of 38)

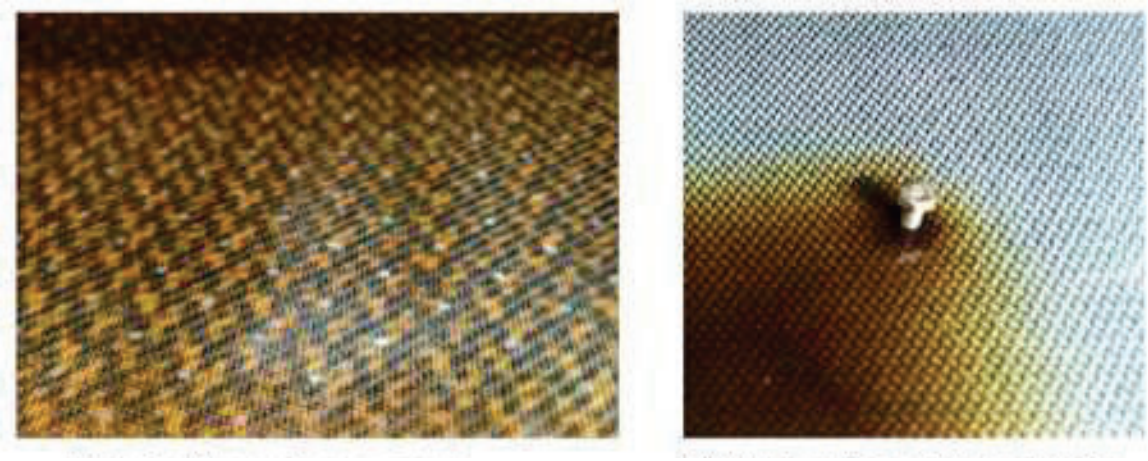
ECCN 9E991 BOEING PROPRIETARY - See title page for details

ETCHED COMPOSITE SURFACE

2480944 S0000582903_V1

Fan Case Module Inspection
Figure 601/72-21-00-990-802-G00 (Sheet 5 of 38)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



WITH GLASS FIBER COATING ON COMPOSITE

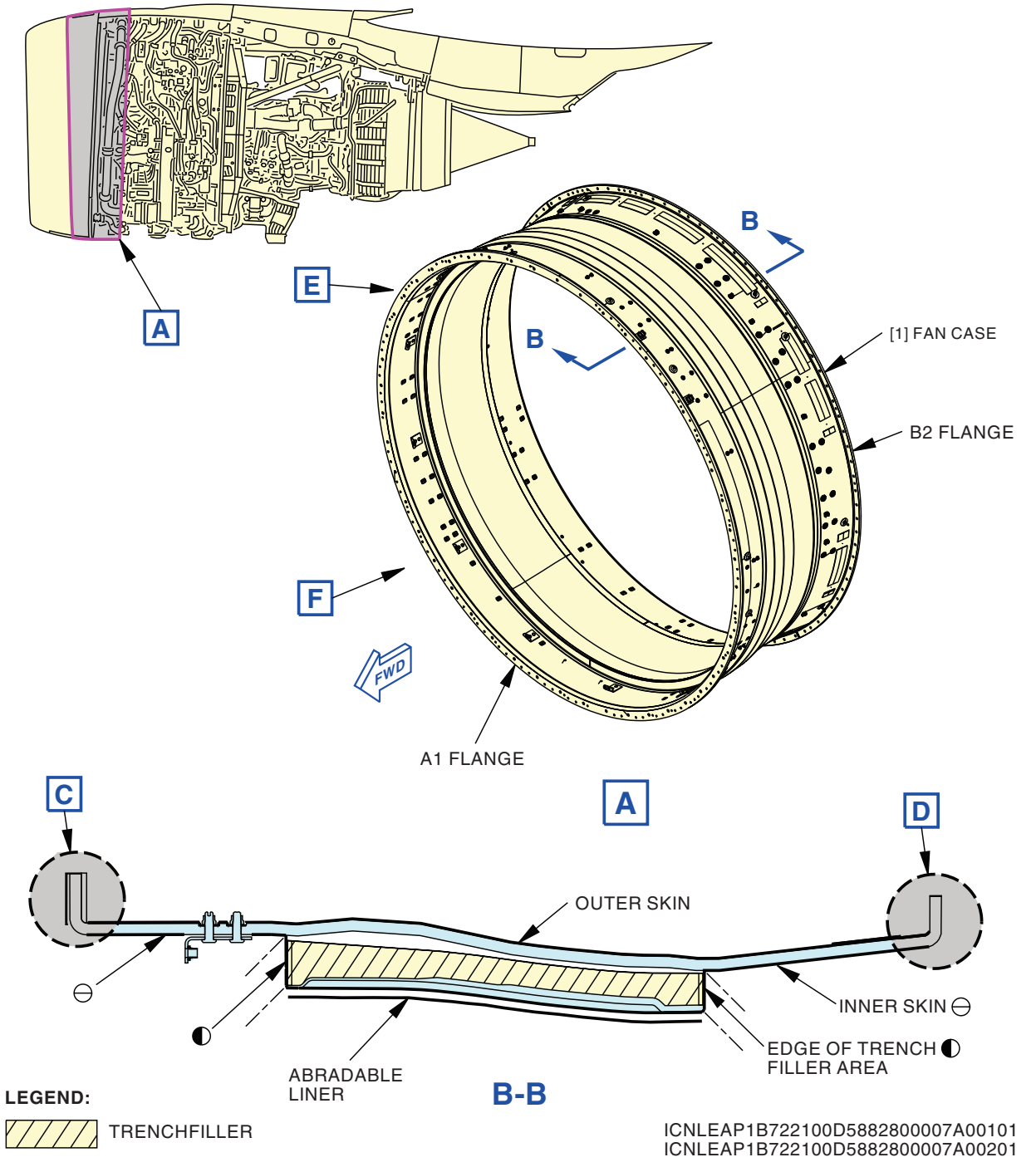
WITHOUT GLASS FIBER COATING ON COMPOSITE

DARK BROWN STAIN ON COMPOSITE

2480947 S0000582904_V1

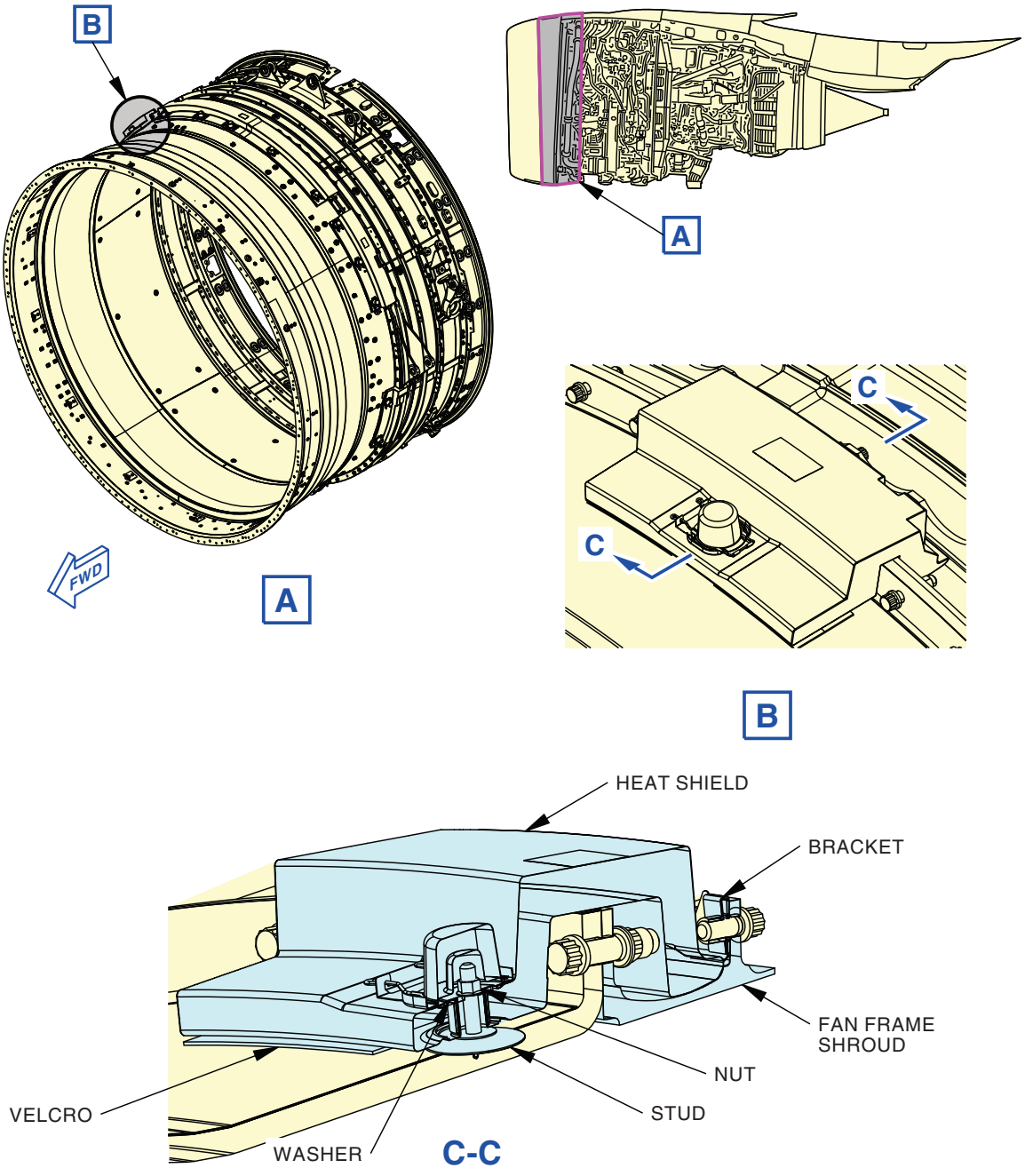
Fan Case Module Inspection
Figure 601/72-21-00-990-802-G00 (Sheet 6 of 38)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fan Case Module Inspection
Figure 601/72-21-00-990-802-G00 (Sheet 7 of 38)

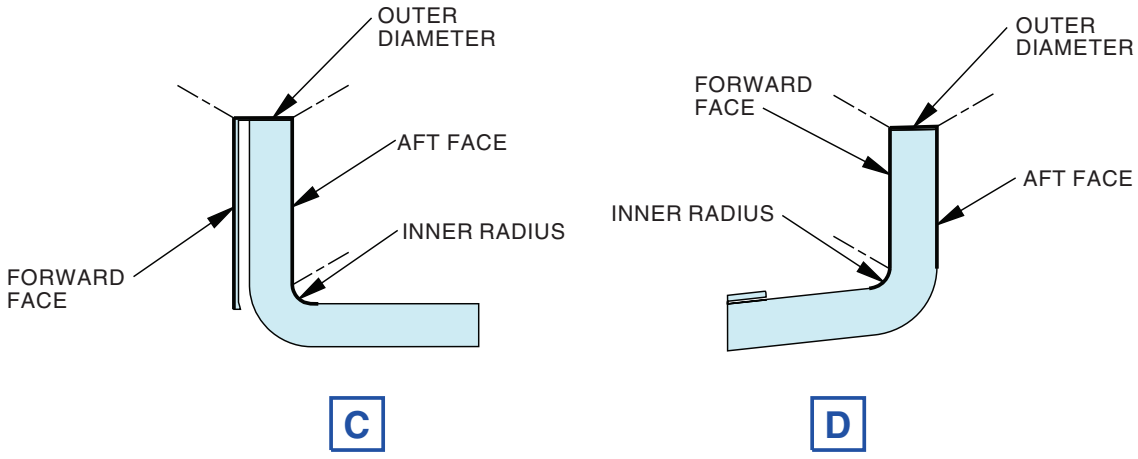
ECCN 9E991 BOEING PROPRIETARY - See title page for details

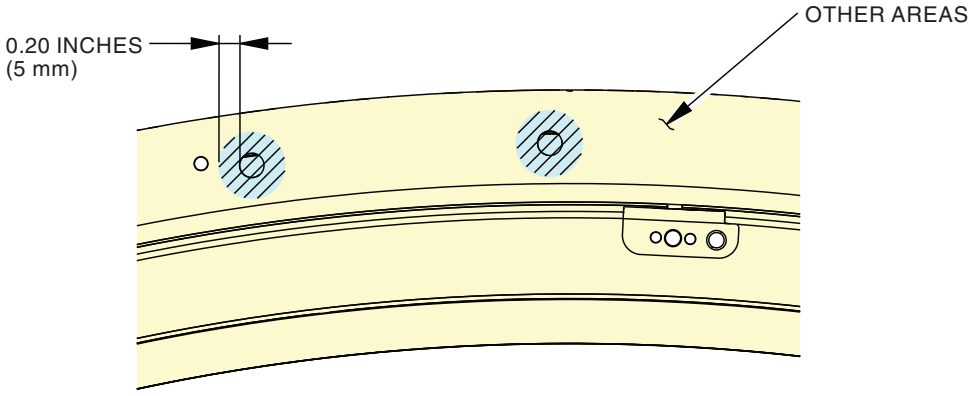


2567751 S0000614819_V1

Fan Case Module Inspection
Figure 601/72-21-00-990-802-G00 (Sheet 8 of 38)

ECCN 9E991 BOEING PROPRIETARY - See title page for details





(TYPICAL VIEW OF FORWARD AND AFT FACE OF A1 AND B2 FLANGES)

E

LEGEND:

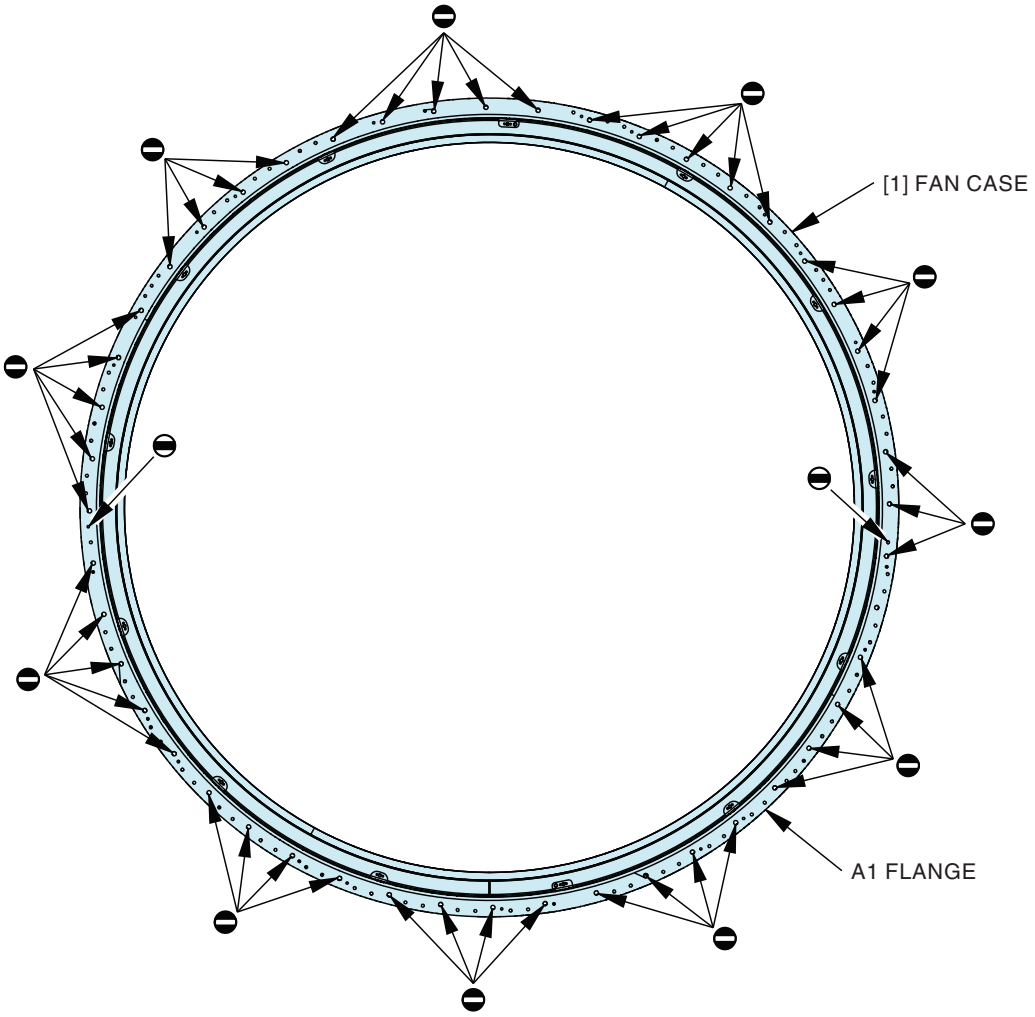
AREA AROUND HOLES

ICNLEAP1B722100D5882800008A00101

2480976 S0000582906_V2

Fan Case Module Inspection
Figure 601/72-21-00-990-802-G00 (Sheet 9 of 38)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



(VIEW IN THE AFT DIRECTION)

F

LEGEND:

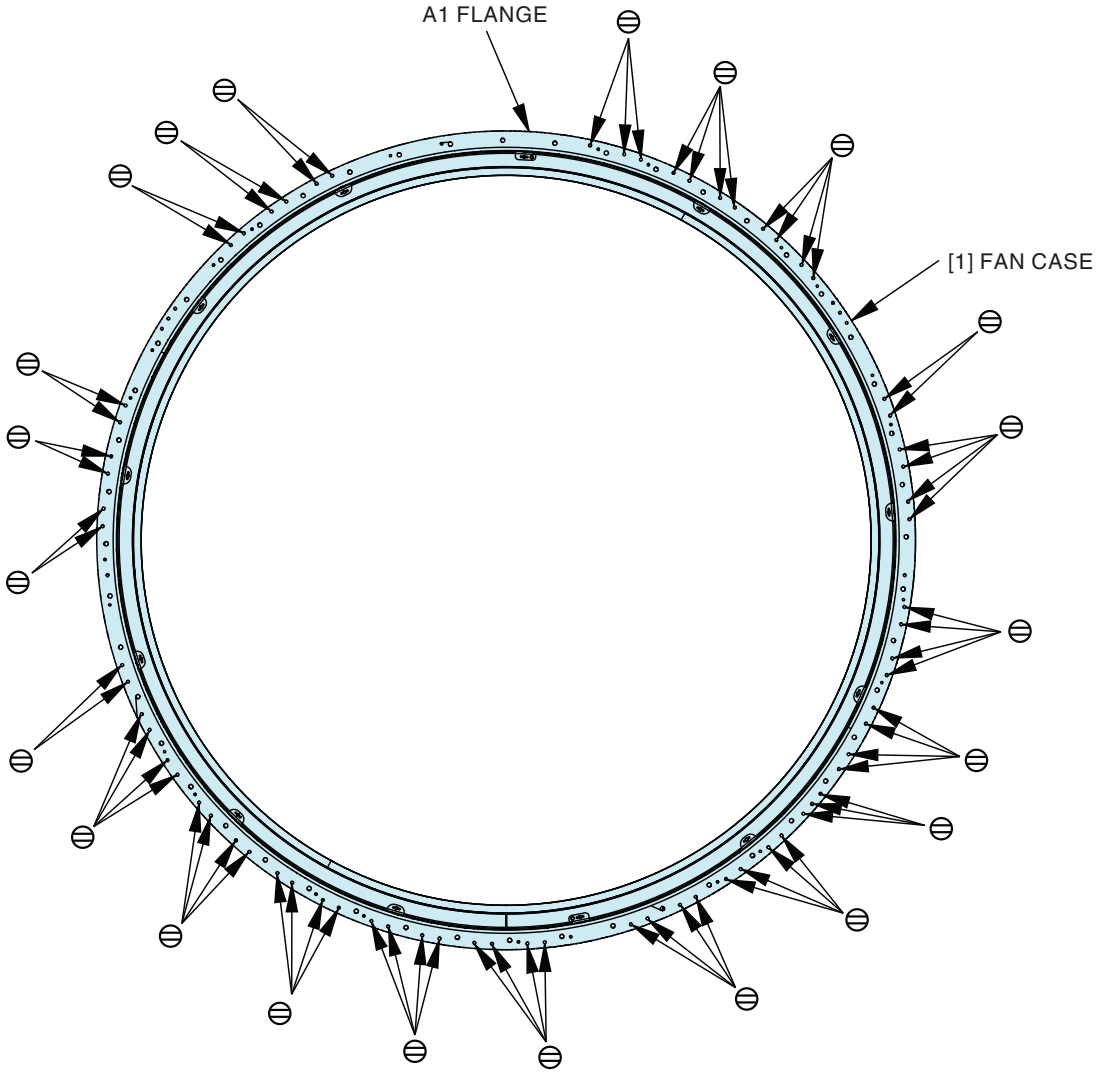
- POSITION OF THE 48 FASTENING HOLES OF A1 FLANGE
- POSITION OF THE TWO LOCATING HOLES OF A1 FLANGE

ICNLEAP1B722100D5882800009A00101

2480982 S0000582907_V2

Fan Case Module Inspection
Figure 601/72-21-00-990-802-G00 (Sheet 10 of 38)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



(VIEW IN THE AFT DIRECTION)

F

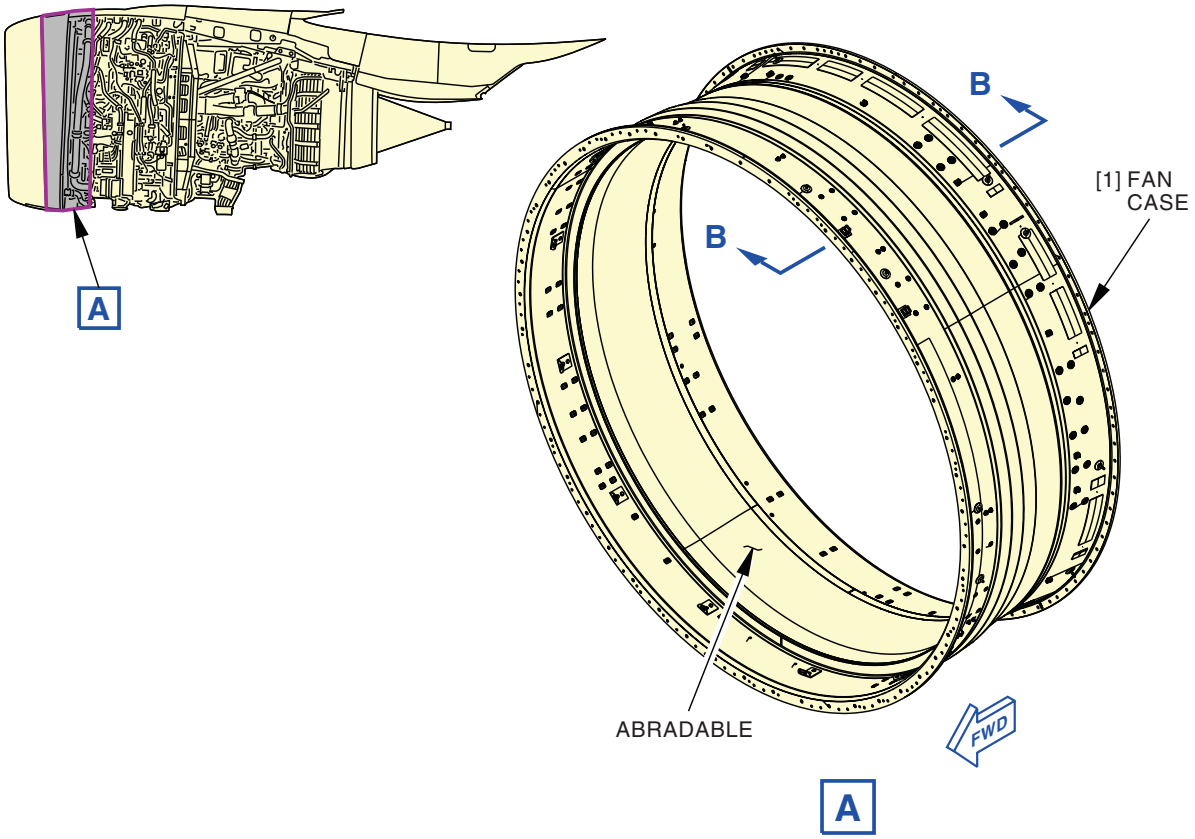
LEGEND:

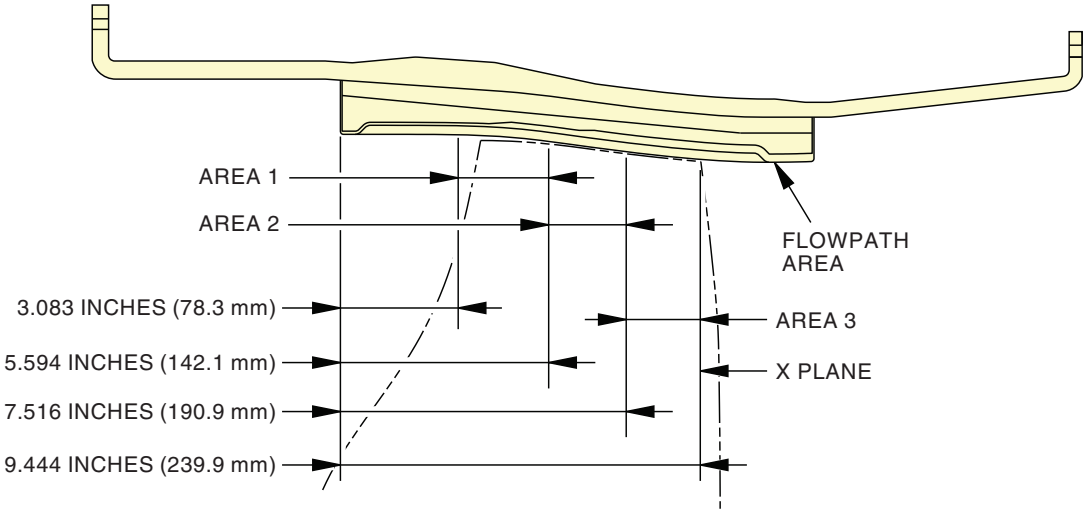
⊖ POSITION OF THE 70 BRACKET ATTACHMENT HOLES

ICNLEAP1B722100D5882800010A00101
2480996 S0000582908_V2

Fan Case Module Inspection
Figure 601/72-21-00-990-802-G00 (Sheet 11 of 38)

ECCN 9E991 BOEING PROPRIETARY - See title page for details





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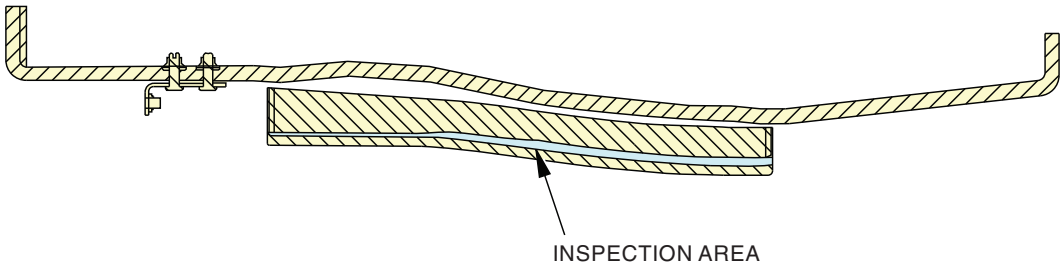
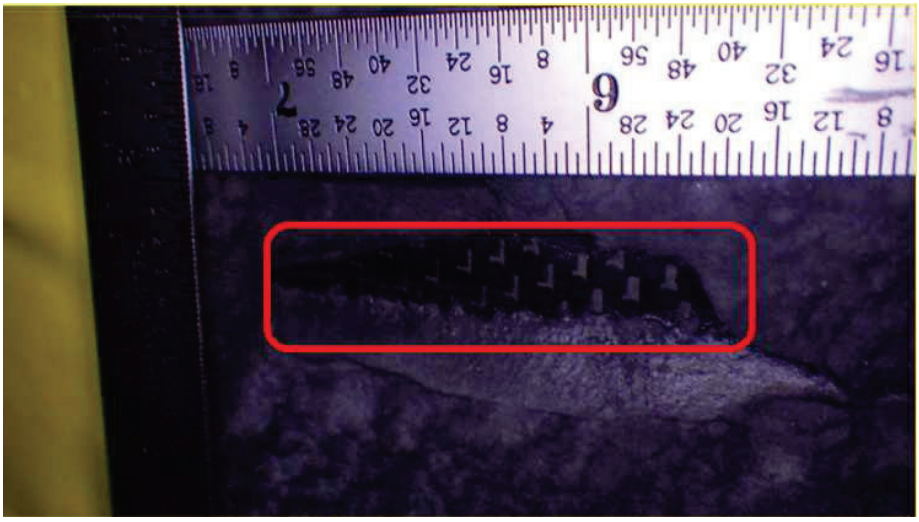
2481001 S0000582909_V7

Fan Case Module Inspection
Figure 601/72-21-00-990-802-G00 (Sheet 12 of 38)

ECCN 9E991 BOEING PROPRIETARY - See title page for details





CARBON PLIES

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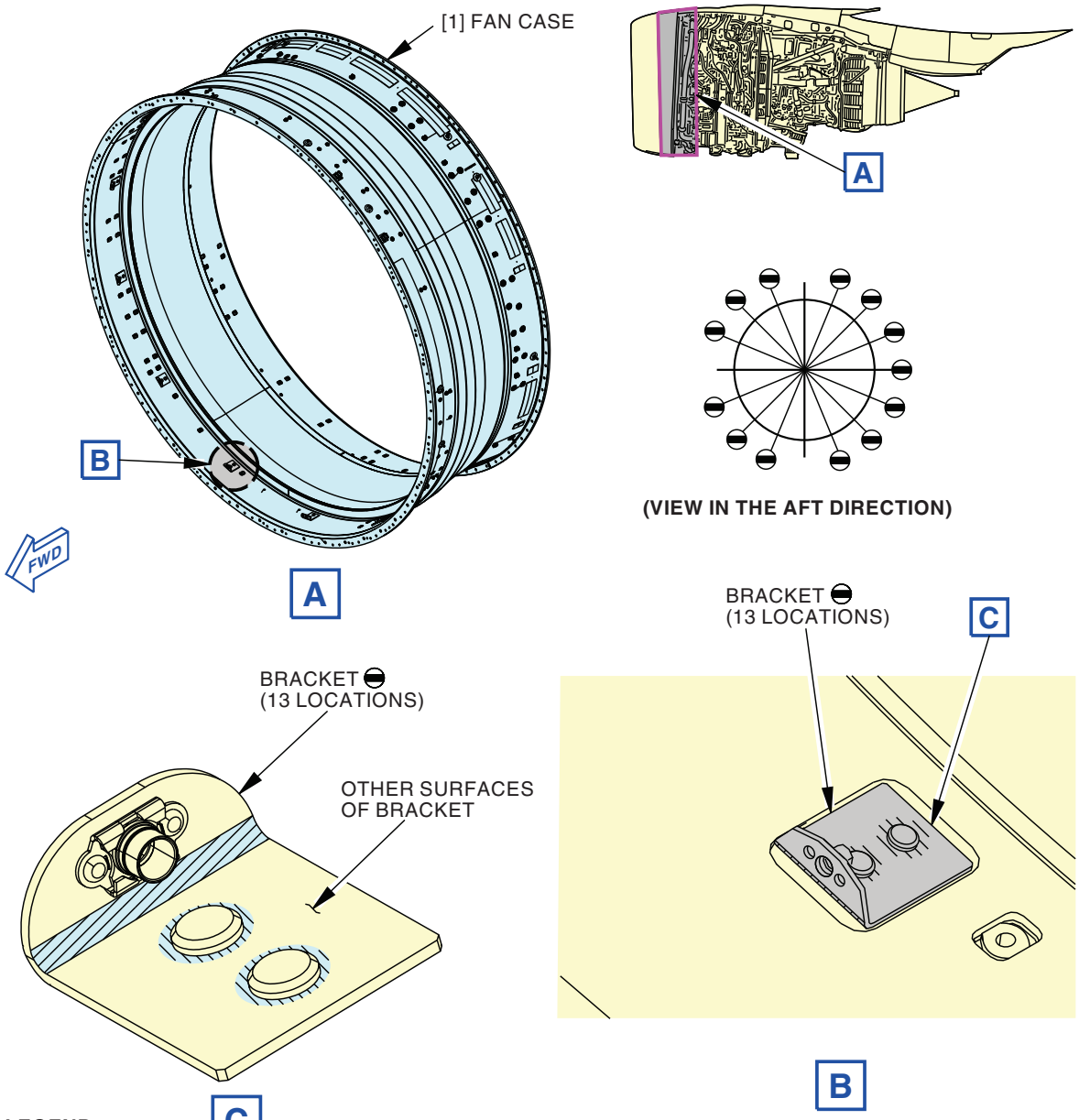
2840142 S0000662503_V1

Fan Case Module Inspection
Figure 601/72-21-00-990-802-G00 (Sheet 13 of 38)

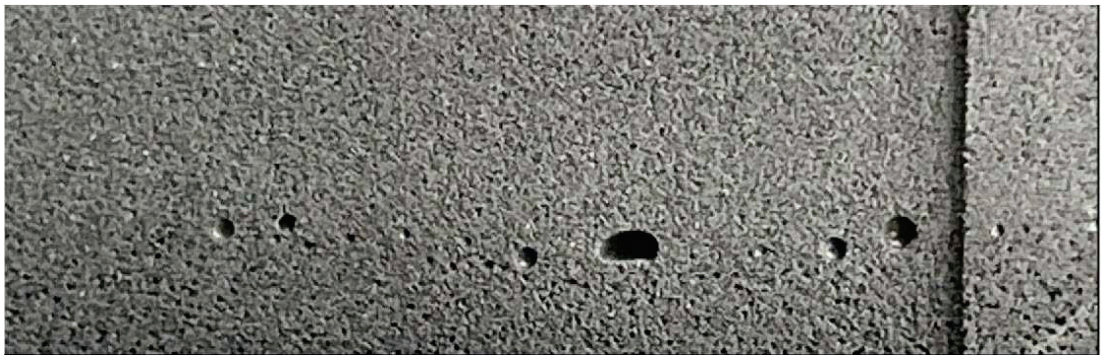
ECCN 9E991 BOEING PROPRIETARY - See title page for details

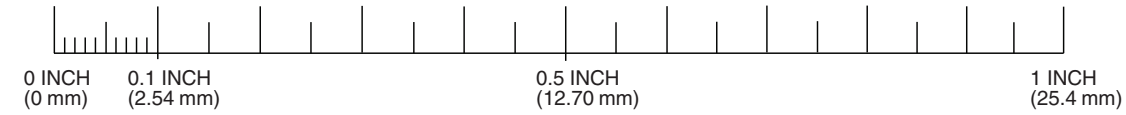


2539531 S0000603195_V2

Fan Case Module Inspection
Figure 601/72-21-00-990-802-G00 (Sheet 14 of 38)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

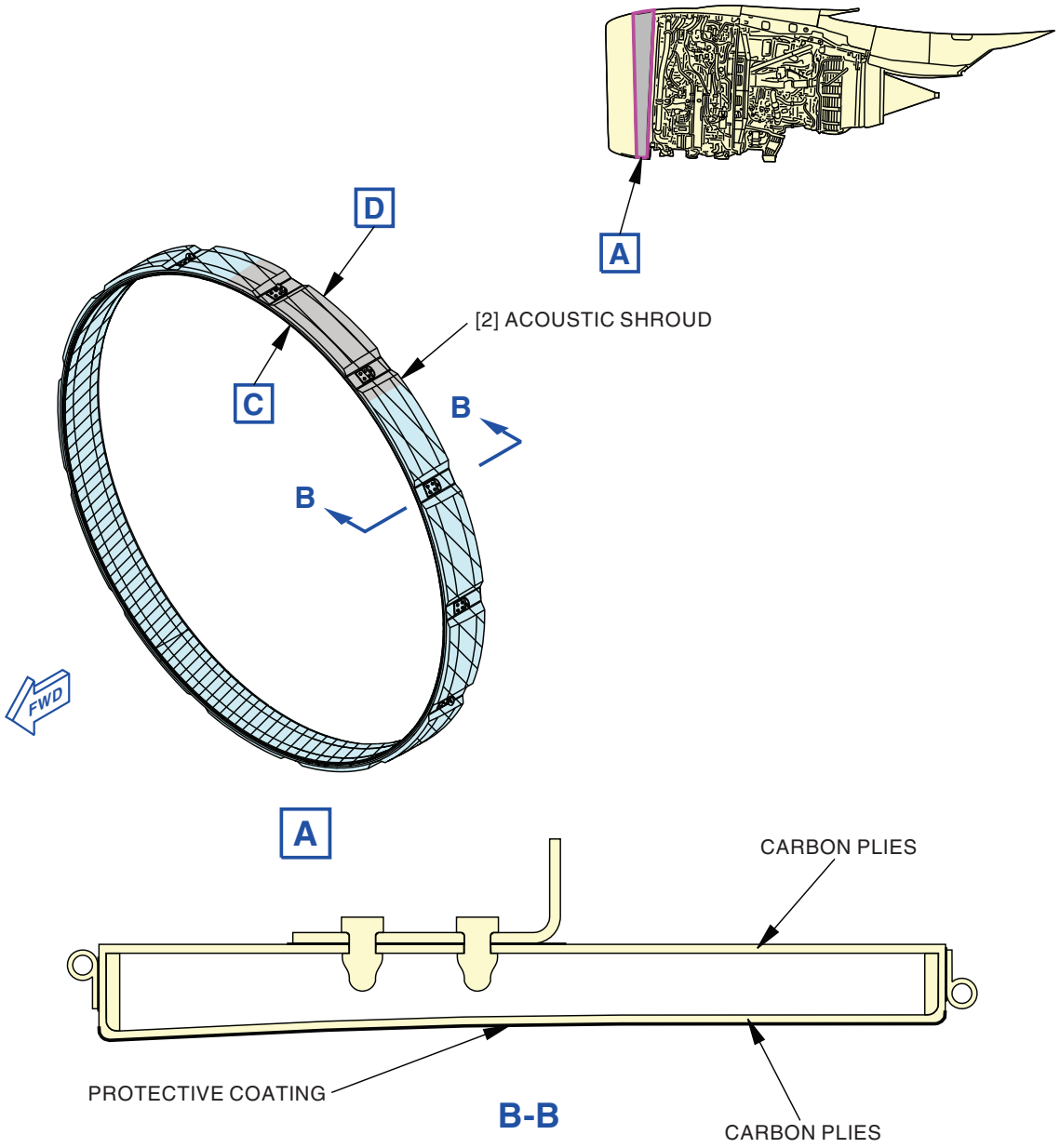




ICNLEAP1B722100D5882800002A00201
2840143 S0000662504_V3

Fan Case Module Inspection
Figure 601/72-21-00-990-802-G00 (Sheet 15 of 38)

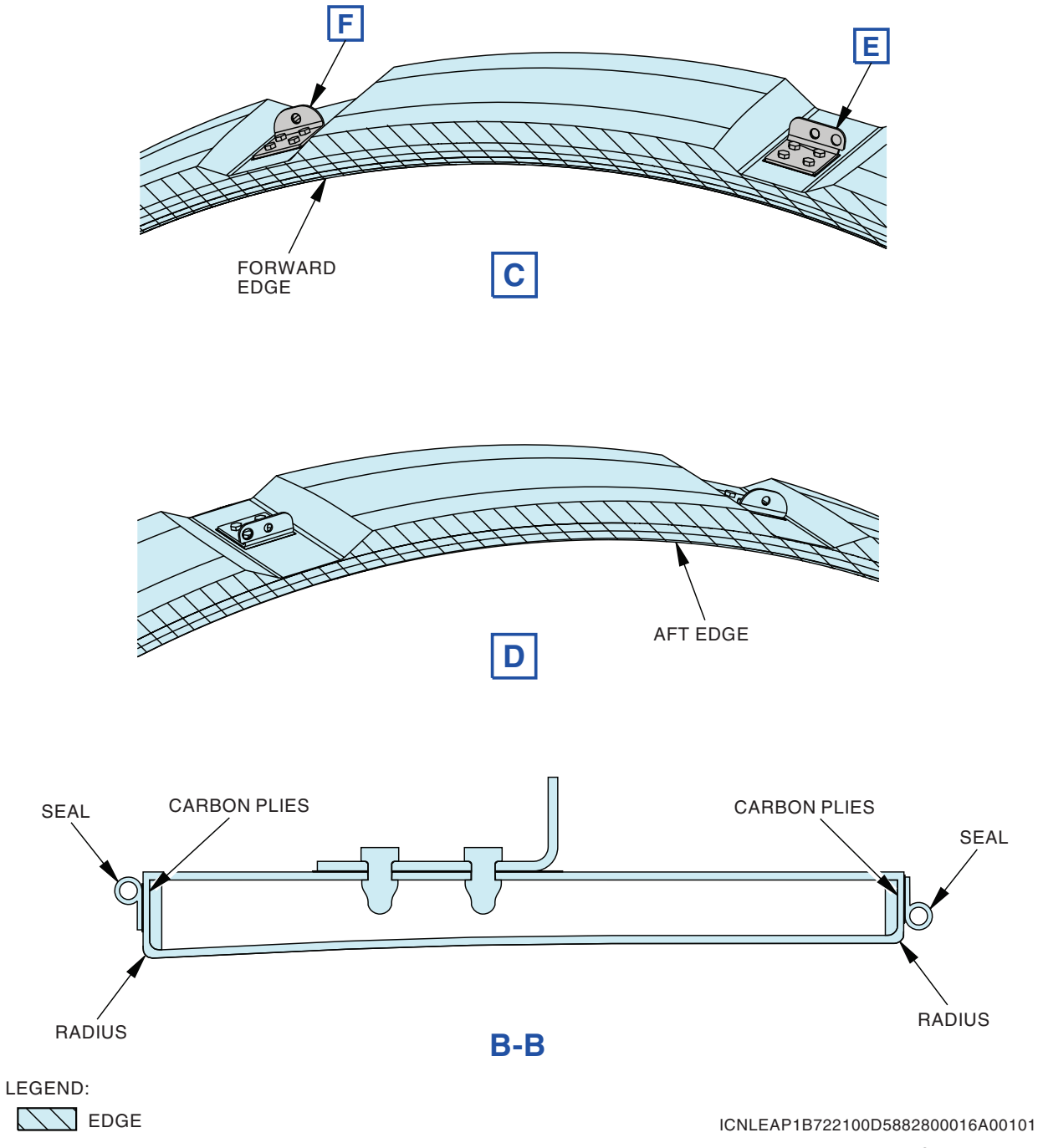
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2539640 S0000603197_V2

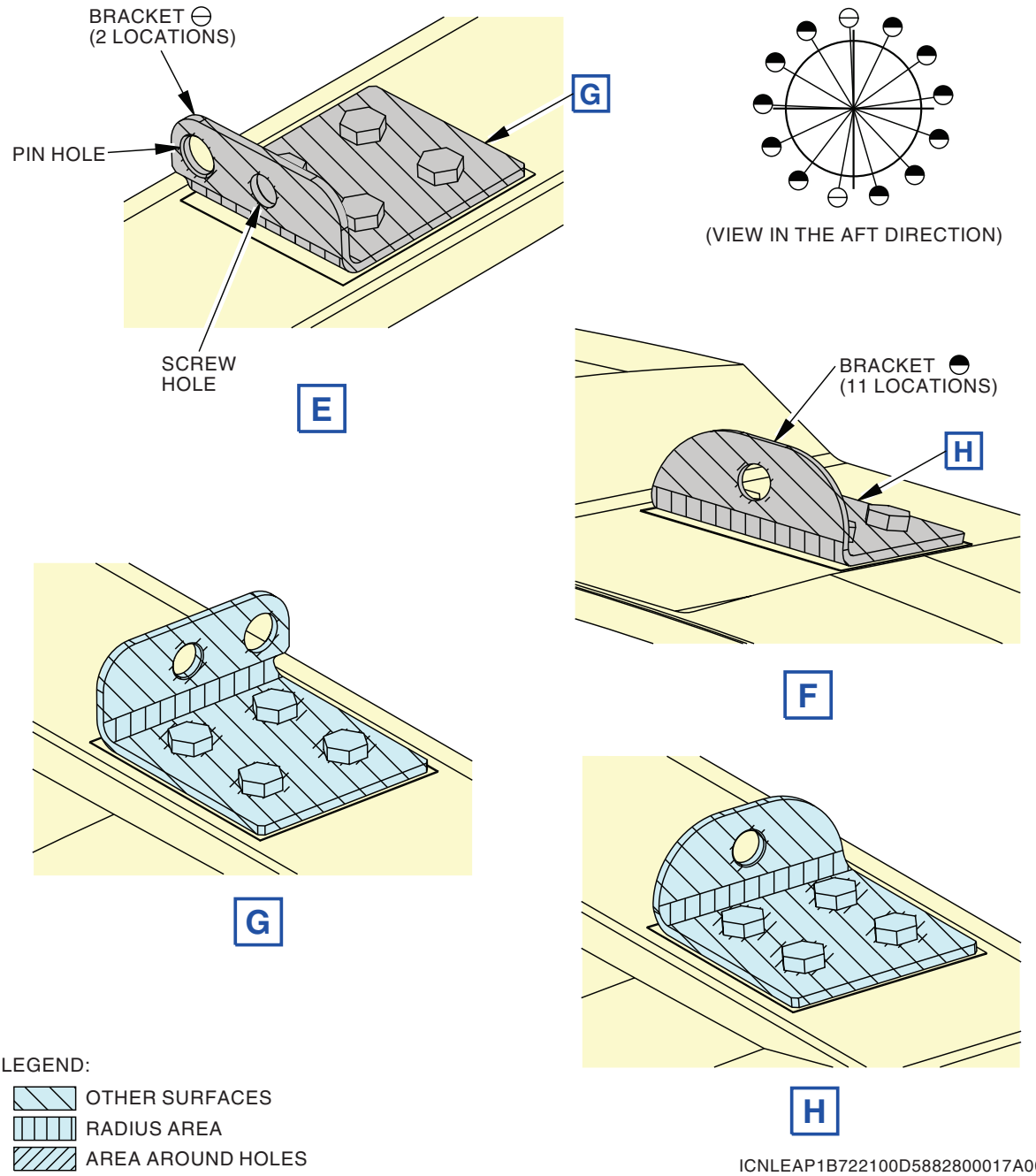
Fan Case Module Inspection
Figure 601/72-21-00-990-802-G00 (Sheet 16 of 38)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fan Case Module Inspection
Figure 601/72-21-00-990-802-G00 (Sheet 17 of 38)

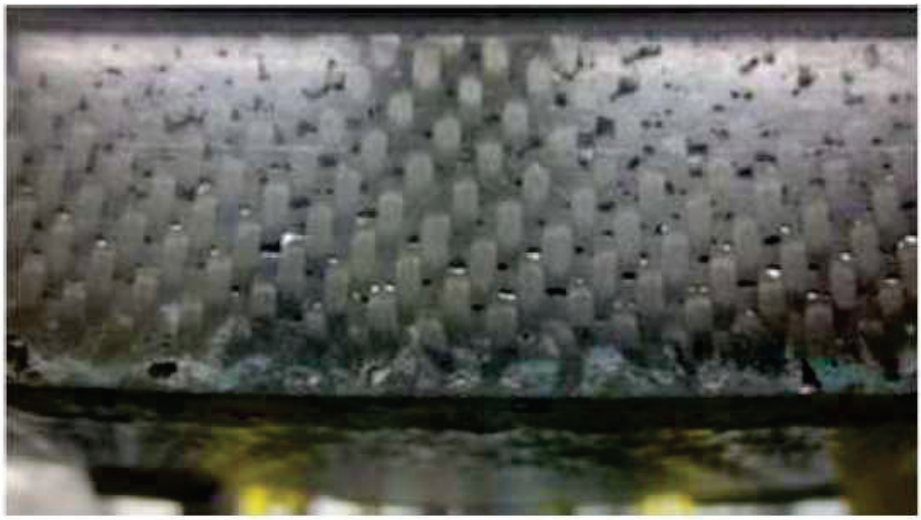
ECCN 9E991 BOEING PROPRIETARY - See title page for details

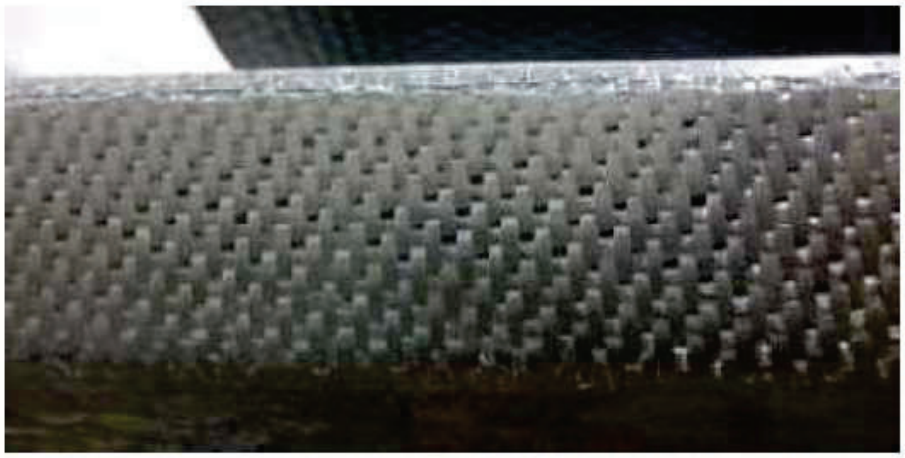


ICNLEAP1B722100D5882800017A00101
2539768 S0000603201_V1

Fan Case Module Inspection
Figure 601/72-21-00-990-802-G00 (Sheet 18 of 38)

ECCN 9E991 BOEING PROPRIETARY - See title page for details





TYPICAL EDGE AND RADIUS OF THE ACOUSTIC SHROUD

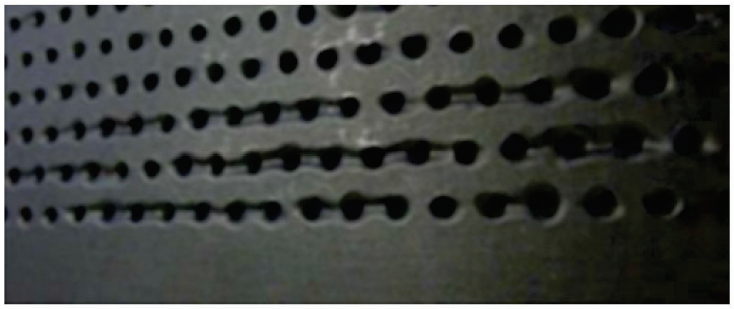
ICNLEAP1B722100D58828A00101

2898927 S0000691680_V1

Fan Case Module Inspection
Figure 601/72-21-00-990-802-G00 (Sheet 19 of 38)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FOLDS ON THE INNER SURFACE OF THE ACOUSTIC PANEL

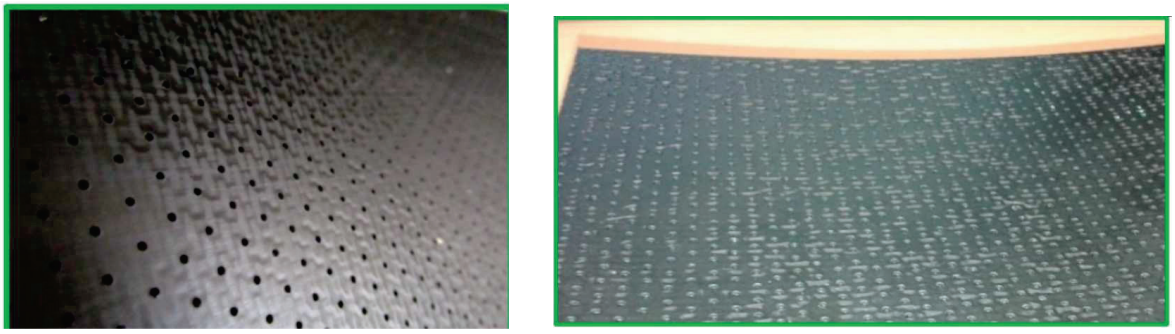


SHRINKAGE ON THE INNER SURFACE OF THE ACOUSTIC PANEL

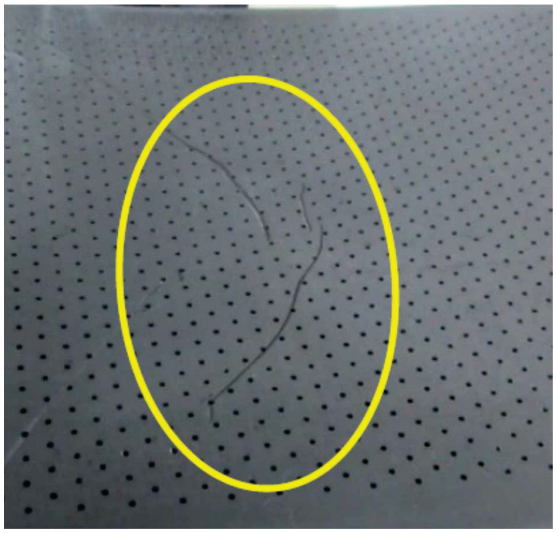
3044261 S0000811258_V1

Fan Case Module Inspection
Figure 601/72-21-00-990-802-G00 (Sheet 20 of 38)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



MICRO-SHRINKAGE (INNER FACE)



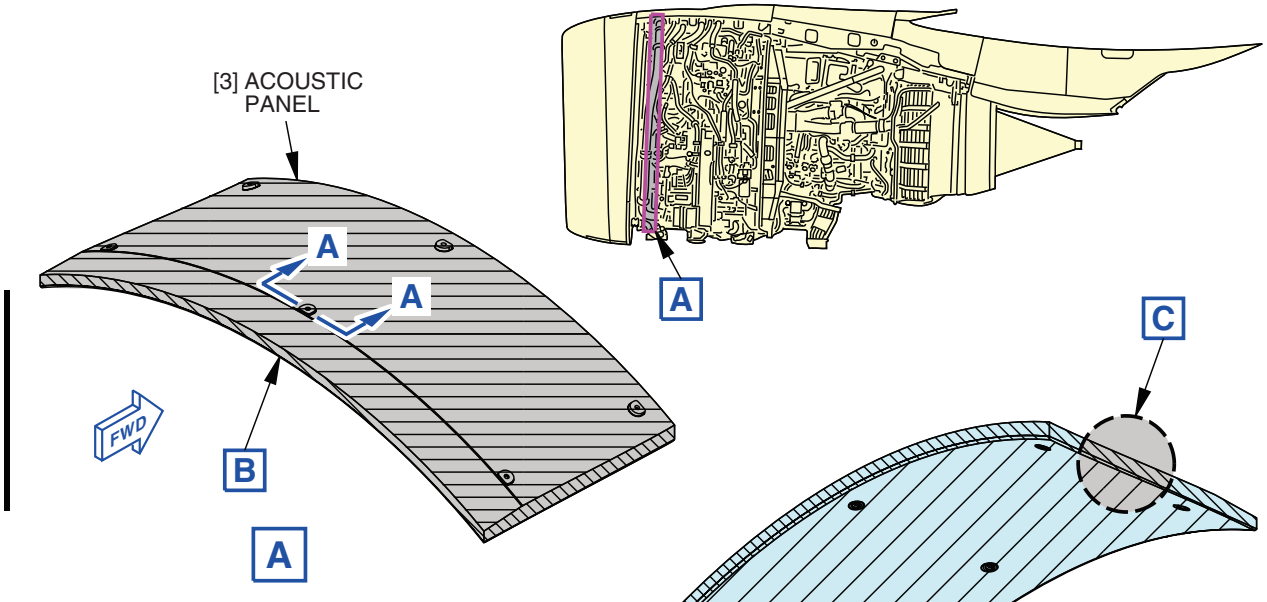
WRINKLES OR FOLDS ON PROTECTIVE COATING (INNER FACE)

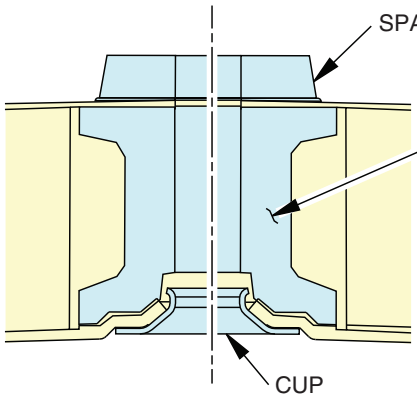
ICNLEAP1B722100D5882800025A00101

3012611 S0000786148_V1

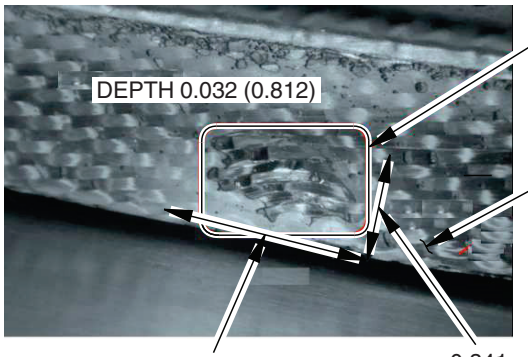
Fan Case Module Inspection
Figure 601/72-21-00-990-802-G00 (Sheet 21 of 38)

ECCN 9E991 BOEING PROPRIETARY - See title page for details





(TYPICAL VIEW OF SPACERS, INSERTS AND CUPS)
A-A



EXAMPLE OF DRY COMPOSITE AREA ON EDGES OF THE ACOUSTIC PANEL

LEGEND:

- OUTER SURFACE
- INNER SURFACE
- EDGE

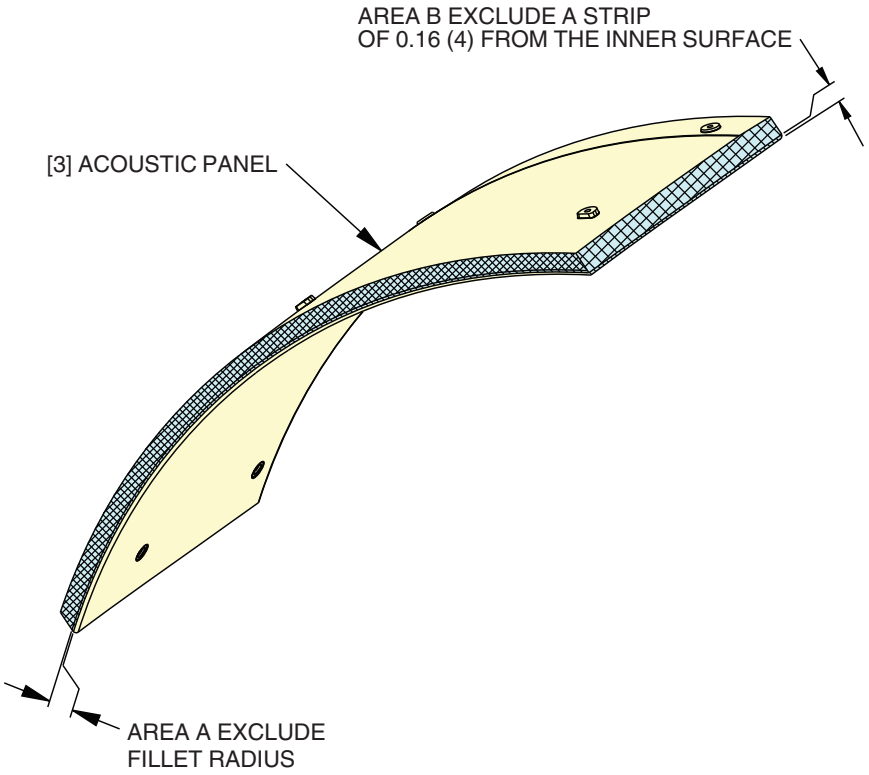
NOTE:

ALL DIMENSIONS ARE IN INCHES (mm).

ICNLEAP1B722100D5882800040A00101
2539802 S0000603204_V2

Fan Case Module Inspection
Figure 601/72-21-00-990-802-G00 (Sheet 22 of 38)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



C

LEGEND:

AREA A
AREA B
AREA THAT IS EXCLUDED

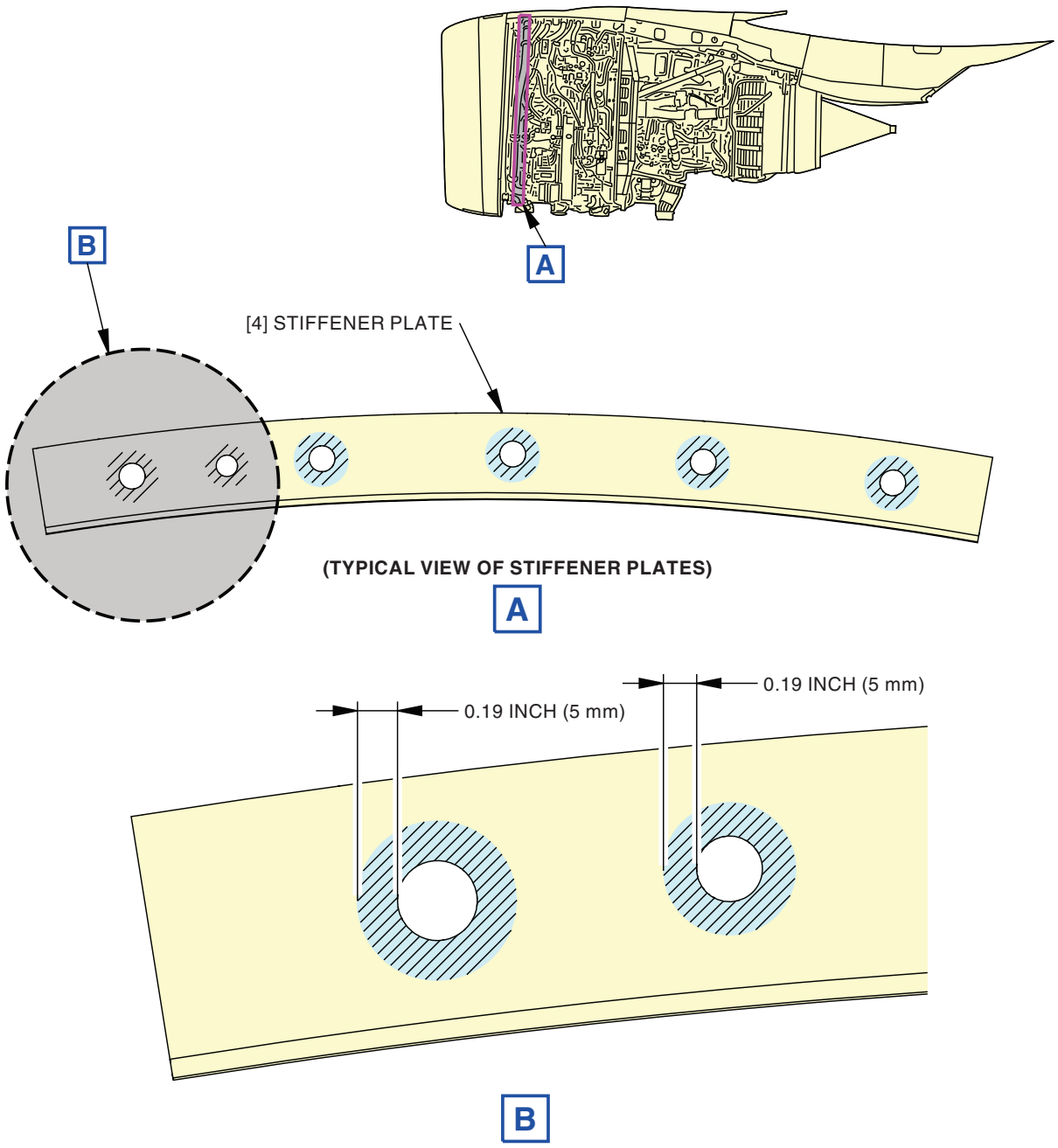
NOTE:

ALL DIMENSIONS ARE IN INCHES (mm).

ICNLEAP1B722100D5882800041A00101
3132419 S0000862313_V1

Fan Case Module Inspection
Figure 601/72-21-00-990-802-G00 (Sheet 23 of 38)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

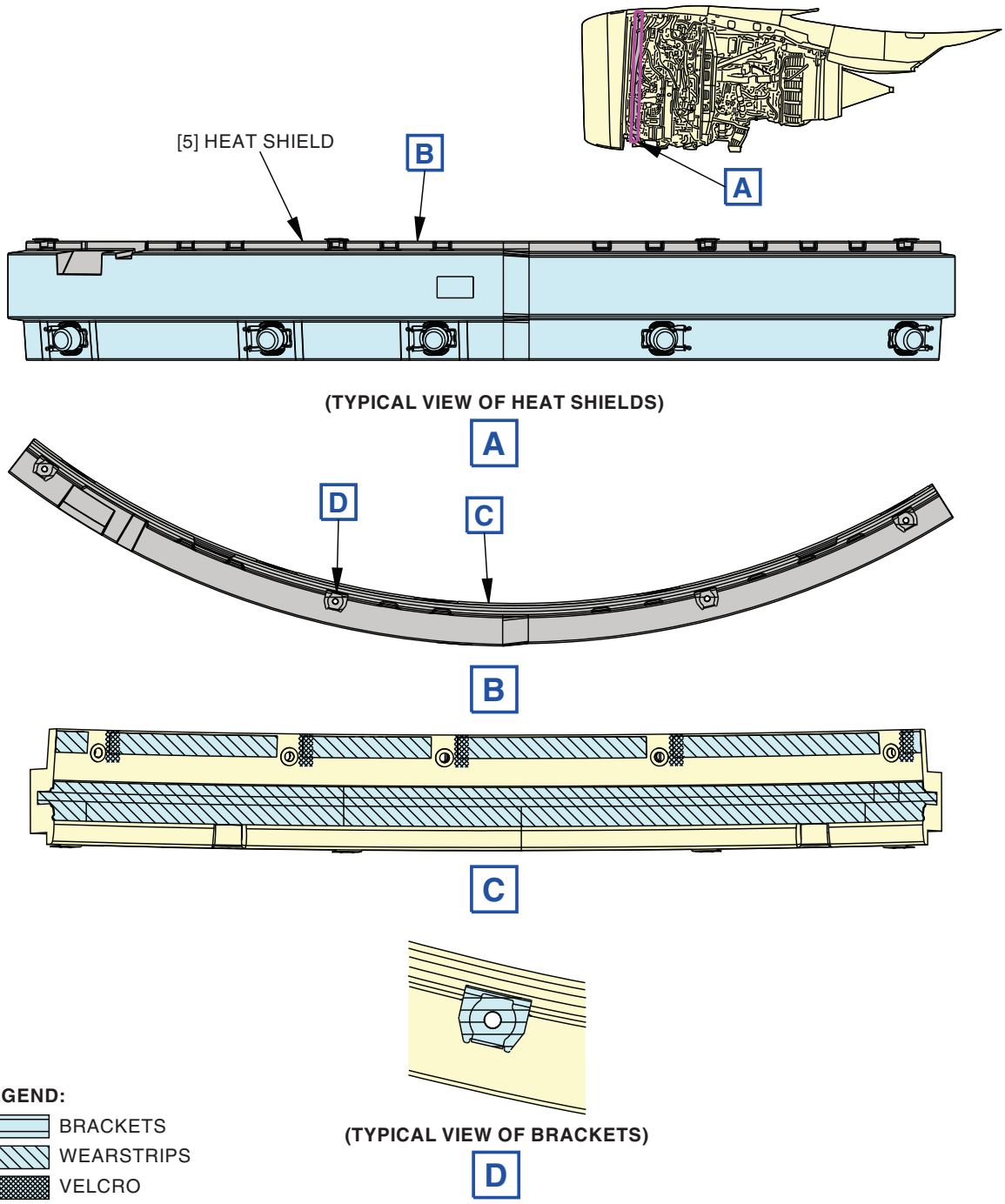


ICNLEAP1B722100D5882800019A00101

2539817 S0000603205_V1

Fan Case Module Inspection
Figure 601/72-21-00-990-802-G00 (Sheet 24 of 38)

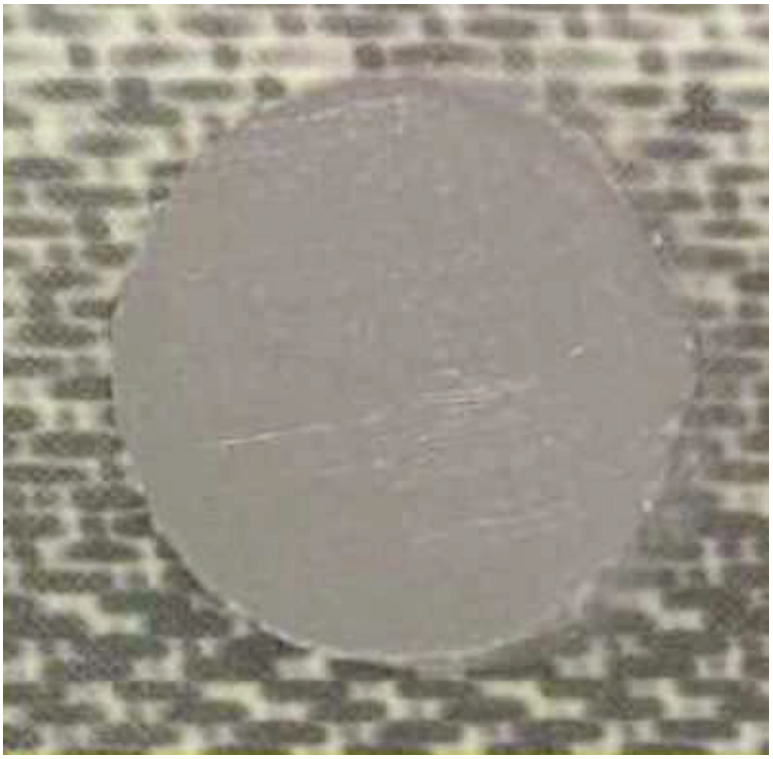
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2539824 S0000603227_V2

Fan Case Module Inspection
Figure 601/72-21-00-990-802-G00 (Sheet 25 of 38)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



GLASS FIBER REWORKED AREA ON OUTER SKIN

ICNLEAP1B722100D5882800022A00101
2924062 S0000704978_V1

Fan Case Module Inspection
Figure 601/72-21-00-990-802-G00 (Sheet 26 of 38)

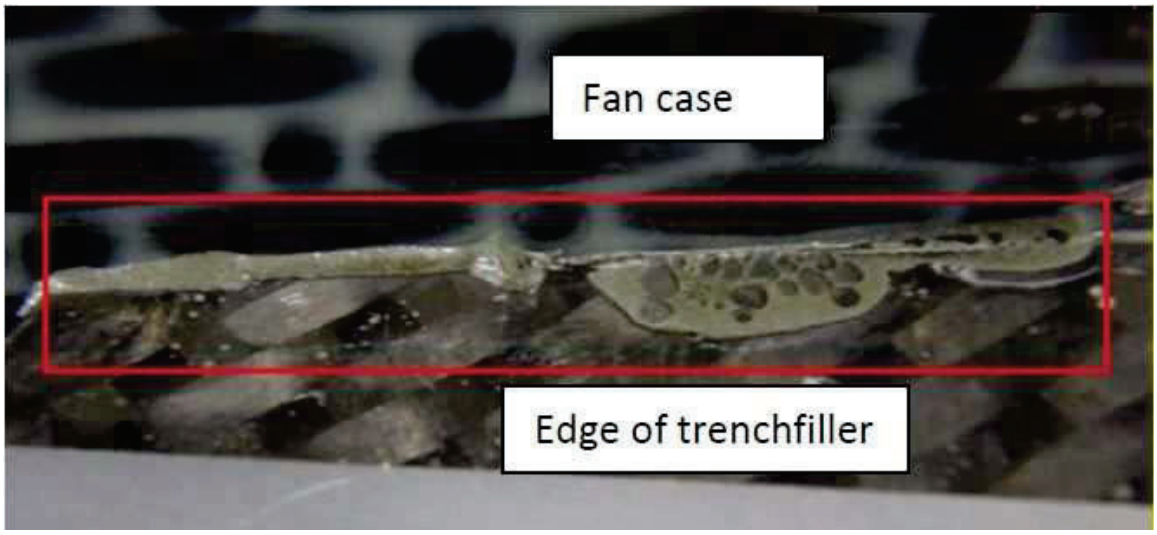
ECCN 9E991 BOEING PROPRIETARY - See title page for details

GLASS FIBER REWORK AREA ON TOP OF THE FAN CASE A1 AND B2 FLANGES

ICNLEAP1B722100D5882800024A00101
2924063 S0000704979_V1

Fan Case Module Inspection
Figure 601/72-21-00-990-802-G00 (Sheet 27 of 38)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



ADHESIVE POROSITIES IN ADHESIVE FLASH LOCATED BETWEEN THE EDGE OF TRENCHFILLER AND FAN CASE

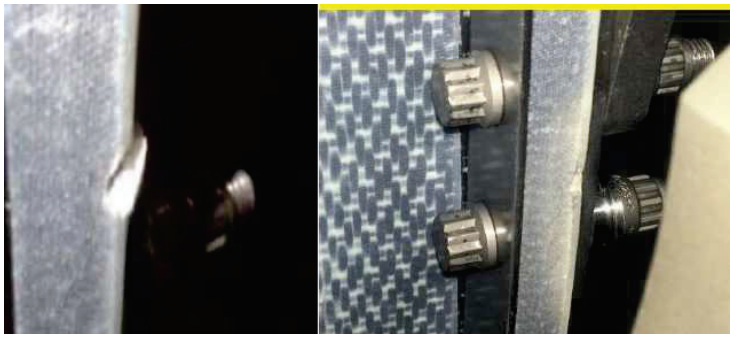
ICNLEAP1B722100D5882800023A00101

2924064 S0000704980_V1

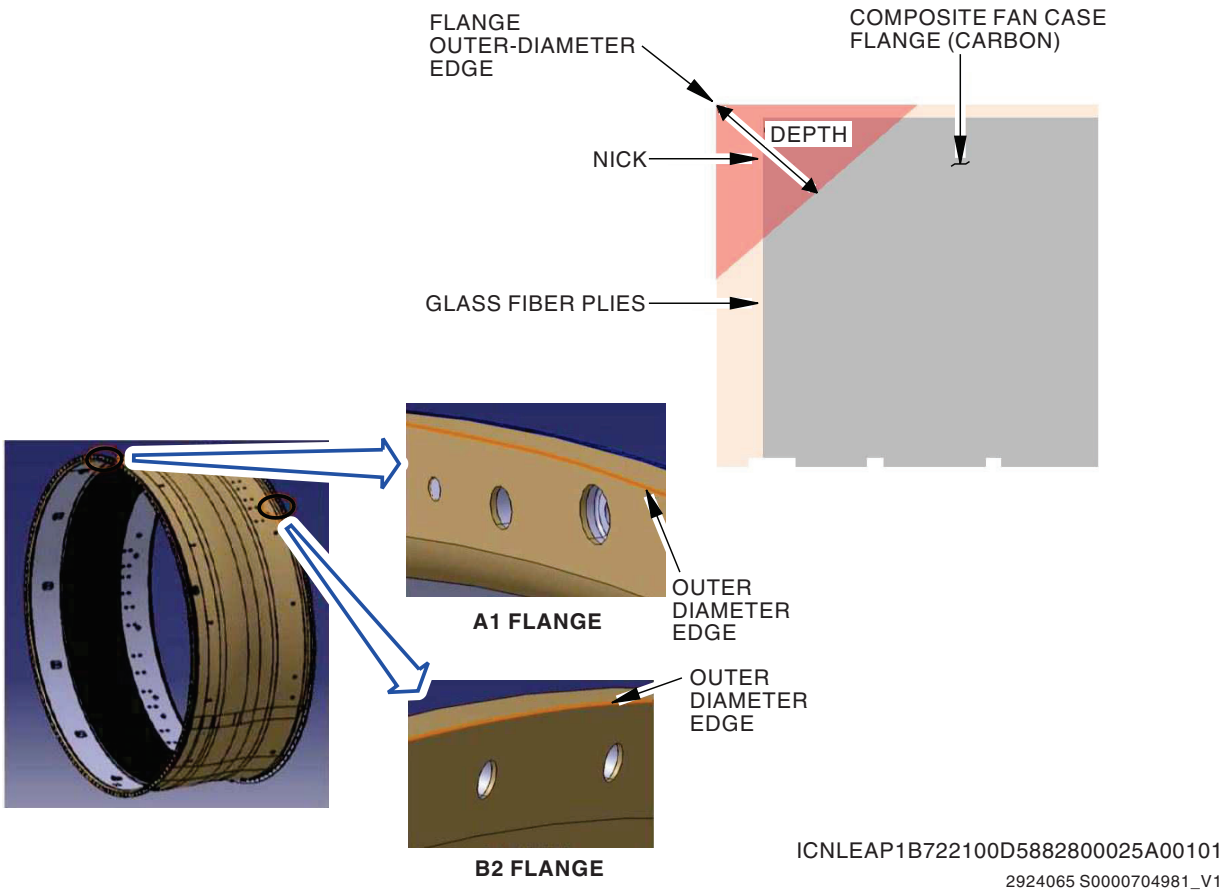
Fan Case Module Inspection
Figure 601/72-21-00-990-802-G00 (Sheet 28 of 38)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

EXAMPLE OF NICK ON THE OUTER DIAMETER EDGE OF THE B2 FLANGE:



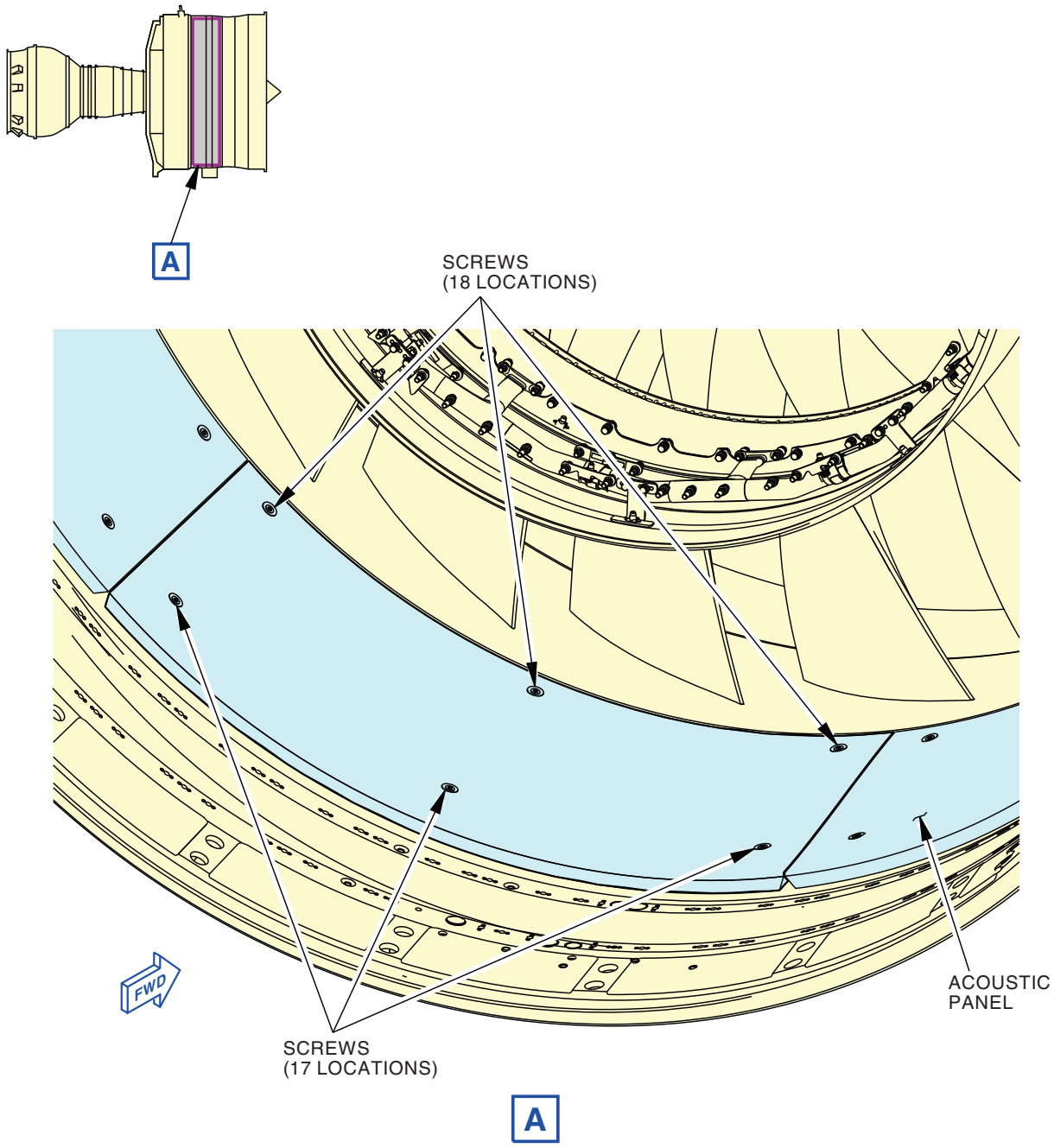
MEASUREMENT METHOD:
THE LENGTH CORRESPONDS TO THE LONGEST DIMENSION OF THE NICK, DENT OR SCRATCH.
THE DEPTH IS MEASURED AS DESCRIBED ON THE DRAWING BELOW:



ICNLEAP1B722100D5882800025A00101
2924065 S0000704981_V1

Fan Case Module Inspection
Figure 601/72-21-00-990-802-G00 (Sheet 29 of 38)

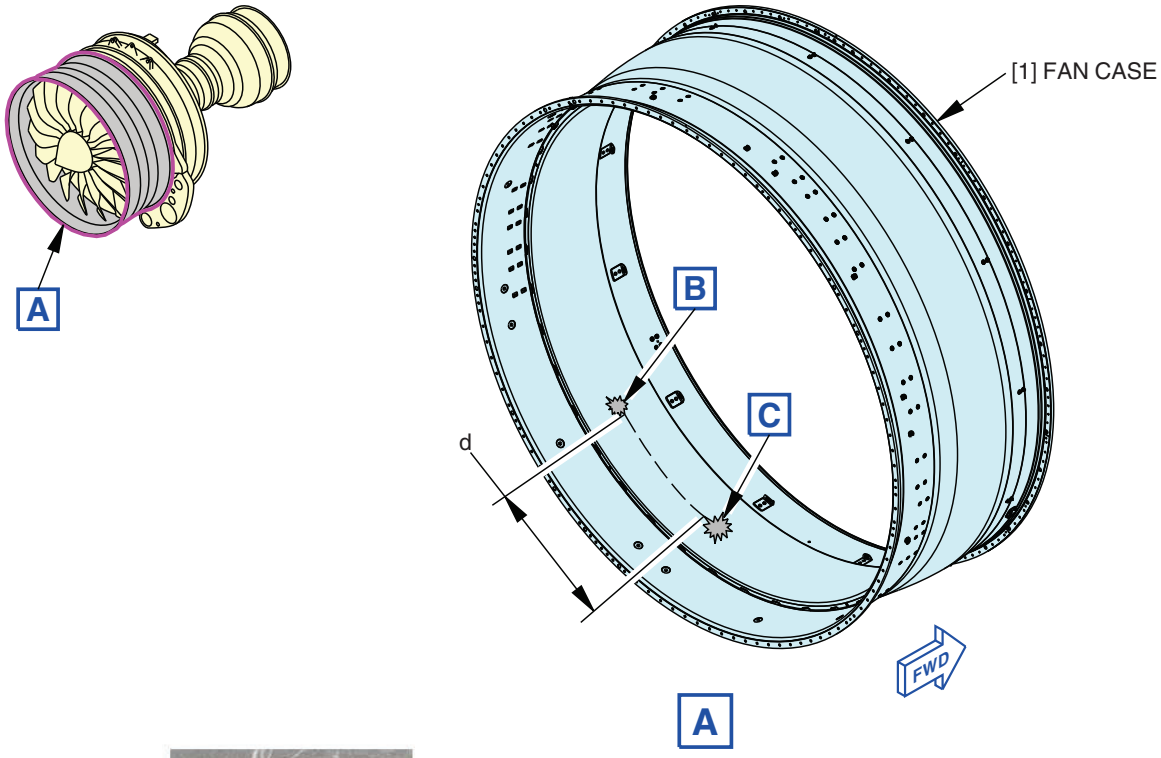
ECCN 9E991 BOEING PROPRIETARY - See title page for details

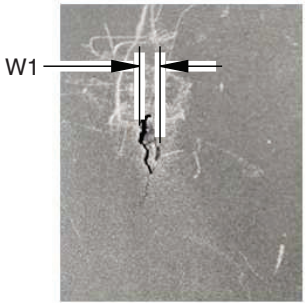


ICNLEAP1B722100D5882800030A00101
2946883 S0000723223_V1

Fan Case Module Inspection
Figure 601/72-21-00-990-802-G00 (Sheet 30 of 38)

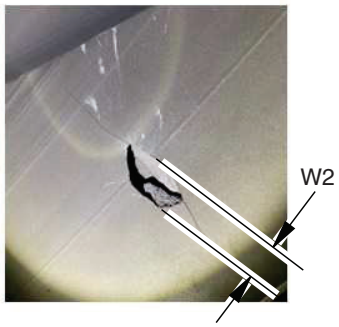
ECCN 9E991 BOEING PROPRIETARY - See title page for details





CRATER NO. 1 ON ABRADABLE (EXAMPLE)

B



CRATER NO. 2 ON ABRADABLE (EXAMPLE)

C

NOTE:

W1 = WIDTH 1

W2 = WIDTH 2

W1 < W2

d≥W2

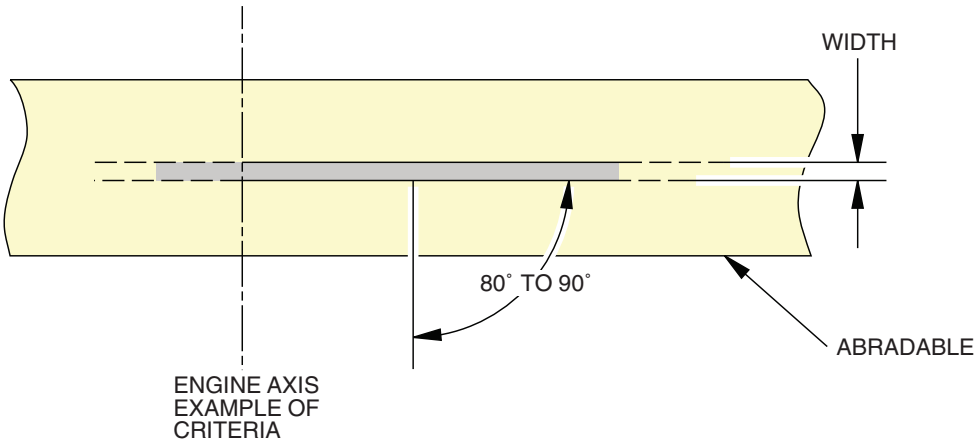
d = THE MINIMUM DISTANCE BETWEEN AFFECTED AREAS IS EQUAL TO THE MAXIMUM WIDTH OF THE NEAREST ONE AFFECTED AREA.

ICNLEAP1B722100D5882800032A00101

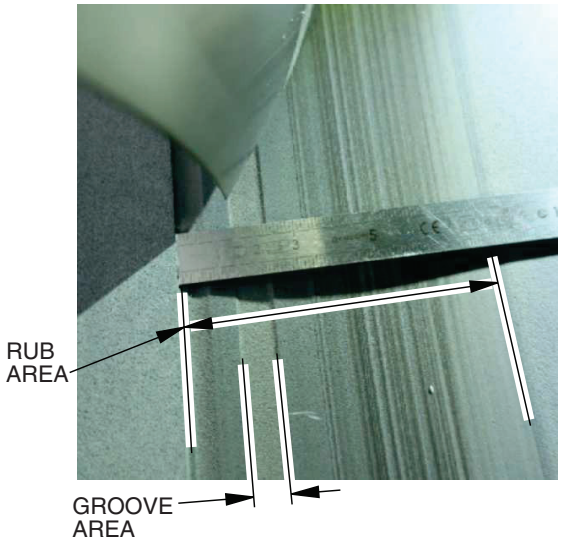
3100958 S0000846255_V2

Fan Case Module Inspection
Figure 601/72-21-00-990-802-G00 (Sheet 31 of 38)

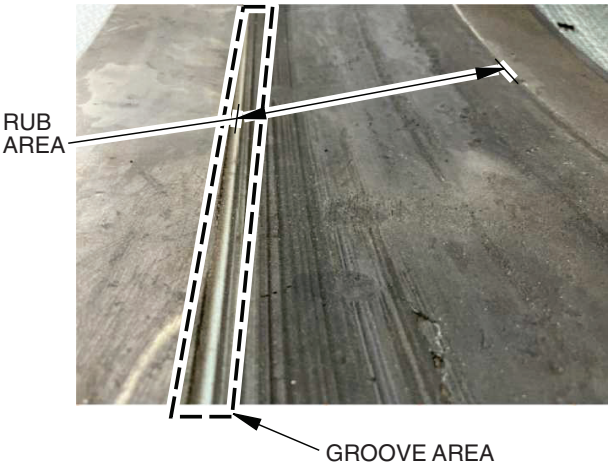
ECCN 9E991 BOEING PROPRIETARY - See title page for details



CIRCUMFERENTIAL GROOVE OR RUB ON ABRADABLE (EXAMPLE)



CIRCUMFERENTIAL GROOVE OR RUB ON ABRADABLE (EXAMPLE 1)

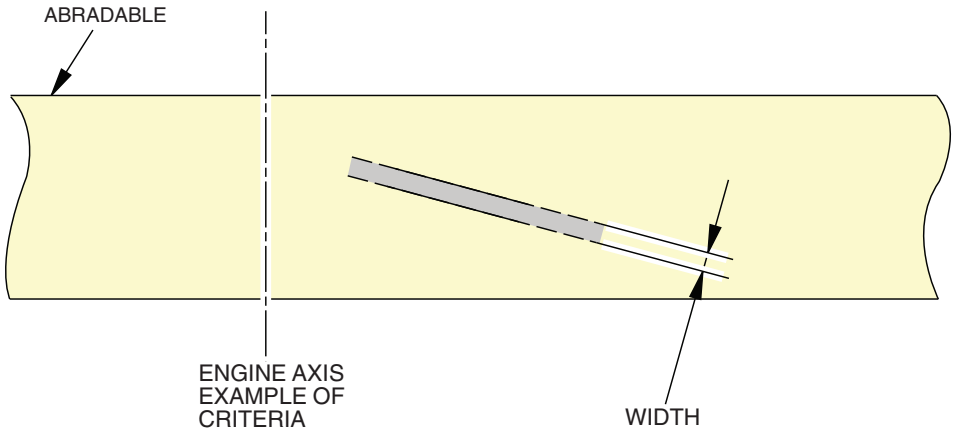


CIRCUMFERENTIAL GROOVE OR RUB ON ABRADABLE (EXAMPLE 2)

ICN-LEAP-1B-722100-D-58828-00033-A-001-01
3101200 S0000846256_V1

Fan Case Module Inspection
Figure 601/72-21-00-990-802-G00 (Sheet 32 of 38)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



OTHER GROOVES ON ABRADABLE (EXAMPLE)

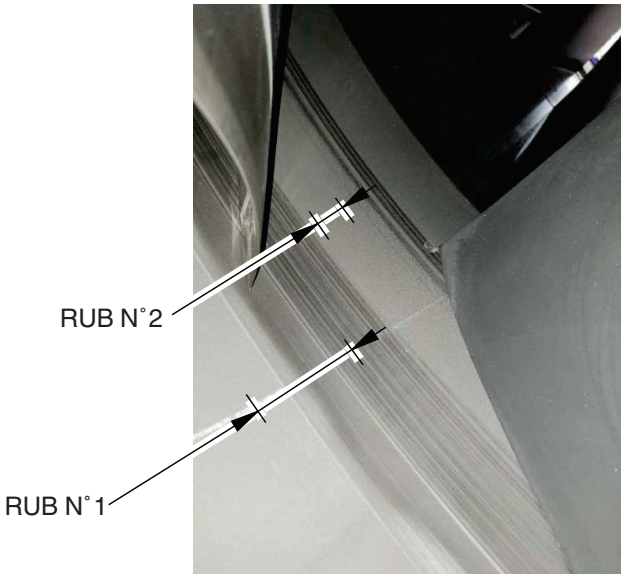
OTHER GROOVES ON ABRADABLE (EXAMPLE 1)

OTHER GROOVES ON ABRADABLE (EXAMPLE 2)

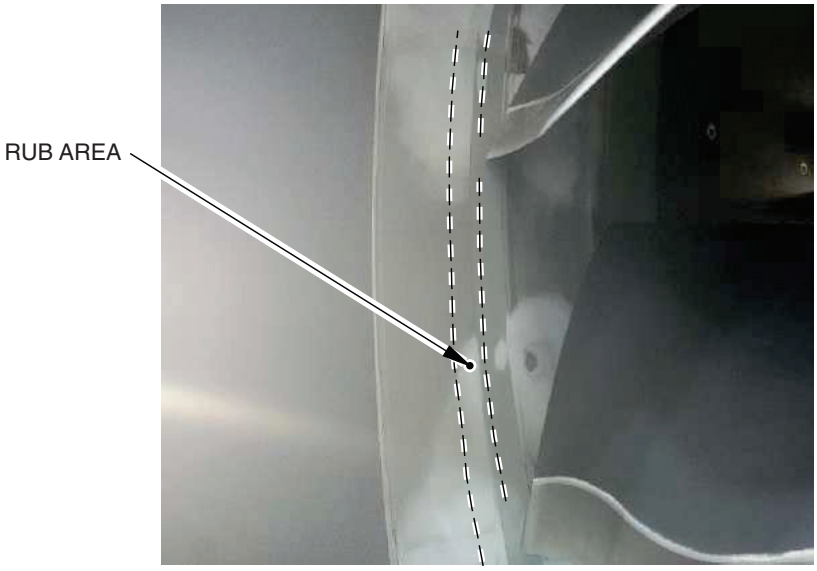
ICNLEAP1B722100D5882800034A00101
3100960 S0000846257_V1

Fan Case Module Inspection
Figure 601/72-21-00-990-802-G00 (Sheet 33 of 38)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



**RUB AREA ON ABRADABLE
(EXAMPLE 1)**

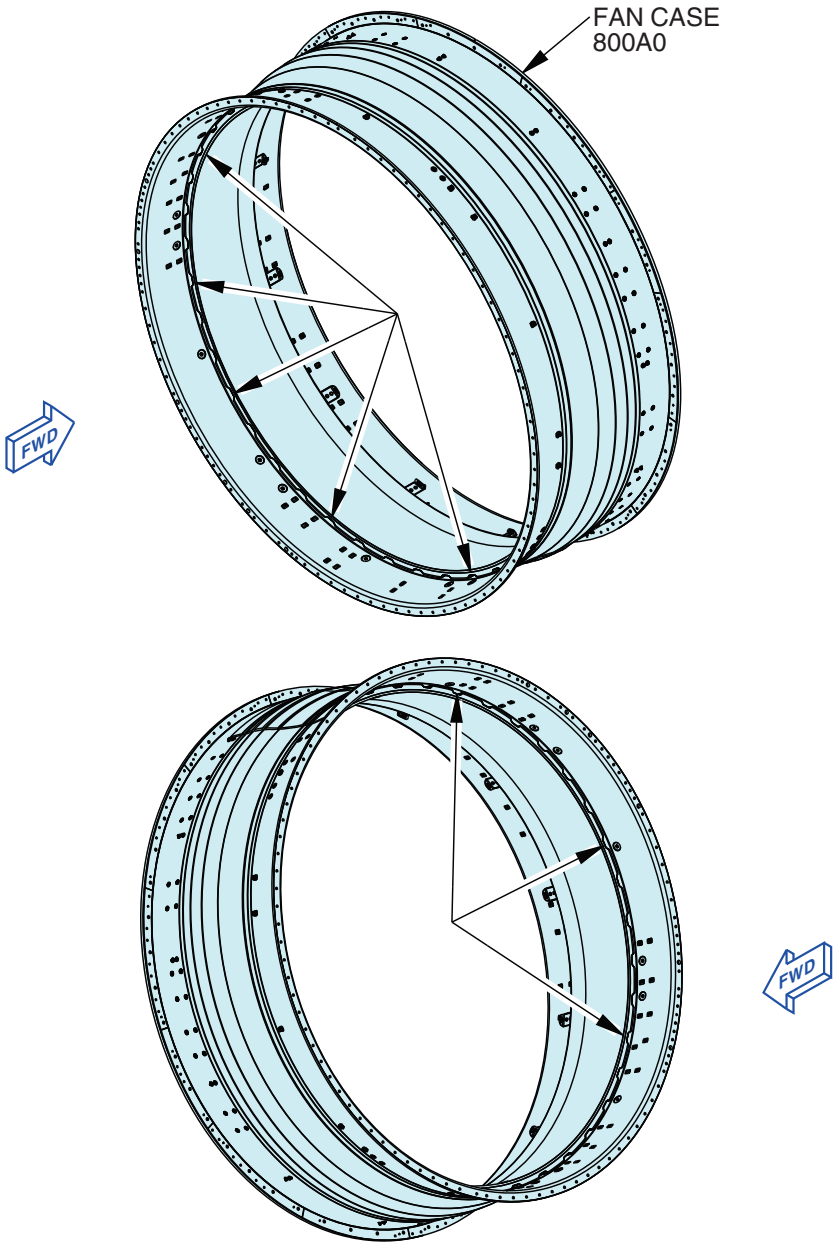


**RUB AREA ON ABRADABLE
(EXAMPLE 2)**

ICNLEAP1B722100D5882800035A00101
3100961 S0000846258_V1

**Fan Case Module Inspection
Figure 601/72-21-00-990-802-G00 (Sheet 34 of 38)**

ECCN 9E991 BOEING PROPRIETARY - See title page for details



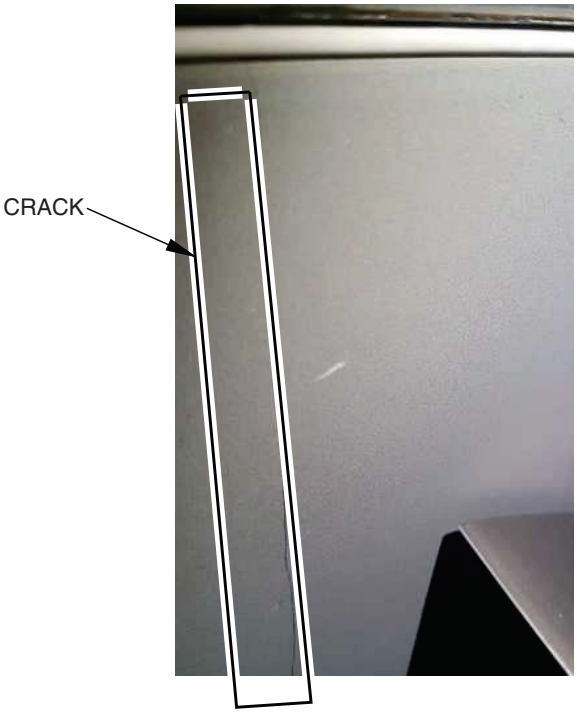
EIGHT POINTS OF MEASUREMENT ON ABRADABLE

ICNLEAP1B722100D5882800036A00101

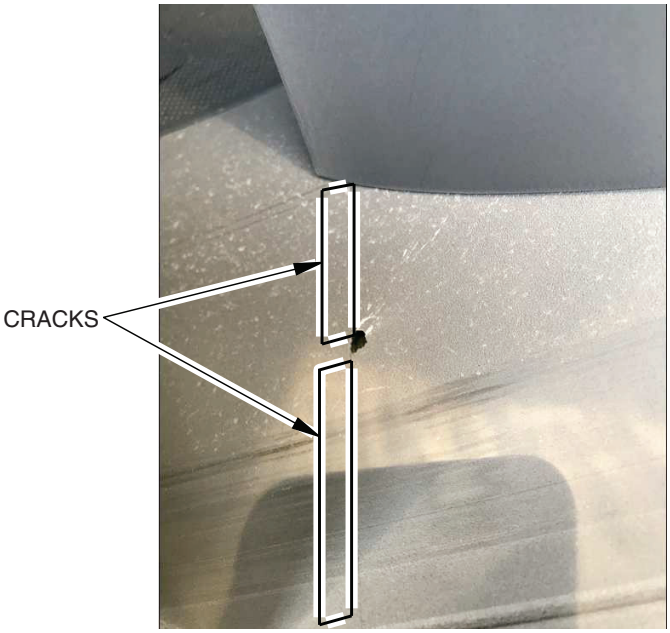
3100962 S0000846260_V1

Fan Case Module Inspection
Figure 601/72-21-00-990-802-G00 (Sheet 35 of 38)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



CRACK ON ABRADABLE (EXAMPLE 1)

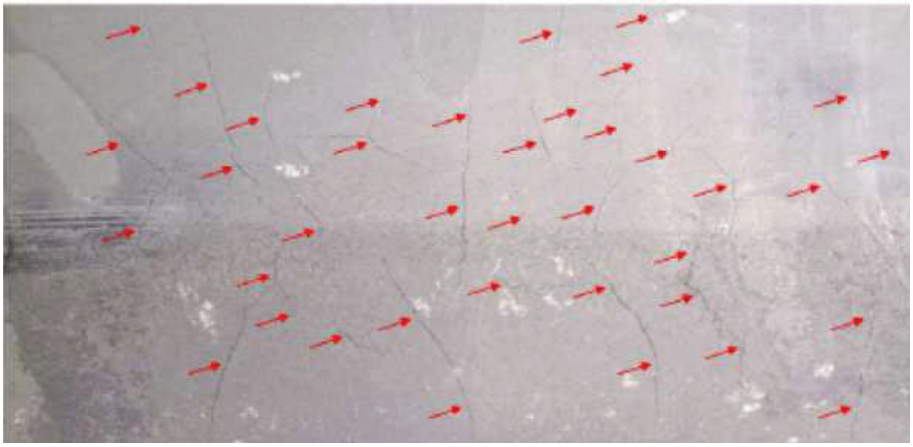


CRACK ON ABRADABLE (EXAMPLE 2)

ICNLEAP1B722100D5882800037A00101
3100969 S0000846261_V1

Fan Case Module Inspection
Figure 601/72-21-00-990-802-G00 (Sheet 36 of 38)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



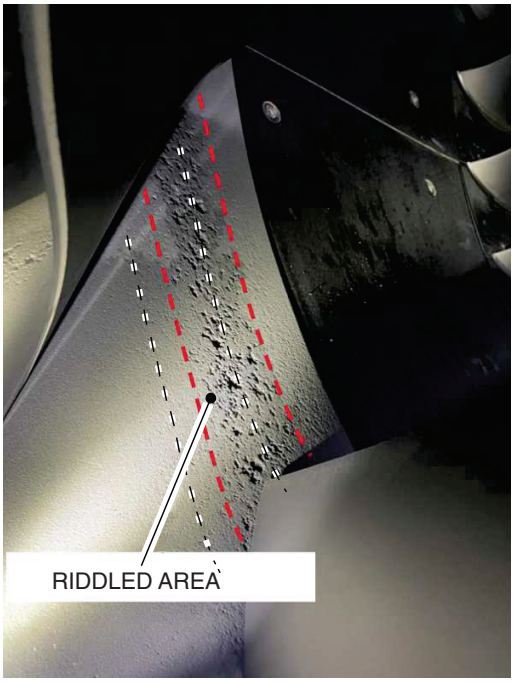
**CRACKS ON ABRADABLE
(EXAMPLE 3)**

ICNLEAP1B722100D5882800038A00101

3100989 S0000846262_V1

**Fan Case Module Inspection
Figure 601/72-21-00-990-802-G00 (Sheet 37 of 38)**

ECCN 9E991 BOEING PROPRIETARY - See title page for details



**RIDDLED AREA ON ABRADABLE
(EXAMPLE 1)**

**RIDDLED AREA ON ABRADABLE
(EXAMPLE 2)**

ICNLEAP1B722100D5882800039A00101
3100990 S0000846263_V1

**Fan Case Module Inspection
Figure 601/72-21-00-990-802-G00 (Sheet 38 of 38)**

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 72-21-00-220-802-G00

3. Confirmed Air Leakage in the Fan Compartment Fan Case Color Inspection

(Figure 602, Figure 603, Figure 604 and Figure 605)

A. General

(1) This task gives the instructions for the fan case color inspection in case of confirmed air leakage in the fan compartment.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-14215	Portable Spectrophotometer Part #: CM-700D Supplier: 4DDU6 Part #: MODEL 6834 Supplier: 1L8C8
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

F. Prepare for the Inspection

SUBTASK 72-21-00-860-013-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-21-00-010-001-G00

(2) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Confirmed Air Leakage in the Fan Compartment Fan Case Color Inspection

SUBTASK 72-21-00-220-003-G00

(1) Do the fan case color inspection in case of confirmed air leakage in fan compartment as follows:

(a) Use a spectrophotometer, COM-14215.

NOTE: Make sure that the device calibration is satisfactory. The device is calibrated at the factory. However, if it is necessary, refer to the factory instruction manual to calibrate the device.

(b) Consider as the area to inspect, area "O", a square area of 7.9 in. (200 mm) X 7.9 in. (200 mm) on the outer skin of the fan case centered on the closest accessible location where you confirm the air leakage (Figure 602, view B).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Consider as area of color reference, area "R", a square area of 7.9 in. (200 mm) X 7.9 in. (200 mm) on the outer skin of the fan case at a distance minimum of 39.37 in. (1000 mm) from the area “O” (Figure 602, view C).

(d) Do the coloration measurement with the spectrophotometer, COM-14215 as follows:

1) Divide the area "O" in to five sub-divisions "O1", "O2", "O3", "O4" and "O5" Figure 602.

NOTE: Make sure that the area "O1" is the closest accessible location to the area of the air leakage and has the area of diameter 3.9 in. (100 mm).

2) Divide the area "R" in to five sub-divisions "R1", "R2", "R3", "R4" and "R5" Figure 602.

NOTE: Make sure the center of area "R1" is at a minimum distance of 39.36 in. (1000 mm) from the area "O1" and has the area of diameter 3.9 in. (100 mm).

3) Do the coloration measurement on the areas "O1", "O2", "O3", "O4" and "O5".

4) Record the value obtained from the spectrophotometer, COM-14215, in the table (Figure 603).

5) Do the coloration measurement on the areas "R1", "R2", "R3", "R4" and "R5".

6) Record the value from the spectrophotometer, COM-14215, in the table (Figure 603).

7) Calculate the delta value from the spectrophotometer, COM-14215, between measurements in area "O" and area "R" (Figure 604).

8) Refer to this figure for the serviceable limit (Figure 605).

H. Put the Airplane Back to Its Usual Condition

SUBTASK 72-21-00-410-001-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 72-21-00-860-014-G00

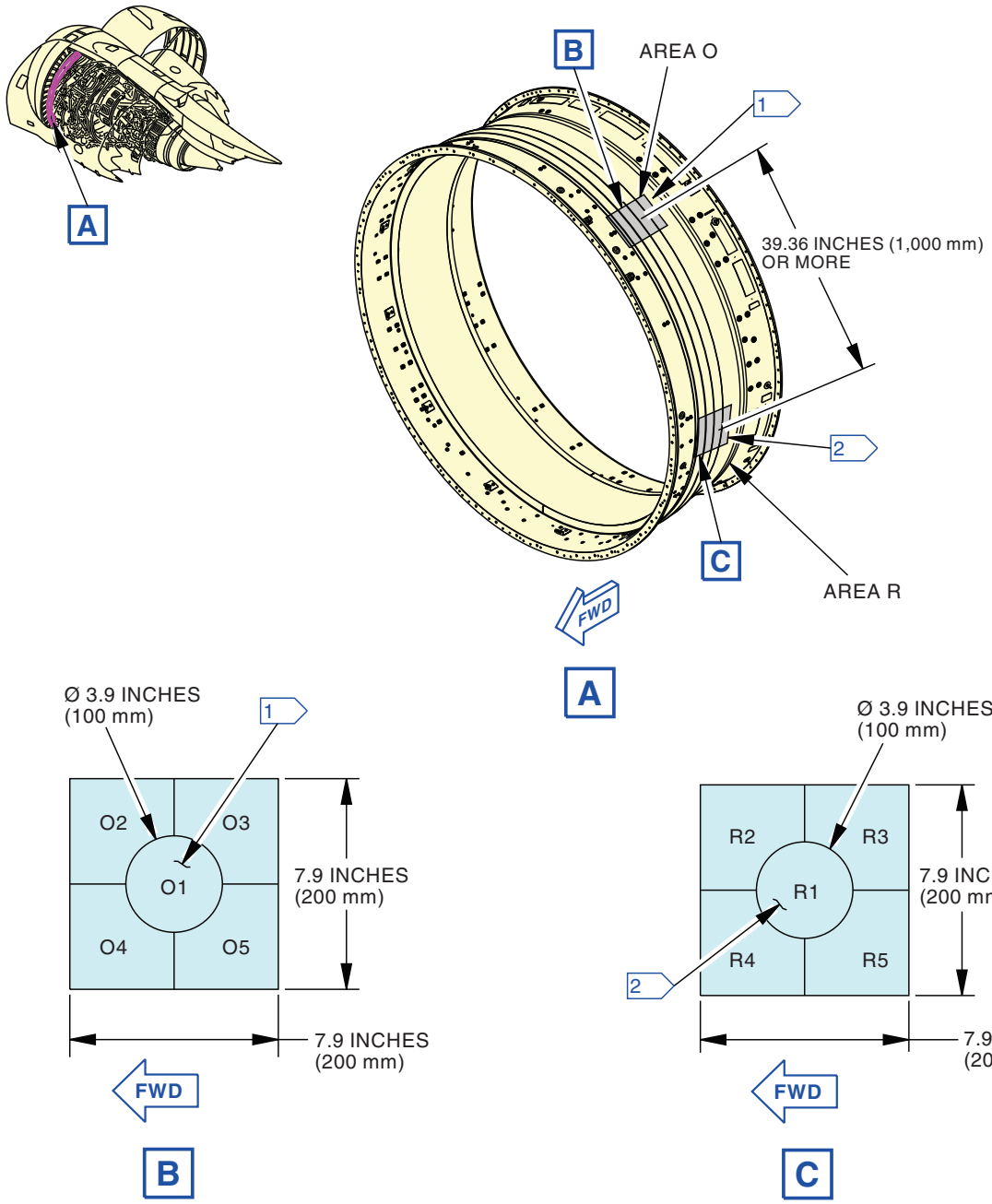
(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



1 ACCESSIBLE LOCATION CLOSEST TO THE AIR LEAK DEFINES THE CENTER OF AREA O

2 ACCESSIBLE LOCATION 39.36 INCHES (1,000 mm) OR MORE FROM THE CENTER OF AREA O DEFINES THE ENTER OF AREA R

ICN-LEAP-1B-722100-B-F0301-A0001-A-001-01
2571600 S0000617249_V1

Confirmed Air Leakage in the Fan Compartment Fan Case Color Inspection
Figure 602/72-21-00-990-848-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

STEP 1 : DO THE 10 MEASUREMENTS AND COMPLETE THE TABLE BELOW

AREA R = REFERENCE, ACCESSIBLE LOCATION AT LEAST 39.36 INCHES (1,000 mm) FROM THE CENTER OF AREA O	L	a	b
AREA R1	(L_R1):	(a_R1):	(b_R1):
AREA R2	(L_R2):	(a_R2):	(b_R2):
AREA R3	(L_R3):	(a_R3):	(b_R3):
AREA R4	(L_R4):	(a_R4):	(b_R4):
AREA R5	(L_R5):	(a_R5):	(b_R5):

AREA O = CLOSEST ACCESSIBLE LOCATION TO THE AIR LEAK	L	a	b
AREA O1	(L_O1):	(a_O1):	(b_O1):
AREA O2	(L_O2):	(a_O2):	(b_O2):
AREA O3	(L_O3):	(a_O3):	(b_O3):
AREA O4	(L_O4):	(a_O4):	(b_O4):
AREA O5	(L_O5):	(a_O5):	(b_O5):

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2571766 S0000617250_V1

Fan Case Color Measurements
Figure 603/72-21-00-990-849-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

STEP 2 : CALCULATE THE DELTA BETWEEN AVERAGE VALUES OF AREA R AND AREA O

$$\Delta L_avg = \frac{(L_R1)+(L_R2)+(L_R3)+(L_R4)+(L_R5)-(L_O1)-(L_O2)-(L_O3)-L_O4)-(L_O5)}{5} = _____$$

$$\Delta a_avg = \frac{(a_R1)+(a_R2)+(a_R3)+(a_R4)+(a_R5)-(a_O1)-(a_O2)-(a_O3)-(a_O4)-(a_O5)}{5} = _____$$

$$\Delta b_avg = \frac{(b_R1)+(b_R2)+(b_R3)+(b_R4)+(b_R5)-(b_O1)-(b_O2)-(b_O3)-(b_O4)-(b_O5)}{5} = _____$$

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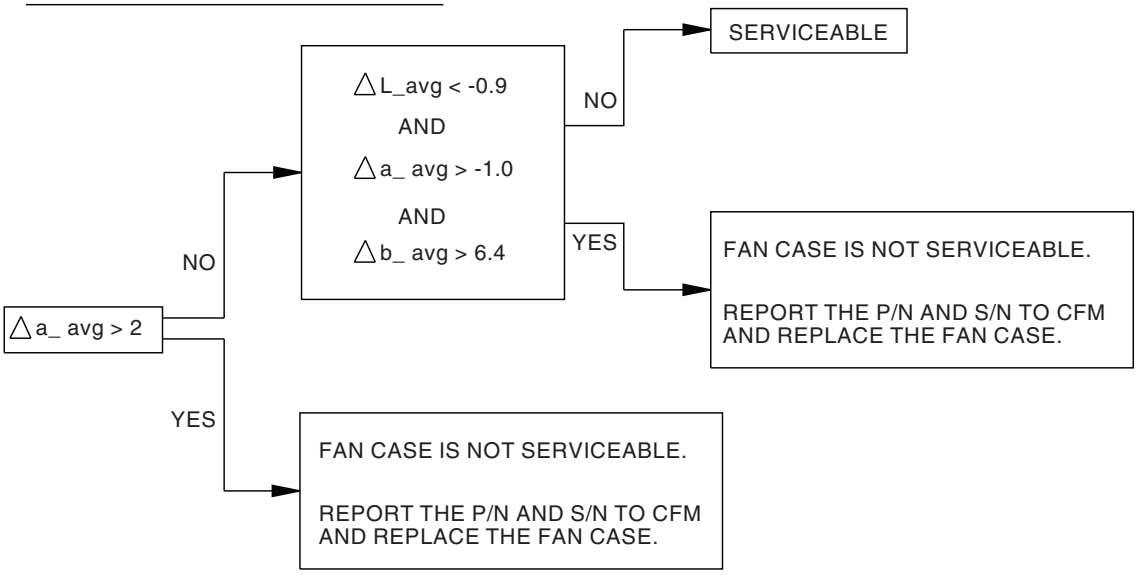
2571785 S0000617251_V1

Delta Between Average Values Calculation
Figure 604/72-21-00-990-850-G00

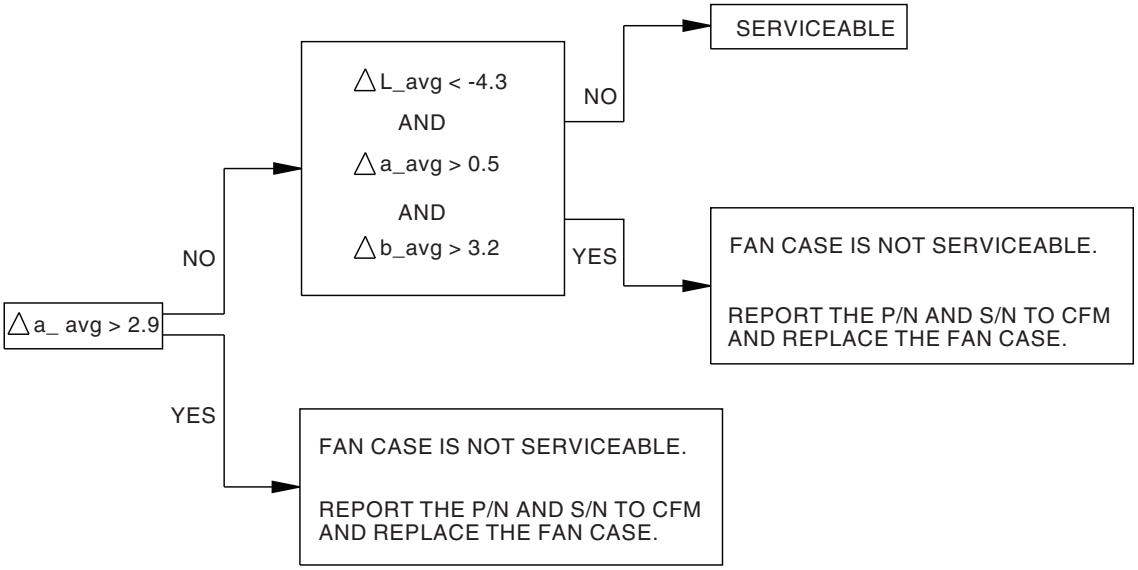
ECCN 9E991 BOEING PROPRIETARY - See title page for details

STEP 3 : USE THE FLOW CHART BELOW IN ORDER TO TAKE A DECISION

KONICA MINOLTA DEVICE (CM-700D) :



BYK DEVICE (BYK 6834 SPECTRO-GUIDE SPHERE GLOSS):



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2571786 S0000617252_V2

Serviceable Limit Flow Chart
Figure 605/72-21-00-990-851-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 72-21-00-220-801-G00

4. Low Pressure Compressor Stage 1 Inlet Guide Vane and Rotating Ring Spacer Inspection

A. General

(1) This task gives the instructions to examine the Low Pressure Compressor (LPC) stage 1 Inlet Guide Vane (IGV) and rotating ring spacer.

B. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

C. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

D. Prepare for the Inspection

SUBTASK 72-21-00-860-015-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-21-00-860-016-G00

(2) Open these applicable circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

E. Low Pressure Compressor Stage 1 Inlet Guide Vane and Rotating Ring Spacer Inspection

SUBTASK 72-21-00-220-004-G00

(1) Examine the LPC stage 1 IGV [1] for these conditions:

(a) Missing pieces

1) Not permitted.

(b) Broken vane

1) Not permitted.

(c) Crack and tears in all areas

1) Not permitted.

(d) Nicks and dents in Area A leading edges

1) Any amount up to 0.004 in. (0.1 mm) in depth. Protrusion is acceptable.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(e) Nicks and dents in Area A

1) Any amount up to 0.010 in. (0.25 mm) in depth. Protrusion is acceptable. The indication must not go through the metal.

(f) Nicks and dents in Area A trailing edges

1) Any amount up to 0.004 in. (0.1 mm) in depth. Protrusion is acceptable.

(g) Nicks and dents in Area B leading edges

1) Any amount up to 0.008 in. (0.2 mm) in depth. Protrusion is acceptable.

(h) Nicks and dents in Area B trailing edges

1) Any amount up to 0.0197 in. (0.5 mm) in depth. Protrusion is acceptable.

(i) Nicks and dents in Area C

1) Any amount up to 0.0197 in. (0.5 mm) in depth. Protrusion is acceptable. The indication must not go through the metal.

(j) Vane looseness

1) Not permitted.

(k) Local distortion in Area A

1) Not permitted.

(l) Local distortion in Area C

1) Not permitted.

(m) Local distortion on the leading edge and trailing edge

1) Any amount, if outside area D and if the length L along the leading (or trailing) edge is not more than 0.295 in. (7.5 mm) and the maximum dimension X along the circumferential direction is not more than 0.029 in. (0.74 mm). Maximum 6 vanes affected.

(n) Deposit

1) Any amount.

(o) Erosion in areas B

1) Any amount.

(p) Deposit in Jet Impigment holes

1) Not permitted.

SUBTASK 72-21-00-220-006-G00

(2) Examine the inner shroud in case of low pressure compressor stage 1 inlet guide vane removal (for blending repair of low pressure compressor stage 2 vane assembly):

(a) Disbounding or flaking of the inner shroud abradable

1) Not permitted.

(b) Rub marks from seal teeth on the inner shroud abradable

1) Any amount, up to 0.026 in. (0.66 mm) for average depth and 0.0335 in. (0.85 mm) in localized areas and not more than 0.1181 in. (3.0 mm) in width all over the circumference.

(c) Dents on the forward or aft edge on the inner shroud

1) Any amount up to 0.05 in. (1.27 mm) in depth and if the length of the indication is at least 10 times greater than its depth.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(d) Crack and tears on the inner shroud

1) Not permitted.

SUBTASK 72-21-00-220-005-G00

(3) Examine the rotating ring spacer [2] for these conditions:

(a) Cracks in area E and F

1) Not permitted.

(b) Nicks, dents, and scratches in area E

1) Not permitted.

(c) Fretting and Wear in area E

1) Any amount up to 0.001 in. (0.025 mm) in depth.

(d) Nicks, dents, and scratches in area F

1) Any amount up to 0.001 in. (0.025 mm) in depth. Minimum distance between indications is 0.47 in. (12 mm).

F. Put the Airplane Back to Its Usual Condition

SUBTASK 72-21-00-860-017-G00

(1) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

SUBTASK 72-21-00-860-018-G00

(2) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

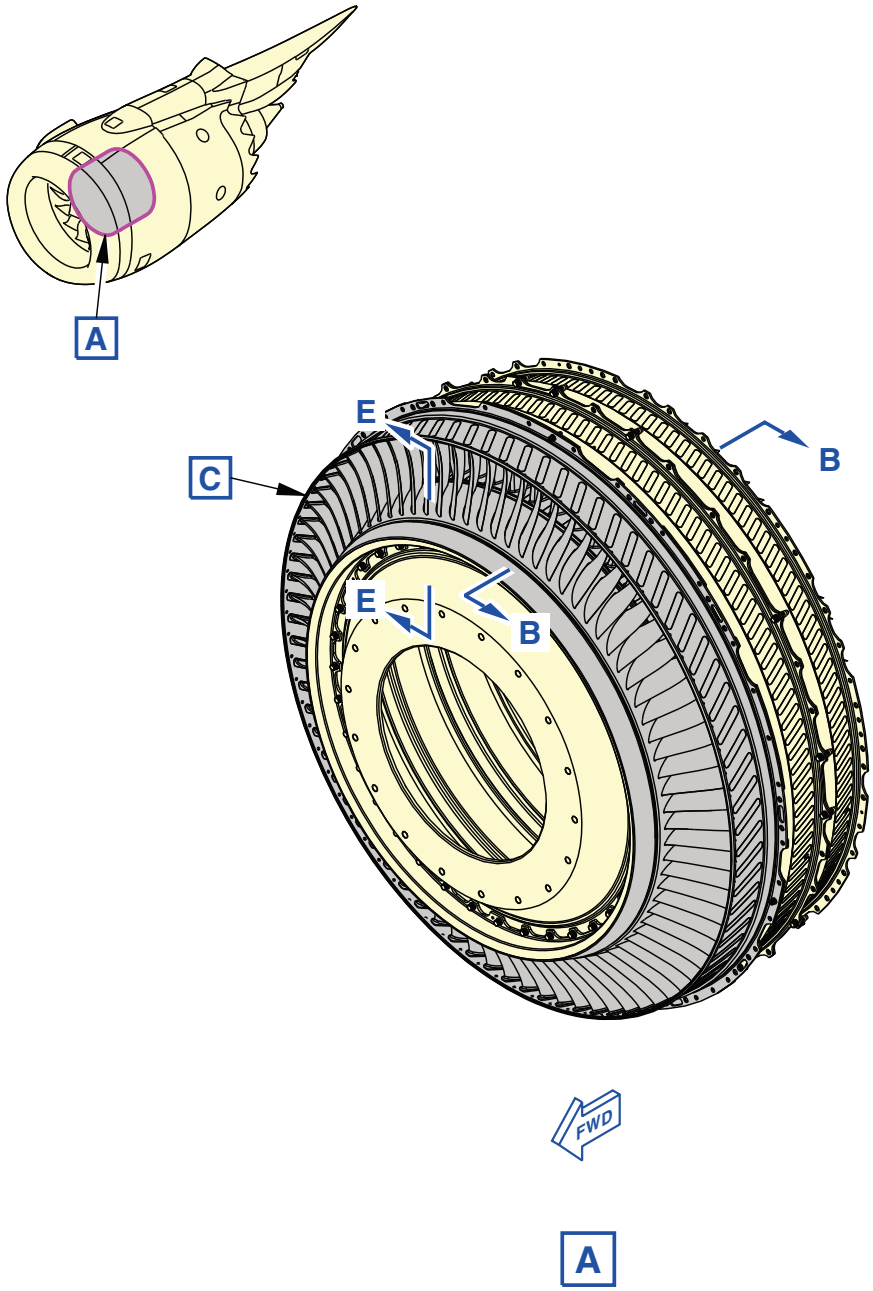
Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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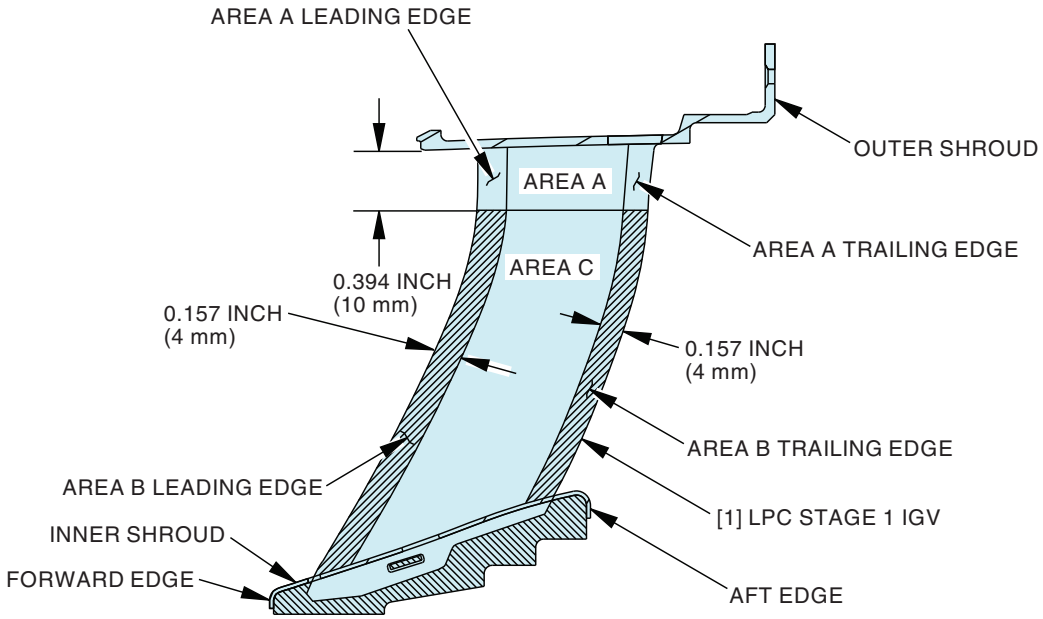
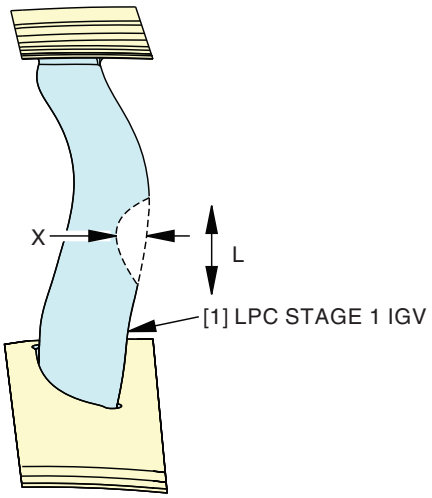
2571092 S0000616714_V1

Low Pressure Compressor Stage 1 Inlet Guide Vane and Rotating Ring Spacer Inspection
Figure 606/72-21-00-990-852-G00 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



B-B



C

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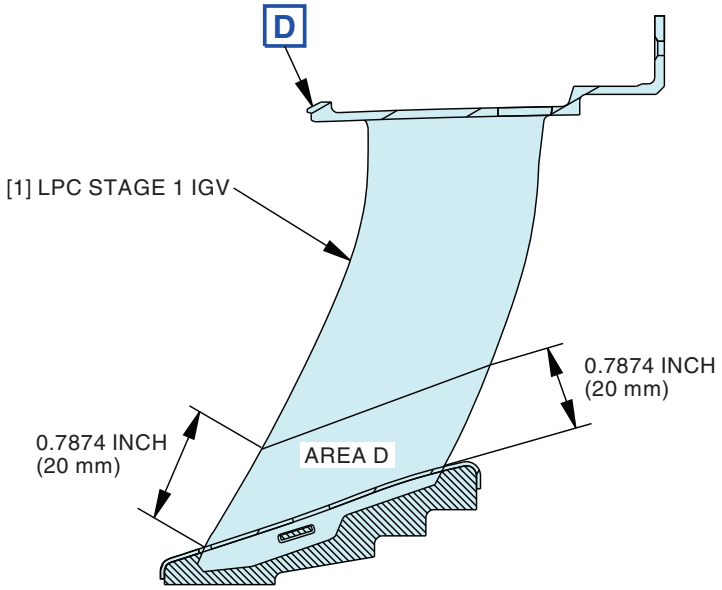
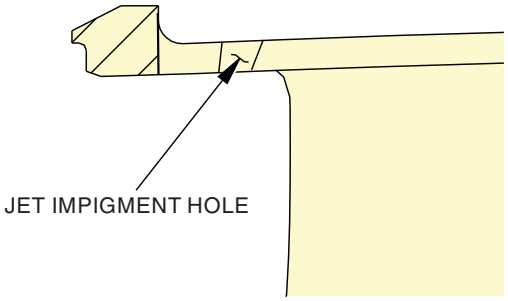
2571104 S0000616715_V1

Low Pressure Compressor Stage 1 Inlet Guide Vane and Rotating Ring Spacer Inspection
Figure 606/72-21-00-990-852-G00 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



B-B



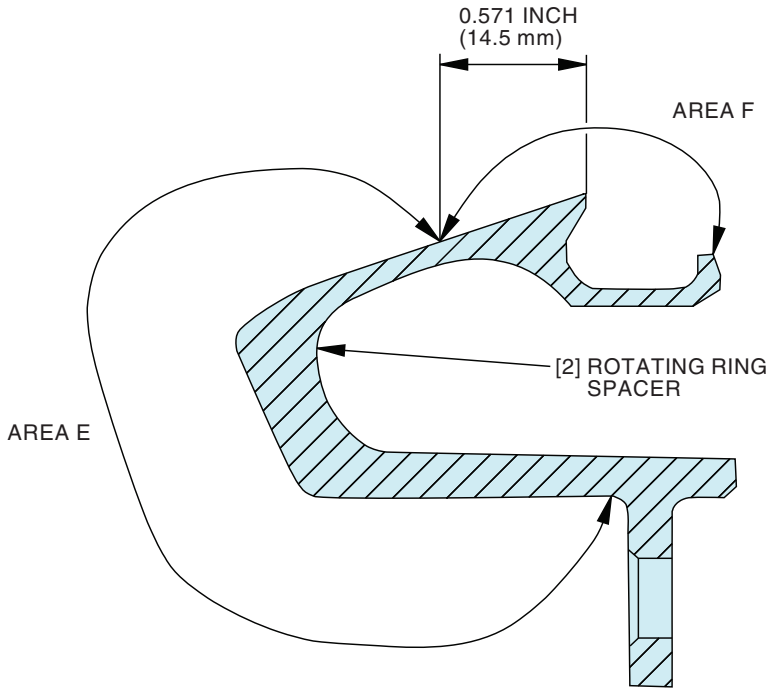
D

ICNLEAP1B722100DF030100005A00101

2571108 S0000616716_V1

Low Pressure Compressor Stage 1 Inlet Guide Vane and Rotating Ring Spacer Inspection
Figure 606/72-21-00-990-852-G00 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



E-E

ICNLEAP1B722100DF030100006A00101

2571117 S0000616717_V1

Low Pressure Compressor Stage 1 Inlet Guide Vane and Rotating Ring Spacer Inspection
Figure 606/72-21-00-990-852-G00 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OUTLET GUIDE VANE - REMOVAL/INSTALLATION

1. General

A. This procedure contains the following tasks:

(1) A removal of the outlet guide vane

(2) An installation of the outlet guide vane.

TASK 72-21-03-000-801-G00

2. Outlet Guide Vane Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove an outlet guide vane.

(2) The outlet guide vane is referred to as the OGV.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
72-21-09-000-801-G00	Outlet Guide Vane Platform Removal (P/B 401)
72-22-05-000-801-G00	Flowpath Fairing Removal (P/B 401)
72-23-04-000-801-G00	Fan Frame Module Segment Shrouds Removal (P/B 401)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
A50472 [CP2734]	Product - Silicone Elastomer Sealing	

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Prepare for the Removal

SUBTASK 72-21-03-210-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-21-03-865-001-G00

(2) Open these applicable circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 72-21-03-010-001-G00

(3) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

H. Outlet Guide Vane Removal

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0181

SUBTASK 72-21-03-210-002-G00

(1) Identify the OGV [1] and fan frame module shroud segment [4].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0181

SUBTASK 72-21-03-210-008-G00

(2) Identify the OGV [1], titanium OGV [23], and fan frame module shroud segment [4].

PTF ALL

SUBTASK 72-21-03-010-003-G00

(3) Remove the applicable OGV platforms [2] (TASK 72-21-09-000-801-G00).

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0181

SUBTASK 72-21-03-020-001-G00

CAUTION

YOU MUST REMOVE A MAXIMUM OF TWO OUTLET GUIDE VANES THAT GO ONE AFTER THE OTHER AT A TIME. IF YOU DO NOT OBEY THIS INSTRUCTION, YOU CAN CAUSE DAMAGE TO THE EQUIPMENT.

(4) Remove the OGV [1] (No. 8, 26, and 31) as follows:

(a) Remove the applicable fan frame module shroud segment [4] (TASK 72-23-04-000-801-G00).

(b) Remove and discard the two inter OGV seals [15].

(c) Remove the three bolts [5] and three cup washers [18] that attach the OGV outer [6] to the fan frame shroud [7].

(d) Remove the bolt [16] and cup washer [20] that attach the OGV outer [6] to the fan frame shroud [7].

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0171

(e) Cut and remove RTV 103 adhesive, A50472 [CP2734], from the counterbore of the two bolts [10] on the OGV inner [11].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0171

(f) Remove the two OGV plugs [22] from the OGV inner [11].

(g) Remove RTV 103 adhesive, A50472 [CP2734], from the two bolts [10].

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0181

(h) Remove the two bolts [10] and two cup washers [19] that attach the OGV inner [11] to the inner shroud [13].

(i) Remove the two bolts [12] that attach the OGV inner [11] to the fan hub [14].

(j) Remove the OGV shim [8] and OGV shim [9] from the OGV outer [6].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0181 (Continued)

(k) Move the OGV [1] carefully forward to disengage from the recess of the inner shroud [13].

(l) Remove the OGV [1] from the engine as follows:

1) If it is necessary, remove the flowpath fairing [3] to remove the OGV [1] (TASK 72-22-05-000-801-G00).

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0181

SUBTASK 72-21-03-020-004-G00

CAUTION YOU MUST REMOVE ONLY ONE OUTLET GUIDE VANE MADE OF TITANIUM AT A TIME. IF YOU DO NOT OBEY THIS INSTRUCTION, YOU CAN CAUSE DAMAGE TO THE EQUIPMENT.

(5) Remove the titanium OGV [23] (No. 1, 2, 37 and 38) as follows (Figure 401 (Sheet 13)):

(a) Remove the applicable fan frame module shroud segment [4] (TASK 72-23-04-000-801-G00).

(b) Remove and discard the two inter OGV seals [15].

(c) Remove the four bolts [16] that attach the OGV outer [24] to the fan frame shroud [7].

(d) Remove the two OGV plugs [22] from the OGV inner [25] and remove the RTV 103 adhesive, A50472 [CP2734] from the two bolts [10].

(e) Remove the two bolts [10] that attach the OGV inner [25] to the inner shroud [13].

(f) Remove the two bolts [12] that attach the OGV inner [25] to the fan hub [14].

(g) Remove the OGV shim [8] and OGV shim [9] from the OGV outer [24].

(h) Move the titanium OGV [23] carefully forward to disengage from the recess of the inner shroud [13].

(i) Remove the titanium OGV [23] from the engine as follows:

1) If it is necessary, remove the flowpath fairing [3] to remove the titanium OGV [23] (TASK 72-22-05-000-801-G00).

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0181

SUBTASK 72-21-03-020-002-G00

CAUTION YOU MUST REMOVE A MAXIMUM OF TWO OUTLET GUIDE VANES THAT GO ONE AFTER THE OTHER AT A TIME. IF YOU DO NOT OBEY THIS INSTRUCTION, YOU CAN CAUSE DAMAGE TO THE EQUIPMENT.

(6) Remove the OGV [1] (No. 9 and 32) as follows:

(a) Remove the applicable fan frame module shroud segment [4] (TASK 72-23-04-000-801-G00).

(b) Remove and discard the two inter OGV seals [15].

(c) Remove the three bolts [5] and three cup washers [18] that attach the OGV outer [6] to the fan frame shroud [7].

(d) Remove the bolt [17] and cup washer [21] that attach the OGV outer [6] to the fan frame shroud [7].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0171

(e) Cut and remove RTV 103 adhesive, A50472 [CP2734], from the counterbore of the two bolts [10] on the OGV inner [11].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0171

(f) Remove the two OGV plugs [22] from the OGV inner [11].

(g) Remove RTV 103 adhesive, A50472 [CP2734], from the two bolts [10].

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0181

(h) Remove the two bolts [10] and two cup washers [19] that attach the OGV inner [11] to the inner shroud [13].

(i) Remove the two bolts [12] that attach the OGV inner [11] to the fan hub [14].

(j) Remove the OGV shim [8] and OGV shim [9] from the OGV outer [6].

(k) Move the OGV [1] carefully forward to disengage from the recess of the inner shroud [13].

(l) If it is necessary, do this step to remove the OGV [1]:

1) Remove the flowpath fairing [3] (TASK 72-22-05-000-801-G00).

(m) Remove the OGV [1] from the engine.

PTF ALL

SUBTASK 72-21-03-020-003-G00

CAUTION

YOU MUST REMOVE A MAXIMUM OF TWO OUTLET GUIDE VANES THAT GO ONE AFTER THE OTHER AT A TIME. IF YOU DO NOT OBEY THIS INSTRUCTION, YOU CAN CAUSE DAMAGE TO THE EQUIPMENT.

(7) Remove the other OGVS [1] as follows:

(a) Remove the applicable fan frame module shroud segment [4] (TASK 72-23-04-000-801-G00).

(b) Remove and discard the two inter OGV seals [15].

(c) Remove the four bolts [5] and four cup washers [18] that attach the OGV outer [6] to the fan frame shroud [7].

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0171

(d) Cut and remove RTV 103 adhesive, A50472 [CP2734], from the counterbore of the two bolts [10] on the OGV inner [11].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0171 OR POST LEAP-1B SERVICE BULLETIN 72-0181

(e) Remove the two OGV plugs [22] from the OGV inner [11].

(f) Remove RTV 103 adhesive, A50472 [CP2734], from the two bolts [10].

PTF ALL

(g) Remove the two bolts [10] and two cup washers [19] that attach the OGV inner [11] to the inner shroud [13].

(h) Remove the two bolts [12] that attach the OGV inner [11] to the fan hub [14].

(i) Remove the OGV shim [8] and OGV shim [9] from the OGV outer [6].

(j) Move the OGV [1] carefully forward to disengage from the recess of the inner shroud [13].

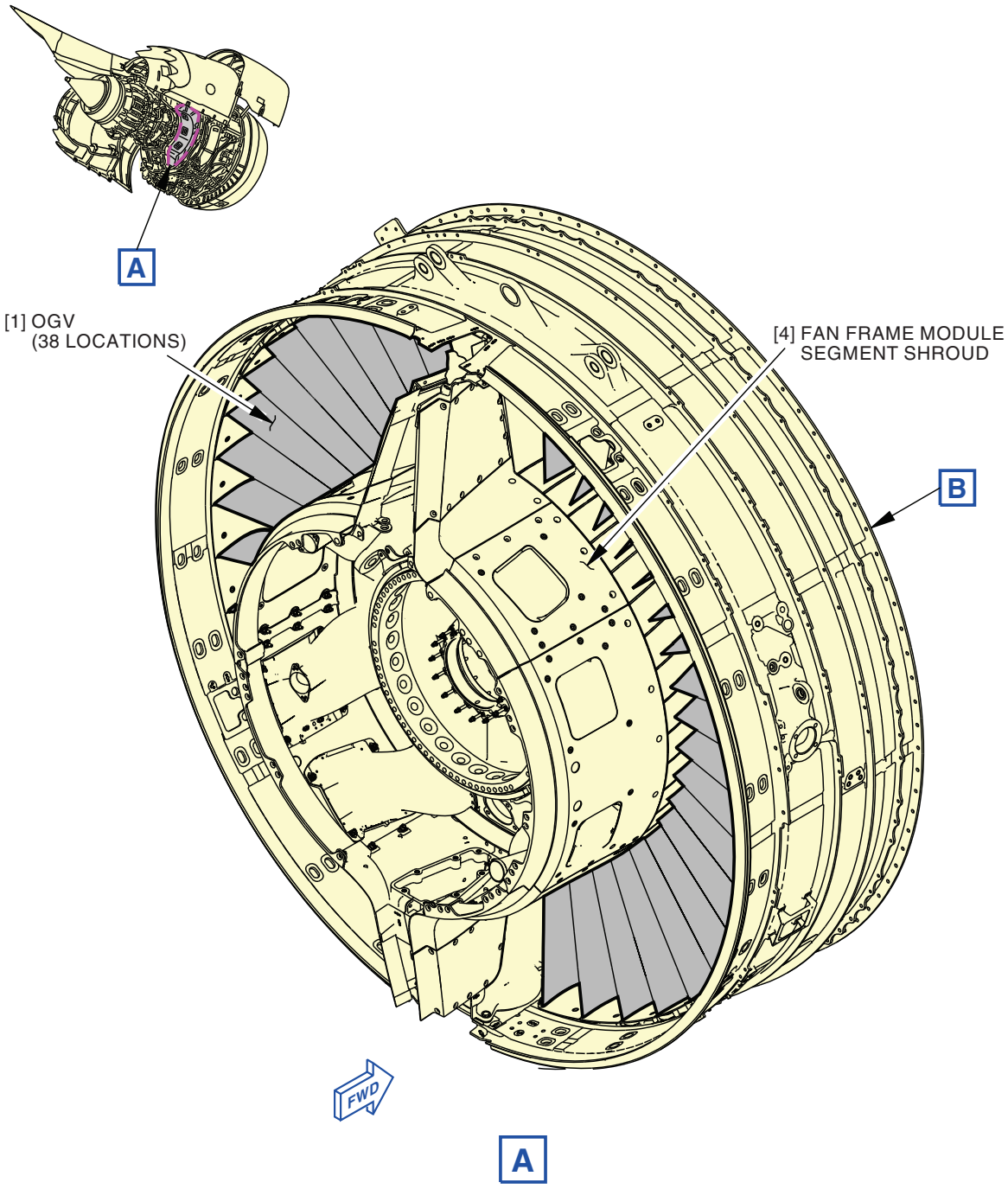
ECCN 9E991 BOEING PROPRIETARY - See title page for details

(k) Remove the OGV [1] from the engine as follows:

1) If it is necessary, remove the flowpath fairing [3] to remove the OGV [1] (TASK 72-22-05-000-801-G00).

——— END OF TASK ———

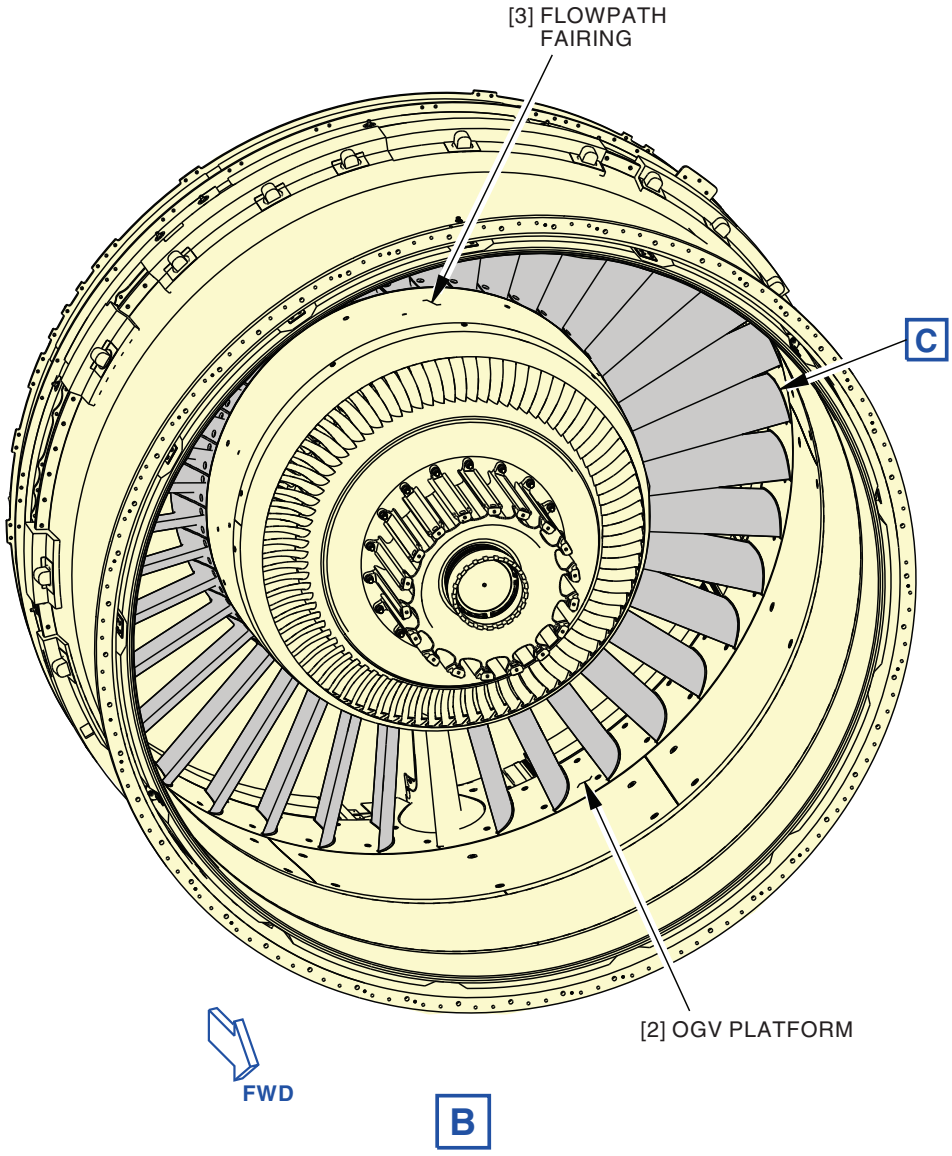
ECCN 9E991 BOEING PROPRIETARY - See title page for details



ICNLEAP1B722103D5882800001A00101
2434356 S0000564075_V4

Outlet Guide Vane Installation
Figure 401/72-21-03-990-803-G00 (Sheet 1 of 16)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



ICNLEAP1B722103D5882800002A00101
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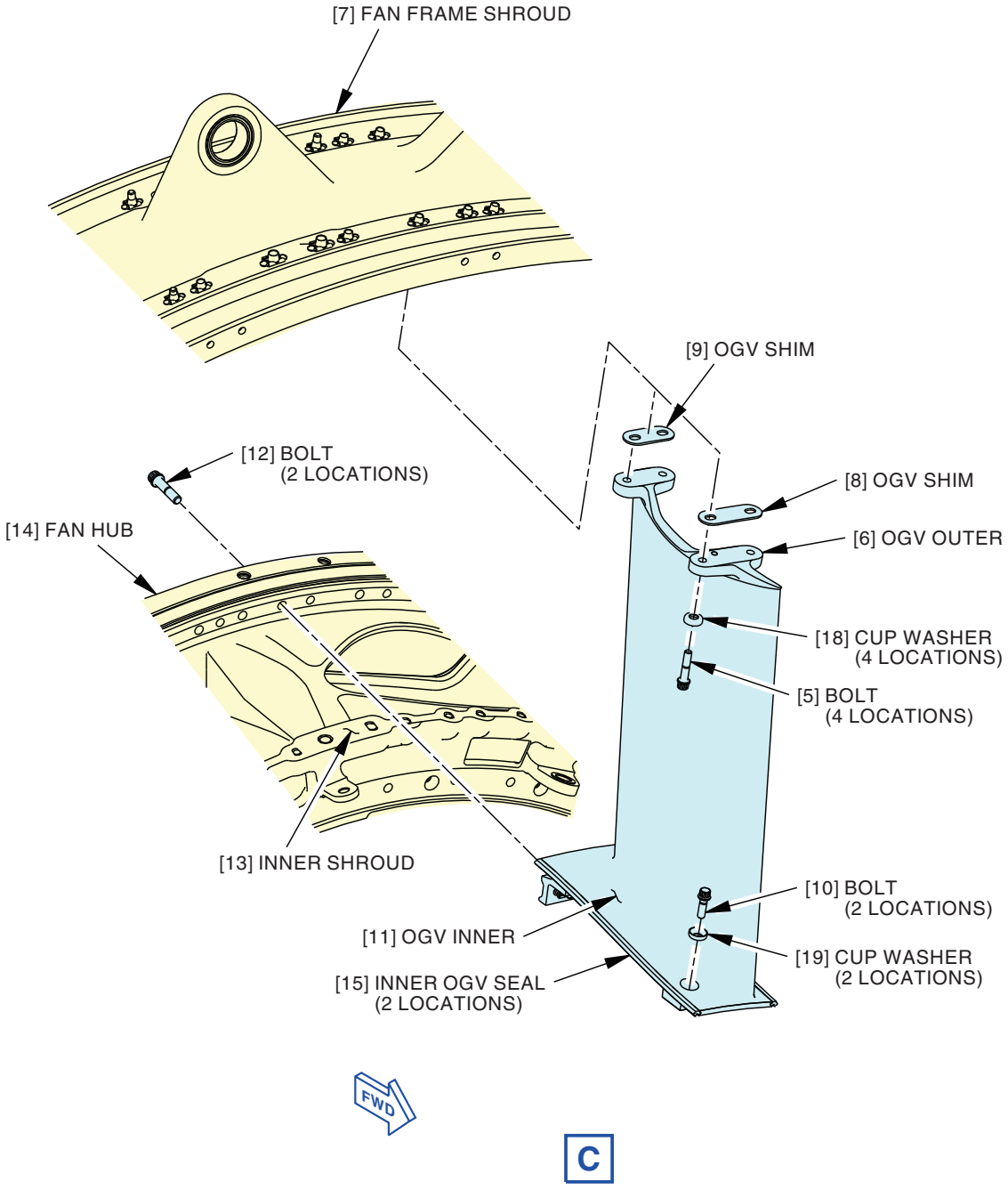
Outlet Guide Vane Installation
Figure 401/72-21-03-990-803-G00 (Sheet 2 of 16)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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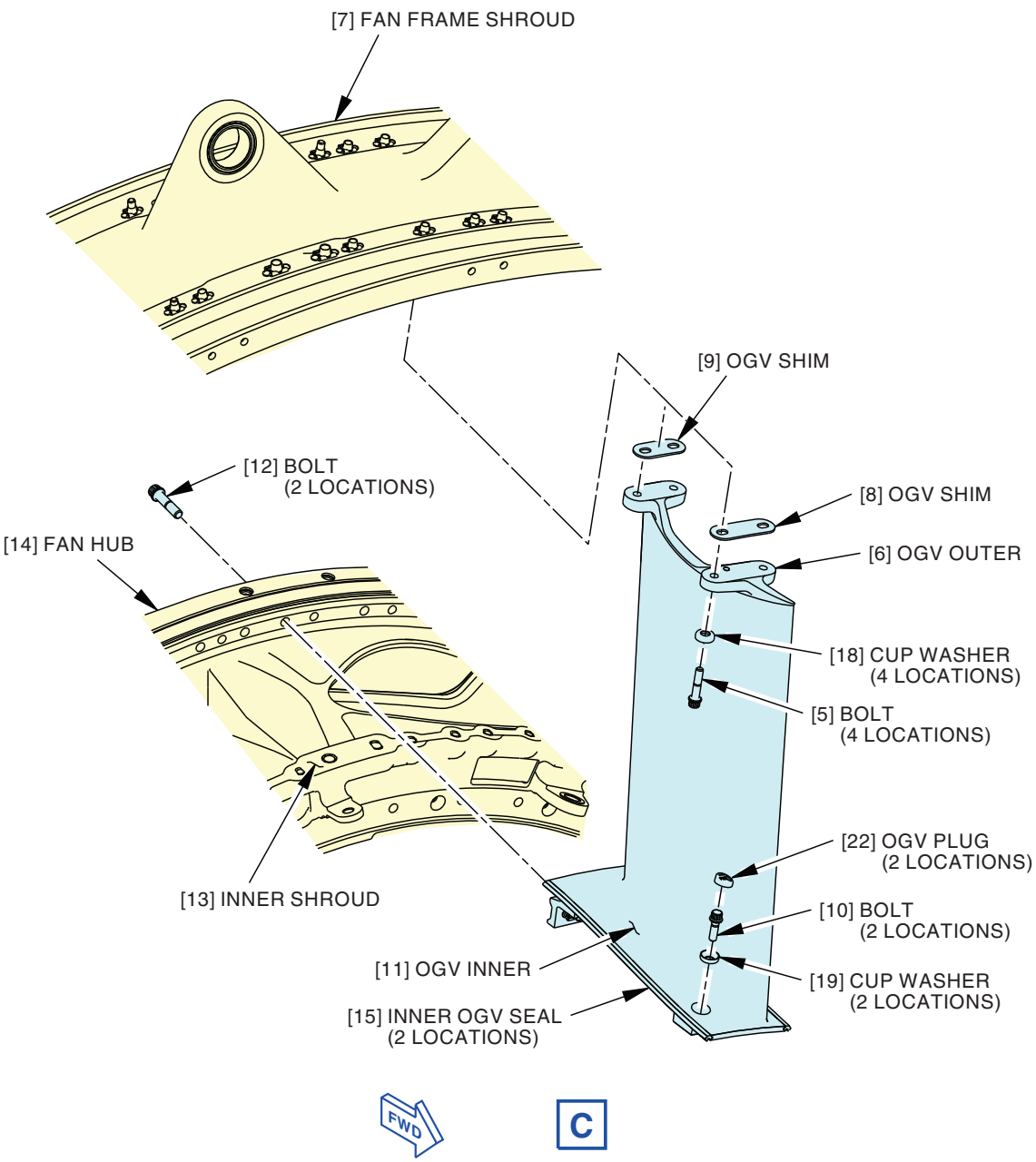
2434358 S0000564077_V4

Outlet Guide Vane Installation
Figure 401/72-21-03-990-803-G00 (Sheet 3 of 16)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

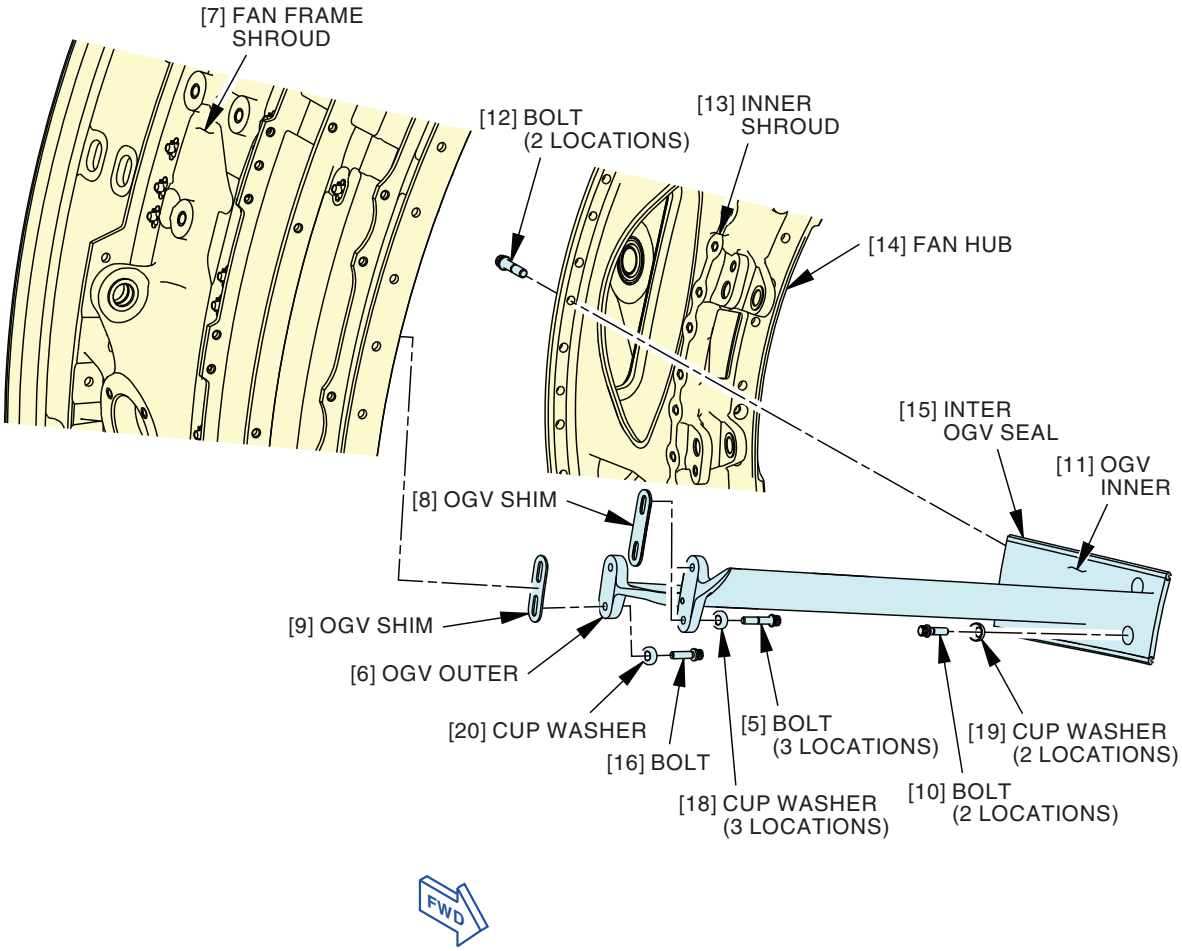


ICNLEAP1B722103D5882800008A00101

2840682 S0000662777_V1

Outlet Guide Vane Installation
Figure 401/72-21-03-990-803-G00 (Sheet 4 of 16)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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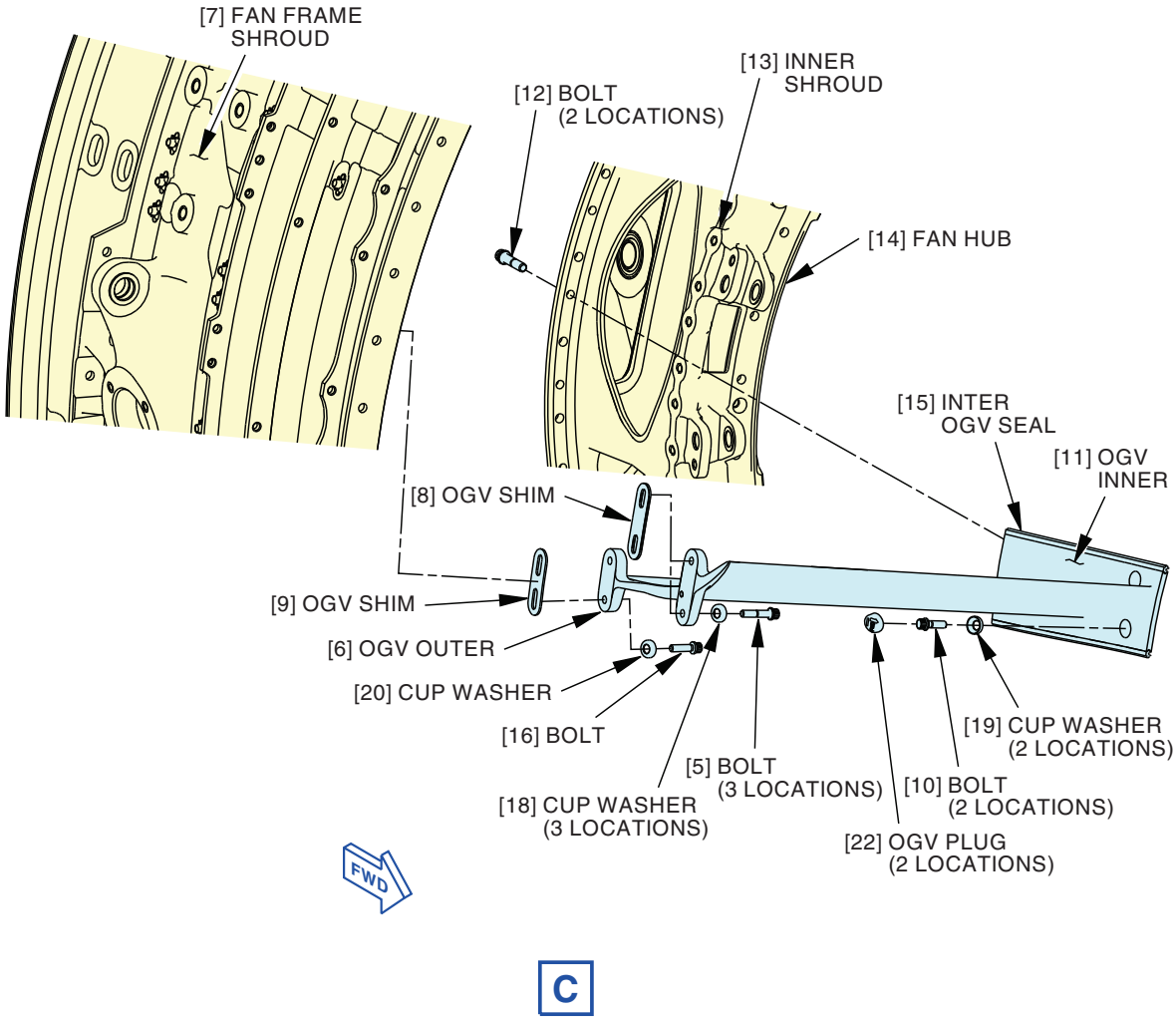
2533513 S0000600217_V4

Outlet Guide Vane Installation
Figure 401/72-21-03-990-803-G00 (Sheet 5 of 16)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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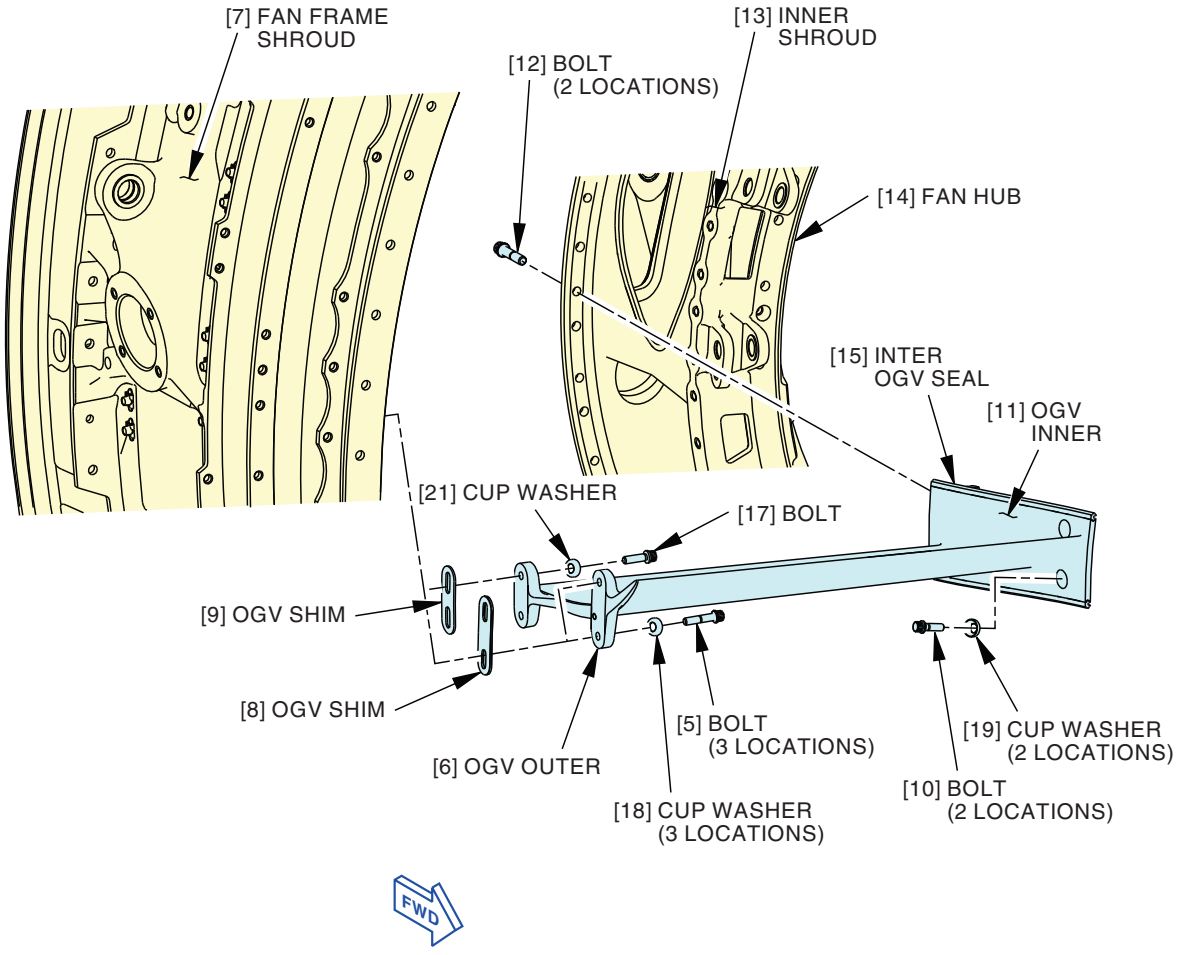
2840700 S0000662778_V1

Outlet Guide Vane Installation
Figure 401/72-21-03-990-803-G00 (Sheet 6 of 16)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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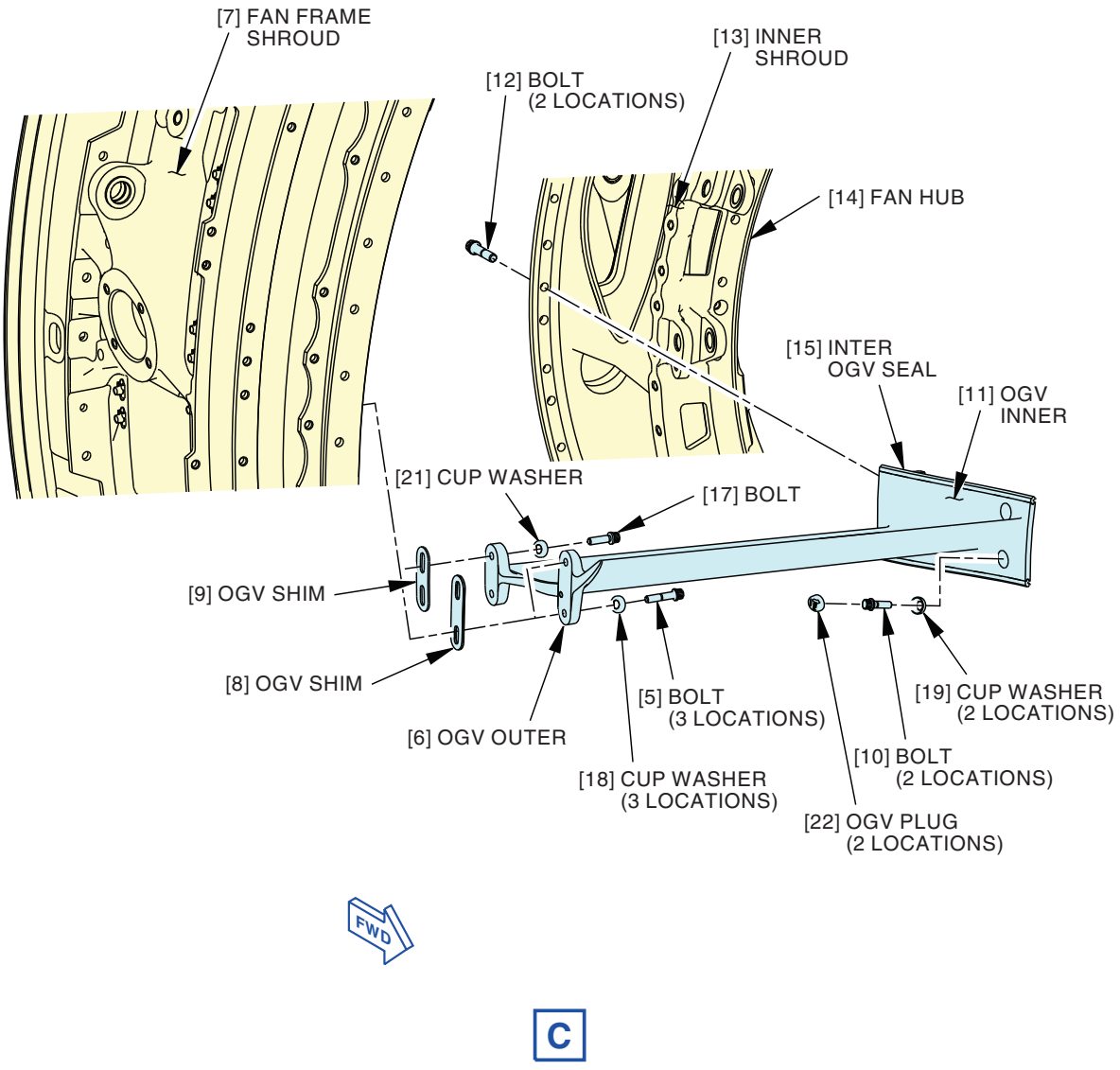
Outlet Guide Vane Installation
Figure 401/72-21-03-990-803-G00 (Sheet 7 of 16)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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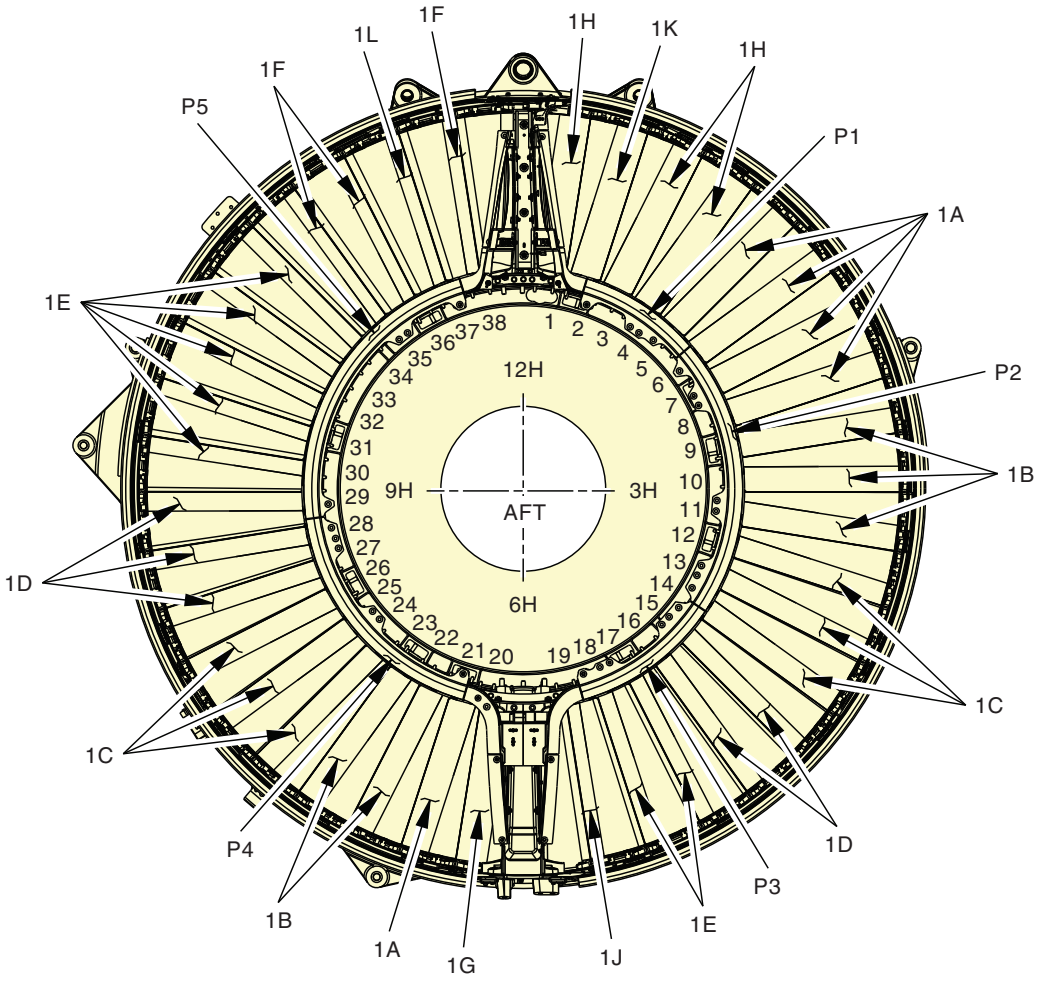
2840716 S0000662779_V1

Outlet Guide Vane Installation
Figure 401/72-21-03-990-803-G00 (Sheet 8 of 16)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



C

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2434360 S0000564078_V3

Outlet Guide Vane Installation
Figure 401/72-21-03-990-803-G00 (Sheet 9 of 16)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OGV NO.	OGV TYPE	FAN FRAME MODULE SHROUD SEGMENT
1	1H	P1
2	1K	P1
3	1H	P1
4	1H	P1
5	1A	P1
6	1A	P1, P2
7	1A	P2
8	1A	P2
9	1B	P2
10	1B	P2
11	1B	P2
12	1C	P2
13	1C	P2
14	1C	P2, P3
15	1D	P3
16	1D	P3
17	1E	P3
18	1E	P3
19	1J	P3
20	1G	P4
21	1A	P4
22	1B	P4
23	1B	P4
24	1C	P4
25	1C	P4
26	1C	P4
27	1D	P4
28	1D	P4
29	1D	P5
30	1E	P5
31	1E	P5
32	1E	P5
33	1E	P5
34	1E	P5
35	1F	P5
36	1F	P5
37	1L	P5
38	1F	P5

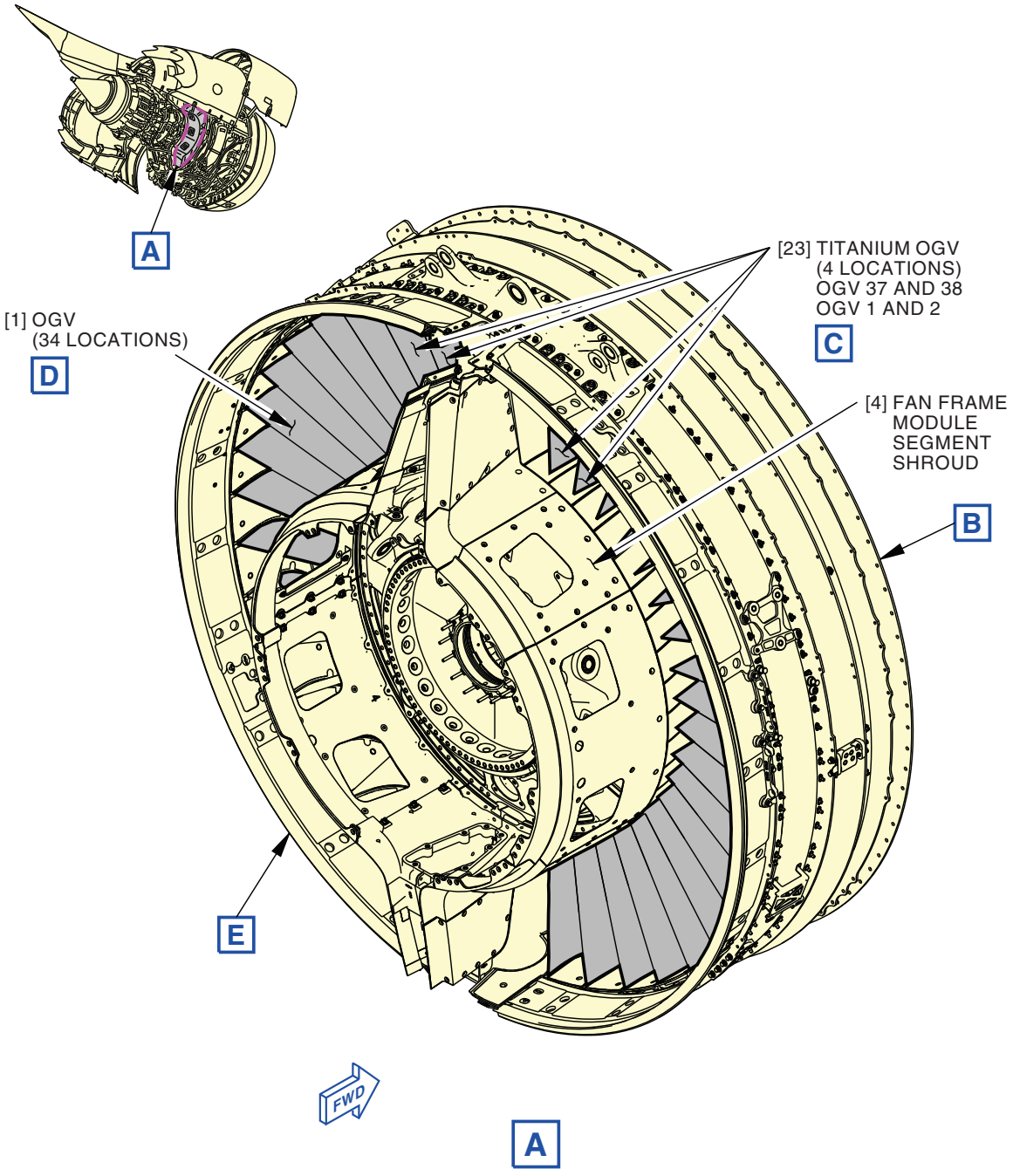
NOTE:

COLUMN REF IS USED TO IDENTIFY TOGETHER WITH SHEET 6; THE POSITION, TYPE, AND FAN FRAME MODULE SHROUD SEGMENT INVOLVED IN A REMOVAL/INSTALLATION OF AN OGV.

2434359 S0000564080_V4

Outlet Guide Vane Installation
Figure 401/72-21-03-990-803-G00 (Sheet 10 of 16)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

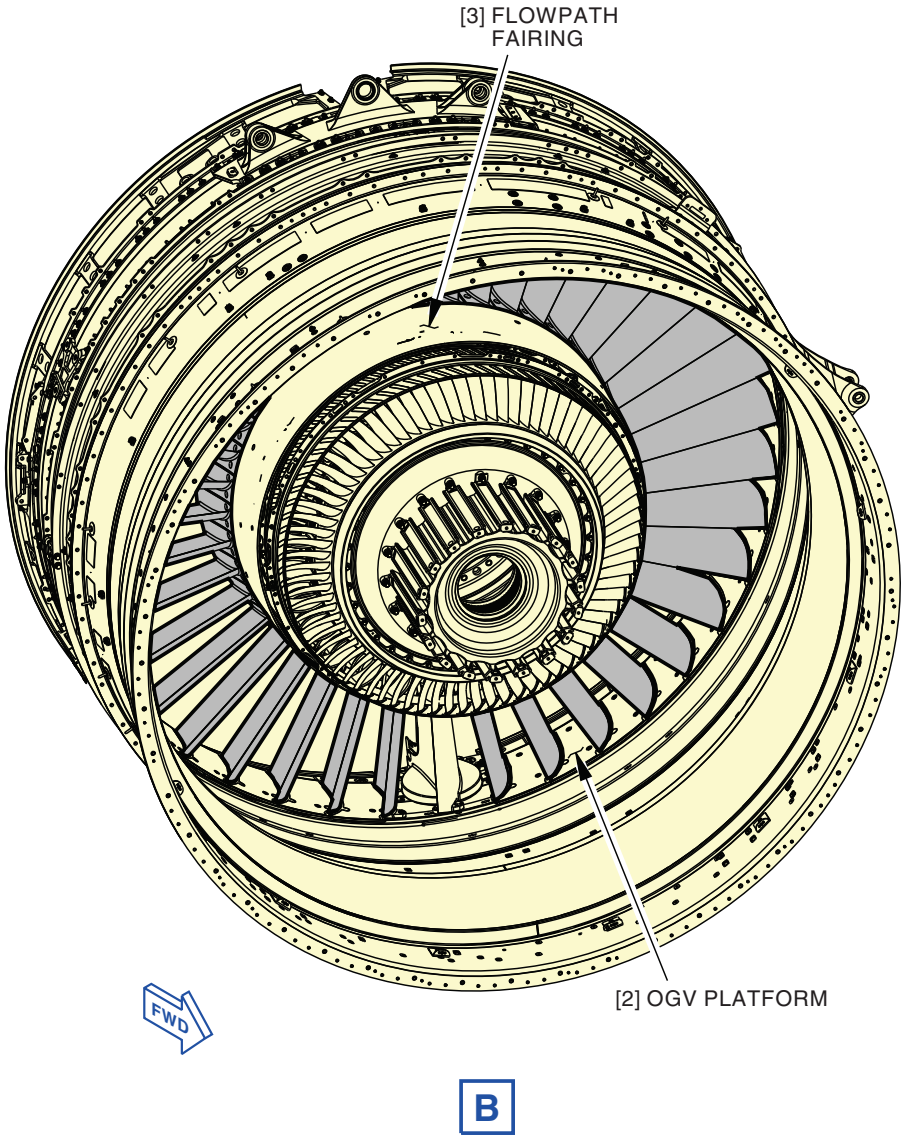


ICNLEAP1B722103D5882800011A00101

2840753 S0000662872_V1

Outlet Guide Vane Installation
Figure 401/72-21-03-990-803-G00 (Sheet 11 of 16)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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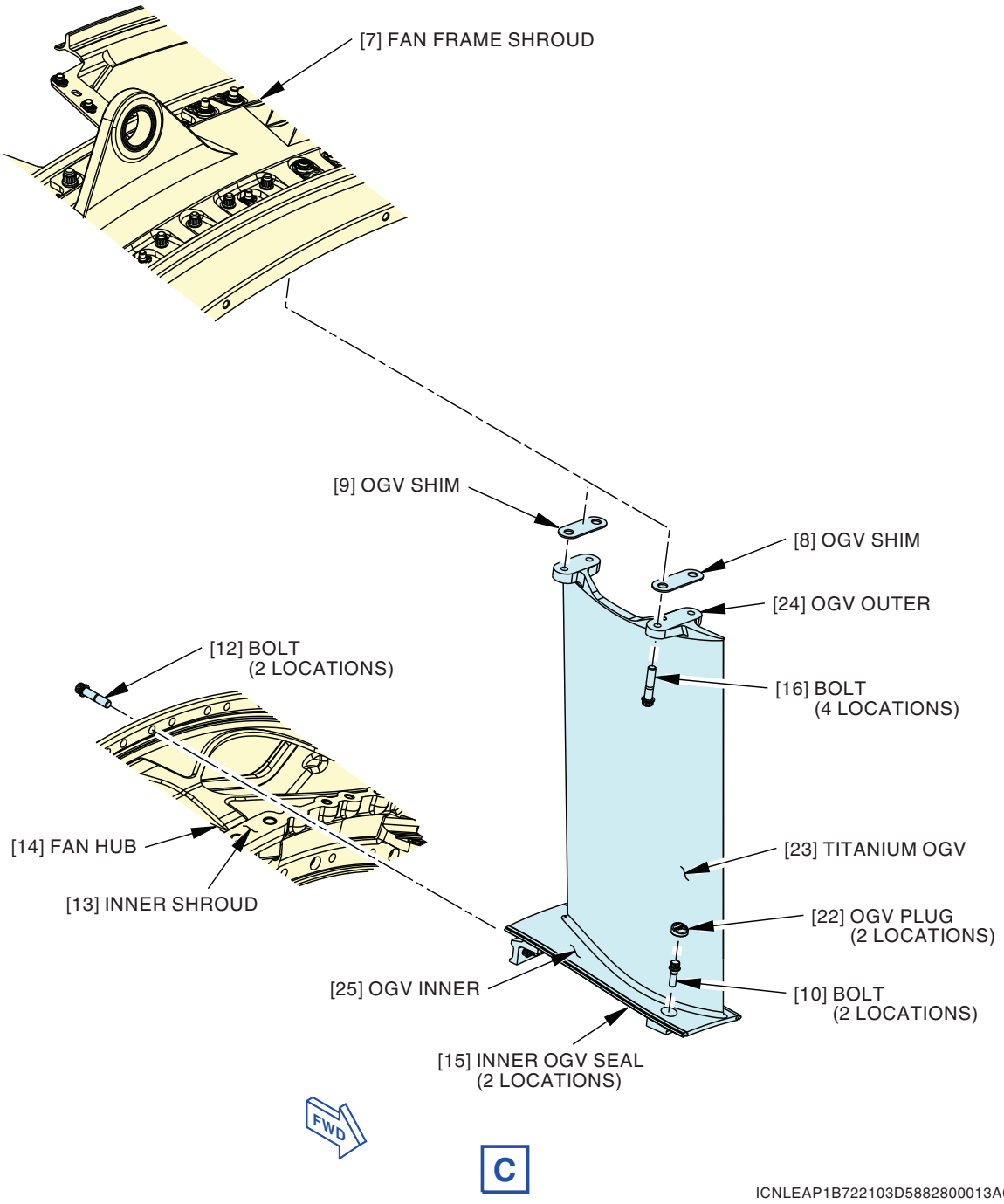
2840755 S0000662873_V1

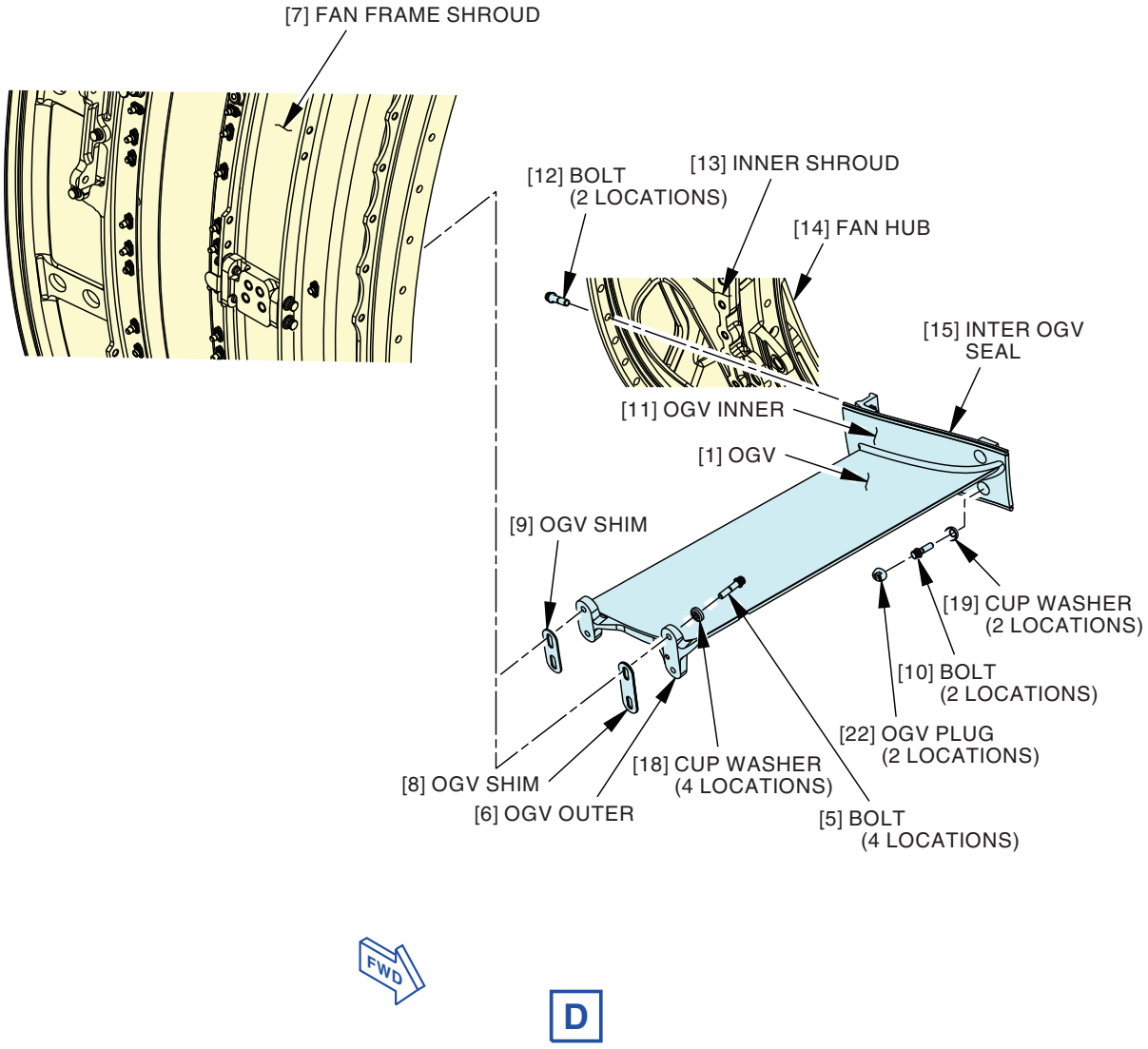
Outlet Guide Vane Installation
Figure 401/72-21-03-990-803-G00 (Sheet 12 of 16)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Outlet Guide Vane Installation
Figure 401/72-21-03-990-803-G00 (Sheet 13 of 16)

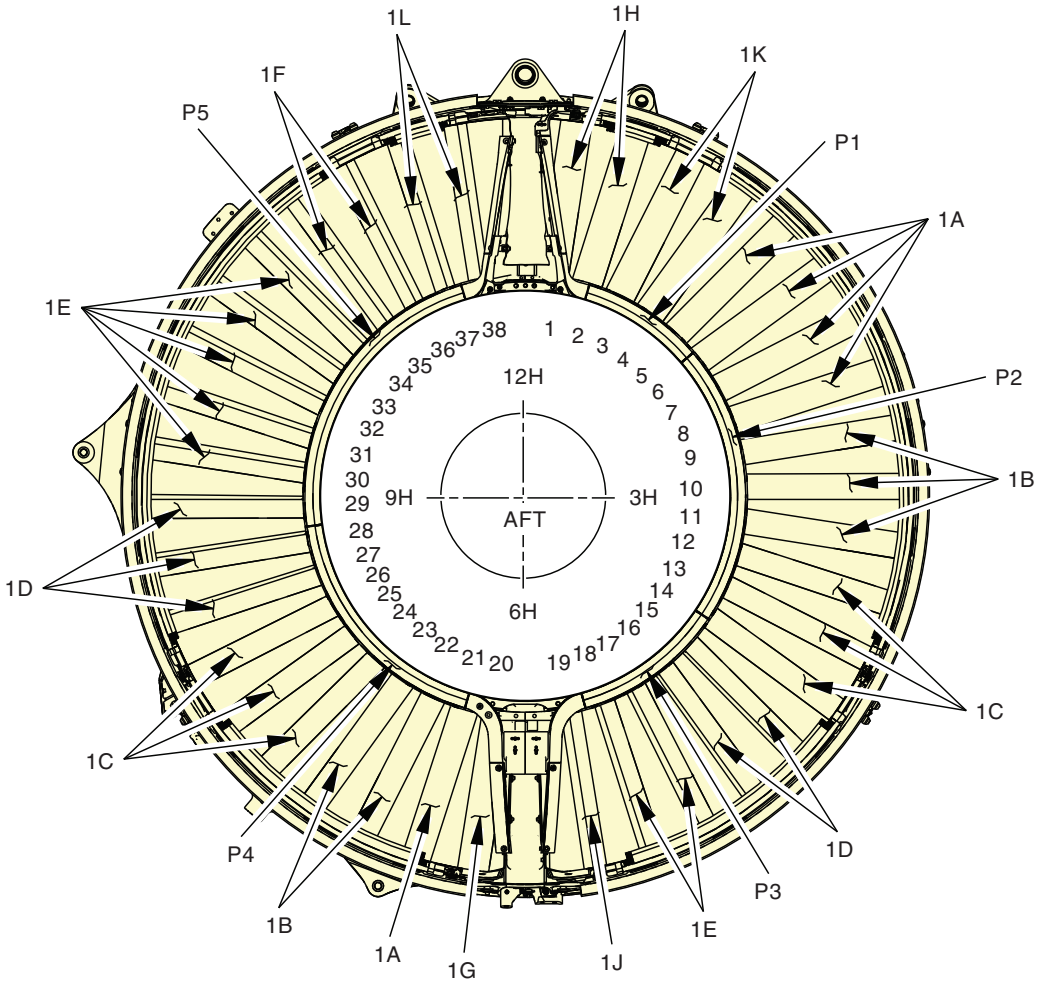
ECCN 9E991 BOEING PROPRIETARY - See title page for details



ICNLEAP1B722103D5882800014A00101
2840897 S0000662875_V1

Outlet Guide Vane Installation
Figure 401/72-21-03-990-803-G00 (Sheet 14 of 16)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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ICNLEAP1B722103D5882800015A00101
2840909 S0000662876_V1

Outlet Guide Vane Installation
Figure 401/72-21-03-990-803-G00 (Sheet 15 of 16)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OGV NO.	REP	FAN FRAME MODULE SEGMENT SHROUD
1	1H	P1
2	1H	P1
3	1K	P1
4	1K	P1
5	1A	P1
6	1A	P1, P2
7	1A	P2
8	1A	P2
9	1B	P2
10	1B	P2
11	1B	P2
12	1C	P2
13	1C	P2
14	1C	P2, P3
15	1D	P3
16	1D	P3
17	1E	P3
18	1E	P3
19	1J	P3
20	1G	P4
21	1A	P4
22	1B	P4
23	1B	P4
24	1C	P4
25	1C	P4
26	1C	P4
27	1D	P4
28	1D	P4
29	1D	P5
30	1E	P5
31	1E	P5
32	1E	P5
33	1E	P5
34	1E	P5
35	1F	P5
36	1F	P5
37	1L	P5
38	1L	P5

ICNLEAP1B722103D5882800016A00101
2840912 S0000662877_V1

Outlet Guide Vane Installation
Figure 401/72-21-03-990-803-G00 (Sheet 16 of 16)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 72-21-03-400-801-G00

3. Outlet Guide Vane Installation

(Figure 401)

A. General

(1) This task gives the instructions to install an outlet guide vane.

(2) The outlet guide vane is referred to as the OGV.

B. References

Reference	Title
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
72-21-09-400-801-G00	Outlet Guide Vane Platform Installation (P/B 401)
72-22-05-400-801-G00	Flowpath Fairing Installation (P/B 401)
72-23-04-400-801-G00	Fan Frame Module Segment Shrouds Installation (P/B 401)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
A50447	Compound - Jointing [CP2773], Corrosion Inhibiting, Chromate Free, Mastinox CA1010	
A50472 [CP2734]	Product - Silicone Elastomer Sealing	
D00672 [CP5070]	Vaseline - Pure Mineral	V V-P-236

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Outlet Guide Vane Installation

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0181

SUBTASK 72-21-03-420-001-G00

(1) Install the OGV [1] (No. 8, 26, and 31) in its position on the engine and do these steps:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0181 (Continued)

(a) Lubricate the two mating faces of the OGV shim [8] and OGV shim [9], threads of the three bolts [5], two bolts [10], two bolts [12], three cup washers [18], cup washer [20], two cup washers [19], and on the threads and under the head of the bolt [16] with jointing compound, A50447.

(b) Lubricate the two new inter OGV seals [15] with pure mineral vaseline, D00672 [CP5070].

(c) Install the two inter OGV seals [15] in the groove of the OGV inner [11].

(d) Install the two bolts [12] that attach the OGV inner [11] to the fan hub [14].

(e) Tighten the two bolts [12] to 201.8 in-lb (22.8 N·m) - 223 in-lb (25.2 N·m).

(f) Install the two bolts [10] and two cup washers [19] that attach the OGV inner [11] to the inner shroud [13].

(g) Tighten the two bolts [10] to 126.12 in-lb (14.25 N·m) - 139.39 in-lb (15.75 N·m).

(h) Measure the gaps between the OGV [1] and shroud to select the shim height.

(i) Install the OGV shim [8] and OGV shim [9] in its position on the OGV outer [6].

NOTE: The selected OGV shim must have the highest thickness possible but under the measured thickness of the gap.

(j) Install the three bolts [5] and three cup washers [18] that attach the OGV outer [6] to the fan frame shroud [7].

(k) Tighten the three bolts [5] to 126.12 in-lb (14.25 N·m) - 139.39 in-lb (15.75 N·m).

(l) Install the bolt [16] and cup washer [20] that attach the OGV outer [6] to the fan frame shroud [7].

(m) Tighten the bolt [16] to 109.25 in-lb (12.34 N·m) - 120.75 in-lb (13.64 N·m).

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0171

(n) Apply RTV 103 adhesive, A50472 [CP2734], in the counterbore of the two bolts [10] on the OGV inner [11].

1) Make sure that the adhesive is smooth along the duct profile.

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0171

(o) Apply RTV 103 adhesive, A50472 [CP2734], on the head of the two bolts [10].

NOTE: Do not fill in counterbore of the OGV inner [11] with RTV 103 adhesive, A50472 [CP2734].

(p) Install the two OGV plugs [22] with the arrow in direction of the front of the engine, on the OGV inner [11].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0181

SUBTASK 72-21-03-420-003-G00

(2) Install the titanium OGV [23] (No. 1, 2, 37 and 38) in its position on the engine and do these steps:

(a) Lubricate the threads of the two bolts [10] and two bolts [12] with jointing compound, A50447.

(b) Lubricate the two new inter OGV seals [15] with pure mineral vaseline, D00672 [CP5070].

(c) Install the two inter OGV seals [15] in the groove of the OGV inner [11].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0181 (Continued)

(d) Install the two bolts [12] that attach the OGV inner [25] to the fan hub [14].

(e) Tighten the two bolts [12] to 201.8 in-lb (22.8 N·m) - 223 in-lb (25.2 N·m).

(f) Install the two bolts [10] attach the OGV inner [25] to the inner shroud [13].

(g) Tighten the two bolts [10] to 126.12 in-lb (14.25 N·m) - 139.39 in-lb (15.75 N·m).

(h) Measure the gaps between the titanium OGV [23] and shroud to select the shim height.

(i) Install the OGV shim [8] and OGV shim [9] in its position on the OGV outer [24].

NOTE: The selected OGV shim must have the highest thickness possible but under the measured thickness of the gap.

(j) Install the four bolts [16] that attach the OGV outer [24] to the fan frame shroud [7].

(k) Tighten the four bolts [16] to 277.4 in-lb (31.34 N·m) - 306.6 in-lb (34.64 N·m).

(l) Apply RTV 103 adhesive, A50472 [CP2734], in the head of the two bolts [10].

NOTE: Do not fill in counterbore of the OGV inner [25] with RTV 103 adhesive, A50472 [CP2734].

(m) Install the two OGV plugs [22] with the arrow in direction of the front of the engine, on the OGV inner [25]

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0181

SUBTASK 72-21-03-420-002-G00

(3) Install the OGV [1] (No. 9 and 32) in its position on the engine and do these steps:

(a) Lubricate the two mating faces of the OGV shim [8] and OGV shim [9], the threads of the three bolts [5], bolt [17], two bolts [10], two bolts [12], three cup washers [18], cup washer [21], and two cup washers [19] with jointing compound, A50447.

(b) Lubricate the two new inter OGV seals [15] with pure mineral vaseline, D00672 [CP5070].

(c) Install the two inter OGV seals [15] in the groove of the OGV inner [11].

(d) Install the two bolts [12] that attach the OGV inner [11] to the fan hub [14].

(e) Tighten the two bolts [12] to 201.8 in-lb (22.8 N·m) - 223 in-lb (25.2 N·m).

(f) Install two bolts [10] and two cup washers [19] that attach the OGV inner [11] to the inner shroud [13].

(g) Tighten the two bolts [10] to 126.12 in-lb (14.25 N·m) - 139.39 in-lb (15.75 N·m).

(h) Measure the gaps between the OGV [1] and shroud to select the shim height.

(i) Install the OGV shim [8] and OGV shim [9] in its position on the OGV outer [6].

NOTE: The selected OGV shim must have the highest thickness possible but under the measured thickness of the gap.

(j) Install the three bolts [5] and three cup washers [18] that attach the OGV outer [6] to the fan frame shroud [7].

(k) Tighten the three bolts [5] to 126.12 in-lb (14.25 N·m) - 139.39 in-lb (15.75 N·m).

(l) Install the bolt [17] and cup washer [21] that attach the OGV outer [6] to the fan frame shroud [7].

(m) Tighten the bolt [17] to 109.25 in-lb (12.34 N·m) - 120.75 in-lb (13.64 N·m).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0171

(n) Apply RTV 103 adhesive, A50472 [CP2734], in the counterbore of the two bolts [10] on the OGV inner [11].

1) Make sure that the adhesive is smooth along the duct profile.

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0171

(o) Apply RTV 103 adhesive, A50472 [CP2734], on the head of the two bolts [10].

NOTE: Do not fill in counterbore of the OGV inner [11] with RTV 103 adhesive, A50472 [CP2734].

(p) Install the two OGV plugs [22] with the arrow in direction of the front of the engine, on the OGV inner [11].

PTF ALL

SUBTASK 72-21-03-410-002-G00

(4) Install the other OGVS [1] in its position on the engine and do these steps:

(a) Lubricate the two mating faces of the OGV shim [8] and OGV shim [9], threads of the four bolts [5], two bolts [10], two bolts [12], four cup washers [18], and two cup washers [19] with jointing compound, A50447.

(b) Lubricate the two new inter OGV seals [15] with pure mineral vaseline, D00672 [CP5070].

(c) Install the two inter OGV seals [15] in the groove of the OGV inner [11].

(d) Install the two bolts [12] that attach the OGV inner [11] to the fan hub [14].

(e) Tighten the two bolts [12] to 201.8 in-lb (22.8 N·m) - 223 in-lb (25.2 N·m).

(f) Install the two bolts [10] and two cup washers [19] that attach the OGV inner [11] to the inner shroud [13].

(g) Tighten the two bolts [10] to 126.12 in-lb (14.25 N·m) - 139.39 in-lb (15.75 N·m).

(h) Measure the gaps between the OGV [1] and shroud to select the shim height.

(i) Install the OGV shim [8] and OGV shim [9] in its position on the OGV outer [6].

NOTE: The selected OGV shim must have the highest thickness possible but under the measured thickness of the gap.

(j) Install the four bolts [5] and four cup washers [18] that attach the OGV outer [6] to the fan frame shroud [7].

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0181

(k) Tighten the four bolts [5] to 126.12 in-lb (14.25 N·m) - 139.39 in-lb (15.75 N·m).

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0181

(l) Tighten the four bolts [5] to 109.25 in-lb (12.34 N·m) - 120.75 in-lb (13.64 N·m).

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0171

(m) Apply RTV 103 adhesive, A50472 [CP2734], in the counterbore of the two bolts [10] on the OGV inner [11].

1) Make sure that the adhesive is smooth along the duct profile.

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PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0171 OR POST LEAP-1B SERVICE BULLETIN 72-0181

(n) Apply RTV 103 adhesive, A50472 [CP2734], on the head of the two bolts [10].

NOTE: Do not fill in counterbore of the OGV inner [11] with RTV 103 adhesive, A50472 [CP2734].

(o) Install the two OGV plugs [22] with the arrow in direction of the front of the engine, on the OGV inner [11].

PTF ALL

SUBTASK 72-21-03-410-003-G00

(5) Install the applicable fan frame module shroud segment [4] (TASK 72-23-04-400-801-G00).

SUBTASK 72-21-03-410-004-G00

(6) If removed, install the flowpath fairing [3] (TASK 72-22-05-400-801-G00).

SUBTASK 72-21-03-410-005-G00

(7) Install the OGV platform [2] (TASK 72-21-09-400-801-G00).

SUBTASK 72-21-03-160-001-G00

(8) Clean the work area and clear the tools and other items.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 72-21-03-410-006-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

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SUBTASK 72-21-03-865-002-G00

(2) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 72-21-03-080-001-G00

(3) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

I. Outlet Guide Vane Installation Test

SUBTASK 72-21-03-700-001-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

——— END OF TASK ———

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OUTLET GUIDE VANE - REPAIRS

1. General

A. This procedure has these tasks:

(1) Aluminum outlet guide vane leading and trailing edge blending repair

(2) Titanium outlet guide vane leading edge blending repair

(3) Outlet guide vane struts 1 and 5 leading edge blending repair

(4) Outlet guide vanes coating local repair

(5) Outlet guide vanes paint primer local repair

(6) Outlet guide vane two-lug nut plate/assembly gang channel nut replacement.

TASK 72-21-03-320-801-G00

2. Aluminum Outlet Guide Vane Leading and Trailing Edge Blending Repair

(Figure 801 and Figure 802)

A. General

(1) This task gives instructions to blend the Leading Edge (LE) and Trailing Edge (TE) of the aluminum Outlet Guide Vane (OGV) with the part removed.

B. References

Reference	Title
70-11-06-230-802-G00	Fluorescent Penetrant Inspection (P/B 201)
70-11-06-230-803-G00	Fluorescent Penetrant Inspection (General) (P/B 201)
70-21-23-100-801-G00	Hand Wipe Solvent Degreasing With Chemical Agent (P/B 701)
70-43-24-100-802-G00	Swab Etching Procedure and Cleaning (P/B 201)
70-43-42-300-802-G00	Blending Procedures - Hand Blending (P/B 201)
70-43-42-300-803-G00	Blending Procedures - Power Blending (P/B 201)
72-21-03-000-801-G00	Outlet Guide Vane Removal (P/B 401)
72-21-03-400-801-G00	Outlet Guide Vane Installation (P/B 401)

C. Consumable Materials

Reference	Description	Specification
B00676 [CP1041]	Alcohol - Isopropyl	
B50333 [CP2687]	Solvent - Degreasing, Diestone DLS	
G01043	Cloth - Lint-free	
G51478 [S1096]	Etchant - Aluminum Alloy	

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Prepare for the Repair

SUBTASK 72-21-03-010-005-G00

(1) Remove the OGV that you will blend repair (TASK 72-21-03-000-801-G00).

NOTE: The table below gives a list of the part numbers that are applicable to this repair.

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Table 801/72-21-03-993-804-G00 OGV Part Numbers

Part Number	Designation
364-068-111-0	Vane, Outlet Guide
364-068-211-0	Vane, Outlet Guide
364-068-311-0	Vane, Outlet Guide
364-068-411-0	Vane, Outlet Guide
364-068-511-0	Vane, Outlet Guide
364-068-611-0	Vane, Outlet Guide
364-068-711-0	Vane, Outlet Guide
364-068-811-0	Vane, Outlet Guide
364-068-911-0	Vane, Outlet Guide

SUBTASK 72-21-03-110-001-G00

WARNING ISOPROPYL ALCOHOL IS TOXIC AND FLAMMABLE. USE PERSONAL PROTECTION EQUIPMENT. USE IN A WELL-VENTILATED AREA.

(2) Clean the part you removed with lint-free cloth, G01043, moisten with alcohol, B00676 [CP1041], or degreasing solvent, B50333 [CP2687].

SUBTASK 72-21-03-220-001-G00

(3) Do a check of the repairability of the part.

(a) Make sure that you can do the blend repair in the reparable areas (Figure 801) (Table 802).

Table 802/72-21-03-993-805-G00 Dimensional Information

Description	Minimum In-Process Dimension	Maximum In-Process Dimension	Minimum Finish Dimension	Maximum Finish Dimension	Reference Dimension
LE					
For 5 OGV*[1]					
DIM. P				0.314 in. (8.0 mm)	
DIM. D			10 x DIM. P if DIM. P less than or equal to 0.237 in. (6.0 mm) 2.37 in. (60 mm) if DIM. P more than 0.237 in. (6.0 mm)		
Total length of blending LE				3.54 in. (90 mm)	
For All OGV					
DIM. P				0.039 in. (1.0 mm)	
DIM. D			10 x DIM. P		

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Table 802/72-21-03-993-805-G00 Dimensional Information (Continued)

Description	Minimum In-Process Dimension	Maximum In-Process Dimension	Minimum Finish Dimension	Maximum Finish Dimension	Reference Dimension
Dim. L				6 x DIM. P	
Dim. H			4.57 in. (116 mm)		
RADIUS R			4 x DIM. P		
RADIUS C			0.04 in. (1 mm)		
Total length of blending LE				1.18 in. (30 mm)	
TE					
DIM. S			4.73 in. (120 mm)		
DIM. U			30 x DIM. V		
RADIUS Y			4 x DIM. P		
RADIUS W			0.04 in. (1 mm)		
DIM. V				0.07 in. (2 mm)	
DIM. X			0.56 in. (14 mm)		
Total length of blending TE				4.72 in. (120 mm)	

*[1] Quantity maximum to respect for each engine OGV set (all OGV Wheel, OGV Struts not included).

(b) Make sure that you can do the blend repair within the limits (Figure 802) (Table 802).

(c) Make sure that the OGV was not blended more than 0.039 in. (1 mm) depth at the leading edge.

1) A blending more than 0.039 in. (1 mm) depth and another blending on the part in this repair is not permitted.

F. Outlet Guide Vane Blending Repair

SUBTASK 72-21-03-320-001-G00

WARNING MAKE SURE THAT YOU USE PROTECTION EQUIPMENT. METAL PARTICLES WHEN YOU DRILL OR MILL CAN CAUSE DAMAGE TO EQUIPMENT AND INJURY TO PERSONS.

CAUTION REMOVE MINIMUM MATERIAL WHEN YOU BLEND TO THE DEPTH OF THE OGV DAMAGE. DAMAGE TO THE ENGINE CAN OCCUR.

(1) Do the blend repair of the OGV as follows (TASK 70-43-42-300-803-G00 or TASK 70-43-42-300-802-G00):

(a) Make sure that the blending is in the limits.

1) Make sure that you do not do the blend repair on the fillets and bottom of the LE and TE. (Figure 801) (Table 802).

(b) Blend the paint and primer in adjacent areas to prepare the surface for etching (TASK 70-43-42-300-803-G00 or TASK 70-43-42-300-802-G00).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 72-21-03-160-002-G00

WARNING BEFORE YOU USE COMPRESSED AIR, PUT ON GOGGLES FOR EYE PROTECTION. DO NOT POINT THE NOZZLE AT OTHER PERSONNEL. IF YOU DO NOT OBEY THESE PRECAUTIONS, INJURIES TO PERSONNEL CAN OCCUR.

(2) Blow the blended areas with dry and oil free shop air.

SUBTASK 72-21-03-230-001-G00

(3) Do a Fluorescent Penetrant Inspection (FPI) as follows:

WARNING ISOPROPYL ALCOHOL IS TOXIC AND FLAMMABLE. USE PERSONAL PROTECTION EQUIPMENT. USE IN A WELL-VENTILATED AREA.

(a) Clean the part with lint-free cloth, G01043, moisten with alcohol, B00676 [CP1041], or degreasing solvent, B50333 [CP2687] (TASK 70-21-23-100-801-G00).

(b) If necessary, mask adjacent areas.

CAUTION DO NOT TOUCH THE REMAINING PAINT WITH THE ETCHANT AND FPI SOLUTIONS. IF YOU DO NOT OBEY, DAMAGE TO THE PART CAN OCCUR.

(c) Locally swab all the blended areas to be etched with aluminum alloy etchant, G51478 [S1096] (TASK 70-43-24-100-802-G00).

(d) Do an FPI of the blended areas (TASK 70-11-06-230-803-G00 or TASK 70-11-06-230-802-G00).

1) Do an FPI sensitivity level 2.

(e) Cracks that you find are not permitted.

(f) Indications of maximum 0.011 in. (0.3 mm) are permitted if they are not aligned.

NOTE: Linear indications are indications that are minimum four times longer than they are wide.

SUBTASK 72-21-03-370-001-G00

(4) Paint the blended areas (TASK 72-21-03-370-801-G00).

G. Put the Airplane Back to Its Usual Condition

SUBTASK 72-21-03-410-010-G00

(1) Install the OGV (TASK 72-21-03-400-801-G00).

END OF TASK

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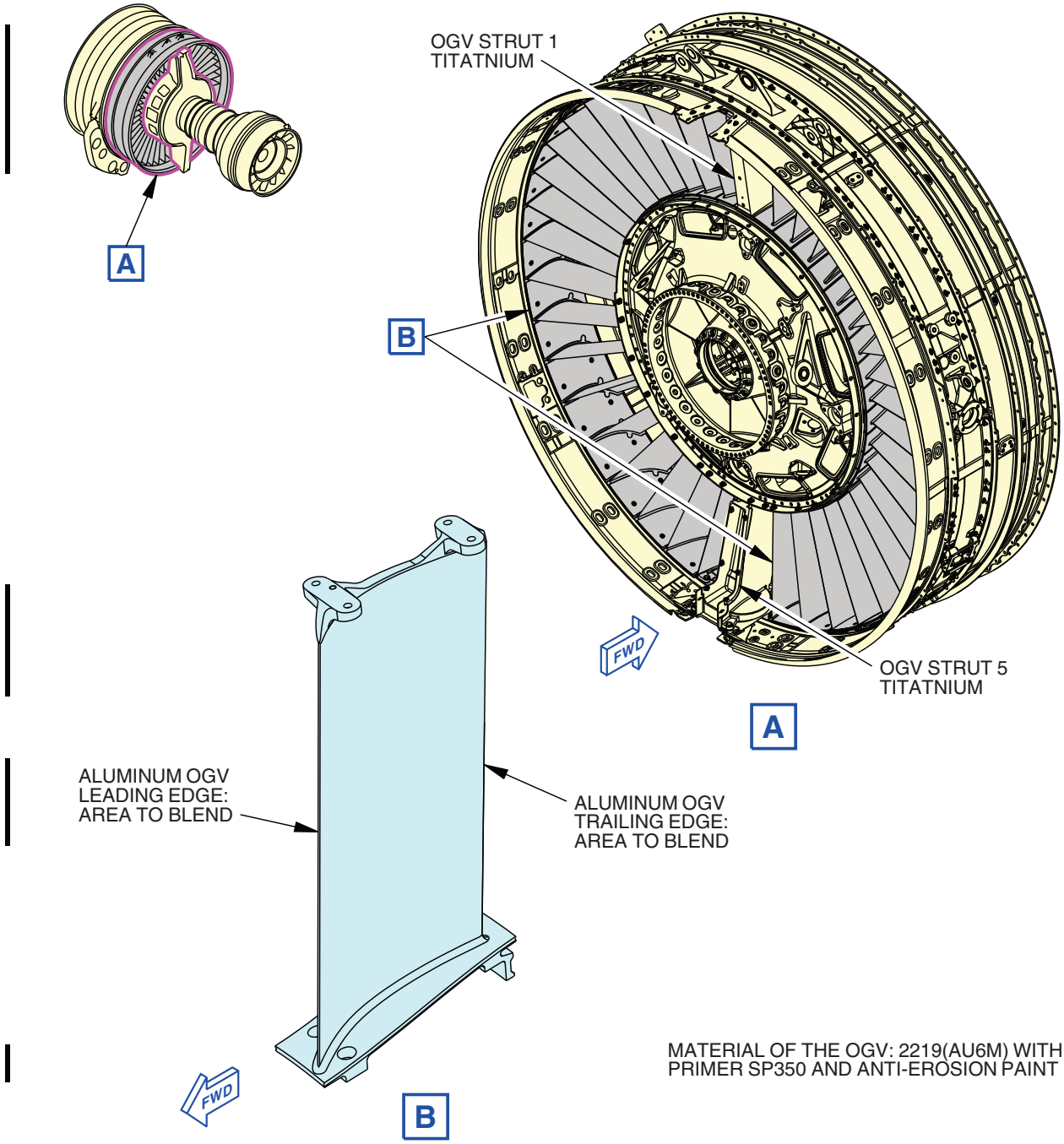

MATERIAL OF THE OGV: 2219(AU6M) WITH
PRIMER SP350 AND ANTI-EROSION PAINT

ICN-LEAP-1 B-722420-C-F0301-00001-A-001-01
2800294 S0000640787_V2

Aluminum Outlet Guide Vane Leading and Trailing Edge Blend Area
Figure 801/72-21-03-990-804-G00 (Sheet 1 of 2)

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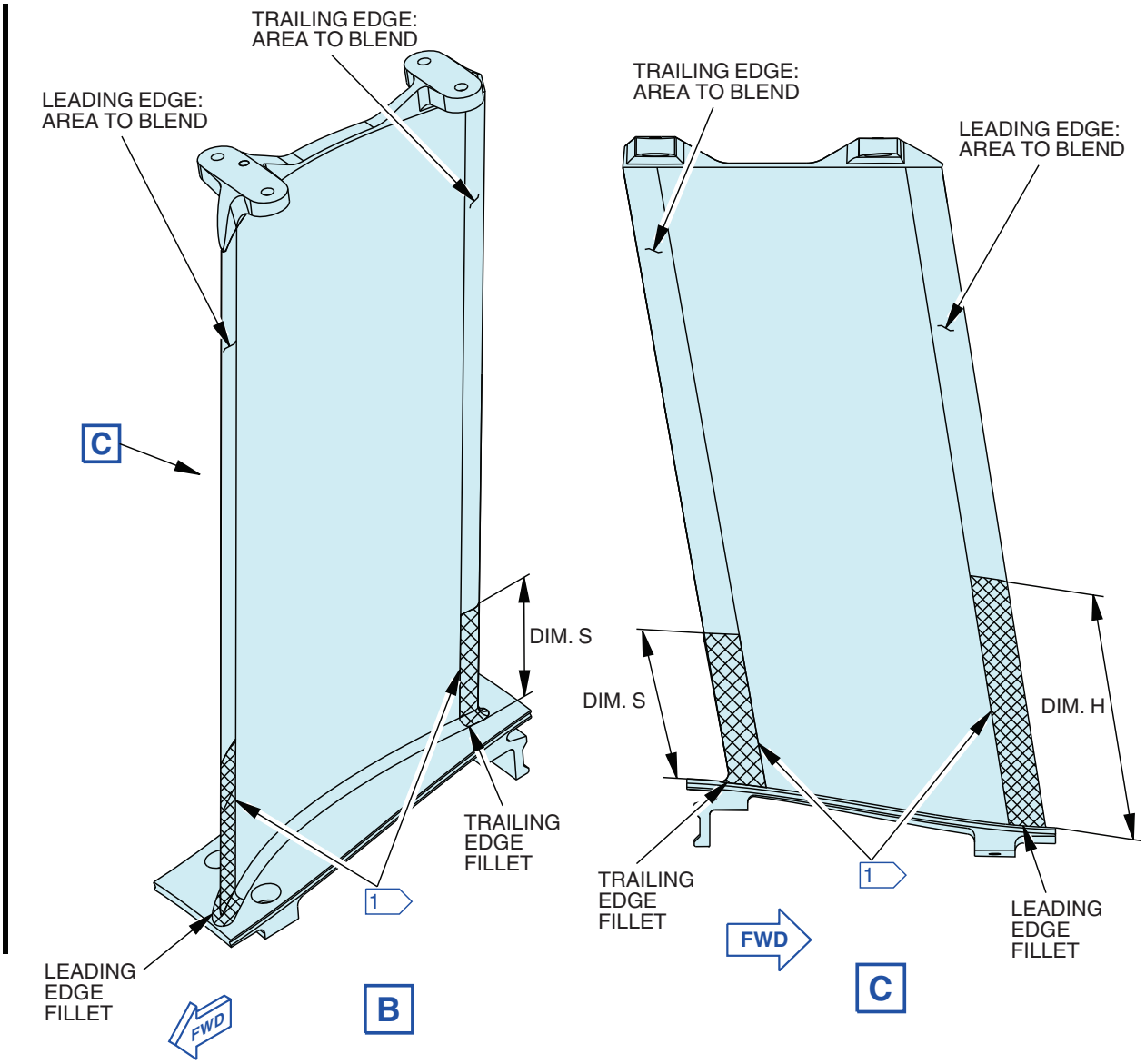
ALUMINUM OGV



NOTE:

1 BOTTOM OF THE LEADING AND TRAILING EDGE AREA NOT TO BLEND

BLENDING IS NOT PERMITTED IN THE BOTTOM OF THE LEADING AND TRAILING EDGE (LEADING EDGE AND TRAILING EDGE FILLETS INCLUDED).

ICN-LEAP-1B-722420-C-F0301-00002-A-001-01

2800663 S0000640791_V2

Aluminum Outlet Guide Vane Leading and Trailing Edge Blend Area
Figure 801/72-21-03-990-804-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details